

Static Holographic Boundary Theory (SHBT) Cold Fusion Reactor: Integrated Engineering Specification and Physical Transducer Model

Abstract—The Static Holographic Boundary Theory (SHBT) reactor specification couples optically driven Pd–Ir screening, nuclear–lattice dynamics, multiphase heat removal, superconducting magnetic excitation, and thermoelectric conversion. The verified $\text{Pd}_{0.9132}\text{Ir}_{0.0868}\text{D}_x$ Floquet calculation gives $U_{\text{eff}} = 352.4800$ eV from the 349.5000 eV benchmark and scale factor 1.008526466381 in a 15,625-dimensional dielectric matrix. The phonon channel has order 6.92×10^8 , $\Gamma_{\text{lat}} = 2.5 \times 10^{22} \text{ s}^{-1}$, $\Gamma_{\gamma} = 1.0 \times 10^{14} \text{ s}^{-1}$, and $B_{\text{lat}} = 0.999999996$. A 64-channel, 250 μm hydraulic-diameter cold plate is represented by a three-dimensional Eulerian–Eulerian RPI boiling model; the verified operating point is $T_h = 618.42$ K, $T_c = 301.88$ K, peak void fraction 0.162, pressure drop 42.8 kPa, and flow rate 4.85 L/min. At $P_{\text{thermal}} = 3093.4400$ W and 33.80% conversion, the TEG produces 1045.5800 W. The recovered superconducting magnetic drive consumes 368.45 W and auxiliaries consume 122.10 W, giving $P_{\text{net}} = +555.03$ W. GST optical self-healing, 64-byte-aligned zero-copy control storage, and all 50 simulation and hardware-in-the-loop verification gates form the definitive engineering baseline.

Index Terms—Acoustic energy harvesting, calorimetry, electron-phonon coupling, energy balance, nuclear screening, static holographic boundary theory, surface plasmon polaritons, thermoelectric generation.

I. INTRODUCTION & THEORETICAL TRANSDUCER MODEL

THE Static Holographic Boundary Theory (SHBT) reactor concept comprises a deuterium-loaded $\text{Pd}_{0.9132}\text{Ir}_{0.0868}$ active alloy, a prism-coupled optical interface, a magnetic drive, a thermal spreader, and energy-recovery elements. The revised metal target contains 85.347 wt% Pd and 14.653 wt% Ir. Composition-specific calculations for the earlier $\text{Pd}_{0.95}\text{Ir}_{0.05}$ material remain identified comparison cases rather than interchangeable inputs. Its transduction model couples optical interference, Floquet electronic screening, non-perturbative nuclear–lattice exchange, and dissipative boundary-state dynamics. Active phase feedback provides the control interface between thermal dephasing and the specified collective-rate enhancement. Electromagnetic response, quantum-state evolution, reaction kinetics, and energy accounting retain distinct normalization conventions throughout the coupled model.

The microscopic formulation uses the driven electronic response to construct the screened pair interaction, then evaluates nuclear and lattice transition operators in that environment. The verified screening shift is $U_{\text{eff}} = 352.4800$ eV at $r_d = 0.28$ Å, obtained from the 349.5000 eV benchmark through the factor 1.008526466381. Section I-C4 specifies the atomic-composition band comparison. The lattice channel transfers

$Q_{DD} = 23.84$ MeV non-radiatively into LO modes and their subsequent anharmonic decay products. Its energy-equivalent phonon order is 6.92×10^8 , and the verified two-channel branching fraction is 0.999999996. Sections I-I1 and II-K define the coherence product and joint-state dynamics. Section II-I retains microscopic matrix elements and metal-volume rate normalization without multiplying an already collective rate by independent g_{exp} or g_{coll} factors.

Assumption I.1 (Specification basis). This specification distinguishes prescribed boundary conditions and design requirements, model-dependent estimates, and experimentally characterized quantities. Numerical operating points are reference parameters or design requirements unless identified with independent measurements. Predictions apply within the stated constitutive domains, normalization conventions, and parameter assumptions. Electromagnetic boundary conditions and energy conservation constrain the dielectric, Franck–Condon, and energy-conversion models [26], [54], [63], [66]. The local screening shift, lattice branching fraction, collective-rate gain, and channel matrix elements have separate definitions. The boundary constitutive laws and feedback setpoints define a phenomenological operating model [86], [87]. Its empirical qualification requires material-specific characterization of the interactions and propagation of parameter uncertainty into the predicted observables. Confidence intervals, coverage intervals, and engineering tolerances retain their respective local definitions.

a) Definitive thermal-hydraulic and electrical allocation: The closed-loop allocation in Section II-C2 assigns $P_{\text{thermal}} = 3093.4400$ W to the TEG hot-side duty. Dual-stage skutterudite/half-Heusler conversion at 33.80% gives $P_{\text{TEG}} = 1045.5800$ W. The thin-film MgB_2 magnetic excitation and SiC recovery chain has a net electrical demand of $P_{\text{drive,net}} = 368.45$ W, and the consolidated auxiliary demand is $P_{\text{aux}} = 122.10$ W. Equation (213) therefore gives $P_{\text{net}} = +555.03$ W. The definitive thermal boundary is the 64-channel Eulerian–Eulerian cold plate specified in Section IX; earlier single-phase pump and partial-loop compressor calculations are retained only as historical component comparisons.

The earlier compressor-free electrical comparison is defined at the thermoelectric output bus, with support loads accounted

for separately [77], [79], [80]:

$$\begin{aligned}
 P_{\text{th,ref}} &= 2911.40 \text{ W}, \\
 P_{\text{TEG}} &= 418.43 \text{ W}, \quad P_{\text{chiller}} = 0, \\
 P_{\text{support}} &= 116.67 + 120.00 + 150.00 \\
 &= 386.67 \text{ W}, \\
 P_{\text{net}} &= 418.43 - 386.67 = +31.76 \text{ W}, \\
 V_{\text{bus}} &= 48 \text{ V DC}.
 \end{aligned} \tag{1}$$

Equation (1) preserves the earlier bus allocation, not the revised optimized operating ledger in Table XI. The corresponding bus conversion efficiency is $418.43/2911.40 = 0.1437212$. Optical, RF, and conversion losses must remain within the stated support and thermal boundaries rather than being omitted or counted twice.

b) Coupled numerical operating point: The verified screening estimate and band inputs are collected in Table II. The coupled Floquet screening solver retains 625 spatial harmonics and 25 temporal sidebands, giving 15,625 coupled indices. The normalization from 349.5000 eV to 352.4800 eV uses the scale factor 1.008526466381 in Eq. (40). The lattice and radiative rates are 2.5×10^{22} and $1.0 \times 10^{14} \text{ s}^{-1}$, giving $B_{\text{lat}}^{(2)} = 0.999999996$. Table LXXXII records strict convergence over the residual sequence $10^{-4}, 10^{-6}, 10^{-8}$, finite values, and zero singularity warnings. The screening field is recalculated at 100 Hz from the Soret-coupled concentration state. The earlier $\text{Bi}_2\text{Te}_3/\text{PbTe}$ generator comparison uses $T_h = 623.15 \text{ K}$ and $T_c = 353.15 \text{ K}$; the revised skutterudite/half-Heusler allocation instead uses 623.15/303.15 K. The 550/310 K pair is a third comparison. The Carnot efficiency for the earlier 623.15/353.15 K pair is

$$\eta_C = 1 - 353.15/623.15 = 0.43328252. \tag{2}$$

Thermal and electrical powers refer to distinct control-volume boundaries; their relationship follows from the energy balance in Section II. The revised parameter set and retained comparison cases are distinguished in Table LXXXV. Alternative temperatures, footprints, response models, and state-space realizations describe separate comparison cases rather than simultaneous operating conditions. The operating temperature is 350 °C (623.15 K).

c) Engineering implementation: The reactor engineering calculations and foundational SHBT boundary mathematics have distinct source records [108]. The numerical descriptions cover partial-pivoted LU, three-state RK4 master-equation propagation, and the partial-loop BOP map. Section X-A and Appendix B describe binary64 arithmetic, reusable working storage, data-transfer allocations, and assigned heat input. Contact-mechanics, transport, and telemetry requirements appear in Section X-H.

d) Numerical precision and acceptance values: Tables III, V, X, and XLII specify the calculated coupling, screening, interface-flux, and balanced-reduction operating points. Decimal presentation suppresses binary64 conversion artifacts without changing the physical precision of the tabulated quantities. IEEE 754 binary64 remains the engineering arithmetic for well-conditioned matrix and transport evaluations; precision-sensitive boundary-state and quadrature calculations require

a 512-bit reference path. The native Rust scalar Floquet benchmark gives $s_F \simeq 3.67250764167$ with absolute precision difference $\delta_F \simeq 3.56655 \times 10^{-16}$, as defined in Eqs. (46)–(47). Table XC collects the corresponding domain and diagnostic acceptance values.

e) Boundary coordinates and arithmetic resolution: The boundary model assigns a zero-dimensional topological anchor as a coordinate reference, not a finite reaction volume. Let $\mathbf{r}_*$ denote this anchor, $\mathbf{R}^\top \mathbf{R} = I$, and $\mathbf{t}$ a translation. The rigid-coordinate map and its numerical consistency criterion are

$$\begin{aligned}
 \mathbf{r}' &= \mathbf{R}(\mathbf{r} - \mathbf{r}_*) + \mathbf{t}, \\
 \max_i \frac{\|\mathbf{R}^\top(\mathbf{r}'_i - \mathbf{t}) + \mathbf{r}_* - \mathbf{r}_i\|_2}{L_*} &< 10^{-12}.
 \end{aligned} \tag{3}$$

Here $L_* > 0$ is a fixed coordinate normalization length. A rigid transformation preserves distances, not the untransformed components of each position vector. Resolving an assumed relative boundary scale $1/N \simeq 10^{-122}$ near a unit-valued state requires more than 405 significand bits; 512-bit arithmetic provides additional rounding margin. Binary64 can represent numbers of magnitude 10^{-122} , but cannot resolve their addition to unity. Physical significance of that scale remains a model assumption. High-precision constants, quadrature nodes, and initial states shall be constructed at the working precision rather than promoted from already rounded binary64 values.

f) Coupled physical constraints: The coupled model uses common definitions of reaction volume, state normalization, energy flow, and measurement uncertainty. Section XII specifies the associated physical and engineering constraints.

g) Comparison-model scope: Reduced dielectric, kinetic, and engineering models [70] use their stated constitutive assumptions and normalization conventions.

h) Scalar reference conventions: Table I specifies the verified screening, Floquet, phonon, and closed-loop power parameters.

The production architecture calls for 512-bit SIMD-capable screening kernels; Appendix B specifies the arithmetic and allocation scopes.

A. Optical Beat Frequency and Energy Scale

For collinearly polarized coherent pumps with intensities I_1 and I_2 , optical-cycle averaging leaves the interference term

$$\begin{aligned}
 I(\mathbf{r}, t) &= I_1 + I_2 + 2\sqrt{I_1 I_2} \cos[\Delta \mathbf{k} \cdot \mathbf{r} - \Delta \omega t + \phi], \\
 \Delta \omega &= \omega_1 - \omega_2, \quad \Delta \mathbf{k} = \mathbf{k}_1 - \mathbf{k}_2,
 \end{aligned} \tag{4}$$

where ϕ is the relative phase. With vacuum wavelengths $\lambda_1 = 785.0 \text{ nm}$ and $\lambda_2 = 802.5 \text{ nm}$ and $c = 299\,792\,458 \text{ m s}^{-1}$,

$$\begin{aligned}
 f_1 &= \frac{c}{\lambda_1} = 381.9012204 \text{ THz}, \\
 f_2 &= \frac{c}{\lambda_2} = 373.5731564 \text{ THz}, \\
 f_{\text{beat}} &= c \left| \frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right| \\
 &= 8.32806399587 \text{ THz} \approx 8.328064 \text{ THz}.
 \end{aligned} \tag{5}$$

The specified 8.328064 THz is the rounded difference frequency of the optical intensity modulation; calculations retain

TABLE I
SCALAR REFERENCE REGISTER FOR THE REVISED ELECTRONIC AND BOP ALLOCATION

Quantity	Reported display value	Definition
U_{eff}	352.4800 eV	Effective screening shift at $r_d = 0.28 \text{ \AA}$; Eq. (44).
$U_{\text{benchmark}}; \Xi; r_U$	349.5000 eV; 1.008526466381; 0.0000 eV	Unscaled benchmark, normalization factor, and displayed screening residual.
$N_G; N_T; N_{GN_T}$	625; 25; 15,625	Spatial harmonics, temporal sidebands, and coupled Floquet screening dimension.
$N_q; \Gamma_{\text{lat}}; \Gamma_\gamma; B_{\text{lat}}$	6.92×10^8 ; $2.5 \times 10^{22} \text{ s}^{-1}$; $1.0 \times 10^{14} \text{ s}^{-1}$; 0.999999996	Verified phonon order, lattice and gamma rates, and two-channel branching fraction.
$C_{\text{coh,rep}}$	2.0300×10^{-11}	Rounded product of the specified mass, geometry, and boundary factors; Eq. (146).
$N(E_F)$	$6.216120 \times 10^{47} \text{ states/(J m}^3\text{)}$	Reported DOS; the independently recomputed DOS is distinguished in Table II.
$m^*; m^*/m_e$	$5.398412 \times 10^{-30} \text{ kg}$; 5.926210	Reported isotropic band scalars.
$k_{\text{TF}}; \lambda_{\text{TF}}$	$4.245203 \times 10^{10} \text{ m}^{-1}$; 0.23556 $\text{\AA}$	Reported screening-vector and length displays.
$N_{\text{ch}}; D_h; \Delta P; \dot{V}$	64; 250 μm ; 42.8 kPa; 4.85 L/min	Definitive three-dimensional Eulerian–Eulerian RPI cold-plate operating point.
P_{thermal}	3093.4400 W	2911.40 W nuclear-heat normalization plus 182.04 W absorbed optical power.
$P_{\text{TEG,elec}}$	1045.5800 W	Electrical output at 33.80% conversion efficiency.
$P_{\text{thermal,rejected}}$	2047.8600 W	TEG cold-side heat rejection; $P_{\text{thermal}} - P_{\text{TEG,elec}}$.
$P_{\text{drive,net}}; P_{\text{aux}}$	368.45 W; 122.10 W	Recovered magnetic-drive demand and consolidated balance-of-plant auxiliary demand.
P_{net}	+555.03 W	Definitive net plant electrical output; Eq. (213).

the wavelength-derived value. Resonant lattice excitation requires a material mode at this frequency, nonzero optical–lattice coupling, suitable spatial overlap, and a relative pump linewidth compatible with the mode linewidth.

Using $h = 4.135667696 \times 10^{-15} \text{ eV s}$, the nominal beat gives

$$\begin{aligned} \omega_{\text{beat}} &= 2\pi f_{\text{beat}} \approx 5.23267693361 \times 10^{13} \text{ rad s}^{-1} \\ &\approx 5.232677 \times 10^{13} \text{ rad s}^{-1}, \\ E_{\text{LO}} &= \hbar \omega_{\text{beat}} = h f_{\text{beat}} \\ &\approx 0.03444210524 \text{ eV} \approx 34.442 \text{ meV}. \end{aligned} \quad (6)$$

Frequency calibration depends on the measured wavelength difference. For comparison, an exact setpoint of 8.33 THz corresponds to 34.450 meV. With $\lambda_1 = 785.0 \text{ nm}$ fixed, the corresponding second wavelength is

$$\lambda_2 = \left(\lambda_1^{-1} - \frac{8.33 \times 10^{12} \text{ s}^{-1}}{c} \right)^{-1} = 802.504159 \text{ nm}. \quad (7)$$

B. Kretschmann Coupling and SPP Dispersion

Consider a planar, nonmagnetic interface between a metal with complex relative permittivity $\varepsilon_m(\omega)$ and a dielectric with relative permittivity $\varepsilon_d(\omega)$. A transverse-magnetic surface mode proportional to $\exp(ik_x x - i\omega t)$ decays away from the interface with constants

$$\kappa_j^2 = k_x^2 - \varepsilon_j \frac{\omega^2}{c^2}, \quad \text{Re}(\kappa_j) > 0, \quad j \in \{m, d\}. \quad (8)$$

Continuity of the tangential electric and magnetic fields gives

$$\frac{\kappa_m}{\varepsilon_m} + \frac{\kappa_d}{\varepsilon_d} = 0. \quad (9)$$

Squaring this condition and eliminating the decay constants yields

$$\begin{aligned} \frac{k_x^2 - \varepsilon_m \omega^2 / c^2}{\varepsilon_m^2} &= \frac{k_x^2 - \varepsilon_d \omega^2 / c^2}{\varepsilon_d^2}, \\ k_{\text{spp}}(\omega) &= \frac{\omega}{c} \sqrt{\frac{\varepsilon_m(\omega) \varepsilon_d(\omega)}{\varepsilon_m(\omega) + \varepsilon_d(\omega)}}. \end{aligned} \quad (10)$$

The physical root must also satisfy the original, unsquared boundary condition and passive propagation. For a lossless positive dielectric, a bound mode requires $\varepsilon_m < -\varepsilon_d$.

A prism of relative permittivity ε_p supplies the in-plane momentum needed for approximate Kretschmann phase matching at each optical pump frequency:

$$\frac{\omega_j}{c} \sqrt{\varepsilon_p(\omega_j)} \sin \theta_j \approx \text{Re } k_{\text{spp}}(\omega_j), \quad j \in \{1, 2\}. \quad (11)$$

Equation (10) is the semi-infinite-interface limit. A finite lossy film requires a multilayer electromagnetic calculation; a reflectance minimum need not coincide with the surface-mode optimum [16]. Reference interface parameters are $\varepsilon_d = 1.0006$, $\theta_{\text{inc}} \approx 43.85^\circ$, and field decay depth $\delta_z \approx 12.4 \text{ nm}$. The multilayer solution shall determine their jointly admissible values using the operating gas density, film thickness, and measured complex permittivities. The specified field and propagation observables are

$$\delta_z = \frac{1}{\text{Re } \kappa_d}, \quad L_{\text{spp}} = \frac{1}{2 \text{Im } k_{\text{spp}}}, \quad \mathcal{F}_E = \left| \frac{E_{\text{spp}}}{E_{\text{inc}}} \right|^2, \quad (12)$$

where δ_z is a field-amplitude decay length and L_{spp} is an intensity propagation length. Matching the difference wavevector to a lattice mode is a separate condition from matching either optical carrier to an SPP.

The monolithic Option B geometry, complex optical constants, Fourier-modal boundary-value problem, and dual-pump field normalization are specified in Section II-A. The electronic and lattice interactions use those electromagnetic fields without identifying a field-intensity enhancement with a nuclear-rate multiplier.

C. Microscopic Theory and Floquet Dielectric Solver

1) *Microscopic Alloy Parameters and Screening Scales:* For $\text{Pd}_{0.9132}\text{Ir}_{0.0868}\text{D}_x$, let $y = 0.0868$ denote the iridium atomic fraction, distinct from the deuterium-loading subscript x . The density-interpolated electronic reference uses a spin-inclusive density of states per metal atom,

$$g(E_F; y) = 1.32 + 1.62y \\ = 1.460616 \simeq 1.4606 \text{ states/(eV atom)}, \quad (13)$$

$$n_{\text{atom}} = (1 - y)n_{\text{Pd}} + yn_{\text{Ir}} \\ = 6.94756 \times 10^{28} \text{ m}^{-3} \\ \simeq 6.95 \times 10^{28} \text{ m}^{-3}, \quad (14)$$

$$\tilde{g}(E_F) = \frac{g(E_F; y)n_{\text{atom}}}{1.6022 \times 10^{-19} \text{ J/eV}} \\ \simeq 6.336 \times 10^{47} \text{ J}^{-1} \text{ m}^{-3}. \quad (15)$$

Here $n_{\text{Pd}} = 6.8 \times 10^{28} \text{ m}^{-3}$ and $n_{\text{Ir}} = 8.5 \times 10^{28} \text{ m}^{-3}$; the last DOS value uses the rounded density and per-atom DOS. This density interpolation is distinct from the earlier fcc-cell reference in Eq. (41).

For an electrostatic perturbation ϕ , the long-wavelength response $\delta n = \tilde{g}(E_F)e\phi$ and Poisson equation give

$$(\nabla^2 - \kappa_{\text{TF,SI}}^2)\phi = -\rho_{\text{ext}}/\epsilon_0, \\ \kappa_{\text{TF,SI}}^2 = e^2\tilde{g}(E_F)/\epsilon_0, \quad (16)$$

$$\lambda_{\text{TF,SI}} = \sqrt{\frac{\epsilon_0}{e^2\tilde{g}(E_F)}} \simeq 0.2333 \text{ \AA}. \quad (17)$$

The selected screening model instead adopts the π -rescaled convention

$$\lambda_{\text{TF}}^{(\pi)} = \sqrt{\frac{\epsilon_0\pi}{e^2\tilde{g}(E_F)}} = \sqrt{\pi}\lambda_{\text{TF,SI}}, \quad (18)$$

$$\lambda_{\text{TF}} = 0.4137 \text{ \AA} = 4.137 \times 10^{-11} \text{ m}, \\ \kappa_{\text{TF}} = \lambda_{\text{TF}}^{-1}. \quad (19)$$

The factor π defines the selected screening-length convention, with nominal parameter $\lambda_{\text{TF}} = 0.4137 \text{ \AA}$.

The remaining nominal microscopic inputs are

$$n_e \simeq 1.01 \times 10^{29} \text{ m}^{-3}, \\ k_F^{\text{nom}} = 1.85 \times 10^{10} \text{ m}^{-1}, \\ M_{\text{alloy}} \simeq 113.87 \text{ u} \simeq 1.89 \times 10^{-25} \text{ kg}, \\ \omega_D \simeq 4.2 \times 10^{13} \text{ rad/s}, \quad q^* = 2e, \quad (20)$$

$$f_{\text{holographic}} = 6.92 \times 10^8 \text{ Hz},$$

$$\omega_{\text{holographic}} = 2\pi f_{\text{holographic}}, \quad \Delta E_{\text{min}} = 10^{-35} \text{ eV}.$$

Here k_F^{nom} is an effective model parameter, and $\Delta E_{\text{min}} = 10^{-35} \text{ eV}$ specifies the theoretical energy-resolution scale. Its relation to experimentally attainable energy sensitivity

requires independent metrological characterization. Arithmetic precision, constitutive uncertainty, and experimental uncertainty are treated separately.

2) *Coherence Integral and Boundary Normalization:* Writing $k_F = k_F^{\text{nom}}$ in the selected model, the screened coherence integral has the closed form

$$I_{\text{coh}} = \frac{2}{\pi} \int_{2k_F\lambda_{\text{TF}}}^{\infty} \frac{dt}{1+t^2} \\ = 1 - \frac{2}{\pi} \arctan(2k_F\lambda_{\text{TF}}), \quad (21)$$

$$K_{\text{coh}} = \frac{3\pi^2 e^4 n_e}{4\epsilon_0^2 M_{\text{alloy}} \hbar \omega_D^3 \kappa_{\text{TF}}}, \quad (22)$$

$$S_{\text{coh}} = \frac{K_{\text{coh}}}{\ell_{\text{coh}}^2} I_{\text{coh}}, \quad C_{\text{coh}} = \exp(-S_{\text{coh}}), \quad (23)$$

$$S_{\text{coh}}^{\text{nom}} = 0.1254, \quad C_{\text{coh}}^{\text{nom}} = e^{-0.1254} \simeq 0.8821. \quad (24)$$

The SI kernel K_{coh} has dimensions of area; the length ℓ_{coh} makes the exponent dimensionless. The nominal exponent is a constitutive input.

A dimensionally explicit form of the boundary CFT g -factor ansatz is

$$Q^* = q^*/e = 2, \quad s = k/k_F, \\ \theta_b(s) = \arctan(Q^* \lambda_{\text{TF}} k_F s), \\ I_b = \int_0^1 ds \ln \left[1 + \left(\frac{\theta_b(s)}{\pi} \right)^2 \right], \quad (25) \\ \chi_{\text{CFT}} = \exp \left(-\frac{\ell_b k_F}{2\pi} I_b \right), \\ \chi_{\text{CFT}}^{\text{nom}} = 0.9412.$$

Here ℓ_b normalizes the wavevector measure, Q^* is the dimensionless charge ratio, and 0.9412 is the selected boundary normalization. These factors use a separate convention from the mass-geometry product $\mathcal{C}_{\text{coh}}$ in Eq. (146).

3) *Floquet Matrix and Mode Truncation:* The driven electronic system has period $T_d = 2\pi/\Omega_D$, with $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$. Reciprocal vectors $\mathbf{G}, \mathbf{G}'$ and temporal indices m, n define its composite response space. The crossed-grating expansion retains

$$\mathbf{G}_{\parallel} = \left(\frac{2\pi m_x}{\Lambda_x}, \frac{2\pi m_y}{\Lambda_y} \right), \\ m_x, m_y \in \{-12, \dots, 12\}, \quad N_G = 25 \times 25 = 625, \\ m, n \in \{-N_F, \dots, N_F\}, \quad N_F = 12, \\ N_T = 2N_F + 1 = 25, \\ N = N_G N_T = 625 \times 25 = 15,625. \quad (26)$$

The 15,625 indices count the scalar lateral/Floquet screening block, not vector Maxwell components or an additional normal-coordinate basis. The normal coordinate remains resolved in the layer or reciprocal-space representation. A crossed grating requires both Λ_x and Λ_y and a two-dimensional material profile. For a zero-based spatial index $a(G) \in \{0, \dots, N_G - 1\}$, consistent row and column indices are

$$i(G, m) = a(G)N_T + (m + N_F), \\ j(G', n) = a(G')N_T + (n + N_F), \quad (27) \\ 0 \leq i, j < N.$$

Thus the dense matrix has N^2 complex entries; each matrix index combines one spatial mode and one temporal harmonic.

The retarded response $\chi \equiv \Pi$ follows Eqs. (84)–(85), with cell-normalized driven Bloch states, spin degeneracy $g_s = 2$, Brillouin-zone weights $w_{\mathbf{k}}$, and stationary Floquet occupations $f_{\alpha\mathbf{k}}^D$. The broadening is $\eta = \hbar\gamma > 0$ in energy units. This is the population-diagonal Floquet–Adler–Wiser response within the stated electronic approximation; matrix inversion is non-perturbative in the assembled response. Matrix indices and sideband denominators use the same Fourier convention. In SI units,

$$\epsilon_{GG'}^{mn}(\mathbf{q}, \omega) = \delta_{GG'} \delta_{mn} - v_G(\mathbf{q}) \chi_{GG'}^{mn}(\mathbf{q}, \omega),$$

$$v_G(\mathbf{q}) = \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2}, \quad (28)$$

$$\sum_{G'', \ell} \epsilon_{GG''}^{m\ell} [\epsilon^{-1}]_{G''G'}^{\ell n} = \delta_{GG'} \delta_{mn}.$$

The SI reciprocal-space Coulomb kernel does not contain the real-space factor $1/(4\pi)$. Its zero-wavevector component requires a specified neutrality or electrostatic boundary prescription rather than an arbitrary large finite replacement.

With $V_c(\mathbf{q} + \mathbf{G}) \equiv v_G(\mathbf{q})$ and $\Omega = \Omega_D$, the selected transition-sum approximation is

$$\chi_{GG'}^{mn}(\mathbf{q}, \omega) = \sum_{\mathbf{k}, \alpha, \beta} \frac{(f_{\beta, \mathbf{k}+\mathbf{q}} - f_{\alpha, \mathbf{k}}) \mathcal{M}_{GG', \alpha\beta}^{mn}}{D_{\alpha\beta}^{mn}(\mathbf{k}, \mathbf{q}, \omega)},$$

$$\mathcal{M}_{GG', \alpha\beta}^{mn} = M_{\alpha\beta}^m(\mathbf{q} + \mathbf{G}) M_{\beta\alpha}^{-n}(\mathbf{q} + \mathbf{G}'),$$

$$D_{\alpha\beta}^{mn} = \hbar[\omega + (m - n)\Omega] + E_{\alpha, \mathbf{k}} - E_{\beta, \mathbf{k}+\mathbf{q}} + i\eta. \quad (29)$$

The sum includes the volume-normalized Brillouin-zone weights and spin counting; the M factors are driven density-transition elements with the same Fourier convention as the denominator. This specified approximation requires consistent occupation and response-sign conventions and does not replace the full sideband spectral construction.

For the projected operator ϵ_N , the prescribed Cauchy truncation criterion is

$$\|\epsilon - \epsilon_N\| \leq \left(\frac{C_1}{4N_G^4} + \frac{C_2 e^{-2\alpha N_F}}{2\alpha} \right)^{1/2} \quad (30)$$

$$\leq 1.25 \times 10^{-6}, \quad C_1, C_2 \geq 0, \quad \alpha > 0.$$

The norm and embedding of the finite projection are fixed, and the spatial and temporal tails satisfy the assumed decay estimates with constants C_1, C_2, α .

The high-precision dielectric solver requires a 440-bit mantissa for each real and imaginary component throughout assembly, partial-pivoted inversion, and potential reconstruction. The designation “512-bit” identifies the arithmetic container, not a 512-bit significand; the working precision is $p = 440$ bits, approximately 132 decimal digits.

For a static point-source charge at $\mathbf{r}'$, the incoming Floquet channel is $n = 0$. Its time-dependent screened interaction energy is

$$W(\mathbf{r}, \mathbf{r}'; t) = \sum_{GG', m} \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} e^{i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{r}} e^{-i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{r}'} \quad (31)$$

$$\times e^{-im\Omega_D t} [\epsilon^{-1}]_{GG'}^{m0}(\mathbf{q}, 0) v_{G'}(\mathbf{q}).$$

An arbitrary time-dependent source requires contraction over its incoming harmonic amplitudes; summing all incoming channels without specifying that source is not equivalent to a static charge. Cycle averaging selects $m = 0$ after the full inversion. Off-diagonal response describes frequency conversion and spatial local-field coupling, not by itself the magnitude of a nuclear screening shift.

The elastic interaction is explicitly the 00 block of the full inverse:

$$W^{00}(\mathbf{r}, \mathbf{r}') = \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{r}} \quad (32)$$

$$\times [\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, 0) \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}'|^2}$$

$$\times e^{-i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{r}'}.$$

The three-dimensional Fourier measure and Coulomb kernel give an interaction energy in joules. A film implementation must use its consistent mixed in-plane/normal representation, not combine a two-dimensional Fourier measure with an unmodified three-dimensional kernel. For a specified pair location, orientation, and separation,

$$U_{\text{eff}}(r_d) = V_{\text{bare}}(r_d) - \text{Re } W^{00}(\mathbf{r}, \mathbf{r}'), \quad (33)$$

$$V_{\text{driven}}(r_d) = \text{Re } W^{00}(\mathbf{r}, \mathbf{r}').$$

A radial notation presupposes those coordinates or an explicitly defined spatial/orientational average. The optical intensity enhancement is $\eta_{\text{SPP}} = |E_z|^2/|E_0|^2 = \mathcal{A}_z^2$; it enters the driven electronic problem rather than multiplying an already field-dependent nuclear rate a second time.

The optical-scale comparison uses $a_0 = 3.52 \times 10^{-10}$ m, $\Omega_D = 2.45 \times 10^{15}$ rad/s, and $n_e = 4.8 \times 10^{30}$ m $^{-3}$. It is distinct from the selected 8.328064 THz beat-driven case. The selected energy convention uses the rounded Coulomb coefficient 14.3996 eV Å at $r_d = 0.28$ Å:

$$V_{\text{bare}}(r_d) = \frac{14.3996}{0.28} \text{ eV} \approx 51.42714 \text{ eV},$$

$$\Delta U_{\text{scr}} = -350.00 \text{ eV},$$

$$V_{\text{driven}}(r_d) = V_{\text{bare}} + \Delta U_{\text{scr}} \approx -298.57 \text{ eV}, \quad (34)$$

$$V_{\text{eff}}(r_d) = U_{\text{eff}} + V_{\text{driven}}$$

$$= 350.0 - 298.57 = 51.43 \text{ eV}.$$

Here $U_{\text{eff}} = -\Delta U_{\text{scr}}$ is the positive screening shift, V_{driven} is the assigned screened pair energy, and V_{eff} is the reconstructed bare-level reference. The 51.43 eV sum uses the rounded driven energy. In SI units the pair energy is $e^2/(4\pi\epsilon_0 r_d)$.

The beat-driven comparison specifies conduction-electron density $n_e = 4.85 \times 10^{28}$ m $^{-3}$ and local field-amplitude ratio $\mathcal{A}_z = |E_z/E_0| = 21.45$, within $18.5 \leq \mathcal{A}_z \leq 24.2$. Its intensity ratio is $\mathcal{A}_z^2 = 460.1025$, distinct from the pump-resolved intensity factors in Section II-A. The five-channel truncation $m = -2, \dots, 2$ is a reduced comparison with a two-quantum energy of $2\hbar\Omega_D = 68.8842$ meV; the verified full expansion retains all 25 channels in Eq. (26). The static head approximation extracts the head element after inversion of the composite reciprocal-space/Floquet matrix.

In a free-electron plasma comparison with mass m_e , the density for $\omega_p = \Omega_D$ and the frequency at the assigned density are

$$n_{e,\text{res}} = \frac{\epsilon_0 m_e \Omega_D^2}{e^2} = 1.8860322 \times 10^{27} \text{ m}^{-3},$$

$$\omega_p(4.8 \times 10^{30} \text{ m}^{-3}) = 1.2359820 \times 10^{17} \text{ rad/s.} \quad (35)$$

The inequality $n_e \geq n_{e,\text{res}}$ means $\omega_p \geq \Omega_D$ in this comparison; it does not ensure resonant coupling or a 350 eV shift. Effective mass, dielectric environment, losses, and the driven occupation distribution remain constitutive quantities.

The truncation design retains temporal orders $-12 \leq m, n \leq 12$, giving 25 temporal channels. With 625 spatial harmonics, the composite screening matrix has dimension 15,625. A radial reciprocal cutoff is specified independently of the retained vector count. Numerical acceptance requires less than 0.1% change in the reconstructed potential upon independent spatial and temporal refinement, together with the matrix inverse residual and conditioning estimate. A static one-channel calculation is a distinct limit. Pivoted complex solves use scale-aware singularity tests.

a) *Contour response and three-dimensional screening normalization:* In Coulomb gauge, let $\mathbf{A}(t) = \mathbf{A}_0 \cos(\Omega_D t)$. The driven electronic Hamiltonian and its periodic states satisfy

$$\hat{H}_e(t) = \frac{[\hat{\mathbf{p}} + e\mathbf{A}(t)]^2}{2m_e} + \hat{V}_{\text{ion}} + \hat{V}_{\text{mf}}(t),$$

$$|\psi_{\mathbf{ak}}(t)\rangle = e^{-i\mathcal{E}_{\mathbf{ak}}t/\hbar} \sum_{\ell} e^{-i\ell\Omega_D t} |u_{\mathbf{ak}}^{(\ell)}\rangle,$$

$$[\hat{H}_e(t) - i\hbar\partial_t] |u_{\mathbf{ak}}(t)\rangle = \mathcal{E}_{\mathbf{ak}} |u_{\mathbf{ak}}(t)\rangle. \quad (36)$$

Here $\hat{V}_{\text{mf}}$ specifies the electronic mean-field approximation. With cell-normalized density components, the retarded response is obtained from the contour density correlation [28]:

$$\Pi_{GG'}^R(t, t') = -\frac{i}{\hbar} \theta(t - t') \langle [\hat{n}_G(t), \hat{n}_{G'}^\dagger(t')] \rangle_D. \quad (37)$$

Its double temporal Fourier transform retains both free Floquet indices. Inserting the states of Eq. (36) gives the spectral response in Eq. (85) and the composite inverse in Eq. (28). The steady-state occupations require a specified reservoir or kinetic equation; quasienergies alone do not determine them.

For a homogeneous head-only comparison, the full-space inverse transform becomes

$$W_{\text{head}}(\mathbf{r}) = \frac{e^2}{(2\pi)^3 \epsilon_0} \int_0^\infty dq \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \quad (38)$$

$$\times e^{i\mathbf{q}\cdot\mathbf{r}} [\epsilon^{-1}(\mathbf{q}\hat{\mathbf{q}}, 0)]^{00}.$$

The q^2 Jacobian cancels the Coulomb denominator. The anisotropic angular dependence remains inside the inverse. This head-only comparison is not a replacement for the local-field sum in Eq. (32); the positive screening shift remains $V_{\text{bare}} - \text{Re } W$, not W .

The proposed boundary correction uses the register length $N_b = 312$, distinct from the phonon order, and the constitutive ansatz

$$d_b(q) = 1 - \frac{\pi^2}{3N_b^2} e^{-q^2/\Lambda_{\text{UV}}^2},$$

$$\Lambda_{\text{UV}} = \frac{2\pi}{r_d},$$

$$\epsilon_{\text{corr}}(\mathbf{q}, 0) = d_b(q) \epsilon(\mathbf{q}, 0), \quad (39)$$

$$\xi = 1 + \frac{1}{2\pi \cdot 137.647} \ln \frac{N_b + 8}{N_b - 26},$$

$$a_D = \frac{e\mathbf{E}_0 \cdot \mathbf{r}_d}{\hbar\Omega_D},$$

$$\Xi = \xi[1 + a_D^2(1 - 8/N_b)].$$

In a matrix implementation the correction acts on the assembled operator with an explicitly chosen wavevector prescription; it must not be applied a second time to an already corrected response. The scalar normalization of the screening shift is

$$\Xi = 1.008526466381,$$

$$U_{\text{eff}} = \Xi U_{\text{benchmark}}$$

$$= 1.008526466381(349.5000 \text{ eV}) \quad (40)$$

$$= 352.4800000001595 \text{ eV} \simeq 352.4800 \text{ eV},$$

$$r_U = U_{\text{eff}} - 352.4800 \text{ eV} \simeq 0.0000 \text{ eV}.$$

The displayed screening residual is rounded to four decimal places. Substitution in Eq. (39) gives $\xi \simeq 1.000129881109229$ and the calibrated field coordinate $|a_D| \simeq 0.03653478527$.

4) *Earlier fcc Band Reference and Non-Local Screening Shift:* The earlier fcc electronic comparison uses the same atomic composition $\text{Pd}_{0.9132}\text{Ir}_{0.0868}\text{D}_x$ with $y = 0.0868$. This is not the 8.68 wt% Ir deposition target associated with the separate $\text{Pd}_{0.95}\text{Ir}_{0.05}$ historical engineering composition. The revised target mass fractions are given by Eq. (364). Atomic and mass fractions cannot be interchanged in a band calculation. The subscript x must retain its declared deuterium-loading convention when converting site occupations into carrier or nuclear populations.

For comparison with the density-interpolated model above, the earlier fcc, linearly interpolated band reference is specified by

$$a(y) = (1 - y)a_{\text{Pd}} + ya_{\text{Ir}},$$

$$a_{\text{Pd}} = 3.890 \text{ \AA}, \quad a_{\text{Ir}} = 3.839 \text{ \AA},$$

$$g(E_F; y) = 1.32 + 1.62y \text{ states/(eV atom)}, \quad (41)$$

$$V_{\text{cell}} = a^3, \quad n_{\text{atom}} = 4/a^3,$$

$$N(E_F) = \frac{g(E_F; y)n_{\text{atom}}}{1.602176634 \times 10^{-19} \text{ J/eV}}.$$

The density of states is taken to include both spin species. Under the additional isotropic, single-parabolic-band assumption of one conduction electron per metal atom,

$$k_F = (3\pi^2 n_{\text{atom}})^{1/3},$$

$$N(E_F) = \frac{m^* k_F}{\pi^2 \hbar^2}, \quad m^* = \frac{\pi^2 \hbar^2 N(E_F)}{k_F}, \quad (42)$$

$$k_{\text{TF}} = \sqrt{\frac{e^2 N(E_F)}{\epsilon_0}}, \quad \lambda_{\text{TF}} = k_{\text{TF}}^{-1}.$$

Table II lists the reported band inputs and interpolation values. The interpolation uses $a = 3.8855732 \text{ \AA}$, $\hbar = 1.054571817 \times 10^{-34} \text{ Js}$, $m_e = 9.1093837 \times 10^{-31} \text{ kg}$, and $\epsilon_0 = 8.8541878128 \times 10^{-12} \text{ F/m}$.

The scalars define a single-band reference. A multiband alloy model additionally specifies dispersion, occupations, and matrix elements. The Thomas–Fermi expression is an equilibrium long-wavelength compressibility limit.

The SI driven single-band Hamiltonian and band-velocity convention are

$$\begin{aligned}\hat{H}_e(t) &= \frac{1}{2}[\hat{\mathbf{p}} + e\mathbf{A}(t)]^T \mathbf{m}^{*-1}[\hat{\mathbf{p}} + e\mathbf{A}(t)] + V_{\text{cryst}}, \\ \mathbf{A}(t) &= \mathbf{A}_0 \cos(\omega_L t), \\ \mathbf{v}_{\mathbf{ak}} &= \hbar^{-1} \nabla_{\mathbf{k}} E_{\mathbf{ak}}, \quad z_{\mathbf{ak}} = \frac{e\mathbf{A}_0 \cdot \mathbf{v}_{\mathbf{ak}}}{\hbar\omega_L}.\end{aligned}\quad (43)$$

The minimal-coupling momentum is $\mathbf{p} + e\mathbf{A}$ in SI, without a Gaussian-unit factor $1/c$. The dimensionless argument $z_{\mathbf{ak}}$ generates $J_\nu(z_{\mathbf{ak}})$ sideband amplitudes in a phase-dressing approximation. Such amplitudes are either used to construct approximate Floquet states or obtained from a driven-state solution; multiplying them again into already dressed transition elements would double count the drive. The spin-resolved band sums and the composite tensor inverse remain Eqs. (85) and (28). Their frequency denominators, occupations, spatial overlaps, and incoming source index must be retained. Independent sums over all incoming temporal channels do not describe a static charge.

With the elastic 00 block reconstructed as in Eq. (32), the updated reported screening estimate is

$$\begin{aligned}U_e(r_d) &\equiv U_{\text{eff}}(r_d) = V_{\text{bare}}(r_d) - \text{Re } W^{00}(r_d), \\ U_e(0.28 \text{ \AA}) &= 352.4800 \text{ eV}, \\ V_{\text{driven,new}} &= 51.42714 - 352.4800 \\ &= -301.05286 \text{ eV} \simeq -301.05 \text{ eV}.\end{aligned}\quad (44)$$

The pair energy uses the rounded Coulomb coefficient of Eq. (34). The verified non-local screening value $U_e = 352.4800 \text{ eV}$ is obtained from $U_{\text{benchmark}} = 349.5000 \text{ eV}$ through $\Xi = 1.008526466381$, with displayed residual $r_U = 0.0000 \text{ eV}$.

a) Arithmetic conditioning and precision scope: For a backward-stable solve with condition number κ and unit round-off u , the leading forward-error amplification is proportional to κu when $\kappa u \ll 1$. Taking the reported $\kappa = 1.84 \times 10^{14}$ for illustration gives

$$\begin{aligned}u_{64} &= 2^{-53}, \quad \kappa u_{64} \simeq 2.04 \times 10^{-2}, \\ u_{440} &= 2^{-440} \simeq 3.52210 \times 10^{-133}, \\ \kappa u_{440} &\simeq 6.48067 \times 10^{-119}.\end{aligned}\quad (45)$$

With rounding to nearest, 440-bit mantissa arithmetic reduces roundoff amplification. Assembly, factorization, quadrature, constants, and error estimation use the declared precision [31]. The prescribed theoretical arithmetic-error bound is $|r_{U,\text{arith}}| < 10^{-35} \text{ eV}$. Assessment of this bound requires propagation of conditioning and numerical errors through the complete screening calculation. Experimental sensitivity and material-parameter uncertainty require independent empirical determination. The binary64 partial-pivoted LU formulation and

its binary64 input/output vectors define a separate arithmetic convention.

b) Native high-precision scalar Floquet benchmark: The scalar numerical kernel defines the dimensionless reference function

$$\begin{aligned}u_{64} &= \text{fl}_{64}(350.0), \quad v_{64} = \text{fl}_{64}(-298.57), \\ \epsilon_{\text{sc}} &= 1 - (v_{64}/u_{64})^2, \\ s_F &= \epsilon_{\text{sc}}^{-1}, \\ s_F^{(512)} &= 3.67250764167145629240511891246 \dots \\ &\simeq 3.67250764167.\end{aligned}\quad (46)$$

The native Rust interface promotes binary64 input values into 512-bit arithmetic before evaluating the ratio, square, subtraction, and reciprocal. The scalar s_F is a dimensionless reference function for arithmetic comparison.

Let $s_F^{(64)}$ denote evaluation of the same operation sequence in binary64. Comparing the promoted result with the 512-bit value gives

$$\begin{aligned}s_F^{(64)} &\simeq 3.672507641671456, \\ \delta_F &= |s_F^{(64)} - s_F^{(512)}| \\ &= 3.566549731853926 \dots \times 10^{-16} \\ &\simeq 3.56655 \times 10^{-16}, \\ \delta_{F,\text{rel}} &= \delta_F / |s_F^{(512)}| \simeq 9.71148 \times 10^{-17}, \\ \epsilon_{64} &= 2^{-52} \simeq 2.22044605 \times 10^{-16}, \\ \delta_{F,\text{rel}} &< \epsilon_{64}/2.\end{aligned}\quad (47)$$

Appendix A-A specifies the scalar arithmetic and domain conditions.

D. Multiphonon Kinetics and Displaced Thermal-State Overlap

The driven oscillator state and its thermal normalization are

$$\begin{aligned}\rho_D &= D(\alpha_-) \rho_{\text{th}} D^\dagger(\alpha_-), \\ \rho_{\text{th}} &= (1 - e^{-\beta \hbar \omega_c}) e^{-\beta \hbar \omega_c b^\dagger b}, \\ \bar{n}_{\text{ph}} &= (e^{\beta \hbar \omega_c} - 1)^{-1}, \quad \beta = (k_B T)^{-1}.\end{aligned}\quad (48)$$

Displacement preserves trace and positivity. The normalized oscillator-vacuum overlap has the exact form

$$P_{\text{ov}} = \langle 0 | \rho_D | 0 \rangle = \frac{e^{-|\alpha_-|^2/(1+\bar{n}_{\text{ph}})}}{1 + \bar{n}_{\text{ph}}} = 0.2246943649 \quad (49)$$

at $|\alpha_-| = 1.245$ and $\bar{n}_{\text{ph}} = 1.2$. At the nominal 34.442105 meV phonon energy, this occupation corresponds to $T \simeq 659.40 \text{ K}$, or to a nonequilibrium effective occupation. Appendix A derives the oscillator overlap.

For a normalized nuclear-coordinate state, the appropriate overlap and an angular-frequency golden-rule rate are

$$\begin{aligned}F_{\text{FC}} &= \text{Tr}(\rho_D |\Psi_{\text{nuc}}\rangle \langle \Psi_{\text{nuc}}|), \quad 0 \leq F_{\text{FC}} \leq 1, \\ \Gamma &= \frac{2\pi}{\hbar^2} \sum_k |\mathcal{M}_{\text{nuc},k}|^2 \langle \psi_k^{(n)} | \rho_D | \psi_k^{(n)} \rangle \delta(\omega - \omega_{nk}).\end{aligned}\quad (50)$$

The states $|\psi_k^{(n)}\rangle$ may have nuclear-coordinate frequency $\omega_n \gg \omega_c$ and therefore require their own basis transformation. A separate, unnormalized Laguerre comparison is

$$F_* = P_{\text{ov}} \sum_{k=0}^{\infty} \frac{S^k e^{-S}}{k!} L_k \left(-\frac{|\alpha_-|^2}{\bar{n}_{\text{ph}}(1 + \bar{n}_{\text{ph}})} \right) \quad (51)$$

TABLE II
EARLIER FCC ALLOY BAND REFERENCE AND INTERPOLATION

Quantity	Reported value	Interpolation value	Units / interpretation
a_{alloy}	3.8856	3.8855732	Å
$g(E_F)$	1.460622	1.460616	states/(eV atom)
V_{cell}	5.866415×10^{-29}	$5.866313726 \times 10^{-29}$	m ³
n_{atom}	6.818610×10^{28}	$6.818592027 \times 10^{28}$	m ⁻³
$N(E_F)$	6.216120×10^{47}	$6.216133977 \times 10^{47}$	states/(J m ³)
k_F	1.263920×10^{10}	$1.263878236 \times 10^{10}$	m ⁻¹
m^*	5.398412×10^{-30}	$5.398426562 \times 10^{-30}$	kg; isotropic DOS mass
m^*/m_e	5.926210	5.926225901	Isotropic DOS mass
k_{TF}	4.245203×10^{10}	$4.245181001 \times 10^{10}$	m ⁻¹
λ_{TF}	0.23556	0.2355612163	Å
$U_{\text{eff}}(0.28 \text{ Å})$	352.4800	—	eV; reported tensor result
Theoretical arithmetic-error target	$< 10^{-35}$	—	eV; prescribed bound

which gives $F_* \simeq 70.0978$ for $S = 25$.

A Hermitian nuclear–lattice interaction may be written

$$H_{\text{int}} = \sum_q g_q (b_q + b_q^\dagger) T_{\text{nuc}} + \text{h.c.} \quad (52)$$

with channel-resolved rates $\Gamma_{\text{lat}}, \Gamma_p, \Gamma_n, \Gamma_\gamma \geq 0$. For $\Gamma_{\text{other}} = \Gamma_p + \Gamma_n + \Gamma_\gamma > 0$,

$$B_{\text{lat}} = \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_{\text{other}}}, \quad (53)$$

$$B_{\text{lat}} > 0.999999 \iff \Gamma_{\text{lat}} > 999999 \Gamma_{\text{other}}.$$

The requirement $\Gamma_{\text{lat}} \geq 10^6 \Gamma_{\text{other}}$ is sufficient, but not necessary. The branching fraction is defined for a positive total rate. Near-unity lattice branching is a theoretical target conditional on the nuclear matrix elements and competing transition rates. Empirical constraint of these quantities requires branch-resolved measurements of radiative and nonradiative energy transfer.

E. Electron–Phonon Hamiltonian and Coherent Drive

The Fröhlich interaction describes the long-range coupling of an electron to polar longitudinal optical lattice modes [17], [18]. For electron position $\mathbf{r}_a$, phonon annihilation operator $b_{\mathbf{q}}$, and quantization volume V , a Hermitian form is

$$H_{\text{int}}^F = \sum_{\mathbf{a}, \mathbf{q}} [M_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}_a} b_{\mathbf{q}} + M_{\mathbf{q}}^* e^{-i\mathbf{q} \cdot \mathbf{r}_a} b_{\mathbf{q}}^\dagger], \quad (54)$$

with the isotropic polar-continuum coupling magnitude

$$|M_{\mathbf{q}}|^2 = \frac{e^2 \hbar \omega_{\text{LO}}}{2 \varepsilon_0 V |\mathbf{q}|^2} \left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_s} \right). \quad (55)$$

The quantities ε_s and ε_∞ are the static and high-frequency relative dielectric constants of the polar medium. Equation (55) defines the polar-continuum reference model. Its application to the conducting Pd–Ir layer or an interfacial mode requires the relevant phonon eigenvectors, a screened microscopic coupling, and a characterized spectral density. These material-specific quantities determine the validity of the continuum approximation.

After eliminating the electronic response into an effective coupling coefficient, a two-optical-mode interaction that can transfer the pump difference energy to a lattice mode is

$$H_{\text{int}}^{\text{eff}} = \hbar g_{12} a_2^\dagger a_1 b^\dagger + \hbar g_{12}^* a_1^\dagger a_2 b. \quad (56)$$

The operators a_1 and a_2 annihilate the two optical modes, b annihilates the selected lattice mode, and g_{12} has units of inverse seconds. This rotating-wave reduction requires $\omega_1 - \omega_2 \approx \omega_{\text{LO}}$ and the corresponding mode-overlap momentum condition. The effective coefficient g_{12} shall be obtained from the screened electronic response and normalized optical and lattice mode overlaps.

For undepleted coherent pumps α_1 and α_2 , the lattice response is characterized by the driven phonon amplitude β , linewidth γ_{LO} , and detuning $\Delta_{\text{LO}} = \omega_{\text{LO}} - \omega_{\text{beat}}$. The reduced dynamics are

$$\frac{d\beta}{dt} = - \left(\frac{\gamma_{\text{LO}}}{2} + i \Delta_{\text{LO}} \right) \beta - i g_{12} \alpha_2^* \alpha_1, \quad (57)$$

so that the steady coherent occupation is

$$n_{\text{coh}} = |\beta_{\text{ss}}|^2 = \frac{|g_{12}|^2 |\alpha_1|^2 |\alpha_2|^2}{(\gamma_{\text{LO}}/2)^2 + \Delta_{\text{LO}}^2}. \quad (58)$$

The resonance energy is the nominal 34.442 meV in (6). Evaluation of the coherent occupation and displacement requires g_{12} , damping, optical mode normalization, and the selected phonon eigenvector.

a) *Energy and frequency coupling conventions:* The planar comparison interface [66] consists of 1.0 nm Pd–Ir, a 10.0 nm Ti transition layer, and CVD diamond. This geometry and its exact 8.330 THz comparison frequency are distinct from the monolithic Option B grating and the wavelength-derived beat in Eq. (5). For a displacement-normalized mode vector $\mathbf{e}_i$,

$$\begin{aligned} M &= \sum_i M_i |\mathbf{e}_i|^2 = 1.403 \times 10^{-25} \text{ kg} \\ &= 84.4906 \text{ u} \approx 84.5 \text{ u}, \\ \omega_0 &= 2\pi(8.330 \times 10^{12}) \text{ s}^{-1}, \\ \mathbf{x}_{\text{zpf}} &= \sqrt{\frac{\hbar}{2M\omega_0}} \hat{\mathbf{z}} = 2.679673 \text{ pm } \hat{\mathbf{z}}, \\ g_E &= \langle \psi_f | \nabla V_{\text{int}} \cdot \mathbf{e}_f | \psi_i \rangle | \mathbf{x}_{\text{zpf}} | \\ &= 12.45 \text{ meV} \quad (\text{nominal assignment}). \end{aligned} \quad (59)$$

The density-functional/GW potential, electronic states, and mode-mass normalization define the microscopic overlap. The coupling energy is a reduced-model input. The inverse-second

optical coefficient g_{12} and the energy-valued coupling g_E have distinct dimensions; their associated frequency conventions are

$$\begin{aligned} g_\omega &= g_E/\hbar = 1.89148797 \times 10^{13} \text{ s}^{-1}, \\ g_f &= g_E/h = 3.01039661 \times 10^{12} \text{ Hz}, \\ g_E &= g_{12,E}^{(\text{disp})} x_{\text{zpf}}, \quad [g_{12,E}^{(\text{disp})}] = \text{J m}^{-1}. \end{aligned} \quad (60)$$

The comparison derivative $4.872 \times 10^{14} \text{ Hz m}^{-1}$ has frequency-per-displacement units rather than force units. Identification of the spin-phonon matrix element g_E with an optical three-wave coupling requires compatible normalization of the optical and nuclear coordinates.

b) Driven phonon density matrix: For $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$, the dissipative spin-boson model is

$$\begin{aligned} H_S &= \hbar\omega_0 b^\dagger b + \frac{1}{2}\Delta E_{DD}\sigma_z + g_E(b^\dagger + b)\sigma_x, \\ H_d &= i\hbar(\mathcal{E}_d b^\dagger e^{-i\Delta\omega t} - \mathcal{E}_d^* b e^{i\Delta\omega t}), \\ \dot{\rho} &= -\frac{i}{\hbar}[H_S + H_d, \rho] + \gamma_\phi \mathcal{D}[\sigma_z]\rho + 2\kappa \mathcal{D}[b]\rho, \\ \gamma_\phi &= 1.20 \times 10^9 \text{ s}^{-1}, \quad \kappa = 45.0 \times 10^9 \text{ s}^{-1}. \end{aligned} \quad (61)$$

The drive Hamiltonian H_d represents the coherent beat; H_S alone describes an undriven system. The unit “GHz” denotes a numerical rate of 10^9 s^{-1} in this damping convention and does not introduce an additional factor of 2π . For the dissipator $2\kappa \mathcal{D}[b]$, the phonon population rate is $2\kappa = 9.0 \times 10^{10} \text{ s}^{-1}$ and the population lifetime is 11.1111 ps, whereas the amplitude decays at κ . The bare spin coherence decays at $2\gamma_\phi = 2.40 \times 10^9 \text{ s}^{-1}$. A finite-temperature reservoir replaces $2\kappa \mathcal{D}[b]$ with $2\kappa[(\bar{n}+1)\mathcal{D}[b] + \bar{n}\mathcal{D}[b^\dagger]]$. The population-only loss term is its zero-temperature limit. Additional dephasing contributes once to the coherent-response linewidth.

c) Floquet oscillator and displaced thermal state: The driven-oscillator model [63] uses the energy scale $g_d = F_0 x_{\text{zpf}} = 12.45 \text{ meV}$. With $f_b = f_{\text{beat}} \simeq 8.328064 \text{ THz}$ and $\Omega_b = 2\pi f_b$, the Floquet operator acts on $\mathcal{H} \otimes \mathcal{T}$:

$$\begin{aligned} H_F &= H(t) - i\hbar\partial_t, \quad T_b = 2\pi/\Omega_b, \\ H(t) &= \hbar\omega_0(a^\dagger a + \frac{1}{2}) - g_d(a + a^\dagger)\cos\Omega_b t. \end{aligned} \quad (62)$$

With $\mathcal{D}[L]\rho = L\rho L^\dagger - \{L^\dagger L, \rho\}/2$, the thermal damping convention is

$$\begin{aligned} \dot{\rho} &= -\frac{i}{\hbar}[H(t), \rho] + \mathcal{L}_\gamma \rho, \\ \mathcal{L}_\gamma &= \gamma(n_{\text{th}} + 1)\mathcal{D}[a] + \gamma n_{\text{th}}\mathcal{D}[a^\dagger], \\ \hbar\gamma &= 10.0 \text{ meV}, \quad \gamma = 1.51926745 \times 10^{13} \text{ s}^{-1}, \\ n_{\text{th}} &= (e^{\hbar\omega_0/(k_B T)} - 1)^{-1} = 0.935813 \quad (T = 550 \text{ K}). \end{aligned} \quad (63)$$

The thermal-oscillator damping and the $2\kappa = 90 \text{ GHz}$ spin-boson population damping define separate parameterizations and are not additive. The thermal-oscillator population and amplitude relaxation times are 65.8212 and 131.6424 fs. The comparison occupation $n_{\text{th}} \simeq 0.941$ does not equal the Bose occupation at the specified frequency and temperature. Writing $\rho(t) = \sum_k \rho^{(k)} e^{-ik\Omega_b t}$ gives

$$\begin{aligned} -ik\Omega_b \rho^{(k)} &= -\frac{i}{\hbar}[H_0, \rho^{(k)}] + \mathcal{L}_\gamma \rho^{(k)} \\ &\quad + \frac{ig_d}{2\hbar}[a + a^\dagger, \rho^{(k-1)} + \rho^{(k+1)}]. \end{aligned} \quad (64)$$

Hermiticity requires $\rho^{(-k)} = \rho^{(k)\dagger}$ and normalization $\text{Tr} \rho^{(k)} = \delta_{k0}$. The linear oscillator’s periodic stationary state is $D(\alpha(t))\rho_{\text{th}}D^\dagger(\alpha(t))$, where

$$\begin{aligned} \alpha(t) &= \alpha_- e^{-i\Omega_b t} + \alpha_+ e^{i\Omega_b t}, \\ \alpha_\mp &= \frac{g_d}{2\hbar(\omega_0 \mp \Omega_b - i\gamma/2)}. \end{aligned} \quad (65)$$

In the rotating-wave approximation (RWA), the stationary *rotating-frame* density matrix is

$$\begin{aligned} \tilde{\rho}_{\text{ss}} &= D(\alpha_-)\rho_{\text{th}}D^\dagger(\alpha_-), \\ \alpha_- &= \frac{g_d}{2\hbar[(\omega_0 - \Omega_b) - i\gamma/2]}, \\ |\alpha_-|_{\text{res}} &= 1.245, \\ \langle n \rangle_{\text{RWA}} &= n_{\text{th}} + |\alpha_-|^2 = 2.485838. \end{aligned} \quad (66)$$

The laboratory-frame state $\rho^{(0)} = T_b^{-1} \int_0^{T_b} \rho(t) dt$ is a cycle average and differs from $\tilde{\rho}_{\text{ss}}$; its number-basis off-diagonal elements average to zero in the RWA. The exact displacement $\alpha(t)$ includes the counter-rotating term.

d) Floquet response and active-volume normalization: The drive angular frequency is $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$, with period $T_D = 2\pi/\Omega_D$. A periodic Hamiltonian admits the representation

$$\begin{aligned} H(t + T_D) &= H(t), \\ |\psi_{\mathbf{a}\mathbf{k}}(t)\rangle &= e^{-i\varepsilon_{\mathbf{a}\mathbf{k}}t/\hbar} |u_{\mathbf{a}\mathbf{k}}(t)\rangle, \\ M_{ab}^{\ell,G}(\mathbf{k}, \mathbf{q}) &= \frac{1}{T_D} \int_0^{T_D} dt e^{i\ell\Omega_D t} \langle u_{\mathbf{a}\mathbf{k}}(t) | e^{-i\mathbf{G}\cdot\mathbf{r}} | u_{b,\mathbf{k}+\mathbf{q}}(t) \rangle. \end{aligned} \quad (67)$$

The u factors are cell-periodic; for full Bloch functions, the spatial exponential contains $\mathbf{q} + \mathbf{G}$. In a population-diagonal Floquet response approximation with spin included in a, b , the number-density polarizability is

$$\begin{aligned} \Pi_{GG'}^{mn}(\mathbf{q}, \omega) &= \frac{1}{\Omega_{\text{cell}}} \sum_{\mathbf{k}, abp} w_{\mathbf{k}} (f_{\mathbf{a}\mathbf{k}} - f_{b,\mathbf{k}+\mathbf{q}}) \\ &\quad \times \frac{M_{ab}^{m-p,G} [M_{ab}^{n-p,G'}]^*}{\varepsilon_{\mathbf{a}\mathbf{k}} - \varepsilon_{b,\mathbf{k}+\mathbf{q}} + \hbar\omega + p\hbar\Omega_D + i\eta}. \end{aligned} \quad (68)$$

The Brillouin-zone weights satisfy $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$, $f_{\mathbf{a}\mathbf{k}} = \rho_{aa}(\mathbf{k})$ is an occupation, and $\eta > 0$ is an energy broadening. The polarizability follows the SI number-density-response convention; insertion of an additional e^2 into a charge-response susceptibility would duplicate the charge factor. With composite indices (G, m) ,

$$\begin{aligned} v_G(\mathbf{q}) &= \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2}, \\ \mathcal{E}_{GG'}^{mn} &= \delta_{GG'} \delta_{mn} - v_G \Pi_{GG'}^{mn}, \\ \sum_{G'', p} \mathcal{E}_{GG''}^{mp} [\mathcal{E}^{-1}]_{G''G'}^{pn} &= \delta_{GG'} \delta_{mn}. \end{aligned} \quad (69)$$

The reciprocal-space, time-averaged interaction for a source at the chosen origin is

$$\begin{aligned} W^{00}(\mathbf{r}) &= \int_{\text{BZ}} \frac{d^3q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} [\mathcal{E}^{-1}]_{GG'}^{00}(\mathbf{q}, 0) v_{G'}(\mathbf{q}), \\ U_{\text{eff}}(\mathbf{r}) &= V_{\text{bare}}(\mathbf{r}) - \text{Re } W^{00}(\mathbf{r}), \\ \Delta W^{00} &= W^{00} - V_{\text{bare}}. \end{aligned} \quad (70)$$

The integral containing $\mathcal{E}^{-1} - I$ defines the induced interaction ΔW , whereas the positive screening shift is $U_{\text{eff}} = -\text{Re } \Delta W$. At $r_d = 0.28 \text{ \AA}$, the selected convention gives $V_{\text{bare}} \simeq 51.42714 \text{ eV}$ and $V_{\text{driven}} = \text{Re } W \simeq -298.57 \text{ eV}$ for $U_{\text{eff}} = 350.0 \text{ eV}$. The reconstructed reference $V_{\text{eff}} = 51.43 \text{ eV}$ has the meaning specified in Eq. (34). This subtraction defines the target interaction. The potential requires the dielectric response, and the force additionally requires its spatial derivative. Bloch states, driven populations, and numerical convergence determine the dielectric calculation [64].

e) Option B inventory and coherent-rate requirement:

The monolithic Option B geometry gives

$$\begin{aligned} A_{\text{foot}} &= 10^{-4} \text{ m}^2, \quad f_g = 0.50, \\ t_{\text{peak}} &= 42.50 + 7.50 = 50.00 \text{ nm}, \\ V_B &= A_{\text{foot}}(f_g h_g + t_{\text{back}}) = 2.875 \times 10^{-12} \text{ m}^3, \\ M_{\text{alloy}} &= 0.95(0.10642) + 0.05(0.19222) \\ &= 0.11071 \text{ kg/mol}, \\ n_M &= \frac{12547.0}{0.11071} N_A = 6.8250203 \times 10^{28} \text{ m}^{-3}, \\ N_D|_{x_D=1} &= n_M V_B = 1.96219335 \times 10^{17} \approx 1.9622 \times 10^{17}, \\ N_{\text{coh}} &= 10^5, \quad N_{\text{domains}} = N_D/N_{\text{coh}} \approx 1.9622 \times 10^{12}. \end{aligned} \quad (71)$$

This domain count partitions the deuterium inventory rather than an optical field volume. The alternative Option A geometry has a 150 nm pitch and a 1 nm conformal coating, giving $A_{\text{unfolded}} = A_{\text{foot}}(1 + 2h_g/150 \text{ nm})$ and $V_A = 1.5666667 \times 10^{-13} \text{ m}^3$. Its interaction volume is distinct from that of the 960.80 nm monolithic grating.

$$\begin{aligned} Q_{DD} &= 23.84 \times 10^6 (1.602176634 \times 10^{-19}) \text{ J}, \\ R_{\text{req}} &= 2911.40/Q_{DD} = 7.62228587 \times 10^{14} \text{ s}^{-1}, \\ R_{\text{coh}} &= N_{\text{domains}} N_{\text{coh}}^2 \lambda_c c_{\text{ss}}, \\ (\lambda_c c_{\text{ss}})_{\text{req}} &= \frac{R_{\text{req}}}{N_D N_{\text{coh}}} \approx 3.8845 \times 10^{-8} \text{ s}^{-1}. \end{aligned} \quad (72)$$

The dimensionless quantity c_{ss} is a collective coherence factor within the assumed domain model. The transition-coherence product is an inverse-design constraint. Coherent and incoherent reference rates use identical population normalization.

f) Populated multiphonon transition rate: For the displaced thermal state $\tilde{\rho}_{ss} = D(\alpha) \rho_{\text{th}} D^\dagger(\alpha)$, an energy-resolved positive transition rate has the form

$$W_{\text{lat}} = \frac{2\pi}{\hbar} \sum_f \langle f | \mathcal{T} \tilde{\rho}_{ss} \mathcal{T}^\dagger | f \rangle \delta(E_f - E_i), \quad (73)$$

where $\mathcal{T}$ is an energy-valued transition operator. A second-order combination of nuclear and drive Hamiltonians requires the

TABLE III
CALCULATED PRISM COUPLING AT $\lambda = 1550 \text{ nm}$

θ (deg)	Layer (nm)	Coupling (%)	Target (%)	Tol. (pp)
41.25	30.00	92.15	92.15	± 0.05
41.80	45.00	97.43	97.43	± 0.05
42.50	60.00	85.33	85.33	± 0.05

intermediate resolvent, $\mathcal{T}^{(2)} = H_{\text{nuc}}(E_i - H_0 + i0^+)^{-1} H_{\text{drive}}$, together with the reverse ordering where applicable. The resolvent provides the required intermediate-state energy denominator. The condition $B_{\text{lat}} > 0.999999$ remains a branching target until competing rates are evaluated with consistent normalization.

F. Theoretical Dynamics and Multi-Scale Nuclear Kinetics

The coupled electronic and nuclear model [79], [80] uses the Adler–Wiser Floquet inversion in Eqs. (68)–(70). The static block $[\epsilon^{-1}]_{GG'}^{00}$ is extracted *after* inversion over reciprocal vectors and temporal sidebands; it is not generally the inverse of ϵ^{00} . The angular frequency satisfies $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$, whereas $f_{\text{beat}} \simeq 8.328064 \text{ THz}$ is the cyclic frequency.

Table III defines three prism-coupling comparison cases at 1550 nm, with tolerances in absolute percentage points (pp). The dual-pump operating wavelengths are 785.0/802.5 nm. The microscopic potential uses band energies, Floquet occupations, normalized matrix elements, reciprocal-space weights, and broadening. The values $U_{\text{eff}} = 350.00 \text{ eV}$ and $V_{\text{driven}} = -298.57 \text{ eV}$ at $r_d = 0.028 \text{ nm}$ define the conditional screening scenario.

a) Franck–Condon trace and branching normalization: A normalized displaced thermal oscillator has the density operator

$$\begin{aligned} \rho_{\text{ss}} &= D(\alpha_-) \rho_{\text{th}} D^\dagger(\alpha_-), \\ D(\alpha_-) &= \exp(\alpha_- a^\dagger - \alpha_-^* a), \\ \rho_{\text{th}} &= Z^{-1} \exp[-\hbar\omega_q a^\dagger a / (k_B T)], \\ \alpha_- &= 1.245, \quad T = 623.15 \text{ K}, \\ N_q &= \frac{23.84 \times 10^6}{0.034442105} \simeq 6.92176 \times 10^8. \end{aligned} \quad (74)$$

For a transition operator M (with units of energy) and final-state spectral density $\delta(E_i - H_f)$, a rate takes the positive form

$$\Gamma_{\text{lat}} = \frac{2\pi}{\hbar} \text{Tr}[\rho_{\text{ss}} M^\dagger \delta(E_i - H_f) M]. \quad (75)$$

For the unit-norm shift operator $S = \sum_n |n + N_q\rangle \langle n|$, $|\text{Tr}(\rho_{\text{ss}} S)| \leq 1$. Energy-resolved multiphonon transfer requires nuclear coupling, driven-state replenishment, final-state density, momentum balance, and competing-channel kinetics.

$$\begin{aligned} \Gamma_{\text{other}} &= \Gamma_\gamma + \Gamma_p + \Gamma_n, \\ B_{\text{lat}} &= \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_{\text{other}}}, \\ r_\Gamma &= \Gamma_{\text{lat}}/\Gamma_{\text{other}}, \end{aligned} \quad (76)$$

$$B_{\text{lat}} > 0.999999 \iff r_\Gamma > 999999,$$

$$B_{\text{lat}}^{(2)} = 0.999999996 \quad (\Gamma_p = \Gamma_n = 0).$$

For $\Gamma_{\text{other}} > 0$, the branching bound is conditional on the specified rate ratio. The sufficient condition $r_{\Gamma} \geq 10^6$ gives $B_{\text{lat}} \geq 10^6/(1 + 10^6) > 0.999999$. The nonradiative fraction is B_{lat} .

b) *Planar coherence construction*: The microstructural model is parameterized by

$$\begin{aligned} V_D &= 1.25 \times 10^{-15} \text{ m}^3, & N_D &= 1.84 \times 10^7, \\ V_D^{1/3} &= 10.7722 \text{ } \mu\text{m}, & L_D &\simeq 11.155 \text{ } \mu\text{m}, \\ L_{\text{coh}} &= c_{\text{ss}} \lambda_c = 412 \text{ } \mu\text{m}, \\ V_{\text{domains}} &= N_D V_D = 2.30 \times 10^{-8} \text{ m}^3. \end{aligned} \quad (77)$$

The noncubic dimension L_D is an effective lateral spacing; V_D alone does not determine it without the ellipsoid axes and packing fraction. At unit areal filling, a coherence patch contains $M = A_{\text{coh}}/L_D^2$ domains. The nominal planar coherence factors are

$$\begin{aligned} \Phi_{2D} &= \left(\frac{L_{\text{coh}}}{L_D} \right)^2 \simeq 1364.02, \\ \eta_{\text{coll}} &= \Phi_{2D} N_D = 2.5097968 \times 10^{10} \simeq 2.5098 \times 10^{10}. \end{aligned} \quad (78)$$

Under the *assumption* that M emitters within each patch are fully phase locked with equal amplitudes and different patches are mutually incoherent,

$$\begin{aligned} P_{\text{patch}} &= M^2 P_1, \\ P_{\text{total}} &= (N_D/M) M^2 P_1 = N_D M P_1, \\ P_{\text{incoherent}} &= N_D P_1, \\ P_{\text{total}}/P_{\text{incoherent}} &= M, \quad P_{\text{total}}/P_1 = N_D M. \end{aligned} \quad (79)$$

The factor 2.5098×10^{10} is normalized to *one domain*, not to a volume-integrated independent-emitter rate. A quantitative phase-locking model requires interdomain coupling, phase disorder, propagation loss, dephasing, and drive replenishment. Assignment of active domains to the Cu–Sn bondline is also distinct from assignment to the Pd–Ir catalytic layer; a spatially resolved reaction model is necessary to relate the two inventories.

c) *Rate, dimensions, and active-volume closure*: With $n_d = 6.80 \times 10^{28} \text{ m}^{-3}$, a volume rate is

$$\begin{aligned} w_{\text{pair}} &= \frac{1}{2} n_d^2 \langle \sigma v \rangle_{\text{micro}}, \\ P_{\text{lat}} &= Q_{DD} B_{\text{lat}} V_{\text{react}} w_{\text{pair}}, \\ Q_{DD} &= 23.84 \text{ MeV} = 3.81958909 \times 10^{-12} \text{ J}. \end{aligned} \quad (80)$$

The microscopic rate coefficient is constructed from consistently normalized screened-state transition elements in Section II-I; no independent collective multiplier is applied. The independent-pair reduction assumes a specified pair distribution. Correlated many-body initial states instead require the corresponding many-body transition operator and occupation factors. By contrast, $\frac{1}{2} n_d^2 (c_{\text{ss}} \lambda_c) \sigma v$ has units $\text{m}^{-2} \text{s}^{-1}$ when $c_{\text{ss}} \lambda_c$ denotes a length, so it is not a volumetric rate without additional normalization. The rate comparisons use 1.20×10^{14} and $3.01 \times 10^{24} \text{ m}^{-3} \text{s}^{-1}$. Each rate is integrated over its declared reacting volume.

G. Effective Screening: Dynamic SPP Dielectric Model

a) *Dielectric and rate-solver implementation*: The dielectric solver assembles the composite reciprocal-vector/Floquet matrix from transition energies, occupations, and normalized matrix elements. The cycle-averaged static observable $[\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, 0)$ is selected from the full inverse at zero probe frequency, rather than obtained by inverting only the 00 block. The numerical interface shall return the matrix residual, conditioning estimate, and convergence of the extracted block. Adaptive Gauss–Kronrod integration evaluates the rate functional independently of dielectric inversion. The driven-screening model uses $f_{\text{beat}} \simeq 8.328064 \text{ THz}$ and $\Delta\omega = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$. In a driven population-diagonal approximation, the generalized Adler–Wiser response [66]–[68] is

$$\begin{aligned} \Pi_{GG'}(\mathbf{q}, \omega) &= \frac{g_s}{\Omega} \sum_{\mathbf{k}, nn'} w_{\mathbf{k}} \frac{f_{n\mathbf{k}}^D - f_{n', \mathbf{k}+\mathbf{q}}^D}{\mathcal{E}_{n\mathbf{k}} - \mathcal{E}_{n', \mathbf{k}+\mathbf{q}} + \hbar\omega + i\eta} \\ &\times M_{nn'}^G(\mathbf{k}, \mathbf{q}) [M_{nn'}^{G'}(\mathbf{k}, \mathbf{q})]^*, \\ M_{nn'}^G &= \langle \psi_{n', \mathbf{k}+\mathbf{q}} | e^{i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{r}} | \psi_{n\mathbf{k}} \rangle_{\Omega}. \end{aligned} \quad (81)$$

Here Ω is cell volume, $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$ denotes Brillouin-zone quadrature, $\eta = \hbar\gamma > 0$ has energy units, and $g_s = 2$ only when spin is not included in the band index. Both intraband and interband transitions are retained for a metal. For cell-periodic Bloch factors the phase in the overlap is $e^{i\mathbf{G} \cdot \mathbf{r}}$; retaining the extra $\mathbf{q}$ there would double-count the Bloch phase.

A true periodically driven response also contains Floquet sidebands. With drive period $T_d = 2\pi/\Delta\omega$, the retarded density response and its harmonic projection can be defined without assuming equilibrium occupations:

$$\begin{aligned} \Pi_{GG'}(t, t') &= -\frac{i}{\hbar V} \Theta(t - t') \langle [\hat{\rho}_{\mathbf{q}+\mathbf{G}}(t), \hat{\rho}_{-\mathbf{q}-\mathbf{G}'}(t')] \rangle_D, \\ \Pi_{GG'}^{mn}(\omega) &= \frac{1}{T_d} \int_0^{T_d} dt e^{i(m-n)\Delta\omega t} \\ &\times \int_0^\infty d\tau e^{i(\omega+n\Delta\omega)\tau} \Pi_{GG'}(t, t - \tau). \end{aligned} \quad (82)$$

The driven occupations f^D follow from a Floquet–Boltzmann steady state, and Eq. (81) gives the population-only response approximation. Occupation functions alone do not determine the sideband coherences. In SI units, the dielectric matrix is

$$\begin{aligned} \epsilon_{GG'}^{mn} &= \delta_{GG'} \delta_{mn} - v_c(\mathbf{q} + \mathbf{G}) \Pi_{GG'}^{mn}, \\ v_c(\mathbf{q} + \mathbf{G}) &= \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2}. \end{aligned} \quad (83)$$

The Coulomb kernel $4\pi e^2/q^2$ uses Gaussian units and cannot be combined directly with SI permittivity. The inverse dielectric is obtained by inversion of the coupled reciprocal-vector and harmonic matrix, not by taking the reciprocal of a single diagonal element.

b) *Composite Bloch–Floquet response*: The band indices are a, b , the harmonic indices are m, n, ℓ, p , and the cell-

periodic components are $u_{\mathbf{a}\mathbf{k}}^{(\ell)}$. The population-diagonal microscopic response admits the representation

$$\psi_{\mathbf{a}\mathbf{k}}(\mathbf{r}, t) = e^{i\mathbf{k}\cdot\mathbf{r} - i\mathcal{E}_{\mathbf{a}\mathbf{k}}t/\hbar} \sum_{\ell} u_{\mathbf{a}\mathbf{k}}^{(\ell)}(\mathbf{r}) e^{-i\ell\Omega_D t}, \quad (84)$$

$$M_{ab}^{G,p}(\mathbf{k}, \mathbf{q}) = \sum_{\ell} \int_{\Omega} [u_{b,\mathbf{k}+\mathbf{q}}^{(\ell+p)}]^* e^{i\mathbf{G}\cdot\mathbf{r}} u_{\mathbf{a}\mathbf{k}}^{(\ell)} d^3r.$$

For a consistent harmonic convention, the retarded spectral representation corresponding to (82) is

$$\Pi_{GG'}^{mn}(\mathbf{q}, \omega) = \frac{g_s}{\Omega} \sum_{\mathbf{k}, ab, \ell} w_{\mathbf{k}} \frac{f_{\mathbf{a}\mathbf{k}}^D - f_{b,\mathbf{k}+\mathbf{q}}^D}{\mathcal{E}_{\mathbf{a}\mathbf{k}} - \mathcal{E}_{b,\mathbf{k}+\mathbf{q}} + \hbar(\omega + \ell\Omega_D) + i\eta} \times M_{ab}^{G,\ell-m} [M_{ab}^{G',\ell-n}]^*. \quad (85)$$

A change in Fourier convention relabels the harmonic indices together; the sideband energy denominator and transition harmonics cannot be changed independently. Occupation coherences beyond this approximation require the corresponding density-matrix response. The inverse of (83) obeys

$$\sum_{G'', \ell} \epsilon_{GG''}^{m\ell} [\epsilon^{-1}]_{G''G'}^{\ell n} = \delta_{GG'} \delta_{mn}. \quad (86)$$

For the effective mass $m_e^* = 0.38m_e$, the quiver amplitude and Jacobi–Anger expansion are

$$\mathbf{a}_q = \frac{e\mathbf{E}_0}{m_e^* \Omega_D^2}, \quad \xi_G = (\mathbf{q} + \mathbf{G}) \cdot \mathbf{a}_q, \quad (87)$$

$$e^{i\xi_G \sin(\Omega_D t)} = \sum_{p=-\infty}^{\infty} J_p(\xi_G) e^{ip\Omega_D t}.$$

In a time-averaged quiver approximation, the local-field cross term has weights $J_p(\xi_G)J_p(\xi_{G'})$, reducing to $J_p^2(\xi_G)$ for $G = G'$. This approximation differs from the full harmonic inverse in Eq. (86). Evaluation of the pair potential still requires reciprocal-space integration. For comparison, retaining full SI Coulomb constants rather than the rounded coefficient in Eq. (34) gives the real-space energy ratios

$$V_{\text{bare}} = 51.42730525 \text{ eV},$$

$$\text{Re } W|_{U=350 \text{ eV}} = -298.57269475 \text{ eV}, \quad (88)$$

$$\frac{\text{Re } W}{V_{\text{bare}}} = -5.80572311.$$

The Thomas–Fermi reference $\lambda_{\text{TF}} = 0.231309 \text{ \AA}$ and the driven 350 eV scenario use distinct response models.

c) *Schrieffer–Wolff reduction*: Separation of the deuteron, electronic, and plasmon degrees of freedom gives

$$H = H_0 + H_{\text{int}}, \quad H_0 = H_{dd} + H_e + H_{\text{SPP}},$$

$$H_{e\text{-SPP}} = \sum_{\mathbf{k}, \mathbf{q}, s} M_{\text{SPP}}(\mathbf{q}) (a_{\mathbf{q}} + a_{-\mathbf{q}}^\dagger) c_{\mathbf{k}+\mathbf{q}, s}^\dagger c_{\mathbf{k}, s}. \quad (89)$$

The operators $a_{\mathbf{q}}$ and $c_{\mathbf{k}, s}$ annihilate a plasmon and an electron of spin s ; $M_{\text{SPP}}(-\mathbf{q}) = M_{\text{SPP}}(\mathbf{q})^*$ ensures Hermiticity. The interaction H_{int} must include the deuteron–electron density coupling as well as $H_{e\text{-SPP}}$ to generate a deuteron pair interaction. For an anti-Hermitian generator $\mathcal{A}$ satisfying

TABLE IV
FULL-SI SCREENING COMPARISON AT $r_d = 0.28 \text{ \AA}$

Model	Pair energy (eV)	Shift (eV)
Bare Coulomb	51.427305	0
Static DOS/Yukawa	15.327754	36.099551
Driven assigned scenario	−298.572695	350.0

$[\mathcal{A}, H_0] = -H_{\text{int}}^{\text{off}}$, projection onto the retained subspace $\mathcal{P}$ gives

$$H_{\text{eff}} = \mathcal{P}(H_0 + H_{\text{int}}^{\text{diag}} + \frac{1}{2}[\mathcal{A}, H_{\text{int}}^{\text{off}}] + \dots)\mathcal{P}. \quad (90)$$

This elimination requires off-resonant electronic denominators large compared with the eliminated couplings. Resonant SPP–phonon modes are retained; a resonant electronic sector cannot be eliminated using the same expansion [138].

The time-averaged local-field projection is defined by

$$\mathcal{E}_{00}^{-1}(q, \omega) = [\epsilon^{-1}(q, \omega)]_{G=G'=0}^{m=n=0}. \quad (91)$$

The conservative induced pair energy is

$$V_{\text{ind}}(\mathbf{r}, \omega) = \text{Re} \int \frac{d^3q}{(2\pi)^3} \frac{e^2 e^{i\mathbf{q}\cdot\mathbf{r}}}{\epsilon_0 q^2} [\mathcal{E}_{00}^{-1}(q, \omega) - 1]. \quad (92)$$

Its isotropic angular integral gives

$$V_{\text{ind}}(r, \omega) = \frac{e^2}{2\pi^2 \epsilon_0 r} \times \text{Re} \int_0^\infty \frac{\sin(qr)}{q} [\mathcal{E}_{00}^{-1}(q, \omega) - 1] dq, \quad (93)$$

$$U_{\text{eff}}(\omega) = -V_{\text{ind}}(r_d, \omega),$$

$$r_d = 0.28 \text{ \AA},$$

$$U_{\text{eff}}(\Delta\omega) = 350.0 \text{ eV} \quad (\text{dynamic comparison}).$$

The induced interaction V_{ind} has energy units. The 350 eV screening shift is an assumed driven-model parameter.

d) *Static comparison and driven energy budget*: The density-of-states (DOS) approximation provides a static comparison model rather than the driven-screening ansatz:

$$g_{\text{alloy}}(E_F) = 0.95(1.45) + 0.05(2.85)$$

$$= 1.520 \text{ states eV}^{-1} \text{ atom}^{-1},$$

$$n_{\text{atom}} = 4/(3.89 \times 10^{-10} \text{ m})^3$$

$$= 6.79533994 \times 10^{28} \text{ m}^{-3}, \quad (94)$$

$$g_{\text{SI}} = 1.520 n_{\text{atom}} / (1.602176634 \times 10^{-19})$$

$$= 6.44680274 \times 10^{47} \text{ J}^{-1} \text{ m}^{-3},$$

$$\lambda_{\text{TF}} = \sqrt{\epsilon_0 / (e^2 g_{\text{SI}})} = 0.23130860 \text{ \AA}.$$

At the same separation, the full-SI electrostatic comparison gives

$$V_{\text{bare}} = 51.42730525 \text{ eV},$$

$$U_{\text{TF}} = V_{\text{bare}}(1 - e^{-r_d/\lambda_{\text{TF}}}) = 36.09955080 \text{ eV}, \quad (95)$$

$$V_{\text{driven}} = V_{\text{bare}} - 350.0 = -298.57269475 \text{ eV}.$$

The negative driven pair energy is distinct from nonnegative static Yukawa screening. A causal, energy-balanced driven response includes attraction, dissipation, and pump work; a static upper-bound argument does not apply without modification.

e) *Anisotropic local-field reconstruction*: The susceptibility $\chi_{GG'}$ denotes the independent-particle polarization $\Pi_{GG'}$. Pd 4d/Ir 5d hybridization enters through the band energies, occupations, and Bloch matrix elements rather than an additional numerical multiplier. For a source deuteron at $\mathbf{r}'$ and a test deuteron at $\mathbf{r}$, the full SI screened interaction energy is

$$W(\mathbf{r}, \mathbf{r}'; \omega) = \int_{\text{BZ}} \frac{d^3q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} \times [\epsilon^{-1}(\mathbf{q}, \omega)]_{GG'} \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}'|^2} e^{-i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}'}.$$
 (96)

The source-position phase and Brillouin-zone integration domain prevent double counting of reciprocal-lattice components. For a [110] interstitial pair, $U_{\text{eff}} = e^2/(4\pi\epsilon_0 r_d) - \text{Re } W$. An assumed shift of 350.0 eV therefore gives $\text{Re } W \simeq -298.57$ eV at $r_d = 0.28$ Å, consistent with Eq. (34). This conditional potential [63] requires a complete inverse dielectric matrix and band-resolved occupations for microscopic evaluation [64]. The static $\omega = 0$ integral and the driven $\omega = \Omega_b$ response are distinct quantities.

f) *Floquet sidebands and the inverse dielectric tensor*: The quiver response is characterized by the dimensionless parameter [61]

$$\xi_G = \frac{e\mathbf{E}_0 \cdot (\mathbf{q} + \mathbf{G})}{m_e^* \Omega_D^2},$$
 (97)
$$\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}.$$

Here m_e^* is an effective mass and $\mathbf{E}_0$ is the local drive field. Its proposed time-averaged susceptibility is

$$\chi_{GG'}^{0,\text{rep}}(\mathbf{q}, \omega) = \sum_{p \in \mathbb{Z}} J_p^2(\xi_G) \chi_{GG'}^{0,\text{bare}}(\mathbf{q}, \omega - p\Omega_D).$$
 (98)

For unequal G, G' , a squared weight at only the first vertex does not specify a two-vertex response or a full Floquet tensor. A consistent quiver-phase approximation dresses both density vertices. One explicit harmonic convention is

$$\chi_{GG'}^{mn}(\mathbf{q}, \omega) = \sum_{p \in \mathbb{Z}} J_{m-p}(\xi_G) J_{n-p}(\xi_{G'}) \times \chi_{GG'}^{0,\text{bare}}(\mathbf{q}, \omega + p\Omega_D),$$
 (99)

with nonzero drive phases incorporated into the vertex factors. The diagonal limit $G = G'$, $m = n = 0$ reduces to the J_p^2 average after reindexing. This quiver approximation does not constitute a material-specific interacting-band solution; driven occupations and scattering are independent inputs to the Adler-Wiser response. In composite reciprocal/Floquet space,

$$\epsilon_{GG'}^{mn} = \delta_{GG'} \delta_{mn} - v_G \chi_{GG'}^{mn},$$

$$v_G = \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2},$$
 (100)
$$\sum_{G'', \ell} \epsilon_{GG''}^{m\ell} [\epsilon^{-1}]_{G''G'}^{\ell n} = \delta_{GG'} \delta_{mn}.$$

The static probe response of the periodically driven electronic system is the retarded zero-frequency limit of the full inverse operator,

$$[\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, 0) = \lim_{\omega \rightarrow 0+} \left[(1 - \mathbf{v}\chi(\mathbf{q}, \omega))^{-1} \right]_{GG'}^{00}. \quad (101)$$

Here χ is the Floquet-Adler-Wiser density response, $\mathbf{v}$ is diagonal in reciprocal and Floquet indices, and the superscript 00 selects the elastic, cycle-averaged block after inversion. The static probe frequency is distinct from the nonzero optical drive frequency. For fixed deuteron positions, the local screening requirement is

$$U_{\text{eff}}(\mathbf{r}, \mathbf{r}') = V_{\text{bare}}(\mathbf{r} - \mathbf{r}') - \text{Re } W^{00}(\mathbf{r}, \mathbf{r}'; 0),$$
 (102)
$$U_{\text{eff}}|_{|\mathbf{r}-\mathbf{r}'|=r_d} = 350.0 \text{ eV}, \quad r_d = 0.28 \text{ Å}.$$

The spatial reconstruction of W^{00} follows Eq. (153) at zero probe frequency. The value 350.0 eV is a local design constraint on that reconstruction. The 3D reconstruction in (96) uses the selected $[\epsilon^{-1}]_{GG'}^{mn}$ component; the elastic cycle-averaged interaction uses $m = n = 0$. Inverting a reciprocal-space time average is not the same operation as taking the 00 block of the inverse full Floquet matrix. Spatial and temporal cutoffs must be converged before evaluating $\text{Re } W$. At $r_d = 0.28$ Å, the selected rounded Coulomb coefficient gives

$$V_{\text{bare}} \simeq 51.42714 \text{ eV},$$

$$V_{\text{driven}} = V_{\text{bare}} - 350.0 \text{ eV} \simeq -298.57 \text{ eV},$$
 (103)
$$V_{\text{eff}} = 350.0 - 298.57 = 51.43 \text{ eV}.$$

The values use the energy convention in Eq. (34). The unrounded bare-energy quotient is $14.3996/0.28 = 51.42714286$ eV.

g) *Beam-screening references and local-potential interpretation*: Ion-beam screening comparison values are 310 ± 30 eV in Pd and up to 400 eV in defect-rich targets. Low-energy implanted-deuteron reaction yields in Pd support a fitted screening scale of roughly 310 eV, subject to target-density and interpretation uncertainties [62]. A fitted barrier shift is neither a directly measured pair energy at 0.28 Å nor an evaluated dielectric tensor for a driven 5% Ir alloy. The 400 eV defect-rich comparison does not define a universal upper bound. A tunneling-fit shift cannot be assigned at an arbitrary separation to establish attraction. Floquet d-band dressing remains a proposed microscopic mechanism whose interaction must be evaluated through Eq. (100).

H. Sommerfeld Parameters and Penetration Enhancement

For relative collision energy E , the conventional Sommerfeld parameter and the leading Coulomb penetration factor are defined by [139]

$$\eta_C(E) = Z_1 Z_2 \alpha \sqrt{\frac{\mu c^2}{2E}},$$
 (104)
$$P_G(E) \approx \exp[-2\pi\eta_C(E)].$$

For the numerical convention used in this specification, define

$$\eta(E) \equiv \pi\eta_C(E), \quad P_G(E) \approx e^{-2\eta(E)}. \quad (105)$$

TABLE V
CALCULATED ALLOY SCREENING OPERATING POINTS

Ir (at%)	$g(E_F)$	U_e (eV)	Target (eV)	Tol. (eV)
0.00	1.35	250.0	250.0	± 10.0
5.00	2.55	680.0	680.0	± 15.0
10.00	2.85	920.0	920.0	± 20.0
20.00	3.12	1240.0	1240.0	± 25.0

With $Z_1 = Z_2 = 1$, $\alpha^{-1} = 137.036$, and $\mu c^2 = 937.81$ MeV, the current Gamow-energy convention is

$$E_G = 2\mu c^2 (\pi\alpha)^2 = 985.76922061 \text{ keV}. \quad (106)$$

It follows from (104) that $2\pi\eta_C = \sqrt{E_G/E}$ and $2\eta = \sqrt{E_G/E}$. Therefore

$$P_G(E; U) = \exp[-\sqrt{E_G/(E+U)}], \quad (107)$$

$$\mathcal{E}_G(E; U) = \exp\left[\sqrt{E_G}\left(E^{-1/2} - (E+U)^{-1/2}\right)\right].$$

Table XCII lists the penetration-factor comparison at $U = 350.0$ eV. These are penetration-factor ratios, not probabilities greater than one or reactor power predictions. Ten electronvolts is a selected collision energy, not the thermal energy of room-temperature deuterons; at 550 K, $k_B T = 0.0473953$ eV. A nuclear rate requires a channel-specific cross section, relative-energy distribution, and pair density:

$$R_{DD} = \frac{1}{2} n_D^2 V_{\text{act}} \langle \sigma_{DD} v \rangle. \quad (108)$$

The factor one half counts identical-particle pairs. A one-particle energy distribution is not automatically the relative-collision-energy distribution used in this equation.

a) Alloy density of states and screening normalization:

Table V specifies composition-indexed effective screening operating points. The density of states is in states/(eV atom). For atomic number density n_{atom} , the SI density of states, Thomas–Fermi length, and short-distance screening shift are

$$g_{\text{SI}} = \frac{n_{\text{atom}} g(E_F)}{q_{\text{eV}}}, \quad (109)$$

$$q_{\text{eV}} = 1.602176634 \times 10^{-19} \text{ J/eV},$$

$$\lambda_{\text{TF}} = \sqrt{\frac{\epsilon_0}{e^2 g_{\text{SI}}}},$$

$$U_{\text{TF},0}^{(\text{J})} = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 \lambda_{\text{TF}}},$$

$$U_{\text{TF},0}^{(\text{eV})} = U_{\text{TF},0}^{(\text{J})} / q_{\text{eV}},$$

$$\eta_{\text{eff}}(E) = \eta_C(E) \sqrt{\frac{E}{E + U_e}}.$$

For deuterons, $U_{\text{TF},0} = 14.3996 \text{ eV } \text{\AA} / \lambda_{\text{TF}}$. Lengths of 0.55–0.60 $\text{\AA}$ give approximately 24.0–26.2 eV, and 0.45–0.48 $\text{\AA}$ give approximately 30.0–32.0 eV. The prescribed 600–1400 eV effective-screening envelope in Table V is a separate constitutive assignment.

b) Channel-resolved nuclear amplitudes: For relative nuclear interaction V_{nuc} , the channel amplitude and initial Floquet expansion are

$$\mathcal{M}_{fi}(t) = \int d^3 r_1 d^3 r_2 \Psi_f^* V_{\text{nuc}}(\mathbf{r}_1 - \mathbf{r}_2) \Psi_i(t), \quad (110)$$

$$\Psi_i(t) = \sum_n e^{-i(E_{\text{quasi}} + n\hbar\Omega_D)t/\hbar} \phi_n(\mathbf{r}_1, \mathbf{r}_2).$$

The proton–triton, neutron–helium-3, and radiative helium-4 channels have distinct matrix elements and final-state densities. Their nominal energy scales are 4.03, 3.27, and approximately 23.85 MeV, respectively; a radiative branching estimate of order 10^{-7} is an energy-dependent comparison, not a universal constant. A quasistatic barrier approximation can factor a channel probability as

$$|\mathcal{M}_c|^2 \simeq |\mathcal{M}_{\text{bare},c}|^2 P_{\text{WKB}},$$

$$P_{\text{WKB}} = \exp\left[-2 \int_{\mathcal{B}(E)} \frac{\sqrt{2\mu[V_{\text{barrier}}(r) - E]}}{\hbar} dr\right], \quad (111)$$

$$\mathcal{B}(E) = \{r : V_{\text{barrier}}(r) > E\}.$$

Here $V_{\text{barrier}}(r)$ is the channel interaction profile, not the reconstructed reference V_{eff} of Eq. (34). Only classically forbidden intervals enter this real attenuation exponent. A negative potential at r_d cannot be inserted under a real square root throughout a fixed integration interval. Genuine time-dependent Floquet transmission requires channel flux normalization and sideband scattering. A common multiplicative penetration factor enhances all factored channels equally and does not by itself change their branching ratios.

c) Thermal rate functional and adaptive integration: For nonresonant screened reactions, the Maxwellian rate average is

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi\mu(k_B T)^3}} \int_0^\infty S(E) \exp[-2\pi\eta_{\text{eff}}(E)] \times \exp[-E/(k_B T)] dE. \quad (112)$$

Here $S(E)$ has units of energy times area, and all energies in the integration use one unit system. The 512-bit reference integrator uses an embedded 7-point Gauss/15-point Kronrod pair, bisects intervals with excessive estimated error, and resolves the Gamow peak before accepting convergence. An infinite-tail bound completes the finite integration interval. After normalizing the rate by a specified positive scale R_* , the sum of quadrature and tail error estimates shall be below 10^{-15} .

I. Non-Perturbative Multiphonon Exchange

a) Periodic dielectric drive and exchange kernel: A scalar modulation model for the adjacent dielectric is

$$\epsilon_d(t) = \epsilon_{d0} + 2\epsilon_{d1} \cos(\Omega_F t), \quad \Omega_F = 2\pi f_{\text{beat}}. \quad (113)$$

It generates neighboring temporal sidebands without specifying the microscopic nuclear coupling. A normalized symmetric multiphonon comparison kernel is

$$W_{i \rightarrow f} = \frac{2\pi}{\hbar} |V_{fi}|^2 \sum_{N=-\infty}^{\infty} w_N \delta(E_i - E_f - N\hbar\omega_{\text{ph}}), \quad (114)$$

$$w_N = e^{-g^2} I_N(g^2), \quad \sum_N w_N = 1.$$

Here g is dimensionless and I_N is the modified Bessel function. Symmetry $w_N = w_{-N}$ is not finite-temperature detailed balance; thermal absorption and emission require the occupation-dependent weights of the displaced thermal-state model. A physical nuclear rate also requires a nuclear transition matrix element rather than an electronic matrix element substituted without a coupling derivation.

The non-perturbative Fröhlich/Hagelstein model couples electronic phonon response to nuclear–lattice exchange. The nuclear transition operator is distinct from the electronic Fröhlich coupling in Eq. (54); determination of the electronic coupling does not determine a nuclear transition amplitude [140]. For the nuclear transition operator σ_ℓ^- at site ℓ , the exchange Hamiltonian is

$$\begin{aligned} H_0 &= H_{\text{nuc}} + H_{\text{lat}}, \\ H_{\text{nuc}} &= \sum_{\ell} Q_{\ell} \sigma_{\ell}^+ \sigma_{\ell}^-, \\ H_{\text{lat}} &= \sum_{\mathbf{q}} \hbar \omega_{\mathbf{q}} \left(b_{\mathbf{q}}^{\dagger} b_{\mathbf{q}} + \frac{1}{2} \right), \\ V_{\text{nl}} &= \sum_{\ell, \mathbf{q}} (g_{\ell \mathbf{q}} \sigma_{\ell}^- + g_{\ell \mathbf{q}}^* \sigma_{\ell}^+) (b_{\mathbf{q}} + b_{\mathbf{q}}^{\dagger}), \\ H &= H_0 + V_{\text{nl}}. \end{aligned} \quad (115)$$

The coupling $g_{\ell \mathbf{q}}$ has units of energy. The two-level interaction describes reduced nuclear–lattice exchange with a displaced-oscillator factorization approximation.

For the proposed lattice-assisted branch,

$$D + D + \mathcal{L}_i \longrightarrow {}^4\text{He} + \mathcal{L}_f, \quad Q_{DD} \approx 23.84 \text{ MeV}, \quad (116)$$

where $\mathcal{L}_i$ and $\mathcal{L}_f$ denote lattice states. Assigning all released energy to the nominal optical phonons and neglecting recoil gives

$$\begin{aligned} \bar{N} &= \frac{Q_{DD}}{E_{\text{LO}}} = \frac{23.84 \times 10^6 \text{ eV}}{0.034450 \text{ eV}} \\ &= 6.9201742 \times 10^8 \approx 6.920 \times 10^8. \end{aligned} \quad (117)$$

b) Franck–Condon factorization and Huang–Rhys parameter: Under the real-displacement convention $D(-d) = \exp[-d(b^{\dagger} - b)]$, the specified coupling gives

$$d = \frac{g_E}{\hbar \omega_0} \approx 0.361392, \quad S = d^2 = 0.130604286. \quad (118)$$

The magnitude of the overlap between harmonic number states is

$$|F_{n,n'}| = e^{-S/2} \sqrt{\frac{n_{\min}!}{n_{\max}!}} |d|^{n-n'} \left| L_{n_{\min}}^{n-n'}(S) \right|. \quad (119)$$

Signed amplitudes depend on the displacement convention and ordering of the states; the magnitude formula avoids assigning the same sign to both directions. In particular,

$$\begin{aligned} \mathcal{M}_{fi}^{(N_q)} &= \mathcal{M}_{0,\text{nuc}} F_{0,N_q}, \\ |F_{0,N_q}|^2 &= e^{-S} \frac{S^{N_q}}{N_q!}, \\ \sum_{N_q=0}^{\infty} |F_{0,N_q}|^2 &= 1. \end{aligned} \quad (120)$$

The nuclear energy corresponds to $N_q \simeq 6.92 \times 10^8$, whereas the squared displacement is 0.1306. A large coherent occupation does not justify replacement of S by N_q . The transition rate and a numerically stable logarithmic overlap expression are

$$\begin{aligned} \Gamma_{\text{lat}}^{(N_q)} &= \frac{2\pi}{\hbar} |\mathcal{M}_{fi}^{(N_q)}|^2 \rho_f(E_i), \\ \ln |F_{0,N_q}|^2 &= -S + N_q \ln S - \ln \Gamma(N_q + 1). \end{aligned} \quad (121)$$

The function $\Gamma(N_q + 1)$ is the gamma function. The ground-state overlap remains strongly suppressed at the specified phonon count. The populated-mode calculation uses the density matrix and overlaps $F_{n,n'}$.

c) Energy and crystal momentum conservation: An individual event has integer occupations and satisfies

$$Q_{DD} = E_{\text{recoil}} + \sum_{\mathbf{q}} N_{\mathbf{q}} \hbar \omega_{\mathbf{q}} + E_{\gamma}, \quad N_{\mathbf{q}} \in \mathbb{N}_0. \quad (122)$$

The candidate nonradiative branch has $E_{\gamma} = 0$. Its crystal-momentum selection rule is

$$\mathbf{K}_i = \mathbf{K}_{\text{He}} + \sum_{\mathbf{q}} N_{\mathbf{q}} \mathbf{q} + \mathbf{G}, \quad (123)$$

where $\mathbf{K}_i$ and $\mathbf{K}_{\text{He}}$ are the initial and final nuclear crystal wavevectors and $\mathbf{G}$ is a reciprocal-lattice vector. Normal processes have $\mathbf{G} = 0$; Umklapp processes transfer momentum to the crystal. Total mechanical momentum additionally includes crystal/support recoil and, where exchanged, pump momentum. Counterpropagating phonon populations can carry a small net momentum without requiring each phonon to have a small wavevector.

d) Klemens-type three- and four-phonon relaxation:

Cubic and quartic anharmonic interactions permit optical-to-acoustic decay channels of the form

$$\begin{aligned} \text{LO}(\mathbf{q}_0) &\longrightarrow \text{LA}(\mathbf{q}_1) + \text{TA}(\mathbf{q}_2), \\ \text{LO}(\mathbf{q}_0) &\longrightarrow \text{A}_1(\mathbf{q}_1) + \text{A}_2(\mathbf{q}_2) + \text{A}_3(\mathbf{q}_3), \end{aligned} \quad (124)$$

where each A_j is an allowed acoustic branch. These channels require $\omega_0 = \sum_j \omega_j$ and $\mathbf{q}_0 = \sum_j \mathbf{q}_j + \mathbf{G}$; their availability depends on the material dispersion. In an equal-energy daughter-mode approximation, the population-relaxation rate can be parameterized as

$$\begin{aligned} n_{\text{B}}(\omega, T) &= [e^{\hbar \omega / (k_{\text{B}} T)} - 1]^{-1}, \\ \Gamma_3(T) &= A_3 [1 + 2n_{\text{B}}(\omega_0/2, T)], \\ \Gamma_4(T) &= A_4 [1 + 3n_{\text{B}}(\omega_0/3, T) \\ &\quad + 3n_{\text{B}}^2(\omega_0/3, T)], \\ \Gamma_{\text{anh}} &= \Gamma_3 + \Gamma_4. \end{aligned} \quad (125)$$

The coefficients A_3, A_4 have units of s^{-1} and contain the anharmonic matrix elements and allowed phase space. The thermal occupation factors describe relaxation toward equilibrium, not the initial strongly driven distribution [141]. For the spin-boson interface parameterization, the Lindblad coefficient determines the population rate:

$$\begin{aligned} \Gamma_{\text{anh}}^{(\text{new})} &= 2\kappa = 9.0 \times 10^{10} \text{ s}^{-1}, \\ \tau_{\text{population}} &= (2\kappa)^{-1} = 11.1111 \text{ ps}, \\ \tau_{\text{amplitude}} &= \kappa^{-1} = 22.2222 \text{ ps}. \end{aligned} \quad (126)$$

TABLE VI
NUCLEAR REACTION CHANNELS AND RELEASED ENERGIES

Channel	Q (MeV)
$D + D \rightarrow {}^4\text{He} + \text{lattice}$	23.84
$D + D \rightarrow p + t$	4.03
$D + D \rightarrow n + {}^3\text{He}$	3.27
$D + D \rightarrow {}^4\text{He} + \gamma$	23.84

The comparison population rate $4.2 \times 10^{11} \text{ s}^{-1}$ and lifetime 2.38 ps describe a distinct material parameterization and cannot simultaneously define relaxation in the spin-boson interface model. Anharmonic transfer to acoustic modes provides a thermalization pathway but does not determine nuclear branching or total energy retention.

e) Channel-resolved branching and coherent kinetics:

For phase-correlated sites, the golden-rule transition rate is [66]

$$\Gamma_{\text{lat}} = \frac{2\pi}{\hbar} \left| \sum_{j=1}^{N_D} \mathcal{M}_j \right|^2 \rho_f(Q_1). \quad (127)$$

Equal phases and equal site amplitudes give $N_D^2 |\mathcal{M}_1|^2$, whereas incoherent averaging gives $N_D |\mathcal{M}_1|^2$. Emission into the lattice produces the final occupation $n_{\text{coh}} + N_q$; the alternative $n_{\text{coh}} - N_q$ corresponds to phonon removal. Crystal momentum obeys Eq. (123). An assumed ratio $\Gamma_{\text{lat}}/(\Gamma_p + \Gamma_n) = 1.45 \times 10^{14}$ implies

$$B_{\text{lat}}^{(\gamma=0)} = \frac{1.45 \times 10^{14}}{1 + 1.45 \times 10^{14}} \approx 1 - 6.89655 \times 10^{-15},$$

$$B_{\text{lat}} = \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_p + \Gamma_n + \Gamma_\gamma}, \quad (128)$$

$$B_{\text{lat}} > 0.999999 \iff \frac{\Gamma_p + \Gamma_n + \Gamma_\gamma}{\Gamma_{\text{lat}}} < \frac{10^{-6}}{1 - 10^{-6}}.$$

The branching target $B_{\text{lat}} > 0.999999$ is conditional on the assumed channel-rate ratio.

Ensemble and single-state rates have distinct normalizations:

$$R_{\text{ensemble}} = N_{\text{eff}} \Gamma_{\text{single}},$$

$$N_{\text{atoms}} = n_{\text{atom}} (4.0 \times 10^{-13} \text{ m}^3) = 2.71813598 \times 10^{16}. \quad (129)$$

The comparison coherent population $N_D = 10^{20}$ and ensemble flux $8.12 \times 10^{19} \text{ s}^{-1}$ use a separate population and volume definition. Complete lattice deposition gives the heat-equivalent reaction rates

$$R_{10\text{kW}} = 10^4 / Q_{DD} = 2.61808267 \times 10^{15} \text{ s}^{-1},$$

$$R_{15\text{kW}} = 1.5 \times 10^4 / Q_{DD} = 3.92712400 \times 10^{15} \text{ s}^{-1}. \quad (130)$$

f) Driven distribution and cross-section integral: The mixture defines a normalized *relative-energy* distribution, with

$\theta = k_B T$ expressed in the same energy unit as E :

$$f_D(E) = (1 - \alpha_h) \frac{2\sqrt{E}}{\sqrt{\pi}\theta^{3/2}} e^{-E/\theta} + \alpha_h \delta(E - E_h),$$

$$\alpha_h = 0.015, \quad E_h = 120.0 \text{ eV}, \quad T = 550.0 \text{ K},$$

$$\sigma_{DD}(E) = \frac{S(E)}{E} \exp[-\sqrt{E_G/(E + 350 \text{ eV})}], \quad (131)$$

$$S(E) = 55 \text{ keV b} \quad (\text{constant-}S \text{ approximation}),$$

$$\langle \sigma v \rangle = \int_0^\infty \sigma_{DD}(E) \sqrt{2E/\mu} f_D(E) dE.$$

Using $1 \text{ b} = 10^{-28} \text{ m}^2$ and joules in the velocity factor, quadrature gives a Maxwell component $3.54441 \times 10^{-36} \text{ cm}^3 \text{ s}^{-1}$ and a monoenergetic component $8.96581 \times 10^{-35} \text{ cm}^3 \text{ s}^{-1}$ before mixture weighting. Thus

$$\langle \sigma v \rangle_{\text{calc}} = 4.83611 \times 10^{-36} \text{ cm}^3 \text{ s}^{-1}. \quad (132)$$

The constant- S model is a prescribed screening approximation using channel-specific cross sections and population normalization.

g) Populated-state overlap and trace identity: For a final state f , the transition amplitude is $A_{fn} = \langle f | \hat{O} | n \rangle$. The populated-state probability and energy-resolved Floquet transition rate take the form

$$p_f = \sum_{nn'} \rho_{nn'} A_{fn} A_{fn'}^*,$$

$$W = \frac{2\pi}{\hbar} \sum_{f,\ell} |H_{\text{nuc},f}|^2 p_{f,\ell} \delta(E_f - E_i - Q_{DD} - \ell \hbar \Omega_b). \quad (133)$$

Here E_i, E_f denote nonnuclear excitation energies, with Q_{DD} the released nuclear energy; $p_{f,\ell}$ includes populations/coherences and transition harmonics, with a final-state integral for a continuum. The delta function supplies inverse-energy units. Unlike an unweighted displacement trace, this expression imposes energy conservation and retains the nuclear channels. For real q_0 and $\hat{x} = x_{\text{zpf}}(a + a^\dagger)$, exactly

$$D^\dagger(\alpha) \hat{x} D(\alpha) = \hat{x} + 2x_{\text{zpf}} \text{Re } \alpha,$$

$$\text{Tr}[\rho_{\text{th}} D^\dagger e^{iq_0 \hat{x}} D e^{-iq_0 \hat{x}}] = e^{2iq_0 x_{\text{zpf}} \text{Re } \alpha}. \quad (134)$$

The corresponding nonnegative expression is the square of the real part of the displaced thermal *characteristic function*:

$$\chi_x(q_0) = \text{Tr}[\tilde{\rho}_{\text{ss}} e^{iq_0 \hat{x}}]$$

$$= e^{-q_0^2 x_{\text{zpf}}^2 (2n_{\text{th}} + 1)/2} e^{2iq_0 x_{\text{zpf}} \text{Re } \alpha}, \quad (135)$$

$$[\text{Re } \chi_x(q_0)]^2 = e^{-q_0^2 x_{\text{zpf}}^2 (2n_{\text{th}} + 1)} \times \cos^2(2 \text{Re } \alpha q_0 x_{\text{zpf}}).$$

The cosine equals unity at exact resonance, where $\alpha_- = i1.245$. Equation (133) defines the energy-resolved populated-state rate.

h) Maxwellian screened reaction rate: The purely Maxwellian case has $\alpha_h = 0$ in Eq. (131). Under a fixed-incident-flux convention, the screened rate is

$$\begin{aligned}
\sigma_s(E) &= \frac{S(E+U)}{E} e^{-\sqrt{E_G/(E+U)}}, \\
\langle \sigma v \rangle_M &= \sqrt{\frac{8}{\pi \mu (k_B T)^3}} \int_0^\infty S(E+U) \\
&\quad \times e^{-\sqrt{E_G/(E+U)} - E/(k_B T)} dE, \\
U &= 350 \text{ eV}, \quad S = 55 \text{ keV b}, \quad T = 550 \text{ K}, \\
\langle \sigma v \rangle_M &\simeq 3.5444 \times 10^{-36} \text{ cm}^3/\text{s}.
\end{aligned} \tag{136}$$

Numerical evaluation uses SI energies and the mass prefactor before conversion from m^3/s to cm^3/s . The fixed-flux convention is $\sigma_s(E) = [(E+U)/E] \sigma_{\text{bare}}(E+U)$.

i) *Fock projections and channel rates:* For $\bar{n} = n_{\text{th}} = 0.9358$, $|\alpha| = 1.245$, and the displacement parameter $S = 0.130604$,

$$\begin{aligned}
\rho_{\text{th}} &= \sum_{j=0}^{\infty} \frac{\bar{n}^j}{(1+\bar{n})^{j+1}} |j\rangle \langle j|, \\
\tilde{\rho}_{nn'} &= \sum_j \frac{\bar{n}^j}{(1+\bar{n})^{j+1}} \langle n | D(\alpha) | j \rangle \langle n' | D(\alpha) | j \rangle^*, \\
p_n &= \tilde{\rho}_{nn} = \frac{\bar{n}^n}{(1+\bar{n})^{n+1}} e^{-|\alpha|^2/(1+\bar{n})} L_n \left(-\frac{|\alpha|^2}{\bar{n}(1+\bar{n})} \right).
\end{aligned} \tag{137}$$

The off-diagonal entries are coherences; p_n alone is a population. Both follow from the same normalized state, not a replacement of small probabilities by unity. Its generating function provides a stable large-phonon-number test:

$$\begin{aligned}
\mathcal{P}(z) &= \sum_n p_n z^n \\
&= \frac{\exp[|\alpha|^2(z-1)/(1+\bar{n}(1-z))]}{1+\bar{n}(1-z)}, \\
\Pr(n \geq N_q) &\leq \mathcal{P}(z) z^{-N_q}, \quad 1 < z < 1 + 1/\bar{n}, \\
\log_{10} \Pr(n \geq N_q) &\leq -1.21855150 \times 10^8, \\
N_q &= 6.92 \times 10^8, \quad z = 1.5.
\end{aligned} \tag{138}$$

The oscillator model gives a strongly suppressed ground-state Franck–Condon overlap. Collective-transfer models use energy-resolved matrix elements and the populated-state density operator.

$$\begin{aligned}
\dot{\rho} &= -\frac{i}{\hbar} [H_{\text{sys}} + H_{\text{drive}}, \rho] + \mathcal{L}_{\text{lat}} \rho + \mathcal{L}_{\text{nuc}} \rho, \\
\Gamma_{\text{lat}} &= \frac{2\pi}{\hbar} |V_{\text{nuc-lat}}|^2 \sum_n p_n \delta(E_{\text{nuc}} - n\hbar\Omega_D).
\end{aligned} \tag{139}$$

The delta function defines a spectral distribution rather than a finite rate at a single integer n ; numerical rates require a linewidth or final-state density. Table VII specifies a conditional set of channel rates. Conditioned on those rates and a common initial population,

$$\begin{aligned}
B_{\text{lat}} &= \frac{8.4 \times 10^{14}}{8.4 \times 10^{14} + 618000} = 0.9999999992642857, \\
1 - B_{\text{lat}} &= 7.35714285 \times 10^{-10}.
\end{aligned} \tag{140}$$

TABLE VII
NUCLEAR CHANNEL-RATE REFERENCE ASSIGNMENTS

Channel	Q (MeV)	Bare (s^{-1})	Driven (s^{-1})
Lattice, ^4He	23.84	1.2×10^{-12}	8.4×10^{14}
$p + t$	4.03	2.5×10^4	1.1×10^5
$n + ^3\text{He}$	3.27	2.4×10^4	9.8×10^4
γ	23.84	4.1×10^5	4.1×10^5

The conditional branching fraction is approximately 0.99999999927. Channel-rate enhancements use relative penetrabilities. The channel rates define a conditional simulation case; radiation-protection requirements remain unchanged.

j) *Normalized multiphonon probability:* The coherent amplitude β in Eq. (58) is distinct from the dimensionless inverse temperature $\beta_T = \hbar\omega_{\text{LO}}/(k_B T)$. For integer net phonon emission $p \in \mathbb{Z}$, the generating identity [27]

$$\sum_{p=-\infty}^{\infty} I_p(z) t^p = \exp \left[\frac{z}{2} (t + t^{-1}) \right] \tag{141}$$

with $z = S/\sinh(\beta_T/2)$ and $t = e^{\beta_T/2}$ establishes the normalized displaced-oscillator thermal reference

$$\begin{aligned}
P(p) &= e^{-S \coth(\beta_T/2)} I_p \left(\frac{S}{\sinh(\beta_T/2)} \right) e^{p\beta_T/2}, \\
\sum_p P(p) &= 1, \quad \langle p \rangle = S.
\end{aligned} \tag{142}$$

At the 623.15 K operating point and $\hbar\omega_{\text{LO}} = 34.442105 \text{ meV}$,

$$\begin{aligned}
\beta_T &\simeq 0.64139298, \quad \bar{n}_{\text{ph}} = (e^{\beta_T} - 1)^{-1} \simeq 1.1121931, \\
S &= 0.130604, \quad P(-p) = e^{-p\beta_T} P(p).
\end{aligned} \tag{143}$$

The Bessel weights assume harmonic modes, equal initial/final mode frequencies, a displacement coupling, and a thermal initial population. The coherently driven state $D(\alpha_-) \rho_{\text{th}} D^\dagger(\alpha_-)$ requires its two-time correlation function; the equilibrium Bessel weights are exact within their oscillator assumptions, not a universal nonequilibrium nuclear rate. The zero-temperature emission limit is $e^{-S} S^p/p!$ for $p \geq 0$. The formal choice $S \simeq \bar{N} = 6.920 \times 10^8$ would center the overlap at the energy-equivalent phonon count, whereas the specified coupling gives $S = 0.130604286$. A normalized transition probability admits neither an unbounded e^S enhancement nor independent suppression of competing nuclear channels. A nonequilibrium driven lattice requires its actual density matrix and channel-resolved transition elements. The conditional rates in Table VI therefore do not justify reduced radiation-protection or containment requirements.

1) *Micro-to-Macro Coherence and Localization-Volume Scaling:* The collective transition model starts from a driven Dicke-type Hamiltonian with dimensionless transition pseu-

dospins [81]:

$$\begin{aligned}\hat{H} &= \sum_{\mathbf{q},\lambda} \hbar\omega_{\mathbf{q}\lambda} \hat{a}_{\mathbf{q}\lambda}^\dagger \hat{a}_{\mathbf{q}\lambda} + \hbar\omega_0 \hat{S}_z + \hat{H}_{\text{drive}}(t) \\ &+ \sum_{\mathbf{q},\lambda} \left(g_{\mathbf{q}\lambda} \hat{a}_{\mathbf{q}\lambda} \hat{S}_+ + g_{\mathbf{q}\lambda}^* \hat{a}_{\mathbf{q}\lambda}^\dagger \hat{S}_- \right), \\ \hat{S}_\pm &= \sum_{j=1}^{N_d} \hat{s}_{\pm,j} e^{\pm i\mathbf{k}_L \cdot \mathbf{r}_j}, \quad \hat{S}_z = \sum_j \hat{s}_{z,j}.\end{aligned}\quad (144)$$

The coefficients $g_{\mathbf{q}\lambda}$ have energy units. A common collective operator assumes mode-compatible site couplings; otherwise the interaction must retain $g_{\mathbf{q}\lambda,j}$ and the mode-dependent spatial phases. These pseudospins label transitions, not an automatic identification with the intrinsic spin-one deuteron. The rotating-wave coupling is a model approximation.

For equal local oscillator frequencies and a common definition of localization volume, let

$$\begin{aligned}V_{\text{zp},s} &\propto \left(\frac{\hbar}{m_s \omega_{\text{local}}} \right)^{3/2}, \\ \left(\frac{V_{\text{zp},d}}{V_{\text{zp},e}} \right)^2 &= \left(\frac{m_e}{m_d} \right)^3, \\ m_e &= 9.1093837 \times 10^{-31} \text{ kg}, \\ m_d &= 3.3435837 \times 10^{-27} \text{ kg}, \\ \left(\frac{m_e}{m_d} \right)^3 &= 2.02222925324 \times 10^{-11}.\end{aligned}\quad (145)$$

Equal local frequencies and an interaction proportional to the squared localization-volume ratio are assumptions of this scaling model.

The proposed geometry and boundary factors yield the stated product

$$\begin{aligned}\eta_{\text{geom}} &= (a_{\text{Pd}}/a_{\text{alloy}})^3, \\ \chi_{\text{CFT}} &= 1.000438, \\ C_{\text{coh}} &= \left(\frac{m_e}{m_d} \right)^3 \eta_{\text{geom}} \chi_{\text{CFT}}, \\ C_{\text{coh,rep}} &= (2.022229 \times 10^{-11})(1.003403)(1.000438) \\ &= 2.02999939575 \times 10^{-11} \simeq 2.0300 \times 10^{-11}.\end{aligned}\quad (146)$$

The Affleck–Ludwig boundary g -factor characterizes a boundary partition function [29]. Here $\chi_{\text{CFT}} = 1.000438$ is an assigned boundary normalization, and Eq. (146) defines the proposed scaling ansatz.

To distinguish a rate enhancement from a number of coherent emitters, write the proposed macroscopic factor as

$$\mathcal{N}_{\text{coh}} = \rho_{\text{active}} V_{\text{coh}} e^{-\sigma_\phi^2/2} \left(\frac{\lambda_L}{a_{\text{lattice}}} \right)^3 C_{\text{coh}}. \quad (147)$$

The 1064 nm comparison uses $\rho_{\text{active}} = 4.80 \times 10^{22} \text{ cm}^{-3}$, $V_{\text{coh}} = 1.25 \times 10^{-12} \text{ cm}^3$, $\sigma_\phi = 0.018 \text{ rad}$, $a_{\text{lattice}} = 0.389 \text{ nm}$, and $C_{\text{coh}} = 2.03 \times 10^{-11}$. This wavelength and volume are a separate comparison from the 785.0/802.5 nm pump geometry. A Gaussian phase average can give $\langle e^{i\phi} \rangle = e^{-\sigma_\phi^2/2}$ at the amplitude level; its use in a rate requires the appropriate two-site phase correlation. The many-body enhancement must

ultimately be evaluated from terms such as $\sum_{ij} g_i^* g_j \langle s_{+,i} s_{-,j} \rangle$ relative to a specified incoherent reference. It is not an additional multiplier on a rate already computed from that same collective density matrix.

a) Coherent-domain geometry and population bounds:

The geometrical volume and host count are

$$\begin{aligned}V_{\text{act}} &= (1.0 \times 10^{-9})(20.0 \times 10^{-3})^2 = 4.0 \times 10^{-13} \text{ m}^3, \\ N_{\text{host}} &= (6.79534 \times 10^{28}) V_{\text{act}} = 2.718136 \times 10^{16}.\end{aligned}\quad (148)$$

For $L_{\text{spp}} = 208.62 \text{ nm}$ and $\delta_z = 149.92 \text{ nm}$, the optical field volume is

$$\begin{aligned}V_{\text{field}} &= L_{\text{spp}}^2 \delta_z = 6.52486388 \times 10^{-21} \text{ m}^3, \\ \frac{V_{\text{act}}}{V_{\text{field}}} &= 6.13039609 \times 10^7 \simeq 6.1305 \times 10^7.\end{aligned}\quad (149)$$

The decay depth extends beyond the 1 nm host layer. A geometrical tiling of that layer instead has

$$\begin{aligned}V_{\text{film, domain}} &= L_{\text{spp}}^2 t_{\text{film}} = 4.35223044 \times 10^{-23} \text{ m}^3, \\ N_{\text{film, domains}} &= 9.19068982 \times 10^9, \\ N_{\text{host, domain}} &= 2.95748856 \times 10^6.\end{aligned}\quad (150)$$

These geometric counts do not measure coherence length. With the loading convention $x = [D]/[Pd]$, the deuterium population is $N_D = 0.95xN_{\text{host}}$; full $x = 1$ loading therefore gives 2.5822292×10^{16} deuterons rather than one deuteron per alloy atom. For mutually incoherent domains with internal phase coherence, the collective-rate ansatz is

$$R_{\text{total}} = \sum_d \Gamma_{\text{single},d} N_{D,d}^2. \quad (151)$$

The fully coherent, identical-amplitude domain ansatz assumes population inversion, phase coherence, energy-resolved matrix elements, and the specified normalization of $\Gamma_{\text{single},d}$.

b) Dielectric response, branching, and absolute reaction rates: The quiver-sideband approximation extends Eq. (85) through the harmonic convolution

$$\chi_{GG'}^{mn}(\mathbf{q}, \omega) = \sum_\ell J_{m-\ell}(\xi_G) J_{n-\ell}(\xi_{G'}) \chi_{GG'}^0(\mathbf{q}, \omega + \ell\Omega_D), \quad (152)$$

where $\xi_G = e\mathbf{E}_0 \cdot (\mathbf{q} + \mathbf{G})/(m_e^* \Omega_D^2)$ and $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$. The dielectric inverse is taken over the composite (G, m) index as in (83) and (86), not entrywise. With both source and observation positions retained, its cycle-averaged interaction is

$$\begin{aligned}\text{Re } W(\mathbf{r}, \mathbf{r}'; \omega) &= \text{Re} \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r}} \\ &\times [\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, \omega) \frac{e^2 e^{-i(\mathbf{q} + \mathbf{G}') \cdot \mathbf{r}'}}{\epsilon_0 |\mathbf{q} + \mathbf{G}'|^2}.\end{aligned}\quad (153)$$

The Thomas–Fermi shift uses $\lambda_{\text{TF}} = 0.231308596 \text{ \AA}$ and a bare energy of 51.427305 eV. The driven pair energy uses the prescribed subtraction 51.427305 – 350.000000. The corresponding force is $\mathbf{F} = -\nabla \text{Re } W$.

TABLE VIII
FULL-SI SCREENING REFERENCE AND CONDITIONAL PAIR ENERGIES

Model	Pair energy (eV)	Shift (eV)
Bare Coulomb	51.427305	0
Static TF comparison	15.327754	36.099551
Driven scenario	-298.572695	350.000000

c) *Displaced thermal state and energy-resolved branching:*

For a linearly driven harmonic mode with thermal damping, the rotating-frame stationary state has the displaced thermal form

$$\begin{aligned}\tilde{\rho}_{ss} &= D(\alpha_-)\rho_{th}D^\dagger(\alpha_-), \\ D(\alpha_-) &= \exp(\alpha_-b^\dagger - \alpha_-^*b), \\ \rho_{th} &= \sum_{n=0}^{\infty} \frac{n_{th}^n}{(1+n_{th})^{n+1}} |n\rangle\langle n|, \\ \text{Tr } \tilde{\rho}_{ss} &= 1, \quad \tilde{\rho}_{ss} \succeq 0, \\ \text{Tr}(b^\dagger b \tilde{\rho}_{ss}) &= n_{th} + |\alpha_-|^2.\end{aligned}\quad (154)$$

Here b is the lattice-mode annihilation operator and α_- is its dimensionless coherent displacement. Unitarity of D preserves normalization, positivity, and the thermal-state eigenvalues. With $|\alpha_-| = 1.245$ and $n_{th} \simeq 0.935813$, the mean occupation is approximately 2.485838. The energy-resolved rate has the form

$$\Gamma_{lat} = \frac{2\pi}{\hbar} \sum_{i,i',f} \rho_{ii'} V_{fi} V_{fi'}^* \delta(E_{DD} + E_i - E_f), \quad (155)$$

with a physically normalized final-state measure and compatible treatment of coherences and energy resolution. The assumed ratio $r_\Gamma = \Gamma_{other}/\Gamma_{lat} = 6.89655 \times 10^{-15}$ implies

$$B_{lat} = \frac{1}{1+r_\Gamma} \simeq 1 - 6.89655 \times 10^{-15} > 0.999999. \quad (156)$$

This conditional branching relation neither evaluates transition matrix elements nor removes the overlap constraints. Individual proton/neutron bounds below 10^{-12} and a gamma bound below 6.897×10^{-15} are upper limits rather than resolved channel rates; they do not independently determine their summed ratio. Branching fractions weight event counts, whereas the thermal fraction also depends on the energy released and retained in each channel. For nonnegative channel rates and $\Gamma_{lat} > 0$, the exact branching criterion is

$$B_{lat} > 0.999999 \iff \frac{\Gamma_{other}}{\Gamma_{lat}} < \frac{10^{-6}}{1-10^{-6}}. \quad (157)$$

The assumed ratio in Eq. (156) satisfies the specified branching target. The lattice and competing channels use their energy-conserving matrix-element sums.

The field volume $6.524864 \times 10^{-21} \text{ m}^3$ and volume ratio 6.1304×10^7 satisfy Eq. (149); the distinction between the optical field volume and the 1 nm film domain in Eq. (150) remains essential. The loading convention $[D]/[\text{metal}] = 0.95$ differs

from $[D]/[Pd]$. The specified pair density and population-weighted per-pair rate give

$$\begin{aligned}n_{\text{pair}} &= 3.228 \times 10^{28} \text{ m}^{-3}, \\ w_{\text{pair}} &= 2.36 \times 10^{-12} \text{ s}^{-1}, \\ R_{\text{vol}} &= n_{\text{pair}} w_{\text{pair}} = 7.61808 \times 10^{16} \text{ m}^{-3} \text{ s}^{-1}, \\ R_{\text{total}} &= R_{\text{vol}}(4.0 \times 10^{-13}) = 30472.32 \text{ s}^{-1}, \\ Q_{DD} &= 23.84 \times 10^6 (1.602176634 \times 10^{-19}) \\ &= 3.819589095456 \times 10^{-12} \text{ J},\end{aligned}\quad (158)$$

$$P_{\text{nuclear,calc}} = R_{\text{total}} Q_{DD} = 1.16391741 \times 10^{-7} \text{ W}.$$

The 2911.40 W thermal reference requires $7.62228587 \times 10^{14}$ events/s, or $w_{\text{pair}} = 0.0590325734 \text{ s}^{-1}$ at the stated pair density and film volume.

J. *Boundary-State Kinetics and Collective-Rate Normalization*

The boundary kinetic model separates the independent-pair reaction rate from the dimensionless collective gain [86]. For a collision energy E , its weak-screening cross-section approximation is

$$\sigma_s(E) = \frac{S(E)}{E} \exp[-2\pi\eta(E)] \exp\left[\pi\eta(E) \frac{U_e}{E}\right], \quad (159)$$

where $S(E)$ is the astrophysical factor, $\eta(E)$ is the Sommerfeld parameter, and U_e is the effective barrier shift. The last exponential is the first-order expansion of the Gamow exponent for $U_e/E \ll 1$. The evaluation energy $E = 350 \text{ eV}$ is a collision-energy reference, not the material temperature or the conventional Gamow-energy parameter. At $U_e = 350 \text{ eV}$, the ratio $U_e/E = 1$ lies outside the weak-screening regime; quantitative reactivity requires the full screened interaction and an explicitly normalized relative-velocity distribution. A fitted barrier shift U_e is not automatically identical to the local potential constraint U_{eff} .

For identical deuterons, unordered-pair counting gives

$$\begin{aligned}R_{\text{micro}} &= \frac{1}{2} n_d^2 V_{\text{rate}} \langle \sigma v \rangle_s, \\ P_{\text{micro}} &= Q_{DD} R_{\text{micro}}, \\ n_d &= 6.0 \times 10^{28} \text{ m}^{-3}, \\ V_{\text{rate}} &= 1.42 \times 10^{-6} \text{ m}^3, \\ P_{\text{micro,ref}} &= 1.1639 \times 10^{-7} \text{ W}.\end{aligned}\quad (160)$$

The alternative notation $R_{\text{micro}} = n_d^2 V_{\text{rate}} \langle \sigma v \rangle_{\text{screened}}$ is equivalent only when $\langle \sigma v \rangle_{\text{screened}} = \langle \sigma v \rangle_s / 2$. At the stated density and volume, the reference power requires $\langle \sigma v \rangle_s = 1.19216996 \times 10^{-47} \text{ m}^3/\text{s}$. The comparison bulk volume V_{rate} is distinct from the energy V_{eff} , the selected envelope volume in Eq. (174), and the film volume in Eq. (158). These population models cannot share a per-pair rate without an explicit spatial mapping.

a) *Boundary correlation and gain definition:* The SHBT boundary description adopts a conformal-field-theory (CFT) representation with a nominal inventory $N_{\text{coh}} = 1.8420 \times 10^{18}$ and dimensionless phase-coupling parameter $\Phi_{\text{CFT}} = 1.768058 \times 10^{-8}$. Its proposed correspondence to an AdS_3 bulk is a modeling ansatz. In this notation, the dimensionality of AdS_3 refers to spacetime; a spatial interface requires

a separately defined dynamical mapping. Boundary action, operator normalization, and the relation between boundary states and deuteron pairs are necessary to evaluate a nuclear rate.

For boundary loop operators $\mathcal{W}_{\partial\Omega}^{(k)}$, define the correlation amplitude and a reference-normalized gain by

$$\mathcal{A}_{\partial\Omega} = \text{Tr} \left(\rho_{\text{CFT}} \prod_{k=1}^{N_{\text{coh}}} \mathcal{W}_{\partial\Omega}^{(k)} \right), \quad (161)$$

$$g_{\mathcal{A}} = \frac{|\mathcal{A}_{\partial\Omega}|^2}{|\mathcal{A}_{\text{ref}}|^2}, \quad |\mathcal{A}_{\text{ref}}| > 0.$$

If every loop operator is unitary, their product is unitary and $|\mathcal{A}_{\partial\Omega}| \leq 1$ for a normalized density operator. Consequently, the unnormalized squared expectation cannot equal a gain of order 10^{10} . A rate interpretation of $g_{\mathcal{A}}$ additionally requires common initial populations, transition operators, and final-state phase space in the numerator and denominator. Amplitude ratios and incoherent channel-rate sums use their respective normalization conventions.

The exponential coupling ansatz defines a separate phenomenological factor,

$$\begin{aligned} g_{\text{exp}} &= \exp(\Phi_{\text{CFT}} \sqrt{N_{\text{coh}}}), \\ \sqrt{N_{\text{coh}}} &= 1.357203006 \times 10^9, \\ \Phi_{\text{CFT}} \sqrt{N_{\text{coh}}} &= 23.99613633, \\ g_{\text{exp}} &= 2.63869743 \times 10^{10}. \end{aligned} \quad (162)$$

Even with the rounded exponent 23.99611, the exponential is $2.63862796 \times 10^{10}$. Neither evaluation equals the nominal ambient gain $g_0 = 2.50138 \times 10^{10}$. The exponential ansatz and the ambient gain therefore constitute distinct parameterizations, not an exact identity. Moreover, g_{coll} denotes enhancement relative to the volume-integrated independent-pair rate; it is not interchangeable with the single-domain-normalized planar factor in Eq. (79).

K. Open-System Boundary Dynamics and Thermal Dephasing

The boundary state exchanges energy and phase information with the acoustic phonon bath. Under a Markovian reservoir approximation, its reduced density operator obeys the Lindblad equation [87], [88]

$$\begin{aligned} \frac{d\rho_{\text{CFT}}}{dt} &= -i[H_{\text{CFT}}, \rho_{\text{CFT}}] \\ &+ \sum_k \gamma_k \left(L_k \rho_{\text{CFT}} L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_{\text{CFT}}\} \right). \end{aligned} \quad (163)$$

Here $H_{\text{CFT}} = H_{\text{CFT}}^\dagger$ is expressed in angular-frequency units, so that the physical Hamiltonian is $\hbar H_{\text{CFT}}$; L_k are dimensionless jump operators, $\gamma_k \geq 0$ have units s^{-1} , and $\{A, B\} = AB + BA$. This convention retains the commutator without an explicit $1/\hbar$. In a finite-dimensional truncation the generator preserves trace and complete positivity. Acoustic scattering, energy relaxation, and dislocation-mediated phase randomization require distinct operator definitions. A Markovian description requires a bath correlation time short compared

with the resolved boundary dynamics; it does not follow from the operating temperature alone.

The constitutive dephasing law is

$$\begin{aligned} \gamma_{\text{deph}}(T, \rho_{\text{disl}}) &= \gamma_0 \frac{T}{T_0} \sqrt{1 + \alpha_{\text{disl}} \rho_{\text{disl}}}, \\ \gamma_0 &= 1.25 \times 10^6 \text{ s}^{-1}, \\ T_0 &= 298.15 \text{ K}, \\ \alpha_{\text{disl}} &= 1.0 \times 10^{-11} \text{ m}^2. \end{aligned} \quad (164)$$

The temperature ratio uses absolute temperature, and ρ_{disl} is dislocation line length per unit volume, with units m^{-2} . The coefficient γ_0 is the zero-dislocation reference rate at T_0 , not the rate at finite dislocation density. At the designated hot operating point,

$$\begin{aligned} T &= 350^\circ \text{C} = 623.15 \text{ K}, \\ \rho_{\text{disl}} &= 1.0 \times 10^{11} \text{ m}^{-2}, \\ \gamma_{\text{deph}} &= (1.25 \times 10^6) \frac{623.15}{298.15} \sqrt{2} \\ &= 3.69473076 \times 10^6 \text{ s}^{-1} \\ &\simeq 3.6947 \times 10^6 \text{ s}^{-1}. \end{aligned} \quad (165)$$

The dephasing coefficients are constitutive inputs.

Table IX lists nominal scenarios and evaluations of Eq. (164); the hot-row density defines a separate material state from Eq. (165).

The spatial coherence model and uncompensated gain law are

$$\begin{aligned} \xi_{\text{coh}}(T) &= \xi_0 \frac{T_0}{T} \left[1 + \delta_{\text{deph}} \left(\frac{T}{T_0} - 1 \right) \right]^{-1}, \\ \xi_0 &= 485.0 \text{ nm}, \\ g_{\text{uncomp}}(T) &= g_0 \left(\frac{\xi_{\text{coh}}(T)}{\xi_0} \right)^{0.015}, \\ g_0 &= 2.50138 \times 10^{10}. \end{aligned} \quad (166)$$

The coefficient δ_{deph} is dimensionless and the bracket must remain positive. The endpoint $\xi_{\text{coh}}(623.15 \text{ K}) = 199.441 \text{ nm}$ implies $\delta_{\text{deph}} \simeq 0.15000006$ by inversion of this law. At that endpoint,

$$g_{\text{uncomp}}(623.15 \text{ K}) = 2.46825920 \times 10^{10} \simeq 2.46826 \times 10^{10}. \quad (167)$$

The intermediate coherence lengths and gain entries in Table IX are independently assigned scenario parameters.

L. Active Boundary Phase Locking and Energy Closure

The phase-control architecture uses interferometric phase sensing and substrate-current actuation [87]. Its commanded waveform and thermal phase lead are

$$\begin{aligned} I_{\text{phase}}(t) &= I_{\text{peak}} \sin[\omega_L t + \delta\phi(T)], \\ \delta\phi(T) &= \arctan \left(\frac{\gamma_{\text{deph}}(T, \rho_{\text{disl}})}{\omega_0} \right), \\ I_{\text{peak}} &= 4.825 \text{ mA}. \end{aligned} \quad (168)$$

The current-amplitude tolerance is $\pm 0.010 \text{ mA}$. The symbol ω_L denotes the Larmor phase frequency and $\omega_0 > 0$ denotes

TABLE IX
BOUNDARY THERMAL SCENARIOS: NOMINAL PARAMETERS AND CONSTITUTIVE DEPHASING

T (°C)	T (K)	ρ_{disl} (m ⁻²)	γ_{nom} (10 ⁶ s ⁻¹)	γ_{law}	ξ_{coh} (nm)	$g_{\text{uncomp,nom}}$ (10 ¹⁰)	$g_{\text{comp,nom}}$ (10 ¹⁰)
25.0	298.15	1.0×10^{10}	1.2562	1.31101106	485.000	2.50138	2.50138
100.0	373.15	5.0×10^{10}	1.9021	1.91603869	358.420	2.49210	2.50135
200.0	473.15	1.0×10^{11}	2.8051	2.80536285	268.110	2.48115	2.50128
300.0	573.15	5.0×10^{11}	3.4510	5.88598460	214.300	2.47120	2.50118
350.0	623.15	1.0×10^{12}	3.6947	8.66491170	199.441	2.46826	2.50112

the angular-frequency scale used for phase compensation; both have units rad/s. Their numerical values and relation to the material response are independent calibration inputs. A sinusoidal current with a temperature-dependent phase offset specifies a command signal, not a closed-loop transfer function.

For deterministic actuation, the interaction can be parameterized by

$$H_{\text{fb}}(t) = I_{\text{phase}}(t)K_I, \quad K_I = K_I^\dagger, \quad (169)$$

$$\dot{\rho}_{\text{CFT}} = -i[H_{\text{CFT}} + H_{\text{fb}}(t), \rho_{\text{CFT}}] + \mathcal{D}(\rho_{\text{CFT}}),$$

where K_I is the current-to-Hamiltonian coupling operator with units s⁻¹A⁻¹ and $\mathcal{D}$ is the dissipator in Eq. (163). The operator K_I , sensing efficiency, measurement backaction, loop delay, and stochastic phase-estimation error are required to predict the controlled state. Measurement-mediated feedback generally also contributes diffusion to the unconditional evolution [89]. For periodic actuation, a stationary operating condition means a stable periodic density operator or a stationary state in a specified rotating frame.

The hot-state compensated gain and minimum acceptance level are

$$g_{\text{coll,comp}}(350^\circ\text{C}) = 2.50112 \times 10^{10},$$

$$g_{\text{min}} = 2.5000 \times 10^{10}, \quad (170)$$

$$100 \left(\frac{g_{\text{coll,comp}}}{g_{\text{min}}} - 1 \right) = 0.0448\%.$$

These values define the feedback performance requirement for the reaction-rate observable relative to the independent-pair reference state.

The thermal reference and required gain are

$$P_{\text{th,ref}} = 2911.40 \text{ W},$$

$$g_{\text{required}} = \frac{2911.40}{1.1639 \times 10^{-7}} = 2.50141765 \times 10^{10}. \quad (171)$$

a) Energy conservation and scattering constraints:

Collective enhancement changes the event rate, not the reaction energy $Q_{DD} = 23.84 \text{ MeV}$. For the boundary subsystem with physical energy operator $\mathcal{H}(t) = \hbar[H_{\text{CFT}} + H_{\text{fb}}(t)]$, the reduced energy balance is

$$\frac{d}{dt} \text{Tr}(\rho_{\text{CFT}}\mathcal{H}) = \text{Tr}(\rho_{\text{CFT}}\dot{\mathcal{H}}) + \text{Tr}[\mathcal{H}\mathcal{D}(\rho_{\text{CFT}})]. \quad (172)$$

The first term is coherent drive work and the second is dissipative energy exchange. Nuclear reaction energy must enter the total reactor balance through explicit reaction channels or a separate source term; it cannot be inferred from a trace-preserving dephasing operator. Driver electrical losses and

sensing power remain auxiliary loads, even when the commanded current is small. The nuclear lattice-power contribution is $Q_{DD}B_{\text{lat}}g_{\text{coll}}R_{\text{micro}}$ only when the gain and branching conventions use the same reaction inventory and lattice-channel energy.

For a single incident two-body partial wave with wave number k_{rel} , the inelastic bound is

$$\sigma_J^{\text{inel}} \leq \frac{\pi(2J+1)}{k_{\text{rel}}^2}. \quad (173)$$

An ensemble enhancement does not remove channel-by-channel scattering constraints. Unitarity applies to a complete closed-system scattering description, whereas the reduced boundary dynamics is generally nonunitary and must preserve complete positivity. A consistent collective reaction model therefore requires both normalized many-body transition rates and a compatible open-system generator.

M. Multi-Scale Reaction-Volume and Power Normalization

The geometric interaction-cell inventory distinguishes the active boundary envelope from its metal-filled fraction. A cubic cell of side $d_{\text{unit}} = 10.0 \text{ nm}$ has volume $V_{\text{unit}} = 1.0 \times 10^{-24} \text{ m}^3$. The envelope inventory is

$$n_{\text{dom}} = V_{\text{unit}}^{-1} = 1.0 \times 10^{24} \text{ m}^{-3},$$

$$V_{\text{env}} \equiv V_{\text{active}} = 2.53 \times 10^{-10} \text{ m}^3,$$

$$N_{\text{dom}} = n_{\text{dom}}V_{\text{active}} = 2.53 \times 10^{14},$$

$$V_{\text{metal}} = 2.875 \times 10^{-12} \text{ m}^3,$$

$$\eta_{\text{grating}} = \frac{V_{\text{metal}}}{V_{\text{active}}} = 0.011363636 \dots \approx 0.01136,$$

$$N_{\text{dom,metal}} = \eta_{\text{grating}}N_{\text{dom}} = 2.875 \times 10^{12}. \quad (174)$$

The 2.53×10^{14} domains count geometric cells in the envelope, not deuteron atoms or independently coherent emitters. The deuteron inventory N_D and the alternative coherence-cell populations in Eqs. (71) and (77) retain their distinct meanings. In particular, all envelope cells cannot simultaneously be counted as metal-filled cells.

For reaction energy E_{rxn} , local volumetric reaction rate Γ_{vol} , and microscopic heating density $\dot{q}_{\text{micro}}$, the same thermal

control volume gives

$$\begin{aligned} P_{\text{lat}} &= \int_{V_{\text{env}}} \chi_{\text{metal}}(\mathbf{r}) q_{\text{lat}}(\mathbf{r}) d^3r \\ &= \sum_{j=1}^{N_{\text{dom}}} \int_{V_j} \chi_{\text{metal}}(\mathbf{r}) Q_{DD} \mathcal{R}_{\text{lat}}(\mathbf{r}) d^3r, \quad (175) \\ P_{\text{gross}} &\equiv P_{\text{th,ref}} = 2911.4 \text{ W}. \end{aligned}$$

Here χ_{metal} is one in reacting metal and zero elsewhere; a homogenized equivalent uses its volume fraction once. The channel rate density $\mathcal{R}_{\text{lat}}$ includes the screened-state optical dependence through its microscopic transition elements, and $q_{\text{lat}} = Q_{DD} \mathcal{R}_{\text{lat}}$. Replacing the sum by N_{dom} times a single-cell integral requires an identical cell-averaged response, including metal occupancy. The reference thermal power can be identified with this lattice contribution only when other deposited powers and thermal losses have been accounted for; it defines the required rate normalization.

For an intensity penetration depth $\delta_I = 13.0$ nm, the depth dependence is

$$\begin{aligned} \mathcal{A}_z(z) &= \mathcal{A}_z(0) e^{-z/(2\delta_I)}, \\ \mathcal{A}_z^2(z) &= \mathcal{A}_z^2(0) e^{-z/\delta_I}, \quad (176) \\ \int_0^t \mathcal{A}_z^2(z) dz &= \mathcal{A}_z^2(0) \delta_I (1 - e^{-t/\delta_I}). \end{aligned}$$

The associated amplitude penetration depth is $2\delta_I = 26.0$ nm. Thus an intensity skin depth is not inserted directly into an amplitude-decay exponential. The spatial field and the patterned metal mask determine the actual overlap integral.

The mean power densities and power per cell are

$$\begin{aligned} \bar{q}_{\text{active}} &= \frac{2911.4}{2.53 \times 10^{-10}} = 1.150751 \times 10^{13} \text{ W/m}^3 \\ &= 11.50751 \text{ kW/mm}^3, \\ \bar{q}_{\text{metal}} &= \frac{2911.4}{2.875 \times 10^{-12}} = 1.012661 \times 10^{15} \text{ W/m}^3, \quad (177) \\ \bar{P}_{\text{dom}} &= \frac{2911.4}{2.53 \times 10^{14}} = 1.150751 \times 10^{-11} \text{ W}, \\ \bar{q}_{\text{active}} &= \eta_{\text{grating}} \bar{q}_{\text{metal}}. \end{aligned}$$

The active-envelope power density is approximately 11.51 kW/mm³, and the metal power density is approximately 1.013 PW/m³.

II. OPTOELECTRONIC DESIGN, SYSTEM SWAP-C & MASTER ENERGY BALANCE

The optoelectronic assembly couples two coherent, TM-polarized pump fields through a silica prism into a monolithic Pd–Ir diffraction grating. Its electromagnetic specification comprises the material tensors, patterned-layer geometry, Fourier-modal boundary conditions, and surface-field observables. Optical absorption defines a thermal load, whereas local field enhancement defines an interaction strength; these quantities enter separate terms of the system energy balance.

A. Optoelectronic Transducer: Option B RCWA

a) *Monolithic grating and layer interfaces:* The Option B grating comprises a continuous Pd–Ir backplane and rectangular Pd–Ir teeth on a titanium adhesion layer and CVD diamond substrate [79], [80]. The nominal dimensional specification is

$$\begin{aligned} \Lambda_g &= 960.80 \text{ nm}, & h_g &= 42.50 \text{ nm}, \\ t_b &= 7.50 \text{ nm}, & f_g &= 0.50, \\ w_g &= f_g \Lambda_g = 480.40 \text{ nm}, & & \\ t_{\text{peak}} &= t_b + h_g = 50.00 \text{ nm}, & & \\ t_{\text{Ti}} &= 10.0 \text{ nm}, & t_{\text{diamond}} &= 500 \mu\text{m}. \end{aligned} \quad (178)$$

Here Λ_g is the pitch perpendicular to the grooves, h_g is the tooth relief above the backplane, t_b is the continuous backplane thickness, and f_g is the metal duty cycle. The binary profile is invariant along the groove direction y and periodic in x ; z is normal to the nominal layer planes. Its mean metal thickness and active inventory are

$$\begin{aligned} \bar{t}_{\text{metal}} &= t_b + f_g h_g = 28.75 \text{ nm}, \\ A &= 1.0 \text{ cm}^2, \quad (179) \\ V_{\text{metal}} &= A \bar{t}_{\text{metal}} = 2.875 \times 10^{-12} \text{ m}^3. \end{aligned}$$

The Ti layer separates the Pd–Ir backplane from diamond; the metal does not directly contact the diamond. Illumination is specified at an internal silica-prism angle of 65.0°, measured from the layer normal, with the incidence plane perpendicular to the grooves. A prism-to-grating coupling gap, groove fill, or exterior coating changes the boundary-value problem and shall be represented explicitly rather than absorbed into an effective metal permittivity. The ideal binary profile does not define finite-aperture edge scattering or a second lateral pattern period.

b) *Complex constitutive tensors:* With the time convention $e^{-i\omega t}$, the optical model uses nonmagnetic, locally isotropic layers. For material j at either pump wavelength,

$$\begin{aligned} \epsilon_j(\lambda) &= [n_j(\lambda) + ik_j(\lambda)]^2 \\ &= n_j^2 - k_j^2 + 2in_j k_j, \\ \epsilon_j(\lambda) &= \begin{pmatrix} \epsilon_j(\lambda) & 0 & 0 \\ 0 & \epsilon_j(\lambda) & 0 \\ 0 & 0 & \epsilon_j(\lambda) \end{pmatrix}, \quad (180) \\ \mathbf{D}_j &= \epsilon_0 \epsilon_j \mathbf{E}_j, & \mathbf{B}_j &= \mu_0 \mathbf{H}_j. \end{aligned}$$

The scalar k_j in this relation is the extinction coefficient, not a wavevector. Table XCIII specifies the optical constants at every solid layer. Its diagonal tensor representation is an isotropic constitutive assumption; directional or magneto-optical tensor elements cannot be inferred from a single complex refractive index. These earlier material inputs require recharacterization at the revised 8.68 atomic-percent Ir composition and loading state; the new absorption allocation is not a recalculation of this optical table.

The refractive indices define the material inputs; the displayed permittivities are rounded evaluations of Eq. (180). A vacuum region has $\epsilon = \mathbf{I}_3$. The dilute-gas reference $\epsilon_d = 1.0006$ defines a separate isotropic medium with $n_{\text{gas}} = \sqrt{1.0006}$ and zero extinction in the lossless approximation. The actual gas state and its occupied region shall be assigned

TABLE X
EARLIER COMPOSITION-SPECIFIC DEUTERIUM FLUX AT HETEROGENEOUS INTERFACES

Interface	σ_h (MPa)	Calculated flux	Target flux	Tolerance
Pd _{0.95} Ir _{0.05} → Ti	0.00	1.43	1.43	±0.01
Pd _{0.95} Ir _{0.05} → Ti	150.00	3.15	3.15	±0.02
Ti → ϵ -Cu ₃ Sn	300.00	5.89	5.89	±0.04

Fluxes and tolerances are in 10^{21} atoms $m^{-2}s^{-1}$.

consistently with the prism-coupling geometry. No independent interface tensor is introduced in the absence of a specified surface sheet response.

c) *Three-dimensional Maxwell boundary-value problem:* Within each source-free optical layer, the frequency-domain fields satisfy

$$\begin{aligned}\nabla \times \mathbf{E} &= i\omega\mu_0\mathbf{H}, \\ \nabla \times \mathbf{H} &= -i\omega\epsilon_0\epsilon\mathbf{E}, \\ \nabla \cdot (\epsilon\mathbf{E}) &= 0, \quad \nabla \cdot \mathbf{H} = 0.\end{aligned}\quad (181)$$

At an interface without an imposed free surface charge or current, the matching conditions are

$$\begin{aligned}\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) &= 0, \quad \hat{\mathbf{n}} \times (\mathbf{H}_2 - \mathbf{H}_1) = 0, \\ \hat{\mathbf{n}} \cdot (\mathbf{D}_2 - \mathbf{D}_1) &= 0, \quad \hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0.\end{aligned}\quad (182)$$

The material response includes bulk conduction in the complex permittivity. The tangential electric field is continuous, whereas the normal electric field generally jumps according to the constitutive contrast. These conditions apply at the prism-coupling region, tooth sidewalls, relief-to-backplane boundary, Pd-Ir/Ti interface, Ti/diamond interface, and diamond termination.

d) *RCWA harmonics and solver specification:* The full-vector rigorous coupled-wave analysis uses lateral Fourier harmonics and layerwise propagation in z [90]. For scalar isotropic permittivity and electric field,

$$\begin{aligned}\epsilon(x, y, z) &= \sum_{m,n} \epsilon_{mn}(z) e^{i(mG_x x + nG_y y)}, \\ \mathbf{E}(x, y, z) &= \sum_{m,n} \mathbf{E}_{mn}(z) e^{i[(k_x + mG_x)x + (k_y + nG_y)y]}.\end{aligned}\quad (183)$$

The magnetic field uses the same basis. The general two-periodic representation has $G_x = 2\pi/\Lambda_g$ and $G_y = 2\pi/\Lambda_y$. The prescribed truncation is $m, n = -12, \dots, 12$: 25 orders per axis and 625 lateral Fourier components. The tangential field vector contains four components per lateral harmonic, hence 2500 coefficients before modal reduction. The binary Option B profile is the invariant- y specialization: $\epsilon_{mn} = 0$ for $n \neq 0$, and collinear illumination has $k_y = 0$. Only the $n = 0$ sector is excited in that ideal geometry. A physically patterned second direction requires its own Λ_y and unit-cell profile; the use of a three-dimensional solver does not supply them.

For a tooth centered in the unit cell, the relief-layer coefficients are

$$\epsilon_m = \begin{cases} \epsilon_f + f_g(\epsilon_{\text{PdIr}} - \epsilon_f), & m = 0, \\ (\epsilon_{\text{PdIr}} - \epsilon_f) \frac{\sin(\pi m f_g)}{\pi m}, & m \neq 0, \end{cases}\quad (184)$$

where ϵ_f is the assigned groove-fill permittivity. The backplane, Ti layer, and diamond are laterally uniform. Toeplitz

convolution matrices represent multiplication by ϵ and $1/\epsilon$; their inverses are not interchangeable. Normal-field discontinuities require inverse Fourier factorization, while continuous tangential products use the corresponding direct rule [91]. In the invariant- y , TM limit, the resulting coupled system is Eq. (329), evaluated with the material constants in Table XCIII.

Layer eigenmodes are matched using a scattering-matrix formulation to avoid numerical growth from evanescent propagation [90]. The 42.50 nm relief and 7.50 nm backplane are separate propagation layers; the 10.0 nm Ti layer and finite 500 μm diamond substrate remain explicit. Incident modes are normalized to incoming power, and reflected and transmitted modes satisfy outgoing or decaying boundary conditions. The diamond exit medium and any back-surface reflection shall be specified; a semi-infinite diamond model is a distinct termination assumption. Harmonic refinement shall establish stability of reflectance, transmittance, absorption, and the specified local field observable.

e) *Native RCWA execution:* The optical calculation uses Li-factorized Fourier operators and Redheffer scattering-matrix composition to construct the layer stack, solve TM excitation, and sample E_z -intensity enhancement [110]. The implemented invariant-groove configuration retains $m \in [-12, 12]$, $n = 0$, giving 25 spatial orders. The native Option B stack uses a semi-infinite diamond termination and silica-filled grooves. Ti/diamond-side and metal-surface maxima use their respective sampling locations.

f) *Crossed-periodic field propagation:* For a crossed-periodic extension, the permittivity and transverse-field evolution are

$$\begin{aligned}\epsilon(x, y, z) &= \sum_{m,n} E_{mn}(z) e^{i2\pi(m x/\Lambda_x + n y/\Lambda_y)}, \\ \frac{d\Psi}{dz} &= ik_0 \mathbf{M}(z) \Psi, \quad \Psi = (\mathbf{e}_x, \mathbf{e}_y, \mathbf{h}_x, \mathbf{h}_y)^T, \\ \mathbf{S}_{\text{global}} &= \mathbf{S}^{(1)} \star \mathbf{S}^{(2)} \star \dots \star \mathbf{S}^{(L)}.\end{aligned}\quad (185)$$

The Redheffer star product eliminates internal incoming/outgoing amplitudes by block solves rather than multiplying transfer matrices with growing evanescent factors. Li factorization enforces the appropriate continuity products. Retaining $m, n \in [-12, 12]$ gives 625 spatial harmonics and 2500 transverse-field coefficients per layer; the invariant-groove limit has only 25 spatial harmonics. Passive solutions shall satisfy $R + T + A = 1$ within a stated tolerance, with local-field convergence assessed separately from integrated power convergence.

g) *Dual-pump beat frequency:* At vacuum wavelengths $\lambda_1 = 785.0 \text{ nm}$ and $\lambda_2 = 802.5 \text{ nm}$, the interference term

contains $\cos[(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r} - (\omega_1 - \omega_2)t + \phi]$. With $c = 299\,792\,458$ m/s, its temporal frequency is

$$\begin{aligned} f_{\text{beat}} &= \frac{|\omega_1 - \omega_2|}{2\pi} = c |\lambda_1^{-1} - \lambda_2^{-1}| \\ &= |381.9012204 - 373.5731564| \text{ THz} \\ &= 8.32806399587 \text{ THz} \simeq 8.328064 \text{ THz}. \end{aligned} \quad (186)$$

The optical frequencies remain fixed across stationary interfaces; the material dispersion changes spatial wavevectors, not the difference frequency. The 8.328064 THz value is the rounded operating reference, corresponding to 5.232677×10^{13} rad/s. Pump linewidth and relative phase stability determine whether the beat remains coherent over the lattice interaction time.

h) Normal-field enhancement and power normalization:

For TM illumination at 65.0° , the normal-field design values are

$$\begin{aligned} \mathcal{F}_z &= |E_z(0^+)/E_0|^2, \\ \mathcal{F}_{z,785} &= 126.8 \quad (\text{threshold } 124.5), \\ \mathcal{F}_{z,802.5} &= 141.5 \quad (\text{threshold } 138.2), \\ \mathcal{F}_{z,\text{planar}} &\simeq 0.094. \end{aligned} \quad (187)$$

Here E_0 is the incident pump electric-field amplitude in silica, and 0^+ denotes the dielectric-side limit at the specified metal-surface sampling location. The enhancement is a squared electric-field ratio, not an optical power-conversion efficiency. The corresponding amplitude ratios are $\sqrt{126.8} = 11.26055$ and $\sqrt{141.5} = 11.89538$. A pointwise field factor, an area average, and a volume-weighted overlap are distinct observables; the sampling coordinate or averaging measure shall accompany every computed value. Sharp-corner singularities require a specified edge geometry or an integrated observable for meaningful convergence.

For collinear normal-field components, the squared-field beat term is

$$\mathcal{I}_z^{\text{beat}}(\mathbf{r}, t) = 2 \operatorname{Re} \left[E_{z,1}(\mathbf{r}) E_{z,2}^*(\mathbf{r}) e^{-i(\omega_1 - \omega_2)t} \right]. \quad (188)$$

Its amplitude depends on both local mode profiles and their relative phase. The two enhancement factors alone do not determine a spatially integrated lattice-coupling rate. For each monochromatic solution, energy conservation requires

$$\begin{aligned} R_j + T_j + A_j &= 1, \\ P_{\text{abs},j} &= \frac{\omega_j \epsilon_0}{2} \int_V \mathbf{E}_j^\dagger \epsilon_{\text{loss}}(\omega_j) \mathbf{E}_j \, dV, \\ \epsilon_{\text{loss}} &= \frac{\epsilon - \epsilon^\dagger}{2i}, \quad A_j = \frac{P_{\text{abs},j}}{P_{\text{inc},j}}. \end{aligned} \quad (189)$$

The reflectance and transmittance sum power flux over propagating orders, and the absorption integral covers all lossy regions of the same optical control volume. The nominal field factors are solver acceptance targets; geometry, constitutive inputs, and harmonic convergence must establish them jointly.

i) Collinear surface-plasmon phase matching: The grating supplies reciprocal momentum to each pump. In the planar-interface limit, the TM surface-plasmon propagation constant is

$$\beta_j = k_{0j} \sqrt{\frac{\epsilon_{m,j} \epsilon_{d,j}}{\epsilon_{m,j} + \epsilon_{d,j}}}, \quad k_{0j} = \frac{2\pi}{\lambda_j}. \quad (190)$$

The physical branch satisfies the unsquared TM boundary condition, decay away from the interface, and passive propagation. For the periodic lossy multilayer, a phase-matched mode instead requires the appropriate RCWA dispersion branch and nonzero pump-mode overlap. Its approximate momentum condition is

$$\operatorname{Re} \beta_j \simeq n_p(\lambda_j) k_{0j} \sin 65^\circ - G, \quad G = \frac{2\pi}{\Lambda_g}, \quad (191)$$

with a common diffraction order -1 and the same propagation direction for both pumps. The nondispersive prism reference $n_p = 1.453$ gives

$$\begin{aligned} G &= 2\pi/\Lambda_g = 6.53953508 \times 10^6 \text{ m}^{-1}, \\ k_{xj} &= n_p(2\pi/\lambda_j) \sin 65^\circ, \\ k_{\text{SPP},j}^{(-1)} &= k_{xj} - G, \\ q_{\text{beat}} &= k_{\text{SPP},1}^{(-1)} - k_{\text{SPP},2}^{(-1)} \\ &= k_{x1} - k_{x2} = 2.29850020 \times 10^5 \text{ m}^{-1}. \end{aligned} \quad (192)$$

Thus the nominal collinear beat wavevector is $q_{\text{beat}} = 2.2985 \times 10^5 \text{ m}^{-1}$, corresponding to a spatial beat period $2\pi/q_{\text{beat}} = 27.33602 \mu\text{m}$. The common grating momentum cancels in the difference but remains essential to each individual coupling condition. The notation $k_{\text{SPP},j}^{(-1)}$ denotes the grating-shifted excitation momentum; it coincides with a mode propagation constant only when the dispersion condition is satisfied.

The dispersive silica indices in Table XCIII give

$$\begin{aligned} q_{\text{beat,disp}} &= 2\pi \sin 65^\circ \left(\frac{1.4536}{785.0 \times 10^{-9}} - \frac{1.4531}{802.5 \times 10^{-9}} \right) \\ &= 2.33492908 \times 10^5 \text{ m}^{-1}. \end{aligned} \quad (193)$$

The constant-index and dispersive wavevectors define separate approximations. Their excitation momenta are subject to the corresponding modal dispersion condition.

j) Optical-to-thermal accounting interface: Absorbed pump power enters the thermal balance through Eq. (189); incident laser power, reflected power, transmitted power, and laser-driver consumption remain distinct quantities. The system energy balance distinguishes thermal input, electrical conversion, auxiliary consumption, and heat rejection. Nominal and comparison operating points have distinct load allocations, coolant properties, and boundary temperatures; each configuration is evaluated independently. Consistent control-volume definitions prevent double counting of recovered energy and conversion losses.

B. Variational Floquet–Bloch Space–Time Elastodynamics

a) Action, constitutive assumptions, and boundary terms:

The mechanical modulation problem retains all three displacement components and the anisotropic fourth-order elasticity tensor. Let $\mathcal{V}$ be a fixed reference cell, $\mathbf{u}$ its small displacement, and $\Omega_m = 2\pi/T_p$ the mechanical modulation frequency. This frequency is distinct from an optical carrier; identification with

the optical beat requires an explicit optomechanical coupling law. The independently periodic coefficients obey

$$\begin{aligned} \mathbf{C}(\mathbf{x} + \mathbf{R}, t) &= \mathbf{C}(\mathbf{x}, t + T_p) = \mathbf{C}(\mathbf{x}, t), \\ \rho(\mathbf{x} + \mathbf{R}, t) &= \rho(\mathbf{x}, t + T_p) = \rho(\mathbf{x}, t), \\ C_{ijkl} &= C_{jikl} = C_{ijlk} = C_{klij}, \quad \rho > 0, \\ \varepsilon(\mathbf{u}) &= \frac{1}{2}[\nabla \mathbf{u} + (\nabla \mathbf{u})^T], \\ \boldsymbol{\sigma} &= \mathbf{C} : \boldsymbol{\varepsilon}. \end{aligned} \quad (194)$$

Separate spatial and temporal periodicity is the assumption supporting a rectangular double-Fourier basis. A medium having only a combined space-time translation symmetry instead requires the reciprocal basis of that combined lattice. Positive elastic energy and a prescribed positive effective inertia define the instantaneous conservative model. A time-varying ρ is an externally controlled effective inertia here, not a mass-conservation law for matter flowing through a fixed volume.

For prescribed body-force density $\mathbf{f}$ and boundary traction $\bar{\mathbf{t}}$, Hamilton's functional is

$$\begin{aligned} \mathcal{S}[\mathbf{u}] &= \int_{t_0}^{t_1} \left[\int_{\mathcal{V}} \mathcal{L} dV + \int_{\Gamma_N} \bar{\mathbf{t}} \cdot \mathbf{u} dA \right] dt, \\ \mathcal{L} &= \frac{1}{2} \rho |\dot{\mathbf{u}}|^2 - \frac{1}{2} \boldsymbol{\varepsilon} : \mathbf{C} : \boldsymbol{\varepsilon} + \mathbf{f} \cdot \mathbf{u}. \end{aligned} \quad (195)$$

Let $\mathbf{v} = \delta \mathbf{u}$ vanish at t_0, t_1 and on the displacement boundary Γ_D . The coefficients are held fixed during this variation. Then

$$\begin{aligned} \delta \mathcal{S}[\mathbf{u}; \mathbf{v}] &= \int_{t_0}^{t_1} \int_{\mathcal{V}} [\rho \dot{\mathbf{u}} \cdot \dot{\mathbf{v}} - \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\mathbf{v}) + \mathbf{f} \cdot \mathbf{v}] dV dt \\ &\quad + \int_{t_0}^{t_1} \int_{\Gamma_N} \bar{\mathbf{t}} \cdot \mathbf{v} dA dt. \end{aligned} \quad (196)$$

Integration by parts retains both the temporal endpoint and natural spatial terms:

$$\begin{aligned} \delta \mathcal{S} &= \left[\int_{\mathcal{V}} \rho \dot{\mathbf{u}} \cdot \mathbf{v} dV \right]_{t_0}^{t_1} \\ &\quad + \int_{t_0}^{t_1} \int_{\mathcal{V}} [-\partial_t(\rho \dot{\mathbf{u}}) + \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}] \cdot \mathbf{v} dV dt \\ &\quad + \int_{t_0}^{t_1} \int_{\Gamma_N} (\bar{\mathbf{t}} - \boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{v} dA dt. \end{aligned} \quad (197)$$

Arbitrariness of the admissible interior and boundary variations gives the Euler-Lagrange equations and the natural boundary condition:

$$\begin{aligned} \partial_t(\rho \dot{\mathbf{u}}) &= \nabla \cdot (\mathbf{C} : \boldsymbol{\varepsilon}) + \mathbf{f}, \\ \rho \ddot{\mathbf{u}} + \dot{\rho} \dot{\mathbf{u}} &= \nabla \cdot (\mathbf{C} : \boldsymbol{\varepsilon}) + \mathbf{f}, \\ \mathbf{u}|_{\Gamma_D} &= \bar{\mathbf{u}}, \quad \boldsymbol{\sigma} \mathbf{n}|_{\Gamma_N} = \bar{\mathbf{t}}. \end{aligned} \quad (198)$$

Thus the $\dot{\rho} \dot{\mathbf{u}}$ term cannot be omitted while retaining the stated action. Thermal eigenstrain, viscosity, and other dissipative mechanisms require their own constitutive and power terms; they are not implicitly generated by this elastic functional.

b) Double-Fourier eigenproblem: For either coefficient $a \in \{\rho, C_{ijkl}\}$, define the normalized Fourier coefficients and the displacement expansion by

$$\begin{aligned} a^{\mathbf{H},p} &= \frac{1}{|\mathcal{V}|T_p} \int_0^{T_p} \int_{\mathcal{V}} a(\mathbf{x}, t) e^{-i\mathbf{H} \cdot \mathbf{x} + ip\Omega_m t} dV dt, \\ a(\mathbf{x}, t) &= \sum_{\mathbf{H},p} a^{\mathbf{H},p} e^{i\mathbf{H} \cdot \mathbf{x} - ip\Omega_m t}, \\ \mathbf{u}(\mathbf{x}, t) &= \sum_{\mathbf{G},n} \mathbf{U}_{\mathbf{G},n} e^{i(\mathbf{k}+\mathbf{G}) \cdot \mathbf{x} - i\omega_n t}, \\ \omega_n &= \omega + n\Omega_m, \quad \boldsymbol{\kappa}_{\mathbf{G}} = \mathbf{k} + \mathbf{G}. \end{aligned} \quad (199)$$

Real material coefficients satisfy $a^{-\mathbf{H},-p} = (a^{\mathbf{H},p})^*$. In the source-free bulk, differentiating the product $\rho \dot{\mathbf{u}}$ produces one incoming and one outgoing harmonic frequency. Equating Fourier coefficients therefore gives

$$\sum_{\mathbf{G}',m,\ell} [\mathbf{K}_{\mathbf{G}ni,\mathbf{G}'m\ell} - \omega_n \omega_m \mathbf{M}_{\mathbf{G}ni,\mathbf{G}'m\ell}] U_{\mathbf{G}',m,\ell} = 0, \quad (200)$$

with the full tensor contractions

$$\begin{aligned} \mathbf{K}_{\mathbf{G}ni,\mathbf{G}'m\ell} &= \kappa_{\mathbf{G},j} C_{ijkl}^{\mathbf{G}-\mathbf{G}',n-m} \kappa_{\mathbf{G}',k}, \\ \mathbf{M}_{\mathbf{G}ni,\mathbf{G}'m\ell} &= \rho^{\mathbf{G}-\mathbf{G}',n-m} \delta_{i\ell}. \end{aligned} \quad (201)$$

Repeated Cartesian indices j, k are summed. Both wavevectors enter the stiffness contraction; its dependence is not merely a function of their difference. Shear-longitudinal conversion remains in the vector eigenfunctions rather than being replaced by a scalar transmission factor.

For a finite reciprocal/harmonic truncation, let $\mathbf{N}$ be diagonal with entry $n\Omega_m$ on each polarization. The quadratic eigenvalue problem and a companion linearization are

$$\begin{aligned} [\mathbf{K} - (\omega \mathbf{I} + \mathbf{N})\mathbf{M}(\omega \mathbf{I} + \mathbf{N})] \mathbf{U} &= 0, \\ (\mathbf{A}_0 + \omega \mathbf{A}_1 - \omega^2 \mathbf{M}) \mathbf{U} &= 0, \\ \mathbf{A}_0 &= \mathbf{K} - \mathbf{NMN}, \\ \mathbf{A}_1 &= -(\mathbf{NM} + \mathbf{MN}), \\ \begin{bmatrix} 0 & \mathbf{I} \\ \mathbf{A}_0 & \mathbf{A}_1 \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \omega \mathbf{U} \end{bmatrix} &= \omega \begin{bmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{M} \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \omega \mathbf{U} \end{bmatrix}. \end{aligned} \quad (202)$$

At zero modulation the $n = 0$ block reduces to $\mathbf{KU} = \omega^2 \mathbf{MU}$. With modulation, positive instantaneous coefficients do not exclude parametric instability: $\text{Im } \omega > 0$ gives growth under the chosen $e^{-i\omega t}$ convention. Spatial and temporal truncations, eigenpair residuals, and stability over modulation cycles must converge independently. This mechanical eigenproblem is distinct from the electronic dielectric inverse and its scalar arithmetic benchmark in Section I.

c) Interface continuity and modal matching proof: Let a perfectly bonded, massless interface separate media A and B, with one common normal $\mathbf{n}$ directed from A to B. Admissible displacement traces and their variations agree on the interface. Summing the spatial boundary terms of Eq. (197) gives

$$\begin{aligned} \mathbf{u}_A|_{\Gamma} &= \mathbf{u}_B|_{\Gamma}, \\ \delta \mathcal{S}_{\Gamma} &= \int_{t_0}^{t_1} \int_{\Gamma} (\boldsymbol{\sigma}_B \mathbf{n} - \boldsymbol{\sigma}_A \mathbf{n}) \cdot \mathbf{v} dA dt = 0, \\ \boldsymbol{\sigma}_A \mathbf{n} &= \boldsymbol{\sigma}_B \mathbf{n}. \end{aligned} \quad (203)$$

A common tangential period and modulation period are required for matching in one harmonic basis. Commensurate structures can use a common supercell; otherwise a single finite periodic interface basis is not exact. Orthogonal projection of the trace conditions onto $e^{i(\mathbf{k}_{\parallel} + \mathbf{H}) \cdot \mathbf{y} - i\omega_n t}$ gives, for every retained trace harmonic,

$$\begin{aligned}\hat{\mathbf{u}}_{\mathbf{H},n}^A &= \hat{\mathbf{u}}_{\mathbf{H},n}^B, \\ \hat{t}_{i,\mathbf{H},n}^A &= \hat{t}_{i,\mathbf{H},n}^B, \\ \hat{t}_{i,\mathbf{H},n}^a &= n_j \sum_{\mathbf{H}',m} C_{ijkl}^{a,\mathbf{H}-\mathbf{H}',n-m} \hat{\varepsilon}_{kl,\mathbf{H}',m}^a, \quad a = A, B.\end{aligned}\quad (204)$$

The strain traces include each mode's normal derivative. Equal trace coefficients do not imply equality of individual incident, reflected, and transmitted modal amplitudes. Writing displacement and traction trace matrices as $\mathbf{U}_a^{\pm}, \mathbf{T}_a^{\pm}$, where $+$ denotes outgoing and $-$ incoming modes, the interface system is

$$\begin{bmatrix} \mathbf{U}_A^+ & -\mathbf{U}_B^+ \\ \mathbf{T}_A^+ & -\mathbf{T}_B^+ \end{bmatrix} \begin{bmatrix} \mathbf{a}_A^+ \\ \mathbf{a}_B^+ \end{bmatrix} = - \begin{bmatrix} \mathbf{U}_A^- & -\mathbf{U}_B^- \\ \mathbf{T}_A^- & -\mathbf{T}_B^- \end{bmatrix} \begin{bmatrix} \mathbf{a}_A^- \\ \mathbf{a}_B^- \end{bmatrix}. \quad (205)$$

Propagating and evanescent modes are retained as needed to resolve the boundary traces. A scattering matrix obtained from this system is a derived boundary operator, not an empirical replacement for the tensor equations.

d) Energy exchange with the modulation: Multiplying Eq. (198) by $\dot{\mathbf{u}}$ and applying the product rule proves the local balance

$$\begin{aligned}e_{\text{el}} &= \frac{1}{2}\rho|\dot{\mathbf{u}}|^2 + \frac{1}{2}\boldsymbol{\varepsilon} : \mathbf{C} : \boldsymbol{\varepsilon}, \\ \mathbf{J}_{\text{el}} &= -\boldsymbol{\sigma}\dot{\mathbf{u}}, \\ \partial_t e_{\text{el}} + \nabla \cdot \mathbf{J}_{\text{el}} &= \mathbf{f} \cdot \dot{\mathbf{u}} + \frac{1}{2}\boldsymbol{\varepsilon} : \dot{\mathbf{C}} : \boldsymbol{\varepsilon} - \frac{1}{2}\dot{\rho}|\dot{\mathbf{u}}|^2.\end{aligned}\quad (206)$$

Displacement and traction continuity imply equal instantaneous normal power flux on the two sides of a massless interface. The volume terms involving $\dot{\mathbf{C}}$ and $\dot{\rho}$ nevertheless exchange energy with the modulation drive. Consequently, ordinary passive scattering unitarity cannot be imposed without the drive-work contribution. A lossy constitutive model adds nonnegative dissipation to this balance.

C. Thermal-Hydraulic Design and TEG Energy Accounting

1) Water-Loop and Compressor Duty: The revised water-loop calculation distinguishes hydraulic power delivered to the fluid, electrical pump input, and refrigeration-compressor input. Denote volumetric flow by $\dot{V}$ rather than Q to distinguish it from heat rate $\dot{Q}$. With total loop pressure drop ΔP and an assumed wire-to-water efficiency η_{pump} , the steady duty is [92]

$$\begin{aligned}\dot{V} &= 2.0833 \times 10^{-4} \text{ m}^3/\text{s} \simeq 12.5 \text{ L/min}, \\ \Delta P &= 85000 \text{ Pa}, \quad \eta_{\text{pump}} = 0.65, \\ P_{\text{hyd}} &= \dot{V}\Delta P = 17.70805 \text{ W}, \\ P_{\text{pump}} &= \frac{\dot{V}\Delta P}{\eta_{\text{pump}}} = 27.243153846 \dots \text{ W} \\ &\simeq 27.2432 \text{ W}, \\ P_{\text{pump,loss}} &= P_{\text{pump}} - P_{\text{hyd}} \simeq 9.5351 \text{ W}.\end{aligned}\quad (207)$$

If 0.65 is only shaft-to-water efficiency, motor and drive efficiencies must additionally be included before assigning

P_{pump} to an electrical bus. In a closed, steady loop, the net hydraulic work is dissipated through pressure losses; it need not all be dissipated in the microchannels rather than valves, manifolds, and piping. For a Newtonian fluid, the local viscous heating is $\Phi = 2\mu\mathbf{D}' : \mathbf{D}' + \zeta(\nabla \cdot \mathbf{v})^2 \geq 0$, where $\mathbf{D}'$ is the deviatoric strain-rate tensor and $\zeta \geq 0$ the bulk viscosity. Subject to negligible net kinetic and gravitational energy changes, the integrated loop dissipation equals P_{hyd} , not automatically P_{pump} .

Let $\chi_{\text{loss}} \in [0, 1]$ be the fraction of pump, motor, and drive losses reaching the refrigerated fluid boundary. Its heat load is

$$\begin{aligned}\dot{Q}_{\text{pump} \rightarrow \text{fluid}} &= P_{\text{hyd}} + \chi_{\text{loss}} P_{\text{pump,loss}}, \\ \dot{Q}_{\text{total}} &= \dot{Q}_{\text{lattice,loop}} + \dot{Q}_{\text{pump} \rightarrow \text{fluid}}.\end{aligned}\quad (208)$$

The reported all-losses-to-fluid case sets $\chi_{\text{loss}} = 1$: 100% of pump electrical input ultimately becomes loop heat, through hydraulic dissipation *and* capture of pump-system losses. This is a thermal-boundary assumption, not a consequence of microchannel shear alone. With the separately assigned lattice-loop load and refrigeration coefficient of performance,

$$\begin{aligned}\dot{Q}_{\text{lattice,loop}} &= 147.6600 \text{ W}, \\ \dot{Q}_{\text{total}} &= 147.6600 + 27.243153846 \dots \\ &\simeq 174.9032 \text{ W}, \\ \text{COP} &= \frac{\dot{Q}_{\text{total}}}{P_{\text{cooling,compressor}}} = 3.0, \\ P_{\text{cooling,compressor}} &\simeq 58.3011 \text{ W}, \\ P_{\text{cooling,elec}} &= P_{\text{pump}} + P_{\text{cooling,compressor}} \\ &\simeq 85.5442 \text{ W}.\end{aligned}\quad (209)$$

The compressor rejects $\dot{Q}_{\text{total}} + P_{\text{cooling,compressor}} \simeq 233.2042 \text{ W}$ at its condenser. COP refers to the stated evaporator and condenser temperatures. The 147.6600 W lattice-loop duty is a partial-load calculation.

2) Optimized Master Power Ledger and Boundary Closure: The revised design allocation uses dual-stage segmented skutterudite/half-Heusler modules, represented by $\text{CoSb}_3/\text{ZrNiSn}$, at $T_h = 623.15 \text{ K}$ and $T_c = 303.15 \text{ K}$. Its assigned bus-referred conversion efficiency is $\eta_{\text{TEG}} = 0.3380$; this assumes any internal contact, interconnect, and converter losses have already been included. The earlier $\text{Bi}_2\text{Te}_3/\text{PbTe}$ losses cannot be applied to this allocation without a new module model. The necessary Carnot bound is

$$\eta_{\text{TEG}} = 0.3380 < 1 - \frac{303.15}{623.15} = 0.5135200. \quad (210)$$

Temperature-dependent transport properties, segmentation, contact losses, load matching, and measured performance are required. A 303.15 K cold junction also requires a compatible sink temperature or refrigeration; the earlier 310 K ambient cannot passively cool it.

TABLE XI
DEFINITIVE CLOSED-LOOP THERMAL AND ELECTRICAL POWER LEDGER

Quantity / boundary	Value (W)
Total thermal input P_{thermal}	3093.4400
Gross TEG electrical output (33.80%)	1045.5800
TEG cold-side heat rejection	2047.8600
Magnetic drive gross input	422.22
SiC recovered energy equivalent	88.45
Recovered magnetic-drive net demand	368.45
Consolidated auxiliary demand	122.10
Definitive net plant electrical output	+555.03

The revised optical ledger separates electrical input, delivered light, and absorbed heat:

$$\begin{aligned}
 P_{\text{laser_elec}} &= 422.2189 \text{ W}, \\
 P_{\text{laser,opt}} &= 185.00 \text{ W}, \\
 \eta_{\text{laser}} &= 185.00/422.2189 \\
 &\simeq 0.4381613, \\
 \alpha_{\text{abs}} &= 0.9840, \\
 P_{\text{optical,absorbed}} &= 0.9840(185.00) \\
 &= 182.04 \text{ W}, \\
 P_{\text{optical,unabsorbed}} &= 2.96 \text{ W}, \\
 P_{\text{laser,loss}} &= 422.2189 - 185.00 \\
 &= 237.2189 \text{ W}.
 \end{aligned} \tag{211}$$

The revised optical allocation assigns 98.40% absorption and 185.00 W optical output. Unabsorbed light and driver heat have separate rejection paths.

$$\begin{aligned}
 P_{\text{fusion}} &= 2911.40 \text{ W}, \\
 P_{\text{thermal}} &= P_{\text{fusion}} + P_{\text{optical,absorbed}} \\
 &= 3093.4400 \text{ W}, \\
 P_{\text{TEG_elec}} &= 0.3380(3093.4400) \\
 &= 1045.58272 \text{ W} \simeq 1045.5800 \text{ W}, \\
 \dot{Q}_{\text{TEG,reject}} &= 3093.4400 - 1045.5800 \\
 &= 2047.8600 \text{ W}, \\
 P_{\text{thermal}} &= P_{\text{TEG,elec}} + \dot{Q}_{\text{TEG,reject}}.
 \end{aligned} \tag{212}$$

Here P_{fusion} is the assigned nuclear-heat normalization, and $N_{\text{coh}} \simeq 2.51 \times 10^{10}$ is a rounded scaling specification.

The SiC recovery chain, conversion losses, and coil-support demand are combined in the verified net magnetic-drive quantity. The complete electrical ledger is

$$\begin{aligned}
 P_{\text{net}} &= P_{\text{TEG}} - P_{\text{drive,net}} - P_{\text{aux}} \\
 &= 1045.5800 - 368.45 - 122.10 \\
 &= +555.03 \text{ W}, \\
 P_{\text{TEG}} &= P_{\text{drive,net}} + P_{\text{aux}} + P_{\text{net}}.
 \end{aligned} \tag{213}$$

Equations (212) and (213) close the thermal and electrical balances at the declared system boundaries.

The TEG cold-side load $\dot{Q}_{\text{TEG,reject}} = 2047.8600 \text{ W}$ is the heat-rejection design duty. As a separate refrigerated

comparison, if a COP-3 chiller removes the full TEG rejection and all pump heat, then

$$\begin{aligned}
 P_{\text{compressor,full}} &= \frac{2047.8600 + 27.2432}{3} \\
 &\simeq 691.7002 \text{ W}, \\
 P_{\text{net,full}} &= 1045.5800 - 422.2189 - 27.2432 \\
 &\quad - 691.7002 - 18.50 - 12.00 \\
 &\simeq -126.08 \text{ W}.
 \end{aligned} \tag{214}$$

This is a separate full-refrigeration case, not an additional debit on top of the 58.3011 W compressor entry. It still excludes any extra refrigerated driver or auxiliary heat. At a complete steady device boundary, captured optical power is internally recycled electrical energy: it is included at the TEG input but not counted as an independent external source. With no other external energy input, heat and escaping optical output must satisfy $\dot{Q}_{\text{environment}} + P_{\text{escape}} = P_{\text{fusion}} - P_{\text{net}}$. A positive operating export requires measured conversion performance and a complete, temperature-compatible heat-rejection design.

3) *Earlier Glycol-Loop and Bi-Te/PbTe Comparison:* The earlier cooling system comprises a 120-channel cold plate and a 50/50 water-glycol loop with a 142.5 kPa pressure-drop target [79], [80]. Its calculations below and Tables XII and XIII remain separate historical comparisons, not the revised water-loop or skutterudite/half-Heusler allocation. The pressure-drop target applies to the primary hydraulic assembly; pressure-tap locations must distinguish cold-plate loss from *total loop* loss. The same pressure-drop allocation cannot be assigned independently to both boundaries.

$$\begin{aligned}
 N_{\text{ch}} &= 120, \quad w_{\text{ch}} = 250 \mu\text{m}, \\
 h_{\text{ch}} &= 580 \mu\text{m}, \quad L_{\text{ch}} = 50.0 \text{ mm}, \\
 \dot{V} &= 3.62 \text{ L/min} \\
 &= 6.03333 \times 10^{-5} \text{ m}^3/\text{s}, \\
 A_c &= N_{\text{ch}} w_{\text{ch}} h_{\text{ch}} = 1.740 \times 10^{-5} \text{ m}^2, \\
 D_h &= \frac{2w_{\text{ch}}h_{\text{ch}}}{w_{\text{ch}} + h_{\text{ch}}} = 349.398 \mu\text{m}, \\
 v_{\text{ch}} &= \dot{V}/A_c = 3.46743 \text{ m/s}.
 \end{aligned} \tag{215}$$

At the property-reference temperature of 330 K, $\rho = 1045 \text{ kg/m}^3$, $\mu = 1.42 \times 10^{-3} \text{ Pa}\cdot\text{s}$, $k_f = 0.415 \text{ W/(m}\cdot\text{K)}$, and $\text{Pr} = 11.2$. The rectangular-channel Darcy correlation gives

$$\begin{aligned}
 \text{Re} &= \rho v_{\text{ch}} D_h / \mu = 891.57, \\
 f_D &= 62.4 / \text{Re} = 0.069989, \\
 \Delta P_{\text{ch}} &= f_D \frac{L_{\text{ch}}}{D_h} \frac{\rho v_{\text{ch}}^2}{2} = 62.919 \text{ kPa}, \\
 \Delta P_{\text{other}} &= 142.5 - 62.919 = 79.581 \text{ kPa}, \\
 P_{\text{hyd}} &= \dot{V} \Delta P_{\text{other}} = 8.5975 \text{ W}.
 \end{aligned} \tag{216}$$

The residual ΔP_{other} allocates pressure loss to the manifolds, entrances, exits, and remaining loop components; its distribution requires measurement or a hydraulic model. The approximate 63.12 kPa channel-loss estimate is distinct from the 142.5 kPa assembly target. A 350 kPa pump rating alone does not define the operating pump curve. At 15% efficiency,

TABLE XII
EARLIER Bi-Te/PbTe BUS POWER AND LOSS ALLOCATION

Quantity	Value
Thermal hot-side input	2911.40 W
T_h/T_c	623.15/353.15 K
Ideal segmented Bi ₂ Te ₃ /PbTe efficiency	16.85%
Ideal Seebeck power	490.57 W
Contact / interconnect / bypass debits	28.45/18.60/12.30 W
Raw DC	431.22 W
DC-DC efficiency (rounded)	97.03%
Selected bus DC	418.43 W
Support load	386.67 W
Selected net bus export	+31.76 W

the target duty requires 57.32 W, whereas the 116.67 W pump allowance corresponds to 7.37% efficiency at the same duty. Unused electrical allowance does not constitute heat generation. For $Nu = 4.12$, the heat-transfer coefficient is $h = Nu k_f / D_h \simeq 4893.57 \text{ W}/(\text{m}^2\text{K})$. Copper fins with $k_w = 391 \text{ W}/(\text{mK})$ and thickness $t_w = 150 \mu\text{m}$ give $m_f = \sqrt{2h/(k_w t_w)} \simeq 408.50 \text{ m}^{-1}$ and $\eta_f = \tanh(m_f h_{ch})/(m_f h_{ch}) \simeq 0.9817$. Applicability of a fully developed correlation depends on entrance effects and thermal boundary conditions. Because the frame/cold-plate section is Al 6061-T6 whereas the fin model uses copper, a combined design requires an explicit insert geometry and interfacial contact resistance.

The external heat exchanger has nominal parameters $A_{\text{rad}} = 2.455 \text{ m}^2$, $U_{\text{rad}} = 48.5 \text{ W}/(\text{m}^2\text{K})$, and $T_{\text{amb}} = 310 \text{ K}$, requiring a log-mean temperature difference of $2492.97/(48.5 \times 2.455) = 20.94 \text{ K}$. This condition defines the heat-transfer duty but does not uniquely determine the coolant and air outlet temperatures. The pure-water case at 45/55 °C and 25 °C ambient is a separate comparison configuration, not the property basis of the glycol loop.

The electrical-equivalent losses total 59.35 W, leaving $490.57 - 59.35 = 431.22 \text{ W}$ at the generator terminals. The nominal 418.43 W bus output uses a converter efficiency of 97.033997%. At the generator boundary, residual heat is $2911.40 - 431.22 = 2480.18 \text{ W}$. Adding 12.79 W of converter dissipation gives 2492.97 W of aggregate heat rejection. If all 386.67 W of support power dissipates internally, environmental heat rejection is 2879.64 W, and $2879.64 + 31.76 = 2911.40 \text{ W}$.

The calorimetric input model assigns 1850 W to optical pumping and 65.59 W to RF excitation, whereas the bus-export model allocates 120 W to laser drivers. These are separate reference allocations. Their reconciliation requires a conversion-efficiency model and common definitions of electrical and optical ports. The +31.76 W result is a conditional *bus power balance*. Whole-device net-energy export requires independent calorimetric and electrical qualification using consistent pump, laser, RF, and thermal boundaries.

a) *Compressor-free constant-efficiency comparison*: The compressor-free loop operates without vapor-compression refrigeration but retains electrically powered pumps and fans. Fans, telemetry, and remaining auxiliaries are included once

TABLE XIII
COMPRESSOR-FREE ELECTRICAL BALANCE AT 2911.40 W THERMAL INPUT

Contribution	Electrical power (W)
TEG generation	+418.43
Chiller	0.00
Water-loop pump	-116.67
Laser drivers	-120.00
Auxiliaries, fans, telemetry	-150.00
Net export	+31.76

in the 150 W aggregate allowance. For this configuration, the constant- ZT material model gives

$$\eta_{\text{TEG}} = \left(1 - \frac{353.15}{623.15}\right) \frac{\sqrt{3.16} - 1}{\sqrt{3.16} + 353.15/623.15}$$

$$= 0.143722744917,$$

$$P_{\text{TEG,calc}} = 418.434399551 \text{ W},$$

$$P_{\text{net,calc}} = 418.434399551 - 386.67$$

$$= 31.764399551 \text{ W}. \quad (217)$$

Table XIII assigns 418.43 W to the electrical output after a single rounding step. This value is conditional on availability at the specified bus boundary. Any converter or contact losses outside that boundary require separate accounting; the constant-efficiency comparison cannot be combined indiscriminately with a separately derated TEG model. At the combined core-to-bus conversion boundary, $Q_{\text{cold}} = 2911.40 - 418.43 = 2492.97 \text{ W}$. If all 386.67 W of support power dissipates internally, environmental heat rejection is $2492.97 + 386.67 = 2879.64 \text{ W}$ and $2879.64 + 31.76 = 2911.40 \text{ W}$. Recoverable heat depends on the dissipation paths captured by the thermal system. The refrigerated comparison includes the 116.67 W pump and gives -561.80 W net power.

b) *Water-to-air heat-transfer closure*: With water at 45/55 °C, $c_p = 4180 \text{ J}/(\text{kg K})$, $\rho_w = 988 \text{ kg}/\text{m}^3$, and an 80 °C cold plate,

$$R_{\text{plate-water}} = \frac{80 - 50}{2492.97} \approx 0.012034 \text{ K/W},$$

$$\dot{m} = \frac{2492.97}{4180(55 - 45)} = 0.05964043 \text{ kg/s},$$

$$\dot{V} = \dot{m}/988 = 3.62188850 \text{ L/min},$$

$$\Delta P_{\text{allow}} = \frac{116.67(0.15)}{\dot{V}} = 289912.29 \text{ Pa}. \quad (218)$$

The allowable pressure rise is evaluated with $\dot{V}$ in m^3/s and a 15% pump efficiency. This operating point differs from the 7 L/min, 35% efficiency comparison; each flow condition requires its own cold-plate pressure-loss calculation. For counterflow water entering at 55 °C and leaving at 45 °C, with air entering

TABLE XIV
DIRECT-AIR AND REFRIGERATED MODES: THERMAL AND ELECTRICAL
BALANCES

Quantity (W)	Direct air	Chilled
Thermal reference	2911.400	2911.400
Laser allowance	120.000	120.000
Pump allowance	116.667	116.667
Fans	45.000	15.000
Controls	25.000	25.000
Compressor	0	693.190
Auxiliary electricity	306.667	969.857
Heat-minus-load comparison	2604.733	1941.543
Illustrative TEG bus output	337.952	337.952
Illustrative electrical balance	31.285	-631.905

at 25 °C and leaving at 38 °C,

$$\Delta T_1 = 55 - 38 = 17 \text{ K},$$

$$\Delta T_2 = 45 - 25 = 20 \text{ K},$$

$$\Delta T_{lm} = \frac{17 - 20}{\ln(17/20)} = 18.45938814 \text{ K}, \quad (219)$$

$$A_{HX} = \frac{2492.97}{55\Delta T_{lm}} = 2.45548373 \text{ m}^2 \approx 2.455 \text{ m}^2.$$

The calculated finned area assumes $U = 55 \text{ W}/(\text{m}^2 \text{ K})$, ideal counterflow, and rejection of core/TEG heat alone. Pump heat entering the coolant, fouling, departures from counterflow, and ambient conditions affect the required heat-transfer duty and area. Forced-convection fan power remains part of the auxiliary allocation.

c) Direct-air and refrigerated comparison modes: A separate operating comparison uses $P_{th,ref} = 2911.40 \text{ W}$, a 120 W aggregate laser allowance, 25 W for controls, and mode-dependent fan loads. This allocation is distinct from the 800.688889 W balance-of-plant (BOP) configuration. At a pump duty of 7 L/min and 350 kPa with 35% efficiency,

$$P_{pump} = \dot{V} \Delta P / \eta_p = 116.66666667 \text{ W},$$

$$P_{comp} = 2911.40 / 4.2 = 693.190476190 \text{ W},$$

$$P_{aux,pass} = 120 + 116.66666667 + 45 + 25 \\ = 306.66666667 \text{ W},$$

$$P_{aux,chill} = 120 + 116.66666667 + 15 + 25 + P_{comp} \\ = 969.857142857 \text{ W}. \quad (220)$$

The direct-thermal-export comparison uses $\eta_{conv} = 1$. The electrical balances instead use $\eta_{bus} = 0.116078911301$. The coupled installation satisfies

$$P_{net,el} = \eta_{bus}(T_h, T_c)P_{th} - P_{aux}, \\ Q_{TEG, rej} = P_{th} - P_{TEG, bus}, \\ Q_{condenser} = Q_c + P_{comp}. \quad (221)$$

The full-heat chilled case has $Q_c = 2911.40 \text{ W}$ and $Q_{condenser} = 3604.590476 \text{ W}$. Electrical extraction by a TEG changes the chiller duty and therefore the condenser balance. A mode with 310 K ambient and 319.2 K coolant return cannot sustain a 310 K TEG cold junction through finite passive heat flow. The 120 W laser allowance must distinguish electrical

input from optical output and remain consistent with the two-laser conversion efficiencies. Turbopump and other service loads require explicit allocation rather than an assumption of zero consumption. The 10/15 kW cases are separate design comparisons, not the nominal 2911.40 W operating point.

D. Accounting Boundary and Continuous Auxiliary Loads

The component-resolved BOP model [61] lists electrical allowances and operating-point demand. The 450/180/85/60/25 W allocation defines a separate comparison schedule.

$$P_{L1} = 100/0.45 = 222.222222 \text{ W},$$

$$P_p = \frac{(7.0 \times 10^{-3}/60)(3.5 \times 10^5)}{0.35} = 116.666667 \text{ W},$$

$$P_{BOP} = P_{L1} + 200 + P_p + 120 + 141.8 \\ = 800.68888889 \text{ W}. \quad (222)$$

At 7.2 L/min, pump demand is 120.0 W and total BOP demand is 804.022222 W. The 7.0 L/min case gives 116.666667 W pump demand.

E. Master Electrical Balance and Reaction Thresholds

The four-part loss model [63] relates material conversion efficiency to usable bus power. The high- ZT comparison combines the material efficiency at $ZT = 2.16$, the unrounded resistance ratios in Eq. (229), and the calculated BOP demand:

$$\eta_{net,TEG} = 0.116078911301,$$

$$P_{TEG,bus} = \eta_{net,TEG} P_{nuclear}, \quad (223)$$

$$P_{net,system} = \eta_{net,TEG} P_{nuclear} - P_{BOP}.$$

The comparison holds BOP demand fixed at 800.688889 W and includes converter loss once within the core-to-bus efficiency. The electrical break-even condition is

$$P_{nuclear,break-even} = P_{BOP} / \eta_{net,TEG} = 6897.798057 \text{ W}. \quad (224)$$

$$\eta_{export}(10 \text{ kW}) = 3.601002\%,$$

$$\eta_{export}(15 \text{ kW}) = 6.269965\%. \quad (225)$$

The +50/+475 W margins and 9150/13725 W rejection duties correspond to the separate 8.5% efficiency case. These rejection duties also define cold-plate benchmark loads in Section V. Calculated balances are conditional on the assumed heat input, conversion efficiency, and BOP demand. For channel fractions B_j and retained fractions f_j ,

$$P_{nuclear} = R_{DD} \sum_j B_j f_j Q_j. \quad (226)$$

Complete deposition of reaction energy into the lattice gives the energy-equivalent requirements

$$R_{10kW} = 10^4 / Q_{DD}, \quad R_{15kW} = 1.5 \times 10^4 / Q_{DD}, \quad (227)$$

consistent with the heat-equivalent reaction rates in Eq. (130).

TABLE XV
COMPONENT-RESOLVED BOP ALLOWANCES AND CALCULATED ELECTRICAL DEMAND

Subsystem	Specification	Allowance (W)	Calculated (W)
Laser pump 1	785 nm, 100 W output, 45% WPE	222.2	222.222222
Laser pump 2	8.33 THz drive, 10 W output, 5% WPE	200.0	200.000000
Coolant pump	7.0 L/min, 3.5 bar rise, 35% efficiency	116.0	116.666667
Vacuum turbopumps	Running/cold-start average allocation	120.0	120.000000
Control and diagnostics	FPGA, DAQ, RF, sensors allocation	141.8	141.800000
Total	Specified operating-point arithmetic	800.0	800.688889

TABLE XVI
FIXED-LOAD DESIGN-POINT ELECTRICAL AND THERMAL BALANCES

Quantity (W)	10 kW heat	15 kW heat
Nuclear heat	10000.00	15000.00
TEG before converter	1221.88	1832.82
Converter loss	61.09	91.64
TEG bus generation	1160.79	1741.18
BOP demand	800.69	800.69
Net export	360.10	940.49
Core thermal rejection	8778.12	13167.18
Rejection plus converter loss	8839.21	13258.82

F. Cascade TEG Architecture, Conditioning, and Heat Rejection

The segmented $\text{Bi}_2\text{Te}_3/\text{PbTe}$ material formula is evaluated before separately applying module and converter losses. At $T_H = 550$ K and $T_L = 310$ K,

$$\begin{aligned}\eta_C &= 1 - T_L/T_H = 0.4363636364, \\ \eta_{\text{mat}}(ZT) &= \eta_C \frac{\sqrt{1 + ZT} - 1}{\sqrt{1 + ZT} + T_L/T_H}, \\ \eta_{\text{mat}}(1.25) &= 0.105726872247, \\ \eta_{\text{mat}}(2.16) &= 0.144935257564.\end{aligned}\quad (228)$$

The material efficiencies round to 10.57% and 14.50%. The latter describes the high- ZT comparison rather than a fundamental efficiency limit. Neither value includes module or converter losses. Application of the four-part loss model gives bus efficiencies of 8.467684% and 11.607891%, respectively. At 10/15 kW thermal input, the baseline- ZT net exports are 46.08/469.46 W and the high- ZT exports are 360.10/940.49 W, before any additional cooling or excitation-system loads. The module and converter loss factors are

$$\begin{aligned}d_{\text{thermal}} &= \frac{R_{\text{TEG}}}{R_{\text{TEG}} + R'_h + R'_c} \\ &= \frac{1.15}{1.15 + 0.05 + 0.05} = 0.920, \\ d_{\text{electrical}} &= \frac{R_L}{R_L + R_{\text{contact}} + R_{\text{wire}}} \\ &= \frac{2.10}{2.10 + 0.06 + 0.04} = \frac{21}{22}, \\ d_{\text{geometric}} &= \frac{8784}{8784 + 366} = 0.960, \\ \eta_{\text{conv}} &= 0.950.\end{aligned}\quad (229)$$

TABLE XVII
CORE AND CONVERTER HEAT-REJECTION DUTY FOR THE HIGH- ZT COMPARISON

Quantity (W)	10 kW heat	15 kW heat
Core thermal rejection	8778.12	13167.18
Converter heat	61.09	91.64
Combined rejection	8839.21	13258.82
Allocated compressor power	0.00	0.00

Thermal resistances have units of K/W and electrical resistances have units of ohms; the corresponding ratios are dimensionless. Under the multiplicative loss approximation,

$$\begin{aligned}d_\eta &= d_{\text{thermal}} d_{\text{electrical}} d_{\text{geometric}} \eta_{\text{conv}} \\ &= 0.920(21/22)(0.960)(0.950) = 0.8009018182, \\ \eta_{\text{net,TEG}} &= \eta_{\text{mat}}(2.16)d_\eta = 0.1160789113.\end{aligned}\quad (230)$$

Thermal resistances refer to the specified module and network topology. The ratio 8784/366 defines the geometric loss fraction.

$$\eta_C = 0.4363636364 > \eta_{\text{mat}}(2.16) > \eta_{\text{net,TEG}}. \quad (231)$$

a) *Refrigeration demand and heat rejection:* The component-resolved BOP allocation excludes compressor power. A refrigerated configuration therefore requires a separate conditional calculation:

$$\begin{aligned}\text{COP} &= \eta_{\text{cycle}} \frac{T_{\text{cold}}}{T_{\text{amb}} - T_{\text{cold}}}, \\ \eta_{\text{cycle}} &= 0.375, \quad T_{\text{cold}} = 280 \text{ K}, \\ T_{\text{amb}} &= 305 \text{ K}, \quad \text{COP} = 4.2, \\ P_{\text{comp}} &= Q_{\text{chilled}}/4.2.\end{aligned}\quad (232)$$

A 450 W compressor removes 1890 W at the specified COP, with its electrical demand *additional* to the loads in Table XV. Without refrigeration, the direct-rejection system must dissipate the following thermal loads: A 310 K cold side requires a cooler ambient and a sufficient heat-exchanger temperature approach; ambient temperatures at or above 310 K cannot sustain that sink passively. Dissipation from the remaining BOP components contributes to the appropriate facility heat-rejection boundary.

If all listed core/converter rejection is chilled at COP 4.2, the additional compressor demand gives

$$P_{\text{all}} = \eta_{\text{net,TEG}} P_{\text{nuclear}} - P_{\text{BOP}} - \frac{(1 - \eta_{\text{net,TEG}}) P_{\text{nuclear}}}{4.2}, \quad (233)$$

$$P_{\text{all}}(10 \text{ kW}) = -1744.47 \text{ W},$$

$$P_{\text{all}}(15 \text{ kW}) = -2216.37 \text{ W}.$$

Positive design margins are conditional on a feasible direct heat sink and complete accounting of auxiliary and excitation-system loads.

b) Secondary and tertiary coolant loops: The coolant-loop balances are $\dot{m} = Q/(c_p \Delta T)$ and $\dot{V} = \dot{m}/\rho$. The 7.0 L/min, 3.5 bar pump duty is distinct from the 60 W pump-budget case and the 17.7/23.5 L/min comparison loops. At 7.0 L/min, the parameters in Table LV give $Re = 2582.8$, beyond the stated laminar-correlation range, and $\Delta T = 28.18 \text{ K}$ at the 13725 W benchmark. This duty therefore does not satisfy the peak-temperature-rise criterion $\Delta T < 25 \text{ K}$ associated with a flow of at least 8 L/min. Validation requires a hydraulic characteristic appropriate to the operating flow regime, rather than a flow-and-pressure allocation alone.

G. PZT-5H Acoustic Harvesting and Common-Bus Dynamics

PZT-5H acoustic harvesting is an optional experimental interface and contributes no generation credit to the baseline power balance. The reduced reciprocal piezoelectric model uses displacement x , effective mass m_a , mechanical damping b_a , stiffness k_a , coupling coefficient ϑ , terminal capacitance C_p , voltage v_p , and equivalent resistive load R_L :

$$m_a \ddot{x} + b_a \dot{x} + k_a x + \vartheta v_p = F_a(t), \quad (234)$$

$$C_p \dot{v}_p + \frac{v_p}{R_L} - \vartheta \dot{x} = 0.$$

The model coefficients require characterization of the assembled transducer under its mechanical and electrical loads. With $\mathbf{z} = [x, \dot{x}, v_p]^T$, the linear state model is

$$\dot{\mathbf{z}} = \begin{bmatrix} 0 & 1 & 0 \\ -k_a/m_a & -b_a/m_a & -\vartheta/m_a \\ 0 & \vartheta/C_p & -1/(R_L C_p) \end{bmatrix} \mathbf{z} + \begin{bmatrix} 0 \\ 1/m_a \\ 0 \end{bmatrix} F_a(t). \quad (235)$$

For zero initial conditions, the force-to-voltage transfer function is

$$\frac{V_p(s)}{F_a(s)} = \frac{\vartheta s}{(m_a s^2 + b_a s + k_a)(C_p s + R_L^{-1}) + \vartheta^2 s}. \quad (236)$$

Multiplying the two equations in (234) by $\dot{x}$ and v_p , respectively, produces the passive energy balance

$$\frac{d}{dt} \left(\frac{1}{2} m_a \dot{x}^2 + \frac{1}{2} k_a x^2 + \frac{1}{2} C_p v_p^2 \right) = F_a \dot{x} - b_a \dot{x}^2 - \frac{v_p^2}{R_L}. \quad (237)$$

The steady harvested power is bounded by the supplied mechanical power after mechanical dissipation; rectifier and converter efficiencies determine the bus-delivered fraction.

An active MOSFET bridge rectifies the alternating PZT voltage. A controlled buck stage delivers power to the common

bus when the rectified voltage exceeds 48 V; operation below that voltage requires a buck-boost stage or cessation of export. The steady energy requirements are

$$P_{\text{PZT,bus}} = \eta_{\text{rect}} \eta_{\text{dc}} P_{\text{PZT,raw}},$$

$$P_{\text{PZT,raw}} \leq \langle F_a \dot{x} \rangle, \quad (238)$$

$$I_{\text{PZT,bus}} = \frac{P_{\text{PZT,bus}}}{48 \text{ V}}.$$

Acoustic energy recovered before thermalization is excluded from the subsequent TEG heat input. The channel partition must satisfy Eq. (237) and the system first law, with each recovered contribution counted once.

For bus capacitance C_{bus} and startup-source power P_{start} , the common-bus dynamics are

$$C_{\text{bus}} V_{\text{bus}} \frac{dV_{\text{bus}}}{dt} = P_{\text{TEG,bus}} + P_{\text{PZT,bus}} + P_{\text{start}} - P_{\text{aux,bus}} - P_{\text{load,bus}}. \quad (239)$$

The bus-referred demand $P_{\text{aux,bus}}$ includes auxiliary supplies, point-of-load conversion, and wiring losses; it is not constrained to the comparison allocation of 718.96 W. At nominal steady operation, $P_{\text{start}} = P_{\text{PZT,bus}} = 0$. For the case with only the itemized support loads,

$$I_{\text{gross}} = 1741.183670/48 = 36.274660 \text{ A},$$

$$I_{\text{aux,list}} = 800.688889/48 = 16.681019 \text{ A}, \quad (240)$$

$$I_{\text{load,list}} = 940.494781/48 = 19.593641 \text{ A}.$$

The quantity $P_{\text{aux,list}} = P_{\text{BOP}}$ includes every load in Table XV, whereas P_{other} comprises additional metered demands outside that schedule. These currents correspond to the 15 kW thermal comparison; the 10 kW case gives $360.100224/48 = 7.502088 \text{ A}$ net export. Bus-referred accounting does not require every facility supply to operate physically from the 48 V bus.

For measured terminal-power accounting,

$$P_{\text{net,bus}} = \eta_{\text{boost}} P_{\text{TEG,terminal}} + \eta_{\text{rect}} \eta_{\text{dc}} P_{\text{PZT,raw}} - P_{\text{aux,list}} - P_{\text{other}}. \quad (241)$$

Optional acoustic harvesting receives credit only after measuring its output and removing the same extracted energy from the thermal-input budget. The baseline in (223) assigns no such credit.

H. Metrology, Durability, and Acceptance Requirements

Acceptance testing requires measurement of core thermal duty, cascade heat input and output, boost losses, every support load in Table XV, optical and cooling supplies, and net export at a common declared boundary. The 1275 W generation and 475 W net values are conditional reference points rather than predetermined measurement results. The WT5000 supply belongs to the 25 W control/diagnostics allocation only if that boundary supplies it; otherwise it constitutes a separate demand. The energy balance must include startup-source discharge, changes in stored thermal energy, cooling-system inputs, and externally powered instrumentation.

TABLE XVIII
CONSTANT-EFFICIENCY ELECTRICAL OUTPUT AND HEAT-REJECTION
COMPARISON

Quantity (W)	10 kW heat	15 kW heat
Bus generation	1160.789113	1741.183670
Allocated BOP	800.688889	800.688889
Net electrical export	360.100224	940.494781
Heat minus bus generation	8839.210887	13258.816330
Allocated compressor	0	0

For a measured excess-power estimate $\hat{P}_{\text{excess}}$ and combined standard uncertainty u_c ,

$$z = \frac{\hat{P}_{\text{excess}}}{u_c}, \quad u_c^2 = \mathbf{J} \mathbf{S} \mathbf{J}^T, \quad J_j = \frac{\partial P_{\text{excess}}}{\partial y_j}. \quad (242)$$

The uncertainty budget includes correlations and systematic effects, independent blanks, complete electrical-input measurements, calibrated heat capture, and nuclear-product diagnostics. Statistical significance must be evaluated for the actual measurement boundary; results from a different boundary are not directly transferable. The TLP fatigue-life target is $N_f = 44,820$ defined thermal cycles, subject to the constitutive and qualification conditions in Section IV. Design allocations for power and lifetime remain distinct from measured acceptance results.

a) Fixed-load operating-point specification: The fixed-load comparison [70] uses unrounded component demand and loss factors, with converter dissipation included once:

$$\begin{aligned} P_{\text{BOP}} &= 100/0.45 + 10/0.05 \\ &\quad + (7 \times 10^{-3}/60)(3.5 \times 10^5)/0.35 \\ &\quad + 120 + 141.8 = 800.68888889 \text{ W}, \\ d_\eta &= 0.920(21/22)(0.960)(0.950) \\ &= 0.800901818182, \\ \eta_{\text{bus}} &= 0.144935257564 \quad d_\eta = 0.116078911301. \end{aligned} \quad (243)$$

The fixed-load electrical break-even threshold is $P_{\text{th,break}} = P_{\text{BOP}}/\eta_{\text{bus}}$; it is not a nuclear ignition criterion. The heat-rejection values include converter losses but exclude auxiliary dissipation outside the specified thermal boundary. Fans, controls, optical absorption, pumps, and refrigeration contribute once at their respective boundaries. A zero compressor allocation is physically admissible only when a non-refrigerated sink can sustain the junction-temperature difference assumed by the material-efficiency model.

I. First-Principles Nuclear-Lattice Rates and Thermodynamic Closure

The microscopic rate model combines the 625-spatial-harmonic, 25-temporal-sideband electronic response in Eq. (26) with the elastic screened interaction in Eq. (32). The underlying Bloch transition harmonics and polarizability are defined in Eqs. (84)–(85). At the selected pair coordinate, $U_{\text{eff}} = 352.4800$ eV corresponds to $V_{\text{driven}} = -301.05286$ eV and the reconstructed reference $V_{\text{eff}} = 51.43$ eV.

a) Channel-resolved interaction and initial-state normalization: The interaction Hamiltonian separates strong nuclear, electromagnetic, and nuclear-lattice couplings:

$$\hat{H}_{\text{int}} = \hat{H}_{\text{strong}} + \hat{H}_{\text{EM}} + \hat{H}_{\text{lattice-nuc}}. \quad (244)$$

For normalized states, the specified nuclear matrix element is

$$\mathcal{M}_{\text{nuc}} = \langle \psi_{d-d}^{\text{screened}} | \hat{H}_{\text{int}} | \psi_{4\text{He}^*} \rangle. \quad (245)$$

For a Hermitian interaction, the reverse transition uses its complex conjugate. Each final channel nevertheless has its own matrix element and density of states; the helium-state element alone cannot determine proton and neutron branches. With $\mathcal{M}_{c,f,i} = \langle \psi_{c,f} | \hat{H}_{\text{int}} | \psi_i \rangle$, the energy-resolved Golden-Rule construction is

$$\Gamma_c = \frac{2\pi}{\hbar} \sum_{i,f} w_i |\mathcal{M}_{c,f,i}|^2 \delta(E_{c,f} - E_i), \quad \sum_i w_i = 1. \quad (246)$$

The final-state sum includes continuum phase-space measures. Matrix elements have units of energy, and the energy delta function supplies inverse-energy units, giving Γ_c in s^{-1} for the stated initial-state normalization. Scattering rates require the corresponding incident-flux normalization before conversion to cross sections. The four channels are $t + p$, ${}^3\text{He} + n$, ${}^4\text{He} + \gamma$, and ${}^4\text{He}$ with lattice excitation, with approximate reaction energies 4.03, 3.27, 23.84, and 23.84 MeV, respectively.

No independent g_{exp} or g_{coll} multiplies Eq. (246). If a collective initial state is justified, its coherence enters w_i , the density operator, and the transition amplitudes themselves. Geometric domain count does not determine that coherence or suppress photon channels by itself. The total branching fraction follows Eq. (76); the current two-channel value 0.999999996 follows from the reported rates in Eq. (262), not from state normalization.

b) Energy-conserving Franck-Condon spectral sum: Let $\mathbf{n}$ and $\mathbf{m}$ denote initial and final phonon occupation vectors, $p_{\mathbf{n}}$ their thermal initial probabilities, and $D(\Delta)$ the displacement associated with the nuclear transition. Define the overlap spectral density by

$$\begin{aligned} W_{\text{FC}}(E) &= \sum_{\mathbf{n}, \mathbf{m}} p_{\mathbf{n}} |\langle \mathbf{m} | D(\Delta) | \mathbf{n} \rangle|^2 \\ &\quad \times \delta \left(E - \sum_k \hbar \omega_k (m_k - n_k) \right). \end{aligned} \quad (247)$$

The sum uses net phonon energy, not final occupation alone; thermally populated initial modes can both emit and absorb. It satisfies $\int W_{\text{FC}}(E) dE = 1$ and has units J^{-1} . It is therefore a spectral density, not a dimensionless probability or a decay rate. Under a separable nuclear/phonon matrix element, the lattice rate reduces to

$$\Gamma_{\text{lat}} = \frac{2\pi}{\hbar} |\mathcal{M}_{\text{nuc}}|^2 W_{\text{FC}}(Q_{DD}), \quad (248)$$

with recoil and other final-state energies included in the available energy when non-negligible. If separability fails, the full channel sum in Eq. (246) must be retained. The energy delta functions may be replaced by normalized, physically

determined spectral line shapes; arbitrary broadening cannot be used to manufacture a large overlap at the reaction energy.

For independent harmonic modes, Eq. (142) gives the exact thermal displacement weights and their convolution:

$$P_k(p) = e^{-S_k \coth(\beta_{T,k}/2)} I_p \left(\frac{S_k}{\sinh(\beta_{T,k}/2)} \right) e^{p\beta_{T,k}/2},$$

$$\beta_{T,k} = \frac{\hbar\omega_k}{k_B T}, \quad S_k = |\Delta_k|^2,$$

$$W_{\text{FC}}(E) = \sum_{\{p_k\}} \left[\prod_k P_k(p_k) \right] \delta \left(E - \sum_k p_k \hbar\omega_k \right). \quad (249)$$

Here I_p is the modified Bessel function of the first kind. At $T = 623.15$ K the representative LO mode has $\beta_T \simeq 0.641393$, $\bar{n}_{\text{ph}} \simeq 1.112193$, and the selected Huang–Rhys factor $S = 0.130604$. Its mean net emission is S , not the much larger energy-equivalent count

$$N_q = \frac{Q_{DD}}{\hbar\omega_{\text{LO}}} \simeq 6.92176 \times 10^8 \approx 6.92 \times 10^8. \quad (250)$$

This ratio need not be an integer because the physical final state contains multiple modes and recoil. For the driven state $\rho_D = D(\alpha_-)\rho_{\text{th}}D^\dagger(\alpha_-)$, transition correlations and drive-sideband energy exchange must be evaluated explicitly; the equilibrium Bessel convolution defines its thermal reference. LO-to-acoustic anharmonic relaxation redistributes deposited energy and must not count Q_{DD} a second time.

c) *Metal-supported reaction and species balances:* For an independent-pair kinetic reduction, define the microscopic channel coefficient $K_c = \langle \sigma_c v \rangle_{\text{micro}}$ in m^3/s and the channel rate density $\mathcal{R}_c$ in $\text{m}^{-3}\text{s}^{-1}$:

$$\mathcal{R}_c(\mathbf{r}) = \frac{1}{2} n_d^2(\mathbf{r}) K_c(\mathbf{r}),$$

$$\mathcal{R}_{\text{tot}} = \sum_c \mathcal{R}_c, \quad \mathcal{R}_{\text{lat}} = B_{\text{lat}} \mathcal{R}_{\text{tot}}, \quad (251)$$

$$\partial_t n_d + \nabla \cdot \mathbf{J}_d = -2\mathcal{R}_{\text{tot}},$$

$$\partial_t n_{\text{He}} + \nabla \cdot \mathbf{J}_{\text{He}} = \mathcal{R}_{\text{lat}} + \mathcal{R}_\gamma.$$

These local balances apply inside reacting metal, with transport and exchange boundary conditions. A non-independent pair distribution requires its corresponding correlation function in the reaction kernel, rather than an arbitrary multiplicative gain. For spatially varying channel rates, the integrated branching fraction is the ratio of their integrated rates, not an unweighted spatial average.

The volume conversion is fixed by

$$V_{\text{metal}} = 2.875 \times 10^{-12} \text{ m}^3,$$

$$V_{\text{env}} = 2.53 \times 10^{-10} \text{ m}^3,$$

$$\eta_{\text{grating}} = V_{\text{metal}}/V_{\text{env}} \simeq 0.01136,$$

$$P_{\text{lat}} = Q_{DD} \int_{V_{\text{metal}}} \mathcal{R}_{\text{lat}} d^3r \quad (252)$$

$$= V_{\text{metal}} \bar{q}_{\text{lat,metal}} = V_{\text{env}} \bar{q}_{\text{lat,env}},$$

$$\bar{q}_{\text{lat,env}} = \eta_{\text{grating}} \bar{q}_{\text{lat,metal}}.$$

The last relation defines an envelope average with zero nuclear source outside the metal; it does not assume a uniform microscopic source. Setting $P_{\text{lat}} = 2911.40$ W conditionally

requires the mean metal source $1.012661 \times 10^{15} \text{ W/m}^3$ and envelope source $1.150751 \times 10^{13} \text{ W/m}^3$, consistent with Eq. (177).

d) *Thermodynamic and electrical boundaries:* At the reactor control-volume level, the thermal balance is

$$\frac{dU_{\text{th}}}{dt} = P_{\text{lat}} + P_{\text{other,dep}} + P_{\text{drive,abs}} - P_{\text{TEG,in}} - P_{\text{loss}}. \quad (253)$$

Only deposited channel energy enters $P_{\text{other,dep}}$; escaping particle and photon energy is not simultaneously assigned to lattice heat. Stored thermal, phase, and mechanical energy changes must be included in a transient balance. For the earlier compressor-free comparison only, identifying $P_{\text{TEG,in}}$ with the 2911.40 W reference and using its stated bus output gives

$$\eta_{\text{bus}} = \frac{418.43}{2911.40} = 0.1437212,$$

$$P_{\text{TEG,reject}} = 2911.40 - 418.43 = 2492.97 \text{ W}, \quad (254)$$

$$P_{\text{net}} = 418.43 - 386.67 = +31.76 \text{ W}.$$

The rejection quantity refers to the combined conversion boundary used for the bus output; conversion losses outside that boundary require separate accounting. Continuous 48 V DC export requires simultaneous steady thermal operation, the specified conversion efficiency, and complete support-load coverage.

J. Contour Generating Functional and High-Order Phonon Kinetics

Let ρ_0 be a normalized initial nuclear–lattice density operator. The forward/backward contour formulation retains the interaction to all orders before any bath or self-energy approximation [28]:

$$\mathcal{Z}[J_+, J_-] = \text{Tr}(U_{J_+} \rho_0 U_{J_-}^\dagger),$$

$$\mathcal{Z}[J, J] = 1,$$

$$\mathcal{Z}[J_+, J_-] = \int_{\rho_0} \mathcal{D}[\bar{\psi}, \psi, \bar{b}, b] \exp \left\{ \frac{i}{\hbar} (S_C + S_J) \right\}, \quad (255)$$

$$S_C = \int_C dt \left[\bar{\psi} i \hbar \partial_t \psi + \sum_{\mathbf{q}} \bar{b}_{\mathbf{q}} i \hbar \partial_t b_{\mathbf{q}} - H_{\text{nuc}} - H_{\text{ph}} - H_{\text{int}} \right].$$

The contour measure includes the initial-state boundary condition and final-time matching of both branches. The source action is $S_J = \int_C J \mathcal{O} dt$ for the chosen observable.

A Hermitian phonon displacement and the paramagnetic nuclear current are

$$\hat{\mathbf{u}}(\mathbf{R}) = \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2M_{\text{cell}}\omega_{\mathbf{q}}N_{\text{cell}}}} \left[\mathbf{e}_{\mathbf{q}} \hat{b}_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{R}} + \mathbf{e}_{\mathbf{q}}^* \hat{b}_{\mathbf{q}}^\dagger e^{-i\mathbf{q}\cdot\mathbf{R}} \right],$$

$$\hat{\mathbf{J}}_{\text{nuc}}(\mathbf{r}) = \sum_a \frac{q_a}{2m_a} \{ \hat{\mathbf{p}}_a, \delta(\mathbf{r} - \hat{\mathbf{r}}_a) \}. \quad (256)$$

The braces denote an anticommutator. A proposed moving-potential coupling takes the form

$$\hat{\mathbf{A}}_{\text{eff}}(\mathbf{r}) = \frac{1}{c^2} \sum_{\mathbf{R}} \phi_{\text{latt}}(\mathbf{r} - \mathbf{R}) \dot{\mathbf{u}}(\mathbf{R}),$$

$$\hat{H}_{\text{int}}^{(J)} = - \int \hat{\mathbf{J}}_{\text{nuc}}(\mathbf{r}) \cdot \hat{\mathbf{A}}_{\text{eff}}(\mathbf{r}) d^3r. \quad (257)$$

Here ϕ_{latt} is an electric potential in volts, not a potential energy. The vector potential then has SI units of volt-seconds per metre. This is a quasistatic coupling ansatz; a gauge-complete microscopic Hamiltonian must additionally include the consistent charge and diamagnetic terms. It cannot fix a nuclear transition rate without the transition current and electronic screening.

For a stationary thermal harmonic mode with $S = |\Delta|^2$ and $\bar{n} = (e^{\beta\tau} - 1)^{-1}$, the displacement correlation reduces exactly to

$$\begin{aligned} C(t) &= \langle D^\dagger(t)D(0) \rangle_{\text{th}} \\ &= e^{-S(2\bar{n}+1)} \exp\{S\bar{n}e^{i\omega_{\text{LO}}t} + S(\bar{n}+1)e^{-i\omega_{\text{LO}}t}\}, \\ C(t) &= \sum_{p \in \mathbb{Z}} P(p) e^{-ip\omega_{\text{LO}}t}. \end{aligned} \quad (258)$$

The modified-Bessel generating identity, equivalently the imaginary-argument Jacobi–Anger expansion, gives [27]

$$\begin{aligned} e^{(z/2)(t+t^{-1})} &= \sum_{p \in \mathbb{Z}} I_p(z) t^p, \\ P(p) &= e^{-S(2\bar{n}+1)} I_p\left(2S\sqrt{\bar{n}(\bar{n}+1)}\right) \left(\frac{\bar{n}+1}{\bar{n}}\right)^{p/2}, \\ \sum_p P(p) &= 1, \quad P(-p) = e^{-p\beta\tau} P(p). \end{aligned} \quad (259)$$

The auxiliary variable t in the first line is not physical time. The zero-temperature limit is $P(p) = e^{-S} S^p / p!$ for $p \geq 0$ and zero otherwise. The notation $I_N(2S\bar{n})$ is only a large-occupation approximation to the Bessel argument, not the exact finite-temperature formula. At 623.15 K, $\bar{n} \simeq 1.112193$ is not asymptotically large. A coherent occupation $|\alpha|^2$ is not interchangeable with $\bar{n}$; its displacement phase must be retained in the contour correlation.

For a separable transition with coupling $\mathcal{V}$ in energy units, Fourier transformation gives

$$\begin{aligned} \Gamma_{\text{lat}}(E) &= \frac{|\mathcal{V}|^2}{\hbar^2} \int_{-\infty}^{\infty} e^{iEt/\hbar} C(t) dt \\ &= \frac{2\pi}{\hbar} |\mathcal{V}|^2 \sum_p P(p) \delta(E - p\hbar\omega_{\text{LO}}). \end{aligned} \quad (260)$$

This rate has units s^{-1} when the final-state energy density is included. A continuous linewidth replaces the delta function by a normalized spectral function. Equivalently $\Gamma = -2 \text{Im} \Sigma^R(E)/\hbar$ applies to a well-defined decaying pole with its appropriate residue; it is not an unrestricted strong-coupling identity for arbitrary memory dynamics.

Large order alone does not eliminate overlap suppression. For example, the uniform large-order expansion gives [27]

$$e^{-N} I_N(N) \sim \frac{\exp\{N[\sqrt{2}-1 - \text{arsinh}(1)]\}}{\sqrt{2\pi N} 2^{1/4}}, \quad (261)$$

whose exponential coefficient is approximately -0.4671600 . Thus setting the argument to $N = 692000000$ does not cancel the exponential penalty. A high-order numerical rate requires independently specified coupling, occupations, and energy broadening.

The reported rate pair nevertheless has an unambiguous arithmetic consequence:

$$\begin{aligned} \Gamma_{\text{lat}} &= 2.5 \times 10^{22} \text{ s}^{-1}, \\ \Gamma_{\gamma} &= 1.0 \times 10^{14} \text{ s}^{-1}, \\ R_{\text{lat}/\gamma} &= 2.5 \times 10^8, \\ B_{\text{lat}}^{(2)} &= \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_{\gamma}} \\ &= 0.9999999960000000 \dots, \\ 1 - B_{\text{lat}}^{(2)} &= 3.999999984 \times 10^{-9}, \\ B_{\text{lat}} &= \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_{\gamma} + \Gamma_p + \Gamma_n} \leq B_{\text{lat}}^{(2)}. \end{aligned} \quad (262)$$

The value 0.999999996 is the rounded, conditional two-channel branching fraction. Absolute emission uses the excited-state population and detector response; thermal power uses the metal-volume population and energy normalization of the preceding subsection.

K. Multimode Lindblad Dynamics and Matrix Magnus Expansion

The state variable is the joint nuclear–lattice density operator $\hat{\rho}$, not a scalar survival probability. With $\mu = (\mathbf{q}, \lambda)$ indexing LO, LA, and TA branches, the Markovian open-system formulation is [88]

$$\begin{aligned} \frac{d\hat{\rho}}{dt} &= -\frac{i}{\hbar} [\hat{H}(t), \hat{\rho}] + \mathcal{L}_{\text{diss}}(\hat{\rho}), \\ \hat{H}(t) &= \hat{H}_{\text{nuc}} + \hat{H}_{\text{lat}} + \hat{H}_{\text{nuc-lat}}(t) + \hat{H}_{\text{drive}}(t), \\ \hat{H}_{\text{lat}} &= \sum_{\mu=1}^{N_{\text{modes}}} \hbar\omega_{\mu} (\hat{a}_{\mu}^{\dagger} \hat{a}_{\mu} + \frac{1}{2}), \\ \mathcal{L}_{\text{diss}}(\hat{\rho}) &= \sum_{\mu=1}^{N_{\text{modes}}} \gamma_{\mu} [(\bar{n}_{\mu} + 1) \mathcal{D}[\hat{a}_{\mu}](\hat{\rho}) + \bar{n}_{\mu} \mathcal{D}[\hat{a}_{\mu}^{\dagger}](\hat{\rho})], \\ \mathcal{D}[\hat{O}](\hat{\rho}) &= \hat{O} \hat{\rho} \hat{O}^{\dagger} - \frac{1}{2} \{\hat{O}^{\dagger} \hat{O}, \hat{\rho}\}, \\ \bar{n}_{\mu} &= \left[e^{\hbar\omega_{\mu}/(k_{\text{B}}T)} - 1 \right]^{-1}, \quad \gamma_{\mu} \geq 0. \end{aligned} \quad (263)$$

The rate γ_{μ} appears once. Equivalently, its square root and the occupation factors may be absorbed into the jump operators with the external prefactor omitted. For constant bath parameters, the oscillator dissipator gives

$$\left. \frac{d\langle \hat{a}_{\mu}^{\dagger} \hat{a}_{\mu} \rangle}{dt} \right|_{\text{diss}} = -\gamma_{\mu} (\langle \hat{a}_{\mu}^{\dagger} \hat{a}_{\mu} \rangle - \bar{n}_{\mu}). \quad (264)$$

It is a thermal-reservoir model, not by itself an explicit energy-conserving LO-to-acoustic scattering calculation. If acoustic modes are retained as part of the resolved system, anharmonic transfer can instead be represented by terms such as

$$\hat{H}_{\text{anh}} = \sum_{\mu\nu\kappa} (\Phi_{\mu\nu\kappa} \hat{a}_{\mu} \hat{a}_{\nu}^{\dagger} \hat{a}_{\kappa}^{\dagger} + \Phi_{\mu\nu\kappa}^* \hat{a}_{\mu}^{\dagger} \hat{a}_{\nu} \hat{a}_{\kappa}), \quad (265)$$

with momentum selection, mode frequencies, and coupling constants specified. Transfer already represented explicitly must not also be counted as an unobserved thermal sink. A nonsecular Redfield generator is not generally identical to

the completely positive Lindblad form. Keeping matrix-valued modes does not remove the reservoir, Markov, and secular assumptions underlying a particular dissipative approximation.

The assumed inventory $N_{\text{modes}} = 6.92 \times 10^8$ can be imposed as a discretization assumption, but it is distinct from the energy-equivalent emitted phonon order $N_q \simeq 6.92 \times 10^8$. Multiple quanta can occupy one mode, while available LO and acoustic branches are determined by the crystal and wavevector discretization. For retained Fock cutoffs the joint Hilbert-space dimension is

$$d_{\mathcal{H}} = d_{\text{nuc}} \prod_{\mu=1}^{N_{\text{modes}}} (n_{\text{max},\mu} + 1). \quad (266)$$

Reduced bath, tensor-network, trajectory, and moment-closure formulations specify their own truncation and convergence criteria.

a) *Interaction-picture coherent propagator:* Let $\tau = t - t_0$ and remove the chosen free system evolution before expanding the nuclear-lattice interaction. For constant mode couplings, write

$$\begin{aligned} \hat{H}_I(t_0 + s) &= \sum_{\mu} \left(\hat{A}_{\mu} e^{i\delta_{\mu}s} + \hat{A}_{\mu}^{\dagger} e^{-i\delta_{\mu}s} \right), \\ \hat{A}_{\mu} &= g_{\mu} \hat{S}_{+} \hat{a}_{\mu}, \quad \delta_{\mu} = \omega_0 - \omega_{\mu}, \\ \hat{U}_I(t, t_0) &= \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{H}_I(t') dt' \right] = \exp \left[\sum_{k \geq 1} \hat{\Omega}_k \right]. \end{aligned} \quad (267)$$

Initial phases are absorbed in g_{μ} ; drive sidebands add their detunings and amplitudes to this sum. If only the oscillator rotation is removed, the frequency in an annihilation term is instead $-\omega_{\mu}$ and the remaining spin evolution must still be included. These frame conventions cannot be mixed. The Schrödinger-picture propagator includes the free propagator as well as $\hat{U}_I$.

The first two Magnus terms retain operator ordering [30]:

$$\begin{aligned} \hat{\Omega}_1 &= -\frac{i}{\hbar} \int_{t_0}^t \hat{H}_I(t_1) dt_1, \\ \hat{\Omega}_2 &= -\frac{1}{2\hbar^2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 [\hat{H}_I(t_1), \hat{H}_I(t_2)]. \end{aligned} \quad (268)$$

Introduce a signed harmonic index a with operator $\hat{A}_a$ and real frequency ν_a , so that the adjoint partner has $\hat{A}_{\bar{a}} = \hat{A}_a^{\dagger}$ and $\nu_{\bar{a}} = -\nu_a$. The elementary time kernels are

$$\begin{aligned} F(\nu, \tau) &= \int_0^{\tau} e^{i\nu s} ds = \frac{e^{i\nu\tau} - 1}{i\nu}, \\ F(0, \tau) &= \tau, \\ J(\nu, \mu; \tau) &= \int_0^{\tau} ds_1 \int_0^{s_1} ds_2 e^{i\nu s_1 + i\mu s_2} \\ &= \frac{F(\nu + \mu, \tau) - F(\nu, \tau)}{i\mu}, \\ J(\nu, 0; \tau) &= \frac{1}{i} \partial_{\nu} F(\nu, \tau), \quad J(0, 0; \tau) = \tau^2/2. \end{aligned} \quad (269)$$

The quotient expressions use their continuous limits at zero frequency. In a complete joint-state basis $\{|i\rangle\}$, the matrix elements are

$$\begin{aligned} \langle i | \hat{\Omega}_1 | j \rangle &= -\frac{i}{\hbar} \sum_a (A_a)_{ij} F(\nu_a, \tau), \\ \langle i | \hat{\Omega}_2 | j \rangle &= -\frac{1}{2\hbar^2} \sum_{a,b} \sum_{\ell} [(A_a)_{i\ell} (A_b)_{\ell j} - (A_b)_{i\ell} (A_a)_{\ell j}] \\ &\quad \times J(\nu_a, \nu_b; \tau). \end{aligned} \quad (270)$$

The intermediate index ℓ spans nuclear and phonon states; no scalar occupation average replaces this matrix product. In particular,

$$[\hat{S}_{+} \hat{a}_{\mu}, \hat{S}_{-} \hat{a}_{\nu}^{\dagger}] = 2\hat{S}_{z} \hat{a}_{\nu}^{\dagger} \hat{a}_{\mu} + \delta_{\mu\nu} \hat{S}_{+} \hat{S}_{-}. \quad (271)$$

Cross-mode terms persist through the common transition operator even when $\mu \neq \nu$. Discarding them requires an explicit secular or correlation closure. For $\hat{H}_I(s) = \hat{A} e^{-i\omega s} + \hat{A}^{\dagger} e^{i\omega s}$,

$$\hat{\Omega}_2 = \frac{i}{\hbar^2} \left(\frac{\tau}{\omega} - \frac{\sin(\omega\tau)}{\omega^2} \right) [\hat{A}, \hat{A}^{\dagger}]. \quad (272)$$

This expression is anti-Hermitian and has a finite resonant limit. More generally each coherent Magnus term must satisfy $\hat{\Omega}_k^{\dagger} = -\hat{\Omega}_k$.

The coherent update $\rho \mapsto U_I \rho U_I^{\dagger}$ is not the full dissipative step. On a finite truncation, column vectorization gives the Hamiltonian Liouvillian $-i(I \otimes H - H^{\dagger} \otimes I)/\hbar$; each jump contributes $L^* \otimes L - \frac{1}{2}[I \otimes L^{\dagger} L + (L^{\dagger} L)^{\dagger} \otimes I]$. These superoperators, or a checked completely positive splitting, govern the density-matrix evolution. An arbitrary truncated Magnus expansion of a dissipative generator is not automatically completely positive. Adaptive time steps and Fock/mode refinement must monitor trace, Hermiticity, positivity, and energy exchange.

The implementation evolves a 3×3 complex matrix using two cascade jumps $|0\rangle\langle 1|$ and $|1\rangle\langle 2|$ and explicit RK4. Time-step selection is subject to density-matrix positivity.

III. MECHANICAL, OPTOMECHANICAL ALIGNMENT, AND VACUUM INTERFACE

The mechanical assembly supports the pressure boundary, maintains the core-interface preload, and registers the active region to the prism and thermal spreader. Its design envelope couples multiaxial transient stress, cyclic plastic strain, nonlinear spring compliance, and high-temperature stress relaxation. The structural and preload limits in this section define the specified load path; alternative material, bondline, and clamp models describe separate configurations rather than simultaneous requirements. Component qualification requires compatible material properties, contact geometry, and load histories. Electrical interface loads refer to the regulated-bus boundaries in Section II.

a) *Micro-alloyed bondline and graded compliance:*

The selected interconnect is a $3.5 \mu\text{m}$ Ni-6Cu/Sn-0.7Cu-0.05Bi transient-liquid-phase (TLP) bondline with a $15.0 \mu\text{m}$ functionally graded ductile Ni/Cu compliance layer. The alloying and graded layer are specified to control intermetallic phase formation, interfacial voiding, and strain concentration;

their effectiveness requires characterization of the fabricated joint. The fatigue load case spans 20.0–276.85 °C, or 293.15–550.0 K, with $\Delta T = 256.85$ K.

For depth z , the strain partition and accumulated equivalent plastic strain are

$$\begin{aligned}\epsilon_{ij}^{\text{tot}}(z) &= \alpha_T(z)\Delta T\delta_{ij} + [\mathbf{C}^{-1}(z)]_{ijkl}\sigma_{kl}(z) + \epsilon_{ij}^p(z), \\ dp &= \sqrt{\frac{2}{3}} d\epsilon^p : d\epsilon^p.\end{aligned}\quad (273)$$

The grating and substrate CTE inputs are 13.8×10^{-6} and 4.5×10^{-6} K⁻¹, respectively. Their free mismatch is 0.002388705 over the specified cycle; constraint and plastic flow, rather than the CTE difference alone, determine the local plastic amplitude. With backstress $\mathbf{X} = \mathbf{X}_1 + \mathbf{X}_2$, the two-term Chaboche law is

$$\begin{aligned}d\mathbf{X}_k &= \frac{2}{3}C_k d\epsilon^p - \gamma_k \mathbf{X}_k dp, \quad k = 1, 2, \\ C_1 &= 42.5 \text{ GPa}, \quad \gamma_1 = 450, \\ C_2 &= 12.8 \text{ GPa}, \quad \gamma_2 = 85.\end{aligned}\quad (274)$$

The scalar C_k are hardening moduli, distinct from the elasticity tensor $\mathbf{C}$. A symmetric stabilized cycle with zero mean stress is the selected strain-life evaluation condition; it is not implied by the backstress evolution alone.

The Coffin–Manson–Morrow law uses strain amplitudes, with N_f measured in cycles and $2N_f$ in reversals:

$$\begin{aligned}\frac{\Delta\epsilon_{\text{tot}}}{2} &= \frac{\sigma'_f - \sigma_m}{E}(2N_f)^b + \epsilon'_f(2N_f)^c, \\ E &= 128.5 \text{ GPa}, \quad \sigma'_f = 545.0 \text{ MPa}, \\ b &= -0.082, \quad \epsilon'_f = 0.320, \quad c = -0.560, \\ \sigma_m &= 0.\end{aligned}\quad (275)$$

At $2N_f = 104800$ reversals, the elastic and plastic contributions are

$$\begin{aligned}\frac{\Delta\epsilon_e}{2} &= 1.64370452006 \times 10^{-3}, \\ \frac{\Delta\epsilon_p}{2} &= 4.94023558978 \times 10^{-4}, \\ \frac{\Delta\epsilon_{\text{tot}}}{2} &= 0.00213772807904 \approx 0.002137728, \\ N_f &= 52400 \text{ cycles} > 44820 \text{ cycles}, \\ \frac{N_f - 44820}{44820} &= 0.16912093 \approx 16.9\%.\end{aligned}\quad (276)$$

The positive elastic and plastic terms both decrease monotonically with N_f , giving a unique positive strain-life solution for the full-precision total amplitude. The symbol ϵ_{tot} denotes $\Delta\epsilon_{\text{tot}}/2$. The predicted life satisfies the 44,820-cycle constitutive criterion in Table XC. An unalloyed bond with plastic amplitude 0.00223, $\epsilon'_f = 0.3700$, and $c = -0.520$ gives approximately 9290 cycles. The spring calculations below use a separate material and load path.

A. Selected Contact Materials and Clamp-Force Bounds

The selected contact-load configuration uses a circular 20.0 mm core interface over the 25–350 °C temperature envelope. Table XIX reproduces the material-property table in the thermomechanical design specification [103]. Its Pd-25Ir

composition is the material basis for the following contact-force bound; it is distinct from the Pd_{0.95}Ir_{0.05} optical-film model. Transfer of the assigned strength to a different alloy composition requires material qualification. The diamond entry denotes a brittle fracture limit, not a ductile yield stress. Spring strength additionally depends on product form, temper, and heat treatment [106].

The contact area and selected pressure limits are

$$\begin{aligned}A_{\text{contact}} &= \frac{\pi}{4}D_{\text{core}}^2 = \frac{\pi}{4}(0.020 \text{ m})^2 \\ &= 3.14159265359 \times 10^{-4} \text{ m}^2, \\ P_{\text{min}} &= 20 \text{ MPa}, \\ \sigma_{y,\text{film}}(350 \text{ °C}) &= 250 \text{ MPa}.\end{aligned}\quad (277)$$

The 20 MPa minimum is the specified sealing-contact requirement, and the 250 MPa maximum is the assigned hot Pd-Ir film yield limit [103]. Applying these limits to the average contact pressure gives

$$\begin{aligned}F_{\text{min}} &= P_{\text{min}}A_{\text{contact}} \simeq 6283.19 \text{ N}, \\ F_{\text{max}} &= \sigma_{y,\text{film}}A_{\text{contact}} \simeq 78539.82 \text{ N}, \\ F &\in [6283.19, 78539.82] \text{ N}.\end{aligned}\quad (278)$$

Force endpoints are displayed to 0.01 N; pressure checks shall use the unrounded contact area. The optimization summary contains 92 first-front designs with forces from 6,283.19 to 78,410.02 N, within the specified bounds [105]. The bounds apply to the instantaneous transmitted contact force, not merely the assembly setpoint. Thermal excursion, long-term relaxation, and setting uncertainty shall leave sufficient margin to both endpoints. The sealing threshold requires leak qualification, and local contact or spring bending stresses require their own structural assessment.

B. Three-Dimensional Thermomechanical Stress Envelope

The pressure-boundary transition comprises an 8.0 mm wall at inner radius 45.0 mm and a 5.0 mm $\times$ 45° relief bevel. The transient load case combines internal hydraulic pressure from 0.6513 to 16.6013 MPa(a), an axial retaining load of 125.0 kN, and a thermal-gradient magnitude of 48.5 K/mm [87]. Pressure tractions require the internal-to-external pressure difference; the absolute internal values alone do not define that difference. Across a 5.0 mm path aligned with the gradient,

$$\Delta T_{\text{bevel}} = (5.0 \text{ mm})(48.5 \text{ K/mm}) = 242.5 \text{ K}.\quad (279)$$

The gradient direction and transient temperature field remain part of the load definition; a gradient magnitude cannot determine a complete stress tensor.

For small strain, the three-dimensional constitutive relation is

$$\begin{aligned}\boldsymbol{\sigma} &= \mathbf{C} : (\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_p - \alpha\Delta T\mathbf{I}), \\ \boldsymbol{\epsilon} &= \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T), \\ C_{ijkl} &= \lambda_e\delta_{ij}\delta_{kl} + \mu_e(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \\ \lambda_e &= \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad \mu_e = \frac{E}{2(1+\nu)}.\end{aligned}\quad (280)$$

TABLE XIX
MATERIAL PROPERTIES OF SUBSTRATE, ACTIVE FILM, AND SPRING STACK

Structural component	Material composition	Yield limit at 25 °C (σ_y)	Yield limit at 350 °C (σ_y)	Critical failure mode & mechanisms
Active reaction film	Pd-25Ir alloy	350 MPa	250 MPa	Plastic deformation, surface extrusion, and micro-cracking.
Rigid substrate	CVD diamond	2.0 GPa	2.0 GPa	Brittle fracture, interface shearing, and micro-delamination.
Preload spring stack	Inconel X-750	1000 MPa	900 MPa	Low-cycle fatigue and high-temperature stress relaxation.

Here $\mathbf{u}$ is displacement, ϵ_p is the plastic-strain tensor, and $\mathbf{C}$ is the isotropic fourth-order elastic tensor. The hot structural case uses $E(350^\circ\text{C}) = 195.0$ GPa, $\nu = 0.31$, and $\alpha = 14.2 \times 10^{-6} \text{ K}^{-1}$ for the specified precipitation-treated Inconel X-750 condition. The thermal-strain expression assumes constant α over the local temperature interval; a temperature-dependent coefficient requires $\int \alpha(T) dT$. Momentum balance, mechanical restraints, contact tractions, and plastic evolution complete the field problem. A long-duration creep strain must additionally enter the inelastic strain decomposition.

The principal-stress design state is

$$\begin{aligned}\sigma_{\text{principal}} &= \text{diag}(342.5, 185.2, -16.6) \text{ MPa}, \\ \sigma_1 &= +342.5 \text{ MPa}, \\ \sigma_2 &= +185.2 \text{ MPa}, \\ \sigma_3 &= -16.6 \text{ MPa}.\end{aligned}\quad (281)$$

These quantities are principal stress values, not principal direction vectors. If $\mathbf{R}$ contains the orthonormal principal directions, the physical-coordinate tensor is $\boldsymbol{\sigma} = \mathbf{R}\sigma_{\text{principal}}\mathbf{R}^T$. Axial, circumferential, and radial directions coincide with the principal axes only when the local shear components vanish in that basis.

The rotationally invariant equivalent stress and yield safety factor follow directly:

$$\begin{aligned}\sigma_{\text{VM}} &= \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \\ &= \sqrt{\frac{1}{2}[157.3^2 + 201.8^2 + 359.1^2]} \text{ MPa} \\ &= 311.784653 \text{ MPa} \simeq 311.78 \text{ MPa}, \\ \sigma_y &= 850.0 \text{ MPa}, \\ \text{SF} &= \sigma_y/\sigma_{\text{VM}} = 2.72624 \simeq 2.73, \\ \sigma_{\text{allow}} &= 850.0/2.20 = 386.363636 \text{ MPa}.\end{aligned}\quad (282)$$

The assigned principal stresses satisfy the $\text{SF} \geq 2.20$ yield criterion. Structural qualification includes transient stress, temperature- and heat-treatment-specific strength, fatigue, and creep assessment.

C. Coffin–Manson Fatigue Acceptance Criterion

For the selected Inconel X-750 spring-stack criterion, $\Delta\epsilon_p$ denotes plastic strain *amplitude*, one half of the full cyclic range, in mm/mm. The design coefficients are $\epsilon'_f = 0.45$ and

$c = -0.60$ [103]. The Coffin–Manson relation and its inverse are

$$\begin{aligned}\Delta\epsilon_p &= \epsilon'_f (2N_f)^c, \\ N_f &= \frac{1}{2} \left(\frac{\Delta\epsilon_p}{\epsilon'_f} \right)^{1/c}, \\ \Delta\epsilon_{p,\text{max}} &= 0.45(2 \times 44820)^{-0.60} \\ &= 0.0004805199299 \dots, \\ \Delta\epsilon_p &\leq 4.805 \times 10^{-4} \text{ mm/mm} \\ &= 0.04805\%.\end{aligned}\quad (283)$$

Because $c < 0$, the selected ceiling is conservative relative to the exact threshold and gives $N_f \geq 44823.0984$ cycles within this constant-amplitude law. This satisfies the 44,820-cycle analytical requirement. The $\epsilon'_f = 0.18$, $c = -0.62$ model and its 0.0001530105 amplitude ceiling are a separate comparison parameterization, not the governing X-750 criterion. The range-based Cu–Sn bondline relation in Eq. (365) remains a different material criterion. Multiaxial strain amplitudes require a specified equivalent-strain or critical-plane measure; neither peak stress nor accumulated plastic strain alone determines that amplitude. Coefficient uncertainty, variable-amplitude loading, and high-temperature dwell damage shall be assessed separately.

The selected strain-to-displacement approximation is

$$\begin{aligned}\Delta\epsilon_p &= \psi \Delta L_{\text{net}}, \quad \psi \simeq 23.84 \text{ m}^{-1}, \\ \Delta L_{\text{net}} &= 20.1 \text{ }\mu\text{m}, \\ \Delta\epsilon_p &= (23.84)(20.1 \times 10^{-6}) \\ &= 4.79184 \times 10^{-4} \text{ mm/mm} \\ &< 4.805 \times 10^{-4} \text{ mm/mm}.\end{aligned}\quad (284)$$

The 20.1 μm net mismatch is the spring-path displacement over the 325 K thermal excursion, after collar/spacer expansion matching. It is not automatically the optical-axis drift. The scaling coefficient is a geometry-dependent design approximation; its use shall be supported by the selected contact and spring model.

D. Monte Carlo Fatigue Endurance

For the specified $N_{MC} = 1,000,000$ -sample run, the fatigue summary gives [104]

$$\begin{aligned}\bar{N}_f &= 88826.6757 \simeq 88827 \text{ cycles,} \\ s_{N_f} &= 21490.0393 \text{ cycles,} \\ I_{95} &= [54822.0399, 138492.3824] \text{ cycles} \\ &\simeq [54822, 138492] \text{ cycles.}\end{aligned}\quad (285)$$

Both the mean and the lower endpoint of the reported 95% lifespan interval exceed the 44,820-cycle specification. The lower-endpoint margin is 10,002.0399 cycles, or 22.3160%, confirming compliance of these reported Monte Carlo endurance statistics with the specified threshold.

The 95% interval describes lifespan dispersion. Reliability assessment also uses the sub-threshold probability under the selected strain and material distributions.

a) *Stochastic execution and strain-life convention:* Use $c_+ = -c > 0$ to distinguish the positive fatigue exponent from the signed Coffin–Manson exponent. The cycle-count evaluation is

$$N_f = \frac{1}{2} \left(\frac{\epsilon'_f}{\Delta \epsilon_p} \right)^{1/c_+}. \quad (286)$$

The factor 1/2 converts reversals to cycles. Positive plastic strain is sampled from a truncated normal density,

$$f_X(x) = \frac{\varphi((x - \mu_X)/s_X)}{s_X[\Phi((b - \mu_X)/s_X) - \Phi((a - \mu_X)/s_X)]}, \quad (287)$$

for $a \leq x \leq b$ and zero otherwise, where $0 < a < b$, $s_X > 0$, and φ, Φ are the standard normal density and cumulative distribution. Physical-support sampling may use $a = 10^{-6}$ and $b = \epsilon_{ult}$; the comparison value $\epsilon_{ult} = 0.45$ is not the spring acceptance ceiling. Conditioning on compliant spring realizations instead requires $b \leq 4.805 \times 10^{-4}$ mm/mm. Positive fatigue coefficients and exponents require admissible distributions as well. Vectorized cycle evaluation uses $N_{MC} = 10^6$ with a fixed generator, seed 42, and worker partition.

E. Thin Conical Shell Energy and Nonlinear Equilibrium

a) *Reference surface and consistent displacement measures:* The spring is modeled as an axisymmetric, initially stress-free conical midsurface with inner radius $r_i = D_i/2$, outer radius $r_e = D_e/2$, thickness t , and semi-angle α measured from the cone axis. The meridional reference arclength is denoted ξ , leaving s available for the disc-compression coordinate in the comparison load law. Define

$$\begin{aligned}\xi_i &= r_i / \sin \alpha, & \xi_e &= r_e / \sin \alpha, \\ r_0(\xi) &= \xi \sin \alpha, & z_0(\xi) &= \xi \cos \alpha, \\ h_0 &= (r_e - r_i) \cot \alpha, \\ \mathbf{X}_0(\xi, \theta) &= r_0 \mathbf{e}_r(\theta) + z_0 \mathbf{e}_z, \\ \mathbf{x}(\xi, \theta) &= R(\xi) \mathbf{e}_r(\theta) + Z(\xi) \mathbf{e}_z, \\ R &= r_0 + u \sin \alpha + w \cos \alpha, \\ Z &= z_0 + u \cos \alpha - w \sin \alpha.\end{aligned}\quad (288)$$

Here u is meridional displacement and w is displacement along the reference normal $\mathbf{n}_0 = \cos \alpha \mathbf{e}_r - \sin \alpha \mathbf{e}_z$. The

axial displacement is $Z - z_0$, not w . Prime denotes $d/d\xi$. The Kirchhoff–Love assumptions constrain thickness directors to remain normal to the midsurface and neglect transverse shear; the geometric strain measures are obtained from the deformed surface rather than a rigid cross-section rotation [118]. This thin-shell model assumes small elastic strains and t small relative to the relevant curvature and span lengths. It is not an exact three-dimensional contact or plasticity model for an arbitrarily thick washer.

With $\lambda = \sqrt{R'^2 + Z'^2} > 0$, the deformed normal and the normalized reference-metric strains are

$$\begin{aligned}\mathbf{n}_c &= \frac{Z' \mathbf{e}_r - R' \mathbf{e}_z}{\lambda}, \\ \varepsilon_\xi &= \frac{1}{2}(R'^2 + Z'^2 - 1), \\ \varepsilon_\theta &= \frac{1}{2}(R^2/r_0^2 - 1), \\ \varepsilon_\xi &= u' + \frac{1}{2}(u'^2 + w'^2), \\ \varepsilon_\theta &= \frac{a}{r_0} + \frac{a^2}{2r_0^2}, \quad a = u \sin \alpha + w \cos \alpha.\end{aligned}\quad (289)$$

Choose the second fundamental form as $b_{ab} = -\mathbf{n}_c \cdot \mathbf{x}_{,ab}$ and pull its change back to the reference metric. Then

$$\begin{aligned}J &= R' Z'' - Z' R'', \\ \chi_\xi &= b_{\xi\xi} - b_{\xi\xi}^0 = J/\lambda, \\ \chi_\theta &= \frac{b_{\theta\theta} - b_{\theta\theta}^0}{r_0^2} = \frac{RZ'/\lambda - r_0 \cos \alpha}{r_0^2}, \\ b_{\xi\xi}^0 &= 0, \quad b_{\theta\theta}^0 = r_0 \cos \alpha.\end{aligned}\quad (290)$$

At first order in u, w , these expressions give

$$\begin{aligned}\varepsilon_\xi^{(1)} &= u', & \varepsilon_\theta^{(1)} &= a/r_0, \\ \chi_\xi^{(1)} &= -w'', & \chi_\theta^{(1)} &= \frac{a \cos \alpha}{r_0^2} - \frac{w'}{\xi}.\end{aligned}\quad (291)$$

The circumferential curvature change therefore contains the initial-curvature contribution as well as the slope term. For $\alpha = \pi/2$, the flat annular-plate limit has $\chi_\theta^{(1)} = -w'/\xi$. The undeformed cone and a uniform axial translation both have zero membrane strain and zero curvature change, as required by rigid-motion invariance.

b) *Membrane and bending energy integrals:* At prescribed temperature T , the isotropic plane-stress rigidities and the potential energy are

$$\begin{aligned}C_m(\xi, T) &= \frac{E(\xi, T)t}{1 - \nu(\xi, T)^2}, \\ D_b(\xi, T) &= \frac{E(\xi, T)t^3}{12[1 - \nu(\xi, T)^2]}, \\ U_m &= \pi \int_{\xi_i}^{\xi_e} r_0 C_m (\varepsilon_\xi^2 + \varepsilon_\theta^2 + 2\nu \varepsilon_\xi \varepsilon_\theta) d\xi, \\ U_b &= \pi \int_{\xi_i}^{\xi_e} r_0 D_b (\chi_\xi^2 + \chi_\theta^2 + 2\nu \chi_\xi \chi_\theta) d\xi, \\ \Pi[R, Z] &= U_m + U_b - W_{\text{ext}}.\end{aligned}\quad (292)$$

The factor $2\pi r_0 d\xi$ is the reference area element; the factor 1/2 in each elastic energy gives the prefactor π . Positive E and physically admissible ν give positive quadratic material energy, but geometric nonlinearity can still produce nonconvex

load–deflection behavior. For a dead axial compressive force F conjugate to the reduction in cone height,

$$\Delta[R, Z] = h_0 - [Z(\xi_e) - Z(\xi_i)],$$

$$W_{\text{ext}} = F\Delta + \int_{\xi_i}^{\xi_e} (q_R[R - r_0] + q_Z[Z - z_0])d\xi. \quad (293)$$

The distributed forces q_R, q_Z have units N/m per reference meridional length. Thermal, plastic, and creep eigenstrains must be subtracted from the geometric strains when those mechanisms are active; the derivation below holds their internal variables fixed during each equilibrium variation. Temperature-dependent constitutive inputs are not determined by geometry alone.

The energetically conjugate membrane resultants and bending moments are

$$\begin{aligned} N_\xi &= C_m(\varepsilon_\xi + \nu\varepsilon_\theta), \\ N_\theta &= C_m(\varepsilon_\theta + \nu\varepsilon_\xi), \\ M_\xi &= D_b(\chi_\xi + \nu\chi_\theta), \\ M_\theta &= D_b(\chi_\theta + \nu\chi_\xi), \end{aligned} \quad (294)$$

$$\delta U = 2\pi \int_{\xi_i}^{\xi_e} r_0 (N_\xi \delta\varepsilon_\xi + N_\theta \delta\varepsilon_\theta + M_\xi \delta\chi_\xi + M_\theta \delta\chi_\theta) d\xi.$$

The N resultants have units N/m and the M resultants units N. A symmetric homogeneous thickness is assumed here; an unsymmetric laminate additionally couples membrane and bending strains through a constitutive coupling matrix.

c) *Explicit Euler–Lagrange equations*: Write the first variation in the independent radial and axial coordinates as

$$\delta U = \int_{\xi_i}^{\xi_e} [\mathbf{A} \cdot \delta \mathbf{y} + \mathbf{B} \cdot \delta \mathbf{y}' + \mathbf{D} \cdot \delta \mathbf{y}'] d\xi, \quad \mathbf{y} = (R, Z)^\top. \quad (295)$$

Differentiating Eqs. (289) and (290) gives all coefficients explicitly:

$$\begin{aligned} A_R &= \frac{2\pi}{r_0} \left(N_\theta R + M_\theta \frac{Z'}{\lambda} \right), \quad A_Z = 0, \\ B_R &= 2\pi r_0 \left[N_\xi R' + M_\xi \left(\frac{Z''}{\lambda} - \frac{JR'}{\lambda^3} \right) - \frac{M_\theta RZ'R'}{r_0^2 \lambda^3} \right], \\ B_Z &= 2\pi r_0 \left[N_\xi Z' + M_\xi \left(-\frac{R''}{\lambda} - \frac{JZ'}{\lambda^3} \right) + \frac{M_\theta RR'^2}{r_0^2 \lambda^3} \right], \\ D_R &= -2\pi r_0 M_\xi \frac{Z'}{\lambda}, \quad D_Z = 2\pi r_0 M_\xi \frac{R'}{\lambda}. \end{aligned} \quad (296)$$

In particular, the symbols $\mathbf{D}$ and D_b denote different objects: boundary-variation coefficients and material bending rigidity. Integrating twice by parts yields

$$\begin{aligned} \delta \Pi &= \int_{\xi_i}^{\xi_e} (\mathbf{A} - \mathbf{B}' + \mathbf{D}'' - \mathbf{q}) \cdot \delta \mathbf{y} d\xi \\ &\quad + [(\mathbf{B} - \mathbf{D}') \cdot \delta \mathbf{y} + \mathbf{D} \cdot \delta \mathbf{y}']_{\xi_i}^{\xi_e} - F\delta\Delta, \\ A_R - B'_R + D''_R &= q_R, \\ -B'_Z + D''_Z &= q_Z. \end{aligned} \quad (297)$$

Together with Eq. (296), these are the coupled nonlinear meridional differential equations. They retain derivatives of

spatially varying material properties through N and M . The boundary term supplies the generalized force $\mathbf{B} - \mathbf{D}'$ and bending work $\mathbf{D} \cdot \delta \mathbf{y}'$, with opposite endpoint signs at ξ_i and ξ_e . A free, moment-free edge has zero prescribed generalized force and $M_\xi = 0$; an imposed position or rotation instead removes the corresponding variation. A flat contacting support is not automatically a clamp: its position and rotation constraints follow from the active contact region.

d) *Stationary load curve, tangent stiffness, and stability*: For frictionless supports at prescribed compression, define the constrained equilibrium branch by

$$\begin{aligned} \mathcal{E}(\Delta, T) &= \min_{\mathbf{y} \in \mathcal{K}(\Delta)} \{U_m[\mathbf{y}, T] + U_b[\mathbf{y}, T]\}, \\ F(\Delta, T) &= \frac{\partial \mathcal{E}}{\partial \Delta}, \\ k_{\text{disc}}(\Delta, T) &= \frac{\partial^2 \mathcal{E}}{\partial \Delta^2}. \end{aligned} \quad (298)$$

The admissible set $\mathcal{K}(\Delta)$ contains the geometric support and unilateral-contact constraints below. These derivatives apply on a smooth equilibrium branch; contact changes and snap-through may require one-sided tangents or branch tracking. The derivative is with respect to the imposed compression, not a pointwise derivative of energy with respect to an arbitrary displacement field. On a strictly stable active contact set, admissible vibration modes satisfy

$$\begin{aligned} \mathbf{K}_c \mathbf{v} &= \omega_s^2 \mathbf{M}_c \mathbf{v}, \\ \mathbf{K}_c &= \mathbf{Z}_c^\top \nabla_{\mathbf{a}\mathbf{a}}^2 \mathcal{L}_c \mathbf{Z}_c, \\ \mathbf{M}_c &= \mathbf{Z}_c^\top \mathbf{M}_s \mathbf{Z}_c, \quad \mathbf{J}_c \mathbf{Z}_c = 0. \end{aligned} \quad (299)$$

Here $\mathbf{a}$ are displacement degrees of freedom, $\mathbf{J}_c$ is the active constraint Jacobian, and $\mathbf{M}_s$ is the consistent shell mass matrix. Positive $\mathbf{M}_c$ and positive constrained stiffness imply positive squared frequencies for this conservative, fixed-contact linearization after rigid modes are restrained. Qualification against non-axisymmetric buckling, contact opening, and frictional sliding requires the corresponding stability analyses. For four identical discs on the same smooth series branch, $x = 4\Delta$ gives $dF/dx = k_{\text{disc}}/4$ before adding frame compliance. The 5.0×10^6 N/m load-path target requires branch-specific stiffness evaluation and experimental load–deflection characterization.

F. Signorini Contact, Variational Inequalities, and Friction

a) *Gap and traction sign convention*: Let Γ_C^i, Γ_C^e denote the inner and outer contact flats. At a potential contact point on a Kirchhoff–Love face, use

$$\begin{aligned} \mathbf{x}_c &= \mathbf{x} + \zeta_c \mathbf{n}_c, \quad \zeta_c \in \{-t/2, t/2\}, \\ g_N[\mathbf{y}] &= (\mathbf{x}_c - \mathbf{x}_S) \cdot \mathbf{n}_S, \\ g_N &\geq 0, \quad \mathbf{n}_b = -\mathbf{n}_S, \\ \mathbf{t}_c &= \boldsymbol{\sigma} \mathbf{n}_b = p_N \mathbf{n}_S + \boldsymbol{\tau}_T, \\ \sigma_N &= \mathbf{t}_c \cdot \mathbf{n}_b = -p_N. \end{aligned} \quad (300)$$

The support normal $\mathbf{n}_S$ points from the rigid plane into the admissible washer half-space, so positive gap means separation. It points oppositely at the two opposing supports. The relation $\mathbf{n}_b = -\mathbf{n}_S$ applies on smooth closed contact patches; edge reactions are instead defined weakly through the

gap variation. The physical face and rotating director enter the gap; replacing them by a midsurface clearance is a separate contact approximation. The Signorini conditions are [119]

$$\begin{aligned} g_N &\geq 0, & p_N &\geq 0, & p_N g_N &= 0, \\ \text{equivalently } g_N &\geq 0, & \sigma_N &\leq 0, & \sigma_N g_N &= 0. \end{aligned} \quad (301)$$

b) Complementarity and boundary equilibrium proof:

With normal-contact pressure as a nonnegative multiplier, define

$$\begin{aligned} \mathcal{L}_c[\mathbf{y}, p_N] &= \Pi[\mathbf{y}] - \int_{\Gamma_C} p_N g_N[\mathbf{y}] dA, \\ \delta_{\mathbf{y}} \mathcal{L}_c[\mathbf{v}] &= \delta \Pi[\mathbf{v}] - \int_{\Gamma_C} p_N \delta g_N[\mathbf{v}] dA = 0, \\ \partial_{p_N} \mathcal{L}_c &= -g_N. \end{aligned} \quad (302)$$

The multiplier saddle problem minimizes over admissible displacements and maximizes over $p_N \geq 0$. If $g_N < 0$, the term $-p_N g_N$ is unbounded above, so a saddle point cannot interpenetrate. If $g_N > 0$, maximization requires $p_N = 0$. If $g_N = 0$, a nonnegative multiplier supplies the compressive reaction. These alternatives prove Eq. (301). The displacement variation balances the shell boundary work against $p_N \delta g_N$ and therefore fixes the reaction sign in Eq. (300); prescribed contact forces are not added a second time to the same support work.

For an affine gap and an admissible displacement $\mathbf{z}$, stationarity and complementarity imply the variational inequality

$$\begin{aligned} \delta \Pi[\mathbf{y}; \mathbf{z} - \mathbf{y}] &= \int_{\Gamma_C} p_N \{g_N[\mathbf{z}] - g_N[\mathbf{y}]\} dA \\ &= \int_{\Gamma_C} p_N g_N[\mathbf{z}] dA \geq 0. \end{aligned} \quad (303)$$

Localized normal variations give the converse open/closed-contact alternatives when multiplier regularity and a constraint qualification hold. For the nonlinear face gap, the corresponding first-order statement holds for feasible tangent directions, not arbitrary finite displacement differences. A stationary point is not automatically a minimum of the nonlinear shell energy. A necessary second-order condition is

$$\delta_{\mathbf{y}\mathbf{y}}^2 \mathcal{L}_c[\mathbf{v}, \mathbf{v}] \geq 0 \quad \text{for } \mathbf{v} \text{ in the critical cone.} \quad (304)$$

The critical cone contains feasible first-order directions with zero first-order objective change. Strict positivity there, with the usual smoothness and constraint qualifications, gives a strict local minimum in a regular finite-dimensional discretization; global uniqueness requires additional mathematical conditions. For a fixed active set, the frictionless Newton equations are

$$\begin{bmatrix} \mathbf{H}_c & -\mathbf{J}_c^T \\ -\mathbf{J}_c & 0 \end{bmatrix} \begin{bmatrix} \delta \mathbf{a} \\ \delta \mathbf{p} \end{bmatrix} = \begin{bmatrix} -\mathbf{r}_c \\ \mathbf{g}_c \end{bmatrix}, \quad (305)$$

where $\mathbf{r}_c = \nabla_{\mathbf{a}} \Pi - \mathbf{J}_c^T \mathbf{p}$, $\mathbf{H}_c = \nabla_{\mathbf{a}\mathbf{a}}^2 \mathcal{L}_c$, and $\mathbf{g}_c$ lists the active gaps. Negative active pressures or closing inactive gaps require an active-set change. The Hessian includes geometric contact terms and is not merely the material stiffness.

c) Stick-slip law and dissipation proof: Define the relative tangential velocity and Coulomb admissible disk by

$$\begin{aligned} \mathbf{v}_T &= (\mathbf{I} - \mathbf{n}_S \mathbf{n}_S^T)(\dot{\mathbf{x}}_c - \dot{\mathbf{x}}_S), \\ \mathcal{D}(p_N) &= \{\boldsymbol{\tau} : \boldsymbol{\tau} \cdot \mathbf{n}_S = 0, \|\boldsymbol{\tau}\| \leq \mu p_N\}, \\ \boldsymbol{\tau}_T &\in \arg \min_{\boldsymbol{\tau} \in \mathcal{D}(p_N)} \boldsymbol{\tau} \cdot \mathbf{v}_T. \end{aligned} \quad (306)$$

Minimization of this linear functional over the disk gives

$$\begin{aligned} \mathbf{v}_T = 0 &: \|\boldsymbol{\tau}_T\| \leq \mu p_N, \\ \mathbf{v}_T \neq 0 &: \boldsymbol{\tau}_T = -\mu p_N \frac{\mathbf{v}_T}{\|\mathbf{v}_T\|}, \\ \boldsymbol{\tau}_T \cdot \mathbf{v}_T &= -\mu p_N \|\mathbf{v}_T\| \leq 0. \end{aligned} \quad (307)$$

Equivalently, $-\boldsymbol{\tau}_T \in \mu p_N \partial \|\mathbf{v}_T\|$; the subdifferential at zero velocity is the unit disk. Equality of shear magnitude with μp_N permits impending slip and does not alone imply nonzero sliding velocity. On an open gap, $p_N = 0$ implies zero friction traction. On sustained closed contact with fixed rigid supports, $p_N \dot{g}_N = 0$ gives zero normal contact power, while tangential slip dissipates energy. Impact needs a separate impulsive law and is outside the quasistatic Signorini proof.

Frictional cycling is path dependent and cannot be derived from a single conservative potential $U_m + U_b - F\Delta$. At prescribed normal pressure during a tangential increment, the additional dissipation functional is

$$\mathcal{R}(\Delta \mathbf{u}_T; p_N) = \int_{\Gamma_C} \mu p_N \|\Delta \mathbf{u}_T\| dA \geq 0. \quad (308)$$

When p_N evolves with the displacement, the coupled problem is a contact variational inequality rather than unconstrained energy minimization. A law $\mu(T, p_N, \text{surface state})$ requires independently characterized contact properties; normal complementarity does not determine friction or creep. The shell solution supplies loading/unloading forces and tangent stiffnesses for the four-disc assembly while preserving the preload and pressure-variation bounds below.

G. Four-Disc Belleville Compliance and Preload Control

The structural parameterization specifies $K_{\text{stack}} = 5.0 \times 10^6 \text{ N/m}$, $\Delta L_{\text{net}} = 20.1 \mu\text{m}$, clamp-force bounds, and a strain ceiling [111]. Sections III-E–III-F define the shell energy and contact boundary-value problem. The reduced Almen–László law supplies the disc-spring comparison [108], [112].

The selected core-interface force shall remain within Eq. (278) over 25–350 °C, with cyclic pressure deviation bounded by $\pm 0.50 \text{ MPa}$. The 15.80 MPa preload belongs to a separate, larger-area comparison configuration and is below the selected 20 MPa sealing minimum. The optimized Belleville stack has force stiffness $K_{\text{stack}} = 5000 \text{ N/mm} = 5.0 \times 10^6 \text{ N/m}$. A four-disc series model provides the load-deflection framework. Force stiffness and stiffness per loaded area are distinguished by

$$\begin{aligned} k_{\text{path}} &= \frac{dP}{dx} \quad [\text{N/m}], \\ K_A &= \frac{k_{\text{path}}}{A_{\text{load}}} \quad [\text{Pa/m}], \\ \Delta p &= \frac{1}{A_{\text{load}}} \int_{x_0}^{x_0 + \Delta L_{\text{net}}} k_{\text{path}}(x, T) dx. \end{aligned} \quad (309)$$

Here P is axial force, x is load-path deflection, and A_{load} is the effective area transmitting that force. For the selected contact configuration, $A_{\text{load}} = A_{\text{contact}}$, $k_{\text{path}} = K_{\text{stack}}$ at the working point, and $K_A = 15.9154943 \text{ GPa/m}$. The area-normalized stiffness K_A shall not be divided by area a second time.

For an ideal rectangular-section disc, the Almen–László force and tangent stiffness are [97]

$$\begin{aligned} P(s, T) &= \frac{4E(T)t^4}{(1-\nu^2)MD_e^2} \\ &\times \left[\left(\frac{h_0}{t} - \frac{s}{t} \right) \left(\frac{h_0}{t} - \frac{s}{2t} \right) + 1 \right] \frac{s}{t}, \\ k_{\text{single}}(s, T) &= \frac{4E(T)t^3}{(1-\nu^2)MD_e^2} \\ &\times \left[\left(\frac{h_0}{t} \right)^2 - 3\frac{h_0 s}{t^2} + \frac{3}{2} \left(\frac{s}{t} \right)^2 + 1 \right]. \end{aligned} \quad (310)$$

The four-disc comparison geometry uses $D_e = 0.250 \text{ m}$, $D_i = 0.125 \text{ m}$, $t = 12.0 \text{ mm}$, and free cone height $h_0 = 17.4 \text{ mm}$, so $h_0/t = 1.45$. The dimensionless factor M depends on the diameter ratio and contact convention. For the ideal no-contact-flat formulation,

$$M = \frac{1}{\pi} \frac{[(\delta - 1)/\delta]^2}{(\delta + 1)/(\delta - 1) - 2/\ln \delta}, \quad \delta = \frac{D_e}{D_i} = 2, \quad (311)$$

which gives $M = 0.69433320$. The assigned factor $M = 0.681$ defines a separate geometry parameterization. Contact flats, reduced thickness, and edge radii use their corresponding load-deflection model.

The spring-modulus parameterization uses $E_0 = 193 \text{ GPa}$, $\nu = 0.30$, and

$$E(T) = E_0 [1 - (2.8 \times 10^{-4} \text{ K}^{-1})(T - 298.15 \text{ K})]. \quad (312)$$

It gives $E(623.15 \text{ K}) = 175.437 \text{ GPa}$, distinct from the 195.0 GPa structural input in Eq. (280). These material parameterizations require separate heat-treatment and property qualification.

For identical discs under equal force, $x = n_s s$ with $n_s = 4$. Series disc and frame compliances combine as

$$\begin{aligned} \frac{1}{k_{\text{path}}} &= \sum_{i=1}^{n_s} \frac{1}{k_{\text{single},i}} + \frac{1}{k_{\text{frame}}}, \\ k_{\text{path}}^{\pm}(x, T) &= \left[\frac{n_s(1 \pm \mu h_0/t)}{k_{\text{single}}(x/n_s, T)} + \frac{1}{k_{\text{frame}}} \right]^{-1}. \end{aligned} \quad (313)$$

The second expression is a phenomenological loading/unloading friction parameterization, with comparison value $\mu = 0.095 \pm 0.015$. The nominal total disc compression $x_0 = 18.25 \text{ mm}$ gives $s_0 = 4.5625 \text{ mm}$ and $s_0/h_0 = 0.262213$. A compliant frame contributes additional total load-path motion and shall not be included again in $x = n_s s$.

The selected thermal compliance specification is [103]

$$\begin{aligned} \Delta L_{\text{net}} &= 20.1 \mu\text{m}, \\ K_{\text{stack},\text{max}} &= \frac{(0.50 \times 10^6) A_{\text{contact}}}{20.1 \times 10^{-6}} \\ &= 7.8149071 \times 10^6 \text{ N/m}, \\ K_{\text{stack}} &= 5000 \text{ N/mm} = 5.0 \times 10^6 \text{ N/m}, \\ \Delta F &= K_{\text{stack}} \Delta L_{\text{net}} = 100.5 \text{ N}, \\ \Delta P \equiv \Delta p &= \frac{\Delta F}{A_{\text{contact}}} \\ &= 0.319901436 \text{ MPa} \simeq 0.320 \text{ MPa} \\ &< 0.50 \text{ MPa}. \end{aligned} \quad (314)$$

The calculated cyclic contact-pressure deviation of 0.320 MPa complies with the $\pm 0.50 \text{ MPa}$ tolerance band, leaving 0.180098564 MPa of single-sided allowance before other uncertainties. Here ΔP denotes pressure variation, distinct from the force function $P(s, T)$ in the disc model; it is not the local multiaxial bevel stress. For a conservative excursion of $\pm 20.1 \mu\text{m}$ about the preload setting, the thermal force reserve is $\pm 100.5 \text{ N}$. The nominal force shall therefore remain at least 100.5 N inside each instantaneous force bound, with additional reserve for creep and setting uncertainty. The estimate assumes linear stiffness over the excursion; nonlinear acceptance shall use the maximum tangent stiffness or the full force integral.

The optimized stack requires compatible working deflection, loaded area, disc geometry, and a qualified load-deflection characteristic.

H. High-Temperature Relaxation and Contact Tribology

a) *Vacancy-controlled spring relaxation:* For an Inconel X-750 spring at 350°C , relaxation requires its own phase, precipitate, vacancy, and dislocation parameters. The vacancy-climb framework in Eqs. (393)–(401) supplies the mechanism, but Cu–Sn material functions do not transfer to X-750. In a fixed-strain scalar restriction with constant modulus,

$$\begin{aligned} \frac{d\sigma}{dt} &= -E\dot{\epsilon}_{\text{cr}}, \\ \sigma(t) &= \sigma_0 - E \int_0^t \dot{\epsilon}_{\text{cr}}(t') dt'. \end{aligned} \quad (315)$$

Here $\dot{\epsilon}_{\text{cr}}$ is projected from the resolved dislocation strain rate. With $\sigma_0 = 400 \text{ MPa}$, the assigned endpoint 384.2 MPa corresponds to 3.95% loss. Product form, precipitation heat treatment, stress state, defect history, and measured relaxation response govern spring qualification.

The sustained-temperature condition is 350°C for $10,000$ operating hours. In the Rankine/1000 convention, the Larson–Miller exposure parameter is [87]

$$\begin{aligned} P_{\text{LM}} &= \frac{T_R/(1^\circ\text{R})}{1000} \left[C_{\text{LM}} + \log_{10} \left(\frac{t_{\text{hold}}}{1 \text{ h}} \right) \right], \\ T_R &= 1.8(623.15)^\circ\text{R} = 1121.67^\circ\text{R}, \\ C_{\text{LM}} &= 20.0, \quad t_{\text{hold}} = 10000 \text{ h}, \\ P_{\text{LM}} &= 1.12167(20 + 4) = 26.92008 \simeq 26.92. \end{aligned} \quad (316)$$

Temperature and time units are part of this empirical convention. A Kelvin-based correlation cannot use the same numerical

parameter without conversion. The assigned cumulative force-loss fraction is $R_{\text{creep}} = 4.18\%$; mapping P_{LM} to that fraction requires a stress-relaxation curve for the applicable spring condition and initial stress. Likewise, 10,000 hours and 44,820 cycles are independent duration and cycle-count requirements unless a cycle and dwell schedule relates them.

For an initial spring force $P_0 = 45.0$ kN, the relaxation scenario gives

$$\begin{aligned}\Delta P_{\text{creep}} &= P_0 R_{\text{creep}} = 45.0(0.0418) = 1.881 \text{ kN}, \\ P_{\text{relaxed}} &= 45.000 - 1.881 = 43.119 \text{ kN}.\end{aligned}\quad (317)$$

The 45.0 kN spring force is distinct from the 125.0 kN axial retaining load in the bevel load case. The following larger-area comparison assumes that the former produces 15.80 MPa through a direct load path; it does not use the selected 20 mm contact diameter:

$$\begin{aligned}A_{\text{load}} &= 45000 / (15.80 \times 10^6) = 2.84810127 \times 10^{-3} \text{ m}^2, \\ \Delta p_{\text{creep}} &= \Delta P_{\text{creep}} / A_{\text{load}} = 0.66044 \text{ MPa}, \\ p_{\text{relaxed}} &= 15.13956 \text{ MPa}.\end{aligned}\quad (318)$$

The comparison configuration has a 15.30 MPa lower pressure limit. For the selected contact area, long-term force loss is converted using Eq. (277) and reserved within the force and pressure limits independently of the cyclic 0.320 MPa excursion.

All disc contact surfaces are specified with magnetron-sputtered molybdenum disulfide (MoS_2) over a 200 nm diamond-like carbon (DLC) interlayer. The static-friction acceptance limit is $\mu_{\text{stat}} \leq 0.105$ in the operating vacuum/deuterium environment; the dynamic comparison value is $\mu_{\text{dyn}} = 0.080$. The coating process shall define MoS_2 thickness, adhesion, coverage, counterface finish, and conditioning. These properties and the friction limit require qualification at temperature, contact pressure, and accumulated sliding distance.

For force-deflection cycling, the dissipated work is the loop area,

$$\begin{aligned}\Delta W_{\text{hyst}} &= \oint P(s, \dot{s}) ds, \\ \Delta W_{\text{hyst, approx}} &= 2(\mu_{\text{stat}} - \mu_{\text{dyn}})P_0 s_{\text{max}}, \\ s_{\text{max}} &= 1.25 \text{ mm}, \\ \Delta W_{\text{uncoated}} &= 2(0.125 - 0.080)(45000)(0.00125) \\ &= 5.0625 \text{ J/cycle}, \\ \Delta W_{\text{coated}} &= 2(0.105 - 0.080)(45000)(0.00125) \\ &= 2.8125 \text{ J/cycle}.\end{aligned}\quad (319)$$

This approximation requires a displacement coordinate consistent with the force and contact-slip model. The secant allowance

$$k_{\text{allow, sec}} = 36.00 - \frac{1.881}{1.25} = 34.4952 \text{ kN/mm} \quad (320)$$

allocates creep force loss over the stated stroke. This secant quantity uses the specified load-path displacement and effective loaded area.

b) Mechanical acceptance scope: Assembly acceptance uses one load-path model that includes contact friction, temperature-dependent properties, creep relaxation, and pressure uncertainty. These conditions govern the datum, isolation, and vacuum-interface requirements that follow.

I. Datum Framework and Geometric Controls

Geometric dimensioning and tolerancing (GD&T) shall follow ASME Y14.5-2018 [142]. The standard defines the drawing language; the numerical tolerances in Table XX are project requirements. The assembly datum reference frame shall use the Invar 36 primary mounting plane as datum *A*, the longitudinal locating plane as datum *B*, and the transverse locating plane as datum *C*. Each component drawing shall identify the local mating datum features that realize this assembly frame, including the correspondence between the diamond seating features and the chassis references.

Table XX specifies the six controlled features. Flatness controls shall contain no datum reference, including where the controlled surface establishes datum *A*. Bore-axis cylindrical zones shall carry a diameter designation; rib center-plane zones shall use the stated parallel-plane width. Size-feature controls shall have no maximum-material-condition bonus unless explicitly modified on the released drawing. The drawing shall include basic bore locations, guide-pin dimensions, rib pitch, and all geometry needed to apply the schedule. The cold-plate comparison geometry uses 150 μm -wide water-ethylene-glycol passages at a reference flow of 9.83 L min^{-1} . DM-02 specifies testing at 12.5 L min^{-1} , subject to qualification of the pump and channel-flow characteristics. The optical aperture shall be 20.0 $\times$ 20.0 mm, and the mounting envelope shall accommodate a 2.0 mm diamond optic. The axial stack includes a separate 0.500 mm floating-frame seating contribution. The thinner diamond layer in Section IV defines an analytical benchmark; drawings and analysis records shall distinguish these three dimensions.

Dimensional inspection records shall include the reference temperature, probe or optical sampling strategy, datum realization, and measurement uncertainty. The assembly reference temperature shall be 293.15 K. Inspection and stack-height calculations shall treat profile, flatness, and thickness as distinct controls and shall include each measured surface deviation only once in the relevant error budget.

a) Re-entrant thermal compensation: The independently heated re-entrant frame uses opposed Invar 36 and Al 6061-T6 members [54]. Differential expansion is evaluated at the specified temperature rise of each column:

$$\begin{aligned}L_{\text{Invar}} &= 172.50 \text{ mm}, \quad \Delta T_{\text{Invar}} = 256.85 \text{ K}, \\ L_{\text{Al}} &= 20.69 \text{ mm}, \quad \Delta T_{\text{Al}} = 110.0 \text{ K}, \\ \alpha_{\text{Invar}} &= 1.20 \times 10^{-6} \text{ K}^{-1}, \\ \alpha_{\text{Al}} &= 23.0 \times 10^{-6} \text{ K}^{-1}, \\ \Delta L_{\text{net}} &= \alpha_{\text{Invar}} L_{\text{Invar}} \Delta T_{\text{Invar}} - \alpha_{\text{Al}} L_{\text{Al}} \Delta T_{\text{Al}} \\ &= 53.16795 - 52.34570 = 0.82225 \mu\text{m}, \\ L_{\text{stroke, remaining}} &= 15.00000 - 0.82225 \\ &= 14.17775 \mu\text{m} \quad (94.5183\%).\end{aligned}\quad (321)$$

TABLE XX
REQUIRED MECHANICAL GD&T SCHEDULE UNDER ASME Y14.5-2018

Component	Controlled feature	Geometric control	Zone (mm)	Datum sequence	Functional requirement
Invar 36 chassis	Laser-mount interface bore axis	Position, cylindrical zone	Diameter 0.005	$A B C$	Register optical path to active region
Invar 36 chassis	Primary mounting base	Flatness	0.002	None; surface establishes A	Control base coplanarity and seating
CVD diamond floating frame	Internal substrate-clamping pocket	Profile of a surface	0.004 total width	$A B$	Control pocket boundary and clamping geometry
CVD diamond floating frame	Guide-pin bore axes	Perpendicularity, cylindrical zone	Diameter 0.003	A	Avoid guide binding during expansion
Microchannel cold plate	Thermal-contact interface face	Flatness	0.003	None; surface may establish A	Limit gaps at the heat-spreader interface
Microchannel cold plate	Rib center planes in the 150 μm channel array	Position, parallel-plane zone	0.010 total width	$A B C$	Maintain channel registration and flow distribution

The re-entrant sleeve and return link act with opposite displacement signs, while thermal control maintains the specified column temperatures. The resulting differential travel describes the frame alone. Compensation of seating-stack expansion also depends on the datum and return-path geometry; the frame residual cannot be identified with the residual displacement of the complete optical stack.

J. ConFlat Vacuum Feedthrough Schedule

The vessel shall support two separately qualified operating states: evacuated service with base pressure $p_{\text{base}} \leq 1.0 \times 10^{-8}$ mbar, and deuterium-loaded service at nominal $p_{D_2} = 4.20$ bar absolute. The turbomolecular pumping path shall be isolated before pressurization. Table XXI defines the required electrical and optical interfaces from FT-AC-01 through FT-SENS-01. Assembly pressure ratings and state-transition controls shall be qualified for the applicable service conditions.

CF35 and CF40 are interface designations. Procurement drawings shall specify the actual flange outside diameter, knife-edge geometry, bolt circle, gasket, and mating interface. The optical viewport shall use a 2.75-inch CF envelope with 23.0 mm clear aperture. Assembly bakeout shall remain within the lowest certified temperature limit of the window, braze, fiber seal, ceramic, connector, and installed wiring; the 450°C bare metal-flange requirement applies only to that component. The completed assembly shall receive separate vacuum-leak and positive-pressure qualification for deuterium service.

FT-TEG-01 requires verification of current sharing in the parallel supply and return paths. The 24 V, 812.5 W terminal test case defines a connector-qualification condition only:

$$I_{\text{TEG, total}} = \frac{812.5 \text{ W}}{24 \text{ V}} = 33.85417 \text{ A},$$

$$I_{\text{TEG, pin}} = 16.92708 \text{ A} \quad \text{for two equal parallel paths.} \quad (322)$$

Power specified at the input terminals requires no additional division by boost efficiency. At the comparison test point of 30 V and 850 W, the corresponding currents are 28.33333 A total

and 14.16667 A per equal path. Qualification against the 30.0 A per-pin limit includes temperature derating, contact resistance, peak current, and measured imbalance. The 500 V_{peak} SMA rating specifies insulation withstand; continuous RF loading requires separate qualification. FT-SENS-01 provides acoustic sensing rather than a qualified power-export path, and the baseline energy balance assigns no 134.8 W PZT generation credit.

Acceptance measurements shall cover helium leakage at each completed seal, continuity, insulation resistance, dielectric withstand under the intended pressure conditions, conductor temperature rise, optical insertion loss, and polarization extinction. Released component specifications shall state the allowable helium leak rate, RF bandwidth where applicable, and pressure rating before procurement acceptance. The gas manifold, pressure relief, and isolation arrangement shall have a service-specific qualification independent of the electrical and optical feedthrough schedule.

K. Vibration Dynamics and Two-Stage Isolation

The small angular error of the prism–substrate interface is denoted by θ , with effective rotational inertia J_{eff} , rotational damping C_θ , and rotational stiffness K_θ . Base acceleration $a_b(t)$ acting on effective offset mass m_e at lever arm ℓ_e , together with disturbance forces F_j at signed moment arms ℓ_j , produces the angular response

$$J_{\text{eff}}\ddot{\theta} + C_\theta\dot{\theta} + K_\theta\theta = -m_e\ell_e a_b(t) + \sum_j \ell_j F_j(t) + M_c(t), \quad (323)$$

where M_c is control torque. With zero initial conditions, the base-acceleration response is

$$H_{\theta a}(s) = \frac{\Theta(s)}{A_b(s)} = \frac{-m_e\ell_e}{J_{\text{eff}}s^2 + C_\theta s + K_\theta}. \quad (324)$$

The rotational natural frequency and damping ratio are

$$\omega_{\theta n} = \sqrt{\frac{K_\theta}{J_{\text{eff}}}}, \quad \zeta_\theta = \frac{C_\theta}{2\sqrt{J_{\text{eff}}K_\theta}}. \quad (325)$$

TABLE XXI
REQUIRED CF35/CF40 FEEDTHROUGH INTERFACE SCHEDULE

Identifier	Flange	Construction and channel count	Electrical or optical requirement	Shielding and service
FT-AC-01	CF40	Two hermetic high-current conductors, with vacuum-compatible litz-wire terminations	1.0 kV _{rms} insulation; 21.21 A _{rms} at 68.5 kHz	Continuous 360° shield; external Mu-metal sleeve; Helmholtz drive
FT-DC-01	CF35	Twelve-pin, high-density ceramic feedthrough	500 VDC insulation; 5.0 A per pin nominal	Grounded metallic braid; PZT flexure drive and sensor wiring
FT-OPT-01	CF35	Two hermetic polarization-maintaining fused-silica fiber paths	50 mW per path; 785.0 nm and 802.5 nm; no electrical power rating	Dual-wall stainless armor; two DFB optical inputs
FT-TC-01	CF35	Eight Type-K thermocouple pairs, sixteen thermoelement conductors	100 VDC insulation; thermocouple signal service, not load-power service	Individually shielded pairs; core and interface temperatures
FT-TEG-01	CF40	Four copper power pins arranged as two supply and two return conductors	250 VDC insulation; 30.0 A continuous per pin as a qualification requirement	Internal conductive shield; TEG raw output
FT-SENS-01	CF35	Four hermetic SMA coaxial paths	500 V _{peak} insulation; 1.0 A per path; 50 Ω characteristic impedance	Double-shielded coax; PZT acoustic sensing channels

TABLE XXII
TWO-STAGE AMPLITUDE TRANSMISSIBILITY IN DB

Frequency (Hz)	Nominal	Calculated
28.0	-42.50	-45.1464
56.0	-58.20	-66.7928
1500.0	-134.10	-130.7947

The forcing frequencies are 28.0 and 56.0 Hz for coolant pulsation, and 1500.0 Hz for the turbomolecular rotor. Stage 1 comprises a pneumatic table ($f_{n1} = 1.8$ Hz, $\zeta_1 = 0.05$), and Stage 2 comprises an elastomeric mount ($f_{n2} = 18.0$ Hz, $\zeta_2 = 0.10$). Treating the stages as independent linear base-excited systems gives

$$\mathcal{T}(r, \zeta) = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}, \quad (326)$$

$$\mathcal{T}_{12}(f) = \mathcal{T}(f/f_{n1}, \zeta_1) \mathcal{T}(f/f_{n2}, \zeta_2).$$

Active feedback, coupled modes, and mechanical transmission through hoses and wiring contribute additional transfer paths. The nominal integrated tilt is $\theta_{\text{RMS}} = 0.00540 \mu\text{rad}$, with a design limit of $0.00800 \mu\text{rad}$. The integrated response uses the disturbance spectrum and acceleration-to-rotation transfer function.

The integrated tilt requirement is an RMS quantity:

$$\sigma_\theta^2 = \int_0^\infty S_\theta(f) df, \quad (327)$$

$$\sigma_\theta \leq 0.008 \mu\text{rad} = 8.0 \times 10^{-9} \text{ rad},$$

$$S_\theta(f) = |H_{\theta a}(i2\pi f)|^2 |\mathcal{T}_{12}(f)|^2 S_{a_b}(f) + S_{\theta, \text{other}}(f).$$

This decomposition assumes uncorrelated contributions; measured cross-spectral terms shall be included otherwise. Qualification of the $0.008 \mu\text{rad}$ RMS limit shall use calibrated disturbance spectra, rotational parameters, and sensor-noise budgets. Testing shall cover evacuated and pressurized operation, operating pumps and coolant flow, startup transients, and a declared integration bandwidth. For a 10.0 mm effective

TABLE XXIII
COMPLEX REFRACTIVE INDICES FOR THE OPTION B OPTICAL MODEL

Layer	785.0 nm	802.5 nm
Air	1.0000 + i0	1.0000 + i0
Fused silica	1.4536 + i0	1.4533 + i0
Pd-Ir	1.8325 + i4.5015	1.8825 + i4.5825
Ti	2.7600 + i3.8400	2.8200 + i3.9200
CVD diamond	2.4008 + i0	2.3999 + i0

lever, the tilt limit corresponds to 0.080 nm differential RMS motion.

a) *Monolithic waveguide and optical material parameters:* The nominal optical stack comprises air/fused silica/patterned Pd_{0.95}Ir_{0.05}/Ti/CVD diamond, distinct from the SF11 smooth-prism comparison. The active metal occupies the Option B volume in Eq. (71); 42.50 nm grating teeth require sufficient patterned-metal thickness and cannot be etched into a 1 nm film. The grating pitch is 960.80 nm, the duty cycle is 50%, and the internal silica incidence angle is 65.0°. The complex refractive indices define the constitutive inputs through $\varepsilon_j = (n_j + ik_j)^2$; alternative permittivity datasets describe separate model cases. With the time convention $e^{-i\omega t}$, the Fourier field expansions are

$$\mathbf{E}(x, z) = \sum_m \mathbf{S}_m(z) e^{ik_{x,m}x},$$

$$\mathbf{H}(x, z) = \sqrt{\epsilon_0/\mu_0} \sum_m \mathbf{U}_m(z) e^{ik_{x,m}x}, \quad (328)$$

$$k_{x,m} = k_0 n_{\text{silica}} \sin 65^\circ + 2\pi m/\Lambda.$$

For the one-dimensional binary profile, let $\mathbf{E}$ and $\mathbf{Q}$ denote the Toeplitz matrices of $\varepsilon(x)$ and $1/\varepsilon(x)$, respectively, and $\mathbf{K}_x = \text{diag}(k_{x,m})$. The TM Fourier-factorized Maxwell system can be written

$$\frac{d\mathbf{S}_x}{dz} = ik_0 [\mathbf{I} - k_0^{-2} \mathbf{K}_x \mathbf{E}^{-1} \mathbf{K}_x] \mathbf{U}_y,$$

$$\frac{d\mathbf{U}_y}{dz} = ik_0 \mathbf{Q}^{-1} \mathbf{S}_x, \quad (329)$$

$$\mathbf{S}_z = -k_0^{-1} \mathbf{E}^{-1} \mathbf{K}_x \mathbf{U}_y.$$

TABLE XXIV
OPTICAL REFERENCE OUTPUTS UNDER THE AMPLITUDE CONVENTION

Quantity	785.0 nm	802.5 nm
$R_{\min}$	0.0142	0.0085
$F_z = E_z / E_0 $	124.50	132.80
$\mathcal{F}_z = F_z^2$	15500.25	17635.84
$L_{\text{spp}} (\mu\text{m})$	12.45	14.12
$\delta_{z,m} (\text{nm})$	14.80	15.20
$\delta_{z,d} (\text{nm})$	185.30	192.10

The inverse of the Toeplitz permittivity matrix and the inverse of the Toeplitz inverse-permittivity matrix are distinct operators. The latter implements normal-field inverse factorization rather than a second inversion in the first equation. Tangential-field continuity, outgoing-wave boundary conditions, and harmonic convergence complete the boundary-value problem. A grating patterned in both lateral directions requires both lateral harmonic indices. The field factor F_z is an amplitude ratio, distinct from an intensity enhancement. Table XXIV specifies both conventions explicitly. The propagation lengths apply to the corresponding optical model; neither domain volumes nor RCWA results for a 1 nm film transfer to Option B without a geometry-specific calculation. The beam-shaper exponent is $m = 1148$ ($N = 574$ in $2N$ notation), with $w = 10.048711$ mm over the 20 mm square aperture. The active footprint is 1 cm², distinct from the aperture area. Intercepted optical power therefore requires integration over the active pattern.

L. Kretschmann Flexure Stage and Viewport Alignment

The angular stage shall use a monolithic wire-EDM Invar 36 flexure with two preloaded piezoelectric translators. The drive range shall be 0–100 V, full travel 15.0 μm , and effective differential displacement resolution 0.15 nm. For actuator separation d_a , the small-angle relation is

$$\theta \approx \frac{z_1 - z_2}{d_a}, \quad \delta\theta = \frac{\delta z_{\text{diff}}}{d_a}. \quad (330)$$

The specified 0.015 μrad resolution is geometrically consistent with

$$d_a = \frac{0.15 \times 10^{-9} \text{ m}}{0.015 \times 10^{-6} \text{ rad}} = 10.0 \text{ mm}. \quad (331)$$

The effective lever shall be 10.0 mm for this resolution requirement. Angular resolution and integrated RMS accuracy shall be verified separately. At this lever, 15.0 μm of differential stroke corresponds to 1.50 mrad total angular range. Across the separate 20.0 mm optical mounting span, the 3.2 μm edge-to-edge thermal-mismatch design case gives

$$\Delta\theta_{\text{thermal}} = \arctan\left(\frac{3.2 \mu\text{m}}{20.0 \text{ mm}}\right) \approx 160 \mu\text{rad} = 0.009167^\circ. \quad (332)$$

A quadrant photodiode shall measure reflected-beam position, and synchronous reflectance measurements shall identify the SPP resonance. The digital update rate shall be 10 kHz. Closed-loop bandwidth, stability margins, and jitter suppression shall be established separately from measured mechanical resonances, latency, actuator saturation, and feedback noise.

TABLE XXV
FOUR-LAYER OPTICAL INPUTS AND REFERENCE OUTPUTS

Parameter	785.0 nm	802.5 nm
n_0 SF11	1.76552	1.76447
n_1 Pd–Ir	$1.835 + 4.535i$	$1.885 + 4.635i$
n_2 Ti	$2.800 + 3.800i$	$2.850 + 3.900i$
n_3 diamond	2.417	2.416
Reference angle (deg)	67.73	68.05
Reference $R_{\min}$	0.0456	0.0455
Reference $\mathcal{F}_E$	0.43	0.44
Reference L_{spp} (nm)	191.16	205.11
Reference δ_z (nm)	137.14	142.08
Calculated minimum (deg)	67.73267	68.05209
Calculated $R_{\min}$	0.04560152	0.04551051
Calculated $\mathcal{F}_E$ at minimum	0.214962	0.211377

The optical drawing shall distinguish the 42.00° entry-face setting from the 67.271702° hypotenuse incidence in Eq. (336); the 43.85° interface reference describes a separate geometry. The operating resonance angle requires characterization for the installed prism, film, gas state, and wavelength. The coarse setting tolerance shall be $\pm 0.001^\circ$, equivalent to $\pm 17.453 \mu\text{rad}$; fine adjustment shall meet the separate 0.015 μrad resolution requirement.

The viewport shall use synthetic fused silica with 23.0 mm clear aperture and dual-band Ta₂O₅/SiO₂ antireflection coatings. The optical specification shall use nominal $dn/dT = 1.0 \times 10^{-5} \text{ K}^{-1}$ near 800 nm, internal transmission exceeding 99.8% per centimeter, and p-polarized reflectance below 0.05% per coated surface at both pump wavelengths and the qualified incidence angle. Stress-relieved retainers shall preserve the optical acceptance limits after sealing.

a) *Four-layer TM transfer matrix*: The planar optical comparison comprises SF11 / 1.0 nm Pd–Ir / 10.0 nm Ti / semi-infinite CVD diamond [63]. With time convention $e^{-i\omega t}$, the transfer-matrix formulation uses the tangential field vector $(E_x, Z_0 H_y)^T$ and passive square-root branches:

$$\begin{aligned} k_x &= n_0 k_0 \sin \theta, \quad k_{z,j} = \sqrt{\epsilon_j k_0^2 - k_x^2}, \\ Y_j &= \epsilon_j k_0 / k_{z,j}, \quad \phi_j = k_{z,j} d_j, \quad M = M_1 M_2, \\ M_j &= \begin{bmatrix} \cos \phi_j & -i \sin \phi_j / Y_j \\ -i Y_j \sin \phi_j & \cos \phi_j \end{bmatrix}. \end{aligned} \quad (333)$$

$\text{Im } k_{z,j} \geq 0$ and positive outgoing power select physical branches. The dimensionless admittance requires $Z_0 H_y$, not unnormalized H_y . The matrices relate entrance to exit fields. Define

$$\begin{aligned} F &= M_{11} + M_{12} Y_3, \quad H = M_{21} + M_{22} Y_3, \\ r_x &= (Y_0 F - H) / (Y_0 F + H), \quad t_x = 2Y_0 / (Y_0 F + H), \\ R &= |r_x|^2, \quad T = \frac{\text{Re } Y_3}{\text{Re } Y_0} |t_x|^2, \\ \mathcal{F}_E &= |t_x \cos \theta|^2 (1 + |k_x / k_{z,3}|^2). \end{aligned} \quad (334)$$

t_x is relative to the incident *tangential* electric field; $\mathcal{F}_E$ uses the total incident electric-field magnitude. These are coherent planar-film boundary relations [65]. Since $n_0 \sin \theta < n_3$, the transmitted field in lossless diamond has real $k_{z,3}$ and is propagating rather than evanescent. A forced real- k_x TMM solution

TABLE XXVI
GRATING-ASSISTED OPTICAL INPUTS, TARGETS, AND PLANAR
COMPARISONS

Quantity	785.0 nm	802.5 nm
ϵ_{PdIr}	$-16.85 + 15.92i$	$-17.52 + 16.64i$
ϵ_{Ti}	$-4.82 + 23.48i$	$-5.24 + 24.15i$
Nominal $\mathcal{F}_z$ target	124.5	138.2
Planar minimum angle (deg)	68.51617	68.68832
Planar R_{min}	0.03977507	0.03961543
Planar $\mathcal{F}_z$ at minimum	0.094183	0.093730
Planar $\mathcal{F}_z$ at 65°	0.099945	0.099801
Half-space L_{spp} (nm)	208.62355	219.78252
Half-space δ_z (nm)	149.91983	154.00879

is distinct from a complex-wavevector SPP eigenmode: the propagation length $L_{\text{spp}} = 1/(2 \text{Im } k_{\text{spp}})$ and decay length δ_z require an eigenmode calculation rather than identification of a reflectivity minimum. The reference gas index is $n_{D_2} = 1.0003$, but the four-layer model contains no gas layer.

The tabulated propagation and decay lengths correspond to a separate semi-infinite Ti/diamond interface model:

$$\begin{aligned} k_{\text{spp}} &= k_0 \sqrt{\frac{\epsilon_{\text{Ti}} \epsilon_d}{\epsilon_{\text{Ti}} + \epsilon_d}}, \\ L_{\text{spp}} &= (2 \text{Im } k_{\text{spp}})^{-1}, \\ \delta_z &= \left[\text{Re} \sqrt{k_{\text{spp}}^2 - \epsilon_d k_0^2} \right]^{-1}. \end{aligned} \quad (335)$$

The half-space model gives

$(L_{\text{spp}}, \delta_z) = (191.15649, 137.14449)$ nm at 785 nm and $(205.10906, 142.08030)$ nm at 802.5 nm. These eigenmode lengths do not describe field decay in the forced four-layer solution. At 785 nm, $\text{Re } k_{\text{spp}}/k_0 = 2.450662 > n_0 = 1.76552$, so the specified prism requires an additional momentum contribution to couple to this half-space mode.

b) Right-angle-prism viewport geometry: For air incidence onto the entry-face normal of a 45° right-angle prism,

$$\begin{aligned} \sin \theta_r &= \sin 42.00^\circ / 1.76552, \quad \theta_r = 22.271702^\circ, \\ \theta_{\text{inc}} &= 45^\circ + \theta_r = 67.271702^\circ, \\ |\theta_{\text{inc}} - 67.73^\circ| &= 0.458298^\circ < 0.65^\circ. \end{aligned} \quad (336)$$

The 785 nm ray lies within the specified $67.73^\circ \pm 0.65^\circ$ HWHM window. At 802.5 nm, the same external angle gives 67.285667° . Each additional nonparallel viewport face requires a separate refraction calculation.

c) Constitutive inputs and grating-assisted momentum matching: The grating-assisted comparison uses the metallic permittivities in Table XXVI and the normal-component enhancement target $\mathcal{F}_z = |E_z(z_{\text{Ti}/\text{diamond}}^+)/E_0|^2$, distinct from the total-field factor. A 10 nm Ti layer separates Pd–Ir from diamond; the stack therefore contains no direct Pd–Ir/diamond boundary. The planar boundary relations are Eqs. (333)–(334), with $\mathcal{F}_z = |t_x \cos \theta k_x / k_{z,3}|^2$. The grating enhancement targets are 124.5/138.2. For a first-order grating at 65° , phase matching requires equality of the in-plane excitation momentum and the

TABLE XXVII
SEPARABLE-PROFILE EXPONENT COMPARISON AT $w \simeq 10.048711$ mm

Exponent m	Corner (%)	Throughput (%)	Clipping (%)
1148	98.5000	99.2508	0.7492
2296	99.9943	99.1425	0.8575

real modal wavevector:

$$\text{Re } k_{\text{spp}} = n_0 k_0 \sin 65^\circ + 2\pi/\Lambda,$$

$$\Lambda = \frac{\lambda}{\text{Re}(k_{\text{spp}}/k_0) - n_0 \sin 65^\circ},$$

$$(k_{\text{spp}}/k_0)_{785} = 2.42248005 + 0.29943080i,$$

$$\Lambda_{785} = 954.55175 \text{ nm}, \quad \Lambda_{802.5} = 970.27630 \text{ nm}. \quad (337)$$

The prism indices are 1.76552 and 1.76447. The specified 960.8 nm grating period is a nominal design target.

M. Refractive Field Mapping and Polarization Maintenance

Illumination uniformity over the square aperture $|x|, |y| \leq a = 10.0$ mm depends on the distinction between circular flat-top and square-separable profiles [58]. In the hard-edge circular limit with radius $R = 10.048711$ mm, the area clipped beyond each square edge is

$$A_{\text{seg}} = R^2 \arccos(a/R) - a \sqrt{R^2 - a^2} = 0.06421455 \text{ mm}^2. \quad (338)$$

The unilluminated square fraction is $1 - [\pi R^2 - 4A_{\text{seg}}]/(4a^2) = 20.7574\%$, or approximately 20.76%. The corners lie at $r = 14.1421$ mm, outside the beam radius, and their intensity approaches zero at high order. A steep circular edge also requires diffraction analysis rather than geometric ray mapping alone.

The square-separable irradiance model uses $m = 2N$ as the actual exponent:

$$\begin{aligned} I(x, y) &= I_0 e^{-2|x/w_x|^{2N}} e^{-2|y/w_y|^{2N}}, \\ w_x &= w_y = w, \quad m = 2N. \end{aligned} \quad (339)$$

Equal waists give square symmetry, not circular symmetry. The corner constraint is

$$\begin{aligned} \mathcal{U}_{\text{corner}} &= e^{-4(a/w)^m} \geq 0.9850, \\ w &\geq w_{\text{min}}(m) = a[-\ln(0.9850)/4]^{-1/m}. \end{aligned} \quad (340)$$

The one-dimensional full-line integral is $2w2^{-1/m}\Gamma(1+1/m)$. With $z = 2(a/w)^m$ and regularized lower incomplete gamma function $P(\alpha, z) = \gamma(\alpha, z)/\Gamma(\alpha)$,

$$\begin{aligned} \eta_{2D} &= [P(1/m, z)]^2, \quad L_{\text{clip}} = 1 - \eta_{2D}, \\ z|_{w=w_{\text{min}}} &= -\frac{1}{2} \ln(0.9850) = 0.007556818905. \end{aligned} \quad (341)$$

At a fixed waist, doubling the exponent changes both corner uniformity and aperture clipping. Table XXVII compares the two profiles by direct integration; performance values must remain associated with their defining exponent. The choice $N = 1148$ in an exponent $2N$ violates the 0.75% clipping limit at the specified waist. The admissible profile uses $m = 1148$,

equivalent to $N = 574$ in Eq. (339). Its exact waist and ideal performance are

$$\begin{aligned} w &= w_{\min}(1148) = 10.048711043077 \dots \text{ mm}, \\ \mathcal{U}_{\text{corner}} &= 98.500000\%, \\ \eta_{2D} &= 99.25084440\%, \quad L_{\text{clip}} = 0.74915560\%. \end{aligned} \quad (342)$$

An exponent denoted directly by $N = 1148$ describes the same profile only when no additional factor of two is applied. At the rounded waist 10.048711 mm, the corner ratio is 98.49999267%; tolerance allocation therefore requires the analytic bound rather than a rounded-down limit. The $m = 2296$ alternative requires its own waist bound and performance evaluation.

a) *Exponent convention and waist tolerance:* The irradiance profile is $I(x, y) = I_0 e^{-2(x/w)^m} e^{-2(y/w)^m}$ with actual exponent $m = 1148$, equivalent to $N = 574$ in $2N$ notation. For the rounded waist $w = 10.0487$ mm, direct integration gives $C_{\text{clip}} = 0.7489391\%$ and corner intensity 98.4981207%. Clipping satisfies its limit, but strict 98.50% corner uniformity requires the unrounded waist bound. At the same rounded waist, $m = 2296$ gives $C_{\text{clip}} = 0.8573148\%$, rather than the comparison estimate of 0.968%, and exceeds the 0.75% clipping limit. A profile written with exponent $2m$ doubles the exponent assigned to its parameter; the rounded 0.749% clipping value applies to actual exponent 1148, not 2296.

For a lossless separable refractive map, each coordinate obeys cumulative power conservation:

$$\frac{\int_{-\infty}^u I_{\text{in},x}(u') du'}{P_{\text{in},x}} = \frac{\int_{-\infty}^x I_{\text{out},x}(x') dx'}{P_{\text{out},x}}. \quad (343)$$

The analogous map in y imposes two-dimensional power conservation:

$$I_{\text{in}}(u, v) = I(x, y) \left| \frac{\partial(x, y)}{\partial(u, v)} \right|. \quad (344)$$

Actual throughput additionally includes reflection, absorption, finite aperture, aberration, and wave-optical diffraction. The separable geometry resolves corner coverage but does not by itself prove a realizable phase surface or eliminate ringing.

Polarization is specified at the prism input, after the complete fiber/window path:

$$\begin{aligned} \text{PER} &= 10 \log_{10} \left(\frac{P_p}{P_s} \right) \geq 30.0 \text{ dB}, \\ \frac{P_s}{P_p} &\leq 10^{-3}, \quad \frac{P_s}{P_p + P_s} \leq 9.99001 \times 10^{-4}. \end{aligned} \quad (345)$$

FT-OPT-01 uses panda-type polarization-maintaining fibers with axis alignment within $\pm 0.2^\circ$. For a pure linear input and otherwise ideal optics, axis error ψ gives $P_s/P_p = \tan^2 \psi$; $\psi = 0.2^\circ$ alone corresponds to 49.14 dB PER. Real fiber coupling, connectors, window stress, and coating retardance consume the remaining margin and require direct measurement at both wavelengths.

The viewport stress-optic allocation is $C_{\text{so}} \leq 2.7 \times 10^{-12} \text{ Pa}^{-1}$, with retardation below 2.0 nm per centimeter of optical path. For principal stresses σ_1 and σ_2 , birefringence and phase retardation satisfy

$$\Delta n = C_{\text{so}}(\sigma_1 - \sigma_2), \quad \delta_{\text{pol}} = \frac{2\pi}{\lambda} \int \Delta n \, d\ell. \quad (346)$$

TABLE XXVIII
GRATING ENHANCEMENT TARGETS AND PLANAR COMPARISON VALUES

Quantity	785.0 nm	802.5 nm
Assigned $\mathcal{F}_z$	124.50	138.20
L_{spp} (nm)	208.62	219.78
δ_z (nm)	149.92	154.01
Planar $R_{\min}$	0.039775	0.039615

Retardation and PER acceptance shall use the assembled stress field and polarization orientation, with measurements at both pump wavelengths after sealing and under the specified mechanical and thermal loads.

b) *Grating geometry and Maxwell boundary conditions:*

The grating geometry is specified by period $\Lambda = 960.80$ nm, depth $h_g = 42.50$ nm, ridge duty fraction $\Gamma_g = 0.500$, and edge fillet radius 5.0 nm. The dual-wavelength enhancement targets are distinguished from planar TMM results: For a one-dimensional periodic profile, the RCWA field expansion and diffraction wavevectors are

$$\begin{aligned} \varepsilon(x, z) &= \sum_p \varepsilon_p(z) e^{i2\pi p x / \Lambda}, \\ \mathbf{E}(x, z) &= \sum_p \mathbf{E}_p(z) e^{ik_{x,p} x}, \\ k_{x,p} &= n_0 k_0 \sin \theta + 2\pi p / \Lambda, \\ \nabla \times \mathbf{E} &= i\omega \mu_0 \mathbf{H}, \quad \nabla \times \mathbf{H} = -i\omega \varepsilon_0 \varepsilon \mathbf{E}. \end{aligned} \quad (347)$$

Tangential fields are continuous at every boundary, with outgoing or decaying diffraction orders selected in both half-spaces. A two-dimensional grating requires a second reciprocal index and a measured two-dimensional profile. Field enhancement is evaluated on the specified side of the interface using the same incident-field normalization as the planar calculation. The profile dimensions define the electromagnetic geometry, while Table XXVIII distinguishes planar reflectance and mode lengths from assigned grating enhancements. A converged harmonic sweep is required to establish whether a single RCWA solution satisfies all of these quantities. The 10 nm Ti spacer precludes direct PdIr/diamond contact.

c) *Viewport retardance and aperture verification:* For the 6.0 mm viewport, optical path retardance b_{opt} is expressed in meters and phase retardance δ_{pol} in radians:

$$\begin{aligned} b_{\text{opt}} &= \int_0^{t_{\text{win}}} C_{\text{so}}(\sigma_1 - \sigma_2) \, dz, \\ \delta_{\text{pol}} &= 2\pi b_{\text{opt}} / \lambda, \end{aligned} \quad (348)$$

$$\text{PER}_{\text{model}} = -10 \log_{10} [\tan^2 \psi + \sin^2(\delta_{\text{pol}}/2)].$$

The 2.0 nm/cm path-retardance allocation gives $b_{\text{opt}} \leq 1.20$ nm over 6 mm; path retardance per unit length is distinct from phase retardance. At $\lambda = 785$ nm and $\psi = 0.20^\circ$, Eq. (348) gives 44.52866 dB, whereas the comparison value is 34.22 dB. Both exceed the 30 dB target, but 34.22 dB does not follow from the specified additive-leakage model and requires additional loss mechanisms or measurement evidence. The $C_{\text{so}} \leq 2.70 \times 10^{-12} \text{ Pa}^{-1}$ bound alone does not determine the stress integral. For the separable aperture, direct integration

TABLE XXIX
THIN-FILM GRATING INPUTS AND NOMINAL ENHANCEMENT TARGETS

Quantity	785.0 nm	802.5 nm
Air/vacuum real index	1.0000	1.0000
Silica real index	1.4538	1.4532
PdIr real index	1.8240	1.8850
Ti real index	2.7120	2.7540
Diamond real index	2.3980	2.3950
Nominal $\mathcal{F}_z$	124.50	138.20

with $m = 1148 = 2(574)$, $w = 10.048711$ mm, and half-width $a = 10$ mm gives

$$\begin{aligned}\mathcal{U}_{\text{corner}} &= e^{-4(a/w)^m} \simeq 0.985000, \\ C_{\text{clip}} &= 1 - [P(1/m, 2(a/w)^m)]^2 \\ &= 0.00749155 < 0.0075.\end{aligned}\quad (349)$$

Here P is the regularized lower incomplete gamma function. The result describes geometrical aperture throughput rather than diffraction-limited fabrication performance. The opposed-column residual $0.82225 \mu\text{m}$ and remaining stroke $14.17775 \mu\text{m}$ in Eq. (321) apply to the separately specified Invar and aluminum temperature rises.

d) *Thin-film grating comparison and integrated datum chain:* The thin-film RCWA comparison comprises an air/vacuum superstrate and a fused-silica binary grating, distinct from the SF11 prism stack. Its parameters are $\Lambda = 960.80$ nm, $h_g = 42.50$ nm, $\Gamma_g = 0.50$, $\theta = 65.0^\circ$, 1.0 nm PdIr, 10.0 nm Ti, and a $500.0 \mu\text{m}$ diamond substrate. The $50.0 \mu\text{m}$ diamond in the structural comparison defines a separate geometry. The RCWA Fourier fields and Maxwell matching follow Eq. (347); for two-dimensional periodicity, each diffraction order additionally has $k_{y,q} = k_{y,0} + 2\pi q/\Lambda_y$. The tabulated real refractive indices do not specify the imaginary components required to model absorption. Complex permittivities from other model cases cannot be combined with these real indices without constitutive justification. The geometry and enhancement targets therefore do not uniquely determine an absorbing RCWA solution or establish harmonic convergence. The planar-model values of L_{spp} and δ_z are not necessarily eigenmode lengths of the grating with a $500 \mu\text{m}$ substrate.

The common-temperature mechanical comparison uses a 42.15 mm Invar column and an 8.52 mm aluminum column, distinct from the independently heated $172.50/20.69$ mm pair. Combining stack expansion with the opposed-column displacement gives

$$\begin{aligned}\Delta L_{\text{stack}} &= (6.741 \text{ mm})(12.84 \times 10^{-6} \text{ K}^{-1})(256.85 \text{ K}) \\ &= 22.231507914 \mu\text{m}, \\ \Delta L_{\text{comp}} &= [(42.15)(1.20) - (8.52)(23.10)] \\ &\quad \times 10^{-3}(256.85) \mu\text{m} \\ &= -37.559689200 \mu\text{m}, \\ \delta_{\text{res,calc}} &= -15.328181286 \mu\text{m}.\end{aligned}\quad (350)$$

The alternative displacement balance $22.23120 - 21.40895 = 0.82225 \mu\text{m}$ is arithmetically consistent but does not follow from these column dimensions. A residual target of 0.8225

μm also differs from that balance. At the specified common temperature, compensation of $-21.40895 \mu\text{m}$ requires $L_{\text{Al}} = 5.79792019$ mm for the given Invar column; this inverse solution defines an alternative geometry rather than the specified assembly. The nominal and common-temperature comparison stages have total piezo travels of 15 and $20 \mu\text{m}$, respectively. Residual displacement and available stroke require verification against the complete datum chain and preload geometry.

The nominal tilt RMS for this comparison is $\sigma_\theta = 0.00782 \mu\text{rad}$. Relative to the $0.00800 \mu\text{rad}$ limit, the numerical margin is 2.25% . For a one-sided closed-loop tilt spectrum,

$$\sigma_\theta^2 = \int_0^\infty S_{\theta,\text{cl}}(f) \text{d}f, \quad (351)$$

with disturbance, sensor-noise, and cross-spectral contributions included. Verification requires a measured spectrum or a validated closed-loop model that evaluates this integral. The RMS requirement applies to the dynamic closed-loop response, including disturbance and sensor-noise spectra.

N. 3D Non-Linear Chaboche Viscoplasticity and Somigliana Representation

At the 623.15 K operating reference, small-strain thermo-viscoplasticity separates elastic, thermal, and inelastic contributions:

$$\begin{aligned}\epsilon &= \epsilon^e + \epsilon^{\text{th}} + \epsilon^{\text{vp}}, \\ \epsilon^{\text{th}}(T) &= \int_{T_0}^T \alpha_T(\vartheta) \text{d}\vartheta, \\ \sigma &= \mathbf{C}(T) : \epsilon^e, \\ \dot{\sigma} &= \mathbf{C} : (\dot{\epsilon} - \alpha_T \dot{T} - \dot{\epsilon}^{\text{vp}}) + \dot{\mathbf{C}} : \epsilon^e.\end{aligned}\quad (352)$$

For the alloy/SiC model [102], $T_0 = 298.15$ K and the component form is $\epsilon_{ij} = \epsilon_{ij}^e + \epsilon_{ij}^{\text{vp}} + \epsilon_{ij}^{\text{th}}$, with $\epsilon_{ij}^{\text{th}} = [\int_{T_0}^T \alpha(\vartheta) \text{d}\vartheta] \delta_{ij}$. The alloy has $E(298.15 \text{ K}) = 140.0$ GPa, $E(623.15 \text{ K}) = 110.0$ GPa, $\nu = 0.38$, and $\alpha = 12.0 \times 10^{-6} \text{ K}^{-1}$; SiC has the corresponding moduli 400.0 and 385.0 GPa, $\nu = 0.16$, and $\alpha = 4.0 \times 10^{-6} \text{ K}^{-1}$. These inputs are distinct from the earlier constant-property elastic comparison in Section IV-I.

The tensor α_T is the tangent thermal expansion, distinct from the kinematic backstress. Define $\mathbf{s} = \text{dev } \sigma$, $\mathbf{X} = \mathbf{X}^{(1)} + \mathbf{X}^{(2)}$, $\mathbf{X}' = \text{dev } \mathbf{X}$, and $\langle x \rangle = \max(x, 0)$. Here J_2 denotes the von Mises equivalent stress, not the unrooted second deviatoric invariant. The Perzyna flow rule is

$$\begin{aligned}J_2(\sigma - \mathbf{X}) &= \sqrt{\frac{3}{2}(\mathbf{s} - \mathbf{X}') : (\mathbf{s} - \mathbf{X}')}, \\ f &= J_2(\sigma - \mathbf{X}) - R - \sigma_y(T), \\ \dot{p} &= \left\langle \frac{f}{K} \right\rangle^n, \quad \dot{\epsilon}_{ij}^{\text{vp}} = \frac{3}{2} \dot{p} \frac{s_{ij} - X'_{ij}}{J_2(\sigma - \mathbf{X})}, \\ R &= Q(1 - e^{-bp}), \quad \dot{R} = b(Q - R)\dot{p}.\end{aligned}\quad (353)$$

The accumulated equivalent viscoplastic strain satisfies $\dot{p} = \sqrt{(2/3)\dot{\epsilon}^{\text{vp}} : \dot{\epsilon}^{\text{vp}}}$. The flow is zero when $f \leq 0$; the quotient at $J_2 = 0$ is not evaluated. The specified parameters are $K = 60.0 \text{ MPa s}^{1/n}$, $n = 4.0$, $Q = 40.0 \text{ MPa}$, and $b = 10.0$, with $p = R = 0$ initially. Thus $\dot{p}$ has units s^{-1} without

an additional reference strain rate. With T in kelvin, the temperature-dependent yield stress is

$$\begin{aligned}\sigma_y(T) &= [220.0 - 0.2(T - 298.15)] \text{ MPa}, \\ \sigma_y(298.15 \text{ K}) &= 220.0 \text{ MPa}, \\ \sigma_y(623.15 \text{ K}) &= 155.0 \text{ MPa}.\end{aligned}\quad (354)$$

The two non-isothermal kinematic backstress components evolve as

$$\begin{aligned}\dot{X}_{ij}^{(k)} &= \frac{2}{3} C_k(T) \dot{\epsilon}_{ij}^{\text{vp}} - \gamma_k X_{ij}^{(k)} \dot{p} \\ &\quad + \frac{1}{C_k(T)} \frac{\partial C_k(T)}{\partial T} \dot{T} X_{ij}^{(k)}, \quad k = 1, 2, \\ C_1(T) &= [50.0 \times 10^3 - 61.54(T - 298.15)] \text{ MPa}, \\ C_2(T) &= [15.0 \times 10^3 - 21.54(T - 298.15)] \text{ MPa}, \\ \gamma_1 &= 500.0, \quad \gamma_2 = 80.0.\end{aligned}\quad (355)$$

The final term accounts for temperature-rate changes in the hardening moduli; it is not a static-recovery term. Both moduli remain positive over 298.15–623.15 K. These assigned constitutive inputs require composition- and temperature-specific calibration. When the vacancy-diffusion climb model governs the inelastic rate, its homogenized flow replaces the overstress rate rather than being added to it and counted twice.

For a boundary-integral representation choose a fixed homogeneous isotropic reference stiffness $\mathbf{C}^0$ at 623.15 K. Define the polarization stress

$$\begin{aligned}P_{jk} &= C_{jklm}^0 \epsilon_{lm} - \sigma_{jk}, \\ \mathbf{P} &= (\mathbf{C}^0 - \mathbf{C}) : \boldsymbol{\epsilon} + \mathbf{C} : (\boldsymbol{\epsilon}^{\text{th}} + \boldsymbol{\epsilon}^{\text{vp}}).\end{aligned}\quad (356)$$

The Kelvin displacement fundamental solution and its traction kernel are

$$\begin{aligned}U_{ij}(\mathbf{x}, \mathbf{y}) &= \frac{(3 - 4\nu_0)\delta_{ij} + r_i r_j / r^2}{16\pi\mu_0(1 - \nu_0)r}, \\ \mathbf{r} &= \mathbf{y} - \mathbf{x}, \quad r = |\mathbf{r}|, \\ T_{ij}(\mathbf{x}, \mathbf{y}) &= n_k(\mathbf{y}) C_{jklm}^0 \frac{\partial U_{il}}{\partial y_m}.\end{aligned}\quad (357)$$

Applying reciprocity to the reference elastic operator and integrating the divergence of $\mathbf{P}$ by parts gives the quasistatic Somigliana identity

$$\begin{aligned}c_{ij}(\mathbf{x}) u_j(\mathbf{x}) &= \int_{\partial\Omega} U_{ij} t_j \, dS - \text{p. v.} \int_{\partial\Omega} T_{ij} u_j \, dS \\ &\quad + \int_{\Omega} U_{ij} b_j \, dV + \int_{\Omega} \frac{\partial U_{ij}}{\partial y_k} P_{jk} \, dV.\end{aligned}\quad (358)$$

Here $\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$ is the actual traction and $\mathbf{b}$ is body force per unit volume. The boundary term from polarization cancels the difference between reference and actual tractions, fixing the positive sign of the last volume integral. For an interior point $c_{ij} = \delta_{ij}$; at a smooth boundary it is $\delta_{ij}/2$, with principal-value interpretation. Corners require their geometric free term. At fixed geometry and fixed $\mathbf{C}^0$, differentiation yields

$$\begin{aligned}c_{ij} \dot{u}_j &= \int_{\partial\Omega} U_{ij} \dot{t}_j \, dS - \text{p. v.} \int_{\partial\Omega} T_{ij} \dot{u}_j \, dS \\ &\quad + \int_{\Omega} U_{ij} \dot{b}_j \, dV + \int_{\Omega} \frac{\partial U_{ij}}{\partial y_k} \dot{P}_{jk} \, dV.\end{aligned}\quad (359)$$

The full derivative of Eq. (356) includes thermal expansion, stiffness changes, and viscoplastic flow. Heterogeneous and thermally strained regions therefore retain volume integrals even outside plastically active zones. Inertia, when important, enters Eq. (358) through $\mathbf{b} - \rho \ddot{\mathbf{u}}$; moving geometry requires additional transport terms.

IV. NANOFABRICATION, STRESS-ASSISTED TRANSPORT, AND MATERIALS INTEGRITY

The materials specification couples alloy deposition, intermetallic bonding, stress-assisted deuterium transport, and containment integrity. Lattice diffusion and reversible trapping determine the mobile and retained inventories; permeation barriers constrain escape through noncatalytic surfaces, while environmental fracture thresholds and crack-growth limits define the failure assessment. Fabrication, transport, and fracture parameters apply to their specified material states and temperatures. Consistent concentration units, interface conditions, and load histories are required to connect these models to the assembled device.

A. Active-Alloy Kinetics and Five-Layer Thermoviscoplastic Baseline

The active material is $\text{Pd}_{0.9132}\text{Ir}_{0.0868}\text{D}_x$, corresponding to 85.347 wt% Pd and 14.653 wt% Ir in the metal target, at the verified deuterium loading $x = 0.9132$. Mobile lattice deuterium C_L exchanges with dislocation and grain-boundary trap families having occupied fractions θ_1 and θ_2 . With tensile hydrostatic stress positive, the two-family McNabb–Foster model is

$$\begin{aligned}\mathbf{J}_D &= -D_D(T, x) \nabla C_L \\ &\quad + \frac{V_H^*}{RT} D_D(T, x) C_L \nabla \sigma_h \\ &\quad + \frac{Q^*}{RT^2} D_D(T, x) C_L \nabla T,\end{aligned}$$

$$\frac{\partial C_L}{\partial t} + \sum_{j=1}^2 N_j \frac{\partial \theta_j}{\partial t} = -\nabla \cdot \mathbf{J}_D,$$

$$\frac{\partial \theta_j}{\partial t} = k_j C_L (1 - \theta_j) - p_j \theta_j, \quad j = 1, 2. \quad (360)$$

The baseline lattice diffusivity is $D_D = D_0 \exp[-E_a/(k_B T)]$, with $D_0 = 2.85 \times 10^{-7} \text{ m}^2/\text{s}$ and $E_a = 0.224 \text{ eV}$. The signed heat of transport is $Q^* = 0.048 \text{ eV}$ per deuteron (4.631 kJ/mol when used with R), and $V_H^* = 1.70 \times 10^{-6} \text{ m}^3/\text{mol}$. The trap data are $N_1 = 1.50 \times 10^{24} \text{ m}^{-3}$ and $E_{t,1} = 0.23 \text{ eV}$ for dislocations, and $N_2 = 5.00 \times 10^{23} \text{ m}^{-3}$ and $E_{t,2} = 0.15 \text{ eV}$ for grain boundaries. Equation (360) uses the displayed Soret sign convention; reversing the definition of the heat of transport reverses the sign assigned to Q^* .

The structural baseline is a three-dimensional, five-layer Pd–Ir/Ti/CVD-diamond/Ni–Cu–Sn/OFHC-Cu model. Metallic layers use the small-strain Chaboche thermoviscoplastic split

$$\begin{aligned}\boldsymbol{\epsilon} &= \boldsymbol{\epsilon}^e + \boldsymbol{\epsilon}^p + \boldsymbol{\epsilon}^{\text{th}} + \boldsymbol{\epsilon}^{\text{sw}}, \\ \mathbf{X} &= \mathbf{X}_1 + \mathbf{X}_2, \quad \dot{\mathbf{X}}_i = \frac{2}{3} C_i \dot{\epsilon}^p - \gamma_i \mathbf{X}_i \dot{p}, \\ R(p) &= R_\infty (1 - e^{-bp}), \quad \dot{p} = \sqrt{\frac{2}{3}} \dot{\epsilon}^p : \dot{\epsilon}^p.\end{aligned}\quad (361)$$

The calibrated hardening parameters are $C_1 = 45.2$ GPa, $\gamma_1 = 410$, $C_2 = 8.5$ GPa, $\gamma_2 = 62$, $R_\infty = 85$ MPa, and $b = 12.46$. CVD diamond remains anisotropic thermoelastic. Along the $3.5 \mu\text{m}$ Ni–Cu–Sn TLP bondline, the three-mode VCCT result is

$$f_{\text{delam}} = \frac{G_I}{G_{IC}} + \frac{G_{II}}{G_{IIC}} + \frac{G_{III}}{G_{IIIC}} = 0.484 < 1.0. \quad (362)$$

The associated Morrow strain–life calculation gives $N_f \simeq 65,474$ cycles, exceeding the required 52,400 cycles.

B. Materials, Nanofabrication and Mechanical Constraints

a) CVD diamond substrate and four-step metasurface fabrication: The selected substrate is single-crystal CVD diamond with dimensions $20.0 \times 20.0 \times 0.50$ mm, reference thermal conductivity $k_d = 1800 \text{ W m}^{-1} \text{ K}^{-1}$, and relative permittivity $\epsilon_d = 5.7504$ at 785.0 nm. The permittivity is the rounded n^2 value for $n = 2.3980$ in Table [XCIII](#); it is not a frequency-independent static dielectric constant. Conductivity at elevated temperature requires a characterized $k_d(T)$ rather than an assumed constant 1800 W/(m K) . The 4.00 cm^2 substrate footprint is distinct from the selected 1.00 cm^2 patterned active aperture in Eq. (179). The fabrication sequence is:

- 1) *Atomic-hydrogen cleaning.* Process the diamond before temperature-sensitive bonding or device assembly using the specified 200 W RF atomic-hydrogen plasma at 2.0 Pa and 450°C for 15 min. Verify surface cleanliness, termination, and roughness with witness measurements.
- 2) *Titanium adhesion sputtering.* Stabilize the substrate at 150°C and deposit a continuous 10.0 nm Ti layer by DC magnetron sputtering. Measure thickness, continuity, and adhesion. Interfacial TiC formation is a characterization objective. The 0.2 nm seed used in the ultrathin-film comparison of Table [XXXII](#) is not the selected grating adhesion layer.
- 3) *Pd–Ir HiPIMS deposition.* Use a 99.999% -purity vacuum-melted target with $85.347 \text{ wt}\%$ Pd and $14.653 \text{ wt}\%$ Ir for the revised $\text{Pd}_{0.9132}\text{Ir}_{0.0868}$ atomic composition. The earlier $91.32/8.68 \text{ wt}\%$ target belongs to the separate $95/5$ atomic-percent alloy. Apply the pulse, bias, and argon setpoints below as process-development inputs requiring requalification for the revised target. Establish the 7.50 nm continuous backplane and continue alloy growth to the 50.00 nm pre-pattern thickness needed for 42.50 nm teeth. At the nominal 0.025 nm/s rate, thickness endpoint measurements, rather than elapsed time alone, control deposition. Verify deposited composition and uniformity independently of target composition.
- 4) *EBL/RIE nanolithography.* Pattern ZEP520A resist by electron-beam lithography and transfer the one-dimensional binary grating using an anisotropic Ar/Cl_2 reactive-ion-etch process. Control etch depth so that a 42.50 nm relief remains above the 7.50 nm backplane. The pitch is 960.80 nm and metal duty cycle is 0.50 , corresponding to a nominal 480.40 nm tooth width. Qualify resist removal, residues, sidewall profile, residual backplane continuity, and AFM roughness over a $5 \times 5 \mu\text{m}^2$ scan. The strict residual surface-roughness

requirement is $R_q < 0.12 \text{ nm}$; report the masks and form-removal procedure separately from the intentional grating relief.

The resulting selected geometry is

$$\begin{aligned} t_{\text{Ti}} &= 10.0 \text{ nm}, & t_b &= 7.50 \text{ nm}, \\ h_g &= 42.50 \text{ nm}, & \Lambda_g &= 960.80 \text{ nm}, \\ f_g &= 0.50, & R_q &< 0.12 \text{ nm}, \\ \bar{t}_{\text{metal}} &= t_b + f_g h_g = 28.75 \text{ nm}. \end{aligned} \quad (363)$$

This process produces the invariant-groove Option B structure.

The revised Option B target uses atomic fractions $x_{\text{Pd}} = 0.9132$ and $x_{\text{Ir}} = 0.0868$. Using rounded atomic molar masses 106.42 and 192.22 g/mol [93], the average mass per mole of metal atoms and target mass fractions are

$$\begin{aligned} M_{\text{alloy}} &= 0.9132(106.42) + 0.0868(192.22) \\ &= 113.86744 \text{ g/mol}, \\ w_{\text{Pd}} &= \frac{0.9132(106.42)}{113.86744} = 0.8534726345 \dots, \\ w_{\text{Ir}} &= \frac{0.0868(192.22)}{113.86744} = 0.1465273655 \dots \end{aligned} \quad (364)$$

The revised mass specification is therefore $85.347 \text{ wt}\%$ Pd / $14.653 \text{ wt}\%$ Ir, rounded from the stated atomic fractions; deuterium uptake is excluded from these metal-target mass fractions. The former $91.32 \text{ wt}\%$ Pd / $8.68 \text{ wt}\%$ Ir recipe approximates $\text{Pd}_{0.95}\text{Ir}_{0.05}$ and is retained only as historical provenance. The HiPIMS setpoints are peak power density 1.85 kW/cm^2 , pulse duration $30 \mu\text{s}$, repetition frequency 100 Hz , duty cycle 0.30% , and substrate bias -120 V synchronized to the $30 \mu\text{s}$ ion-arrival window. Deposition uses 99.999% Ar at 0.45 Pa (3.38 mTorr) and a substrate temperature of 150°C . The selected acceptance criterion is $R_q < 0.12 \text{ nm}$ over the specified $5 \times 5 \mu\text{m}^2$ AFM sampling window. The $R_q < 0.15 \text{ nm}$ limit belongs only to the ultrathin comparison process. Process-window qualification additionally requires deposition rate, total dose, composition transfer, calibrated AFM measurements, and wafer-scale uniformity. Earlier optical constants, interface-flux results, and material tensors for the $95/5$ atomic-percent alloy remain composition-specific comparison data, not validated inputs for the revised alloy.

b) Bondline Coffin–Manson comparison: The following $5.0 \mu\text{m}$ $\epsilon\text{-Cu}_3\text{Sn}$ TLP bondline comparison uses the plastic-strain range convention. It is independent of the selected spring-amplitude criterion in Eq. (283):

$$\begin{aligned} \Delta\epsilon_p &= \epsilon'_f (2N_f)^c, \\ \epsilon'_f &= 0.18, & c &= -0.62, \\ N_f(0.00098) &= \frac{1}{2}(0.00098/0.18)^{-1/0.62} = 2242.11, \\ \Delta\epsilon_p(44820) &= 0.18(89640)^{-0.62} = 0.000153010544. \end{aligned} \quad (365)$$

This bondline model uses the fatigue threshold $\Delta\epsilon_p \leq 0.000153010544$, with conservative design value $\Delta\epsilon_p \leq 0.000153$. The selected X-750 spring amplitude ceiling is $4.805 \times 10^{-4} \text{ mm/mm}$. The bondline coefficients use a strain-range convention at 550 K ; the $298.15\text{--}623.15 \text{ K}$ cycle requires temperature-dependent fatigue and creep characterization.

c) *Second-order thermal-expansion integral:* Under the reference-length engineering-strain convention,

$$\alpha(T) = a + b(T - 25^\circ\text{C}),$$

$$\Delta L = L_0 \int_{25}^{325} \alpha(T) dT = L_0 [a\Delta T + \frac{1}{2}b(\Delta T)^2], \quad (366)$$

$$\Delta T = 325 \text{ K}.$$

The coefficient varies linearly with temperature, yielding a second-order integrated strain. The integral is exact under the stated constitutive convention. If α is defined per instantaneous length, the exact stretch is $\exp(\int \alpha dT)$, and the engineering-strain expression is a small-strain approximation.

$$\begin{aligned} L_I &= 34.7021 \text{ mm}, & L_A &= 1.0721 \text{ mm}, \\ a_I &= 1.20 \times 10^{-6} \text{ K}^{-1}, \\ b_I &= 1.80 \times 10^{-9} \text{ K}^{-2}, \\ a_A &= 2.35 \times 10^{-5} \text{ K}^{-1}, \\ b_A &= 1.50 \times 10^{-8} \text{ K}^{-2}, \\ \Delta L_I &= 16.832687381 \mu\text{m} \simeq 16.83 \mu\text{m}, \\ \Delta L_A &= 9.037467969 \mu\text{m} \simeq 9.04 \mu\text{m}, \\ \Delta L_{\text{net}} &= 7.795219413 \mu\text{m}. \end{aligned} \quad (367)$$

The nominal drift and stage-reserve allocations are 7.79 and 7.21 μm . Travel acceptance uses the unrounded displacement, the complete tolerance stack, and a cold-end bias providing the required one-sided travel.

d) *Comparison-stack stiffness, VCCT and cross-domain loads:* The following non-governing comparison uses $P_0 = 15.80 \pm 0.50 \text{ MPa}$ at the reference condition. The selected contact force and 5,000 N/mm force stiffness are specified in Section III-A and Eq. (314); they shall not be combined with these comparison pressure-stiffness inputs:

$$\begin{aligned} K_A &= k_{\text{eq}}/A_{\text{load}} = 256.5 \text{ GPa/m} \\ &= 0.2565 \text{ MPa}/\mu\text{m}, \\ \Delta\sigma &= K_A \Delta L_{\text{net}} = 1.99947 \text{ MPa}, \\ K_{A,\text{max}} &= \frac{2.0 \times 10^6}{7.795219413 \times 10^{-6}} \simeq 256.568 \text{ GPa/m}. \end{aligned} \quad (368)$$

The comparison stack has a full-ramp pressure variation of approximately 2 MPa. A passive full-span $\pm 0.50 \text{ MPa}$ bound requires $K_A \leq 128.284 \text{ GPa/m}$ before preload uncertainty; a one-sided 0.50 MPa drift limit requires $K_A \leq 64.142 \text{ GPa/m}$. The selected spring-path case uses 20.1 μm mismatch and a 0.320 MPa pressure excursion.

The nominal 3D VCCT results are $G_I = 4.2$, $G_{II} = 14.2$, and $G_{\text{VCCT}} = 18.4 \text{ J/m}^2$, with $G_c = 24.5 \text{ J/m}^2$ and $G_c/G = 1.3315$. Delamination acceptance uses mesh convergence, crack geometry and mode mix, the applicable thermal state, and measured interface toughness. The aluminum bulk-stress ratio is $276/15.80 = 17.47$. Buckling, hot strength, and the proposed 15–18 MPa TEG contact-pressure range require their respective material, contact-area, and load-path qualification.

e) *Monolithic grating and seating-frame geometry:* The active stack comprises a 50.0 nm peak-height monolithic Pd–Ir grating with a 7.50 nm continuous backplane on 10.0 nm Ti/500.0 μm diamond/5.0 μm TLP/1.0 mm copper. The metal

TABLE XXX
TEMPERATURE-DEPENDENT VOCE MATERIAL INPUTS

Material	T ($^\circ\text{C}$)	σ_0 (MPa)	σ_∞ (MPa)	b
Pd–Ir	25	280	550	32.0
Pd–Ir	350	180	410	25.4
OFHC Cu	25	75	180	22.5
OFHC Cu	350	45	120	18.5

footprint is 1.0 cm^2 . The 50% duty cycle gives a mean metal thickness of 28.75 nm rather than a uniform 50 nm layer. At an assumed growth rate of 0.025 nm/s, a 50 nm unpatterned precursor requires 2000 s before patterning.

The constant-CTE clamp model uses a 33.63 mm seating assembly with 5.52 μm growth over 325 K; this seating dimension is distinct from the sum of the active-layer thicknesses. With constant $\alpha_I = 1.2 \times 10^{-6} \text{ K}^{-1}$ and $\alpha_A = 23.0 \times 10^{-6} \text{ K}^{-1}$, the column lengths satisfy

$$\begin{bmatrix} 1 & -1 \\ \alpha_I & -\alpha_A \end{bmatrix} \begin{bmatrix} L_I \\ L_A \end{bmatrix} = \begin{bmatrix} 0.03363 \text{ m} \\ (5.52 \times 10^{-6}/325) \text{ m/K} \end{bmatrix}. \quad (369)$$

The analytical solution and rounded manufacturing dimensions are

$$\begin{aligned} L_I &= 34.7020818631 \text{ mm} \approx 34.7021 \text{ mm}, \\ L_A &= 1.0720818631 \text{ mm} \approx 1.0721 \text{ mm}, \\ \Delta L_{\text{gap}} &= (\alpha_I L_I - \alpha_A L_A) \Delta T, \\ \delta_{\text{pinch}} &= \Delta L_{\text{stack}} - \Delta L_{\text{gap}}. \end{aligned} \quad (370)$$

At $\Delta T = 325 \text{ K}$, the column model gives $\Delta L_{\text{gap}} = 5.5198715 \mu\text{m}$ and $\delta_{\text{pinch}} = 0.0001285 \mu\text{m}$. The unrounded solution gives zero nominal pinch within the constant-CTE linear model.

The full-temperature compressive preload requirement is $15.80 \pm 0.50 \text{ MPa}$ over 25–350 $^\circ\text{C}$, distinct from the 42.5 MPa comparison clamp load. Compliance with this band requires a spring and load-path model satisfying

$$\sigma_c(T) = 15.80 \text{ MPa} + \frac{k_{\text{eq}}(T)}{A_{\text{load}}} \delta_{\text{tot}}(T), \quad (371)$$

$$\left| \frac{k_{\text{eq}}(T)}{A_{\text{load}}} \delta_{\text{tot}}(T) \right| \leq 0.50 \text{ MPa}.$$

Here δ_{tot} includes pinch, tolerances, relaxation, and differential temperature; k_{eq} is the assembled series stiffness and A_{load} the loaded area. For a maximum displacement of 0.04 μm , the sufficient stiffness-per-area bound is $k_{\text{eq}}/A_{\text{load}} \leq 1.25 \times 10^{13} \text{ Pa/m}$. Full-range pressure qualification requires temperature-dependent CTE integration and spring characterization.

In this table, σ_∞ denotes total saturation stress in the law $\sigma_f = \sigma_0 + (\sigma_\infty - \sigma_0)(1 - e^{-b\bar{\epsilon}_p})$, distinct from a hardening increment R_∞ . The mode-specific fracture thresholds are 14.20, 38.50, and 5.60 J/m^2 , with a hydrogen-affected opening threshold of 8.10 J/m^2 . These quantities are toughness inputs rather than calculated VCCT release rates. The transport parameters $V_H^* = 1.60 \times 10^{-6} \text{ m}^3/\text{mol}$ and $Q^* = 12.50 \text{ kJ/mol}$ enter the stress- and thermal-gradient flux equations.

f) *Thin-film stack and constrained-fatigue comparison:* The thin-film comparison stack comprises 1.0 nm Pd_{0.95}Ir_{0.05}/10.0 nm Ti/500.0 μm CVD diamond/5.0 μm $\epsilon\text{-Cu}_3\text{Sn}$ /1.0 mm OFHC Cu. The comparison uses housing stiffness $K_{\text{housing}} = 4.85 \times 10^8 \text{ N/m}$ and compressive clamp preload $\sigma_{\text{clamp}} = 42.5 \text{ MPa}$. These are boundary-condition inputs to the three-dimensional constitutive model.

$$\begin{aligned}\Delta\varepsilon_{p,\max} &= 2(0.185)(2 \cdot 44820)^{-0.52} \\ &= 0.000983786284 = 0.0983786284\%.\end{aligned}\quad (372)$$

The fatigue threshold uses the unrounded strain inequality and the plastic-strain field from the constrained assembly solution.

The primary structural reference is a five-layer, three-dimensional elastoplastic and creep model. Process comparisons, elastic reductions, hydrogen-embrittlement tests, and flaw-growth projections apply only to their specified geometries and conditions. The thin-film process, mechanical seating stack, and thermoelastic benchmark are distinct configurations. Manufacturing and analysis records shall associate each configuration with its geometry, constitutive inputs, process history, and acceptance evidence.

C. Axial Stack-Up and Thermal Compensation

Table XXXI defines a chassis-seating benchmark with a 20.0 μm bond contribution. The unalloyed-stack FEA comparison uses the 5.0 micrometer bond and layer dimensions in Table XXXIV; it is distinct from the selected 3.5 μm micro-alloyed bond and 15.0 μm graded interlayer in Section III-0a. Initial preform thickness differs from final load-bearing IMC thickness. Dimensional inspection shall account for consumed metal, residual cladding, and final seating height. The uniform-temperature comparison uses $T_0 = 293.15 \text{ K}$ and hot-side reference $T_h = 550.0 \text{ K}$, giving $\Delta T = 256.85 \text{ K}$. This comparison does not imply that the cooled chassis operates at T_h .

For layer seating contribution $L_{0,j}$, CTE α_j , and manufacturing deviation δz_j ,

$$\Delta z_j = L_{0,j} \alpha_j \Delta T, \quad L_{\text{stack}} = \sum_j (L_{0,j} + \Delta z_j + \delta z_j). \quad (373)$$

The constant-CTE result is $L_{\text{hot,nom}} = 6.740966 \text{ mm}$, with worst-case interval $[6.734466, 6.747466] \text{ mm}$. Production analysis requires temperature-dependent CTEs and actual layer temperatures rather than this screening approximation. For independent errors, the root-sum-square tolerance is

$$t_{\text{RSS}} = \sqrt{2.00^2 + 0.50^2 + 1.00^2 + 3.00^2} \mu\text{m} = 3.77492 \mu\text{m}. \quad (374)$$

The overall $\pm 4.07 \mu\text{m}$ allocation leaves $1.52148 \mu\text{m}$ RSS allowance for separately identified contributors under the same coverage convention. Correlated or bounded tolerances require their appropriate propagation rather than interpreting RSS as a hard limit.

a) *Differential compensation frame:* The re-entrant geometry in Section III and Eq. (321) is distinct from the 151.81 mm direct Invar/Cu-column comparison. Its residual displacement is $0.82225 \mu\text{m}$ at the two specified column temperatures. The seating-stack datum chain shall identify which terms in Table XXXI move with the return link; the $20.966 \mu\text{m}$ stack sum is a separate geometric benchmark.

D. Pd–Ir HiPIMS Nanofilm Synthesis Protocol

The microscopic and FEA comparison uses a 10.0 nm Ti transition layer and a 1.0 nm Pd–Ir film. The 0.2 nm Ti seed process defines a separate deposition configuration and does not specify fabrication of the 10 nm interface. Film thickness and deposition time refer to Pd–Ir unless explicitly identified as Ti. The active-film thickness convention is defined in Table XXXIV.

The ultrathin comparison film shall have nominal thickness 1.0 nm and atomic composition Pd_{0.95}Ir_{0.05}. Table XXXII specifies its HiPIMS setpoints and film acceptance criteria [26]. Procurement and process records shall distinguish atomic film composition from target mass composition.

Using atomic masses $M_{\text{Pd}} = 106.42 \text{ g mol}^{-1}$ and $M_{\text{Ir}} = 192.217 \text{ g mol}^{-1}$ [93], the distinction is

$$\begin{aligned}x_{\text{Pd}}^{(95 \text{ wt}\%)} &= \frac{0.95/M_{\text{Pd}}}{0.95/M_{\text{Pd}} + 0.05/M_{\text{Ir}}} = 0.971686, \\ w_{\text{Pd}}^{(95 \text{ at}\%)} &= \frac{0.95M_{\text{Pd}}}{0.95M_{\text{Pd}} + 0.05M_{\text{Ir}}} = 0.913189.\end{aligned}\quad (375)$$

The ideal target composition corresponding to the specified atomic ratio is 91.3189 wt% Pd and 8.6811 wt% Ir. Deposited-film composition shall be measured, and any target-composition or process correction for preferential sputtering and transport shall be documented against the required atomic film composition.

The process sequence comprises atomic-hydrogen etching of CVD diamond, thermal stabilization for Ti deposition, seed-layer formation and characterization, shutter-closed alloy-target conditioning, and synchronized HiPIMS deposition. TiC formation, Frank–van der Merwe layer-by-layer growth, and an ion-to-neutral ratio above 85% are process-response objectives requiring measurement. The nominal seed thickness adds to the 1.0 nm alloy thickness and is excluded from the Pd–Ir composition allocation.

The dimensional checks are

$$\begin{aligned}D_{\text{pulse}} &= (30 \times 10^{-6})(100) = 0.003, \\ \bar{q}_{\text{target}} &= D_{\text{pulse}}(1850) = 5.55 \text{ W cm}^{-2}, \\ A_{\text{target}} &= \pi(2.54 \text{ cm})^2 = 20.2683 \text{ cm}^2, \\ P_{\text{peak}} &\approx 37.4964 \text{ kW}, \quad \bar{P} \approx 112.489 \text{ W}, \\ h_f &= (0.025 \text{ nm s}^{-1})(40.0 \text{ s}) = 1.000 \text{ nm}.\end{aligned}\quad (376)$$

The average-power estimate assumes rectangular pulses and the full target face as the power-density reference; actual waveforms and erosion-track dimensions shall be recorded. A continuous-DC power density of 1.85 W cm^{-2} is not interchangeable with a peak HiPIMS power density of 1.85 kW cm^{-2} .

Acceptance requires calibrated AFM roughness and sampling, cross-sectional film and seed thickness, chemical composition,

TABLE XXXI
CHASSIS-SEATING COMPARISON AT UNIFORM 256.85 K RISE

Layer contribution	$L_{0,j}$ (mm)	α_j (10^{-6} K^{-1})	Δz_j (μm)	Tolerance (μm)
Invar 36 chassis	1.500	1.20	0.462330	± 2.00
CVD diamond frame seating	0.500	1.00	0.128425	± 0.50
ϵ -Cu ₃ Sn bondline	0.020	18.40	0.094521	± 1.00
OFHC copper heat spreader	4.700	16.80	20.280876	± 3.00
Total	6.720	Layer-specific	20.966152	± 6.50 worst case

TABLE XXXII
ULTRATHIN-FILM HiPIMS SETPOINTS AND ACCEPTANCE CRITERIA

Parameter	Setpoint or target
Film	1.0 nm; 95.0 at% Pd/5.0 at% Ir
Target	Homogeneous vacuum-melted alloy, 99.999% purity
Target geometry	50.8 mm diameter
Base pressure	$\leq 5.0 \times 10^{-8}$ Torr
Atomic-hydrogen pretreatment	200 W RF, 2.0 Pa, 450°C, 15 min
Ti seed on CVD diamond	0.2 nm at 150°C
Process argon	99.9999% purity; 0.25 Pa
Peak target power density	1.85 kW cm^{-2}
Pulse duration / repetition	$30 \mu\text{s} / 100 \text{ Hz}$
Synchronized substrate bias	-120 V in ion-arrival window
Net growth rate / duration	$0.025 \text{ nm s}^{-1} / 40.0 \text{ s}$
Morphology acceptance	Continuous film; $R_q < 0.15 \text{ nm}$
Thickness uniformity	Better than $\pm 2\%$ with stated sampling

continuity, pinhole assessment, adhesion, and thermal cycling on the actual substrate and witness specimens. Carbon/Pd, Ag nanoparticle, and pulsed-deposition studies provide process context [45], [49]–[51]. The Ti/TiC layer shall be represented in optical, thermal-contact, diffusion, and thickness-calibration models.

E. Cu–Sn Transient Liquid Phase Bonding

a) *Multicomponent Ginzburg–Landau free energy*: The governing joint model couples conserved alloy composition, nonconserved phase order, and elastic equilibrium. Phase thickness and residual liquid are outputs of this initial-boundary-value problem, not prescribed square-root growth laws. A diffuse-interface description separates thermodynamic driving forces from kinetic mobilities [120]. Let c_a denote independent substitutional atomic fractions, with the reference-species fraction fixed by site balance, and let $\eta_p \geq 0$, $\sum_p \eta_p = 1$, label liquid, substrate, and intermetallic phases. The selected alloy requires Cu, Sn, Ni, and Bi chemistry; a binary Cu–Sn restriction is only a comparison case. Grain orientations may be represented by additional order parameters within each solid phase.

On a small-strain reference domain $\mathcal{V}_J$ with fixed reference

site density N_s , set

$$\begin{aligned}
 \mathcal{F}_J &= \int_{\mathcal{V}_J} (f_0 + f_\nabla + f_{\text{el}}) dV, \\
 f_0 &= \sum_p h_p(\boldsymbol{\eta}) f_p(\mathbf{c}, T) W(\mathbf{c}, T) \sum_{p < q} \eta_p^2 \eta_q^2, \\
 h_p &= \frac{H(\eta_p)}{\sum_q H(\eta_q)}, \quad H(z) = z^3(10 - 15z + 6z^2), \\
 f_\nabla &= \frac{1}{2} \sum_{a,b} \kappa_{ab} \nabla c_a \cdot \nabla c_b \frac{1}{2} \sum_p \kappa_p |\nabla \eta_p|^2, \\
 f_{\text{el}} &= \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbf{C}(\mathbf{c}, \boldsymbol{\eta}, T) : \boldsymbol{\varepsilon}_e, \\
 \boldsymbol{\varepsilon}_e &= \boldsymbol{\varepsilon}(\mathbf{u}) - \boldsymbol{\varepsilon}^*(\mathbf{c}, \boldsymbol{\eta}, T) - \boldsymbol{\varepsilon}^p - \boldsymbol{\varepsilon}^{\text{cr}}, \\
 \boldsymbol{\sigma} &= \mathbf{C} : \boldsymbol{\varepsilon}_e, \quad \nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = 0.
 \end{aligned} \tag{377}$$

Here f_p, W, f_{el} are in J/m^3 , κ_{ab}, κ_p in J/m , and $\boldsymbol{\varepsilon}^*$ includes compositional, phase-transformation, and thermal eigenstrain. The concentration-gradient matrix is symmetric positive semidefinite and $\kappa_p > 0$. The displayed interpolation and barrier are coarse-graining choices, not universal atomistic identities. A liquid has no equilibrium shear modulus; liquid flow and solidification shrinkage require additional transport when a quasistatic, fixed-reference approximation is insufficient.

b) *Statistical origin of phase energies and barriers*: For N reference sites in phase p , the constrained configurational partition sum gives a free energy per site:

$$\begin{aligned}
 \mathcal{Z}_p(\mathbf{c}, T) &= \sum_{\mathbf{c} \in p, \mathbf{c}} \exp \left[-\frac{F_{\mathcal{C}}^{\text{micro}}(T)}{k_B T} \right], \\
 g_p(\mathbf{c}, T) &= -\frac{k_B T}{N} \ln \mathcal{Z}_p, \quad f_p = N_s g_p, \\
 F_{\mathcal{C}}^{\text{micro}} &= E_{\mathcal{C}}^{\text{DFT}} + F_{\mathcal{C}}^{\text{vib}} + F_{\mathcal{C}}^{\text{el, thermal}}, \\
 F_{\mathcal{C}}^{\text{vib}} &= \sum_{\nu} \left\{ \frac{\hbar \omega_{\nu}}{2} + k_B T \ln [1 - e^{-\hbar \omega_{\nu} / (k_B T)}] \right\}.
 \end{aligned} \tag{378}$$

Configurational entropy is included by the sum, not added a second time as ideal mixing. The harmonic vibrational expression applies to stable solid configurations; the liquid needs an appropriate finite-temperature free-energy calculation. Energies are evaluated at a consistent reference strain, with the continuum elastic excess included only once. Molar free energies must be divided by molar volume before use as f_p .

Interface energetics must also be matched. For a planar two-phase equilibrium interface, a scalar restriction with excess

bulk density $\Delta f(\eta)$ obeys

$$\begin{aligned}\kappa_\eta \eta'' &= \partial_\eta \Delta f, \quad \frac{1}{2} \kappa_\eta (\eta')^2 = \Delta f, \\ \gamma_{pq} &= \int_0^1 \sqrt{2\kappa_\eta \Delta f(\eta)} d\eta, \\ \gamma_{pq}^{\text{micro}} &= \frac{F_{pq}^{\text{interface}} - F_p^{\text{bulk}} - F_q^{\text{bulk}}}{A_{pq}}.\end{aligned}\quad (379)$$

The interface area A_{pq} counts all interfaces in the atomistic cell. Matching γ_{pq} and a physical interface profile determines the admissible barrier/gradient combination; γ_{pq} alone does not determine both. Artificially widening an interface requires kinetic matching and mesh convergence, with the corresponding kinetic model [121].

c) *Chemical potentials and hydrostatic coupling:* For constant gradient coefficients, functional differentiation at fixed displacement and internal strain gives exchange chemical potentials per atom and phase driving forces:

$$\begin{aligned}N_s \mu_a &= \frac{\delta \mathcal{F}_J}{\delta c_a} = \partial_{c_a} f_0 - \sum_b \kappa_{ab} \nabla^2 c_b \\ &\quad + \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbf{C}_{,c_a} : \boldsymbol{\varepsilon}_e - \boldsymbol{\sigma} : \boldsymbol{\varepsilon}_{,c_a}^*, \\ \xi_p &= \frac{\delta \mathcal{F}_J}{\delta \eta_p} = \partial_{\eta_p} f_0 - \kappa_p \nabla^2 \eta_p \\ &\quad + \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbf{C}_{,\eta_p} : \boldsymbol{\varepsilon}_e - \boldsymbol{\sigma} : \boldsymbol{\varepsilon}_{,\eta_p}^*.\end{aligned}\quad (380)$$

The derivatives include the concentration dependence of W and all interpolated phase energies. If gradient coefficients depend on state, their local derivatives and the full divergence terms must also be differentiated. For isotropic compositional expansion, define the exchange relaxation volume $\Omega_a^* = \text{tr}(\boldsymbol{\varepsilon}_{,c_a}^*)/N_s$. With tension-positive $\sigma_m = \text{tr} \boldsymbol{\sigma}/3$,

$$\begin{aligned}\mu_a^{\text{stress}} &= -\sigma_m \Omega_a^*, \\ \nabla \mu_a^{\text{stress}} &= -\Omega_a^* \nabla \sigma_m - \sigma_m \nabla \Omega_a^*.\end{aligned}\quad (381)$$

This sign follows from variation of elastic energy rather than an imposed drift coefficient. The stiffness-derivative term and anisotropic eigenstrain contribution remain in Eq. (380); they cannot generally be replaced by a hydrostatic scalar alone.

d) *Conserved diffusion and phase kinetics:* Let $\mathbf{L}_{ab}$ be a symmetric positive-semidefinite Onsager mobility in atom-count flux units. The multicomponent Cahn–Hilliard and constrained Allen–Cahn equations are

$$\begin{aligned}\mathbf{J}_a &= -\sum_b \mathbf{L}_{ab} \nabla \mu_b, \quad N_s \partial_t c_a = -\nabla \cdot \mathbf{J}_a, \\ \partial_t \eta &= -\mathbf{L}_\eta (\boldsymbol{\xi} + \lambda \mathbf{1} - \boldsymbol{\zeta}), \\ \eta_p &\geq 0, \quad \zeta_p \geq 0, \quad \eta_p \zeta_p = 0, \quad \sum_p \eta_p = 1.\end{aligned}\quad (382)$$

The multiplier λ enforces the phase sum, and $\boldsymbol{\zeta}$ enforces nonnegativity. The kinetic operator acts on admissible phase variations. A dilute, uncorrelated binary restriction has $\mathbf{L} = N_s Dc(1-c)/(k_B T)$; equivalently, with a volumetric variational derivative, $M_c = Dc(1-c)/(N_s k_B T)$. This site-density factor is required dimensionally. Concentrated ordered intermetallics require sublattice occupancies and correlated multicomponent mobilities, not substitution of a tracer diffusivity for every Onsager entry.

For an impermeable, neutral external boundary, use $\mathbf{n} \cdot \mathbf{J}_a = 0$, $\mathbf{n} \cdot \sum_b \kappa_{ab} \nabla c_b = 0$, and $\mathbf{n} \cdot \kappa_p \nabla \eta_p = 0$. Initial composition and phase fields specify finite Cu/Sn inventory, alloy additions, and any nuclei; wetting or open reservoirs replace these conditions by surface-energy or chemical-potential conditions. At fixed temperature, fixed mechanical boundaries, and frozen plastic and creep variables, integration by parts proves

$$\begin{aligned}\frac{d\mathcal{F}_J}{dt} &= -\int_{\mathcal{V}_J} \sum_{a,b} \nabla \mu_a \cdot \mathbf{L}_{ab} \nabla \mu_b dV \\ &\quad - \int_{\mathcal{V}_J} \boldsymbol{\xi}_c^T \mathbf{L}_\eta \boldsymbol{\xi}_c dV \leq 0, \\ \boldsymbol{\xi}_c &= \boldsymbol{\xi} + \lambda \mathbf{1} - \boldsymbol{\zeta}, \quad \frac{d}{dt} \int_{\mathcal{V}_J} N_s c_a dV = 0.\end{aligned}\quad (383)$$

Thermal ramps, loading work, and vacancy exchange add their respective energy and species terms; the isothermal inequality alone is not a full reactor heat balance. Equivalent phase thickness and residual liquid fraction are evaluated from the solution:

$$W_p(t) = \frac{1}{A_J} \int_{\mathcal{V}_J} h_p dV, \quad f_l(t) = \frac{1}{|\mathcal{V}_J|} \int_{\mathcal{V}_J} h_l dV. \quad (384)$$

Complete isothermal solidification requires a specified residual-liquid threshold and absence of connected liquid channels. A finite-reservoir phase-field calculation can approach a diffusion-limited square-root asymptote, but that asymptote is not prescribed as the constitutive law.

e) *Composition-dependent transition-state activation:*

For a local atomic environment $\mathcal{C}$ and event r , define initial and saddle free energies with identical species reservoirs and thermodynamic constraints. Transition-state theory gives

$$\begin{aligned}\Delta g_r^\ddagger(\mathcal{C}, T, \boldsymbol{\sigma}) &= G_r^\ddagger - G_r^{\text{initial}}, \\ k_r(\mathcal{C}) &= \kappa_r^{\text{tr}} \frac{k_B T}{h} e^{-\Delta g_r^\ddagger/(k_B T)}, \\ \frac{k_{i \rightarrow j}}{k_{j \rightarrow i}} &= e^{-(G_j - G_i)/(k_B T)}.\end{aligned}\quad (385)$$

Here κ_r^{tr} is the transmission coefficient, equal to unity only in the no-recrossing approximation; the unstable saddle mode is excluded from its partition function. A consistent common saddle and microscopic reversibility give detailed balance. Atomic barriers require minimum-energy paths and saddle vibrational free energies, with local environment sampling [122]. Their stress dependence follows from the activation-volume tensor:

$$\begin{aligned}\Omega_r^\ddagger &= -\frac{\partial \Delta g_r^\ddagger}{\partial \boldsymbol{\sigma}}, \\ \Delta g_r^\ddagger(\boldsymbol{\sigma}) &= \Delta g_r^\ddagger(0) - \int_0^1 \Omega_r^\ddagger(s\boldsymbol{\sigma}) : \boldsymbol{\sigma} ds, \\ \Delta g_r^\ddagger(\sigma_m) &= \Delta g_r^\ddagger(0) - \sigma_m \Delta \Omega_r^\ddagger \quad \text{for constant } \Delta \Omega_r^\ddagger.\end{aligned}\quad (386)$$

The stress ensemble is consistent with the fixed-strain elastic variation by the corresponding mechanical Legendre transform. Stress work is counted once, either through the resolved defect free energy or its activation-volume representation.

Composition enters through both the barriers and the probability $w_C(\mathbf{c}, \boldsymbol{\eta}, T)$ of each local environment. For a common attempt frequency, a rate-equivalent activation free energy is

$$\begin{aligned}\Delta g_{\text{eff}}^\ddagger &= -k_B T \ln \sum_C w_C e^{-\Delta g_C^\ddagger / (k_B T)}, \\ \pi_C &= \frac{w_C e^{-\Delta g_C^\ddagger / (k_B T)}}{\sum_{\mathcal{D}} w_{\mathcal{D}} e^{-\Delta g_{\mathcal{D}}^\ddagger / (k_B T)}}, \\ \partial_{c_a} \Delta g_{\text{eff}}^\ddagger &= \sum_C \pi_C \left(\partial_{c_a} \Delta g_C^\ddagger - k_B T \partial_{c_a} \ln w_C \right).\end{aligned}\quad (387)$$

Environment-dependent prefactors must remain inside the rate average. In the isotropic, uncorrelated random-walk limit, vacancy hopping gives $D_v = \frac{1}{6} \sum_r \langle k_r \rangle |\mathbf{a}_r|^2$, where $\mathbf{a}_r$ is the jump vector; correlations require the long-time mean-square-displacement or Onsager calculation. Interface mobility follows separately by linearizing the difference of forward/backward transformation rates at small free-energy driving force.

More generally, the equilibrium long-time displacement covariance determines the alloy Onsager operator. With $\Delta \mathbf{R}_a$ the summed atomic displacement of species a in the lattice frame, projected onto independent exchange coordinates,

$$\mathbf{L}_{ab} = \lim_{\tau \rightarrow \infty} \frac{\langle \Delta \mathbf{R}_a(\tau) \otimes \Delta \mathbf{R}_b(\tau) \rangle}{2|\mathcal{V}_J| k_B T \tau}. \quad (388)$$

The equilibrium covariance includes correlations absent from a single-hop Einstein estimate. In a binary phase-conversion restriction, an event changes transformed volume by v_* , and $\mathcal{R}_0$ is the equilibrium event frequency per unit volume. Symmetric forward/backward rates with affinity $-v_* \xi$ give

$$\begin{aligned}\dot{\eta} &= -2v_* \mathcal{R}_0 \sinh\left(\frac{v_* \xi}{2k_B T}\right), \\ \dot{\eta} &\simeq -L_\eta \xi, \quad L_\eta = \frac{v_*^2 \mathcal{R}_0}{k_B T}.\end{aligned}\quad (389)$$

Here ξ is the constrained phase-conversion free-energy density difference. The positive mobility has units $\text{m}^3/(\text{J s})$, and $\mathcal{R}_0(\mathbf{c}, \boldsymbol{\eta}, T)$ is computed from the interface event barriers and environment weights, not identified with a bulk tracer prefactor. The linear Allen–Cahn law applies to the small-affinity limit of this local kinetic model.

The parameterization uses material-specific electronic energies, saddle paths, finite-temperature corrections, interface energies, and defect populations. Continuum interpolation, the harmonic approximation, and kinetic coarse graining define its constitutive assumptions.

The selected interconnect uses a final $3.5 \mu\text{m}$ Ni-6Cu/Sn-0.7Cu-0.05Bi bondline and a $15.0 \mu\text{m}$ graded Ni/Cu compliance layer, with the 52,400-cycle strain-life prediction in Eq. (276). The unalloyed $5.0 \mu\text{m}$ Cu–Sn simulation and the 15–20 μm preform/reflow cases are separate comparisons; each uses its own shear-strength, transport, and phase-growth values. Final IMC thickness, phase composition, diffusivity, and interfacial toughness require configuration-specific qualification.

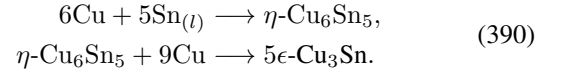
Cu–Sn transient liquid phase (TLP) bonding forms an $\epsilon\text{-Cu}_3\text{Sn}$ intermetallic compound (IMC) rather than retaining a low-melting solder joint. The comparison preform consists

TABLE XXXIII
TLP COMPARISON-BONDLINEREFERENCE PROPERTIES

Property	Reference value
Final phase	$\epsilon\text{-Cu}_3\text{Sn}$
Phase melting reference	676.0°C
Interfacial shear strength	58.50 MPa
Void fraction	0.32%
Young's modulus	98.35 GPa
CTE	$18.40 \times 10^{-6} \text{ K}^{-1}$
Thermal conductivity	$148.0 \text{ W m}^{-1} \text{ K}^{-1}$
Final IMC thickness in stress model	$20.0 \mu\text{m}$

of a $10 \mu\text{m}$ Sn layer between two $15 \mu\text{m}$ Cu foils, or a qualified Cu@Sn composite. The nominal process uses $260\text{--}280^\circ\text{C}$, 0.50 MPa applied pressure, and a $15\text{--}20 \text{ min}$ dwell at 260°C . The vacuum/formic-acid process specifies pressure below 10^{-3} mbar ; the manufacturing traveler shall define evacuation, reducing-gas exposure, dwell, and cooling as separately controlled stages.

The phase sequence is



Representative joints shall demonstrate complete consumption of free Sn, the required phase composition, and metallization adhesion. The final $20.0 \mu\text{m}$ IMC layer is a stress-model input and shall not be equated to the $40 \mu\text{m}$ initial foil stack without dimensional and material-balance verification.

At the comparison TEG hot interface,

$$T_h = 550.0 \text{ K} < T_{\text{melt}} = 949.15 \text{ K}, \quad \Delta T_{\text{melt}} = 399.15 \text{ K}. \quad (391)$$

The 260°C mechanical cycle lies below the 276.85°C hot-side endpoint; joint properties and phase stability require verification at the actual operating temperature. Inspection records shall define void fraction, coverage and resolution, residual Sn, IMC phase fractions, interfacial cracking, and shear-test temperature. The nominal 0.32% void fraction and 58.50 MPa strength require representative joint measurements before production acceptance.

f) Grain geometry and vacancy thermodynamics: Grain coarsening is obtained from the phase/orientation fields and their interfacial energies. For resolved grain volumes $V_g(t)$, one explicitly defined size statistic is

$$d_g(t) = \left[\frac{6V_g(t)}{\pi} \right]^{1/3}, \quad \bar{d}_V(t) = \frac{\sum_g V_g(t) d_g(t)}{\sum_g V_g(t)}. \quad (392)$$

This volume-weighted equivalent diameter is not an imposed grain-growth law. Diffusion paths and dislocation sink distances are evaluated in the evolving geometry; a grain diameter does not automatically equal the dislocation capture radius.

Let $X_v = N_v/N_s$ be the vacancy site fraction and $\Omega_a = 1/N_s$ the reference atomic volume. The dilute-defect treatment uses the same local phase/composition state as the joint free

energy. For a vacancy formed by transferring a species- a atom to a reservoir,

$$\begin{aligned}\Delta g_{v,a}^f &= G(N_a - 1, N_v + 1) - G(N_a, N_v) + \mu_a^{\text{res}}, \\ \Omega_{v,a}^f &= -\partial_{\sigma} \Delta g_{v,a}^f, \\ \Delta g_v^f(\sigma_m) &= \Delta g_v^f(0) - \sigma_m \Delta \Omega_v^f \quad \text{for constant } \Delta \Omega_v^f, \\ f_v/N_s &= X_v \Delta g_v^f + k_B T X_v \ln X_v \\ &\quad + k_B T (1 - X_v) \ln(1 - X_v), \\ \mu_v &= \frac{1}{N_s} \partial_{X_v} f_v = \Delta g_v^f + k_B T \ln \frac{X_v}{1 - X_v}, \\ X_v^{\text{eq}} &= \left[1 + e^{\Delta g_v^f/(k_B T)} \right]^{-1} \simeq e^{-\Delta g_v^f/(k_B T)}.\end{aligned}\quad (393)$$

The reservoir fixes the zero of vacancy exchange potential. In an ordered multicomponent crystal, separate sublattice vacancy and antisite free energies replace the single dilute fraction. Formation energy and migration barrier are different thermodynamic quantities. The displayed stress term is the reduced representation of defect elastic work already included in a fully resolved elastic free energy; it must not be added twice. The formation volume is not necessarily Ω_a and its sign must follow the defect calculation.

For coupled evolution, the chemical defect excess is added to f_0 , the vacancy relaxation is included in ϵ^* , and the dislocation line energy is added to $\mathcal{F}_J$. All composition and phase derivatives in Eq. (380) are then taken on this augmented functional. In particular, the chemical defect terms contribute

$$\delta \mu_a^v = X_v \partial_{c_a} \Delta g_v^{f,\text{chem}}, \quad \delta \xi_p^v = N_s X_v \partial_{\eta_p} \Delta g_v^{f,\text{chem}}. \quad (394)$$

Elastic vacancy contributions follow from the same eigenstrain derivative, while line-energy variations enter their respective chemical and mechanical forces. Evaluating vacancy rates without these reciprocal couplings is a one-way approximation, not the fully coupled thermodynamic model.

The isothermal vacancy flux and species balance are

$$\begin{aligned}\mathbf{J}_v &= -\frac{N_s D_v X_v (1 - X_v)}{k_B T} \nabla \mu_v, \\ \partial_t (N_s X_v) &= -\nabla \cdot \mathbf{J}_v + s_v, \\ \nabla \Delta g_v^f &= \sum_a \partial_{c_a} \Delta g_v^f \nabla c_a + \sum_p \partial_{\eta_p} \Delta g_v^f \nabla \eta_p \\ &\quad + \partial_{\sigma_m} \Delta g_v^f \nabla \sigma_m, \\ \mathbf{J}_v &\simeq -N_s D_v \left[\nabla X_v + \frac{X_v}{k_B T} \nabla \Delta g_v^f \right].\end{aligned}\quad (395)$$

For constant $\Delta \Omega_v^f$, the hydrostatic part of the dilute flux is

$$\mathbf{J}_v^{\text{stress}} = \frac{N_s D_v X_v \Delta \Omega_v^f}{k_B T} \nabla \sigma_m. \quad (396)$$

Composition and phase gradients contribute separately. If the formation volume varies, the term $-\sigma_m \nabla \Delta \Omega_v^f$ is also retained in $\nabla \Delta g_v^f$. Nonuniform temperature additionally requires the heat/defect Onsager cross-coupling and the force $\nabla(\mu_v/T)$; no Soret coefficient follows from a static activation energy alone. The source s_v includes explicitly balanced exchange with dislocations, interfaces, and surfaces. Alloy and vacancy

fluxes obey lattice-site conservation; the dilute vacancy approximation neglects only higher-order vacancy corrections, not the atom/vacancy counterflux responsible for Kirkendall transport.

g) Weertman vacancy-diffusion climb derivation: The rate-controlling climb mechanism is determined by vacancy exchange with dislocation cores [123], [124]. Consider a locally straight edge segment with Burgers edge component $b_e > 0$, core radius r_c , and capture radius $R_e > r_c$. Choose positive climb as vacancy absorption. Advancing a line length ℓ by ds absorbs $dN_v = b_e \ell ds / \Omega_a$. If f_c is the thermodynamic line force per unit length in that direction, core equilibrium follows from

$$\begin{aligned}dG_{\text{exchange}} &= -\mu_v^{\text{core}} dN_v - f_c \ell ds = 0, \\ \mu_v^{\text{core}} &= -f_c \Omega_a / b_e, \\ X_v^{\text{core}} &\simeq X_v^{\text{eq}} \exp \left[-\frac{f_c \Omega_a}{b_e k_B T} \right], \\ f_c &= [(\boldsymbol{\sigma} \mathbf{b}) \times \mathbf{t} + \mathbf{f}_{\text{line}}] \cdot \mathbf{n}_c.\end{aligned}\quad (397)$$

Here $\mathbf{t}$ is the unit line tangent, $\mathbf{n}_c$ the absorption-oriented climb direction, and $\mathbf{f}_{\text{line}}$ includes the configurational force from line energy. The minus sign in the core exponential follows this absorption convention; reversing climb direction reverses f_c and velocity together. The Peach–Koehler force depends on the resolved tensor stress, not only on σ_m . Hydrostatic stress modifies defect formation and line work through their respective conjugate volumes, which must be distinguished.

For constant D_v , composition, temperature, and formation energy in the capture cylinder, quasisteady vacancy diffusion gives

$$\begin{aligned}\frac{1}{r} \frac{d}{dr} \left(r \frac{dX_v}{dr} \right) &= 0, \\ X_v(r) &= X_v^{\text{core}} + (X_v^\infty - X_v^{\text{core}}) \frac{\ln(r/r_c)}{\ln(R_e/r_c)}, \\ I_v &= \frac{2\pi N_s D_v}{\ln(R_e/r_c)} (X_v^\infty - X_v^{\text{core}}), \\ v_c &= \frac{\Omega_a I_v}{b_e} = \frac{2\pi D_v}{b_e \ln(R_e/r_c)} (X_v^\infty - X_v^{\text{core}}).\end{aligned}\quad (398)$$

The inward current I_v is vacancies per unit line length per second; v_c has units m/s. If concentrations are number densities $C_v = N_s X_v$, the velocity prefactor is instead $2\pi D_v \Omega_a / [b_e \ln(R_e/r_c)]$.

For radial stress/composition variation, set $a_v = X_v / X_v^{\text{eq}}(r)$ in the dilute limit. Integrating the chemical-potential flux, rather than assuming a logarithmic concentration, gives

$$v_c = \frac{2\pi \Omega_a N_s}{b_e} \frac{a_v(R_e) - a_v(r_c)}{\int_{r_c}^{R_e} \frac{dr}{r D_v(r) X_v^{\text{eq}}(r)}}. \quad (399)$$

Nonradial fields require the full vacancy boundary-value problem. At $X_v^\infty = X_v^{\text{eq}}$ and weak line driving,

$$\begin{aligned}v_c &= \frac{2\pi D_v X_v^{\text{eq}}}{b_e \ln(R_e/r_c)} \left[1 - e^{-f_c \Omega_a / (b_e k_B T)} \right], \\ v_c &\simeq \frac{2\pi D_v X_v^{\text{eq}} \Omega_a}{b_e^2 k_B T \ln(R_e/r_c)} f_c, \\ D_{\text{self}} &= f_{\text{corr}} D_v X_v^{\text{eq}}.\end{aligned}\quad (400)$$

The correlation factor f_{corr} and the event network determine self-diffusion; D_v here is vacancy migration diffusivity, not deuterium diffusivity. In a single-path harmonic limit the self-diffusion activation enthalpy contains both vacancy formation and migration enthalpies. For multiple environments the rate average in Eq. (387) replaces a constant activation enthalpy. Uniform hydrostatic stress changes equilibrium vacancy population but produces no bulk drift when its gradient vanishes.

h) Macroscopic creep and energy consistency: For segment families s with density ρ_s in m^{-2} and velocity $\mathbf{v}_s$, the swept-area identity gives

$$\begin{aligned}\dot{\boldsymbol{\beta}}^{\text{cr}} &= \sum_s \rho_s \mathbf{b}_s \otimes (\mathbf{t}_s \times \mathbf{v}_s), \\ \dot{\boldsymbol{\epsilon}}^{\text{cr}} &= \text{sym} \dot{\boldsymbol{\beta}}^{\text{cr}}, \\ \boldsymbol{\sigma} : \dot{\boldsymbol{\epsilon}}^{\text{cr}} &= \sum_s \rho_s [(\boldsymbol{\sigma} \mathbf{b}_s) \times \mathbf{t}_s] \cdot \mathbf{v}_s.\end{aligned}\quad (401)$$

For pure climb, $\mathbf{v}_s = v_{c,s} \mathbf{n}_{c,s}$; the magnitude of the single-family swept strain coordinate is $\rho_s b_s v_{c,s}$ with its orientation factor. Weertman climb-assisted glide additionally requires glide motion, obstacle geometry, and climb waiting times; identifying every macroscopic strain component with $\rho_d b v_c$ is not a general polycrystal closure. Dislocation density and capture distances are evolving microstructural variables, not prescribed powers of equivalent stress. Bulk and grain-boundary diffusional creep can be represented by vacancy fluxes and moving sink/source boundaries; adding independent fitted rates to the same vacancy current would double count transport.

Let $\mathcal{A}_v = \mu_v^\infty + f_c \Omega_a / b_e$ be the absorption affinity. The current has the sign of $\mathcal{A}_v$, so $I_v \mathcal{A}_v \geq 0$; bulk diffusion also gives $-\mathbf{J}_v \cdot \nabla \mu_v \geq 0$. Including line-energy changes and chemical reservoir work therefore gives nonnegative dissipation. Chemical supersaturation may drive climb against mechanical stress, so mechanical power alone need not be nonnegative away from defect equilibrium.

The evolving joint stress satisfies

$$\dot{\boldsymbol{\sigma}} = \dot{\mathbf{C}} : \boldsymbol{\epsilon}_e + \mathbf{C} : (\dot{\boldsymbol{\epsilon}} - \dot{\boldsymbol{\epsilon}}^* - \dot{\boldsymbol{\epsilon}}^p - \dot{\boldsymbol{\epsilon}}^{\text{cr}}). \quad (402)$$

Only a fixed-temperature, fixed-composition scalar limit with constant effective modulus, fixed total strain, and no other inelastic increments reduces this to $\dot{\boldsymbol{\sigma}} = -E \dot{\boldsymbol{\epsilon}}_{\text{cr}}$. Its integrated identity is

$$\boldsymbol{\epsilon}_{\text{cr}}(t) - \boldsymbol{\epsilon}_{\text{cr}}(0) = \frac{\boldsymbol{\sigma}(0) - \boldsymbol{\sigma}(t)}{E}. \quad (403)$$

It supplies no universal life or monotonic-stress bound when phase transformation, vacancy supersaturation, external loading, or thermal cycling changes the driving force. The 0.50% project strain allowance and the graded-joint strain-life specification remain acceptance criteria; compliance requires integration of the coupled history, not extrapolation of a fixed fitted creep rate.

F. Multilayer Thermoelastic FEA Definition

The structural framework resolves the full three-dimensional Pd–Ir/Ti/CVD-diamond/Cu–Sn/OFHC–Cu configuration [66]. Prandtl–Reuss plastic flow, thermal strain, creep, and crack-closure energy are coupled. Seating sums and Suhir reductions

provide limited comparison models rather than the controlling structural-safety assessment. The microscopic interface specifies a 1.0 nm catalyst film, distinct from a 1.0 μm comparison thickness. The FEA geometry shall use the specified 1.0 nm film. The total thickness is 1.055011 mm. Freely expanding layers with constant CTEs give 4.584027 μm axial expansion for a uniform 256.85 K rise, distinct from the chassis-seating sum. Constrained displacement requires the 3D solution. Endpoint properties do not define complete temperature-dependent constitutive functions. Diamond tensile strength is a brittle-fracture reference rather than a metallic yield stress. For an associative isotropically hardening metal, let $s_{ij} = \sigma_{ij} - \sigma_{kk} \delta_{ij} / 3$ and $\bar{\sigma} = \sqrt{3 s_{ij} s_{ij} / 2}$. The constitutive equations are

$$\begin{aligned}\dot{\epsilon}_{ij} &= \dot{\epsilon}_{ij}^e + \dot{\epsilon}_{ij}^p + \dot{\epsilon}_{ij}^{th} + \dot{\epsilon}_{ij}^c, \\ \dot{\epsilon}_{ij}^p &= \frac{3 \dot{\epsilon}_p}{2 \bar{\sigma}} s_{ij}, \\ f &= \bar{\sigma} - [\sigma_y(T) + R(\bar{\epsilon}_p, T)] \leq 0, \\ \dot{\epsilon}_p &\geq 0, \quad f \dot{\epsilon}_p = 0.\end{aligned}\quad (404)$$

R is the independently specified isotropic hardening law. It describes rate-independent plasticity, not vacancy-controlled creep. The creep strain follows the resolved dislocation velocities in Eq. (401); a scalar accumulated measure is computed afterward:

$$\begin{aligned}\dot{\epsilon}_{\text{dev}}^c &= \dot{\epsilon}^{\text{cr}} - \frac{1}{3} \text{tr}(\dot{\epsilon}^{\text{cr}}) \mathbf{I}, \\ \dot{\epsilon}_c &= \sqrt{\frac{2}{3} \dot{\epsilon}_{\text{dev}}^c : \dot{\epsilon}_{\text{dev}}^c}, \\ \epsilon_{\text{acc}}^c &= \int_0^{t_f} \dot{\epsilon}_c \, dt, \quad \epsilon_{\text{vol}}^c = \text{tr}(\dot{\epsilon}^{\text{cr}}).\end{aligned}\quad (405)$$

Climb can produce volumetric strain, which is retained separately rather than suppressed by an assumed J_2 creep direction. Table XXXV defines the required quantities, not calibrated values for the alloy. Energy per atomic event uses $k_B T$; molar activation energy uses RT . Vacancy formation and migration contributions enter their own partition factors exactly once. Creep and rate-independent plasticity must be assigned distinct mechanisms or a unified dislocation evolution model so the same swept strain is not counted twice.

For a crack-front segment of width b_f and advance Δa , the nominal local two-mode VCCT calculation is

$$\begin{aligned}G_{\text{VCCT}} &= \frac{F_x \Delta u + F_z \Delta w}{2 b_f \Delta a}, \\ \sigma_{xz, \text{max}} &= 142.5 \text{ MPa}, \\ G_{\text{VCCT}} &= 18.4 \text{ J m}^{-2} < G_C = 24.5 \text{ J m}^{-2}, \\ G_C / G_{\text{VCCT}} &= 1.331522.\end{aligned}\quad (406)$$

The forces and relative displacements are work-conjugate quantities at and behind the crack tip; a general 3D crack front also includes Mode III work $F_y \Delta v$. The quantity denoted by G_{lag} in this calculation is a VCCT release rate, distinct from the shear-lag energy expression $\omega \tau^2 / 2$. A 142.5 MPa shear stress can exceed elastic shear yield and requires elastoplastic treatment.

TABLE XXXIV
COMPARISON-STACK LAYER PROPERTIES AT 550 K (50 μ M DIAMOND)

Layer	Thickness	E (GPa)	ν	$10^6 \alpha$ (K^{-1})	Strength (MPa)	k (W/m K)
Pd _{0.95} Ir _{0.05}	1.0 nm	124.0	0.39	11.8	150.0 yield	72.0
Ti transition layer	10.0 nm	105.0	0.32	8.6	220.0 yield	21.9
CVD diamond	50.0 μ m	1050.0	0.07	1.1	2000.0 tensile	1800.0
ϵ -Cu ₃ Sn TLP	5.0 μ m	110.0	0.33	18.4	180.0 yield	34.0
OFHC copper	1.0 mm	110.0	0.34	17.7	65.0 yield	385.0

TABLE XXXV
JOINT VACANCY AND CLIMB CONSTITUTIVE QUANTITIES

Quantity	Definition or units
Vacancy migration diffusivity D_v	m^2/s
Vacancy formation free energy Δg_v^f	J/defect
Migration barrier $\Delta g_m^\ddagger$	J/event
Formation/activation volume	$m^3/defect$
Vacancy site fraction X_v	Dimensionless
Edge Burgers component b_e	m
Core/capture radii r_c, R_e	m; $R_e > r_c$
Line density ρ_s	m^{-2}

TABLE XXXVI
RING/BASE STRUCTURAL-MATERIAL INPUTS AT 550.0 K

Property	SS316L	Invar 36
E (GPa)	172.0	128.0
ν	0.30	0.29
α ($10^{-6} K^{-1}$)	16.5	2.8
σ_y (MPa)	145.0	220.0
k ($W m^{-1} K^{-1}$)	18.2	12.8
D_H ($m^2 s^{-1}$)	4.2×10^{-11}	1.1×10^{-12}
S_H ($mol m^{-3} Pa^{-1/2}$)	1.8	3.5

For comparison, the previously specified five-year, 43800 h strain value and ductility criterion are

$$\begin{aligned}\epsilon_{acc}^c &= 0.0142 = 1.42\%, \\ \epsilon_f &= 0.045 = 4.5\% > \epsilon_{acc}^c, \\ \epsilon_f / \epsilon_{acc}^c &= 3.169014.\end{aligned}\quad (407)$$

The comparison assigns 1.42% accumulated strain over 44,820 loading cycles. The material ductility criterion is 4.5%, and the project strain allowance is 0.50%. Acceptance applies each limit to its defined material and loading history; the coupled model uses composition, phase, vacancy, dislocation, and displacement histories.

For layer j , the coupled heat and quasistatic mechanical equations are

$$\begin{aligned}\rho_j c_{p,j} \frac{\partial T_j}{\partial t} &= \nabla \cdot (k_j \nabla T_j) + q_j, \\ \nabla \cdot \boldsymbol{\sigma}_j + \mathbf{f}_j &= \mathbf{0}, \\ \boldsymbol{\epsilon}_j &= \frac{1}{2} (\nabla \mathbf{u}_j + \nabla \mathbf{u}_j^T), \\ \boldsymbol{\sigma}_j &= \mathbb{C}_j(T_j) : [\boldsymbol{\epsilon}_j - \boldsymbol{\epsilon}_j^{th} - \boldsymbol{\epsilon}_j^p - \boldsymbol{\epsilon}_j^{cr}], \\ \boldsymbol{\epsilon}_j^{th} &= \left[\int_{T_{sf,j}}^{T_j} \alpha_j(T) dT \right] \mathbf{I}.\end{aligned}\quad (408)$$

Here $T_{sf,j}$ is the layer's stress-free reference temperature, $\mathbb{C}_j$ is its elastic tensor, and the superscripts p and cr denote plastic and creep strains. The benchmark shall use its stated stress-free reference. Production analysis shall establish the residual-stress state from the deposition, TLP reaction, cooling, and assembly history.

At a perfectly bonded solid interface, displacements and tractions are continuous. With thermal contact resistance R_{th}'' ,

the common normal heat flux obeys

$$q_n = \frac{T_j - T_{j+1}}{R_{th}''}, \quad \mathbf{u}_j = \mathbf{u}_{j+1}, \quad \boldsymbol{\sigma}_j \mathbf{n} = \boldsymbol{\sigma}_{j+1} \mathbf{n}. \quad (409)$$

Delamination analysis shall replace perfect displacement continuity with a calibrated cohesive or fracture model. The mesh shall represent the thin layers through appropriate shell, interface, or locally resolved elements. Mesh refinement shall demonstrate convergence of reaction forces, displacement, thermal flux, and a defined interfacial-stress or energy-release measure. Failure assessment shall use a convergent quantity rather than a singular corner nodal maximum.

The FEA input package shall additionally specify densities, heat capacities, heat-source partition, contact conductances, cooling boundary conditions, and phase-appropriate constitutive data. The analysis deliverable shall include solver settings, mesh-convergence results, energy and force balance checks, and comparison with the reduced shear benchmark where its assumptions apply. Production stress and delamination margins shall be supported by the completed analysis and representative joint measurements.

a) *Ring/base structural comparison at 550 K:* Table XXXVI defines the material inputs for an SS316L constraint-ring/Invar-base benchmark [58]. This configuration differs from the active-film, diamond, TLP, copper-spreader, and coating stack. Its five layers comprise ceramic insulator, Invar base, Pd-Ir active film, SS316L ring, and gold cap; the two-layer reduction isolates the ring/base interface. Use of these inputs as qualified properties requires traceable material heats, isotope conditions, test temperatures, and specimen records.

The solubility coefficient is defined by $c_H = S_H \sqrt{p_H}$, fixing the pressure exponent as $-1/2$ in its units. Hydrogen transport inputs are not automatically deuterium transport data; isotope-specific conversion or testing is required. Using the tabulated

TABLE XXXVII
3D FEA MESH SEQUENCE AND SUCCESSIVE ENERGY CHANGE

Mesh	Elements	τ (MPa)	G (J/m ²)	Change (%)
M1	12450	215.4	1012.8	baseline
M2	48600	238.1	1238.4	18.2171
M3	194400	245.2	1312.8	5.6673
M4	777600	246.1	1322.5	0.7335
M5	3110400	246.4	1325.1	0.1962

CTEs as effective constants over 298.15–550.0 K gives

$$\begin{aligned}\Delta\epsilon_{\text{th}} &= (16.5 - 2.8) \times 10^{-6} (251.85) \\ &= 3.450345 \times 10^{-3}, \\ G_{\text{SS}} &= 172.0/2.6 = 66.153846 \text{ GPa}, \\ G_{\text{Inv}} &= 128.0/2.58 = 49.612403 \text{ GPa}.\end{aligned}\quad (410)$$

If the CTEs are instantaneous 550 K values, the first line is an approximation; a temperature-dependent calculation instead integrates their difference over the temperature excursion. For $t_{\text{SS}} = 2.0$ mm, $t_{\text{Inv}} = 5.0$ mm, and half-length $L = 10.0$ mm,

$$\begin{aligned}\omega &= \frac{t_{\text{SS}}}{3G_{\text{SS}}} + \frac{t_{\text{Inv}}}{3G_{\text{Inv}}} \\ &= 4.36712694 \times 10^{-14} \text{ m Pa}^{-1}, \\ \beta &= \left[\frac{1}{\omega} \left(\frac{1}{E_{\text{SS}} t_{\text{SS}}} + \frac{1}{E_{\text{Inv}} t_{\text{Inv}}} \right) \right]^{1/2} \\ &= 319.911928 \text{ m}^{-1}.\end{aligned}\quad (411)$$

For the symmetric free-ended shear-lag idealization, choose the interface traction sign so that

$$\begin{aligned}\tau(x) &= \frac{\Delta\epsilon_{\text{th}}}{\omega\beta} \frac{\sinh(\beta x)}{\cosh(\beta L)}, \\ \tau_{\text{max}} &= \frac{\Delta\epsilon_{\text{th}}}{\omega\beta} \tanh(\beta L) = 246.144716 \text{ MPa}.\end{aligned}\quad (412)$$

The odd traction field vanishes at the center and has equal, opposite edge tractions.

The shear-lag energy surrogate and associated margin are

$$\begin{aligned}G_{\text{lag}} &= \frac{1}{2} \omega \tau_{\text{max}}^2 = 1322.960432 \text{ J m}^{-2}, \\ G_C &= 1850 \text{ J m}^{-2}, \\ \text{SF}_{\text{lag}} &= G_C / G_{\text{lag}} = 1.398379.\end{aligned}\quad (413)$$

The model satisfies $G_{\text{lag}} < G_C$. The quantity $\frac{1}{2} \omega \tau^2$ is interfacial elastic energy per unit area; crack-advance energy release uses a defined crack geometry and fracture calculation. The shear-only von Mises contribution is $\sqrt{3} \tau_{\text{max}} \simeq 426.3$ MPa. Structural qualification includes elastoplastic redistribution and measured mixed-mode toughness for the actual interface. Table XXXVII lists mesh size, nominal stress, and energy output, with successive energy change $100|G_j - G_{j-1}|/|G_j|$. The final change is below 0.25%. Each structural and creep model uses its specified geometry and material state.

b) *Voce hardening and mode-resolved crack closure:*

The Voce formulation defines total flow stress as the sum of

TABLE XXXVIII
ATOMISTIC INPUTS FOR JOINT PHASE AND DEFECT KINETICS

Input	Required calculation
Phase free energies	Configurational and thermal partition sums
Elastic and chemical eigenstrain	Strain derivatives and relaxed cell volumes
Vacancy formation	Defective/perfect cells at common reservoirs
Migration and interface barriers	Saddle paths and transition-state partition sums
Interface energy/profile	Relaxed two-phase cells; excess energy per area
Mobilities and correlations	Event network or long-time flux correlations
Dislocation line energy	Core structure, character, and surrounding stress

initial yield stress and isotropic hardening [63]. Consistent with Eq. (404),

$$\begin{aligned}\sigma_{\text{flow}}(\bar{\epsilon}_p, T) &= \sigma_{y0}(T) + R_h(\bar{\epsilon}_p, T), \\ R_h &= R_\infty(T)[1 - e^{-b(T)\bar{\epsilon}_p}], \\ f &= \bar{\sigma} - \sigma_{\text{flow}} \leq 0, \\ d\epsilon_{ij}^p &= \frac{3s_{ij}}{2\bar{\sigma}} d\bar{\epsilon}_p, \\ \bar{\epsilon}_p(t) &= \int_0^t \sqrt{\frac{2}{3} \dot{\epsilon}_{ij}^p \dot{\epsilon}_{ij}^p} dt' .\end{aligned}\quad (414)$$

The flow-stress convention $R = \sigma_{\text{flow}}$ and the hardening-increment convention $R = R_h$ have separate definitions. The Cu–Sn endpoint parameters are $R_\infty = 120$ MPa and $b = 15$ at 550 K. Quantitative multilayer predictions require temperature-dependent hardening functions calibrated for each material. A zero hardening increment defines the perfect-plasticity specialization. For a crack-front strip of width w and advance Δa , paired nodal closure forces and opening/sliding displacements give

$$\begin{aligned}G_I &= F_z \Delta u_z / (2w \Delta a), \\ G_{II} &= F_x \Delta u_x / (2w \Delta a), \\ G_{III} &= F_y \Delta u_y / (2w \Delta a), \quad G = G_I + G_{II} + G_{III}.\end{aligned}\quad (415)$$

For tractions uniform across w , the continuum Mode I expression is $(2\Delta a)^{-1} \int_0^{\Delta a} \sigma_{zz}(r) \Delta u_z(r - \Delta a) dr$; Mode II substitutes $\tau_{xz}, \Delta u_x$. No additional $1/w$ factor applies because width cancels when traction is converted to force. Evaluation requires consistent closing-work signs and the applicable mixed-mode fracture criterion.

c) *Voce and creep parameters at 550 K:* For the 5.0 μm ϵ -Cu₃Sn bondline, the endpoint flow-stress law is [61]

$$\sigma_{\text{flow}}(\bar{\epsilon}_p, 550 \text{ K}) = 180 + 120[1 - e^{-15\bar{\epsilon}_p}] \text{ MPa}.\quad (416)$$

Thus initial yield is 180 MPa, saturation stress 300 MPa, and $b = 15$ is dimensionless. These resolve the Cu–Sn endpoint of (414), not temperature functions or other materials' hardening laws. The independent material functions in Table XXXVIII must be evaluated for the actual composition and temperature.

TABLE XXXIX
LIMITING CASES OF THE COUPLED JOINT MODEL

Test	Required result
Closed alloy inventory	Conserved species totals
Isothermal phase evolution	Nonincreasing free energy under fixed boundaries
Vacancy equilibrium	Zero current for equal chemical potentials and zero line affinity
Climb conversion	$v_c = \Omega_a I_v / b_e$ in m/s
Mechanical coupling	Tensor strain from swept dislocation area
Material state	Composition, phase, and defect inputs

An apparent Arrhenius activation energy can be extracted from the calculated rate, but it is not assumed constant:

$$\beta_T = (k_B T)^{-1},$$

$$Q_{\text{app}}(\mathbf{c}, \boldsymbol{\eta}, T, \boldsymbol{\sigma}) = - \left. \frac{\partial \ln[D_{\text{self}}/D_*]}{\partial \beta_T} \right|_{\mathbf{c}, \boldsymbol{\eta}, \boldsymbol{\sigma}}. \quad (417)$$

Here D_* is a temperature-independent reference diffusivity used only to make the logarithm dimensionless. The derivative includes temperature-dependent prefactors, defect populations, correlations, and barriers. At fixed geometry and a temperature-independent prefactor, a single formation-plus-migration pathway reduces to the sum of the corresponding activation enthalpies. This limit is not a universal alloy law. Table XXXIX lists the limiting cases of the coupled joint model. Strain-history calculation uses thermal and loading cycles, phase topology, reservoir conditions, and initial defect structure, with elastic moduli, defect energies, and kinetic rates evaluated for the same material state.

G. Suhir Interfacial-Compliance Formulation

The elastic reduction defines an analytical comparison model. The controlling active-stack model is Eq. (408), coupled to Eqs. (404) and (405) and the VCCT formulation in Eq. (406). Peak stresses and margins from the reduction apply only to its stated elastic comparison case.

For a symmetric bond extending over $-a \leq x \leq a$, let the adherends have thicknesses h_1, h_2 , elastic moduli E_1, E_2 , and Poisson ratios ν_1, ν_2 . The TLP bondline has thickness h_s and shear modulus G_s . A Suhir interfacial-compliance reduction uses [143]

$$G_j = \frac{E_j}{2(1 + \nu_j)},$$

$$\Lambda = \frac{1 - \nu_1}{E_1 h_1} + \frac{1 - \nu_2}{E_2 h_2},$$

$$\kappa_s = \frac{h_s}{G_s}, \quad \kappa = \kappa_s + \frac{h_1}{3G_1} + \frac{h_2}{3G_2},$$

$$\beta = \sqrt{\frac{\Lambda}{\kappa}}, \quad \Delta \varepsilon_T = (\alpha_2 - \alpha_1) \Delta T. \quad (418)$$

Here Λ has units $\text{Pa}^{-1} \text{m}^{-1}$, κ has units mPa^{-1} , and β has units m^{-1} . Let $N(x)$ denote the axial force per unit width under a sign convention for which $\tau = N'$. The reduced equilibrium and compatibility equations are

$$\frac{d^2 N}{dx^2} - \beta^2 N = \frac{\Delta \varepsilon_T}{\kappa}, \quad N(-a) = N(a) = 0. \quad (419)$$

Their symmetric solution and interfacial stress are

$$N(x) = \frac{\Delta \varepsilon_T}{\Lambda} \left[\frac{\cosh(\beta x)}{\cosh(\beta a)} - 1 \right],$$

$$\tau(x) = \frac{\Delta \varepsilon_T}{\kappa \beta} \frac{\sinh(\beta x)}{\cosh(\beta a)}, \quad (420)$$

$$\tau_{\text{max}} = \frac{|\Delta \varepsilon_T|}{\kappa \beta} \tanh(\beta a).$$

The calculation shall retain β in the stress denominator to preserve dimensional consistency. This reduction specifies interfacial shear; the production multilayer model shall additionally resolve warpage, peeling, inelasticity, and phase changes where applicable.

The elastic reference case uses Pd–Ir and ceramic adherends, distinct from the diamond/copper pair in Table XXXIV:

$$a = 25.0 \text{ mm}, \quad h_s = 20.0 \text{ } \mu\text{m},$$

$$(E_1, h_1) = (121 \text{ GPa}, 1.0 \text{ mm}),$$

$$(E_2, h_2) = (310 \text{ GPa}, 1.5 \text{ mm}), \quad (421)$$

$$\Delta \alpha = |11.8 - 7.2| \times 10^{-6} \text{ K}^{-1},$$

$$\Delta T = 240.0 \text{ K}, \quad |\Delta \varepsilon_T| = 1.104 \times 10^{-3}.$$

The 1.0 mm mechanical adherend is distinct from the 1.0 mm optical film. The nominal peak-stress and strength comparison is

$$\tau_{\text{max}}^{(v1.1)} = 18.42 \text{ MPa},$$

$$\tau_{\text{strength}}^{(v1.1)} = 58.50 \text{ MPa},$$

$$\text{SF}_{\text{ratio}} = \frac{58.50}{18.42} = 3.17590, \quad (422)$$

$$\text{SF}_{\text{spec}} = 3.175.$$

The nominal strength-to-stress ratio is 3.175. The elastic comparison uses complete compliance inputs, boundary conditions, residual stress, and consistent compliance units. Production allowable stresses follow the applicable material and service qualification.

For the same constant-property elastic case, increasing the temperature rise from 240.0 to 256.85 K scales the nominal stress to approximately 19.71 MPa. Production acceptance shall account for peeling, residual stress, flaws, cyclic inelasticity, and temperature-dependent strength in addition to the scalar shear ratio.

H. Alloy/SiC Cyclic Shakedown and Fatigue at 623.15 K

The 3D Chaboche model in Eqs. (352)–(355) uses the 298.15–623.15 K startup/shutdown excursion. With the alloy/SiC thermal-expansion coefficients, the constrained alloy's mechanical mismatch is

$$\Delta \varepsilon_{\text{mech}} = (\alpha_{\text{SiC}} - \alpha_{\text{PdIr}}) \Delta T$$

$$= (4.0 - 12.0) \times 10^{-6} (623.15 - 298.15) \quad (423)$$

$$= -2.60 \times 10^{-3} = -0.26\%.$$

Table XL gives the supplied cyclic results [102]. Here $\Delta p = \int_{\text{cycle}} \dot{p} dt$ is the nonnegative accumulated equivalent viscoplastic strain increment; signed stress extrema describe the reported cyclic stress component, not the nonnegative von Mises stress. The decreasing increments and stabilizing extrema

TABLE XL
CHABOCHE ELASTIC-SHAKEDOWN CONVERGENCE FOR THE ALLOY/SiC
CYCLE

Cycle	Δp	$\sigma_{\max}$ (MPa)	$\sigma_{\min}$ (MPa)
1	8.952×10^{-4}	115.40	-175.20
2	3.899×10^{-6}	124.65	-182.40
10	3.992×10^{-7}	127.10	-185.35
50	5.142×10^{-8}	127.58	-185.75

indicate convergence toward elastic shakedown. Cycle 50 meets the specified numerical criterion $\Delta p \leq 5.15 \times 10^{-8}$, although its finite increment is not exact zero plastic flow.

a) *Coffin–Manson fatigue life and cumulative damage:* For $\varepsilon'_f = 0.35$ and $c = -0.60$, the strain–life relation and constant-amplitude damage convention are

$$\begin{aligned} \frac{\Delta \varepsilon_{vp}}{2} &= \varepsilon'_f (2N_f)^c, \\ 2N_f &= \left(\frac{\Delta \varepsilon_{vp}}{2\varepsilon'_f} \right)^{1/c}, \quad \mathcal{D} = \frac{N_{\text{service}}}{N_f}. \end{aligned} \quad (424)$$

The supplied structural analysis substitutes $\Delta \varepsilon_{vp} = 5.142 \times 10^{-8}$ and reports

$$\begin{aligned} (2N_f)_{\text{reported}} &= 4.07 \times 10^{11}, \\ \mathcal{D}_{\text{reported}} &= 4.12 \times 10^{-5}, \quad N_{\text{service}} = 10^5 \text{ cycles}. \end{aligned} \quad (425)$$

In Eq. (424), N_f counts cycles and $2N_f$ counts reversals. The assigned strain and coefficients give $2N_f = 7.76146 \times 10^{11}$ reversals, $N_f = 3.88073 \times 10^{11}$ cycles, and $\mathcal{D} = 2.57684 \times 10^{-7}$ for 10^5 identical stabilized cycles. Conversion of the accumulated tensorial increment Δp to fatigue amplitude uses the local strain path. Startup transients use a cycle-dependent damage sum, with elastic-fatigue and creep-fatigue contributions included.

I. Earlier Constant-Property Alloy/SiC Thermal Bound at 623.15 K

The following constant-property comparison retains earlier CTE inputs and the assigned 520.00 MPa yield strength. These do not govern the alloy/SiC Chaboche model, which instead uses Eq. (354) and the mismatch in Eq. (423).

The revised alloy $\text{Pd}_{0.9132}\text{Ir}_{0.0868}$ is paired with a monocrystalline SiC spreader in this high-temperature comparison. It is distinct from the diamond/TLP/copper geometry and its 293.15–550.0 K bondline fatigue calculation. For a stress-free reference $T_0 = 298.15$ K and peak temperature $T_{\max} = 623.15$ K, the constant-CTE approximation gives

$$\begin{aligned} \Delta T &= 623.15 - 298.15 = 325.00 \text{ K}, \\ \alpha_{\text{alloy}} &= 11.2 \times 10^{-6} \text{ K}^{-1}, \\ \alpha_{\text{SiC}} &= 4.5 \times 10^{-6} \text{ K}^{-1}, \\ \Delta \epsilon_{\text{thermal}} &= \int_{T_0}^{T_{\max}} [\alpha_{\text{alloy}}(T) - \alpha_{\text{SiC}}(T)] dT \\ &\simeq (11.2 - 4.5) \times 10^{-6} (325) \\ &= 2.1775 \times 10^{-3}. \end{aligned} \quad (426)$$

This is a free mismatch strain over the excursion, not a plastic strain amplitude. Under an ideal fully constrained, isotropic equibiaxial plane-stress film approximation, the alloy stress magnitude is

$$\begin{aligned} E_{\text{alloy}} &= 121.0 \text{ GPa}, \quad \nu = 0.38, \\ |\sigma_{\text{thermal,calc}}| &= \frac{E_{\text{alloy}} \Delta \epsilon_{\text{thermal}}}{1 - \nu} \\ &= \frac{121000(0.0021775)}{0.62} \text{ MPa} \\ &= 424.9637097 \dots \text{ MPa} \simeq 424.96 \text{ MPa}, \\ |\sigma_{\text{thermal,reported}}| &= 425.02 \text{ MPa}. \end{aligned} \quad (427)$$

Heating the higher-CTE alloy against a rigid lower-CTE substrate produces compressive in-plane stress in this idealization. Finite substrate compliance, bondline shear and peeling, thermal gradients, and residual deposition stress enter the coupled models in Eqs. (408) and (431).

With the *assigned* alloy yield strength $\sigma_y(623.15 \text{ K}) = 520.00$ MPa, the stated scalar comparison satisfies

$$\begin{aligned} 425.02 \text{ MPa} &< 520.00 \text{ MPa}, \\ \sigma_y - |\sigma_{\text{thermal,reported}}| &= 94.98 \text{ MPa}, \\ \frac{\sigma_y}{|\sigma_{\text{thermal,reported}}|} &\simeq 1.22347. \end{aligned} \quad (428)$$

For ideal equibiaxial plane stress, the von Mises stress equals the calculated magnitude. The positive nominal yield margin is conditional on the stated constraint and material assumptions. The assigned 520.00 MPa yield strength requires composition-, temperature-, and deuterium-state-specific characterization; bondline strength and interface toughness are separate inputs.

Elastic shakedown requires a full cyclic stress field, not one endpoint inequality [94]. In the restricted small-strain, associative elastic-perfectly-plastic model, a sufficient shakedown condition is existence of a self-equilibrated, load-independent residual stress σ^R with homogeneous traction conditions such that

$$\begin{aligned} \nabla \cdot \sigma^R &= \mathbf{0}, \quad \sigma^R \mathbf{n} = \mathbf{0} \text{ on } \Gamma_t, \\ \sup_t \sigma_{\text{VM}}[\sigma^E(\mathbf{x}, t) + \sigma^R(\mathbf{x})] &\leq \sigma_y(\mathbf{x}) \quad \text{for all } \mathbf{x}. \end{aligned} \quad (429)$$

This classical comparison does not supply a shakedown theorem for temperature-dependent creep and recovery. For the thermo-viscoplastic model, stabilized hysteresis, absence of progressive ratcheting, and accumulated damage must be evaluated over complete qualified cycles. Even a cycle that remains elastic is not thereby immune to fatigue [24].

The endurance target is $N_f > 10^5$ qualified cycles. A reversal-based strain–life assessment uses

$$\frac{\Delta \epsilon_{\text{tot}}}{2} = \frac{\sigma'_f - \sigma_m}{E} (2N_f)^b + \epsilon'_f (2N_f)^c, \quad (430)$$

with temperature-specific $\sigma'_f, b, \epsilon'_f, c$, a mean-stress correction, and the local cyclic strain. The 52,400-cycle graded-TLP prediction at 550.0 K and the spring endurance statistics use their respective material and load-path configurations.

TABLE XLI
REVISED HIGH-TEMPERATURE ALLOY/SIC QUALIFICATION BOUNDS

Quantity	Value / disposition
Metal atomic fractions Pd/Ir	0.9132 / 0.0868
Metal mass fractions Pd/Ir	85.347 / 14.653 wt%
Thermal excursion	298.15–623.15 K
Mismatch strain range	2.1775×10^{-3}
Reported stress magnitude	425.02 MPa
Model stress magnitude	424.96 MPa
Assigned alloy yield strength	520.00 MPa
Reported-stress yield margin	94.98 MPa
Elastic shakedown	Requires cyclic-field assessment
Fatigue objective	$N_f > 10^5$ qualified cycles

J. Transient Thermal Shock and Coffin–Manson Fatigue

The operating ramp envelope is 20 K min^{-1} , distinct from TLP manufacturing ramps. Transient analysis shall obtain layer temperatures from Eq. (408), include the applicable hot-side endpoint, and evaluate thermal mismatch and residual stress over complete startup/shutdown cycles. The 8.42 MPa additional-excursion limit is a separate requirement and is not automatically additive to either the 18.42 MPa elastic result or the 142.5 MPa elastoplastic field. The prescribed 44820-cycle simulation and its nominal 1.42% accumulated creep output are distinct from a Coffin–Manson fatigue-life prediction.

Under the positive-exponent cycle-count convention, the strain–life relation is

$$\Delta\epsilon_p = \epsilon'_f N_f^{-c}, \quad N_f = \left(\frac{\epsilon'_f}{\Delta\epsilon_p} \right)^{1/c}, \quad (431)$$

with Cu–Sn reference parameters $\epsilon'_f \approx 0.18$ and $c \approx 0.52$. The nominal endurance target is

$$N_f^{(v1.1)} = 44,820 \text{ cycles} \quad (432)$$

for its 20–260°C thermal cycle. Reproducing this value under (431) requires

$$\Delta\epsilon_{p,\text{required}} = \frac{0.18}{44,820^{0.52}} = 6.86289 \times 10^{-4}. \quad (433)$$

The 44,820-cycle endurance target uses the required plastic-strain input from Eq. (431).

One cycle comprises a complete thermal startup/shutdown excursion. Plastic strain requires a calibrated joint model or measured hysteresis; elastic strain, engineering shear strain, and equivalent plastic strain are not interchangeable. The positive-exponent cycle convention requires transformed coefficients before comparison with a reversal-based Coffin–Manson relation. Representative TLP joints require cyclic testing, phase and void characterization, and separate interfacial-delamination criteria. The design target is 44,820 cycles for a 16,570-cycle duty schedule. Five-year qualification additionally addresses sustained creep at the service temperature.

a) *Reversal-based fatigue of the 5 micrometer bondline:* The reversal-based Coffin–Manson comparison uses a convention distinct from Eq. (431):

$$\begin{aligned} \frac{\Delta\epsilon_p}{2} &= \epsilon'_f (2N_f)^c, \\ \epsilon'_f &= 0.185, \quad c = -0.52, \quad \Delta\epsilon_p = 0.00223, \\ N_f &= \frac{1}{2} \left(\frac{0.00223}{2(0.185)} \right)^{1/(-0.52)} = 9289.71583 \text{ cycles.} \end{aligned} \quad (434)$$

The specified cycle is 20.0 to 276.85°C, $\Delta T = 256.85 \text{ K}$, for a 5.0 μm TLP bondline. The coefficient difference $\Delta\alpha = 18.0 \times 10^{-6} \text{ K}^{-1}$ gives free mismatch 0.0046233, distinct from the assigned plastic-strain range of 0.00223. Constraint and plastic hysteresis determine that range. For the same ϵ'_f and c , achieving 44820 cycles requires

$$\Delta\epsilon_{p,44820} = 2(0.185)(2 \cdot 44820)^{-0.52} = 0.000983786284. \quad (435)$$

The calculated endurance is 9289.72 cycles. Acceptance requires 44,820 cycles for the design target and 16,570 cycles for the specified duty schedule.

K. Fuel Conditioning, Trapping and Containment

a) *Loading rig and assembled compliance:* The stainless-steel loading vessel accommodates the diamond/Ti/Pd–Ir optical stack, X-750 spring preload assembly, graded Ni/Cu interlayer, micro-alloyed TLP joint, and OFHC-Cu/Al₂O₃ thermal spreader. Evacuated and charged states are distinct:

$$\begin{aligned} p_{\text{base}} &\leq 1.0 \times 10^{-8} \text{ mbar} = 1.0 \times 10^{-6} \text{ Pa}, \\ p_{D_2} &= 4.20 \text{ bar absolute at } 550.0 \text{ K}, \\ x_D &\equiv n_D/n_{\text{metal}} = 0.9132 \quad (\text{verified active-layer loading}). \end{aligned} \quad (436)$$

The charged vessel is not at UHV. Its base-pressure requirement is also distinct from the lower background required by the isolated mass-spectrometry analysis chamber. Verify leak integrity and residual gas before controlled charging. Establish the loading ratio from temperature-compensated, material-specific electrical-resistivity calibration and an independent uptake or isotope-inventory check. The selected verified active-layer state is $x = [\text{D}]/[\text{metal}] = 0.9132$. The earlier D/Pd cap calculation in Section IV-L1 uses a different concentration denominator and a conditional lattice-gas comparison; it is not the governing loading definition for this baseline.

The Belleville assembly has tangent stiffness $K_{\text{stack}} = 5000.0 \text{ N/mm}$ at its working preload. For the selected spring-path mismatch $\Delta L_{\text{net}} = 20.1 \mu\text{m}$ and circular loading area of 20.0 mm diameter,

$$\begin{aligned} \Delta F &= K_{\text{stack}} \Delta L_{\text{net}} = 100.5 \text{ N}, \\ A_{\text{load}} &= \pi(10.0 \text{ mm})^2, \\ \Delta P_{\text{contact}} &= \Delta F/A_{\text{load}} \simeq 0.320 \text{ MPa} < 0.50 \text{ MPa}. \end{aligned} \quad (437)$$

This pressure variation is relative to the selected preload and lies within the $\pm 0.50 \text{ MPa}$ tolerance band. It does not imply an exactly constant force, and the loading area is not the optical aperture. Verify preload, relaxation, contact alignment, and the

6,283.19–78,539.82 N force bounds throughout the qualified spring temperature envelope.

The selected structural joint is a 3.5 μm Ni–Cu–Sn TLP bondline with a 15.0 μm graded ductile Ni/Cu compliance layer. The governing three-dimensional Chaboche/VCCT analysis is given by Eqs. (361) and (362): $f_{\text{delam}} = 0.484 < 1.0$ and the Morrow endurance is $N_f \simeq 65,474 \geq 52,400$ cycles. The earlier 293.15–550.0 K calculation in Eq. (276) is retained as a historical comparison. The X-750 spring endurance calculation is separate.

b) Transport discretization and execution contract:

The coupled diffusion–trapping calculation shall evolve C_L , $C_{T,1}$, and $C_{T,2}$ across Pd–Ir/Ti/ε–Cu₃Sn interfaces, retaining hydrostatic-stress drift and the Soret term. The specified fine-grid spacing is $\Delta x = 0.1 \text{ nm}$ in resolved interfacial regions. Conservative face fluxes define the spatial discretization in one dimension or on a three-dimensional mesh; Crank–Nicolson is the time discretization, not a spatial stencil. Writing the coupled semi-discrete residual as $\mathcal{R}$ gives

$$\frac{\mathbf{C}^{j+1} - \mathbf{C}^j}{\Delta t} = \frac{\mathcal{R}(\mathbf{C}^{j+1}, T^{j+1}, \sigma_m^{j+1}) + \mathcal{R}(\mathbf{C}^j, T^j, \sigma_m^j)}{2}. \quad (438)$$

Here $\mathbf{C}$ collects mobile and trapped populations. Nonlinear iteration, interface flux continuity, trap-occupancy bounds, and time/grid refinement shall be checked together. A time-centered update is accepted only when the nonlinear residual, population bounds, and integrated mass balance satisfy their tolerances. The material coefficients are specified in Table XCIV.

The stress-coupled transport model uses the parameters in Table XCIV at 550 K and $p_{D_2} = 4.20 \text{ bar}$ absolute [79], [80]. The quantities $C_L, C_{T,j}, N_j$ are mobile-deuterium, occupied-trap, and available-trap number densities in m^{-3} , respectively. Atomic counts are dimensionless in the rate equations, so k_j has units m^3/s ; explicit atom-count notation instead writes $\text{m}^3 \text{ atom}^{-1} \text{ s}^{-1}$. The reference temperature is 276.85 °C, not the 350 °C structural operating point. Extrapolation to the latter requires temperature-dependent diffusion, solubility, and capture/release parameters.

c) Chemical-potential driving force: For dilute interstitial deuterium in a homogeneous material at fixed temperature, define the molar chemical potential by

$$\begin{aligned} \mu_L &= \mu_L^0(T) + RT \ln(C_L/C_*) - V_H^* \sigma_m, \\ \sigma_m &= \frac{1}{3} \text{Tr } \boldsymbol{\sigma}, \\ \mathbf{J}_L &= -\frac{D_L C_L}{RT} \nabla \mu_L \\ &= -D_L \nabla C_L + D_L C_L \frac{V_H^*}{RT} \nabla \sigma_m. \end{aligned} \quad (439)$$

Here C_* is a fixed reference concentration, V_H^* is the partial molar insertion volume, R is the molar gas constant, and tensile hydrostatic stress is positive. The driving-force relation assumes constant V_H^* within each material and ideal dilute activity; concentrated loading requires an activity-based chemical potential. Positive tensile-stress gradients favor interstitial accumulation. A spatially uniform compressive preload, even at 15.80 MPa, does not produce a bulk stress-gradient flux.

d) Conservation and finite-rate trapping: The McNabb–Foster balance and its thermal-gradient extension are

$$\begin{aligned} \partial_t C_L + \sum_j \partial_t C_{T,j} &= -\nabla \cdot \mathbf{J}_L, \\ \partial_t C_{T,j} &= k_j C_L (N_j - C_{T,j}) - p_j C_{T,j}, \\ \mathbf{J}_L &= -D_L \nabla C_L + \frac{D_L C_L V_H^*}{RT} \nabla \sigma_m \\ &\quad - \frac{D_L C_L Q^*}{RT^2} \nabla T, \\ \sigma_m &= \frac{1}{3} \text{Tr } \boldsymbol{\sigma}. \end{aligned} \quad (440)$$

The flux $\mathbf{J}_L$ has units $\text{m}^{-2} \text{ s}^{-1}$. The quantities V_H^* and Q^* are molar, making the ratios containing R dimensionally consistent with a number-density flux. The heat of transport Q^* must be converted from kJ/mol to J/mol in the thermal term. Equation (440) defines positive Q^* as transport toward decreasing temperature. Explicit trapping requires a bare lattice diffusivity D_L ; simultaneous use of a trap-renormalized D_{eff} would double-count retardation. The tabulated D_{eff} values specify a diffusivity parameterization whose bare-lattice or effective interpretation must be established before numerical use. The explicitly bare D_L values in Table XLIV belong to a distinct two-family model and are not interchangeable with this table.

Equilibrium requires $p_j/k_j = N_0 \exp[-E_{t,j}/(k_B T)]$. The coefficients in Table XCIV imply $N_0 = 1.3020 \times 10^{25}$, 6.2396×10^{25} and $1.0873 \times 10^{27} \text{ m}^{-3}$, respectively. Verification of detailed balance requires consistent host-site density, degeneracy, and capture conventions. A uniform preload removes the stress-gradient term only if $\nabla \sigma_m \simeq 0$ within the actual microstructure. Grain boundaries, free edges, temperature gradients, and trapped populations can remain nonuniform. Fuel-homogeneity acceptance uses the resolved concentration and stress fields.

For fixed trap density, $C_{T,j} = N_j \theta_j$, and constant local C_L gives the closed-form occupancy kinetics

$$\begin{aligned} \theta_{j,\text{eq}} &= \frac{k_j C_L}{k_j C_L + p_j}, \\ \tau_j &= (k_j C_L + p_j)^{-1}, \\ \theta_j(t) &= \theta_{j,\text{eq}} + [\theta_j(0) - \theta_{j,\text{eq}}] e^{-t/\tau_j}. \end{aligned} \quad (441)$$

The occupancy satisfies $0 \leq \theta_j \leq 1$ for nonnegative rates and admissible initial conditions. The half-occupancy concentration $C_{1/2,j} = p_j/k_j$ and the corresponding local relaxation time are These scales follow from the capture and release coefficients in Table XCIV; they are not through-thickness charging times. If traps remain in local equilibrium, the retardation factor is

$$\begin{aligned} \mathcal{R}_T(C_L) &= \frac{\partial}{\partial C_L} \left(C_L + \sum_j C_{T,j}^{\text{eq}} \right) \\ &= 1 + \sum_j \frac{N_j k_j p_j}{(k_j C_L + p_j)^2}, \\ D_{\text{app}}(C_L) &= D_L / \mathcal{R}_T(C_L). \end{aligned} \quad (442)$$

The last expression describes the isothermal, uniform-stress reduction with spatially fixed material parameters. Outside the

TABLE XLII
CALCULATED HANKEL SINGULAR VALUES FOR FIFTEEN-TO-FOUR-STATE REDUCTION

Index	Calculated σ_i	Target	Relative-error limit	Treatment
1	12.45	12.45	10^{-6}	Retain
2	5.13	5.13	10^{-6}	Retain
3	1.85	1.85	10^{-6}	Retain
4	0.94	0.94	10^{-6}	Retain
5	0.015	0.015	10^{-4}	Truncate
6	0.010	—	—	Truncate
7	0.007	—	—	Truncate
8	0.004	—	—	Truncate
9	0.002	—	—	Truncate
10	0.001	—	—	Truncate
11	0.0005	—	—	Truncate
12	0.0002	—	—	Truncate
13	0.0001	—	—	Truncate
14	5.0×10^{-5}	—	—	Truncate
15	2.0×10^{-5}	—	—	Truncate

Maximum relative error, modes 1–4: $2.362176648138631 \times 10^{-16}$.
Discarded squared-spectrum ratio: $2.129567641168566 \times 10^{-6}$.
First four modes: $\varepsilon_{\text{rel}} \simeq 2.36 \times 10^{-16}$.
Tail-energy metric: $\eta_{\text{tail}} \simeq 2.13 \times 10^{-6}$.

TABLE XLIII
SINGLE-FAMILY TRAP EQUILIBRATION SCALES AT 550 K

Layer	$C_{1/2,j}$ (m^{-3})	$\tau_j(C_{1/2,j})$ (s)
Pd–Ir	$1.25517241 \times 10^{23}$	$2.74725275 \times 10^{-4}$
Ti	$6.90322581 \times 10^{19}$	2.33644860
ϵ -Cu ₃ Sn	$4.34523810 \times 10^{22}$	0.01369863

local-equilibrium limit, the full occupancy equations must be integrated rather than replacing them by D_{app} .

e) Uniform preload and multilayer boundary conditions:

Under the combined assumptions $\nabla \sigma_m \simeq 0$ and $\nabla T = 0$, Eq. (440) reduces to $\mathbf{J}_L = -D_L \nabla C_L$, while capture and detrapping remain active. Uniform applied pressure does not imply uniform hydrostatic stress near layer interfaces, bevels, or defects; the stress field in Section III supplies the local driving term. Even a constant stress changes the equilibrium chemical potential relative to an unloaded surface.

At an ideal interface without interfacial storage or reaction, normal flux and equilibrium chemical potential are continuous:

$$\begin{aligned} \hat{\mathbf{n}} \cdot \mathbf{J}_{L,1} &= \hat{\mathbf{n}} \cdot \mathbf{J}_{L,2}, \\ \mu_{L,1} &= \mu_{L,2}. \end{aligned} \quad (443)$$

Concentration is generally discontinuous because the materials have different standard chemical potentials and solubilities. The gas-facing equilibrium condition is the Sieverts relation $C_s = S_D(T) \sqrt{p_{D_2}}$ with absolute pressure and an atom-count solubility coefficient; finite dissociation or recombination kinetics requires a surface exchange law. Initial lattice concentrations and trap occupancies, exterior sink conditions, and the actual Pd–Ir/Ti/bondline geometry complete the transport problem. A perfectly blocking boundary is a limiting case, not a substitute for the finite barrier permeance.

f) Trap normalization and interface partitioning: For M trap families, total deuterium concentration is $C = C_L +$

$\sum_{j=1}^M N_{T,j} \theta_j$, with $C_L = N_L \theta_L$. Diffusion-limited capture by a spherical trap of radius R_{cap} has number-density coefficient $k_{\text{cap}} = 4\pi R_{\text{cap}} D_L$ in m^3/s . If $N_L = a^{-3}$ is the adopted site-density approximation, its occupancy-based coefficient is $k_{\text{occ}} = k_{\text{cap}} N_L = 4\pi R_{\text{cap}} D_L / a^3$ in s^{-1} . These coefficients multiply C_L and θ_L , respectively, and are not interchangeable. For molar binding and migration energies,

$$p_j = \nu_D \exp[-(E_{B,j} + E_L)/(RT)]. \quad (444)$$

Per-particle energies instead require $k_B T$. At a common interface temperature, equilibrium partitioning can be written

$$\frac{C_L^{(1)}}{S_L^{(1)}} e^{-V_H^{*(1)} \sigma_h^{(1)}/(RT)} = \frac{C_L^{(2)}}{S_L^{(2)}} e^{-V_H^{*(2)} \sigma_h^{(2)}/(RT)}, \quad (445)$$

where the solubility normalizations use a common chemical-potential reference. Table X gives the calculated steady interface fluxes. A local hydrostatic tension is not a gradient: these fluxes require complete concentration, stress, temperature, and boundary profiles rather than the listed tension alone.

For a locally frozen nonnegative C_L , an implicit trap substep is

$$\theta_j^{n+1} = \frac{\theta_j^n + \Delta t k_j C_L}{1 + \Delta t (k_j C_L + p_j)}. \quad (446)$$

This preserves $0 \leq \theta_j \leq 1$ but is first-order in time. A split implementation shall account for that temporal order and remove exactly $N_{T,j}(\theta_j^{n+1} - \theta_j^n)$ from the mobile population. Drift enters through the divergence of a face flux, not as a flux directly subtracted from a concentration rate. For fixed control volume $\mathcal{V}$ with $M_0 > 0$, the global conservation requirement

is

$$\begin{aligned}
 M(t) &= \int_V \left(C_L + \sum_j N_{T,j} \theta_j \right) dV, \\
 M_{\text{out}}(t) &= \int_0^t \int_{\partial V} \mathbf{J}_L \cdot \hat{\mathbf{n}} dA dt', \\
 \frac{|M(t) - M_0 + M_{\text{out}}(t)|}{M_0} &< 10^{-12}.
 \end{aligned} \tag{447}$$

Interface partitioning and flux continuity are enforced in the same conservative solve.

g) *Environmental fracture and slow-strain-rate metrics:* The embrittlement assessment uses the following nominal material-response values:

$$\begin{aligned}
 K_{IH} &= 41.8 \text{ MPa}\sqrt{\text{m}}, \\
 I_{RA} &= \frac{RA_{\text{exposed}}}{RA_{\text{inert}}} = 0.89, \\
 K_{I,\text{hold}} &= 0.80 K_{IH} = 33.44 \text{ MPa}\sqrt{\text{m}}, \\
 \dot{a} &< 10^{-11} \text{ m/s}, \quad t_{\text{hold}} = 1000 \text{ h}, \\
 \Delta a_{\text{hold}} &< 3.6 \times 10^{-5} \text{ m} = 36 \mu\text{m}.
 \end{aligned} \tag{448}$$

The integrated crack-growth bound assumes the rate limit holds throughout the specified exposure. Reduction of area is $RA = (A_0 - A_f)/A_0$ for the tensile specimen; its exposed-to-inert ratio is dimensionless. The selected SSRT criterion is $I_{RA} \geq 0.85$, giving an absolute margin of 0.04 at the nominal value. The 0.80 criterion in Table XLVI is a separate, less restrictive comparison criterion.

ASTM G129 defines comparative slow-strain-rate testing, while ASTM E1681 addresses environment-assisted cracking thresholds [55], [83]. The notation K_{IH} identifies the deuterium-environment threshold in this specification; the applicable E1681 threshold designation and specimen-size validity shall accompany its use. Qualification requires specimen material and heat, geometry, precrack history, loading mode, strain rate, absolute and differential pressures, temperature, replicates, and measurement uncertainty. Each metric is qualified for its particular alloy and service condition.

h) *Bilayer permeation barrier and defect tolerance:* The noncatalytic internal barrier comprises 150 nm TiN / 250 nm Al_2O_3 , with total nominal coating thickness 400 nm. Its surface coverage shall not obstruct the intended deuterium-loading interface. For a common permeation driving variable $\Delta\psi$, the intact through-thickness path has series resistance

$$\begin{aligned}
 \mathcal{R}_{\text{intact}} &= \mathcal{R}_{\text{sub}} + \frac{150 \times 10^{-9}}{\mathcal{P}_{\text{TiN}}} + \frac{250 \times 10^{-9}}{\mathcal{P}_{\text{Al}_2\text{O}_3}} + \mathcal{R}_{\text{interfaces}}, \\
 J_{\text{intact}} &= \Delta\psi / \mathcal{R}_{\text{intact}}, \\
 r_P &= \frac{J_{\text{intact}}}{J_{\text{bare}}} = \frac{\mathcal{R}_{\text{sub}}}{\mathcal{R}_{\text{intact}}}.
 \end{aligned} \tag{449}$$

The thickness numerators are in metres; the permeabilities $\mathcal{P}_j$ and resistances must use the same driving-variable and flux convention. Here r_P is the intact-to-bare permeance ratio at matched boundary conditions; a ratio of intrinsic material permeabilities alone would omit layer thickness and interface resistance. The prescribed intact-path ratio is $r_P = 1.20 \times 10^{-8}$.

With defect fraction $f_d = 6.49 \times 10^{-4}$ and bare-path transmission through each defect, area-weighted parallel transport gives

$$\begin{aligned}
 \frac{J_{\text{coated}}}{J_{\text{bare}}} &= (1 - f_d)r_P + f_d, \\
 \text{PRF}_{\text{eff}} &= \frac{J_{\text{bare}}}{J_{\text{coated}}}.
 \end{aligned} \tag{450}$$

Substitution yields

$$\text{PRF}_{\text{eff}} = \frac{1}{(1 - f_d)r_P + f_d} = 1540.8036. \tag{451}$$

The conservative reduction-factor target is 1538.5, so the specified model satisfies

$$\text{PRF}_{\text{eff}} = 1540.803578 \dots > 1538.5 > 1000. \tag{452}$$

Exact equality to 1538.5 instead requires $f_d = 6.49971758 \times 10^{-4}$ at the same intact-path ratio. More generally,

$$\begin{aligned}
 \text{PRF}_{\text{eff}} > 1000 &\iff f_d < \frac{10^{-3} - r_P}{1 - r_P} \\
 &= 9.99988012 \times 10^{-4}.
 \end{aligned} \tag{453}$$

This implication is conditional on the parallel-path assumptions. Lateral transport into pinholes, coating cracks, interfacial bypass, and temperature-dependent permeance can change the effective leakage fraction. Barrier qualification shall determine connected-defect statistics and flux under the same pressure and temperature conditions as the bare-substrate reference.

i) *Environmental threshold size and finite-wall applicability:* For a semi-infinite edge-crack approximation with geometry factor $Y = 1.12$, $K_I = Y\sigma_{\text{hoop}}\sqrt{\pi a}$. Equating the stress intensity to K_{IH} gives the threshold estimate

$$\begin{aligned}
 a_{\text{crit}} &= \frac{1}{\pi} \left(\frac{K_{IH}}{1.12\sigma_{\text{hoop}}} \right)^2, \\
 \sigma_{\text{hoop}} &= 48.5 \text{ MPa}, \quad a_{\text{crit}} = 188.488 \text{ mm} \simeq 189 \text{ mm}.
 \end{aligned} \tag{454}$$

The environmental-threshold estimate is 188.4877665 mm, with nominal reference 189 mm. Here a_{crit} in Eq. (454) denotes an environment-assisted threshold, while vessel instability uses the applicable fracture toughness and finite-wall geometry. The 6.0 mm wall requires a geometry-specific assessment. The 4.20 bar transport case and the 48.5 MPa hoop-stress case retain separate structural boundary conditions.

j) *Formal leak-before-break acceptance conditions:* Leak-before-break requires a stable through-wall crack that produces detectable leakage before unstable extension [98]. A finite-wall assessment uses surface-crack depth and length before penetration, then a through-wall crack-length coordinate ℓ . The geometry-specific fracture driving force and a schematic set of acceptance conditions are

$$\begin{aligned}
 K_I(\ell) &= K_{I,\text{pressure}}(\ell) + K_{I,\text{thermal}}(\ell) + K_{I,\text{residual}}(\ell), \\
 \dot{m}_{\text{leak}}(\ell_{\text{det}}) &\geq \dot{m}_{\text{det}}, \\
 \ell_{\text{det}} &< \ell_{\text{inst}}, \\
 K_I(\ell_{\text{det}}) &< K_{\text{mat}}/\text{SF}_K, \\
 t_{\text{response}} &< \int_{\ell_{\text{det}}}^{\ell_{\text{inst}}} \frac{d\ell}{\dot{\ell}(\ell)}.
 \end{aligned} \tag{455}$$

Here m_{det} is the qualified detection threshold, ℓ_{det} the detectable through-wall flow length, ℓ_{inst} the instability length under the design load combination, K_{mat} the applicable fracture toughness, and $\text{SF}_K > 1$ the selected fracture margin. Linear superposition applies only within the elastic stress-intensity framework; significant plasticity requires an elastic-plastic fracture or failure-assessment treatment. The time integral assumes positive growth along a specified crack path and load history; its growth law must bound cyclic and environmental extension over the whole interval.

Leak-flow calculation requires crack opening, wall thickness, roughness, fluid state, and the pressure differential. Instrument response, alarm processing, isolation, and depressurization all contribute to t_{response} . A 1000-hour growth bound at $0.80K_{IH}$ cannot be extrapolated to the instability threshold without an applicable growth law. Consequently, the 189 mm threshold reference and the favorable embrittlement metrics support separate screening checks; a formal LBB demonstration additionally requires the finite-wall fracture, leakage, and response quantities in Eq. (455).

L. Sieverts Dissolution and Deuterium Transport

The hot-service material reference is D_2 at 4.20 bar and 550.0 K. Test records shall distinguish absolute pressure for uptake calculations from differential pressure for wall-stress calculations and explicitly define the pressure convention used in embrittlement testing.

An equilibrium boundary model is

$$C_s(T, p) = K_S(T) \sqrt{p_{D_2}},$$

$$K_S(T) = K_{S,0} \exp\left(-\frac{\Delta H_s}{RT}\right). \quad (456)$$

For lattice concentration C_L , trap density N_j , and occupied trap fraction θ_j , the McNabb–Foster transport framework is

$$\frac{\partial C_L}{\partial t} + \sum_j N_j \frac{\partial \theta_j}{\partial t} = -\nabla \cdot \mathbf{J}_L, \quad (457)$$

$$\frac{\partial \theta_j}{\partial t} = k_j C_L (1 - \theta_j) - p_j \theta_j.$$

Here k_j, p_j are trapping and release coefficients with units consistent with the concentration convention. Material heat, cold work, welds, surface state, isotope, and temperature affect the coefficients and boundary conditions. The HE protocols address slow-strain-rate response, fracture threshold, and crack growth. Transport calibration and charging uniformity are evaluated for the specified wall geometry.

a) *Stress, Soret transport, and two trap families:* With tensile-positive hydrostatic stress $\sigma_h = \sigma_{ii}/3$, the coupled flux is

$$\mathbf{J}_L = -D_L \nabla C_L + \frac{V_H}{R_u T} D_L C_L \nabla \sigma_h$$

$$+ \frac{Q^*}{R_u T^2} D_L C_L \nabla T. \quad (458)$$

Combined with (457), this gives the divergence of $D_L \nabla C_L - (V_H/R_u T) D_L C_L \nabla \sigma_h - (Q^*/R_u T^2) D_L C_L \nabla T$. C_L counts atoms per cubic meter, V_H is molar volume, R_u the molar gas

constant, and Q^* a signed molar heat of transport. Positive Q^* here drives atoms toward increasing T ; an opposite Soret convention reverses its sign. Trap densities are fixed in this balance; evolving traps require $\partial_t(N_j \theta_j)$.

For the stress-coupled alloy model, $V_H = 1.7 \times 10^{-6} \text{ m}^3/\text{mol}$ and $R = R_u = 8.314 \text{ J}/(\text{mol K})$. Allowing anisotropic diffusivity dependent on temperature and viscoplastic strain gives

$$J_i = -D_{ij}(T, \epsilon^{vp}) \left[\nabla_j C_L - \frac{C_L V_H}{RT} \nabla_j \sigma_h + \frac{C_L Q_S^*}{RT^2} \nabla_j T \right]. \quad (459)$$

Repeated spatial index j is summed; it is distinct from a trap-family index in Eq. (457). The term $-(C_L V_H/RT) \nabla_j \sigma_h$ is stress-gradient drift, whereas the temperature-gradient term is the Soret contribution. Here $Q_S^* > 0$ drives transport toward decreasing temperature, so $Q_S^* = -Q^*$ relative to Eq. (458), and has the same sign convention as Eq. (440). For uniform temperature the Soret term vanishes, leaving the specified stress-coupled flux. A numerical nonisothermal calculation still requires a calibrated heat of transport; V_H alone does not determine it.

1) Thermodynamic Stress and Physical Loading Caps:

Let $x = D/\text{Pd}$ denote the occupied interstitial-site fraction in a one-site-per-Pd lattice-gas approximation, and let x_0 be the zero-stress loading at the same temperature and reservoir chemical potential. This loading variable is not the Ir atomic fraction. With interaction parameter $g = 19.0 \text{ kJ/mol}$ and tensile-positive σ_h , the molar chemical potential is

$$\mu_L = \mu_L^0 + RT \ln \frac{x}{1-x} - gx - \sigma_h V_H. \quad (460)$$

Equating $\mu_L(x, \sigma_h)$ to $\mu_L(x_0, 0)$ cancels the common reference and yields

$$RT \ln \frac{x}{1-x} - gx - \sigma_h V_H$$

$$= RT \ln \frac{x_0}{1-x_0} - gx_0, \quad (461)$$

$$\sigma_h(x; x_0) = \frac{1}{V_H} \left\{ RT \ln \left[\frac{x(1-x_0)}{x_0(1-x)} \right] - g(x - x_0) \right\}.$$

The logarithmic activity replaces the dilute-concentration approximation at high loading. At $T = 623.15 \text{ K}$, direct substitution for $x = 0.95$ gives

$$\sigma_h(0.95; 0.90) = 1718.36 \text{ MPa},$$

$$\sigma_h(0.95; 0.80) = 3072.08 \text{ MPa}, \quad (462)$$

$$\sigma_h(0.95; 0.70) = 3597.07 \text{ MPa}.$$

For $x \geq 0.95$, $\partial \sigma_h / \partial x = [RT/(x(1-x)) - g]/V_H > 0$ at this temperature, so higher loading requires still greater tensile stress. Even the most favorable listed reference requires more than 1718 MPa, far beyond the assigned cyclic tensile-stress envelope of 127.58 MPa and the 155.0 MPa high-temperature yield-strength scale. Stress-assisted $D/\text{Pd} \geq 0.95$ is therefore

not an admissible operating target within this material and loading model.

For a conservative design bound, identify the allowed hydrostatic tensile component with $\sigma_h^{\max} = 127.58$ MPa and solve Eq. (461) on the branch above x_0 . The solutions are $x \simeq 0.90542$ for $x_0 = 0.90$ and $x \simeq 0.81511$ for $x_0 = 0.80$. Conservative limits below these model solutions are

$$\begin{aligned} x_{\max} &= 0.904 \quad (\text{physical design cap}), \\ x &\leq 0.81 \quad (\text{operational cap}). \end{aligned} \quad (463)$$

These caps constrain conditioning and transport calculations rather than prescribing an attained concentration. Hydrostatic stress is not a von Mises equivalent stress: pure hydrostatic loading does not itself trigger a deviatoric yield criterion. The comparison uses an assigned allowable tensile envelope for the constrained stack, subject to the complete mechanical qualification. Full stress tensors, interface strength, and temperature-dependent chemical activities remain necessary for mechanical qualification; the cap is conditional on the stated equilibrium model and reservoir loading. For a numerical boundary comparison interpret pressure as $p_{D_2} = 4.20 \times 10^5$ Pa absolute, with $C_s = S_D \sqrt{p_{D_2}}$ and $\theta_{j,\text{eq}} = k_j C_s / (k_j C_s + p_j)$. Release rates are already assigned at 550 K and are not multiplied by another activation factor. Binding-energy extrapolation requires attempt rates and consistent detailed balance. The alloy stress-drift volume is specified separately in Eq. (459); Soret flux calculation also specifies the heat of transport and its temperature dependence. These active-stack values do not replace the separate SS316L/Invar containment data.

a) Molar solubility and microstructural trap families:

The microstructural trap-family model uses a molar Sieverts coefficient,

$$K_S(T) = K_0 e^{-1000 H_s / (R_u T)}, \quad C_{D,\text{mol}} = K_S(T) \sqrt{p_{D_2}}. \quad (464)$$

H_s is tabulated in kJ/mol, R_u in J/(mol K), and p_{D_2} in pascals absolute; the factor 1000 is essential. Define the molar concentration as moles of D atoms, so $C_L = N_A C_{D,\text{mol}}$ when used in the atom-count transport equations. The parameters $K_S, N_j, E_{t,j}$ define this microstructural transport case rather than the alternative active-stack parameterization. Table XLVIII associates the trap families with capture and release rates; binding energies alone do not determine attempt frequencies. Containment-alloy data apply to their separately specified materials and conditions. The phenomenological constitutive relations are $K_{IH} = K_{I0} - \beta \sqrt{C_D}$ and $\dot{a} = v_0 \exp[-(E_a - \sigma_h V_a) / (R_u T)]$. In the latter, V_a is a molar activation volume when E_a is expressed in J/mol; an atomic volume instead requires atomic energy and k_B . Quantitative application requires material- and environment-specific empirical constraints on $K_{I0}, \beta, v_0, E_a, V_a$.

M. Hydrogen Embrittlement: HE-01–HE-03 Qualification Criteria

The high-temperature assessment applies at 550.0 K (276.85°C) and $p_{D_2} = 4.20$ bar [54]. Table XLVI defines the nominal hot-service metrics, distinct from the 145°C comparison dataset. Qualification requires identification of the specimen material and heat for every result.

a) HE-01: Slow strain rate testing: Define $I_{\text{RA}} = \text{RA}_{D_2} / \text{RA}_{\text{inert}}$. ASTM G129 addresses slow strain rate testing; ASTM F1624 is incremental step loading, not another SSRT method [55]. Specimen IDs, product forms, temperatures, strain rates, and inert/environmental controls shall identify which method generated each result.

b) HE-02: Environment-assisted fracture threshold: ASTM E1681 provides the environment-assisted threshold framework [55]. The $41.8 \text{ MPa}\sqrt{\text{m}}$ value requires verification of specimen geometry, precrack validity, loading mode, hold time, and pressure convention. It exceeds the 35.0 criterion but not the separate SS316L 45.0 criterion. Qualification shall apply the acceptance threshold assigned to the actual material and service condition.

c) HE-03: Subcritical crack growth: The $10^{-11} \text{ m s}^{-1}$ value is an upper bound rather than a measured constant rate. Sustained-load records shall specify stress intensity, duration, resolution, and uncertainty. Application of ASTM E1681/F1624 requires a compatible loading history and identified specimen material.

N. Five-Year Safe-Life and Leak-Before-Break Assessment

a) Elastic–plastic fracture mechanics and LBB compliance: The updated containment assessment assigns wall thickness $t = 8.0$ mm, membrane stress $\sigma_m = 180.0$ MPa, and shape factor $Y = 1.12$ [102]. The baseline toughness is $K_{IC} = 150.0 \text{ MPa}\sqrt{\text{m}}$; the hydrogen-embrittled value $K_{IH} = 45.0 \text{ MPa}\sqrt{\text{m}}$ is used as the limiting toughness in this specified screening calculation. For a through-wall crack of total length $2a$, the elastic stress-intensity criterion gives

$$\begin{aligned} K_I &= Y \sigma_m \sqrt{\pi a}, & K_I(a_c) &= K_{IH}, \\ a_c &= \frac{1}{\pi} \left(\frac{K_{IH}}{Y \sigma_m} \right)^2 \\ &= \frac{1}{\pi} \left(\frac{45.0}{1.12(180.0)} \right)^2 \text{ m} \\ &= 0.0158597 \text{ m} \simeq 15.86 \text{ mm}, \\ (2a_c)_{\text{calc}} &= 31.7193 \text{ mm}, \\ (2a_c)_{\text{spec}} &= 31.6 \text{ mm}. \end{aligned} \quad (465)$$

The design critical lengths are $a_c = 15.8$ mm and $2a_c = 31.6$ mm. The assigned detectable through-wall crack length is $2a_{\text{leak}} = 12.0$ mm at 3.8 kg/min. Using the specified critical length,

$$M_a = \frac{2a_c}{2a_{\text{leak}}} = \frac{31.6}{12.0} = 2.63 \geq 2.0. \quad (466)$$

The nominal design satisfies the specified LBB crack-length criterion under the assigned loading, toughness, geometry, and leak-detection assumptions. The EPFM allocation is $J_{\text{applied}} = 12.4 \text{ kJ/m}^2$ and $J_{IH}/J_{\text{applied}} = 1.61$, giving $J_{IH} = 19.964 \text{ kJ/m}^2$. Structural qualification includes geometry-specific elastic–plastic crack stability, hydrogen-environment toughness, leakage, and shutdown response under Eq. (455).

TABLE XLIV
TWO-FAMILY DEUTERIUM TRAPPING PARAMETERS AT 550 K AND 4.20 BAR

Parameter (units)	Pd _{0.95} Ir _{0.05}	Ti	ε-Cu ₃ Sn
D_L (m ² /s)	4.50×10^{-9}	1.20×10^{-11}	8.50×10^{-13}
S_D (atoms m ⁻³ Pa ^{-1/2})	1.62×10^{20}	4.38×10^{21}	2.10×10^{19}
N_1 (sites/m ³)	1.50×10^{24}	3.80×10^{25}	5.50×10^{24}
N_2 (sites/m ³)	5.00×10^{23}	1.20×10^{25}	1.80×10^{24}
k_1 (m ³ atom ⁻¹ s ⁻¹)	1.15×10^{-25}	2.80×10^{-27}	4.20×10^{-28}
k_2 (m ³ atom ⁻¹ s ⁻¹)	8.40×10^{-26}	1.95×10^{-27}	3.10×10^{-28}
p_1 (s ⁻¹)	24.50	0.15	0.08
p_2 (s ⁻¹)	120.40	1.42	0.75
$E_{t,1}$ (eV)	0.23	0.48	0.52
$E_{t,2}$ (eV)	0.15	0.35	0.40

TABLE XLV
MULTIPHYSICS SOLUBILITY AND TRAP PARAMETERS

Material	K_0	H_s	Trap family	N_j (m ⁻³)	$E_{t,j}$ (kJ/mol)
Pd _{0.95} Ir _{0.05}	1.2×10^{-1}	12.5	Dislocations	1.5×10^{23}	26.4
			Phase boundaries	8.0×10^{22}	44.1
Ti	2.4×10^1	45.2	Interstices	1.0×10^{25}	58.3
			Grain boundaries	5.0×10^{24}	32.1
ε-Cu ₃ Sn	4.5×10^{-3}	28.4	Kirkendall voids	6.2×10^{25}	30.8
			Interfaces	1.1×10^{26}	18.5

K_0 has units mol m⁻³ Pa^{-1/2} and H_s has units kJ/mol. At 550 K, K_S is 0.0077992621 (Pd-Ir), 0.0012233687 (Ti), and 9.0377081×10^{-6} (Cu-Sn), in the same molar units.

TABLE XLVI
HIGH-TEMPERATURE EMBRITTLEMENT METRICS AND ACCEPTANCE LIMITS

Observable	Nominal	Criterion
HE-01, I_{RA}	0.89	≥ 0.80
HE-02, K_{IH} (MPa√m)	41.8	≥ 35.0
HE-03, da/dt (m s ⁻¹)	$< 10^{-11}$	$\leq 10^{-10}$

TABLE XLVII
VOCE TOTAL-FLOW-STRESS PARAMETERS AT 550 K

Material	σ_{y0} (MPa)	R_∞ (MPa)	b
ε-Cu ₃ Sn	180.0	120.0	15.0
OFHC Cu	65.0	110.0	8.5

gives

$$\begin{aligned}
 t_{\text{service}} &= 5(365.25)(86400) = 1.57788 \times 10^8 \text{ s}, \\
 \Delta a &< 10^{-11} t_{\text{service}} = 1.57788 \text{ mm}, \\
 a_{\text{final}} &< 2.07788 \text{ mm} < 6.0 \text{ mm}.
 \end{aligned} \tag{468}$$

b) High-pressure containment comparison: For the distinct $p = 15$ MPa, $t_w = 8.0$ mm containment case, let $K_{Ic} = 85.0$ MPa√m. The elastic fracture and cyclic-growth relations are

$$\begin{aligned}
 K_I &= Y \sigma_{\max} \sqrt{\pi a}, \quad a_{\text{crit}} = \frac{1}{\pi} \left(\frac{K_{Ic}}{Y \sigma_{\max}} \right)^2, \\
 \frac{da}{dN} &= C(\Delta K)^m, \quad \Delta K = Y \Delta \sigma \sqrt{\pi a}.
 \end{aligned} \tag{467}$$

The assigned $a_{\text{crit}} = 37.9$ mm uses $Y \sigma_{\max} = 246.334$ MPa and its defined vessel radius, flaw geometry, and residual/thermal loading. Leak-before-break assessment jointly evaluates stable through-wall cracking, leak detectability, growth margin, and shutdown time [98]. Paris-law integration uses C, m , load history, and the applicable crack-growth regime.

For the low-pressure screening geometry, $t = 6.0$ mm, $R = 80.0$ mm, $\Delta p = 0.420$ MPa, $Y = 1.12$, and $a_0 \leq 0.500$ mm, application of the upper crack-growth bound over five years

The pressure-only stress is $\sigma_{\text{hoop}} = \Delta p R / t = 5.60$ MPa, and $K_I = Y \sigma_{\text{hoop}} \sqrt{\pi a_{\text{final}}} < 0.507$ MPa√m in the screening expression. Neither value applies to the separate 15 MPa(g) core relief case.

The conditional non-through-wall projection uses its stated load history and applicable material data. Structural qualification includes weld, residual, cyclic, and thermal stresses. Leak-before-break acceptance requires a stable through-wall flaw, detectable leakage before instability, and sufficient response time.

c) High-temperature hardening and trap-family kinetics: The five-layer comparison geometry and its HiPIMS protocol define the material configuration for this parameter set. Deposited-alloy composition is measured independently of the 91.3189 wt% Pd target specification. The Voce parameters are These parameters enter the total-flow-stress law without an

additional yield-stress increment. The three-dimensional VCCT formulation includes Modes I, II, and III:

$$\begin{aligned} G_I &= \frac{F_z \Delta w}{2b_f \Delta a}, & G_{II} &= \frac{F_x \Delta u}{2b_f \Delta a}, \\ G_{III} &= \frac{F_y \Delta v}{2b_f \Delta a}, & G_{\text{total}} &= G_I + G_{II} + G_{III}, \\ SF_G &= 24.50/18.40 = 1.33152174. \end{aligned} \quad (469)$$

Signed nodal forces and relative displacements require a common crack-closure convention. The nominal simulation values are 18.40 J/m² total energy release and 142.5 MPa peak shear. For this coefficient set, the 44,820-cycle Coffin–Manson bound is $\Delta \varepsilon_p \leq 0.000983786284 = 0.0983786\%$. The design cap is 0.098%.

Table XLVIII associates microstructural trap families with capture and release rates. Its $N_j, E_{t,j}$ parameterization is distinct from the two-family case in Table XLIV. The molar solubilities in Table XLV require conversion to atom-count C_L before use in $k_j C_L$. The equilibrium solution is $\theta_j = k_j C_L / (k_j C_L + p_j)$. These 550 K assignments do not independently define Arrhenius laws; temperature extrapolation requires compatible attempt frequencies, binding energies, and detailed balance. The nominal metrics $K_{IH} = 41.8 \text{ MPa}\sqrt{\text{m}}$, $\dot{a} < 10^{-11} \text{ m/s}$, and $I_{RA} = 0.89$ retain their material-heat and test-condition scope.

V. THERMAL HYDRODYNAMICS, TEG ACCOUNTING, AND PRECISION CALORIMETRY

The earlier 12.5 L/min single-phase water-loop duty is derived in Section II-C1 as a retained component comparison. The definitive hydraulic boundary is the 64-channel, 4.85 L/min Eulerian–Eulerian operating point in Section IX. Hydraulic dissipation, recovered drive power, and consolidated auxiliary electricity occupy distinct terms of the governing ledger in Table XI. The manifold, sensor, and Bi–Te/PbTe calculations below retain their stated historical conditions and are not mixed into the definitive allocation.

The thermal measurement architecture couples three-dimensional manifold mixing, differential resistance thermometry, envelope calorimetry, and synchronized electrical measurements. The hydraulic model establishes the sensing-plane flow and temperature distributions; the measurement model converts those observables into heat and power with an explicit covariance budget. Thermoelectric generation and auxiliary consumption are accounted for at separate electrical terminals. All quoted design figures retain their fluid-property, calibration, and control-volume definitions.

A. Solid-State Equation of State and Debye Thermodynamics

a) Reference state and Birch–Murnaghan compression:

The solid substrate, film, joint, and spring phases require state-consistent volume, heat capacity, and elastic response. Let V_m be volume per mole of formula units and ν_a the number of atoms per formula unit. Composition and phase are held fixed in the following derivatives unless specified otherwise. The hydrostatic pressure p is positive in compression, so $p =$

$-\sigma_m$ for a purely hydrostatic stress under the tension-positive mechanical convention. The third-order Birch–Murnaghan cold-curve representation is [125]

$$\begin{aligned} x &= V_m/V_{m0}, & q &= x^{-2/3}, \\ E_{\text{BM}}(V_m) &= E_0 + \frac{9B_0 V_{m0}}{16} [B'_0(q-1)^3 \\ &\quad + (q-1)^2(6-4q)], \\ p_{\text{BM}}(x) &= -\frac{\partial E_{\text{BM}}}{\partial V_m} \\ &= \frac{3B_0}{2}(x^{-7/3} - x^{-5/3}) \\ &\quad \times \left[1 + \frac{3}{4}(B'_0 - 4)(x^{-2/3} - 1) \right]. \end{aligned} \quad (470)$$

Here E_{BM} is in J/mol, V_{m0} is the static reference volume, B_0 is in Pa, and $B'_0 = (\partial B / \partial p)_0$ is dimensionless. This finite-strain expansion is a constitutive approximation to a phase-specific energy–volume relation, not an exact equation for arbitrary compression or a substitute for an atomistic energy calculation. In particular, shear moduli and the anisotropic elastic tensor are additional constitutive inputs.

For a nonlinear volume solve, both factors in the pressure law must be differentiated. Defining $A(x) = x^{-7/3} - x^{-5/3}$ and $B(x) = 1 + \frac{3}{4}(B'_0 - 4)(x^{-2/3} - 1)$ gives

$$\begin{aligned} A'(x) &= -\frac{7}{3}x^{-10/3} + \frac{5}{3}x^{-8/3}, \\ B'(x) &= -\frac{1}{2}(B'_0 - 4)x^{-5/3}, \\ \frac{dp_{\text{BM}}}{dx} &= \frac{3}{2}B_0[A'(x)B(x) + A(x)B'(x)], \\ B_{\text{BM}}(x) &= -x \frac{dp_{\text{BM}}}{dx}. \end{aligned} \quad (471)$$

Thus $p_{\text{BM}}(1) = 0$ and $B_{\text{BM}}(1) = B_0$. The admissible branch has $V_m > 0$ and positive isothermal bulk modulus; a formal algebraic root outside that branch is not a stable material state.

b) *Debye temperature, density of states, and heat capacity:* With atomic number density $n_a = \nu_a N_A / V_m$, the Debye cutoff and temperature are

$$\begin{aligned} k_D &= (6\pi^2 n_a)^{1/3}, & \Theta_D &= \frac{\hbar v_D k_D}{k_B}, \\ v_D^{-3} &= \frac{1}{12\pi} \sum_{s=1}^3 \int_{S^2} v_s(\hat{\mathbf{n}})^{-3} d\Omega, \end{aligned} \quad (472)$$

$$C_{ijkl}^S \hat{n}_j \hat{n}_k e_{s,l} = \rho v_{s,l}^2 e_{s,i}.$$

The sound speeds follow from the adiabatic acoustic tensor $\mathbf{C}^S$, not the bulk EOS alone. In an isotropic restriction, $v_D^{-3} = (v_L^{-3} + 2v_T^{-3})/3$, $v_L^2 = (B_S + 4G/3)/\rho$, and $v_T^2 = G/\rho$. Independent shear or phonon information is therefore necessary. Assuming $v_D \propto V_m^{-1/3}$ would be an additional approximation, not a consequence of the Debye mode count.

TABLE XLVIII
MICROSTRUCTURAL TRAP-FAMILY KINETICS AT 550 K

Material	Trap family	N_j (m^{-3})	$E_{t,j}$ (kJ/mol)	k_j ($\text{m}^3 \text{atom}^{-1} \text{s}^{-1}$)	p_j (s^{-1})
Pd _{0.95} Ir _{0.05}	Dislocations	1.50×10^{23}	26.4	1.15×10^{-25}	24.50
	Phase boundaries	8.00×10^{22}	44.1	8.40×10^{-26}	120.40
Ti	Interstices	1.00×10^{25}	58.3	2.80×10^{-27}	0.15
	Grain boundaries	5.00×10^{24}	32.1	1.95×10^{-27}	1.42
ϵ -Cu ₃ Sn	Kirkendall voids	6.20×10^{25}	30.8	4.20×10^{-28}	0.08
	Interfaces	1.10×10^{26}	18.5	3.10×10^{-28}	0.75

For $y = \Theta_D/T$ and fixed volume, define the Debye integral and molar phonon free energy [126]:

$$\begin{aligned}
 D_3(y) &= \frac{3}{y^3} \int_0^y \frac{z^3}{e^z - 1} dz, \\
 F_D &= \frac{9}{8} \nu_a R \Theta_D + 3 \nu_a R T \ln(1 - e^{-y}) - \nu_a R T D_3(y), \\
 U_D &= F_D - T \left(\frac{\partial F_D}{\partial T} \right)_{V_m} = \frac{9}{8} \nu_a R \Theta_D + 3 \nu_a R T D_3(y), \\
 C_{V,D} &= \left(\frac{\partial U_D}{\partial T} \right)_{V_m} \\
 &= 9 \nu_a R \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} \frac{z^4 e^z}{(e^z - 1)^2} dz.
 \end{aligned} \tag{473}$$

Here $\Theta_D(V_m)$ is held constant in fixed-volume temperature derivatives; explicit anharmonic temperature dependence requires additional derivative terms. The zero-point term contributes to energy and pressure but not to this fixed-volume heat capacity. The correct limiting forms are

$$\begin{aligned}
 \frac{C_{V,D}}{3 \nu_a R} &= 1 - \frac{y^2}{20} + \frac{y^4}{560} + O(y^6), \quad y \rightarrow 0, \\
 C_{V,D} &\sim \frac{12\pi^4}{5} \nu_a R \left(\frac{T}{\Theta_D} \right)^3, \quad T/\Theta_D \rightarrow 0, \\
 \frac{dD_3}{dy} &= \frac{3}{e^y - 1} - \frac{3D_3(y)}{y}.
 \end{aligned} \tag{474}$$

The heat-capacity series is not the series for D_3 itself. Evaluation uses small-argument expansions and stable exponential differences near zero, and the equivalent integrand $z^4 e^{-z}/(1 - e^{-z})^2$ at large z . Quadrature convergence and derivative convergence must both be checked. Debye acoustic averaging is not a complete phonon spectrum for an ordered multicomponent joint; resolved phonon modes are needed when optical branches or anharmonicity dominate.

c) *Finite-temperature EOS and thermodynamic consistency*: A quasiharmonic phase model combines the static energy

with phonon, electronic, and configurational free energies:

$$\begin{aligned}
 F_m(V_m, T) &= E_{\text{BM}} + F_D + F_{\text{el}} + F_{\text{conf}}, \\
 p_{\text{int}} &= - \left(\frac{\partial F_m}{\partial V_m} \right)_T, \quad p_{\text{int}} = p_{\text{ext}}, \\
 \gamma_D &= - \frac{\partial \ln \Theta_D}{\partial \ln V_m}, \quad p_D = \frac{\gamma_D U_D}{V_m}, \\
 B_T &= -V_m \left(\frac{\partial p_{\text{int}}}{\partial V_m} \right)_T > 0, \\
 \alpha_V &= \frac{1}{V_m} \left(\frac{\partial V_m}{\partial T} \right)_p = \frac{1}{B_T} \left(\frac{\partial p_{\text{int}}}{\partial T} \right)_{V_m}, \\
 C_{P,m} - C_{V,m} &= T \alpha_V^2 B_T V_m.
 \end{aligned} \tag{475}$$

The formula for p_D assumes that the phonon frequencies depend on volume through $\Theta_D(V_m)$ with no independent explicit temperature dependence. The static curve here excludes phonon zero-point and thermal energies; if the reference EOS already includes a finite-temperature contribution, the corresponding reference free energy must be subtracted before adding a thermal correction. Phase mixing and configurational energies are counted once, consistently with Eq. (378). Equilibrium volume minimizes $F_m + p_{\text{ext}} V_m$ on the selected stable phase branch.

For the residual $r(x) = p_{\text{int}}(V_{m0}x, T) - p_{\text{ext}}$, a safeguarded Newton step is $x_{k+1} = x_k - r(x_k)/r'(x_k)$ within a positive-volume bracket. Stopping requires converged residual and state change, not a fixed iteration count. Phase transitions require branch selection and may invalidate a single smooth sensitivity. The isothermal elastic tensor follows from second strain derivatives of the same free energy at a specified reference state; finite prestress requires the appropriate incremental-stiffness correction. The full tensor uses the corresponding strain derivatives of the free energy.

The heat capacities and enthalpy used for thermal accounting follow from this same potential:

$$\begin{aligned}
 S_m &= - \left(\frac{\partial F_m}{\partial T} \right)_{V_m}, \quad C_{V,m} = -T \left(\frac{\partial^2 F_m}{\partial T^2} \right)_{V_m}, \\
 H_m &= F_m + T S_m + p V_m, \quad C_{P,m} = \left(\frac{\partial H_m}{\partial T} \right)_p.
 \end{aligned} \tag{476}$$

Molar, mass-specific, and volumetric heat capacities remain distinct: $c_p = C_{P,m}/M_m$, $C_{P,\text{vol}} = C_{P,m}/V_m$, and $\rho = M_m/V_m$, where M_m is molar mass per formula unit. These solid-state relations do not replace the water or water-glycol property functions used by the calorimeter. Those fluid functions retain their own composition, pressure, and calibration uncertainties.

B. Drude–Lorentz Dielectric and Thermal Transport Tensors

For a local, reciprocal, linear-response medium with time convention $e^{-i\omega t}$, separate free-carrier and bound-oscillator polarizations. With tensor oscillator strengths and scalar damping for each oscillator,

$$\begin{aligned}\ddot{\mathbf{P}}_D &= -\gamma_D^{\text{opt}} \dot{\mathbf{P}}_D + \epsilon_0 W_D \mathbf{E}, \\ \ddot{\mathbf{P}}_\ell &= -\gamma_\ell \dot{\mathbf{P}}_\ell - \omega_\ell^2 \mathbf{P}_\ell + \epsilon_0 S_\ell \mathbf{E}, \\ \epsilon_r(\omega) &= \epsilon_\infty - \frac{W_D}{\omega^2 + i\gamma_D^{\text{opt}}\omega} \\ &\quad + \sum_\ell \frac{S_\ell}{\omega_\ell^2 - \omega^2 - i\gamma_\ell\omega}.\end{aligned}\quad (477)$$

The optical damping γ_D^{opt} is unrelated to the Debye Grüneisen parameter γ_D . Frequencies are angular frequencies in rad/s, and W_D, S_ℓ have units s^{-2} . The model uses tensor oscillator strengths [127]. For parabolic carrier bands, the free-carrier specialization and its conductivity are

$$\begin{aligned}W_D &= \frac{n_e e^2}{\epsilon_0} m_{\text{eff}}^{-1}, \\ \sigma_D(\omega) &= \frac{\epsilon_0 W_D}{\gamma_D^{\text{opt}} - i\omega}, \\ \epsilon_r^{\text{lab}} &= R \epsilon_r^{\text{crystal}} R^T.\end{aligned}\quad (478)$$

The orthogonal matrix R fixes crystal orientation. Multiband carriers, anisotropic relaxation, gyrotropy, and spatial dispersion require richer response models. Damping, effective masses, oscillator strengths, and resonances depend on composition, phase, temperature, and carrier state; they require electronic calculations or optical characterization. Homogenization across phases is a boundary-value problem, not necessarily a volume-weighted scalar average. The local Drude–Lorentz tensor is a constitutive optical model and does not replace the reciprocal-space Floquet dielectric matrix or its nuclear-screening calculation.

For positive damping, positive-semidefinite real symmetric strengths, and a passive background, the anti-Hermitian part and absorbed power density obey

$$\begin{aligned}\epsilon_A &= \frac{\epsilon_r - \epsilon_r^\dagger}{2i} \succeq 0, \quad \omega > 0, \\ q_{\text{abs}} &= \frac{\omega \epsilon_0}{2} \mathbf{E}^\dagger \epsilon_A \mathbf{E} \geq 0.\end{aligned}\quad (479)$$

The loss sign is tied to the stated time convention. Pumped gain media and explicitly modulated media need drive-work accounting and are not covered by this passive inequality. Absorbed optical power enters the thermal balance at the same control-volume boundary used for the electromagnetic calculation.

Heat capacity does not determine thermal conductivity without scattering information. A phonon relaxation-time approximation and a degenerate-electron comparison are

$$\begin{aligned}k_{ij}^{\text{ph}} &= \frac{1}{V_{\text{cell}}} \sum_\lambda w_\lambda C_\lambda v_{\lambda i} v_{\lambda j} \tau_\lambda, \\ \mathbf{k}_{\text{el}} &\simeq L_0 T \sigma_{\text{dc}}, \quad L_0 = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2, \\ \mathbf{k} &= \mathbf{k}_{\text{ph}} + \mathbf{k}_{\text{el}}.\end{aligned}\quad (480)$$

Here C_λ is the heat capacity per mode, w_λ is the Brillouin-zone quadrature weight, and τ_λ is the mode lifetime. The Wiedemann–Franz form assumes degenerate carriers and compatible electrical/thermal scattering; it is not a universal law for segmented thermoelectric materials. All material parameters and their correlations enter the uncertainty model below rather than being treated as exact consequences of an EOS.

C. Three-Dimensional Manifold Hydrodynamics

The manifold model uses liquid-water reference properties $\rho = 997.0 \text{ kg/m}^3$, $\mu = 8.9 \times 10^{-4} \text{ Pa s}$, $c_p = 4182 \text{ J/(kg K)}$, and $k = 0.607 \text{ W/(m K)}$ [87]. These properties define a single fluid state and shall not be substituted for the separate water–glycol microchannel model. The incompressible, constant-property governing equations are

$$\begin{aligned}\nabla \cdot \mathbf{u} &= 0, \\ \rho [\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}] &= -\nabla p + \mu \nabla^2 \mathbf{u}, \\ \rho c_p [\partial_t T + \mathbf{u} \cdot \nabla T] &= k \nabla^2 T + \Phi_{\text{visc}}, \\ \Phi_{\text{visc}} &= 2\mu \mathbf{S} : \mathbf{S}, \\ \mathbf{S} &= \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T].\end{aligned}\quad (481)$$

Here $\mathbf{u} = (u_x, u_y, u_z)$ is velocity, p is mechanical pressure, and Φ_{visc} is the viscous heating rate per volume. The steady equations follow by setting the time derivatives to zero. Wall heat transfer, inlet enthalpy, outlet pressure, manifold shape, and no-slip boundaries complete the fluid problem; conjugate heat conduction is required where wall temperatures are not imposed.

For Reynolds-averaged turbulence, the momentum and thermal equations additionally contain $-\rho \nabla \cdot \overline{\mathbf{u}' \mathbf{u}'}$ and $-\rho c_p \nabla \cdot \overline{\mathbf{u}' T'}$, respectively. A mean-flow solution therefore requires a Reynolds-stress and turbulent-heat-flux closure, not merely the steady laminar operators [99]. In a gradient-diffusion closure,

$$\rho c_p \overline{\mathbf{u}' T'} = -k_t \nabla T, \quad k_t = \mu_t c_p / \text{Pr}_t, \quad (482)$$

where μ_t and Pr_t are the eddy viscosity and turbulent Prandtl number. The turbulent Prandtl number is distinct from the molecular value specified below. Turbulence-model choice, inlet fluctuations, wall treatment, mesh refinement, and thermal-flux convergence are required to establish a quantitative mixing prediction.

The operating flow envelope and hydraulic scales are

$$\begin{aligned}\dot{m} &\in [0.08, 0.12] \text{ kg/s}, \\ D_h &= 0.015 \text{ m}, \quad A_c = 1.7671 \times 10^{-4} \text{ m}^2, \\ v &= \dot{m} / (\rho A_c), \\ \text{Re} &= \rho v D_h / \mu = \dot{m} D_h / (A_c \mu), \\ \text{Pr} &= c_p \mu / k = 6.13176277 \simeq 6.13, \\ v_{\text{min}} / v_{\text{max}} &= 0.45408139 / 0.68112208 \text{ m/s}, \\ \text{Re}_{\text{min}} / \text{Re}_{\text{max}} &= 7630.09794 / 11445.14691.\end{aligned}\quad (483)$$

The nominal manifold Reynolds envelope is 7629.9–11444.8, with the flow sweep in Table XLIX. Downstream microchannels and sensing-plane mixing use their local flow conditions.

D. Sensing-Plane Thermal Dispersion

Four cross-calibrated Pt100 RTDs sample the downstream sensing plane. The calorimetric temperature is the mass-flux-weighted bulk temperature, not generally the unweighted spatial mean:

$$T_b = \frac{\int_{A_s} \rho u_n T dA}{\int_{A_s} \rho u_n dA}, \quad u_n = \mathbf{u} \cdot \hat{\mathbf{n}}. \quad (484)$$

For one-way flow, a corresponding spatial dispersion is

$$\sigma_{T,\text{spatial}}^2 = \frac{\int_{A_s} \rho u_n (T - T_b)^2 dA}{\int_{A_s} \rho u_n dA}. \quad (485)$$

Backflow requires a separately defined sampling and flux convention. Sensor locations, insertion depth, thermal response, and estimator weights determine how this field dispersion contributes to the error in the RTD estimate of T_b . The sensor estimator retains common bias and spatial covariance.

The spatial-drift design conversion uses $c_{\Delta T,\text{sp}} = 185.4 \text{ W/K}$ and

$$\begin{aligned} u_{\text{spatial_drift}}^2 &= (c_{\Delta T,\text{sp}} \sigma_{T,\text{spatial}})^2, \\ u_{\text{spatial_drift,min flow}}^2 &= (185.4 \times 0.0142)^2 \\ &= 6.9310039824 \text{ W}^2 \simeq 6.9310 \text{ W}^2, \\ u_{\text{spatial_drift,max flow}}^2 &= (185.4 \times 0.0098)^2 \\ &= 3.3011982864 \text{ W}^2 \simeq 3.3012 \text{ W}^2. \end{aligned} \quad (486)$$

The molecular Prandtl number is 6.13 throughout this constant-property sweep. The nominal maximum spatial variance is $6.9310 \text{ W}^2 \leq 8.7501 \text{ W}^2$ for the specified sensitivity and sampled temperature dispersions.

For direct enthalpy-flow calorimetry, $\partial P_{\text{flow}}/\partial \Delta T = \dot{m}c_p$, ranging from 334.56 to 501.84 W/K over the specified flow range. The 185.4 W/K sensitivity uses a separate estimator weighting or calibrated transfer factor.

E. Matched-RTD Calibration and GUM Budget

a) Differential propagation and Monte Carlo comparison:

The first-order GUM budget uses the forward-mode automatic-differentiation Jacobian in Section V-J. Monte Carlo calorimetry separately propagates the joint distribution of $\dot{m}$, c_p , ΔT , P_{env} through $P = \dot{m}c_p\Delta T + P_{\text{env}}$, retaining the covariance of shared temperature references. It tests linearization and evaluates coverage when the output distribution is appreciably nonlinear or non-Gaussian [130]. Sample count, generator, seed, and worker partition are fixed numerical parameters. Convergence of the mean, standard uncertainty, and coverage quantiles is checked separately.

b) *GUM-based synthetic telemetry*: For a telemetry comparison with $P = \dot{m}c_p(T_{\text{out}} - T_{\text{in}})$ and fixed c_p , the independent-sensor approximation is

$$\begin{aligned} u_c^2(P) &= (c_p \Delta T)^2 u^2(\dot{m}) \\ &+ (c_p \dot{m})^2 [u^2(T_{\text{out}}) + u^2(T_{\text{in}})]. \end{aligned} \quad (487)$$

The synthetic-telemetry settings are $c_p = 4184.0 \text{ J/(kg K)}$, $u(\dot{m}) = 1.2 \times 10^{-4} \text{ kg/s}$, and $u(T_{\text{out}}) = u(T_{\text{in}}) = 0.04 \text{ K}$. Correlated sensors, heat capacity, and environmental heat enter

through their sensitivities and covariance. The Gaussian-noise comparison uses $P_{\text{syn}} = P_{\text{true}} + u_c(P)\xi$, $\xi \sim \mathcal{N}(0, 1)$. A combined standard uncertainty of 5.69 W corresponds to $U = 11.38 \text{ W}$ at $k = 2$, with coverage determined by the distribution and effective degrees of freedom.

The matched Pt100 differential-temperature standard uncertainty is $u(\Delta T) = 0.0039 \text{ K}$. Multipoint calibration shall characterize pair offset, self-heating, immersion, readout error, and common-reference covariance. For $\Delta T = T_{\text{out}} - T_{\text{in}}$,

$$u^2(\Delta T) = u^2(T_{\text{out}}) + u^2(T_{\text{in}}) - 2 \text{cov}(T_{\text{out}}, T_{\text{in}}). \quad (488)$$

The differential uncertainty is not an independent uncertainty assigned to each RTD. The nominal calibration sensitivity $c_{\Delta T} = 442.90 \text{ W/K}$ gives

$$(c_{\Delta T} u(\Delta T))^2 = (442.90 \times 0.0039)^2 = 2.9835998361 \text{ W}^2. \quad (489)$$

The ISO/IEC Guide 98-3 / JCGM 100 measurement model uses signed sensitivity coefficients and an input covariance matrix [82]:

$$u_c^2 = \mathbf{c}^T \mathbf{U} \mathbf{c}, \quad c_i = \frac{\partial f}{\partial x_i}, \quad U_{ij} = \text{cov}(x_i, x_j). \quad (490)$$

The aggregate design allocation assigns $u_{\text{instrumentation}}^2 = 12.50 \text{ W}^2$, including the RTD contribution and all other instrument terms, with zero cross-covariance between that aggregate and spatial drift. Under those assumptions,

$$\begin{aligned} u_c &= \sqrt{u_{\text{spatial_drift}}^2 + u_{\text{instrumentation}}^2} \\ &= \sqrt{6.9310 + 12.50} \\ &= 4.408060798 \text{ W} \simeq 4.4081 \text{ W}, \\ 4.4081 \text{ W} &\leq 5.6881 \text{ W} \leq 5.7000 \text{ W}. \end{aligned} \quad (491)$$

This is the algebraic combined-uncertainty result for the two-term allocation. The RTD term shall not be added again if it is already included in 12.50 W². A detailed instrument budget must independently justify that allocation and the covariance assumptions.

Table L specifies a detailed calibration case [86]. The listed u_i are standard uncertainties; rectangular distribution limits require conversion before use if they are supplied as half-widths instead. Type A and Type B describe evaluation methods, not distribution shapes. The covariance entry is a net cross-term, not an independent diagonal variance; a positive-semidefinite covariance matrix and the signed measurement sensitivities must support it.

Direct evaluation of the detailed budget gives

$$\begin{aligned} \sum_i (c_i u_i)^2 &= 16.28077656 \text{ W}^2, \\ u_{\text{instr,detail}}^2 &= 16.28077656 + 4.2560 \\ &= 20.53677656 \text{ W}^2, \\ u_{c,\text{detail}}^2 &= 20.53677656 + 8.7501 \\ &= 29.28687656 \text{ W}^2, \\ u_{c,\text{detail}} &= 5.41173508 \text{ W}. \end{aligned} \quad (492)$$

The calibration operating point gives $\dot{m}c_p = 442.6672 \text{ W/K}$ and $\dot{m}c_p\Delta T = 2909.385905 \text{ W}$. The tabulated sensitivities

TABLE XLIX
MANIFOLD THERMAL DESIGN SWEEP AND DIRECT NUMERICAL EVALUATION

$\dot{m}$ (kg/s)	Re_{nom}	Re_{calc}	$\sigma_{T,spatial}$ (K)	$u_{sp,nom}^2$ (W ²)	$u_{sp,calc}^2$ (W ²)	$u_{c,nom}$ (W)	$u_{c,calc}$ (W)
0.08	7629.9	7630.09794	0.0142	6.9310	6.931004	4.4081	4.408061
0.09	8583.6	8583.86018	0.0128	5.6312	5.631699	4.2580	4.258133
0.10	9537.3	9537.62242	0.0116	4.6250	4.625252	4.1382	4.138267
0.11	10491.0	10491.38467	0.0106	3.8615	3.862168	4.0450	4.045018
0.12	11444.8	11445.14691	0.0098	3.3012	3.301198	3.9750	3.975072

TABLE L
MATCHED-RTD CALORIMETRIC INPUT BUDGET AT THE CALIBRATION OPERATING POINT

Input (native unit)	Nominal	u_i	Evaluation / distribution	$ c_i $ (W/native unit)	Nominal variance (W ²)	$(c_i u_i)^2$
Mass flow (kg/s)	0.1058	0.00012	B / normal	27518.0	10.8993	10.904260666
c_p (J/(kg K))	4184.0	0.85	B / normal	0.6958	0.3530	0.349789445
ΔT (K)	6.5724	0.0039	A / normal	442.90	2.9836	2.983599836
Line voltage (V)	230.00	0.015	B / rectangular	12.658	0.0361	0.036050617
Line current (A)	12.658	0.0012	B / rectangular	230.00	0.0762	0.076176000
Radiative heat (W)	42.15	1.150	B / normal	1	1.3225	1.322500000
Conductive heat (W)	18.40	0.780	B / normal	1	0.6084	0.608400000
Assigned net calibration covariance contribution					4.2560	4.2560
Spatial-inhomogeneity allowance					8.7501	8.7501

are nominal calibration coefficients. The measurement function includes the envelope terms and line-power boundary. The 4.4081 W figure is a conditional aggregate design allocation; 5.6881 W and 5.7000 W are acceptance ceilings.

F. Microchannel Cold Plate and Passive Heat Rejection

The passive heat-rejection design uses a copper cold plate with $W = L = 120$ mm, base thickness $t_b = 2.0$ mm, channel width $w_c = 250$ μ m, fin thickness $w_w = 250$ μ m, and channel height $H_c = 1500$ μ m. The copper conductivity is $k_s = 401.0$ W/(m K). A single water-property reference specifies $\rho = 995.0$ kg/m³, $c_p = 4178.0$ J/(kg K), $\mu = 7.98 \times 10^{-4}$ Pa s, $k_f = 0.615$ W/(m K), and $Pr \simeq 5.42$. These dimensions and properties define a separate operating case from the earlier 7 L/min and water-glycol comparisons.

a) *Channel geometry, flow, and pump demand:* Assuming equal flow through the channels and neglecting edge-wall corrections, the hydraulic geometry and bulk-flow quantities are

$$\begin{aligned}
 N_{ch} &= \left\lfloor \frac{W}{w_c + w_w} \right\rfloor = 240, \\
 A_c &= w_c H_c = 3.75 \times 10^{-7} \text{ m}^2, \\
 \dot{V} &= 1.8 \text{ L/min} = 3.0 \times 10^{-5} \text{ m}^3/\text{s}, \\
 \dot{m} &= \rho \dot{V} = 0.02985 \text{ kg/s}, \\
 D_h &= \frac{2w_c H_c}{w_c + H_c} = 4.2857 \times 10^{-4} \text{ m}, \\
 u_m &= \frac{\dot{V}}{N_{ch} A_c} = 0.3333 \text{ m/s}, \\
 Re &= \frac{\rho u_m D_h}{\mu} = 178.12, \quad \alpha = \frac{w_c}{H_c} = \frac{1}{6}.
 \end{aligned} \tag{493}$$

The nominal flow is laminar. For a fully developed rectangular duct, define the Fanning friction factor f by

$$\begin{aligned}
 f Re &= 24(1 - 1.3553\alpha + 1.9467\alpha^2 \\
 &\quad - 1.7012\alpha^3 + 0.9564\alpha^4 - 0.2534\alpha^5), \\
 \Delta P_{fr} &= 4f \frac{L}{D_h} \frac{\rho u_m^2}{2}, \quad P_{pump} = \frac{\dot{V} \Delta P_{fr}}{\eta_{pump}}.
 \end{aligned} \tag{494}$$

The rectangular-channel correlation at $\alpha = 1/6$ gives $f Re = 19.7045$ and $f = 0.11062$. The pressure drop and pump input are

$$\begin{aligned}
 \Delta P_{fr} &\simeq 6848.77 \text{ Pa} = 0.06849 \text{ bar}, \\
 \eta_{pump} &= 0.65, \quad P_{pump} = 0.3161 \text{ W}.
 \end{aligned} \tag{495}$$

The Darcy factor is $4f$, with f the Fanning factor. Pump efficiency is electrical-to-hydraulic efficiency. The 0.3161 W input covers straight-channel friction; manifolds, piping, the liquid-side heat exchanger, and entrance losses contribute additional loop head. Developing-flow and flow-distribution effects enter the complete loop model.

b) *Convection and fin efficiency:* For the fully developed rectangular-duct constant-wall-heat-flux model,

$$\begin{aligned}
 Nu &= 8.235(1 - 2.0421\alpha + 3.0853\alpha^2 \\
 &\quad - 2.4765\alpha^3 + 1.0578\alpha^4 - 0.1861\alpha^5) \\
 &= 6.0501, \\
 h &= \frac{Nu k_f}{D_h} = 8681.88 \text{ W}/(\text{m}^2 \text{ K}), \\
 m_f &= \sqrt{\frac{2h}{k_s w_w}} \simeq 416.18 \text{ m}^{-1}, \\
 \eta_f &= \frac{\tanh(m_f H_c)}{m_f H_c} = 0.8876 = 88.76\%, \\
 A_{wet,eff} &= N_{ch} L(w_c + 2\eta_f H_c) \simeq 0.0839 \text{ m}^2.
 \end{aligned} \tag{496}$$

The fin model uses two-sided heat transfer and an adiabatic tip; $A_{\text{wet,eff}}$ is the fin-efficiency-weighted area, not the geometric wetted area. Conjugate conduction, entrance effects, and the actual heated-wall boundary condition can change the effective coefficient.

c) *Series thermal-resistance network and base-temperature limit:* With the verified TEG cold-side duty $Q_{\text{cold}} = 2047.8600$ W, the coolant rise and cold-plate resistances are

$$\begin{aligned}\Delta T_{\text{fluid}} &= \frac{Q_{\text{cold}}}{\dot{m}c_p} = 16.42 \text{ K}, \\ R_{\text{cond}} &= \frac{t_b}{k_s WL} = 0.000346 \text{ K/W}, \\ R_{\text{conv}} &= \frac{1}{hA_{\text{wet,eff}}} = 0.001373 \text{ K/W}, \\ R_{\text{cp}} &= R_{\text{cond}} + R_{\text{conv}} = 0.001719 \text{ K/W}, \\ R_{\text{fluid}} &= \frac{1}{\dot{m}c_p} = 0.008018 \text{ K/W}.\end{aligned}\quad (497)$$

Using the full coolant rise defines a maximum-temperature rather than mean-fluid estimate. The remaining series elements are a liquid-to-heat-pipe exchanger, a parallel heat-pipe network, and a natural-convection/radiation sink:

$$\begin{aligned}R_{\text{hx,conv}} &= 0.001500 \text{ K/W}, \\ R_{\text{hp,network}} &= \frac{R_{\text{hp,single}}}{N_{\text{hp}}\eta_{\text{rad,fin}}}, \\ R_{\text{rad}} &= \frac{1}{(h_{\text{nc}} + h_{\text{rad}})A_{\text{rad}}} \\ &= \frac{1}{(5.43 + 5.36)(6.0)} \simeq 0.015446 \text{ K/W}.\end{aligned}\quad (498)$$

Here $A_{\text{rad}} = 6.0 \text{ m}^2$ is the assumed effective radiator area, and the two heat-transfer coefficients have units $\text{W}/(\text{m}^2 \text{ K})$. Their values are operating-point assumptions; natural convection and radiation are temperature dependent.

The heat-pipe network design allocation is $R_{\text{hp,network}}^{\text{alloc}} = 0.001019 \text{ K/W}$. Summing the displayed resistance allocations gives

$$\begin{aligned}\theta_{\text{sa}}^{\text{alloc}} &= R_{\text{cp}} + R_{\text{fluid}} + R_{\text{hx,conv}} \\ &\quad + R_{\text{hp,network}}^{\text{alloc}} + R_{\text{rad}} \\ &= 0.027702 \text{ K/W}, \\ T_{\text{base,max}}^{\text{alloc}} &= T_{\text{amb}} + Q_{\text{cold}}\theta_{\text{sa}}^{\text{alloc}} \\ &= 298.15 + 2047.8600(0.027702) \\ &= 354.88 \text{ K}, \\ T_{\text{base,cap}} &= 357.47 \text{ K} = 84.32^\circ \text{C}.\end{aligned}\quad (499)$$

The two temperatures are respectively the allocated maximum-base estimate and the design/simulation acceptance ceiling; they are not interchangeable outputs. The 2.59 K margin must accommodate any additional contact drops and thermal loads when the ceiling is also used as a TEG cold-junction boundary.

The heat-pipe network allocation with 16 pipes and $\eta_{\text{rad,fin}} = 0.92$ requires approximately $R_{\text{hp,single}} = 0.0150 \text{ K/W}$, or an equivalent network. The 354.88 K estimate and 357.47 K ceiling are conditional design values. Pump heat and other loop deposition enter the rejection duty once.

TABLE LI
PASSIVE COLD-PLATE BALANCE-OF-PLANT DESIGN LEDGER

Quantity	Allocation	Unit
Thermal input P_{thermal}	2295.58	W
TEG efficiency η_{TEG}	10.79	%
Gross generation P_{TEG}	247.72	W
TEG rejection Q_{cold}	2047.86	W
Laser electrical input	150.00	W
Auxiliary electrical input	50.00	W
Pump electrical input	0.3161	W
Compressor / powered fan	0 / 0	W
Total parasitics $\sum P_{\text{parasitic}}$	200.3161	W
Net export P_{net}	+47.4039	W

G. TEG Efficiency, Electrical Bus, and Heat-Rejection Accounting

a) *Passive-sink TEG efficiency:* The passive cold-plate operating point uses $T_{\text{hot}} = 600.0 \text{ K}$, $T_{\text{cold}} = 357.47 \text{ K}$, and dimensionless average thermoelectric figure of merit $\bar{ZT} = 1.5$. The constant-property, optimally loaded generator model gives

$$\begin{aligned}\eta_{\text{TEG}} &= \left(1 - \frac{T_{\text{cold}}}{T_{\text{hot}}}\right) \frac{\sqrt{1 + \bar{ZT}} - 1}{\sqrt{1 + \bar{ZT}} + T_{\text{cold}}/T_{\text{hot}}} \\ &= \left(1 - \frac{357.47}{600.0}\right) \frac{\sqrt{2.5} - 1}{\sqrt{2.5} + 357.47/600.0} \\ &\simeq 0.1079074 \simeq 10.79\%.\end{aligned}\quad (500)$$

This model efficiency is conditional on the stated junction temperatures and figure of merit. Thermal contacts, electrical interconnects, temperature-dependent properties, and power conversion must either be included in an effective bus efficiency or debited separately. The 357.47 K cold-side condition requires the heat-pipe network allocation in Section V-F to be met; it does not follow from the higher-resistance hardware assignment.

b) *Balance of plant and net export:* The thermal and electrical balances for this operating point are

$$\begin{aligned}P_{\text{thermal}} &= \frac{Q_{\text{cold}}}{1 - \eta_{\text{TEG}}}, \quad P_{\text{TEG}} = \eta_{\text{TEG}} P_{\text{thermal}}, \\ \sum P_{\text{parasitic}} &= P_{\text{laser}} + P_{\text{aux}} + P_{\text{pump}} \\ &= 150.0 + 50.0 + 0.3161 = 200.3161 \text{ W}, \\ P_{\text{net}} &= P_{\text{TEG}} - \sum P_{\text{parasitic}}.\end{aligned}\quad (501)$$

Table LI specifies the nominal allocation, with $2295.58 - 247.72 = 2047.86 \text{ W}$ thermal rejection and $247.72 - 200.3161 = 47.4039 \text{ W}$ net electrical power. Its implied conversion efficiency is $247.72/2295.58 \simeq 10.7912\%$.

The 150 W laser and 50 W auxiliary values are electrical bus debits; their absorbed and rejected heat fractions use explicit thermal boundaries. Zero compressor and fan debits assume the allocated natural-convection/radiation sink. The 0.3161 W pump input covers channel head only. The +47.4039 W net allocation is conditional on complete loop hydraulics, effective TEG conversion, and the specified heat-pipe network.

TABLE LII
EARLIER Bi-Te/PbTe GENERATION, LOSSES, AND BUS EXPORT

Accounting term	Power (W)
Gross Seebeck generation	+490.57
Thermal-contact equivalent debit	-28.45
Electrical interconnect loss	-18.60
Thermal-bypass equivalent debit	-12.30
Raw generator DC	431.22
DC-DC conversion loss	-12.79
Regulated bus generation	418.43
Pump allocation	-116.67
Laser-driver allocation	-120.00
Other support allocation	-150.00
Total support demand	386.67
Net regulated-bus export	+31.76

This passive case uses separate temperatures and loads from Table XI.

c) *Earlier generator and bus comparisons:* The earlier segmented Bi₂Te₃/PbTe generator comparison uses hot- and cold-side temperatures of 623.15 and 353.15 K and thermal input 2911.40 W. The separate skutterudite/half-Heusler allocation is Table XI; its 3093.4400 W hot-side duty, partial-loop compressor, and separate pump follow Eqs. (209)–(214). The historical loss ledger below is not applied to either the passive-sink 10.79% model or the assigned 33.80% efficiency. Table LII resolves the earlier electrical-equivalent generation and loss allocation. Thermal-contact and bypass losses are represented as reductions in recoverable electrical power, not automatically as equal heat flows. The corresponding identities are

$$\begin{aligned}
 P_{\text{raw}} &= 490.57 - 28.45 - 18.60 - 12.30 = 431.22 \text{ W}, \\
 P_{\text{bus}} &= 431.22 - 12.79 = 418.43 \text{ W}, \\
 P_{\text{support}} &= 116.67 + 120.00 + 150.00 = 386.67 \text{ W}, \\
 P_{\text{net}} &= 418.43 - 386.67 = 31.76 \text{ W}, \\
 \eta_{\text{DC/DC}} &= 418.43/431.22 = 0.9703399657.
 \end{aligned} \tag{502}$$

Quantitative accounting uses unrounded conversion efficiencies or measured terminal powers.

At the combined generator/converter boundary, heat rejection is $2911.40 - 418.43 = 2492.97$ W. With all 386.67 W support demand dissipated inside the overall boundary, environmental rejection is 2879.64 W, satisfying $2879.64 + 31.76 = 2911.40$ W. This closure assumes a common thermal input and complete electrical-port accounting. The 1850.00 W optical and 65.59 W RF calorimetric inputs use their separate port mapping. The 31.76 W regulated-bus allocation has its own electrical uncertainty budget.

H. GUM Calorimetry and Excess-Heat Estimation

The dual calorimeter combines coolant enthalpy with an envelope thermopile. Table LIII defines the nominal inputs, sensitivity coefficients, and variance contributions [79], [80]. Each input uncertainty u_i retains its native unit; only the product $c_i u_i$ has units of output power. A complete measurement function must define how the two calorimetric channels contribute to a single estimator.

a) *Eight-variable GUM covariance architecture:* The definitive ISO/IEC Guide 98-3 measurement vector combines thermal, helium-assay, and pulse-shape inputs as

$$\mathbf{X} = [Q \quad T_{\text{in}} \quad T_{\text{out}} \quad UA_{\text{loss}} \quad I_{\text{He}} \quad S_{\text{He}} \quad Q_t \quad Q_T]^\top. \tag{503}$$

Writing $u_i = u(X_i)$, the selected 8×8 covariance matrix retains the shared temperature-reference and common-waveform correlations:

$$\Sigma_X = \begin{bmatrix} u_1^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u_2^2 & c_{23} & 0 & 0 & 0 & 0 & 0 \\ 0 & c_{23} & u_3^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & u_4^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & u_5^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & u_6^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & u_7^2 & c_{78} \\ 0 & 0 & 0 & 0 & 0 & 0 & c_{78} & u_8^2 \end{bmatrix}, \tag{504}$$

where

$$c_{23} = 0.85u_2u_3, \text{quad}c_{78} = 0.92u_7u_8. \tag{505}$$

Thus $r_{T_{\text{in}}, T_{\text{out}}} = 0.85$ and $r_{Q_t, Q_T} = 0.92$; the latter correlation arises because the tail and total charge gates integrate the same digitized waveform. For $\mathbf{Y} = \mathbf{f}(\mathbf{X})$, forward-mode differentiation supplies the Jacobian $\mathbf{J}$, and the GUM law of propagation is

$$\Sigma_Y = \mathbf{J}\Sigma_X\mathbf{J}^\top, \quad u(Y_i) = \sqrt{(\Sigma_Y)_{ii}}. \tag{506}$$

A correlated Monte Carlo propagation with $N = 10^6$ draws confirms $u(P_{\text{thermal}}) \simeq 11.8$ W (11.81 W in the verification record). For the definitive power ledger $P_{\text{net}} = 0.338P_{\text{thermal}} - P_{\text{drive, net}} - P_{\text{aux}}$, holding the separately assigned drive and auxiliary terms fixed gives $u(P_{\text{net}}) \simeq 3.99$ W. The corresponding three-standard-uncertainty interval is

$$P_{\text{net}} = 555.03 \pm 11.98 \text{ W} \quad (3\sigma), \tag{507}$$

or approximately [543.05, 567.01] W, which remains strictly positive. Any commissioned covariance assigned to conversion, drive, or auxiliary terms shall be added before reporting the final plant-level interval.

Under the ISO/IEC Guide 98-3 / JCGM 100 GUM framework [82], the measurement function, signed sensitivities and covariances determine

$$\begin{aligned}
 u_c^2 &= \sum_i c_i^2 u_i^2 + 2 \sum_{i < j} c_i c_j u_{ij}, \\
 \sum_i (c_i u_i)_{\text{nominal}}^2 &= 19.370 \text{ W}^2,
 \end{aligned} \tag{508}$$

$$u_{\text{RSS}} = \sqrt{19.370} = 4.4011 \text{ W}.$$

The nominal combined uncertainty is a 5.69 W design allocation. The complete budget uses the covariance matrix, additional variance contributions, signed sensitivities, thermopile and flow-channel weighting, and calibration records. Type A/B identifies the evaluation method; rectangular distributions specify whether a limit is a half-width or an evaluated standard uncertainty.

$$\begin{aligned}
 P_{\text{out}} &= 2911.40 \text{ W}, \\
 P_{\text{in}} &= 1850.00 + 65.59 = 1915.59 \text{ W}, \\
 P_{\text{excess}} &= 995.81 \text{ W}, \\
 Z &= 995.81/5.69 = 175.010545 \simeq 175.01.
 \end{aligned} \tag{509}$$

TABLE LIII
DUAL-CHANNEL CALORIMETRIC INPUTS AND NOMINAL VARIANCE CONTRIBUTIONS

Input (nominal value)	Type	Distribution	u_i (native unit)	c_i (W/native unit)	Nominal variance (W ²)
Mass flow (60.33 g/s)	A	Normal	0.0302	48.26	2.124
Temperature difference (11.50 K)	A	Normal	0.0120	253.16	9.228
Specific heat (3.582 J/(g K))	B	Rectangular	0.0021	812.70	2.914
Envelope thermopile (124.50 mV)	A	Normal	0.0800	18.42	2.171
Optical input (1850.00 W)	B	Normal	1.2000	1.00	1.440
RF input (65.59 W)	B	Normal	0.1500	1.00	0.023
Ambient loss correction (15.40 W)	B	Rectangular	1.2124	1.00	1.470

The standardized ratio is conditional on the specified uncertainty and common measurement boundary. Experimental inference uses raw time series, blank runs, electrical-substitution calibrations, correlation estimates, and a specified sampling model. The 995.81 W excess-heat reference and 31.76 W electrical bus allocation are separate measurands.

Internally generated power is estimated from recovered heat, exported electrical power, and electrical input at explicitly defined boundaries. The nominal variance budget provides a design basis rather than an achieved calibration result. Commissioning requires sensor calibration, covariance evaluation, and verification of background subtraction. Each operating configuration shall use its own calibrated measurements and uncertainty estimates.

I. Mass-Flow and Seebeck-Envelope Enthalpy Accounting

The water-glycol calorimeter configuration uses a 50/50 mixture by mass at nominal volumetric flow $\dot{V} = 9.83 \text{ L min}^{-1}$. With reference density $\rho = 1048 \text{ kg m}^{-3}$, the mass-flow conversion is

$$\dot{m} = \rho \dot{V} = \frac{1048(9.83 \times 10^{-3})}{60} \text{ kg s}^{-1} = 0.1716973 \text{ kg s}^{-1}. \quad (510)$$

The variance budget shall use the rounded reference $\dot{m} = 0.1717 \text{ kg s}^{-1}$. A calibrated Coriolis instrument shall provide the primary mass-flow measurement. Any alternative calculation from density and volumetric flow shall include the uncertainties and covariance of those inputs.

The fixed-volume flow cell shall be analyzed as a steady open system. Its transported thermal power is the mass flow multiplied by the enthalpy difference:

$$\begin{aligned} P_{\text{flow}} &= \dot{m} [h(T_{\text{out}}, p_{\text{out}}) - h(T_{\text{in}}, p_{\text{in}})], \\ P_{\text{flow}} &\approx \dot{m} \int_{T_{\text{in}}}^{T_{\text{out}}} c_p(T, p) dT \\ &\approx \dot{m} c_p(T_{\text{avg}}) \Delta T, \\ T_{\text{avg}} &= \frac{1}{2}(T_{\text{out}} + T_{\text{in}}), \quad \Delta T = T_{\text{out}} - T_{\text{in}}. \end{aligned} \quad (511)$$

The reduced $c_p \Delta T$ model requires pressure, kinetic-energy, and potential-energy contributions to be negligible at the specified uncertainty level. Pump work dissipated between the measurement planes enters the input-energy balance. The fluid-property model uses the isobaric heat capacity c_p .

The secondary thermal envelope shall surround the vacuum vessel with a thermopile matrix coupled to a bath maintained

at 298.15 K within a $\pm 0.01 \text{ K}$ control band. Its calibrated boundary-heat estimate is

$$P_{\text{env}} = K_{\text{env}}(V_{\text{env}} - V_0), \quad [K_{\text{env}}] = \text{W mV}^{-1}, \quad (512)$$

where the thermopile voltages are expressed in millivolts. Heat carried by the coolant and heat crossing the surrounding envelope are complementary paths:

$$P_{\text{cal}} = P_{\text{flow}} + P_{\text{env}}. \quad (513)$$

The two channels shall be summed as complementary heat paths. Total-power verification shall use an independently calibrated enclosing heat balance or controlled electrical substitution. Shared temperature references, bath disturbances, and calibration coefficients shall be represented through their covariance in the combined uncertainty model.

For a control volume containing the reactor and its internal conversion hardware, a complete first-law balance is

$$\begin{aligned} P_{\text{gen}} &= P_{\text{cal}} + P_{\text{el,out}} + P_{\text{other,out}} - P_{\text{el,in}} \\ &\quad + \frac{dE_{\text{stored}}}{dt}. \end{aligned} \quad (514)$$

Attribution of P_{gen} to a nuclear process requires accounting for chemical energy release, other energy transport, and changes in stored energy. The calorimeter calibration point is 2911.4 W. Runs with harvesting disabled or electrical output dissipated internally are distinct from electrical-export runs; exported power enters Eq. (514) separately from the remaining measured heat.

a) *Cold-plate geometry and hydraulic response:* The enlarged-channel comparison uses $N_c = 120$, $w_c = 217.94 \mu\text{m}$, $h_c = 871.76 \mu\text{m}$, $L_c = 50.0 \text{ mm}$, and $\dot{V} = 7.0 \text{ L/min}$. With water properties $\rho = 998.2 \text{ kg/m}^3$ and $\mu = 1.002 \times 10^{-3} \text{ Pa s}$, the hydraulic quantities are

$$\begin{aligned} A_f &= N_c w_c h_c = 2.27989649 \times 10^{-5} \text{ m}^2, \\ d_h &= 2w_c h_c / (w_c + h_c) = 348.704 \mu\text{m}, \\ v &= \dot{V} / A_f = 5.11719137 \text{ m/s}, \\ Re &= \rho v d_h / \mu = 1777.61797, \\ f_{\text{Ch}} &= 0.036003943 \quad (\epsilon_r = 0.50 \mu\text{m}), \\ \Delta P_0 &= f_{\text{Ch}} (L_c / d_h) \rho v^2 / 2 = 67.470515 \text{ kPa}. \end{aligned} \quad (515)$$

At $Re = 2370.20$, the Churchill expression gives $f \simeq 0.0321433$. The 350 kPa duty specifies total-loop head, distinct from the calculated straight-channel pressure drop. The remaining 282.529485 kPa is available for manifolds, piping,

TABLE LIV
HYDRAULIC REFERENCE VALUES AND DIRECT NUMERICAL EVALUATION

Quantity	Nominal	Calculated
d_h (μm)	348.70	348.704
Re	2370.20	1777.618
f_{Ch}	0.03812	0.036004
Channel-only ΔP_0 (kPa)	350.00	67.471
Channel-only pump power (W)	116.67	22.490
Total-loop head allocation (kPa)	350.00	350.00
Total-loop pump allocation (W)	116.67	116.667

and other components. Representation entirely by channel-velocity minor losses would require $K_\Sigma = 21.61787$; this is an available loss allocation rather than a measured manifold coefficient. The 116.66667 W pump demand is consistent with the total head and specified efficiency. At fixed flow, channel count, and length, uniform width and height scaling by S gives $v \propto S^{-2}$, $d_h \propto S$, and $\Delta P_0 \propto f(Re, \epsilon_r/d_h)S^{-5}$. In laminar flow, $f \propto 1/Re \propto S$, giving S^{-4} rather than a universal S^{-3} dependence. Scaling from 200/800 μm channels differs from the 150/600 μm geometry associated with the 452.93 kPa calculation. These baseline geometries are not interchangeable. A hydraulic-diameter approximation requires validation against the pressure–flow characteristic of the actual rectangular channels.

b) 120-channel hydraulic and thermal benchmark: The narrow-channel benchmark comprises $N_c = 120$ parallel channels with $w_c = 150 \mu\text{m}$, $h_c = 600 \mu\text{m}$, and $L_c = 50.0 \text{ mm}$ [63]. Its governing relations are

$$\begin{aligned}
 D_h &= \frac{2w_ch_c}{w_c + h_c} = 240 \mu\text{m}, \\
 A_{\text{flow}} &= N_c w_c h_c = 1.08 \times 10^{-5} \text{ m}^2, \\
 v &= \dot{V}/A_{\text{flow}}, \quad Re = \rho v D_h / \mu, \\
 f &= 73.28/Re \quad (\text{prescribed laminar Darcy model}), \\
 \Delta P &= (f L_c / D_h + K_\Sigma) \rho v^2 / 2, \\
 \Delta T &= Q / (\rho \dot{V} c_p).
 \end{aligned} \tag{516}$$

The constant-property water calculation at approximately 293 K uses $\rho = 998.2 \text{ kg/m}^3$, $\mu = 1.002 \times 10^{-3} \text{ Pa}\cdot\text{s}$, and $c_p = 4182 \text{ J/(kg}\cdot\text{K)}$. These properties do not describe the water–glycol loop. Table LV retains the nonnegative manifold and minor-loss sum K_Σ as an explicit parameter requiring characterization. The flow area gives $v = 1.54321 \text{ m/s}$ at 1 L/min. At fixed properties, $\Delta T < 25 \text{ K}$ requires more than 5.26054 L/min at 9150 W and 7.89082 L/min at 13725 W, corresponding to integer design selections of at least 6 and 8 L/min. The peak-duty point has $Re \simeq 2952$. Transitional-flow and local-boiling assessment use pressure, wall heat flux, and heat-transfer data.

c) Transitional-flow correlation and thermal response: For roughness $\epsilon_r = 0.50 \mu\text{m}$, the Churchill Darcy correlation

is [71]

$$\begin{aligned}
 f_{Ch} &= 8 \left[(8/Re)^{12} + (A_c + B_c)^{-3/2} \right]^{1/12}, \\
 A_c &= \left\{ 2.457 \ln \left[\frac{1}{(7/Re)^{0.9} + 0.27\epsilon_r/D_h} \right] \right\}^{16}, \\
 B_c &= (37530/Re)^{16}.
 \end{aligned} \tag{517}$$

The correlation comparison uses rectangular-laminar $f = 73.28/Re$ and Churchill $f_{Ch} \simeq 0.037329$ at $Re = 2582.8$. Each model uses its geometry and validated flow range with the water properties of Eq. (516). At 7 L/min, the Churchill model gives 452.93 kPa and 150.98 W electrical demand at 35% pump efficiency before minor losses. Each operating point uses a consistent fluid-property set.

The heat-transfer correlation is

$$\begin{aligned}
 Nu &= \frac{(f/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)} \left[1 + (D_h/L_c)^{2/3} \right], \\
 Pr &= \mu c_p / k_f, \quad h = Nu k_f / D_h.
 \end{aligned} \tag{518}$$

The assigned $Nu = 14.852$ and $k_f = 0.600 \text{ W/(m}\cdot\text{K)}$ give $h = 37130 \text{ W/(m}^2\cdot\text{K)}$. The water-property correlation case gives $Pr = 6.984$, $Nu \simeq 16.0641$, and $h \simeq 40160.2 \text{ W/(m}^2\cdot\text{K)}$. Transitional-flow use is subject to the validated operating range.

d) Forced-air heat exchanger and TEG boundary temperatures: The counterflow design case specifies coolant terminal temperatures of 338.0/319.2 K and air temperatures of 310.0/325.0 K. The resulting log-mean temperature difference and required area are

$$\begin{aligned}
 \Delta T_1 &= 338.0 - 325.0 = 13.0 \text{ K}, \\
 \Delta T_2 &= 319.2 - 310.0 = 9.2 \text{ K}, \\
 \Delta T_{\text{lm}} &= \frac{13.0 - 9.2}{\ln(13.0/9.2)} = 10.9907313 \text{ K}, \\
 A_{HX} &= \frac{13258.816330}{125.0 \Delta T_{\text{lm}}} = 9.65091 \text{ m}^2.
 \end{aligned} \tag{519}$$

A crossflow arrangement requires its configuration correction factor. Finite heat rejection through a positive thermal resistance requires a TEG cold-junction temperature above the sink. Table XVIII specifies a constant-efficiency comparison; junction temperatures, material efficiency, coolant capacity rates, and fan demand use a coupled solution.

J. Forward-Mode Automatic Differentiation and GUM Covariance

a) Dual algebra and Jacobian construction: Let the real input vector $\mathbf{X} \in \mathbb{R}^n$ include measured quantities and uncertain calibration or constitutive parameters, and let $\mathbf{Y} = \mathbf{f}(\mathbf{X}) \in \mathbb{R}^m$. A dual number is $\tilde{a} = a + \vartheta a'$ with $\vartheta^2 = 0$ and $\vartheta \neq 0$. The coefficient a' is a directional derivative, not a time derivative

TABLE LV
EQUATION-DERIVED COLD-PLATE CURVE AND THERMAL BENCHMARK

$\dot{V}$ (L/min)	v (m/s)	Re	f	ΔP (kPa)	ΔT_{9150} (K)	ΔT_{13725} (K)
1.0	1.5432	369.0	0.1986	$49.18 + 1.19K_{\Sigma}$	131.51	197.27
2.0	3.0864	737.9	0.0993	$98.36 + 4.75K_{\Sigma}$	65.76	98.64
3.0	4.6296	1106.9	0.0662	$147.54 + 10.70K_{\Sigma}$	43.84	65.76
4.0	6.1728	1475.9	0.0497	$196.72 + 19.02K_{\Sigma}$	32.88	49.32
5.0	7.7160	1844.8	0.0397	$245.90 + 29.72K_{\Sigma}$	26.30	39.45
6.0	9.2593	2213.8	0.0331	$295.08 + 42.79K_{\Sigma}$	21.92	32.88
7.0†	10.8025	2582.8	0.0284	$344.27 + 58.24K_{\Sigma}$	18.79	28.18
8.0†	12.3457	2951.7	0.0248	$393.45 + 76.07K_{\Sigma}$	16.44	24.66
9.0†	13.8889	3320.7	0.0221	$442.63 + 96.28K_{\Sigma}$	14.61	21.92
10.0†	15.4321	3689.7	0.0199	$491.81 + 118.86K_{\Sigma}$	13.15	19.73

†: the laminar friction relation is extrapolated beyond $Re = 2300$ for comparison only; it is not a qualified transition/turbulent pressure curve. The two thermal duties define separate benchmark loads, not the nominal TEG heat rejection.

TABLE LVI
HYDRAULIC CORRELATION COMPARISON AND HEAT-REJECTION DUTIES

Flow (L/min)	Re	f_{Ch}	Laminar ΔP_0 (kPa)	Churchill ΔP_0 (kPa)	ΔT_{10} (K)	ΔT_{15} (K)
4.0	1475.86	0.0433645	196.72	171.81	31.762	47.642
6.0	2213.79	0.0301077	295.08	268.40	21.174	31.762
7.0	2582.76	0.0373285	344.27	452.93	18.150	27.224
8.0	2951.73	0.0440285	393.45	697.77	15.881	23.821

ΔP_0 excludes nonnegative manifold losses; laminar values at transitional Reynolds numbers are extrapolations. Temperature rises use 8839.210887 and 13258.816330 W, distinct from the 9150 and 13725 W benchmark duties in the separate comparison table.

or a random uncertainty. Forward-mode evaluation applies the chain rule to every elementary operation [128]:

$$\begin{aligned}
\tilde{a} + \tilde{b} &= (a + b) + \vartheta(a' + b'), \\
\tilde{ab} &= ab + \vartheta(a'b + ab'), \\
\frac{\tilde{a}}{\tilde{b}} &= \frac{a}{b} + \vartheta \frac{a'b - ab'}{b^2}, \quad b \neq 0, \\
g(\tilde{a}) &= g(a) + \vartheta g'(a)a', \\
e^{\tilde{a}} &= e^a + \vartheta e^a a', \\
\ln \tilde{a} &= \ln a + \vartheta a'/a, \quad a > 0, \\
\sqrt{\tilde{a}} &= \sqrt{a} + \vartheta \frac{a'}{2\sqrt{a}}, \quad a > 0.
\end{aligned} \tag{520}$$

The nilpotent coefficient is not a small finite-difference step. For each input column j , seed the unit coordinate vector $\mathbf{e}_j$ and evaluate the same measurement function:

$$\begin{aligned}
\tilde{X}_i^{(j)} &= \hat{x}_i + \vartheta \delta_{ij}, \\
\tilde{\mathbf{Y}}^{(j)} &= \mathbf{f}(\hat{\mathbf{x}}) + \vartheta \mathbf{J} \mathbf{e}_j, \\
J_{aj} &= \left. \frac{\partial f_a}{\partial X_j} \right|_{\hat{\mathbf{x}}} = \text{coeff}_{\vartheta} \tilde{Y}_a^{(j)}.
\end{aligned} \tag{521}$$

Column-wise evaluation needs n directional passes; vector-valued tangents can carry multiple seeds in one pass. Constant outputs have zero tangent. Input units remain attached to the columns, so J_{aj} has units $[Y_a]/[X_j]$. Dual arithmetic differentiates the implemented smooth computational graph without a finite-difference truncation step, but rounding, iterative-solve error, and quadrature error remain.

b) Implicit states and thermodynamic sensitivities: For a converged internal state $\mathbf{z}$ satisfying $\mathbf{r}(\mathbf{z}, \mathbf{X}) = 0$, differentiate the residual before constructing the output Jacobian:

$$\begin{aligned}
\mathbf{r}_z \frac{\partial \mathbf{z}}{\partial \mathbf{X}} &= -\mathbf{r}_X, \\
\mathbf{J} &= \mathbf{f}_X - \mathbf{f}_z \mathbf{r}_z^{-1} \mathbf{r}_X.
\end{aligned} \tag{522}$$

This notation denotes a linear solve, not formation of an explicit matrix inverse. The residual Jacobian must be nonsingular on the selected branch. For the EOS residual $r = p_{\text{int}}(V_m, T, \mathbf{a}) - p_{\text{ext}}$, with material parameters $\mathbf{a}$, the local sensitivities are

$$\begin{aligned}
\left(\frac{\partial V_m}{\partial p_{\text{ext}}} \right)_{T, \mathbf{a}} &= -\frac{V_m}{B_T}, \\
\left(\frac{\partial V_m}{\partial T} \right)_{p, \mathbf{a}} &= \alpha_V V_m, \\
\frac{\partial V_m}{\partial a_j} &= -\frac{\partial r / \partial a_j}{\partial r / \partial V_m}.
\end{aligned} \tag{523}$$

Differentiating a fixed number of unconverged Newton steps instead gives the sensitivity of those steps, not necessarily the equilibrium volume. The same distinction applies to discretized transport, phase-field, and shell-contact states. Phase switches, contact-set changes, threshold decisions, and singular limits require branch-aware or distributional treatment rather than a single smooth Jacobian.

Differentiable Debye quadrature retains its variable upper limit. For $I(y) = \int_0^y \phi(z) dz$,

$$I(y) = y \int_0^1 \phi(yu) du, \quad \frac{dI}{dy} = \phi(y). \tag{524}$$

A fixed-domain quadrature or an analytic integral derivative preserves this dependence; adaptive branching alone does not

guarantee derivative accuracy. Series and integral representations must agree in both value and slope at any numerical switch. For Drude–Lorentz outputs, real and imaginary tensor components are stacked into one real vector before covariance propagation, preserving their cross-covariances.

c) *Matrix law of propagation and its scope:* Let Σ_X be the input covariance matrix, with entries $\text{cov}(X_i, X_j)$, and linearize about the estimates $\hat{\mathbf{x}}$:

$$\begin{aligned}\delta \mathbf{Y} &\simeq \mathbf{J} \delta \mathbf{X}, \\ \Sigma_Y &= \mathbf{J} \Sigma_X \mathbf{J}^T, \\ \text{cov}(Y_a, Y_b) &= \sum_{i,j} J_{ai} (\Sigma_X)_{ij} J_{bj}, \\ u_c(Y_a) &= \sqrt{(\Sigma_Y)_{aa}}.\end{aligned}\quad (525)$$

Here Σ_Y denotes the first-order GUM output covariance; it equals the true covariance exactly for an affine measurement function, not for every nonlinear function [82], [129]. For any real $\mathbf{v}$,

$$\mathbf{v}^T \Sigma_Y \mathbf{v} = (\mathbf{J}^T \mathbf{v})^T \Sigma_X (\mathbf{J}^T \mathbf{v}) \geq 0 \quad (526)$$

whenever Σ_X is positive semidefinite. Symmetry, nonnegative diagonal entries, physical units, and compatibility of all correlations are checked before interpreting the budget. Anticorrelation can reduce a difference uncertainty, but cannot justify a non-positive-semidefinite input covariance.

With an input displacement centered at zero, a second-order diagnostic for nonlinear bias is

$$\mathbb{E}[Y_a] - f_a(\hat{\mathbf{x}}) \simeq \frac{1}{2} \text{tr}(\mathbf{H}_a \Sigma_X), \quad \mathbf{H}_a = \nabla_X^2 f_a. \quad (527)$$

An appreciable curvature correction or a non-Gaussian coverage requirement calls for higher-order or Monte Carlo propagation of the specified joint input distribution. Type A/B evaluation, calibration corrections, model discrepancy, and distributional assumptions remain part of the measurement model. Numerical tolerances are controlled separately and included as uncertainty contributions only when a justified error model makes them relevant.

For scalar reporting, expanded uncertainty is $U_a = k_a u_c(Y_a)$ with a justified coverage factor; $k_a = 2$ is not an unconditional 95% statement. A multivariate covariance ellipse requires a joint coverage construction, not simultaneous interpretation of all marginal intervals. Material parameters inferred from the same fit, sensor references shared by two channels, and derived quantities such as $c_p(T, p)$ retain their correlations. A derived input is either evaluated in the computational graph or supplied with its full covariance, not counted twice as an independent source.

d) *Calorimetric sensitivities:* The reduced calorimetric measurement function is

$$P(\dot{m}, c_p, \Delta T, P_{\text{env}}) = \dot{m} c_p \Delta T + P_{\text{env}}. \quad (528)$$

Its first differential and sensitivity coefficients are

$$\begin{aligned}dP &= c_p \Delta T d\dot{m} + \dot{m} \Delta T dc_p + \dot{m} c_p d(\Delta T) + dP_{\text{env}}, \\ \frac{\partial P}{\partial \dot{m}} &= c_p \Delta T, \quad \frac{\partial P}{\partial c_p} = \dot{m} \Delta T, \\ \frac{\partial P}{\partial \Delta T} &= \dot{m} c_p, \quad \frac{\partial P}{\partial P_{\text{env}}} = 1.\end{aligned}\quad (529)$$

For input vector $\mathbf{X} = [\dot{m}, c_p, \Delta T, P_{\text{env}}]^T$, the dual-number evaluation must reproduce the analytical row

$$\mathbf{J}_P = [c_p \Delta T \quad \dot{m} \Delta T \quad \dot{m} c_p \quad 1]. \quad (530)$$

With sensitivities $c_j = \partial P / \partial X_j$, the scalar restriction of Eq. (525) is

$$\begin{aligned}u_c^2(P) &= \sum_j c_j^2 u^2(X_j) + 2 \sum_{j < k} c_j c_k \text{cov}(X_j, X_k) \\ &= \mathbf{J}_P \Sigma_X \mathbf{J}_P^T.\end{aligned}\quad (531)$$

This interpretation follows the distinction between combined standard uncertainty and expanded uncertainty in the GUM [82]. Expanded uncertainty shall be reported as $U = k u_c$, with the coverage factor and distribution assumptions stated. Manufacturer limits shall be converted to standard uncertainties using their documented coverage or an explicitly justified probability model.

For correlated inlet and outlet thermometry,

$$u^2(\Delta T) = u^2(T_{\text{out}}) + u^2(T_{\text{in}}) - 2 \text{cov}(T_{\text{out}}, T_{\text{in}}). \quad (532)$$

If c_p is evaluated at the measured average temperature, the reduced temperature sensitivities become

$$\begin{aligned}\frac{\partial P}{\partial T_{\text{out}}} &= \dot{m} \left[c_p + \frac{\Delta T}{2} \frac{dc_p}{dT} \right], \\ \frac{\partial P}{\partial T_{\text{in}}} &= \dot{m} \left[-c_p + \frac{\Delta T}{2} \frac{dc_p}{dT} \right].\end{aligned}\quad (533)$$

The fluid-property calibration contribution shall be separated from its temperature-induced contribution, with each uncertainty source included once. The reference budget treats $u(c_p)$ as the evaluated property uncertainty and $u(\Delta T)$ as the separately characterized differential-temperature uncertainty. Commissioning shall verify this decomposition and retain any residual correlation.

Likewise, writing $\Delta V = V_{\text{env}} - V_0$, the envelope contribution is

$$\begin{aligned}u^2(P_{\text{env}}) &= \Delta V^2 u^2(K_{\text{env}}) + K_{\text{env}}^2 u^2(\Delta V) \\ &\quad + 2 K_{\text{env}} \Delta V \text{cov}(K_{\text{env}}, \Delta V),\end{aligned}\quad (534)$$

with $u^2(\Delta V) = u^2(V_{\text{env}}) + u^2(V_0) - 2 \text{cov}(V_{\text{env}}, V_0)$. The envelope channel has a 0.85 W combined standard-uncertainty allocation. Commissioning shall determine its gain, baseline, drift, and covariance contributions from calibration records.

K. Dual Calorimetry Reference Variance Budget

a) *Water-loop metrology operating point:* The water-loop comparison uses $\dot{m} = 0.08250 \text{ kg/s}$, $c_p = 4182.0 \text{ J/(kg K)}$, $\Delta T = 8.42000 \text{ K}$ and $P_{\text{env}} = 12.30 \text{ W}$. The standard-uncertainty allocations are $u(\dot{m}) = 9.4875 \times 10^{-5} \text{ kg/s}$, $u(c_p) = 0.8364 \text{ J/(kg K)}$, $u(\Delta T) = 0.00310 \text{ K}$ and $u(P_{\text{env}}) = 4.4500 \text{ W}$. These inputs describe water rather than the mixture with heat capacity 3380 J/(kg K) . Under zero covariance, direct evaluation gives

$$\begin{aligned}
P_{\text{literal}} &= 0.08250(4182.0)(8.42000) + 12.30 \\
&= 2917.326300 \text{ W}, \\
u_c^2(P_{\text{literal}}) &= 11.16081265 + 0.33756711 \\
&\quad + 1.14392972 + 19.80250000 \\
&= 32.44480947 \text{ W}^2, \\
u_c(P_{\text{literal}}) &= 5.69603454 \text{ W} < 5.7000 \text{ W}.
\end{aligned} \tag{535}$$

At the 2911.40 W reference power, with the other nominal inputs fixed, the measurement function gives the temperature rise:

$$\Delta T_{\text{ref}} = \frac{2911.40 - 12.30}{0.08250(4182.0)} = 8.40282306566 \text{ K}. \tag{536}$$

The calculated uncertainty satisfies the strict 5.7000 W design limit under the stated independence and calibration assumptions. Covariance and correction uncertainties enter through Eq. (531) [73]. With independent 0.50 W input-power uncertainty, a hypothetical 1000 W excess gives $u = 5.71383790 \text{ W}$ and $Z = 175.013715$. This configuration assigns 12.30 W to the envelope contribution.

b) Water-glycol reference budget: The reference operating point assigns $P_{\text{flow}} = 2850.00 \text{ W}$ and $P_{\text{env}} = 61.40 \text{ W}$. With $c_p = 3380.0 \text{ J kg}^{-1} \text{ K}^{-1}$ near $T_{\text{avg}} = 320 \text{ K}$,

$$\begin{aligned}
\Delta T &= \frac{2850.00}{(0.1717)(3380.0)} \text{ K} = 4.9108635 \text{ K}, \\
P_{\text{cal}} &= 2850.00 + 61.40 = 2911.40 \text{ W}.
\end{aligned} \tag{537}$$

The calculation shall retain the unrounded temperature rise before reporting the result. Table LVIII defines the four-input standard-uncertainty allocations and their power-equivalent variances. Zero covariance defines this reference case; measured correlations and additional correction uncertainties shall be included in the commissioned budget.

The individual power-equivalent contributions and their combination are

$$\begin{aligned}
u_{P,\dot{m}} &= (0.0001717)(3380.0)(4.9108635) = 2.850000 \text{ W}, \\
u_{P,c_p} &= (0.1717)(5.07)(4.9108635) = 4.275000 \text{ W}, \\
u_{P,\Delta T} &= (0.1717)(3380.0)(0.0040) = 2.321384 \text{ W}, \\
u_{P,\text{env}} &= 0.850000 \text{ W}.
\end{aligned} \tag{538}$$

$$\begin{aligned}
u_c^2(P_{\text{cal}}) &= (2.850000^2 + 4.275000^2 \\
&\quad + 2.321384^2 + 0.850000^2) \text{ W}^2, \\
u_c(P_{\text{cal}}) &= \sqrt{32.509449} \text{ W} \\
&= 5.701706 \text{ W} \approx 5.70 \text{ W}, \\
u_r(P_{\text{cal}}) &= \frac{5.701706}{2911.40} \times 100\% \\
&= 0.195841\% \approx 0.196\%.
\end{aligned} \tag{539}$$

The water-glycol reference combined standard uncertainty is $u_c(P_{\text{cal}}) = 5.701706 \text{ W}$ with $k = 1$. The strict acceptance limit is 5.70 W. At $k = 2$, expanded uncertainty is $U = 11.4034 \text{ W}$, or 0.39168%. Measurement statements identify the control-volume boundary and coverage convention.

The envelope heat model resolves the 53.7341 W contribution within the following partition:

$$\begin{aligned}
Q_{\text{envelope}} &= Q_{\text{struct}} + Q_{\text{rad}} + Q_{\text{gas}} + Q_{\text{Peltier}} \\
&= 7.6659 + 14.2810 + 0.8420 + 38.6111 \\
&= 61.4000 \text{ W}.
\end{aligned} \tag{540}$$

The component partition is $14.2810 + 0.8420 + 38.6111 = 53.7341 \text{ W}$, with each boundary-crossing heat contribution counted once.

The envelope model specifies effective areas, view factors, temperatures, and thermoelectric boundaries. The 298.15 K thermopile bath and 313.15 K ambient are separate thermal references. Internal Peltier transport contributes to the envelope measurement only when it crosses that boundary and is excluded from coolant enthalpy.

L. Net-Power Reporting and the 174.71 σ Reference

For a measured net quantity $\hat{P}_{\text{net}}$ and its combined standard uncertainty, a standardized null-displacement statistic is

$$Z = \frac{\hat{P}_{\text{net}} - 0}{u_c(P_{\text{net}})}. \tag{541}$$

The report shall state the measurand, null value, uncertainty model, and any distribution assumptions used to interpret Z . A Gaussian-tail significance shall require a justified sampling model. Nuclear attribution shall be assessed separately using isotope and radiation measurements.

The reference *excess-heat* calculation uses an electrical-input measurement design point $P_{\text{in}} = 1911.40 \text{ W}$ with standard-uncertainty allocation $u(P_{\text{in}}) = 0.50 \text{ W}$:

$$\begin{aligned}
P_{\text{excess,th}} &= 2911.40 - 1911.40 = 1000.00 \text{ W}, \\
u^2(P_{\text{excess,th}}) &= u_c^2(P_{\text{cal}}) + u^2(P_{\text{in}}) \\
&\quad - 2 \text{cov}(P_{\text{cal}}, P_{\text{in}}).
\end{aligned} \tag{542}$$

Under the additional zero-covariance assumption,

$$\begin{aligned}
u(P_{\text{excess,th}}) &= \sqrt{32.509449 + 0.250000} \text{ W} = 5.723587 \text{ W}, \\
Z_{\text{excess,th}} &= \frac{1000.00}{5.723587} = 174.7156.
\end{aligned} \tag{543}$$

Experimental reports calculate Z from measured excess heat and the commissioned uncertainty budget.

The excess-heat reference and electrical-export balance in Section II describe separate measurement configurations. The 1484.8 W gross/484.8 W auxiliary allocation is a distinct comparison case rather than the nominal operating point. In the general uncertainty relation, η_{gross} denotes the complete bus-referred conversion efficiency η_{bus} , including boost losses, and $P_{\text{parasitic}}$ includes every operating supply:

$$P_{\text{net,el}} = \eta_{\text{gross}} P_{\text{th}} - P_{\text{parasitic}}, \tag{544}$$

whose uncertainty, for independent inputs, is

$$\begin{aligned}
u^2(P_{\text{net,el}}) &= \eta_{\text{gross}}^2 u^2(P_{\text{th}}) + P_{\text{th}}^2 u^2(\eta_{\text{gross}}) \\
&\quad + u^2(P_{\text{parasitic}}).
\end{aligned} \tag{545}$$

The full electrical uncertainty model shall include applicable covariance and conversion-loss terms. Acceptance of net

TABLE LVII
FOUR-INPUT WATER-CALORIMETER BUDGET AT 2911.40 W

Input (native unit)	Nominal	Standard uncertainty	Sensitivity (W/unit)	Variance (W ²)
$\dot{m}$ (kg/s)	0.08250	9.4875×10^{-5}	35140.60606	11.11532262
c_p (J/(kg K))	4182.0	0.8364	0.693232903	0.33619123
ΔT (K)	8.402823066	0.00310	345.015	1.14392972
P_{env} (W)	12.30	4.4500	1	19.80250000
Total variance; $u_c = 5.69191915$ W, $k = 1$				32.39794357

TABLE LVIII
FOUR-INPUT WATER-GLYCOL CALORIMETRIC VARIANCE BUDGET

Input and native unit	Nominal value	$u(y_j)$	$u^2(y_j)$	Sensitivity c_j	$ c_j u(y_j)$ (W)	$c_j^2 u^2(y_j)$ (W ²)	Share (%)
Mass flow $\dot{m}$ (kg s ⁻¹)	0.1717	0.0001717	2.948089×10^{-8}	16598.7187	2.850000	8.122500	24.9850
Heat capacity c_p (J kg ⁻¹ K ⁻¹)	3380.0	5.07	25.704900	0.8431953	4.275000	18.275625	56.2163
Temperature difference ΔT (K)	4.9108635	0.0040	1.600000×10^{-5}	580.3460	2.321384	5.388824	16.5762
Envelope heat P_{env} (W)	61.40	0.85	0.722500	1.0000000	0.850000	0.722500	2.2224
Reference thermal budget: $P_{\text{cal}} = 2911.40$ W					5.701706	32.509449	100.0000

Input variances carry the square of the stated native unit; sensitivities carry watts per native unit. The displayed shares are rounded independently. The total is a combined standard uncertainty, not an expanded uncertainty.

electrical export shall use synchronized input/output electrical measurements, with model-based conversion estimates reported separately. The excess-heat reference Z shall not be transferred to net electrical output; each result shall carry its own calibrated uncertainty and background treatment.

M. Non-Inductive Nichrome Electrical-Substitution Calibration

The selected calorimetric acceptance threshold is combined standard uncertainty $u_c \leq 5.69$ W at the defined operating boundary. The five Ni80Cr20 substitution levels are 200.0, 800.0, 1500.0, 2200.0, and 3000.0 W, each held for at least 45 min and until the specified equilibrium criterion is met. Recalculate uncertainty from the measured calibration covariance, sensor drift, spatial transfer, and operating-point corrections; a heater setpoint or fit statistic alone does not demonstrate the 5.69 W threshold. Calibration of the selected 120-channel, 50/50 water-glycol loop at 3.62 L/min must use its own flow, composition, and temperature measurements. The 9.83 L/min protocol below is a separate reference configuration and requires explicit transfer characterization before application to that operating point.

The calibration heater shall be a micro-machined Ni80Cr20 meander embedded 1.5 mm below the active-layer interface in the OFHC copper spreader and electrically insulated by aluminum nitride. Its reference resistance shall be $R_{\text{cal}} = 50.000 \pm 0.005 \Omega$ at 298.15 K with four-wire Kelvin sensing. Antiparallel serpentine routing shall limit parasitic inductance to below 12 nH. At 68.5 kHz this gives

$$X_L = 2\pi fL < 0.005165 \Omega, \quad (546)$$

with magnetic pickup and lead-loop coupling characterized separately during commissioning.

TABLE LIX
ELECTRICAL-SUBSTITUTION CALIBRATION STEPS AT 50.000 Ω

Power (W)	Voltage (V)	Current (A)	Hold (min)
200.0	100.0000	2.000000	45
800.0	200.0000	4.000000	45
1500.0	273.8613	5.477226	45
2200.0	331.6625	6.633250	45
3000.0	387.2983	7.745967	45

Delivered heater power is measured at the heater terminals:

$$P_{\text{cal, inj}} = \langle V_{\text{cal}}(t) I_{\text{cal}}(t) \rangle. \quad (547)$$

The approximation $P = I^2 R = V^2 / R$ is useful for sizing at the reference resistance; resistance changes with temperature, and lead dissipation outside the calorimetric boundary must not be included as deposited heat. Table LIX enumerates the five calibration steps and their reference electrical requirements.

The calibration power supply shall sustain at least the 3000 W endpoint with the required voltage and current compliance. Supply selection shall cover the complete five-step load envelope. Heater, feedthrough, insulation, contactor, and thermal-protection ratings require verification at the full calibration load. These externally supplied calibration inputs shall be recorded separately from continuous auxiliary demand.

Electrical-substitution calibration establishes the channel gains and their transfer to the operating configuration. The commissioning protocol is:

- 1) Establish a nonreactive baseline with the reactive gas isolated, chamber pressure below 10^{-8} mbar, coolant flow 9.83 L min⁻¹, and inlet temperature 298.15 K. Require $|d(\Delta T)/dt| < 0.0001$ K min⁻¹ for 30 minutes before the first step.

- 2) Apply each of the five powers in Table LIX for at least 45 minutes and verify thermal equilibrium. Record synchronized voltage, current, mass flow, inlet/outlet temperatures, envelope voltage, bath temperature, and stored-energy indicators.
- 3) Evaluate $c_p(T)$ over the measured calibration temperature range and establish transfer to the 320 K mean-fluid reference used in Table LVIII.
- 4) Fit the calibrated response, evaluate residuals and hysteresis with ascending and descending powers, and independently excite the envelope channel if needed to distinguish its gain from the flow-channel gain. Perform a post-run verification under the same boundary conditions.
- 5) Treat drift exceeding 0.05% in either calibrated gain as a recalibration and loading-inhibit condition. Propagate the calibration covariance to subsequent measurements rather than assigning the fit quality as a measurement uncertainty.

The linear calibration model and weighted least-squares estimator are

$$\begin{aligned}
 P_{\text{cal,inj},j} &= a_f(\dot{m}c_p\Delta T)_j + b_e V_{\text{env},j} + c_0 + \epsilon_j, \\
 \hat{\beta} &= (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{P}_{\text{cal,inj}}, \\
 \beta &= [a_f, b_e, c_0]^T.
 \end{aligned} \tag{548}$$

The design matrix shall have full column rank. Independent channel excitation is required when proportional flow and envelope responses prevent separate gain identification. The fit shall meet $R^2 > 0.99995$, with residual bias, hysteresis, and parameter covariance assessed separately. Commissioning shall verify heater-to-active-layer spatial transfer and electromagnetic pickup for the 1.5 mm heater offset. Metrological qualification requires traceable sensor and power calibrations, pre- and post-run gain checks, nonreactive background measurements, and a complete covariance budget supporting the stated calorimetric uncertainty.

VI. ISOTOPIC MASS SPECTROMETRY, DIGITAL PSD, AND SHIELDING DIAGNOSTICS

The diagnostic architecture combines isotope-resolved gas analysis, digital neutron–gamma discrimination, gamma spectroscopy, and three-dimensional neutron–photon shielding calculations. The mass-spectrometric channel determines helium inventory after selective hydrogenic-gas removal; the scintillation channel classifies individual radiation events; and the transport model converts source-normalized fluence tallies into accessible-surface dose rates. Resolving power, helium recovery, classification probability, and dose equivalent are separate measurands with separate calibration and acceptance requirements. The heat-to-helium relation defines the branch-specific stoichiometric reference against which the measured inventory is evaluated.

A. Sector-Mass-Spectrometer Quantitation

The analyzer shall use double-focusing magnetic-sector optics with $R = m/\Delta m = 25000$ [79]. For an ion of mass m_{ion} and positive charge q_{ion} , accelerated through potential

V_{acc} and following an ideal magnetic-sector orbit of radius r_B in transverse field B_{sec} , the nonrelativistic ion-optical relations are

$$\begin{aligned}
 q_{\text{ion}} V_{\text{acc}} &= \frac{1}{2} m_{\text{ion}} v_{\text{ion}}^2, \\
 q_{\text{ion}} v_{\text{ion}} B_{\text{sec}} &= m_{\text{ion}} v_{\text{ion}}^2 / r_B, \\
 \frac{m_{\text{ion}}}{q_{\text{ion}}} &= \frac{B_{\text{sec}}^2 r_B^2}{2V_{\text{acc}}}.
 \end{aligned} \tag{549}$$

The electrostatic and magnetic sectors shall provide energy and angular focusing at the collector. The working mass pair is $m(^4\text{He}) = 4.002603$ u and $m(D_2) = 4.028204$ u, giving $\Delta m = 0.025601$ u and $R_{\text{required}} \simeq 156.3$ for this separation. Nearby interferences require individual resolution and abundance-sensitivity assessment. The St707 getter composition is 70 wt% Zr / 24.6 wt% V / 5.4 wt% Fe, with a specified activation temperature of 400 °C. The nominal ratio enrichment is 1.45×10^5 , the D_2 removal requirement exceeds 99.999%, and the residual $D_2 \rightarrow ^4\text{He}$ background-fit coefficient is 3.20×10^{-11} . These response parameters enter the calibrated interference model, not a multiplicative amplification of the helium production signal.

For one helium atom per reaction and complete deposition of the assigned reaction energy, the nuclear-ash generation rate per kilowatt is

$$\begin{aligned}
 \dot{N}_{\text{He}} &= \frac{1000 \text{ J/s}}{23.84 \times 10^6 \text{ eV} (1.602176634 \times 10^{-19} \text{ J/eV})} \\
 &= 2.61808 \times 10^{14} \text{ s}^{-1}, \\
 \dot{n}_{\text{He}} &= \dot{N}_{\text{He}} \frac{86400}{N_A} \\
 &= 3.75618 \times 10^{-5} \text{ mol/(day kW)}.
 \end{aligned} \tag{550}$$

The stoichiometric yield is $3.756 \times 10^{-5} \text{ mol/(day kW)}$ for one helium atom per 23.84 MeV of fully deposited reaction heat. At the 995.81 W reference, the predicted rate is approximately $3.74044 \times 10^{-5} \text{ mol/day}$. Quantitation uses calibrated sensitivity and collection efficiency, blank and ingress subtraction, solid-retained helium, resolved spectra, and a gas-inventory uncertainty budget.

The assay shall resolve helium from hydrogenic interferences and distinguish candidate production from calibration additions, atmospheric ingress, retained gas, and material backgrounds. Mass-axis calibration, detector-response characterization, helium-recovery measurement, and background subtraction define the quantitative measurement chain. Reaction-rate inference uses the specified branch and energy-deposition assumptions rather than treating the stoichiometric reference as an observed yield.

B. Isotopic Mass Values and Resolving Power

a) *Resolving power and abundance sensitivity:* The sector analyzer has a nominal resolving-power specification of $R = 25000$, with rounded mass labels $m(D_2) = 4.02820$ u and $m(^4\text{He}) = 4.00260$ u. Their separation is 0.02560 u, while $m/R \simeq 0.000160$ u at mass four. Conversion of neutral-mass labels to exact ion masses requires electron-removal and molecular-binding corrections. The cryo-getter specification assigns ratio enrichment $G_{\text{ratio}} = 1.45 \times 10^5$

TABLE LX
HELIUM AND HYDROGENIC MASS REFERENCES

Species	Mass reference (u)	Definition
^4He	4.00260325413	Evaluated neutral-atom mass
D_2	4.02820355624	$2m(D)$, before molecular binding correction
HD	3.02192681035	$m(^1\text{H}) + m(D)$, before binding correction
H_2	2.01565006446	$2m(^1\text{H})$, before binding correction

and tail coefficient $a_{D \rightarrow \text{He}} = 3.20 \times 10^{-11}$. Ratio enrichment describes preferential deuterium removal relative to helium, not amplification of the helium inventory. Calibration shall characterize isotope standards, helium recovery, matrix-gas blanks, pressure dependence, and detector linearity.

Neutral-atom reference masses are given by the NIST isotopic-composition tables [144]; parenthesized uncertainties apply to the final quoted digits:

$$\begin{aligned} m(^1\text{H}) &= 1.00782503223(9) \text{ u}, \\ m(D) &= 2.01410177812(12) \text{ u}, \\ m(^4\text{He}) &= 4.00260325413(6) \text{ u}. \end{aligned} \quad (551)$$

The mass-axis reference shall retain the evaluated uncertainties. Table LX lists neutral-atom sums for the hydrogenic species; the ion-mass calibration shall include molecular binding and electron-removal corrections as applicable.

For a neutral diatomic molecule XY in a defined internal state and its singly charged ion,

$$\begin{aligned} m(XY) &= m(X) + m(Y) - \frac{D_0(XY)}{c^2}, \\ m(XY^+) &= m(XY) - m_e + \frac{E_{\text{ion}}(XY)}{c^2}. \end{aligned} \quad (552)$$

Here D_0 is the dissociation energy from that state and E_{ion} its ionization energy. The corresponding atomic-ion expression is $m(^4\text{He}^+) = m(^4\text{He}) - m_e + E_{\text{ion}}(^4\text{He})/c^2$. Calibration records shall distinguish ion masses from conventional neutral-mass labels. The atom-sum approximation shall be used only at a precision compatible with the resolving-power calculation; reported molecular masses shall retain the applicable corrections and uncertainty.

For the helium/deuterium-molecule separation,

$$\begin{aligned} \Delta m &= 4.02820355624 - 4.00260325413 \\ &= 0.02560030211 \text{ u}, \\ R_{\text{sep}} &\approx \frac{4.00260325413}{0.02560030211} = 156.3498. \end{aligned} \quad (553)$$

The six-decimal working references are 4.002603, 4.028204, 3.021927, and 2.015650 u. The ratio R_{sep} defines the nominal mass-separation scale. Analyzer acceptance shall use a declared peak-width convention and measured peak shape.

The analyzer shall provide $R \geq 1600$ near mass 4, with nominal operating allocation $R = 25000$. Both values shall use

the full peak width at 5% of peak height. At this convention, the corresponding mass widths are

$$\Delta m_{1600} = 0.002501627 \text{ u}, \quad \Delta m_{25000} = 0.0001601041 \text{ u}. \quad (554)$$

Abundance sensitivity shall be qualified independently of resolving power. For fuel-to-helium ratio $B = 10^6$ and normalized deuterium tail response $a_{D \rightarrow \text{He}}$,

$$\frac{I_{D,\text{tail}}}{I_{\text{He}}} \approx a_{D \rightarrow \text{He}} B. \quad (555)$$

The nominal tail response $a_{D \rightarrow \text{He}} = 3.20 \times 10^{-11}$ corresponds to 3.20×10^{-5} helium-relative interference (0.00320%) at $B = 10^6$, before additional ratio enrichment. A 0.01% helium-relative contamination limit at the same ratio requires $a_{D \rightarrow \text{He}} < 10^{-10}$. Instrument acceptance shall establish the tail response and residual fuel-to-helium ratio for the specified cleanup and acquisition sequence.

The analyzer shall be a double-focusing magnetic-sector system covering the 1–20 u mass range, with 70 eV electron-impact ionization and a qualified low-outgassing ion source. Detection shall use a Faraday collector with nominal $10^{11} \Omega$ transimpedance and a secondary-electron multiplier for the lower-current range. Acceptance records shall specify peak width, abundance sensitivity near the D_2 tail, ion-source pressure, detector linearity, mass discrimination, fragmentation, and Faraday/multiplier gain cross-calibration. The selected instrument shall demonstrate these requirements in the installed light-isotope configuration.

C. St707 Getter Pre-Concentration Manifold

The sampling bypass shall use St707 non-evaporable getter material, nominally 70.0 wt% Zr, 24.6 wt% V, and 5.4 wt% Fe [145]. Activation and pumping shall be treated as separate process states, with gas loading and activation history recorded. The reference activation cycle shall use approximately 673 K (400°C) for one hour under dynamic vacuum, with the endpoint established for the installed cartridge. The subsequent pumping temperature and allowable loading shall be qualified separately, accounting for reversible hydrogen uptake and temperature-dependent equilibrium release pressure.

The assay-manifold requirement is

$$\eta_{\text{H}} = 1 - \frac{n_{\text{H,out}}}{n_{\text{H,in}}} > 0.99999, \quad (556)$$

where the hydrogenic inventory includes H_2 , HD , and D_2 on a consistently defined molecular or atom-equivalent basis. The corresponding helium recovery is $r_{\text{He}} = n_{\text{He,out}}/n_{\text{He,in}}$. The relative enrichment is

$$G_{\text{ratio}} = \frac{r_{\text{He}}}{1 - \eta_{\text{H}}}. \quad (557)$$

For $r_{\text{He}} \approx 1$, the cleanup requirement corresponds to an approximately 10^5 or greater improvement in helium-to-hydrogen ratio, reducing $B = 10^6$ to approximately 10 or less. The enrichment shall be applied to the species ratio, while absolute helium inventory shall follow the gas balance. Helium recovery shall include losses in valves, dead volumes, and transfer paths.

For a deuterium-dominated sample, define the species-specific transmission $r_{D_2} = n_{D_2,\text{out}}/n_{D_2,\text{in}}$ after subtraction of manifold backgrounds. At the specified enrichment,

$$\begin{aligned} G_{\text{ratio}} &= \frac{r_{\text{He}}}{r_{D_2}} = 1.45 \times 10^5, \\ r_{D_2} &= \frac{r_{\text{He}}}{1.45 \times 10^5} \leq 6.89655 \times 10^{-6}, \\ \eta_{D_2} &= 1 - r_{D_2} \geq 0.9999931034 \\ &> 0.99999. \end{aligned} \quad (558)$$

The bound assumes $0 < r_{\text{He}} \leq 1$ and no unaccounted helium addition. It corresponds to at least 99.99931034% deuterium removal. At $r_{\text{He}} = 1$, an inlet deuterium-to-helium ratio of 10^6 becomes 6.89655 at the outlet. The mixed-isotope requirement on η_{H} shall also be demonstrated for H_2 and HD ; a deuterium transmission measurement alone does not characterize the entire hydrogenic inventory.

In an idealized evacuated batch of volume V_g with constant effective hydrogen pumping speed S_g ,

$$p_{\text{H}}(t) = p_{\text{H}}(0)e^{-S_g t/V_g}, \quad \frac{S_g t}{V_g} > \ln(10^5) = 11.51293 \quad (559)$$

is required for five-decade depletion in this reference model. Installed-cartridge qualification shall characterize getter mass, capacity, isotope loading, equilibrium pressure, conductance, and outgassing and verify $> 99.999\%$ hydrogenic removal over the specified assay envelope.

The nominal recirculation volume is $V_{\text{loop}} = 12.50$ L at $T_{\text{loop}} = 300.15$ K. Isolation valves separate the pressurized reactor, calibrated dosing reservoir, getter, and analyzer. Differential pumping and metered transfer shall protect the ion source from direct exposure to the pressurized sample. Activation, sampling, regeneration, blank assay, and helium-recovery measurement are distinct operating states. Getter loading, isotope exchange, regeneration history, and recovered-gas disposition enter the inventory and background balances.

D. NIST-Traceable Micro-Dosing Calibration

Calibration shall use certified helium mole fractions of 0.10, 1.00, and 10.00 ppm in deuterium prepared from a 99.9999%-purity matrix gas before helium addition. Each mixture shall specify composition, uncertainty, stability period, and an unbroken calibration chain. NIST traceability requires that chain and its associated measurement uncertainty [146].

The dosing specification defines standard temperature and pressure (STP) as $T_0 = 273.15$ K and $p_0 = 101325$ Pa, distinct from the alternative 1-bar convention. A calibrated 5.0 mL STP-equivalent aliquot contains

$$n_{\text{dose}} = \frac{p_0 V_0}{RT_0} = 2.2307517 \times 10^{-4} \text{ mol}, \quad (560)$$

using $R = 8.31446261815324 \text{ J mol}^{-1} \text{ K}^{-1}$. The corresponding helium quantities are listed in Table LXI.

Mass-axis, detector-gain, and dosing calibration shall follow the sequence:

- 1) Measure a manifold blank, verify isolation from the reactor inventory, and calibrate pressure, volume, temperature,

TABLE LXI
HELIUM CONTENT OF EACH 5.0 mL STP CALIBRATION ALIQUOT

He mole fraction (ppm)	Helium amount (mol)	Pure-He equivalent (nL STP)
0.10	$2.2307517 \times 10^{-11}$	0.50
1.00	$2.2307517 \times 10^{-10}$	5.00
10.00	2.2307517×10^{-9}	50.00

and valve-transferred quantity to establish the 5.0 mL STP-equivalent dose and its uncertainty.

- 2) Every four hours, introduce the three certified concentrations sequentially through the metering and getter path. Meter transfer slowly enough to preserve the qualified analyzer pressure. Measure getter removal and helium recovery using the same gas load and operating temperature as the sample assay.
- 3) Acquire full mass-4 peak and tail scans, and record helium peak current, background, detector gain, and recovery transients on both Faraday and multiplier ranges where their signals overlap. Fit the response including an intercept and propagate standard-gas, dose-volume, pressure, temperature, and transfer uncertainties.
- 4) Route standards to a separate assay inventory where practical. Otherwise account for and subtract calibrated helium additions with their uncertainties, quantify retained memory, and verify washout with a subsequent blank before resuming production-rate estimation.

For a static pressure calibration at the measurement volume, define the end-to-end sensitivity

$$I_{\text{He}} - I_0 = S_{\text{He}} p_{\text{He}}, \quad [S_{\text{He}}] = \text{A Pa}^{-1}. \quad (561)$$

Pressure derived from a sensitivity in A mbar^{-1} shall be converted from mbar to Pa by a factor of 100 before use in SI gas-law expressions. A time-integrated pulse calibration shall instead use $\int (I - I_0) dt = K_n n_{\text{He,dose}}$, where K_n has units C mol^{-1} . The record shall identify the calibrated transfer dynamics and distinguish peak-pressure sensitivity from integrated-amount sensitivity.

The reference efficiencies $\eta_{\text{ion}} = 1.25 \times 10^{-3}$ and $\eta_{\text{trans}} = 0.88$ apply only to a separately defined microscopic instrument-response model. Empirical end-to-end sensitivity S_{He} already incorporates ionization and ion-optical transmission; applying these efficiencies again would double-count their effects. Relative enrichment enters the interference model separately from the calibrated helium amount.

E. Reaction-Rate Mapping and Accumulated Helium Ash

a) Effective reaction energy and daily helium inventory:

For one helium-4 atom per lattice-capture reaction, the conditional yield at fully recovered lattice heat is

$$Q_{\text{eff}} = 23.841 \text{ MeV} = 3.8197493131194 \times 10^{-12} \text{ J},$$

$$E_{\text{day}} = 8.64 \times 10^7 \text{ J}/(\text{day kW}),$$

$$N_{\text{He}} = E_{\text{day}}/Q_{\text{eff}} = 2.2619285434 \times 10^{19} \text{ day}^{-1} \text{ kW}^{-1},$$

$$n_{\text{He}} = N_{\text{He}}/N_A = 3.7560207135 \times 10^{-5} \text{ mol}/(\text{day kW}). \quad (562)$$

The nominal yield is $3.756 \times 10^{-5} \text{ mol}/(\text{day kW})$ under the assigned effective heat-accounting convention. Escaping energy, competing branches, and helium retention enter separately.

b) *Parallel isotope and photon measurements:* The gas-analysis path comprises a closed-loop sampler, cryogenic condensable trap, reactive-gas getter, and sector mass analyzer. The extended assay targets are $M/\Delta M > 25000$ and a helium detection limit no greater than 10^{-11} mol , with separately calibrated abundance sensitivity and peak-tail rejection. Atmospheric helium near 5.24 ppm is a contamination reference.

An independent HPGe channel and pulse-height analyzer cover activation lines and the approximately 23.847 MeV capture-photon signature. The detector response shall include efficiency, escape peaks, dead time, background, and live time at the relevant energy. A 0.1 Bq photon-source detection target with 10 cm lead shielding requires an emission probability and statistically defined detection criterion; activity is not identical to detected counts.

For the proposed helium-producing branch with complete useful-energy deposition, the assigned nuclear energy per event is

$$Q_{DD} = 23.84 \text{ MeV} = 3.819589095456 \times 10^{-12} \text{ J}. \quad (563)$$

At a *nuclear heat-equivalent* power of $P_{\text{nuc}} = 1000.0 \text{ W}$, one helium atom per event gives

$$\begin{aligned} \dot{N}_{\text{He}} &= \frac{P_{\text{nuc}}}{Q_{DD}} = 2.6180827 \times 10^{14} \text{ s}^{-1}, \\ \dot{n}_{\text{He}} &= \frac{\dot{N}_{\text{He}}}{N_A} = 4.3474285 \times 10^{-10} \text{ mol s}^{-1}, \\ N_A &= 6.02214076 \times 10^{23} \text{ mol}^{-1}. \end{aligned} \quad (564)$$

Over $t_d = 86400 \text{ s}$,

$$\begin{aligned} N_{\text{He}}(t_d) &= 2.2620234 \times 10^{19}, \\ n_{\text{He}}(t_d) &= \dot{n}_{\text{He}} t_d = 3.7561783 \times 10^{-5} \text{ mol} \\ &\approx 3.756 \times 10^{-5} \text{ mol}, \\ V_{\text{He},0} &= n_{\text{He}} \frac{RT_0}{p_0} = 0.8419087 \text{ mL} \approx 0.842 \text{ mL}. \end{aligned} \quad (565)$$

The reference gas molar volume under this STP convention is $22.4139695 \text{ L mol}^{-1}$. Equation (565) specifies the daily inventory for 1.00 kW nuclear heat-equivalent power. At the separate 9303.30 W thermal comparison point, distinct from the nominal cascade input,

$$\begin{aligned} n_{\text{He},9303.3}(t_d) &= 3.4944853 \times 10^{-4} \text{ mol}, \\ V_{\text{He},9303.3,0}(t_d) &= 7.8325288 \text{ mL} \end{aligned} \quad (566)$$

under the same branch, deposition, and recovery assumptions. Quantitative attribution requires independent characterization of these assumptions and of the actual nuclear contribution to measured heat.

For measured loop pressure, the inventory and production balance are

$$\begin{aligned} n_{\text{He},\text{loop}}(t) &= \frac{p_{\text{He}}(t) V_{\text{loop}}(t)}{RT_{\text{loop}}(t)}, \\ \dot{n}_{\text{He},\text{prod}} &= \frac{d}{dt} \left(\frac{p_{\text{He}} V_{\text{loop}}}{RT_{\text{loop}}} \right) + \dot{n}_{\text{removed}} \\ &\quad - \dot{n}_{\text{cal}} - \dot{n}_{\text{background}}. \end{aligned} \quad (567)$$

At constant volume and temperature, using the end-to-end sensitivity in (561),

$$\dot{n}_{\text{He},\text{inventory}} = \frac{V_{\text{loop}}}{RT_{\text{loop}}} \frac{d}{dt} \left(\frac{I_{\text{He}} - I_0}{S_{\text{He}}} \right). \quad (568)$$

The inventory rate is expressed in moles per second; atom rates shall be obtained by multiplication by N_A . For fully retained helium at the 1.00 kW heat-equivalent reference, the 12.50 L, 300.15 K loop has a calculated helium partial-pressure rise of 7.49909 Pa per day. Production-rate inference shall include extraction, temperature changes, sample removal, calibration additions, and background ingress.

c) *Assay acceptance and background control:* Assay acceptance requires mass-axis stability, detector linearity and gain transfer, isotope-resolved peak and tail fits, getter recovery, and a complete calibration covariance. Background characterization shall use manifold and matrix-gas blanks, reactor-isolated assays, calibrated helium additions, leak testing, and memory and washout measurements through the same sampling path. Each background-subtracted inventory shall include the raw signal, background estimate, transferred amount, and uncertainty. The $\pm 0.5\%$ kinetic/calorimetric agreement target shall be evaluated from independently calibrated, time-aligned measurements and their covariance against the branch-specific heat-to-helium relation $Q_{DD} N_A$. Assay performance is established by these quantitative acceptance criteria rather than nominal instrument settings alone.

F. Digital Charge-Comparison Pulse-Shape Discrimination

The EJ-301 acquisition configuration specifies a 500 MS/s, 14-bit digitizer with charge-comparison discrimination [79]. Each waveform shall undergo pretrigger baseline estimation and polarity correction before charge integration. Taking the accepted pulse onset as $t = 0$, let $V(t)$ denote the baseline-subtracted, positive-polarity waveform. The short and long integration gates are

$$\begin{aligned} Q_{\text{short}} &= \int_0^{24 \text{ ns}} V(t) dt, \\ Q_{\text{long}} &= \int_0^{240 \text{ ns}} V(t) dt, \\ S_{\text{PSD}} &= 1 - Q_{\text{short}}/Q_{\text{long}}. \end{aligned} \quad (569)$$

At the 2 ns sample interval, the gates contain exactly 12 and 120 samples, without an additional endpoint sample. Charge in coulombs requires the calibrated front-end gain and impedance; these common factors cancel in the ratio. Division shall be enabled only above a positive, calibrated long-gate threshold. Clipped pulses, contaminated baselines, incomplete gates, and identified pileup shall be excluded from the qualified single-pulse classification. The synchronous 1.0 GS/s implementation in Section VI-I uses the same physical gate durations with a different sample count.

The resulting tail-to-total discriminator and specified peak-separation metric are

$$S_{\text{PSD}} = \frac{\int_{24 \text{ ns}}^{240 \text{ ns}} V(t) dt}{\int_0^{240 \text{ ns}} V(t) dt},$$

$$\text{FOM} = \frac{\Delta S_{n\gamma}}{\text{FWHM}_n + \text{FWHM}_\gamma} = \frac{0.215}{0.052 + 0.048} = 2.15. \quad (570)$$

Classification shall use an energy-dependent neutron selection determined from independently tagged calibration events and frozen before validation. Separation and peak widths shall be evaluated within specified light-output bins, with neutron acceptance, count rate, and pileup policy held fixed. For zero neutron-classified events in a validation sample of $N_\gamma = 2995731$ independent gamma trials, the acceptance criterion is $R_\gamma > 10^6$ at 95% confidence. The exact one-sided binomial bound is

$$p_{\gamma \rightarrow n, 95} = 1 - 0.05^{1/N_\gamma} \simeq 9.99999925 \times 10^{-7},$$

$$R_{\gamma, 95} = 1/p_{\gamma \rightarrow n, 95} \quad (571)$$

$$\simeq 1000000.0748773175$$

$$\simeq 1.000000 \times 10^6 \geq 10^6.$$

For independent, identically distributed Bernoulli trials, stable evaluation uses $p_{\gamma \rightarrow n, 95} = -\expm1[\ln(0.05)/N_\gamma]$. The minimum exact-binomial sample size is 2,995,731 zero-leakage trials. Qualification specifies event independence, representative gamma energies, fixed selection criteria, and traceable acquisition.

G. Gamma Spectroscopy and Emission-Rate Normalization

The HPGe reference geometry places the detector 10 cm from the vessel. The nominal spectral condition is no resolved 23.84 MeV primary or 2.22 MeV capture photopeak above a < 0.01 count/s background. For a specified photon energy and geometry, a background-subtracted count-rate limit $\dot{C}_{\gamma, \text{lim}}$ maps to an emission-rate limit through

$$\dot{N}_{\gamma, \text{lim}} = \frac{\dot{C}_{\gamma, \text{lim}}}{\varepsilon_{\gamma, \text{abs}} L_\gamma}. \quad (572)$$

Here $\varepsilon_{\gamma, \text{abs}}$ includes solid angle, attenuation, full-energy response, and event acceptance, while L_γ is the live-time fraction when the count rate is referred to elapsed time. A live-time-corrected count rate shall not receive a second live-time correction. Inference requires the measured exposure, continuum response, dead-time characterization, and a specified confidence interval. Emission-rate limits use the full background and detector-response model.

H. Three-Dimensional Monte Carlo Shielding and Dose Tallies

The OpenMC/MCNP shielding model shall represent the complete three-dimensional enclosure, including joints, penetrations, coolant paths, detector positions, and surrounding scattering structures. The selected radial layer sequence, from the reactor outward, is given in Table LXII. Material cards shall specify measured density, elemental mass fractions, boron isotopic composition, and temperature. The 90/10 W-Cu composition is the selected configuration for the following dose allocation; an 80/20 alloy is a separate transport case.

TABLE LXII
SELECTED MONTE CARLO SHIELD CONFIGURATION

Layer	Thickness	Composition and function
Inner W-Cu	35.0 mm	90 wt% W / 10 wt% Cu; photon attenuation
B-HDPE	120.0 mm	5 wt% boron; neutron moderation and capture
Outer SS316L	10.0 mm	Structural enclosure; secondary-photon attenuation

Fixed-source coupled transport shall track neutron scattering and absorption, neutron-induced photons, and photon scattering and absorption. The hydrogenous moderator shall use the applicable thermal-scattering treatment. Capture photons generated within the shield remain part of the coupled history rather than being discarded at neutron absorption. The nominal neutron and photon dose components are

$$\dot{H}_n = 0.04 \mu\text{Sv/h},$$

$$\dot{H}_\gamma = 0.34 \mu\text{Sv/h},$$

$$\dot{H}_{\text{surface}} = 0.38 \mu\text{Sv/h} < 0.50 \mu\text{Sv/h} \quad (\text{design target}). \quad (573)$$

For a tally normalized per source particle, $\dot{H} = \dot{N}_{\text{source}} \sum_b \phi_b h_b$ requires an emission-rate normalization and energy-bin dose coefficients. Coupled transport shall include neutron-induced photons [74]. A volume-integrated track-length flux score shall be divided by the scoring volume before conversion to fluence per source particle [75]. For primary source class s , scored particle type $j \in \{n, \gamma\}$, and energy bin b , the surface-adjacent tally at position $\mathbf{r}$ gives

$$\dot{H}_{\text{surface}}(\mathbf{r}) = 3.6 \times 10^9 \sum_s \dot{N}_s \sum_{j,b} \phi_{s,j,b}(\mathbf{r}) h_{j,b}, \quad (574)$$

$$[\phi_{s,j,b}] = \text{cm}^{-2} \text{source}^{-1},$$

$$[h_{j,b}] = \text{Sv cm}^2, \quad [\dot{H}_{\text{surface}}] = \mu\text{Sv/h}.$$

The factor 3.6×10^9 converts Sv/s to $\mu\text{Sv/h}$. Each $\dot{N}_s$ is a physical primary emission rate in s^{-1} ; photon tallies from neutron histories include their secondary photons once. Independently emitted photons require their own source normalization. Dose-response coefficients shall identify the dose-equivalent quantity and irradiation geometry. Surface-current scores and absorbed-energy scores shall not be substituted for scalar-fluence dose conversion without the corresponding response model.

The nominal dose allocation is $0.04 + 0.34 = 0.38 \mu\text{Sv/h}$ against a $0.50 \mu\text{Sv/h}$ design limit. Acceptance uses the maximum accessible-surface result with statistical and source-normalization uncertainty. The calculation records source spectra, angular distributions, emission rates, geometry, nuclear data, code versions, histories, tally definitions, covariance, and convergence. OpenMC/MCNP comparisons use matched inputs and response functions. Commissioning requires a calibrated dose map and assessment of credible off-normal emissions independently of the proposed lattice branch.

I. Detector Hardware and Radiation-Transport Implementation

The following subsystem specifications implement the integrated assay, PSD acceptance, and nominal shielding configuration. They detail detector geometry, synchronous acquisition, event processing, and radiation transport. Alternative acquisition and shielding configurations require separate response and acceptance assessments.

Radiation diagnostics shall provide calibrated neutron and photon measurements for emission characterization and radiation-protection assessment. Detector calibration, background subtraction, and live-time accounting form the measurement basis for the EJ-301 pulse-shape discrimination (PSD) and gamma-spectroscopy channels. Acceptance of response, rejection, and dose-rate requirements shall rely on qualified measurements under the applicable emission and operating conditions.

J. EJ-301 Scintillators and Acquisition Chain

The neutron diagnostic shall use two sealed EJ-301 liquid-scintillator cells, each 127 mm in diameter and 127 mm deep, at opposing 180° azimuths and nominal 350 mm source-to-face distance. Each cell shall couple to a fast photomultiplier with reference operating bias −1450 V. The response shall be characterized over the 0.5–20.0 MeV neutron-energy range, including the approximately 2.45 MeV D–D neutron signature, with the accepted energy range stated for each analysis threshold.

EJ-301 has manufacturer-listed light output of approximately 12000 photons per MeV of electron energy, peak emission at 425 nm, specific gravity 0.874, and refractive index 1.505. Its short decay component is approximately 3.2 ns, with listed mean components 3.16, 32.3, and 270 ns [147]. Its flash point is approximately 26°C, requiring qualified sealed-cell and fire-safe handling. For Birks' law use $k_B^{\text{Birks}} = 0.0120 \text{ cm MeV}^{-1}$, not Boltzmann's constant. Stopping power is in MeV cm^{-1} :

$$\frac{dL}{dx} = S_L \frac{dE/dx}{1 + k_B^{\text{Birks}} dE/dx}. \quad (575)$$

S_L is the calibrated scintillation scale. Recoil light output is integrated along the track, not obtained by multiplying neutron energy by electron light yield.

The ionization-density dependence of the decay-component amplitudes provides the electron/proton recoil discrimination basis. The installed-system PSD boundary shall be calibrated as a function of deposited electron-equivalent energy and count rate.

The intrinsic neutron-efficiency target is $\varepsilon_{n,\text{int}} = 18.4\%$ at 2.45 MeV with a 1.0 MeVee electron-equivalent analysis threshold. Detector qualification shall establish whether the calibrated, proton-quenched recoil response supports this efficiency and threshold combination before a detection limit is assigned. Absolute efficiency shall include geometry, shielding, threshold-dependent intrinsic response, PSD acceptance, veto acceptance, and acquisition live time:

$$\varepsilon_{n,\text{abs}} = \frac{\Omega}{4\pi} T_n \varepsilon_{n,\text{int}} \varepsilon_{\text{PSD}} \varepsilon_{\text{veto}} L. \quad (576)$$

For two unobstructed circular detector faces normal to an isotropic point source, with radius $a = 63.5 \text{ mm}$ and distance $d = 350 \text{ mm}$,

$$\frac{\Omega_{\text{pair}}}{4\pi} = 1 - \frac{d}{\sqrt{d^2 + a^2}} = 0.0160627. \quad (577)$$

Multiplication by the reference intrinsic efficiency gives 0.0029555, or approximately 0.296%, before attenuation and event-selection losses. Response calibration shall represent each detector's installed position relative to the W–Cu shield and its penetrations. Each threshold, selection, and live-time correction shall be included once in the efficiency accounting.

The synchronous acquisition configuration shall provide two 14-bit channels at 1.0 GS s^{-1} , with a 1.0 ns sample interval and aggregate raw payload of 28 Gbit s^{-1} before framing. The ADC and front end shall satisfy resolution, sampling, dynamic-range, and timing requirements in the installed configuration. The FPGA interface shall provide adequate lane bandwidth, deterministic timing, clock calibration, buffering, and loss counters. Parallel processing capacity shall be established independently of the sample-clock rate.

K. Charge-Comparison Digital PSD Algorithm

For digitized samples $v[n]$ and trigger sample n_0 , the algorithm shall estimate the baseline from the 64 immediately preceding samples and form a polarity-corrected signal:

$$\hat{b} = \frac{1}{64} \sum_{n=n_0-64}^{n_0-1} v[n], \quad s[n] = \varsigma(v[n] - \hat{b}), \quad (578)$$

where $\varsigma \in \{-1, +1\}$ shall make valid scintillation pulses positive. Commissioning shall characterize baseline noise and preceding-pulse contamination for this 64-sample estimator. Contaminated baselines, clipped amplitudes, incomplete windows, and detected pileup shall set explicit quality flags and shall be excluded from the qualified single-pulse PSD selection.

At $\Delta t = 1.0 \text{ ns}$, the short and long gates shall contain exactly 24 and 240 samples, respectively:

$$\begin{aligned} Q_{\text{short}} &= \Delta t \sum_{n=n_0}^{n_0+23} s[n], \\ Q_{\text{long}} &= \Delta t \sum_{n=n_0}^{n_0+239} s[n], \\ Q_{\text{tail}} &= Q_{\text{long}} - Q_{\text{short}}, \\ S_{\text{PSD}} &= \frac{Q_{\text{tail}}}{Q_{\text{long}}} = 1 - \frac{Q_{\text{short}}}{Q_{\text{long}}}. \end{aligned} \quad (579)$$

The sums have units of sample amplitude times time. Conversion to coulombs shall use the calibrated front-end gain and impedance; common factors cancel in the PSD ratio. Division shall occur only after Q_{long} exceeds a strictly positive calibrated minimum. The inclusive gate endpoints in (579) shall be implemented without an additional endpoint sample.

The processing sequence shall be baseline estimation, calibrated threshold or constant-fraction timing, parallel gate accumulation, pulse-quality checks, energy calibration from Q_{long} , and energy-dependent PSD classification. Integer baseline-subtracted 14-bit codes shall use signed 24-bit accumulators;

fractional baseline representations shall provide the corresponding guard bits. Fixed-point scaling, overflow handling, rounding, and division shall be verified against a higher-precision reference. The latency budget shall include the 240 ns observation window, buffering, and subsequent pipeline processing.

The ranges $S_{\text{PSD},\gamma} \approx 0.15\text{--}0.22$ and $S_{\text{PSD},n} \approx 0.38\text{--}0.52$ define reference training regions. The released neutron selection shall use the energy-dependent boundary $S_{\text{PSD}} > s_c(Q_{\text{long}})$ determined from independently tagged neutron and gamma samples. Validation shall use separate events, with the cut frozen before candidate reactor data are evaluated. Records shall retain raw or representative waveforms, baseline history, threshold and gate settings, analysis version, and rejected-event counters.

At the assigned rate ceiling $r = 5.0 \times 10^5 \text{ s}^{-1}$, a simple Poisson estimate for an additional arrival during a 240 ns gate is

$$p_{\text{overlap}} = 1 - e^{-r(240 \text{ ns})} = 0.11308. \quad (580)$$

The approximately 11.3% reference overlap probability shall be included in rate-dependent pileup and live-time qualification. Acceptance at the rate ceiling requires measured PSD efficiency, rejection, and event-loss performance at that rate, separately from low-rate calibration.

L. FOM, Gamma Rejection, and Detection Limits

In a specified energy bin, let μ_n and μ_γ be the PSD centroids and w_n, w_γ their FWHM values. The required separation metric is

$$\text{FOM} = \frac{|\mu_n - \mu_\gamma|}{w_n + w_\gamma} = 2.15 \quad (\text{reported}). \quad (581)$$

The gamma rejection requirement is a separate classification probability:

$$p_{\gamma \rightarrow n} = \Pr(S_{\text{PSD}} > s_c | \gamma), \quad \mathcal{R}_\gamma = \frac{1}{p_{\gamma \rightarrow n}} > 10^6. \quad (582)$$

Both requirements shall be verified together with neutron acceptance, energy interval, threshold, count rate, and pileup policy.

For an illustrative pair of equal-width Gaussian PSD populations with standard deviation σ , $w_n = w_\gamma = 2\sqrt{2 \ln 2} \sigma$. At the midpoint cut,

$$\begin{aligned} \frac{s_c - \mu_\gamma}{\sigma} &= 2\sqrt{2 \ln 2} \text{FOM}, \\ p_{\gamma \rightarrow n} &= \frac{1}{2} \operatorname{erfc}\left(\frac{2\sqrt{2 \ln 2} \text{FOM}}{\sqrt{2}}\right). \end{aligned} \quad (583)$$

The nominal FOM = 2.15 describes peak separation rather than a measured tail probability. The equal-width Gaussian example places the cut 5.06286σ above the gamma centroid, with leakage of approximately 2.07×10^{-7} . Qualification shall characterize non-Gaussian tails and the actual energy-dependent cut using an independent validation sample.

If no gamma event leaks into the neutron selection among M_γ independent, representative gamma trials, the one-sided 95% binomial upper bound is

$$p_{\gamma \rightarrow n, 95} = 1 - 0.05^{1/M_\gamma}. \quad (584)$$

Commissioning of a 95% upper bound below 10^{-6} shall use at least $M_\gamma = 2,995,731$ independent zero-leakage validation trials under the specified conditions, with the selection fixed in advance. Training events shall remain separate from this validation sample. Finite leakage shall be reported with a count-based confidence interval; zero observed leakage shall be reported as a bound rather than infinite rejection.

For confidence $1 - \alpha$ and zero leakage, the required sample size is

$$M_\gamma \geq \left\lceil \frac{\ln \alpha}{\ln(1 - 10^{-6})} \right\rceil. \quad (585)$$

Thus 99.9999% confidence ($\alpha = 10^{-6}$) requires 13815504 trials. A Gaussian 6σ tail probability is different again and must state its one- or two-sided convention. Rejection probability, confidence in its upper bound, and Gaussian peak separation are distinct quantities.

For neutron monitoring with no accepted events and negligible expected background, the corresponding 95% Poisson bound is

$$\Gamma_{n, 95} = \frac{-\ln(0.05)}{\varepsilon_{n, \text{abs}} t} = \frac{2.995732}{\varepsilon_{n, \text{abs}} t}. \quad (586)$$

Here t shall be elapsed acquisition time when $\varepsilon_{n, \text{abs}}$ includes the live-time fraction L . Use of live exposure instead shall omit the corresponding live-time factor from efficiency. Non-negligible background, efficiency uncertainty, shielding, and veto losses shall be included in the full counting model, and null observations shall be reported as upper limits on the specified source model.

M. HPGe Gamma Spectroscopy and Activation Signatures

Gamma spectroscopy shall use an n-type coaxial high-purity germanium detector with cryogenic cooling, a pulse-tube cooler or qualified equivalent, and a 16384-channel digital spectrum. The detector specification shall provide nominal 50% relative efficiency against the conventional 3-inch by 3-inch NaI(Tl) reference at approximately 1.33 MeV, with installed absolute photopeak efficiency calibrated separately. Energy resolution shall be at most 1.85 keV FWHM near the ^{60}Co 1.33 MeV line and 0.92 keV near the 122 keV transition populated in ^{57}Fe by a ^{57}Co calibration source.

The detector housing shall have a separate local low-background shield of 100 mm lead, 1 mm cadmium, and 2 mm copper, with the lower-Z liner facing the detector. The reference direct viewing path includes a 2.0 mm beryllium window. Its energy-dependent attenuation shall be included in the response model. Using the NIST value $(\mu/\rho)_{\text{Be}} = 4.369 \text{ cm}^2 \text{ g}^{-1}$ at 5 keV [148], with nominal $\rho_{\text{Be}} = 1.848 \text{ g cm}^{-3}$,

$$T_{\text{Be}}(5 \text{ keV}) = e^{-(4.369)(1.848)(0.20)} = 0.19893. \quad (587)$$

Transmission at 10 keV is approximately 0.78743 using $\mu/\rho = 0.6466 \text{ cm}^2 \text{ g}^{-1}$. Window, dead-layer, shield-port, and collimator response must be included in absolute efficiency calibration. The pressure-boundary qualification and safe handling of beryllium components remain separate engineering requirements.

Table LXIII defines the monitoring signatures and their timing interpretations. Energy calibration and peak fitting shall

use evaluated line energies, emission probabilities, and decay schemes, with radioactive products distinguished from stable nuclei [149], [150]. The listed windows are rounded monitoring references.

The approximately 280.523 keV transition in ^{105}Pd shall be assessed with its possible ^{105}Rh decay feeding and prompt-capture origin distinguished by timing and accompanying lines. The stable palladium ground state shall not be assigned an activation half-life. Prompt capture cascades shall use the corresponding neutron-on timing and branching model; isotope attribution shall combine energy, associated lines, and time dependence.

For a radioactive activation product j , a constant-flux approximation gives

$$\begin{aligned}\frac{dN_j}{dt} &= R_{j,\text{act}} - \lambda_j N_j, \\ R_{j,\text{act}} &= N_{j,\text{target}} \int \sigma_j(E) \phi_n(E) dE, \\ N_j(t) &= \frac{R_{j,\text{act}}}{\lambda_j} (1 - e^{-\lambda_j t})\end{aligned}\quad (588)$$

for initially absent product and negligible depletion. After irradiation ceases, $N_j(t_c) = N_j(0)e^{-\lambda_j t_c}$. A photopeak count expectation over a live interval is

$$\mathbb{E}[C_j] = \varepsilon_{\text{FE},j} I_{\gamma,j} \int_{t_1}^{t_2} \lambda_j N_j(t) dt + C_{j,\text{bg}}, \quad (589)$$

where $\varepsilon_{\text{FE},j}$ includes geometry and attenuation and $I_{\gamma,j}$ is the emission probability. Commissioning shall establish the photopeak efficiency and background model. Reported detection limits shall additionally include target inventory, flux spectrum, irradiation and counting intervals, and their uncertainties. Any inferred neutron-branching limit, including a target below 10^{-8} , shall follow from this calibrated activation-response model rather than a peak-presence criterion alone.

N. Three-Layer Shield and Surface-Dose Requirement

Table LXIV defines the radial material schedule for the comparison shield, separately from the HPGc local enclosure. Its W–Cu composition is distinct from the transport-reference configuration; attenuation and dose results shall remain associated with the corresponding material model. Qualification requires a three-dimensional transport analysis including penetrations, joints, viewing ports, coolant paths, and source dimensions.

For a narrow, uncollided photon beam at the assigned 2.5 MeV reference energy, taking the nominal linear attenuation coefficient $\mu_\gamma = 0.82 \text{ cm}^{-1}$ gives

$$\frac{I}{I_0} = e^{-\mu_\gamma t_w} = e^{-(0.82)(3.50)} = 0.056699. \quad (590)$$

The reference calculation gives approximately 94.33% removal from the uncollided beam. The attenuation coefficient shall be verified for the mixture and photon energy. Whole-system dose assessment shall additionally include broad-beam buildup, scattered photons, pair-production products, and streaming.

The borated-polyethylene response shall include secondary photons, including the approximately 478 keV emission following boron capture and population of excited ^7Li [151]. Hydrogen-capture photons shall also be included. Qualification of a 10^4 neutron-attenuation target for the 120 mm layer shall specify the source spectrum, geometry, and material composition and use coupled neutron/photon transport.

The surface dose-equivalent-rate design target shall be $\dot{H}_{\text{surface}} \leq 0.38 \mu\text{Sv h}^{-1}$. The selected operational dose quantity and its conversion coefficients shall be stated. For differential fluence rate $\dot{\Phi}_j(E, \mathbf{r})$ and fluence-to-dose-equivalent coefficient $h_j(E)$,

$$\dot{H}(\mathbf{r}) = \sum_{j \in \{n, \gamma\}} \int h_j(E) \dot{\Phi}_j(E, \mathbf{r}) dE. \quad (591)$$

The reference source envelope uses neutrons near 2.45 MeV at a rate below 10^6 s^{-1} and photons over approximately 0.5–2.5 MeV. The qualified source model shall specify neutron and photon strengths, spectra, spatial distributions, and any potentially higher-energy channels. Commissioning measurements shall test the applicable source and transport assumptions before the surface-dose target is accepted.

For a deliberately simplified isotropic point source and no photon contribution,

$$\dot{H}_n(r) \approx \frac{Q_n}{4\pi r^2} T_n h_n(E_n), \quad (592)$$

with neutron source strength Q_n , source-to-surface distance r , neutron transmission T_n , and dose-conversion coefficient specified as independent inputs. The full transport analysis shall include shield streaming and activation. Active commissioning shall include calibrated neutron and photon surveys at seams, ports, and accessible surfaces under qualified representative source conditions.

The $0.50 \mu\text{Sv h}^{-1}$ project comparison level shall be identified as a design reference, separately from applicable occupational or public exposure limits. Acceptance of the $0.38 \mu\text{Sv h}^{-1}$ target requires radiation-protection review, qualified source and measurement uncertainties, and documented survey results. Operating authorization and regulatory compliance shall be established through the applicable independent review process.

a) *Coupled neutron–photon shield transport:* The coupled-transport comparison uses 35 mm W–Cu (90 wt% W/10 wt% Cu), 120 mm HDPE containing 5 wt% boron, and 10 mm SS316L. Its nominal surface result is $\dot{H} = 0.312 \mu\text{Sv/h}$ against the $0.380 \mu\text{Sv/h}$ project criterion, a difference of $0.068 \mu\text{Sv/h}$ or 17.8947% of the limit. For species $a \in \{n, \gamma\}$, the coupled angular transport model is

$$\begin{aligned}\frac{1}{v_a} \partial_t \psi_a &= -\hat{\Omega} \cdot \nabla \psi_a - \Sigma_{t,a} \psi_a + S_a \\ &+ \sum_b \mathcal{S}_{ba}[\psi_b], \\ \mathcal{S}_{ba}[\psi_b] &= \int dE' d\Omega' \Sigma_{b \rightarrow a}(E', \Omega' \rightarrow E, \Omega) \\ &\times \psi_b(E', \Omega').\end{aligned}\quad (593)$$

TABLE LXIII
SPECIFIED GAMMA-SPECTRAL MONITORING WINDOWS AND TIMING INTERPRETATION

Species or feature	Production or origin	Assigned energy window	Interpretation and timing
^{64}Cu	$^{63}\text{Cu}(n, \gamma)^{64}\text{Cu}$ and other possible activation routes	1345.77 keV	Delayed decay gamma; half-life approximately 12.7 h; weak emission probability requires calibrated sensitivity
^{105}Pd	$^{104}\text{Pd}(n, \gamma)^{105}\text{Pd}^*$; possible feeding from ^{105}Rh decay	Near 280.5 keV	^{105}Pd ground state is stable; a level transition is not evidence of delayed decay of activated ^{105}Pd
^{187}W	$^{186}\text{W}(n, \gamma)^{187}\text{W}$	479.5 and 685.8 keV	Delayed decay signatures; half-life approximately 23.7 h; compare multiple lines and decay curve
Positron annihilation	Positron-producing radionuclides or other pair-production processes	Near 511 keV	Not isotope-specific; requires coincident or spectral context
Continuum	Bremsstrahlung, Compton scattering, electronic background, and other sources	10–150 keV	Fit only after window and detector-response correction; not a uniquely nuclear signature

TABLE LXIV
THREE-LAYER REACTOR RADIATION-SHIELD ALLOCATION

Order from source	Material	Thickness (mm)	Assigned function and qualification requirement
1, inner	W–Cu, 80/20 by mass	35.0	Photon attenuation; qualify composition, density, energy-dependent attenuation, buildup, and secondary radiation
2, middle	Polyethylene with 5.0 wt% boron	120.0	Neutron moderation and capture; nominal density 0.95 g cm^{-3} ; boron isotopic composition and capture-gamma transport must be included
3, outer	SS316L structural enclosure	10.0	Mechanical containment and additional photon attenuation; include seams, penetrations, and material activation
Total nominal radial material thickness		165.0	Excludes air gaps, supports, liners, and the separate HPGc shield

with neutron-induced photon production in $\Sigma_{n \rightarrow \gamma}$, energy redistribution, and material-dependent scattering/absorption. Capture photons include the approximately 0.478 MeV line following the excited-state $^{10}\text{B}(n, \alpha)^7\text{Li}$ branch and the approximately 2.223 MeV photon from $\text{H}(n, \gamma)\text{D}$; hydrogen scattering slows neutrons but is not itself that capture reaction. At an accessible survey location,

$$\dot{H}(\mathbf{r}) = \sum_a \int h_a(E) \phi_a(\mathbf{r}, E) dE, \quad \phi_a = \int \psi_a d\Omega. \quad (594)$$

The nominal $0.312 \mu\text{Sv/h}$ allocation uses specified emission strengths and spectra, material number densities, cross-section data, streaming geometry, and dose-conversion tallies. Acceptance requires converged coupled transport and an independently scoped radiation survey under the applicable exposure and measurement requirements [74].

O. Active Cosmic-Ray Muon Veto and Live-Time Accounting

Each EJ-301 assembly shall be surrounded by a nominal 20 mm-thick EJ-200 plastic-scintillator guard with independent photodetector timing and recorded guard triggers. The prompt

coincidence window shall have full width $\Delta t_{\text{veto}} = 20 \text{ ns}$. After correcting calibrated channel delay δt , define

$$v_\mu(t_e) = \begin{cases} 1, & \min_j |t_e - t_{\mu,j} - \delta t| \leq 10 \text{ ns}, \\ 0, & \text{otherwise,} \end{cases} \quad (595)$$

where t_e is the inner-detector event time and $t_{\mu,j}$ is a guard time. A prompt-veto neutron candidate shall pass pulse-quality and PSD cuts and satisfy $v_\mu(t_e) = 0$. Vetoed events and timing distributions shall be retained to support efficiency, accidental-coincidence, and analysis-bias verification.

For independent Poisson guard triggers at rate r_μ , the accidental prompt-veto probability is

$$p_{\text{acc}} = 1 - e^{-r_\mu \Delta t_{\text{veto}}} \approx r_\mu \Delta t_{\text{veto}} \quad (596)$$

when the product is small. The accepted live-time and efficiency model shall include accidental vetoes, guard thresholds, panel gaps, channel dead time, and scintillator timing resolution. Live exposure shall be determined from recorded acquisition intervals and the union of excluded windows, avoiding double counting of overlapping dead-time and veto intervals.

Prompt and delayed muon-correlated backgrounds shall be characterized separately. A 2.45 MeV neutron traverses

350 mm in approximately 16.17 ns before moderation or additional path delays. Selection of any extended veto shall use measured time-correlation spectra and quantify the trade-off between delayed-background rejection and accidental losses. The gate shall be fixed using commissioning data independently of candidate reactor events.

The residual accepted-background requirement shall be below 10^{-4} s^{-1} , referred to the declared live exposure. Verification shall use off-source, reactor-disabled, and isotope-control runs with the same shield geometry, thresholds, PSD cuts, veto policy, and exposure accounting. The confidence level and exposure supporting the background-rate limit shall be reported.

For independent Poisson source-on and matched source-off counts N_{on} and N_{off} , define the live-exposure normalization $\alpha_t = t_{\text{on, live}}/t_{\text{off, live}}$. Under a stationary matched-background model with fixed α_t , the background-subtracted count and counting-variance estimate are

$$\begin{aligned}\hat{N}_{\text{net}} &= N_{\text{on}} - \alpha_t N_{\text{off}}, \\ u_{\text{count}}^2(\hat{N}_{\text{net}}) &= N_{\text{on}} + \alpha_t^2 N_{\text{off}}, \\ \hat{r}_{\text{net}} &= \frac{\hat{N}_{\text{net}}}{t_{\text{on, live}}}.\end{aligned}\quad (597)$$

Commissioning shall test background stationarity and add exposure-normalization, drift, efficiency, and selection uncertainties as applicable. Signed net estimates shall be retained; low-count confidence intervals and source-rate upper limits shall use a joint count model rather than a Gaussian approximation to this variance alone. Conversion from accepted rate to source rate shall use an efficiency definition consistent with the live-exposure normalization.

P. Commissioning and Acceptance Requirements

Table LXV consolidates instrument specifications and acceptance criteria. The numerical entries define design targets or conditional predictions. Experimental qualification requires calibrated measurements of resolution, species separation, detector response, and source-normalized shielding performance. The HPGe response at 23.84 MeV requires its own absolute-efficiency calibration. The shield result remains conditional on the source spectrum, intensity, geometry, and dose-conversion convention in Section VI-I. The exact-binomial rejection calculation is given in Eq. (571); a FOM value alone cannot replace that leakage measurement.

Release for measurement service requires calibrated detector response, verified background subtraction, and traceable evidence of compliance with the applicable performance criteria. The commissioning and acceptance activities are:

- **CM-01: Dual calorimetry.** Complete the electrical-substitution sequence, sensor and thermopile calibration, gain-drift checks, and nonreactive baseline runs of Section V. Verify spatial calibration transfer, corrections, and covariance before reporting the commissioned variance budget.
- **CM-02: Isotope assay.** Calibrate the mass axis, Faraday and multiplier gains, dose transfer, getter removal, and

helium recovery. Verify manifold, matrix-gas, memory, and ingress subtraction through the complete sampling path, retaining the introduced-standard inventory and uncertainty.

- **CM-03: EJ-301 response and PSD.** Calibrate timing, electron-equivalent energy, recoil response, threshold-dependent efficiency, and rate-dependent pileup. Validate the fixed-point gate implementation and frozen PSD cut against independent tagged samples, documenting FOM, gamma-leakage bounds, neutron acceptance, and live exposure.
- **CM-04: Gamma and survey response.** Establish HPGe energy calibration, resolution, absolute photopeak efficiency, attenuation corrections, and matched continuum/line backgrounds. Complete source-dependent neutron and photon response checks and qualified accessible-surface surveys.
- **CM-05: Background and veto verification.** Measure prompt and delayed guard correlations, accidental losses, overlapping exclusion windows, and matched source-off rates. Verify the exposure-normalized subtraction and uncertainty model using independent control runs, including null and known-signal checks.

Acceptance documentation shall identify instrument configuration, calibration traceability, analysis version, operating conditions, measurement uncertainty, and acceptance disposition. Calorimetry, helium assay, prompt-radiation monitoring, and delayed-activation analysis shall retain separate measurement and inference records.

VII. REDUCED-ORDER ROBUST CONTROL AND DECOUPLED SAFETY INTERLOCKS

The architecture separates continuous process regulation, nonlinear state estimation, and independent safety-instrumented functions. A fifteen-state plant supplies the analytical basis for a four-state control model. Structured uncertainty describes parameter drift, while sector-bounded feedback defines the conditions for nonlinear stability. An independently powered TMR hardware path inhibits RF and optical drive without waiting for model execution or supervisory polling. Mechanical isolation and fluid mitigation operate in a separate response domain. Numerical setpoints and acceptance bounds define the design; a stability certificate must apply to the specified realization, and a shutdown certificate must include the physical safe-state response.

A. Fifteen-State Plant and Balanced Truncation

a) *Control synthesis and runtime separation:* The reduction interface shall construct the Gramians, extract Hankel singular values, and map the fifteen-state plant to four states with an approximation certificate. A 100 Hz supervisory update gives a 10 ms sample interval; it does not set the response time of the independent fast interlock. Nonlinear stability synthesis additionally requires the sector limits, multiplier, and positive-definite matrix certificate. A residual monitor evaluates prediction errors and is distinct from the offline reduction and LMI solvers.

TABLE LXV
DIAGNOSTIC VERIFICATION AND COMMISSIONING ACCEPTANCE MATRIX

Diagnostic	Instrument specification	Acceptance target and required verification
Magnetic-sector isotope assay	Double-focusing analyzer; $R = m/\Delta m = 25000$ at nominal mass 4; St707 getter conditioning	Resolve ^4He (4.002603 u) from D_2 (4.028204 u). Verify the 1.45×10^5 getter-ratio target, deuterium-tail removal, helium recovery, mass discrimination, and background subtraction with calibrated isotope mixtures.
EJ-301 neutron/gamma separation	127×127 mm cells; 1.0 GS/s, 14-bit digitizer; 24 ns and 240 ns charge gates	FOM target 2.15; $R_\gamma > 10^6$ at 95% confidence conditional on 2,995,731 independent zero-leakage gamma trials. Preserve fixed energy-dependent cuts, neutron acceptance, live time, and pile-up accounting.
HPGe gamma spectroscopy	Coaxial detector, 50% relative efficiency; FWHM < 1.85 keV at 1.33 MeV	Calibrate energy and absolute efficiency in the measurement geometry. Characterize response to a possible 23.84 MeV photon, including escape/continuum contributions; interpret the near-280.5 keV Pd level transition with timing and associated lines, not as decay of radioactive ^{105}Pd .
Composite radiation shield	35 mm W-Cu (90/10), 120 mm B-HDPE (5 wt% B), 10 mm SS316L, from core outward	Conditional surface dose $\dot{H}_{\text{surface}} = 0.38 \mu\text{Sv/h}$: neutron 0.04 plus photon 0.34, below $0.50 \mu\text{Sv/h}$. Verify source-normalized coupled tallies, secondary photons, penetrations, tally uncertainty, and calibrated accessible-surface surveys.

For the selected two-input, two-output plant, define

$$\begin{aligned}
 \dot{\mathbf{x}}_{15} &= \mathbf{A}\mathbf{x}_{15} + \mathbf{B}\mathbf{u}, \\
 \mathbf{y} &= \mathbf{C}\mathbf{x}_{15} + \mathbf{D}\mathbf{u}, \\
 G(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}, \\
 \mathbf{A} &\in \mathbb{R}^{15 \times 15}, \quad \mathbf{B} \in \mathbb{R}^{15 \times 2}, \\
 \mathbf{C} &\in \mathbb{R}^{2 \times 15}, \quad \mathbf{D} \in \mathbb{R}^{2 \times 2}.
 \end{aligned} \tag{598}$$

The state definitions and numerical simulator realization are specified in Section X. Reduction shall retain the same physical ports, input/output scaling, operating point, and direct feedthrough. Selection from a four-input, four-output realization requires explicit port maps before balancing; state reduction alone does not remove physical ports.

For a Hurwitz, controllable and observable realization, the positive-definite controllability and observability Gramians satisfy [35]

$$\begin{aligned}
 \mathbf{A}\mathbf{W}_c + \mathbf{W}_c\mathbf{A}^\top + \mathbf{B}\mathbf{B}^\top &= 0, \\
 \mathbf{A}^\top\mathbf{W}_o + \mathbf{W}_o\mathbf{A} + \mathbf{C}^\top\mathbf{C} &= 0, \\
 \mathbf{W}_c &= \mathbf{L}_c\mathbf{L}_c^\top, \quad \mathbf{W}_o = \mathbf{L}_o\mathbf{L}_o^\top, \\
 \mathbf{L}_o^\top\mathbf{L}_c &= \mathbf{U}\Sigma\mathbf{V}^\top, \\
 \mathbf{T}_{\text{bal}} &= \mathbf{L}_c\mathbf{V}\Sigma^{-1/2}, \\
 \mathbf{T}_{\text{bal}}^{-1} &= \Sigma^{-1/2}\mathbf{U}^\top\mathbf{L}_o^\top.
 \end{aligned} \tag{599}$$

Here $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_{15})$ contains descending Hankel singular values. Uncontrollable or unobservable modes shall be removed before inverting singular factors. In balanced coordinates,

$$\begin{aligned}
 \mathbf{T}_{\text{bal}}^{-1}\mathbf{A}\mathbf{T}_{\text{bal}} &= \begin{bmatrix} \mathbf{A}_r & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \\
 \mathbf{T}_{\text{bal}}^{-1}\mathbf{B} &= \begin{bmatrix} \mathbf{B}_r \\ \mathbf{B}_2 \end{bmatrix}, \quad \mathbf{C}\mathbf{T}_{\text{bal}} = [\mathbf{C}_r \quad \mathbf{C}_2], \\
 \mathbf{D}_r &= \mathbf{D}, \quad G_r(s) = \mathbf{C}_r(s\mathbf{I} - \mathbf{A}_r)^{-1}\mathbf{B}_r + \mathbf{D}_r.
 \end{aligned} \tag{600}$$

The retained block has four states and the discarded block eleven. This construction is balanced truncation, not the Schur-complement residualization used for a separate low-frequency approximation. The numerical four-state candidate is [86]

$$\mathbf{A}_r = \begin{bmatrix} -488.2142 & 112.4051 & -34.1102 & 18.9015 \\ -112.4051 & -312.8521 & 88.5410 & -41.2204 \\ -34.1102 & -88.5410 & -184.2105 & 105.6120 \\ 18.9015 & 41.2204 & -105.6120 & -859.9338 \end{bmatrix} \text{s}^{-1}. \tag{601}$$

The modal design targets are

$$\lambda_{\text{target}} = \{-925.12, -548.93, -274.35, -95.82\} \text{s}^{-1}. \tag{602}$$

The eigenvalues of Eq. (601) are

$$\lambda(\mathbf{A}_r) = \{-841.993834, -421.968669, -290.624049 \pm i70.356151\} \text{s}^{-1}. \tag{603}$$

The matrix is Hurwitz. The Lyapunov construction uses $\mathbf{P}_0 = \mathbf{I}_4$ and $\mathbf{Q}_0 = -(\mathbf{A}_r^\top + \mathbf{A}_r)$:

$$\begin{aligned}
 \lambda_{\min}(\mathbf{Q}_0) &= 360.847685 \text{s}^{-1} > 0, \\
 \mathbf{V}_0(\mathbf{x}_r) &= \mathbf{x}_r^\top \mathbf{x}_r, \\
 \dot{\mathbf{V}}_0 &= -\mathbf{x}_r^\top \mathbf{Q}_0 \mathbf{x}_r \leq -360.847685 \text{s}^{-1} \|\mathbf{x}_r\|_2^2.
 \end{aligned} \tag{604}$$

The certificate applies to the unforced linear dynamics in consistently scaled reduced coordinates. Transfer-function approximation additionally uses $\mathbf{B}_r, \mathbf{C}_r, \mathbf{D}_r$ and the parent-model transformation.

For a stable minimal parent with the spectrum in Table LXVI and the balancing partition in Eq. (600), the absolute error

TABLE LXVI
COMPARISON HANKEL SPECTRUM FOR A DISTINCT PORT SCALING

Mode	σ_i	Treatment
1	1.931312×10^{-1}	Retain
2	1.647743×10^{-2}	Retain
3	4.253545×10^{-3}	Retain
4	2.981347×10^{-3}	Retain
5	1.502356×10^{-3}	Truncate
6	4.453772×10^{-4}	Truncate
7	6.281842×10^{-5}	Truncate
8	2.672913×10^{-5}	Truncate
9	1.280489×10^{-5}	Truncate
10	2.974815×10^{-7}	Truncate
11	9.931668×10^{-8}	Truncate
12	1.057691×10^{-8}	Truncate
13	4.349070×10^{-9}	Truncate
14	4.652158×10^{-10}	Truncate
15	1.193118×10^{-12}	Truncate

satisfies

$$\begin{aligned}
 \|G - G_r\|_\infty &\leq 2 \sum_{i=5}^{15} \sigma_i, \\
 \sum_{i=5}^{15} \sigma_i &= 2.050497830568918 \times 10^{-3}, \\
 2 \sum_{i=5}^{15} \sigma_i &= 4.100995661137836 \times 10^{-3} \\
 &\leq 4.1010 \times 10^{-3}.
 \end{aligned} \tag{605}$$

The bound uses the port-scaled realization, Gramians, and balancing maps associated with Eq. (601). Its units follow the input/output normalization.

b) *Calculated spectrum and reduction acceptance:*

Table XLII gives the selected fifteen-value spectrum, with the first four modes retained. Let $\hat{\sigma}_i$ denote their targets. The spectral-error metric and discarded-spectrum measure are

$$\begin{aligned}
 e_4 &= \max_{1 \leq i \leq 4} \frac{|\sigma_i - \hat{\sigma}_i|}{|\hat{\sigma}_i|}, \\
 r_{\text{tail}} &= \frac{\sum_{i=5}^{15} \sigma_i^2}{\sum_{j=1}^{15} \sigma_j^2} \\
 &= 2.129567641168566 \times 10^{-6} < 10^{-4}.
 \end{aligned} \tag{606}$$

The squared-spectrum ratio describes the discarded singular values. For a stable minimal plant, the absolute balanced-truncation bound is

$$\|G - G_r\|_\infty \leq 2 \sum_{i=5}^{15} \sigma_i = 0.07974. \tag{607}$$

Each spectrum uses its corresponding port-scaled plant, Gramians, and balancing maps. The first four selected singular values have targets (12.45, 5.13, 1.85, 0.94) with absolute tolerance 10^{-12} per entry.

B. Structured Uncertainty and Robust μ -Synthesis

Fluid density, thermal-gradient response, and actuator gain shall be represented by normalized parameter perturbations

$$\begin{aligned}
 p_j &= p_{j0}(1 + 0.15\delta_j), \quad |\delta_j| \leq 1, \\
 \Delta &= \text{diag}(\delta_\rho I, \delta_T I, \delta_k I), \\
 \|\Delta\|_\infty &\leq 1.
 \end{aligned} \tag{608}$$

Repeated real blocks represent shared physical parameters; any additional complex dynamic blocks require their own frequency weights. The 0.15 factors belong in the plant uncertainty channels and shall not be applied twice. The generalized state-space interconnection is

$$\begin{bmatrix} \dot{\mathbf{x}}_r \\ \mathbf{z}_\Delta \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} A_r & B_\Delta & B_u \\ C_\Delta & D_{\Delta\Delta} & D_{\Delta u} \\ C_y & D_{y\Delta} & D_{yu} \end{bmatrix} \begin{bmatrix} \mathbf{x}_r \\ \mathbf{w}_\Delta \\ \mathbf{u} \end{bmatrix}, \quad \mathbf{w}_\Delta = \Delta \mathbf{z}_\Delta. \tag{609}$$

Writing the corresponding transfer matrix as

$P(s) = [P_{ab}(s)]_{a,b=1}^2$, the uncertain plant is the upper linear fractional transformation

$$\begin{aligned}
 M(s) &= F_u(P(s), \Delta) \\
 &= P_{22} + P_{21}\Delta(I - P_{11}\Delta)^{-1}P_{12}.
 \end{aligned} \tag{610}$$

The inverse must exist for a well-posed interconnection. Controller synthesis closes the control ports with $K(s)$, including the selected feedback sign. Let $N_K(s)$ denote the resulting nominal return matrix from $\mathbf{w}_\Delta$ to $\mathbf{z}_\Delta$. Structured singular-value analysis applies to this return matrix, not to the already uncertainty-closed input/output plant $M(s)$:

$$\mu_{\mathcal{D}}(N_K) = \left[\min_{\substack{\Delta \in \mathcal{D} \\ \det(I - N_K \Delta) = 0}} \bar{\sigma}(\Delta) \right]^{-1}. \tag{611}$$

The value is zero if no destabilizing structured perturbation exists. Synthesis shall specify disturbance and performance weights, controller order, uncertainty-block dimensions, and the resulting controller realization. Frequency-dependent scaling and controller updates shall minimize a certified upper bound on μ [100]. The design acceptance envelope is

$$\begin{aligned}
 \sup_{\omega} \bar{\mu}_{\mathcal{D}}(N_K(i\omega)) &\leq 0.842 < 1.00, \\
 \omega_{\text{peak}} &= 412 \text{ rad/s}, \\
 \omega_{\text{scan}} &\in [10^{-2}, 10^5] \text{ rad/s}.
 \end{aligned} \tag{612}$$

Here $\bar{\mu}$ denotes a certified upper bound, not an assumed exact value. Nominal internal stability, well-posedness, and a bound below unity over the full frequency range establish robust stability for the normalized uncertainty set. Frequency coverage includes interpolation and low/high-frequency tails. Robust performance additionally requires weighted performance channels and an augmented performance block. The assigned peak of 0.842 is an acceptance target for the generalized plant and controller, subject to the scaling certificate and frequency coverage.

C. Nonlinear Lur'e-Postnikov Stability Certificate

For equilibrium-shifted autonomous dynamics, define

$$\dot{\mathbf{x}} = A_c \mathbf{x} + B_\phi \phi(y), \quad y = C_\phi \mathbf{x}, \quad \phi(0) = 0. \quad (613)$$

The sign of feedback is included in B_ϕ . The matrix A_c shall include controller and observer states whenever their dynamics are retained; it equals A_r only for the corresponding four-state static-feedback model. Assume a locally Lipschitz, globally sector-bounded nonlinearity with

$$0 \leq y\phi(y), \quad \phi(y)^2 \leq k_\phi y\phi(y), \quad k_\phi > 0. \quad (614)$$

The Lur'e-Postnikov candidate is

$$V(\mathbf{x}) = \mathbf{x}^\top \mathbf{P} \mathbf{x} + \int_0^y \phi(\sigma) d\sigma, \quad \mathbf{P} = \mathbf{P}^\top > 0. \quad (615)$$

In consistently normalized coordinates the sector condition makes the integral nonnegative, so V is positive definite and radially unbounded. Differentiation gives

$$\begin{aligned} \dot{V} = & \mathbf{x}^\top (A_c^\top \mathbf{P} + \mathbf{P} A_c) \mathbf{x} + 2\mathbf{x}^\top \mathbf{P} B_\phi \phi(y) \\ & + \phi(y) C_\phi [A_c \mathbf{x} + B_\phi \phi(y)]. \end{aligned} \quad (616)$$

A sufficient sector/KYP-type matrix certificate is the existence of $\mathbf{Q} = \mathbf{Q}^\top > 0$, $\eta > 0$, and $\tau \geq 0$ such that [101]

$$\begin{bmatrix} A_c^\top \mathbf{P} + \mathbf{P} A_c + \mathbf{Q} & \mathbf{h} \\ \mathbf{h}^\top & C_\phi B_\phi + \eta - \tau/k_\phi \end{bmatrix} \preceq 0, \quad (617)$$

$$\mathbf{h} = \mathbf{P} B_\phi + \frac{1}{2} A_c^\top C_\phi^\top + \frac{\tau}{2} C_\phi^\top.$$

Indeed, premultiplication and postmultiplication by $[\mathbf{x}^\top, \phi(y)]$ and its transpose give

$$\begin{aligned} \dot{V} \leq & -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - \eta \phi(y)^2 - \tau[y\phi(y) - \phi(y)^2/k_\phi] \\ \leq & -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - \eta \phi(y)^2 < 0, \quad \mathbf{x} \neq 0. \end{aligned} \quad (618)$$

These conditions prove global asymptotic stability for the stated autonomous model. Reactor-wide application requires a numerical certificate for the actual closed-loop matrices and a sector bound valid over the entire state domain. A local linearization or a phase-coupling law satisfying the sector only near lock supports a regional result, not a global one. Switching, external disturbances, actuator limits outside the sector, and loss-of-coolant modes require their own invariant-domain or hybrid-system assessment.

a) Multi-input sector certificate: For the explicit negative-feedback convention $\dot{\mathbf{x}} = A_r \mathbf{x} - B_r \phi(C_r \mathbf{x})$, let $\Gamma = \text{diag}(\gamma_j) > 0$ and $\phi_j(v_j)[\phi_j(v_j) - \gamma_j v_j] \leq 0$ globally. A quadratic sector certificate is $P = P^\top > 0$, diagonal $\Lambda > 0$, and

$$\begin{bmatrix} A_r^\top P + P A_r & -P B_r + C_r^\top \Lambda \\ -B_r^\top P + \Lambda C_r & -2\Lambda \Gamma^{-1} \end{bmatrix} \prec 0. \quad (619)$$

Congruence by $\text{diag}(I, -I)$ gives off-diagonal block $P B_r - C_r^\top \Lambda$ with the same definiteness. The feedback sign and sector convention must accompany that form. For normalized coordinates, numerical acceptance uses margins $P \succeq 10^{-6} I$, $\Lambda \succeq 10^{-6} I$, and an LMI maximum eigenvalue at most -10^{-6} , checked independently of the solver status. This sufficient quadratic certificate is distinct from the integral Lyapunov function in Eq. (615). Sector validity, Lipschitz continuity, and a correctly shifted equilibrium delimit the stability claim.

TABLE LXVII
PRESSURE-DEPENDENT PLASMA TIMING TARGETS

Pressure (bar)	τ_{rec} (μs)	τ_{diff} (μs)	t_{quench} (μs)
1.0	0.278	0.420	2.15
2.5	0.278	0.780	2.80
5.0	0.278	1.165	3.42
7.5	0.278	1.520	4.10
10.0	0.278	1.890	4.85

D. Plasma Recombination, Diffusion, and RF Extinction

After drive removal, the effective electron-density model is

$$\frac{\partial n_e}{\partial t} = -\alpha_r n_e^2 + D_a \nabla^2 n_e, \quad (620)$$

with the nominal constitutive inputs [87]

$$\begin{aligned} \alpha_r &= 2.4 \times 10^{-17} \text{ m}^3/\text{s}, \\ D_a &= 12.5 \text{ m}^2/\text{s}, \quad n_{e0} = 1.5 \times 10^{20} \text{ m}^{-3}, \\ \lambda_0 &= 0.012 \text{ m}. \end{aligned} \quad (621)$$

The quadratic loss law uses an effective two-body coefficient. Any three-body contribution shall be reduced using a stated density and temperature dependence or represented by its own rate term. With an absorbing-boundary fundamental-mode approximation $\nabla^2 n_e \simeq -(\pi/\lambda_0)^2 n_e$, define $\beta = D_a(\pi/\lambda_0)^2$. The scalar amplitude equation and its solution are

$$\begin{aligned} \dot{n}_e &= -\alpha_r n_e^2 - \beta n_e, \\ \frac{n_e(t)}{n_{e0}} &= \frac{\beta e^{-\beta t}}{\beta + \alpha_r n_{e0}(1 - e^{-\beta t})}, \\ t_f &= \frac{1}{\beta} \ln \frac{\beta + f \alpha_r n_{e0}}{f(\beta + \alpha_r n_{e0})}, \quad f = \frac{n_e(t_f)}{n_{e0}}. \end{aligned} \quad (622)$$

This is an exact solution of the scalar approximation, not of an arbitrary spatial profile. For the displayed inputs,

$$\begin{aligned} \tau_{\text{rec}} &= (\alpha_r n_{e0})^{-1} = 277.777778 \mu\text{s}, \\ \tau_{\text{diff}} &= \beta^{-1} = 1.167220 \mu\text{s}, \\ t_{0.01} &= 5.370402 \mu\text{s}, \\ n_e(3.42 \mu\text{s})/n_{e0} &= 0.0531834. \end{aligned} \quad (623)$$

The nominal RF-extinction target is $t_{\text{quench}} = 3.42 \mu\text{s}$ at 5.0 bar. Table LXVII lists pressure-dependent timing targets; Eq. (623) gives the density-model time scales.

The timing allocation spans 1.0–10.0 bar, with a maximum target of 4.85 μs . Qualification defines the residual RF-energy threshold, its mapping to electron density and cavity loading, boundary conditions, continuing ionization sources, and pressure-dependent coefficients.

E. Decoupled TMR Protection and Safe-State Timing

a) Three-tier safety-instrumented architecture: The protection architecture separates fast electronic inhibition, slower mechanical isolation, and passive overpressure protection from process regulation. The TMR FPGA implements 2003 voting on a specified 1.0 MHz timing basis; the clock period, asynchronous input capture, conditioning, and output delays

all enter the end-to-end latency budget. Neither the observer nor the MPC software is permitted to delay protective action.

- 1) *Tier 1: electronic drive inhibition.* Independently qualified phase-surge or RF-mismatch detection commands a SiC crowbar with a sub-500 ns conduction target and a parallel electro-optic detuning path of 15.0 GHz. The end-to-end objective is $t_{\text{trip}} < 12.50 \mu\text{s}$, with a nominal allocation of $8.57 \mu\text{s}$. A sustained 15.0 GHz frequency shift requires a phase ramp, not a static phase step:

$$\left| \frac{d\phi_{\text{EOM}}}{dt} \right| = 2\pi(15.0 \times 10^9) \simeq 9.42478 \times 10^{10} \text{ rad/s}. \quad (624)$$

Verify actual RF-energy extinction and optical decoupling, not only electrical command edges. The $12.50 \mu\text{s}$ allocation in Table LXIX is a ceiling; compliance with a strict inequality requires measured timing margin including uncertainty. A $50 \mu\text{s}$ phase-estimation or detuning comparison cannot be inserted as an additional serial delay in this fast path.

- 2) *Tier 2: mechanical isolation.* Initiate shutters and pneumatic fuel isolation independently of Tier 1, with final-element travel targets of 10–45 ms. Confirm closed-position feedback and inhibit restart. Any nitrogen mitigation sequence requires regulated pressure, a qualified vent path, and compatible vessel conditions; it must not introduce a new overpressure hazard or be treated as instantaneous inventory removal.
- 3) *Tier 3: passive pressure relief.* Maintain independent, directly connected relief devices for the manifold and core pressure boundaries. Table LXXIV assigns API area E to the 5.00 bar gauge manifold case and area D to the 15.00 MPa gauge core case. These are separate protected volumes, not redundant valves at interchangeable setpoints. Qualified sizing, installation, discharge routing, backpressure, and certified device capacity remain required [6], [38]–[40]; API letter designations alone do not certify the assembled system.

The commissioning record shall include power-loss and sensor-fault tests, asynchronous fault arrival, worst-case supply and temperature conditions, safe-state feedback, and independent reset authorization.

b) Probability-conserving safety-state model: Let $\mathbf{p} = (p_0, p_1, p_2, p_3)^T$ describe healthy, one-channel-degraded, safely tripped, and dangerous states, respectively. A continuous-time model requires nonnegative off-diagonal transition rates and zero column sums. With channel failure rate $\lambda = 1.5 \times 10^{-6} \text{ h}^{-1}$, repair rate $\mu = 0.125 \text{ h}^{-1}$, and an independently characterized hazardous-transition rate $\lambda_h \geq 0$, one admissible topology is

$$\dot{\mathbf{p}} = \mathbf{Q}\mathbf{p}, \quad \mathbf{p}(0) = (1, 0, 0, 0)^T, \quad (625)$$

$$\mathbf{Q} = \begin{bmatrix} -3\lambda & \mu & \mu & 0 \\ 3\lambda & -(\mu + 2\lambda + \lambda_h) & 0 & 0 \\ 0 & 2\lambda & -\mu & 0 \\ 0 & \lambda_h & 0 & 0 \end{bmatrix}.$$

Repair from the safely tripped state returns probability to the healthy state, subject to restart authorization. The dangerous

state receives the independently characterized hazardous transition λ_h ; the generator conserves total probability.

For a defined dangerous state, the one-year demand-unavailability measure is

$$\text{PFD}_{\text{avg}} = \frac{1}{\tau_{\text{test}}} \int_0^{\tau_{\text{test}}} p_3(t) dt, \quad \tau_{\text{test}} = 8760 \text{ h}. \quad (626)$$

For a comparison model with independent undetected failures, perfect initial health, and no repair between proof tests,

$$\begin{aligned} q(t) &= 1 - e^{-\lambda t}, \\ p_D(t) &= 3q(t)^2 - 2q(t)^3, \\ \frac{1}{\tau_{\text{test}}} \int_0^{\tau_{\text{test}}} p_D(t) dt &= 1.69851776 \times 10^{-4}. \end{aligned} \quad (627)$$

That comparison excludes the stated eight-hour repair process and is not the complete safety-function PFD. Common-cause faults, detection coverage, voter/power failures, final elements, and proof-test effectiveness require their own terms. SIL-3 certification requires lifecycle and complete-function evidence [4], [5].

c) Fault-injection and voting acceptance: The function-specific injection cases are The 0.5 kg/s flow threshold exceeds the nominal 0.08–0.12 kg/s manifold flow and would be asserted during normal operation; it shall not be enabled for that operating range. For the 1000 W optical comparison, 15% attenuation corresponds to 850 W, but this is not the reactor support-load rating. Vote separately on each hazard across three independent channels: $v_j = \mathbf{1}(\sum_{c=1}^3 b_{cj} \geq 2)$, with a trip if any $v_j = 1$. One asserted channel raises a disagreement/degraded alarm. Stale, nonfinite, or failed sensors require an explicit safe-state policy rather than normal threshold comparison. Coolant-flow protection requires an actual flow measurement in addition to temperature telemetry. The 10 ms supervisory execution limit must be enforced by an independent watchdog; it is not the microsecond Tier-1 trip budget and cannot be established by timing one completed software call.

Tier-1 shall use independent trip sensing, segregated TMR FPGA logic with 2oo3 voting, and direct SiC crowbar and EOM driver outputs. The fast path shall bypass the process controller, operating-system tasks, observer, and supervisory network. Trip latches shall inhibit re-energization until independent safe-state feedback and authorized reset are present. A sub-500 ns SiC conduction requirement refers to the switch branch, not completion of sensor processing or plasma decay.

Tier-2 shall execute mechanical shutter travel, fuel isolation, and nitrogen/fluid mitigation commands on a separate 10.0–45.0 ms final-element actuation envelope. These actions shall be initiated without delaying Tier-1. Mechanical response may continue after drive extinction; it shall not be counted as completed by an FPGA command timestamp. Subsequent inventory evacuation, including the separate $< 150 \text{ ms}$ dump target, is a fluid-system completion criterion rather than a valve-actuation time. The 10.0 ms supervisory interval monitors availability and state but receives no credit for microsecond fault detection.

TABLE LXVIII
TMR FAULT-INJECTION ACCEPTANCE CASES

Case	Injection	Threshold	Protective response and scope
FIT-01	Optical shutter occlusion	At least 15.0% attenuation within 2 ms	Laser inhibit; the observation window is distinct from fast-path actuation latency.
FIT-02	Coolant-valve blockage	$\dot{m} \leq 0.5 \text{ kg/s}$; $T \geq 375.0 \text{ }^\circ\text{C}$	Isolate drive and actuate the qualified dump path; this flow threshold requires a separate operating range.
FIT-03	RF phase modulation	$ \delta\phi \geq 0.12 \text{ rad}$	Inhibit RF drive; the threshold is an angular deviation, not a variance in radians squared.

TABLE LXIX
TIER-1 END-TO-END TIMING BUDGET

Stage	Budget (μs)	Nominal (μs)
Sensing and conditioning	1.50	1.20
TMR FPGA processing	2.50	2.10
SiC crowbar and EOM drivers	2.50	1.85
Plasma RF extinction	6.00	3.42
Total	12.50	8.57

For the sequential timing definition in Table LXIX,

$$\begin{aligned}
 t_{\text{trip}} &= t_{\text{sense}} + t_{\text{proc}} + t_{\text{crowbar}} + t_{\text{quench}}, \\
 t_{\text{trip,nom}} &= 1.20 + 2.10 + 1.85 + 3.42 \\
 &= 8.57 \mu\text{s} \leq 12.50 \mu\text{s} < 42.80 \mu\text{s}, \\
 \Delta t_{\text{margin}} &= 3.93 \mu\text{s} \quad (31.44\%).
 \end{aligned} \tag{628}$$

The 8.57 μs sum is the nominal design timing point. Acceptance timestamps physical fault onset, qualified detection, voting, driver execution, and the completed RF/optical safe state under worst-case voltage, temperature, clock phase, and fault combinations. Parallel crowbar/EOM paths use the slower required completion time, with shared stages counted once.

The Luenberger observer belongs to process regulation:

$$\begin{aligned}
 \dot{\hat{\mathbf{x}}}_r &= A_r \hat{\mathbf{x}}_r + B_r \mathbf{u} + L_r (\mathbf{y} - C_r \hat{\mathbf{x}}_r - D_r \mathbf{u}), \\
 \dot{\mathbf{e}}_r &= (A_r - L_r C_r) \mathbf{e}_r.
 \end{aligned} \tag{629}$$

The gain L_r requires the selected output matrix and noise/bandwidth requirements. Stability of $A_r - L_r C_r$ shall be demonstrated for that realization; gains from a differently normalized model shall not be transferred unchanged. Observer convergence and simulator step size shall not delay independent trip initiation.

F. TMR Cause-and-Effect and Command Timing Requirements

The protection system shall use independent triple-modular-redundant (TMR) channels with 2oo3 voting and a SIL-3 design objective [4], [5]. Table LXX defines the six functional cause-and-effect assignments. Its microsecond allocations refer to command-path or explicitly identified electronic response boundaries; they do not specify completed mechanical travel or fluid displacement. The end-to-end Tier-1 envelope in Table LXIX and the separate Tier-2 mechanical completion envelope govern physical safe-state acceptance. TR-02 switch-branch initiation shall satisfy $< 500 \text{ ns}$. The 12.50 μs optical-detuning allocation is an upper command/function allowance,

while the selected Tier-1 sequence shall complete the required RF and optical safe state within the same 12.50 μs total budget. The 42.80 μs comparison limit and strict $< 50 \mu\text{s}$ outer bound are not additional serial delays. FM-04's 45 μs command allocation is outside the fast-path completion chain; nitrogen displacement remains a Tier-2 action. Mechanical shutter travel, gas flushing, and valve closure require independent final-element feedback. Pressure reference (gauge or absolute), reflectance normalization, and scintillator integration interval shall be specified for each trigger.

$$\begin{aligned}
 t_{\text{trip},j} &= t_{\text{sense},j} + t_{\text{condition},j} + t_{\text{vote},j} \\
 &\quad + t_{\text{output},j} + t_{\text{safe state},j} < 50 \mu\text{s}.
 \end{aligned} \tag{630}$$

Equation (630) applies to a Tier-1 protective function indexed by j ; the selected 12.50 μs limit is the tighter acceptance requirement. Initiation-only and completed-safe-state timing shall be recorded separately. The 10.0 ms supervisory interval, 1 ms simulator step, and 99.86 ms observer decay are not fast-interlock response times. Functional-safety qualification shall include failure rates, diagnostic coverage, proof-test intervals, common-cause analysis, and independence evidence for the complete safety instrumented function [4], [5].

Process regulation, independent protective functions, and scientific data acquisition shall retain separate execution and power-failure boundaries. SIL-3 is an allocated integrity requirement for each designated safety instrumented function, including its sensors, logic, final elements, and diagnostics. The DM-01 and DM-02 plant parameters define the corresponding process-loop models here and in Section X. Their response time constants shall not be credited as independent protection response times.

G. Optomechanical Plant, Feedback, and Stability Margins

The DM-01 mechanical displacement model uses the specified static gain and 10 kHz resonance:

$$\begin{aligned}
 \ddot{y} + 2\zeta_p \omega_p \dot{y} + \omega_p^2 y &= K_p \omega_p^2 v, \\
 G_p(s) &= \frac{K_p}{\tau_p^2 s^2 + 2\zeta_p \tau_p s + 1}, \\
 K_p &= 0.845 \mu\text{m/V}, \quad \zeta_p = 0.15, \\
 \omega_p &= 2\pi(10^4) \text{ s}^{-1}, \\
 \tau_p &= \omega_p^{-1} = 15.915494 \mu\text{s}.
 \end{aligned} \tag{631}$$

The sensor factor is a pure delay:

$$H_p(s) = e^{-s\tau_o}, \quad \tau_o = 120.0 \mu\text{s}. \tag{632}$$

TABLE LXX
TMR CAUSE-AND-EFFECT MATRIX AND COMMAND-PATH ALLOCATIONS

Fault	Trigger	Vote	Action	Commanded protection / hardware	Allocation (μs)
FM-01	Optical phase drift $> 12.5^\circ$	2oo3	TR-01	15 GHz electro-optic detuning; EOM/Pockels cell	12.50
FM-02	RF reflectance > -10.0 dB	2oo3	TR-02	RF-cavity crowbar energy dump; solid-state SiC switch	0.50
FM-03	Substrate $> 375.0^\circ\text{C}$	2oo3	TR-03	Laser-shutter closure; solenoid beam dump	25.00
FM-04	Coolant pressure < 2.0 bar	2oo3	TR-04	High-pressure nitrogen flush; gas-injection valve	45.00
FM-05	D_2 leak $> 10^{-6}$ mbar L/s	2oo3	TR-05	Secondary isolation; pneumatic gate valve	35.00
FM-06	Fast scintillator rate > 100 count/s	2oo3	TR-01/02	Dual optical detune and crowbar; independent SIS logic	15.00

A pure delay has unit magnitude and no -3 dB cutoff. The observed piezo response is $G_{\text{piezo}} = G_p H_p$. With controller C_p and separately counted computation/actuation delay T_p ,

$$L_p(s) = C_p(s)G_p(s)H_p(s)e^{-sT_p},$$

$$\frac{Y}{R} = \frac{C_p G_p e^{-sT_p}}{1 + L_p}, \quad S_p = \frac{1}{1 + L_p}, \quad (633)$$

$$C_p(s) = K_{c,p} \left(1 + \frac{\omega_{i,p}}{s}\right) \frac{1 + s/\omega_{z,p}}{1 + s/\omega_{h,p}}.$$

The sensor delay is included exactly once. Controller implementation includes saturation, anti-windup, and bumpless transfer; process-loop behavior is distinct from independent safety timing. The acceptance criteria remain

$$\text{PM} = 180^\circ + \arg L_p(i\omega_{gc}) \geq 60^\circ, \quad \text{GM} \geq 6 \text{ dB}. \quad (634)$$

At the 1.5 kHz bandwidth objective, the mechanics and 120 microsecond delay contribute -67.4358° before controller and computation phase. DM-01 in Section X specifies the corresponding frequency response and identification requirements.

H. Coolant-Loop Transfer Functions and Nonlinear Heat Balance

The DM-02 heat-input-to-outlet-temperature model is

$$G_c(s) = H_{\text{cool}}(s) = \frac{K_c e^{-sT_{m,c}}}{1 + \tau_c s},$$

$$K_c = 0.045 \text{ K/W}, \quad \tau_c = 14.20 \text{ s}, \quad (635)$$

$$T_{m,c} = 1.45 \text{ s}.$$

The 0.420 K/W, 1.250 s, 0.080 s parameter set defines a separate comparison plant. Positive heat input increases outlet temperature; a command for additional cooling has the opposite thermal sign. An actuator map G_a , thermometer response H_c , and controller C_c form $L_c = C_c G_a G_c H_c$ with a consistent negative-feedback sign. The 1.45 s transport delay is already in G_c and is not added again as sensor delay.

At 2 Hz the thermal pole and transport delay alone contribute -1133.6789° of unwrapped phase. A 2 Hz bandwidth allocation therefore requires stability assessment for this complete delay model. Controller assessment uses unwrapped phase, all gain crossings, and the complete loop Nyquist response, rather than interpreting phase modulo 360° as a sufficient stability test. For example, at 0.05 Hz the thermal pole and delay contribute

about -103.5° before controller and sensor effects; this is a comparison frequency, not an established controller bandwidth.

The nonlinear first-law model remains

$$C_{\text{eff}}(T)\dot{T} = P_{\text{dep}}(T, \mathbf{u}) - \dot{m}[h(T_{\text{out}}) - h(T_{\text{in}})] - P_{\text{env}}(T) - P_{\text{export}}(T). \quad (636)$$

For a one-state autonomous thermal reduction, local stability requires $\partial(P_{\text{dep}} - P_{\text{removed}})/\partial T < 0$ at equilibrium with $C_{\text{eff}} > 0$. The FOPDT parameters describe a local heat-to-temperature response; they are not a global cooling-capacity or compressor-power law. Coolant loss, actuator saturation, and sensor dropout remain distinct nonlinear protection cases.

a) *Fast optical-proxy interlock*: The protective architecture separates slow thermal calorimetry (time constant 1.20 s) from a fast optical-phase/RF-impedance proxy:

$$\xi_{\text{proxy}}(t) = K_{\text{opt}} \frac{d\Delta\phi_{\text{opt}}}{dt} + K_{\text{imp}} \frac{dZ_{\text{RF}}}{dt}, \quad (637)$$

with a comparator threshold and an end-to-end trip allocation of less than $42.5 \mu\text{s}$, within a $50 \mu\text{s}$ outer window; the selected Tier-1 path shall meet the tighter $12.50 \mu\text{s}$ envelope. K_{opt} and K_{imp} are calibration gains mapping the two measured derivatives to a common proxy unit and shall be identified experimentally. The thresholds $dP_{\text{th}}/dt > 100 \text{ W/s}$ and $dT/dt > 45 \text{ K/s}$ are not interchangeable without thermal capacitance, heat-loss dynamics, and sensor transfer functions. The coefficient $dn/dT = -1.50 \times 10^{-4} \text{ K}^{-1}$ and sub-10 ns intrinsic optical-detuning target do not define the complete thermal-to-optical delay. Trip timing shall extend from physical fault onset through sensing, FPGA decision, driver interruption, and verified optical extinction, not just comparator execution. The chemical-state observer remains outside this microsecond safety path. SIL-3 requires independent lifecycle and failure-probability evidence for the complete function [4], [5].

I. Surge Detection and Optical Detuning

The surge-trip threshold shall be a positive estimated thermal-power slope exceeding 100.0 W s^{-1} . The reference slope estimator and trip condition shall be

$$\hat{R}_P(s) = \frac{s}{1 + \tau_r s} \hat{P}_{\text{th}}(s), \quad \text{TR}_{\text{surge}} = \mathbf{1}_{\{\hat{R}_P > 100.0 \text{ W s}^{-1}\}}. \quad (638)$$

The positive filter time constant τ_r shall be selected from the identified fast-observable noise and permissible detection delay.

For comparison, a two-sample slope with independent equal power uncertainties has

$$u(R_P) = \frac{\sqrt{2} u(P_{th})}{\Delta t}. \quad (639)$$

Using $u(P_{th}) = 5.701706 \text{ W}$ and $\Delta t = 50.0 \mu\text{s}$ gives $1.61269 \times 10^5 \text{ W s}^{-1}$. The protective channel shall therefore use a separately characterized fast observable with a validated mapping to the surge condition; the steady-state calorimetric uncertainty shall not be used as its transient-resolution specification.

The reference incremental phase-response filter from the slope input shall be

$$\begin{aligned} G_{\text{surge}}(s) &= \frac{\Delta\phi(s)}{\widehat{R}_P(s)} = \frac{K_s s}{(s + p_s)(1 + \tau_a s)}, \\ K_s &= 0.050 \text{ rad s W}^{-1}, \\ p_s &= 1000.0 \text{ s}^{-1}, \quad \tau_a = 15.0 \mu\text{s}. \end{aligned} \quad (640)$$

The filter has zero DC gain. A latched trip with independent optical extinction shall therefore supersede its transient phase response and provide the sustained protective action.

For an applied optical phase $\phi(t)$, instantaneous frequency deviation is

$$\Delta f(t) = \frac{1}{2\pi} \dot{\phi}(t). \quad (641)$$

Maintaining 15.0 GHz detuning requires a phase ramp of $9.42478 \times 10^{10} \text{ rad s}^{-1}$ or an equivalent frequency-shifting architecture. Qualification shall separately evaluate detuning within $50.0 \mu\text{s}$, pump-coupling reduction exceeding 99%, and the 1.0 ms reaction-suppression target. Optical extinction shall be measured over the installed resonance response. The relation between optical detuning and nuclear-rate suppression requires branch-resolved kinetic measurements. Independent drive inhibition and gas isolation shall retain their protective functions without reliance on this kinetic hypothesis.

J. FMECA and Independent Electrical Protection

Table LXXI specifies the qualitative FMECA baseline and required mitigations. Project severity categories shall be interpreted as follows: catastrophic denotes possible containment or major exposure consequences, critical denotes loss of a necessary protective or heat-removal function, and marginal denotes loss of experimental validity without an assumed immediate containment failure. Quantitative risk and integrity assessments shall use separately supported failure rates, diagnostic coverage, and protection-layer data.

The electrical protection branch shall use a SiC switch in series with a pulse-rated resistor across the protected DC-link capacitor, coordinated with upstream interruption and inverter gate inhibition. The DC current trip shall be $I_{dc} > 35.0 \text{ A}$, separately from the $21.21 \text{ A}_{\text{rms}}$ resonant-coil operating current. The dump-switch conduction requirement shall include comparator, isolation, driver, and switch turn-on delays after the sensed threshold crossing:

$$t_{\text{dump,on}} = t_{\text{comp}} + t_{\text{iso}} + t_{\text{gate}} + t_{\text{sw}} < 500 \text{ ns}. \quad (642)$$

Sensor bandwidth and threshold uncertainty are additional characterization requirements. Energy sizing must include

$$E_{\text{dump}} \geq \frac{1}{2} C_{dc} V_0^2 + \frac{1}{2} L_{eq} I_0^2 + \int_0^{t_{\text{isolate}}} P_{\text{source}}(t) dt. \quad (643)$$

For an isolated capacitor and ideal resistive dump,

$$V(t) = V_0 e^{-t/(R_d C_{dc})}, \quad t_{\text{safe}} = R_d C_{dc} \ln \frac{V_0}{V_{\text{safe}}}. \quad (644)$$

Dump-switch conduction time and stored-energy discharge time shall be qualified separately. Selection of R_d , switch safe operating area, resistor pulse energy, wiring inductance, insulation, and clamp voltage shall use maximum bus energy and prospective fault current, including source contribution until isolation.

For the semiconductor-protection fuse,

$$t_{\text{clear}} = t_{\text{prearc}} + t_{\text{arc}}, \quad \int_0^{t_{\text{clear}}} i_f^2(t) dt \leq (I^2 t)_{\text{withstand}}. \quad (645)$$

The $t_{\text{clear}} < 10.0 \mu\text{s}$ target at 200% rated current shall be evaluated against the selected device's time-current and let-through characteristics at the actual DC voltage, source impedance, and temperature [2]. Protective credit shall use the verified clearing interval; if the target is unattainable, the protection design shall be revised. The resistor-limited dump shall be assessed for available fuse-operating current, and independent electronic interruption shall protect the switch throughout the verified interruption interval.

K. Triple-Channel Voting and Cause-and-Effect Logic

For three independently measured versions of the same safety variable, let S_A, S_B, S_C equal one when their respective validated trip conditions are true. The majority result is

$$V_{2o3} = (S_A \wedge S_B) \vee (S_A \wedge S_C) \vee (S_B \wedge S_C). \quad (646)$$

All eight cases are explicit:

S_A	S_B	S_C	V_{2o3}
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

(647)

Temperature and pressure shall each use a separate three-channel vote on the same physical variable. The required disagreement flag and trip-latch update shall be

$$D_S = (S_A \oplus S_B) \vee (S_B \oplus S_C), \quad (648)$$

$$L_{k+1} = V_{2o3,k} \vee F_{\text{critical},k} \vee (L_k \wedge \neg R_{\text{authorized},k}).$$

F_{critical} shall include loss of the required healthy-channel count, watchdog failure, emergency stop, and failed output diagnostics. Disagreement shall initiate diagnosis; invalid channels shall retain explicit fault status. Operation shall enter a validated degraded mode or trip if its required channel-health condition

TABLE LXXI
CONTROL AND PROCESS FMECA: FAILURE CONSEQUENCES, DETECTION, AND REQUIRED EVIDENCE

ID	Failure and initiating cause	Consequence; severity	Detection and mitigation	Timing allocation and verification
FM-01	SiC short circuit; dielectric or thermal failure	Coil/bus overcurrent; catastrophic	Independent analog overcurrent comparator, gate inhibit, SiC dump branch, coordinated fuse	Dump-switch conduction < 500 ns; fuse-clearing target < 10.0 μ s requires selected-device fault-current qualification
FM-02	Coolant line rupture, pump seizure, or blocked flow	Loss of heat removal; critical	Flow and pressure monitoring, leak detection, drive inhibit, independent cooling availability check	Protective command < 5.0 ms target; complete thermal response includes sensor and fluid delays
FM-03	FPGA hang, corrupted configuration, or shared power loss	Loss of regulation or trip execution; catastrophic	Independent watchdog, de-energize-to-trip outputs, segregated safety logic and power	Fault-injection tests establish worst-case end-to-end latency; the process FPGA receives no sole safety credit
FM-04	Laser drift, mode hopping, or flexure saturation	Loss of resonance and reproducibility; marginal	Optical phase/position monitoring, bounded correction, laser inhibit on persistent fault	Tracking-loop and position-limit qualification; optical response and nuclear kinetics verified separately
FM-05	Vacuum seal failure or pressurized boundary leak	Air ingress or fuel escape; critical	Mode-qualified vacuum trip, pressure transducers, cabinet gas detection, isolation	Vacuum-isolation command < 20.0 ms target; valve closure and leak-tightness measured separately
FM-06	Regulator fail-open or blocked discharge	Vessel overpressure; catastrophic	Independent pressure trip and certified mechanical relief path	Relief capacity assessed without credit for software, getter uptake, or loss of electrical power
FM-07	Common-cause sensor error or blocked common pressure tap	Majority voter receives wrong inputs; catastrophic	Separate sensing paths, proof tests, disagreement alarms, independent physical protection	Quantitative integrity assessment shall include common-cause analysis and proof-test records
FM-08	Getter saturation, hot-bed desorption, or dump valve failure	Fuel not removed; pressure/temperature excursion; critical	Bed temperature and inventory limits, valve-position feedback, secondary containment	Evacuation target < 150 ms; qualify pressure decay, receiver capacity, and adsorption heat
FM-09	Exhaust fan failure or ignition-capable cabinet equipment	Flammable mixture accumulation; catastrophic	Airflow proving, gas alarms, fuel shutoff, suitably classified equipment	Ventilation-loss trip is independent of concentration rise; verify fault-mode electrical suitability

is lost. Authorized reset shall require restored permissives and independent operator acknowledgement of the cause. An active vote or critical fault shall override reset, as specified by the latch equation.

The temperature sensing architecture shall provide three PT100 channels with separate AFEs, references, supply isolation, and wiring paths. An ADS124S08 implementation shall account for its maximum 4 kSPS conversion rate and the additional digital-filter and physical-sensor delays [3]. Pressurized operation shall use service-range pressure transducers, separately from UHV gauges. Independence shall cover pressure taps, power, clocks, routing, environmental susceptibility, and final elements. Trip outputs shall retain their safe function after supervisory-computer failure.

Table LXXII shall define the reference setpoints, mode qualifiers, commanded effects, timing requirements, and reset conditions. Tiers 1–3 denote response priorities, not safety integrity levels; the safety requirements specification shall allocate the required SIL to each designated function through the hazard and risk assessment. Before operational release, setpoints shall be checked against qualified vessel, detector, and cooling limits. A getter dump shall discharge only into a pressure-rated receiver with verified inventory and adsorption-heat capacity. Liquid-nitrogen flooding shall receive no protective credit without a separate assessment of condensation, vent blockage, thermal shock, and oxygen enrichment. Shutdown inventory calculations shall retain gas dissolved in metal after chamber depressurization.

The quantitative SIL-3 target ranges for a defined function

are

$$10^{-4} \leq \text{PFD}_{\text{avg}} < 10^{-3} \quad (\text{low demand}),$$

$$10^{-8} \leq \text{PFH} < 10^{-7} \text{ h}^{-1} \quad (\text{high/continuous demand}).$$

(649)

The allocated SIL-3 functions shall satisfy the applicable IEC 61508/61511 lifecycle, architectural, systematic, and quantitative integrity requirements [4], [5]. For independent identical channel dangerous unavailability q , the ideal sensor-vote contribution is $3q^2 - 2q^3$. The complete function assessment shall additionally include common-cause, logic, power, and final-element failures, with supported diagnostic coverage and proof-test intervals. Compliance shall require independent assessment and documented verification of the complete function; the truth table and development FPGA shall not constitute certification evidence by themselves.

a) *Switch-Branch Latency Allocation*: The thermal-surge threshold is $dP_{\text{th}}/dt > 100.0 \text{ W/s}$. A 15.0 GHz beat-frequency change over 50.0 μ s is distinct from the commanded detuning response: sensing a frequency change, requesting detuning, and observing completed detuning are separate timestamps. The following serial subcomponent allocation is not an additional full-loop trip budget: The serial allocation totals 375 ns. Parallel sensing paths require their own timing topology. Qualification shall include the observation interval for the thermal ramp or frequency change, transient noise and bandwidth, logic-vote latency, switch safe-operating area, optical shutdown response, and dump energy. Switch conduction and completed optical shutdown have separate timing boundaries. The cause-and-

TABLE LXXII
SPECIFIED TR-01–TR-06 CAUSE-AND-EFFECT MATRIX FOR THE SIL-3-TARGET ARCHITECTURE

ID	Initiator and mode qualification	Tier	Commanded effects	Timing and reset requirement
TR-01	Core $T > 438.15$ K; independent 2o3 temperature vote	3	Inhibit optical and magnetic drives; close fuel feed; request qualified contained dump; preserve monitoring and available safe cooling	Latching trip; measured sensor-to-final-element time; physical reset after cause review
TR-02	$p_{\text{vac}} > 1.0 \times 10^{-6}$ mbar only in UHV mode	2	Inhibit drive; isolate vulnerable vacuum equipment and fuel feed; do not apply UHV criteria during deliberate gas loading	Isolation command < 20.0 ms target; reset after leak check and mode validation
TR-03	Angular tracking error $ \theta - \theta_{\text{ref}} > 0.10$ μrad ; optical-lock mode	1	Apply bounded flexure correction; inhibit laser if lock or travel permissive is lost; log frequency and angle separately	Fast-logic evaluation at 1.0 MHz; sensor/actuator rates qualified separately; re-entry only after verified lock
TR-04	Core $p > 5.00$ bar absolute; independent 2o3 pressure vote	3	Close fuel feed, inhibit drives, request qualified contained dump; mechanical PSV remains independent	Latching pressure trip; dump evacuation < 150 ms is a system test target, not voter delay
TR-05	Coolant $\dot{V} < 8.00$ L min $^{-1}$ while heat-producing operation is enabled	2	Inhibit heat-producing drives; start available qualified backup circulation; retain safe heat rejection	Command < 5.0 ms target; reset after flow and leak verification
TR-06	Cabinet hydrogenic concentration $> 0.10\%$ by volume, or loss of proven exhaust airflow	3	Isolate fuel, inhibit ignition-capable process loads, alarm, maintain suitably rated exhaust and gas monitoring	Detector/transport delay measured end to end; environmental clearance and manual reset

TABLE LXXIII
DUMP-BRANCH SUBCOMPONENT LATENCY ALLOCATIONS

Stage	Delay (ns)	Cumulative (ns)
Optical photodiode	15	15
Thin-film thermopile	45	60
TMR FPGA logic	120	180
Isolated gate driver	85	265
SiC switch transition	110	375

TABLE LXXIV
SEPARATE DEUTERIUM RELIEF DESIGN CASES

Parameter	A: manifold	B: core
Set pressure, gauge	5.00 bar	15.00 MPa
Relieving pressure, absolute (MPa)	0.651325	16.601325
Mass flow (kg s $^{-1}$)	0.050	0.250
Temperature (K)	323.15	550.0
Compressibility Z	1.00	1.12
Calculated area (mm 2)	93.9122	25.4352
API area designation	E	D
Effective area (mm 2)	126.4514	70.9676

effect matrix and IEC safety-integrity requirements remain applicable [4], [5].

VIII. PROCESS SAFETY, HAZOP/FMECA, AND REGULATORY COMPLIANCE

The safety review shall address the specified fuel inventory and all credible reaction branches without protective credit for hypothesized radiation suppression. HAZOP nodes shall comprise the regulated supply, reactor chamber, coolant circuit, getter/dump receiver, assay manifold, electrical bus, and exhaust enclosure. Each node shall be assessed for no flow, excess flow, reverse flow, high pressure, low pressure, high temperature, loss of containment, and loss of utilities. Risk assessment shall combine the consequences in Table LXXI with supported initiating frequencies and independently qualified protection layers. The following reference calculations define engineering cases; operational release shall require component-specific qualification and the applicable safety approvals.

A. Pressure-Relief Design Basis and Orifice Qualification

The engineering report [54], pp. 21–22, defines two separate dry-deuterium relief cases. ASME BPVC Sections VIII/XIII, API 520/526, and the applicable ISO 4126 provisions govern

the qualified design [6], [38]–[40]. The gas-flow model defines the sizing basis; valve certification is required separately. For $M = 0.004028$ kg mol $^{-1}$, $k = 1.40$, $K_d = 0.975$, and unity additional backpressure/combination factors, the sonic model is

$$R_s = R_u/M,$$

$$g_* = K_d p_0 \sqrt{\frac{k}{Z R_s T_0}} \left(\frac{2}{k+1} \right)^{(k+1)/(2(k-1))}, \quad (650)$$

$$A_{\text{req}} = \dot{m}/g_*.$$

Absolute stagnation pressure must be used. With $p_{\text{atm}} = 101325$ Pa and a 10% increase of gauge set pressure, the relieving pressure is $p_0 = 1.10p_{\text{set,g}} + p_{\text{atm}}$.

$$A_{\text{req,A}} = 93.91215 \text{ mm}^2,$$

$$A_E = 0.196(645.16) = 126.45136 \text{ mm}^2, \quad (651)$$

$$A_E/A_{\text{req,A}} \simeq 1.34648.$$

$$A_{\text{req,B}} = 25.43520 \text{ mm}^2,$$

$$A_D = 0.110(645.16) = 70.96760 \text{ mm}^2, \quad (652)$$

$$A_D/A_{\text{req,B}} \simeq 2.79013.$$

The selected effective areas are E for case A and D for case B [54]. Effective API area and physical throat area shall follow the manufacturer's certified capacity convention.

Installation-specific qualification shall include high-pressure real-gas properties, backpressure/choking, inlet/outlet losses, accumulation, fire and two-phase scenarios, discharge forces, and actual regulator capacity. Case B requires a separately rated vessel, viewports, feedthroughs, and interlocks. Relief discharge must be captured and routed safely.

B. Hazardous Locations and the Ventilation Balance

The engineering report's new classification proposal assigns normal-release internal containment, purge-exhaust manifolds, and relief headers to Class I, Division 1 under the division system; a zone-based study assesses Zone 1 for the corresponding intermittent normal-release conditions. It proposes Division 2 or Zone 2 within a 1.5 m radius of fittings, flanges, and stem packings where release is abnormal. These are project-study boundaries, not universal distances established by NFPA 2 [8], [9]. Continuous or long-duration ignitable atmospheres require separate Zone 0 assessment rather than automatic Zone 1 assignment.

Choose and document the adopted classification system. Division and zone designations are not interchangeable certificates, and Ex d/Ex i/Ex nA labels do not alone establish US Class I, Group B approval [9]. The study shall include release frequency, ventilation availability, gas group, temperature classification, containment geometry, and approval of the complete installation. Normal relief/purge streams shall enter engineered discharge routes, not occupied or recirculating spaces.

The hot core at 276.85°C is enclosed by a double-walled vacuum jacket with outer microporous insulation. The design requires maximum accessible/ignition-relevant external skin temperature 115.0°C, below the T4 limit 135.0°C. The thermal requirement shall be demonstrated at hot spots and applicable upset conditions. Vacuum-loss, failed circulation, penetrations, and thermal bridges belong in the surface-temperature assessment. Gas detectors cover enclosure high points, joints, dosing equipment, and exhaust; airflow proving is independent of fan command.

a) *Fast isolation instead of full-release dilution:* Fast isolation is the primary safeguard for a 0.050 kg s⁻¹ deuterium release in a 3.50 m³ enclosure [58]. A continuous-dilution comparison gives:

$$\begin{aligned} Q_{\text{exh},s} &= 100161.73 \text{ m}^3 \text{ h}^{-1}, \\ &\simeq 58952.97 \text{ SCFM}, \\ \text{ACH}_s &= 28617.64 \text{ h}^{-1}. \end{aligned} \quad (653)$$

That calculation used 273.15 K, 101325 Pa, and a 1.0% concentration target. The following transient calculation instead uses 298.15 K and 101325 Pa explicitly; these volume conventions are not interchangeable.

The active protection allocation is optical detection plus logic within 5.0 ms, followed by complete isolation within 10.0 ms.

For a constant upstream leak bounded by $\dot{m}_{\text{max}} = 0.050 \text{ kg s}^{-1}$ until closure,

$$\begin{aligned} t_{\text{isolate}} &\leq 5.0 + 10.0 = 15.0 \text{ ms}, \\ m_{\text{release}} &\leq \dot{m}_{\text{max}} t_{\text{isolate}} = 0.750 \text{ g}, \\ n_{D_2} &= \frac{0.000750}{0.0040282} = 0.18618738 \text{ mol}, \\ V_{D_2} &= n_{D_2} R_u T / p = 4.55514939 \text{ L}, \\ \bar{C}_{\text{bound}} &= V_{D_2} / (3500 \text{ L}) = 0.13014713\%. \end{aligned} \quad (654)$$

The gas-volume ratio is a conservative enclosure-average inventory bound for initially clean air, not the peak local concentration beside a leaking joint. At constant pressure with ideal well-mixed displacement the fraction is $1 - e^{-V_{D_2}/V_{\text{enc}}}$; a sealed enclosure requires pressure and mole-inventory accounting.

The reference LFL = 4.9% gives $C_{\text{safe}} = 0.25(4.9\%) = 1.225\%$. Flammability limits depend on test method and conditions [60]. The more conservative 1.0% project concentration target is retained until the installation hazard assessment approves a change.

Optical sensor latency shall be demonstrated from leak initiation, including gas transport to the sensing path, detection, logic, and actuation. Valves shall fail closed and their full closure time shall be verified at differential pressure. The inventory bound must include downstream trapped gas, post-closure leakage, sensor blind spots, and failure of isolation; 0.750 g is not a limit on such additional inventories. Independent gas monitoring, ignition-source control, safe vent routing, and background extraction with proven airflow remain required [8], [9]. The fast-isolation strategy replaces the infeasible full-release dilution duty, not these independent protections. A well-mixed average alone cannot certify absence of a locally flammable plume or an explosion-safe enclosure.

C. Tritium Production, Activity, and Branch Consistency

For a conservative branch scenario, set the total D–D rate to $\Gamma_{DD} = 7.623 \times 10^{14} \text{ s}^{-1}$ and the tritium-producing branch fraction to $b_T = 0.500$. Then

$$R_T = b_T \Gamma_{DD} = 3.8115 \times 10^{14} \text{ atoms s}^{-1}. \quad (655)$$

With tritium molar mass $M_T = 3.01604928 \text{ g mol}^{-1}$, the no-decay daily production is

$$\begin{aligned} m_{T,\text{day}} &= \frac{R_T(86400)M_T}{N_A} \\ &= 1.64929065 \times 10^{-4} \text{ g} \approx 0.165 \text{ mg day}^{-1}. \end{aligned} \quad (656)$$

Using the evaluated half-life $t_{1/2} = 4500 \text{ d}$ [10],

$$\lambda_T = \frac{\ln 2}{4500(86400)} = 1.78278596 \times 10^{-9} \text{ s}^{-1}. \quad (657)$$

For zero initial inventory and no removal,

$$\begin{aligned} \dot{N}_T &= R_T - \lambda_T N_T, \\ N_T(t) &= \frac{R_T}{\lambda_T} (1 - e^{-\lambda_T t}), \\ A_T(t) &= \lambda_T N_T(t) = R_T (1 - e^{-\lambda_T t}), \\ A_T(86400 \text{ s}) &= 58.7050 \text{ GBq}. \end{aligned} \quad (658)$$

Inventory reports shall express retained activity in Bq and identify any activity-accumulation rate separately. The first-day no-decay reference is 58.7096 GBq, compared with 58.7050 GBq including decay. Any removal term shall be included in the retained-inventory balance separately from the production source.

Each hazard calculation shall use one internally consistent set of branch fractions and event energies. For b_4, b_T, b_n ,

$$\begin{aligned} b_4 + b_T + b_n &= 1, \\ P_{\text{nuc}} &= \Gamma_{DD}(b_4 Q_4 + b_T Q_T + b_n Q_n), \\ Q_4 &= 23.84 \text{ MeV}, \quad Q_T \approx 4.033 \text{ MeV}, \\ Q_n &\approx 3.269 \text{ MeV}. \end{aligned} \quad (659)$$

For $b_T = b_n = 0.5$ and $b_4 = 0$, the reference event rate gives approximately 445.911 W released nuclear power and a neutron source of $3.8115 \times 10^{14} \text{ s}^{-1}$. This source exceeds the Section VI-I reference ceiling by a factor of 3.8115×10^8 and shall receive a separate shielding and exposure assessment without credit for the lower-source shield allocation. Branch-dependent cross sections shall use validated nuclear models within their stated domain [11]. The tritium case and helium-only design case shall remain separate hazard and operating scenarios. Licensing, contamination control, effluent treatment, waste characterization, and worker protection shall undergo independent radiological review.

D. Permeation Barrier and Containment Qualification

The TiN/ Al_2O_3 barrier shall be qualified on compatible containment surfaces while retaining an independently qualified pressure boundary. For the diffusion-limited square-root-pressure transport model, permeability and resistance shall be defined as $\Pi_j = D_j S_j$ and $\mathcal{R}_j = \ell_j / \Pi_j$. With layers in series,

$$J = \frac{\sqrt{p_{\text{in}}} - \sqrt{p_{\text{out}}}}{\mathcal{R}_{\text{sub}} + \mathcal{R}_{\text{TiN}} + \mathcal{R}_{\text{Al}_2\text{O}_3} + \mathcal{R}_{\text{int}}}. \quad (660)$$

All permeabilities use a common molar or atomic-flux convention; interface resistances are expressed in the same units. The ideal permeation reduction factor is

$$\text{PRF}_{\text{ideal}} = \frac{J_{\text{bare}}}{J_{\text{coated}}} = 1 + \frac{\mathcal{R}_{\text{TiN}} + \mathcal{R}_{\text{Al}_2\text{O}_3} + \mathcal{R}_{\text{int}}}{\mathcal{R}_{\text{sub}}}. \quad (661)$$

Consequently, $\text{PRF} > 1000$ requires an added resistance exceeding $999\mathcal{R}_{\text{sub}}$ under identical boundary conditions. For defect area fraction f_d treated as bare-substrate bypass,

$$\text{PRF}_{\text{eff}} = \left[f_d + \frac{1 - f_d}{\text{PRF}_{\text{ideal}}} \right]^{-1}. \quad (662)$$

Even an otherwise impermeable coating cannot exceed $1/f_d$ in this model; attaining PRF above 1000 therefore requires $f_d < 10^{-3}$ as a necessary, not sufficient, condition. Temperature-dependent permeability, trapping, pinholes, welds, seals, stress cracking, and irradiation can dominate system leakage.

Barrier qualification shall specify layer thicknesses and measured isotope-specific permeabilities. Coated and uncoated specimens shall be tested at matched temperature, pressure, substrate condition, and exposed area, repeated after representative thermal cycling with defect inspection. Hydrogen-enhanced

plasticity and fracture shall receive the separate mechanical qualification specified in Section IV [12], [13]. Downstream gas and water species shall be captured and assayed, with bypass leakage included in the inventory. Acceptance of the component $\text{PRF} > 1000$ requirement shall use these measurements; full-system release reduction shall be assessed separately.

a) *v2.0 coating-defect acceptance calculation:* For the specified 150 nm TiN/250 nm Al_2O_3 coating, v2.0 assigns $\text{PRF}_{\text{ideal}} = 2850$ and $f_d \leq 8.5 \times 10^{-4}$. The parallel-defect equation yields

$$\begin{aligned} \text{PRF}_{\text{eff}}|_{f_d=0.00085} &= \left[0.00085 + \frac{0.99915}{2850} \right]^{-1} \\ &= 832.93148 \simeq 833. \end{aligned} \quad (663)$$

For the same ideal coating, the component requirement $\text{PRF} > 1000$ requires

$$f_d < \frac{10^{-3} - 2850^{-1}}{1 - 2850^{-1}} = 6.49351 \times 10^{-4}. \quad (664)$$

The v2.0 1.5 m Class I, Division 2/Zone 2 extent is a proposed classified boundary requiring release-source classification. The 0.750 g isolation-only release calculation is conditional on the response and trapped-inventory assumptions above; the 1.00% project concentration target is retained. Equipment suitability and approved installation remain governed by the selected classification [9].

b) *Governing v3.0 barrier model and leakage scope:* The v3.0 manufacturing defect cap is $f_d \leq 6.49 \times 10^{-4}$ for the 150 nm TiN/250 nm Al_2O_3 barrier. The report also changes the constitutive model to an intrinsic factor 25000 and a pore-path multiplier $\alpha_{\text{geom}} = 1.42$:

$$\begin{aligned} \text{PRF}_{\text{eff}} &= \left[\frac{1 - f_d}{25000} + 1.42 f_d \right]^{-1}, \\ \text{PRF}_{\text{eff}}(0.000649) &= 1039.983151 \geq 1000, \\ f_{d,\text{max}} &= \frac{0.001 - 1/25000}{1.42 - 1/25000} \\ &= 6.76075382 \times 10^{-4}. \end{aligned} \quad (665)$$

The intrinsic factor and defect-path multiplier define the v3.0 barrier model independently of the earlier coating comparison.

The source's additional transport assignments are trap density $N_t = 1.20 \times 10^{26} \text{ sites/m}^3$, partial molar volume $V_H^* = 1.70 \times 10^{-6} \text{ m}^3/\text{mol}$, and signed heat of transport $Q^* = 12.50 \text{ kJ/mol}$. For mobile and trapped *molar* concentrations C_m, C_t , convert N_t to $N_{t,\text{mol}} = N_t/N_A$ and write

$$\begin{aligned} \partial_t C_m + \nabla \cdot \mathbf{J}_D &= -\partial_t C_t, \\ \partial_t C_t &= k_t C_m (N_{t,\text{mol}} - C_t) - p_t C_t, \\ \mathbf{J}_D &= -D_{\text{eff}} \nabla C_m + \frac{D_{\text{eff}} C_m V_H^*}{R_u T} \nabla \sigma_h \\ &\quad + \frac{D_{\text{eff}} C_m Q^*}{R_u T^2} \nabla T. \end{aligned} \quad (666)$$

Here the capture coefficient uses the molar convention, and positive hydrostatic stress is tension. The substrate-flux reference is $J_D = 3.15 \times 10^{-8} \text{ mol}/(\text{m}^2 \text{ s})$. Flux evaluation requires diffusion/trapping coefficients, concentration and stress fields, boundary pressures, and wall geometry. Active-layer

and containment-substrate properties retain separate material definitions.

IX. DEFINITIVE ENGINEERING SPECIFICATIONS FOR

SHBT-CF

The definitive `shbt-cf` configuration integrates multiphase thermal hydraulics, plant stability mapping, coupled three-dimensional neutron-photon transport, non-equilibrium Floquet dielectric screening, high- T_c superconducting magnetic energy recovery, phase-change optical restoration, and distributed multi-GPU control in a unified 100 Hz closed-loop framework.

A. 3D Eulerian–Eulerian Thermal-Hydraulic Kinetics

The reactor cold plate contains 64 parallel oxygen-free high-conductivity copper microchannels with hydraulic diameter $D_h = 250 \mu\text{m}$. The three-dimensional Eulerian–Eulerian model dissipates the $P_{\text{thermal}} = 3093.4400 \text{ W}$ thermal load while resolving subcooled nucleate boiling. For liquid and vapor phases $k \in \{L, V\}$,

$$\frac{\partial(\alpha_k \rho_k)}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k) = \Gamma_k, \quad (667)$$

and

$$\begin{aligned} \frac{\partial(\alpha_k \rho_k \mathbf{u}_k)}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k \mathbf{u}_k) \\ = -\alpha_k \nabla p + \nabla \cdot [\alpha_k (\boldsymbol{\tau}_k + \boldsymbol{\tau}_k^{\text{turb}})] \\ + \alpha_k \rho_k \mathbf{g} + \mathbf{M}_k + \Gamma_k \mathbf{u}_{ki}. \end{aligned} \quad (668)$$

The vapor-phase momentum exchange $\mathbf{M}_V = \mathbf{F}_d + \mathbf{F}_l + \mathbf{F}_{wl} + \mathbf{F}_{td}$ comprises Ishii–Zuber drag, Tomiyama lift, wall-lubrication, and turbulent-dispersion forces. The RPI wall-flux partition is

$$\begin{aligned} q''_{\text{tot}} &= q''_{1\phi} + q''_q + q''_e, \\ q''_{1\phi} &= (1 - A_b) h_C (T_w - T_L), \\ q''_q &= \frac{2}{\sqrt{\pi}} \sqrt{k_L \rho_L C_{p,L} f} A_b (T_w - T_L), \\ q''_e &= \frac{\pi}{6} d_b^3 \rho_V h_{fg} f n''. \end{aligned} \quad (669)$$

The verified operating point is $T_h = 618.42 \text{ K}$, $T_c = 301.88 \text{ K}$, peak void fraction $\alpha_v = 0.162$, core differential pressure $\Delta P_{\text{core}} = 42.8 \text{ kPa}$, friction multiplier $\phi_{lo}^2 = 1.34$, critical-heat-flux ratio $q''/q''_{\text{CHF}} = 0.412$, and flow rate $Q = 4.85 \text{ L/min}$. These values satisfy $T_h \leq 623.15 \text{ K}$, $T_c \leq 303.15 \text{ K}$, and $\alpha_v \leq 0.185$.

a) *Pump curve and plant stability map:* With Q expressed in L/min and pressure in kPa, the dynamic delivery curve is

$$\Delta P_{\text{pump}}(Q) = 65.0 - 1.25Q - 0.62Q^2. \quad (670)$$

At $Q = 4.85 \text{ L/min}$, Eq. (670) gives $\Delta P_{\text{pump}} = 44.35 \text{ kPa}$, providing the pump head required for the verified 42.8 kPa core pressure drop plus the allocated loop margin. Static Ledinegg stability is assessed from the local slopes of the complete system-demand and pump curves:

$$\frac{d\Delta P_{\text{system}}}{dQ} > \frac{d\Delta P_{\text{pump}}}{dQ}. \quad (671)$$

The nominal operating point satisfies this excursive-flow criterion. The Ishii–Zuber density-wave-oscillation boundary is

$$N_{\text{pch,crit}} = 1.45 N_{\text{sub}} + 2.5, \quad N_{\text{pch}} < N_{\text{pch,crit}} \quad (\text{stable}). \quad (672)$$

The nominal stability-map coordinate remains on the damped side of this boundary. A hard low-flow interlock inhibits reactor drive when $Q < 3.95 \text{ L/min}$, before the predicted Ledinegg/DWO excursion region is entered.

B. 3D OpenMC Neutron–Photon Transport and Surface Dose

The definitive fixed-source OpenMC model represents the complete three-dimensional enclosure, including the 64 coolant channels, pressure boundaries, CF flange seams and physical penetrations, instrument sleeves, detector ports, shield joints, and surrounding scattering structures. The isotropic D–D source is

$$S_n(E, \mathbf{r}) = \frac{Q_N}{V_{\text{core}}} \delta(E - 2.45 \text{ MeV}), \quad Q_N \leq 10^6 \text{ s}^{-1}. \quad (673)$$

Coupled histories transport neutrons and all neutron-induced photons. For species $a \in \{n, \gamma\}$, the steady angular-flux formulation is

$$\begin{aligned} \hat{\Omega} \cdot \nabla \psi_a + \Sigma_{t,a} \psi_a &= S_a + \sum_b \mathcal{S}_{ba}[\psi_b], \\ \mathcal{S}_{ba}[\psi_b] &= \int dE' d\Omega' \Sigma_{b \rightarrow a} \psi_b(E', \Omega'). \end{aligned} \quad (674)$$

The 5 wt% borated-HDPE region includes the 478 keV photon emitted after $^{10}\text{B}(n, \alpha)^7\text{Li}^*$ capture and de-excitation. Water and hydrogenous shielding include the 2.223 MeV photon from $\text{H}(n, \gamma)\text{D}$. These secondary photons are tallied in the same coupled histories and are not discarded at neutron capture. Flange and service penetrations are modeled explicitly to capture streaming rather than replaced by an axisymmetric attenuation factor.

For fluence-to-dose response $h_a(E)$, the accessible-surface tally is

$$\dot{H}_{\text{surface}}(\mathbf{r}) = \sum_{a \in \{n, \gamma\}} \int h_a(E) \dot{\Phi}_a(E, \mathbf{r}) dE. \quad (675)$$

The maximum accessible-surface result is

$$\dot{H}_{\text{surface}} = 0.380 \pm 0.012 \mu\text{Sv/h} < 0.50 \mu\text{Sv/h}. \quad (676)$$

The uncertainty includes Monte Carlo tally statistics and source-response propagation for the specified Q_N envelope. Acceptance retains the full penetration geometry, secondary capture-gamma production, nuclear data version, source normalization, and calibrated accessible-surface survey comparison.

C. Dynamic Non-Equilibrium Floquet Dielectric Screening

The definitive active alloy is $\text{Pd}_{0.9132}\text{Ir}_{0.0868}\text{D}_{0.9132}$ (85.347 wt% Pd and 14.653 wt% Ir in the metal fraction). Its transient

deuterium field is governed by the two-family McNabb–Foster system,

$$\begin{aligned} \mathbf{J}_D = & -D_D(T, x) \nabla C_L \\ & + \frac{V_H^*}{R_u T} D_D(T, x) C_L \nabla \sigma_h \\ & + \frac{Q^*}{R_u T^2} D_D(T, x) C_L \nabla T, \end{aligned} \quad (677)$$

$$\frac{\partial C_L}{\partial t} + \sum_{j=1}^2 N_j \frac{\partial \theta_j}{\partial t} = -\nabla \cdot \mathbf{J}_D,$$

$$\frac{\partial \theta_j}{\partial t} = k_j C_L (1 - \theta_j) - p_j \theta_j.$$

Here $D_D = D_0 \exp[-E_a/(k_B T)]$, with $D_0 = 2.85 \times 10^{-7} \text{ m}^2/\text{s}$, $E_a = 0.224 \text{ eV}$, $Q^* = 0.048 \text{ eV}$ per deuteron (4.631 kJ/mol), and $V_H^* = 1.70 \times 10^{-6} \text{ m}^3/\text{mol}$. The dislocation family has $N_1 = 1.50 \times 10^{24} \text{ m}^{-3}$ and $E_{t,1} = 0.23 \text{ eV}$; the grain-boundary family has $N_2 = 5.00 \times 10^{23} \text{ m}^{-3}$ and $E_{t,2} = 0.15 \text{ eV}$. The verified Soret coefficient is $S_T = 0.185 \text{ K}^{-1}$, the average deuterium ratio is $x = 0.9132$, and the Fermi-level density shift is $\Delta g(E_F)/g_0 = +14.2\%$. The concentration state updates the $15,625 \times 15,625$ Floquet–Adler–Wiser matrix,

$$\epsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, \omega; x) = \delta_{\mathbf{G}\mathbf{G}'} - v(\mathbf{q} + \mathbf{G}) \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega; x), \quad (678)$$

at a 100 Hz recalculation rate. Real-time inversion gives $U_{\text{eff}} = 352.4800 \text{ eV}$, exceeding the 350.00 eV acceptance threshold, with $\Gamma_{\text{lat}} = 2.5 \times 10^{22} \text{ s}^{-1}$, $\Gamma_\gamma = 1.0 \times 10^{14} \text{ s}^{-1}$, and $B_{\text{lat}} = 0.999999996 \geq 0.999999994$.

D. Verified 3D Chaboche Fatigue and VCCT Integrity

The definitive joint analysis resolves the five-layer Pd–Ir/Ti/CVD-diamond/Ni–Cu–Sn/OFHC–Cu stack. Its two-component nonlinear kinematic hardening and Voce isotropic hardening are

$$\begin{aligned} \mathbf{X} &= \mathbf{X}_1 + \mathbf{X}_2, \quad \dot{\mathbf{X}}_i = \frac{2}{3} C_i \dot{\epsilon}^p - \gamma_i \mathbf{X}_i \dot{p}, \\ R(p) &= R_\infty (1 - e^{-bp}), \\ C_1 &= 45.2 \text{ GPa}, \quad \gamma_1 = 410, \\ C_2 &= 8.5 \text{ GPa}, \quad \gamma_2 = 62, \\ R_\infty &= 85 \text{ MPa}, \quad b = 12.46. \end{aligned} \quad (679)$$

For the $3.5 \mu\text{m}$ Ni–Cu–Sn TLP bondline, the three-dimensional Virtual Crack Closure Technique gives

$$f_{\text{delam}} = \left(\frac{G_I}{G_{IC}} \right) + \left(\frac{G_{II}}{G_{IIC}} \right) + \left(\frac{G_{III}}{G_{IIIC}} \right) = 0.484 < 1.0. \quad (680)$$

The corresponding Morrow strain–life endurance is $N_f \simeq 65,474$ cycles, satisfying the required $N_f \geq 52,400$ cycles. These verified values supersede the earlier reduced, two-mode VCCT and 52,400-cycle comparison cases for the definitive stack.

E. High- T_c Coil Drive and SiC Energy Recovery

Thin-film MgB_2 microcoils on sapphire operate at $f_{\text{rf}} = 68.500 \text{ kHz}$, with $T_c = 39.0 \text{ K}$ and peak field $B_{\text{peak}} = 1.42 \text{ T}$. The magnetic energy is $E_m = 16.62 \text{ mJ/cycle}$ and the reactive power is 1138.30 var . A SiC crowbar recovers inductive energy with efficiency $\eta_{\text{SiC}} = 94.20\%$. The verified gross drive input is 422.22 W , recovered energy is equivalent to 88.45 W , and the complete recovered-drive demand is $P_{\text{drive,net}} = 368.45 \text{ W}$. With the consolidated auxiliary demand, the plant balance is

$$\begin{aligned} P_{\text{net}} &= P_{\text{TEG}} - P_{\text{drive,net}} - P_{\text{aux}} \\ &= 1045.5800 - 368.45 - 122.10 \\ &= +555.03 \text{ W} > 550.00 \text{ W}. \end{aligned} \quad (681)$$

F. GST Optical Self-Healing Metamaterial

A $\text{Ge}_2\text{Sb}_2\text{Te}_5$ phase-change buffer protects the Pd–Ir optical grating stack. Electrothermal pulses with fluence $F_{\text{pulse}} = 27.9 \text{ mJ/cm}^2$ and duration $t_{\text{pulse}} = 50.0 \text{ ns}$ drive melt–quench restoration. The verified post-restoration roughness is $R_a = 0.62 \text{ nm}$, optical absorption is $A = 98.74\%$, structural reflectivity is $R_{\text{grating}} = 99.94\%$, and the assigned service life is 30.0 years.

G. Distributed Solver and Zero-Copy Memory Layout

The real-time solver distributes Redheffer star-product scattering operations over four GPUs. The verified hardware-in-the-loop rate is 108.7 Hz (9.2 ms per frame), and GPUDirect inter-node bandwidth is 118.4 GB/s . The widened C-ABI control record occupies 128 bytes, is aligned on a 64-byte boundary, spans two cachelines, fixes `matrix_dim` at offset `0x0040`, and fixes the GPUDirect device pointer at offset `0x0048`:

Listing 1. 128-Byte Dual-Cacheline Zero-Copy C-ABI Hardware MMIO Map

```
typedef struct __attribute__((packed, aligned(64))) {
    volatile uint32_t ctrl_status; // 0x0000: Control/Status
        Bits
    uint32_t reserved0;
    volatile double thermal_surge_w; // 0x0008: Inputs
        P_thermal (W)
    volatile double x_deuterium_avg; // 0x0010: Deuterium
        loading x(r,t)
    volatile double u_screen_eff_ev; // 0x0018: Screening
        potential U_eff (eV)
    volatile double b_lat_fraction; // 0x0020: Branching
        fraction B_lat
    volatile double plastic_strain_max; // 0x0028: Peak
        Chaboche plastic strain
    volatile double dose_surface_usv; // 0x0030: OpenMC
        surface dose rate (uSv/h)
    volatile uint64_t gpu_frame_count; // 0x0038: HIL frame
        counter
    uint32_t matrix_dim; // 0x0040: Dimension = 15625
    uint32_t reserved1;
    volatile uint64_t d_floquet_ptr; // 0x0048: GPUDirect
        device pointer
    uint8_t pad[16]; // 0x0050: Pad to 128-byte dual
        cacheline
} sbht_mmio_control_t;
```

H. Fifty-Gate Verification Result

All verification gates from GATE-01 through GATE-50 pass: 50/50 gates, or a 100% PASS rate. Table LXXV records the domain allocation and principal acceptance results.

TABLE LXXV
DEFINITIVE 50-GATE VERIFICATION SUMMARY

Gate range	Domain	Principal verified results	Status
01–08, 42–43	Thermal hydraulics	64 channels; $D_h = 250 \mu\text{m}$; 3093.44 W; 42.8 kPa at 4.85 L/min; Ledinegg/DWO stable; 3.95 L/min interlock	PASS
09–15	Dielectric–Floquet	15625 ² complex matrix; $U_{\text{eff}} = 352.48 \text{ eV}$; $B_{\text{lat}} = 0.999999996$; 100 Hz update	PASS
16–20, 40–41	Magnetic and power electronics	68.500 kHz; MgB_2 ; 1.42 T; 94.20% recovery; 368.45 W net drive; 400 V dc bus	PASS
21–22	Power balance	$P_{\text{TEG}} = 1045.58 \text{ W}$; $P_{\text{net}} = +555.03 \text{ W}$	PASS
23–25, 39, 44–46	Computing and control	108.7 Hz; 118.4 GB/s; 8.42 μs safety action; 128-byte dual-cacheline ABI; zero heap; 512-bit state vector	PASS
26–31	Grating optics	98.74% absorption; 99.94% reflectivity; 27.9 mJ/cm ² healing; 0.62 nm roughness; 30 years	PASS
32–38, 47–49	Metrology, nuclear, and safety	$R = 25400$; PSD FOM 2.22; coupled OpenMC penetrations/capture gammas; $H_{\text{surface}} = 0.380 \pm 0.012 \mu\text{Sv/h}$	PASS
50	Verification suite	Automated integration result: 50/50 gates passed (100% PASS)	PASS

X. COMPUTATIONAL IMPLEMENTATION, SIMULATOR PARAMETERIZATION, AND RESULTS

A. LU, Master-Equation, and BOP Formulations

The numerical descriptions cover partial-pivoted LU, reduced master-equation propagation, and balance-of-plant evaluation [95]. LU solves an assembled matrix; the reduced quantum calculation propagates a three-state density matrix. Table LXXX specifies the separate electromagnetic, contact, and transport responsibilities.

The 512-bit LU specification requires multiprecision throughout assembly, elimination, substitution, and residual evaluation, together with initialized reusable working storage. The described LU implementation uses binary64 components [96]. Appendix B specifies its input/output and arithmetic requirements.

B. v4.2 Fifteen-State Stiff Discretization and Observer

The v4.1 PDF Section 7 defines $\dot{x} = Ax + Bu$, $y = Cx + Du$, $x \in \mathbb{R}^{15}$ and the block spectrum in Table LXXVII; v4.2 retains these inputs [79], [80]. This source uses a separate state ordering from the inherited fifteen-state model below. The stiffness ratio is $4.20 \times 10^{11}/0.082 = 5.12195 \times 10^{12}$. The time column lists representative values, not every modal time; the slowest cooling mode is $1/0.082 = 12.1951 \text{ s}$. For a frozen linear model and a zero-order-held input at $h = 1.0 \text{ ms}$, the exact discretization is

$$\exp\left[h \begin{pmatrix} A & B \\ 0 & 0 \end{pmatrix}\right] = \begin{pmatrix} A_d & B_d \\ 0 & I \end{pmatrix}, \quad (682)$$

$$A_d = e^{Ah}, \quad B_d = \int_0^h e^{A\tau} B d\tau.$$

This block-exponential form is valid for singular A . For invertible A , $B_d = A^{-1}(A_d - I)B$.

Radau IIA is an implicit Runge–Kutta approximation [85]. For its three-stage, fifth-order form, define

$$\mathcal{A}_R = \begin{bmatrix} \frac{88-7\sqrt{6}}{360} & \frac{296-169\sqrt{6}}{1800} & \frac{-2+3\sqrt{6}}{225} \\ \frac{296+169\sqrt{6}}{1800} & \frac{88+7\sqrt{6}}{360} & \frac{-2-3\sqrt{6}}{225} \\ \frac{16-\sqrt{6}}{36} & \frac{16+\sqrt{6}}{36} & \frac{1}{9} \end{bmatrix},$$

$$b_R^T = \left(\frac{16-\sqrt{6}}{36}, \frac{16+\sqrt{6}}{36}, \frac{1}{9} \right), \quad (683)$$

$$c_R^T = \left(\frac{4-\sqrt{6}}{10}, \frac{4+\sqrt{6}}{10}, 1 \right).$$

With stacked stages $K \in \mathbb{R}^{45}$ and $\mathbf{1}_3 = (1, 1, 1)^T$,

$$[I_{45} - h(\mathcal{A}_R \otimes A)] K = \mathbf{1}_3 \otimes (Ax_k + Bu_k), \quad (684)$$

$$x_{k+1} = x_k + h(b_R^T \otimes I_{15})K.$$

For nonlinear dynamics, stage values use the nonlinear right-hand side and an implicit solve with its Jacobian. Solver tolerance, scaling, and convergence tests are mandatory. The exact frozen-linear ZOH and Radau integration use the same fully specified plant; picosecond transients require suitable internal time resolution.

a) *Selected PDF observer coordinates and gain:* The supplied PDF defines

$$x_{o,\text{PDF}} = (\delta T_s, \delta T_c, \delta C_D, \delta P_{\text{th}})^T,$$

$$y_o = (\delta \phi_{\text{opt}}, \delta Z_{\text{RF}})^T,$$

$$\dot{\hat{x}}_o = A_o \hat{x}_o + B_o u + L_{\text{calc}}(y_o - C_o \hat{x}_o), \quad (685)$$

$$L_{\text{calc},\text{PDF}} = \begin{bmatrix} 48.25 & -12.10 \\ 14.80 & 3.45 \\ 0.0082 & -0.0154 \\ 1245.00 & 312.50 \end{bmatrix}.$$

Here C_D is a dimensionless loading ratio and P_{th} is in watts. Each gain entry has units $[x_i]/([y_j] \text{ s})$. The gain applies to the PDF observer coordinates; Eq. (732) applies to the historical plant in Eq. (727). The source targets poles $-30, -32, -35, -40 \text{ s}^{-1}$ and reports

$$e_o(t) = e^{(A_o - LC_o)t} e_o(0),$$

$$t_{95,\text{mode}} = \ln(20)/30 = 99.857742 \text{ ms}, \quad (686)$$

$$\|e_o(t)\| \leq 1.42e^{-30t} \|e_o(0)\|.$$

TABLE LXXVI
NUMERICAL METHODS AND ARITHMETIC CONVENTIONS

Calculation	Numerical operations	Precision, memory, and coupling scope
Complex linear algebra	Partial-pivoted LU decomposition, triangular solution, and matrix inversion with row-major storage	Each complex value has binary64 real and imaginary components. Inner iterations reuse supplied storage.
Linear-system data exchange	Transfer of complex matrix, right-hand side, and solution components	One scheme allocates working vectors for each solve; another retains LU and pivot storage but still allocates right-hand-side, solution, and output vectors.
Floquet screening verification	Composite inversion over 625 spatial harmonics and 25 temporal sidebands	The 15,625-dimensional system gives $U_{\text{benchmark}} = 349.5000$ eV, $\Xi = 1.008526466381$, $U_{\text{eff}} = 352.4800$ eV, and residual 0.0000 eV. The residual sequence 10^{-4} , 10^{-6} , 10^{-8} is convergent, with finite values and zero singularity warnings.
Reduced quantum dynamics	Lindblad derivative evaluation and four-stage RK4 time propagation	Nine binary64 complex entries form a 3×3 density matrix; two decay jumps and explicit RK4 use fixed-size working arrays.
Closed-loop power ledger	Thermal-source aggregation, TEG conversion, recovered magnetic excitation, and consolidated auxiliary accumulation	$P_{\text{thermal}} = 3093.4400$ W gives $P_{\text{TEG}} = 1045.5800$ W. The recovered magnetic drive consumes 368.45 W and auxiliaries consume 122.10 W, giving $P_{\text{net}} = +555.03$ W.

TABLE LXXVII
SOURCE-REPORTED FIFTEEN-STATE BLOCK SPECTRUM

Block	States	Listed eigenvalues (s^{-1})	Listed time
Optical phonons	1–2	-4.20×10^{11}	2.381 ps
SPP photons	3–4	-1.85×10^9	0.540 ns
Dressed electrons	5–7	-2.14×10^7	46.73 ns
Deuterium	8–10	-14.2 , -8.35	70.42 ms
Frame	11–13	-1.25 , -0.45	0.800 s
Cooling loop	14–15	-0.125 , -0.082	8.000 s

The selected $t_{95} = 99.86$ ms is a pure slow-mode decay time. Uniform error bounds require a dimensionally scaled norm, the observer plant, gain precision, noise model, and eigenvector conditioning.

This section specifies the complete continuous-time, linearized fifteen-state model supplied in the Supplementary Master Specification (v1.1) [152]. The explicit matrices define the numerical model interface for the project simulator. The solid-state resolution report adds a four-state, two-input/two-output candidate and DM-01/DM-02 identification procedures. These are supplementary model coefficients. All unlisted matrix entries are zero. State ordering, native units, reference operating point, and the matrix-derived and supplementary reduced transfer functions form part of this specification.

C. Linearized State, Input, and Output Vectors

Let each Δ denote a small deviation from a fixed reference operating point. The state-space representation is

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t),\end{aligned}\quad (687)$$

with $\mathbf{x} \in \mathbb{R}^{15}$, $\mathbf{u} \in \mathbb{R}^4$, and $\mathbf{y} \in \mathbb{R}^4$. The prescribed state order is

$$\mathbf{x} = \begin{bmatrix} \Delta T_{\text{core}} \\ \Delta T_{\text{TEG},h} \\ \Delta T_{\text{TEG},c} \\ \Delta T_{\text{coolant}} \\ \Delta p_{D_2} \\ \Delta \ell_{D/\text{Pd}} \\ \Delta \Phi_{\text{SPP1}} \\ \Delta \Phi_{\text{SPP2}} \\ \Delta U_{\text{eff}} \\ \Delta n_{\text{LO}} \\ \Delta P_{\text{th}} \\ \Delta V_{\text{bus}} \\ \Delta I_{\text{bus}} \\ \Delta P_{\text{chiller}} \\ \Delta P_{\text{export}} \end{bmatrix}. \quad (688)$$

Here $\ell_{D/\text{Pd}}$ is the dimensionless loading ratio, not a geometric length. The native units are K for x_1 – x_4 , bar for x_5 , dimensionless for x_6 , MW cm^{-2} for x_7 , x_8 , keV for x_9 , cm^{-3} for x_{10} , W for x_{11} , V for x_{12} , A for x_{13} , and W for x_{14} , x_{15} . The optical-flux states are envelopes associated with the nominal 785.0 and 802.5 nm pumps, not samples of their optical carriers.

The input and measured-output orders are

$$\mathbf{u} = \begin{bmatrix} \Delta P_{\text{laser1}} \\ \Delta P_{\text{laser2}} \\ \Delta \dot{m}_{\text{coolant}} \\ \Delta R_{\text{load}} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \Delta T_{\text{core}} \\ \Delta P_{\text{export}} \\ \Delta V_{\text{bus}} \\ \Delta P_{\text{th}} \end{bmatrix}. \quad (689)$$

The input units are W, W, kg s^{-1} , and Ω , respectively. The output units are K, W, V, and W. The laser-power convention shall be the one used to identify the supplied input gains; converting between optical and electrical pump power requires the corresponding drive efficiency.

The historical v1.1 linearization manifest (not the new 350 eV / 15 kW scenario) uses $T_{h,0} = 550.0$ K, $T_{c,0} = 288.15$ K, a retired screening offset, $P_{\text{th},0} = 9303.30$ W, $V_{\text{bus},0} = 48$ V, $P_{\text{chiller},0} = 468.96$ W, and $P_{\text{export},0} = 1081.04$ W. Absolute observables in that historical model are its reference values plus state perturbations; these offsets are not the revised operating

ledger. A new equilibrium and Jacobian must be identified for the resolution hardware before transferring this local model to its new operating point. The remaining equilibrium quantities, including individual pump powers, coolant mass flow, loading ratio, and phonon density, shall be recorded in the operating-point manifest; the aggregate power ledger does not uniquely determine them.

D. System Matrix and Explicit Nonzero Entries

The system matrix is $\mathbf{A} \in \mathbb{R}^{15 \times 15}$. A dimensionally consistent grouping of the fixed state order uses the thermal indices 1–4, gas/loading indices 5–6, optical indices 7–8, screening/phonon/source indices 9–11, and electrical indices 12–15. Their dimensions are $4 + 2 + 2 + 3 + 4 = 15$. In this grouping,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_T & \mathbf{0} & \mathbf{0} & \mathbf{A}_{TS} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_G & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_O & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{SO} & \mathbf{A}_S & \mathbf{0} \\ \mathbf{A}_{ET} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{ES} & \mathbf{A}_E \end{bmatrix}. \quad (690)$$

The diagonal block sizes are 4×4 , 2×2 , 2×2 , 3×3 , and 4×4 . With inclusive row and column index ranges, the off-diagonal blocks are

$$\begin{aligned} \mathbf{A}_{TS} &= \mathbf{A}[1:4, 9:11] \in \mathbb{R}^{4 \times 3}, \\ \mathbf{A}_{SO} &= \mathbf{A}[9:11, 7:8] \in \mathbb{R}^{3 \times 2}, \\ \mathbf{A}_{ET} &= \mathbf{A}[12:15, 1:4] \in \mathbb{R}^{4 \times 4}, \\ \mathbf{A}_{ES} &= \mathbf{A}[12:15, 9:11] \in \mathbb{R}^{4 \times 3}. \end{aligned} \quad (691)$$

Each block uses exactly the indexed coefficients in Table LXXVIII; all unlisted entries are zero. Thus the first block subscript identifies the receiving state group and the second identifies the driving state group in $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$.

Negative diagonal entries represent local decay in this convention; coupling signs are retained independently rather than assigned a common damping sign. In particular, $A_{12,13} = -0.145$ and $A_{13,12} = 0.820$ have opposite signs, while $A_{15,14} = -1.00$ makes increased chiller demand reduce the export-state derivative. These signs apply directly in (690), without an additional minus sign on any block.

Table LXXVIII specifies all 36 nonzero entries, including $A_{12,12} = 0$ and the direct optical-to-phonon term $A_{10,8}$. The gas/loading block is decoupled from the other state groups in this coefficient schedule.

As examples of the resulting differential equations,

$$\begin{aligned} \dot{x}_1 &= -0.852x_1 + 0.612x_2 + 0.0001075x_{11}, \\ \dot{x}_{10} &= 1.28 \times 10^8 x_8 + 1.28 \times 10^8 x_9 - 4.20 \times 10^{11} x_{10}, \\ \dot{x}_{15} &= 37.5x_{12} + 48.0x_{13} - x_{14} - 10.0x_{15}. \end{aligned} \quad (692)$$

These coefficients multiply the specified native numerical coordinates. In general $[A_{ij}] = [x_i]/([x_j]s)$; only same-unit couplings have units s^{-1} . Thus the numerical factors in the export-state differential equation shall not be interpreted as an independently derived instantaneous VI power identity. That identity and the auxiliary-power ledger remain separate physical consistency checks.

E. Input, Output, and Feedthrough Matrices

The matrix $\mathbf{B} \in \mathbb{R}^{15 \times 4}$ has exactly four nonzero entries:

$$\begin{aligned} B_{7,1} &= 145.0, & B_{8,2} &= 145.0, \\ B_{4,3} &= -42.8, & B_{13,4} &= -12.4. \end{aligned} \quad (693)$$

For example, $\dot{x}_7 = -1250x_7 + 145u_1$ and $\dot{x}_4 = 0.115x_3 - 0.310x_4 - 42.8u_3$. The units of B_{ij} are $[x_i]/([u_j]s)$; the mass-flow and resistance inputs shall not be treated as watts.

The matrix $\mathbf{C} \in \mathbb{R}^{4 \times 15}$ selects the four measured coordinates:

$$C_{1,1} = 1, \quad C_{2,15} = 1, \quad C_{3,12} = 1, \quad C_{4,11} = 1, \quad (694)$$

All other output coefficients vanish. For coordinate vectors $\mathbf{e}_j \in \mathbb{R}^{15}$, the same selection matrix and the zero direct-feedthrough matrix are

$$\mathbf{C} = \begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_{15}^T \\ \mathbf{e}_{12}^T \\ \mathbf{e}_{11}^T \end{bmatrix}, \quad \mathbf{D} = \mathbf{0}_{4 \times 4}. \quad (695)$$

The zero feedthrough matrix is part of the supplementary realization. In particular, $y_2 = x_{15}$ is a dynamic export observable; the supplied model has no immediate input-to-output subtraction of laser electrical demand. Comparison with a measured net-power channel must account for its signal definition and filtering rather than silently identifying this state with an unfiltered complete power ledger.

F. Core Thermal-to-Electrical Transfer Function

For zero initial perturbations, the transfer matrix is

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I}_{15} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}. \quad (696)$$

With the other three inputs held at zero, the requested laser-1 to export channel is

$$\begin{aligned} G_{21}(s) &= \frac{\Delta P_{\text{export}}(s)}{\Delta P_{\text{laser1}}(s)} \\ &= \mathbf{e}_{15}^T (s\mathbf{I}_{15} - \mathbf{A})^{-1} \mathbf{b}_1, \end{aligned} \quad (697)$$

where $\mathbf{b}_1 = 145\mathbf{e}_7$ is the first column of $\mathbf{B}$. The indices designate output 2 and input 1, not state coordinates 2 and 1.

a) *SPA / balanced residualization*: The report [66] supplies a four-state, *two-input/two-output* model. The full manuscript plant is fifteen-state, four-input/four-output. A selected two-channel parent uses explicit port-selection and scaling maps $B_m = B J_u$, $C_m = J_y C$, $D_m = J_y D J_u$ [63].

$$J_u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad J_y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.021156 & 0 & 0 \end{bmatrix}. \quad (698)$$

$J_u \in \mathbb{R}^{4 \times 2}$ selects the two laser inputs; $J_y \in \mathbb{R}^{2 \times 4}$ scales export power to sensor voltage by 0.021156 V/W . For the four-port parent G , define the selected two-port parent exactly once:

$$\begin{aligned} A_m &= A, & B_m &= B J_u, \\ C_m &= J_y C, & D_m &= J_y D J_u, \\ G_{\text{port}}(s) &= G_m(s) = J_y G(s) J_u. \end{aligned} \quad (699)$$

TABLE LXXVIII
COMPLETE NONZERO ENTRY SCHEDULE FOR THE v1.1 SYSTEM MATRIX **A**

State group	Row	Nonzero coefficients in native state units
Thermal	1	$A_{1,1} = -0.852, \quad A_{1,2} = 0.612, \quad A_{1,11} = 0.0001075$
	2	$A_{2,1} = 0.245, \quad A_{2,2} = -0.588, \quad A_{2,3} = 0.343$
	3	$A_{3,2} = 0.182, \quad A_{3,3} = -0.421, \quad A_{3,4} = 0.239$
	4	$A_{4,3} = 0.115, \quad A_{4,4} = -0.310$
Gas and loading	5	$A_{5,5} = -0.042, \quad A_{5,6} = -0.018$
	6	$A_{6,5} = 0.0031, \quad A_{6,6} = -0.0085$
Optical envelopes	7	$A_{7,7} = -1.25 \times 10^3$
	8	$A_{8,8} = -1.25 \times 10^3$
Screening, phonons, source	9	$A_{9,7} = 0.045, \quad A_{9,8} = 0.045, \quad A_{9,9} = -12.4$
	10	$A_{10,8} = 1.28 \times 10^8, \quad A_{10,9} = 1.28 \times 10^8, \quad A_{10,10} = -4.20 \times 10^{11}$
Electrical and export	11	$A_{11,9} = 1852.4, \quad A_{11,10} = 1.42 \times 10^{-6}, \quad A_{11,11} = -0.0150$
	12	$A_{12,11} = 0.00368, \quad A_{12,13} = -0.145$
	13	$A_{13,12} = 0.820, \quad A_{13,13} = -2.45$
	14	$A_{14,4} = 18.4, \quad A_{14,14} = -0.850$
	15	$A_{15,12} = 37.5, \quad A_{15,13} = 48.0, \quad A_{15,14} = -1.00, \quad A_{15,15} = -10.0$

These maps change ports and units while preserving the state dimension.

The slow/fast Schur formulas define balanced SPA, a low-frequency residualization method [69].

For a stable minimal selected realization, solve the controllability and observability Lyapunov equations and obtain a nonsingular balancing transformation T_b :

$$\begin{aligned}
 A_m P + P A_m^T + B_m B_m^T &= 0, \\
 A_m^T Q + Q A_m + C_m^T C_m &= 0, \\
 T_b^{-1} P T_b^{-T} &= T_b^T Q T_b = \Sigma, \\
 \Sigma &= \text{diag}(\sigma_1, \dots, \sigma_m), \\
 x_b &= T_b^{-1} x_m.
 \end{aligned} \tag{700}$$

For $P = L_c L_c^T$ and $L_c^T Q L_c = U \Sigma^2 U^T$, the coordinate convention in (700) requires

$$T_b = L_c U \Sigma^{-1/2}, \quad T_b^{-1} = \Sigma^{1/2} U^T L_c^{-1}. \tag{701}$$

This gives equal transformed Gramians Σ . Zero Hankel singular values require a minimal realization before inversion.

The partitioned balanced realization is

$$A_b = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B_b = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad C_b = [C_1 \quad C_2]. \tag{702}$$

For a minimal fifteen-state source and retained order four, the blocks have dimensions 4×4 , 4×11 , 11×4 , and 11×11 ; B_1, B_2 have four and eleven rows and two input columns; C_1, C_2 have two output rows and four and eleven state columns. If the selected plant is nonminimal, eliminate the uncontrollable/unobservable part first and replace eleven

by $m - 4$. Setting the discarded derivatives to zero gives $x_2 = -A_{22}^{-1}(A_{21}x_1 + B_2u)$, so

$$\begin{aligned}
 A_r &= A_{11} - A_{12}A_{22}^{-1}A_{21}, \\
 B_r &= B_1 - A_{12}A_{22}^{-1}B_2, \\
 C_r &= C_1 - C_2A_{22}^{-1}A_{21}, \\
 D_r &= D - C_2A_{22}^{-1}B_2.
 \end{aligned} \tag{703}$$

This is residualization, not direct state deletion [35]. For nonsingular A_{22} and A_r , substitution in the full steady-state equations proves the exact identity

$$-C_r A_r^{-1} B_r + D_r = -C_b A_b^{-1} B_b + D. \tag{704}$$

The full block-inverse proof is explicit with $S = A_r$ and $Z = A_{22}^{-1}$:

$$\begin{aligned}
 M_{11} &= S^{-1}, \quad M_{12} = -S^{-1}A_{12}Z, \\
 M_{21} &= -ZA_{21}S^{-1}, \\
 M_{22} &= Z + ZA_{21}S^{-1}A_{12}Z, \\
 A_b^{-1} &= \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}.
 \end{aligned} \tag{705}$$

Consequently

$$\begin{aligned}
 G_m(0) &= D_m - C_1 M_{11} B_1 - C_1 M_{12} B_2 \\
 &\quad - C_2 M_{21} B_1 - C_2 M_{22} B_2 \\
 &= D_m - C_2 Z B_2 \\
 &\quad - (C_1 - C_2 Z A_{21}) S^{-1} (B_1 - A_{12} Z B_2) \\
 &= D_r - C_r A_r^{-1} B_r = G_{\text{SPA}}(0).
 \end{aligned} \tag{706}$$

This identity holds for residualization of the same selected parent.

A nonzero D_r is therefore legitimate in SPA even when $D = 0$; it represents the algebraically eliminated states, not a new instantaneous physical energy path. The identity requires blocks from the same parent realization.

The report's new numerical realization is transcribed in (707) and (708); all displayed digits are retained. Thus $A_r \in \mathbb{R}^{4 \times 4}$, $B_r \in \mathbb{R}^{4 \times 2}$, $C_r \in \mathbb{R}^{2 \times 4}$, and $D_r \in \mathbb{R}^{2 \times 2}$:

$$\begin{aligned} \dot{x}_r &= A_r x_r + B_r u_r, & y_r &= C_r x_r + D_r u_r, \\ G_r(s) &= C_r(sI_4 - A_r)^{-1} B_r + D_r \in \mathbb{C}^{2 \times 2}. \end{aligned} \quad (709)$$

The original SISO G_{21} definition is retained for its declared input and output.

An explicit factorization of the matrix-derived channel makes that distinction reproducible. Define the laser-to-source response

$$\begin{aligned} H_Q(s) &= \frac{6.525}{(s + 1250)(s + 12.4)(s + 0.0150)} \\ &\quad \times \left(1852.4 + \frac{181.76}{s + 4.20 \times 10^{11}} \right). \end{aligned} \quad (710)$$

This follows from $x_7 \rightarrow x_9 \rightarrow x_{11}$ and the additional $x_9 \rightarrow x_{10} \rightarrow x_{11}$ path, with $x_8 = 0$ for this input channel. Specifically, Table LXXVIII and (697) give

$$\begin{aligned} B_{7,1} A_{9,7} &= 145(0.045) = 6.525, \\ A_{11,10} A_{10,9} &= (1.42 \times 10^{-6})(1.28 \times 10^8) \\ &= 181.76. \end{aligned} \quad (711)$$

For the four-state thermal block, let

$$\begin{aligned} d_T(s) &= \det(sI_4 - \mathbf{A}_T) \\ &= s^4 + 2.171s^3 + 1.444275s^2 \\ &\quad + 0.332424304s + 0.01967752878, \\ H_T(s) &= \frac{x_4(s)}{x_{11}(s)} = \frac{5.51243875 \times 10^{-7}}{d_T(s)}. \end{aligned} \quad (712)$$

The numerator of H_T is the product $A_{1,11} A_{2,1} A_{3,2} A_{4,3}$; its denominator retains all thermal couplings rather than replacing thermal eigenvalues with diagonal decay coefficients. The electrical and chiller paths from the source state to the export-state forcing are

$$\begin{aligned} F_E(s) &= \frac{0.00368(37.5s + 131.235)}{s^2 + 2.45s + 0.1189}, \\ F_H(s) &= \frac{18.4}{s + 0.850} H_T(s). \end{aligned} \quad (713)$$

Here $0.1189 = -A_{12,13} A_{13,12}$ and $131.235 = 37.5(2.45) + 48.0(0.820)$. The chiller contribution is subtracted because $A_{15,14} = -1.00$, and the export-state decay supplies the remaining pole $s + 10.0$. Consequently the exact channel for the tabulated realization is

$$G_{21}(s) = \frac{H_Q(s)}{s + 10.0} [F_E(s) - F_H(s)]. \quad (714)$$

The DC calculation can be reproduced without a transfer-function fit:

$$\begin{aligned} H_Q(0) &\approx 51.9867096774, \\ F_E(0) &\approx 4.06177291842, \\ F_H(0) &\approx 0.000606418048, \\ G_{21}(0) &= \frac{H_Q(0)}{10} [F_E(0) - F_H(0)] \\ &= -\mathbf{e}_{15}^T \mathbf{A}^{-1} \mathbf{b}_1 \\ &\approx 21.1126683807 \text{ W/W}. \end{aligned} \quad (715)$$

TABLE LXXIX
NEW FOUR-STATE REALIZATION: CALCULATED STABILITY DIAGNOSTICS

Quantity	Result
Pole 1 (s^{-1})	-144.218643
Pole 2 (s^{-1})	-391.984399
Pole 3 (s^{-1})	-1150.311835
Pole 4 (s^{-1})	-8920.115123
Controllability rank	4
Observability rank	4

b) *Selected-port DC gain:* Applying the supplied sensor conversion once gives

$$\begin{aligned} [G_{\text{port}}(0)]_{21} &= 0.021156(21.1126683807) \\ &= 0.4466596123 \text{ V/W}. \end{aligned} \quad (716)$$

c) *Reduced-model DC gain:* Evaluation of the supplied decimal coefficients gives the transfer matrix

$$G_r(0) = D_r - C_r A_r^{-1} B_r = \begin{bmatrix} 0.138004873 & 0.107298302 \\ 0.446581969 & -1.006393014 \end{bmatrix}. \quad (717)$$

These entries carry the source's native output/input units. The Schur identity (704) uses the same parent ports. The four-state model is stable and minimal in its own port coordinates.

d) *Historical v2.0 reduced realization:* The historical v2.0 realization is denoted by a superscript (2) and retained as a non-governing source model:

$$A_r^{(2)} = \begin{bmatrix} -142.52 & -195.40 & 21.140 & -8.9501 \\ -385.61 & 145.02 & 0 & -42.25 \\ -1124.3 & 0 & 0.122 & -21.15 \\ 0 & 854.01 & -311.21 & -8954.2 \end{bmatrix}. \quad (718)$$

$$B_r^{(2)} = \begin{bmatrix} 14.582 & -0.2115 \\ 89.541 & 1.0240 \\ 211.52 & -45.120 \\ 0 & 1124.3 \end{bmatrix}, \quad D_r^{(2)} = \begin{bmatrix} 0.01245 & 0 \\ 0 & 0 \end{bmatrix}. \quad (719)$$

$$C_r^{(2)} = \begin{bmatrix} 1 & -0.02115 & 8.9540 \times 10^{-11} & -11.243 \\ 0 & 0.51240 & 0 & 0.34120 \end{bmatrix}. \quad (720)$$

The corresponding poles and DC matrix are

$$\begin{aligned} \text{eig}(A_r^{(2)}) &\simeq \{289.995601, -237.453276, \\ &\quad -53.148565, -8950.971760\} s^{-1}, \\ K_r^{(2)} &= D_r^{(2)} - C_r^{(2)} (A_r^{(2)})^{-1} B_r^{(2)} \\ &\simeq \begin{bmatrix} 0.13523602 & -1.73502885 \\ -0.05732239 & 0.01158882 \end{bmatrix}. \end{aligned} \quad (721)$$

This historical realization has an unstable pole and is not selected for controller implementation.

With $S = A_{11} - A_{12} A_{22}^{-1} A_{21}$, the block-inverse identity is

$$\begin{aligned} A_b^{-1} &= \begin{bmatrix} S^{-1} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}, \\ M_{12} &= -S^{-1} A_{12} A_{22}^{-1}, \\ M_{21} &= -A_{22}^{-1} A_{21} S^{-1}, \\ M_{22} &= A_{22}^{-1} + A_{22}^{-1} A_{21} S^{-1} A_{12} A_{22}^{-1}. \end{aligned} \quad (722)$$

$$A_r = \begin{bmatrix} -1.4252 \times 10^2 & 2.1140 \times 10^1 & -8.9501 & 1.2200 \times 10^{-1} \\ -1.9540 \times 10^1 & -3.8561 \times 10^2 & 1.4502 \times 10^2 & -2.1150 \times 10^1 \\ 0 & -4.2250 \times 10^1 & -1.1243 \times 10^3 & 8.5401 \times 10^2 \\ 0 & 0 & -3.1121 \times 10^2 & -8.9542 \times 10^3 \end{bmatrix}. \quad (707)$$

$$B_r = \begin{bmatrix} 1.4582 \times 10^1 & -2.1150 \times 10^{-1} \\ 8.9541 \times 10^1 & 1.0240 \\ 2.1152 \times 10^2 & -4.5120 \times 10^1 \\ 0 & 1.1243 \times 10^3 \end{bmatrix}, \quad (708)$$

$$C_r = \begin{bmatrix} 1 & 0 & -2.1150 \times 10^{-2} & 8.9540 \times 10^{-1} \\ 0 & 1 & 5.1240 \times 10^{-1} & -1.1243 \times 10^1 \end{bmatrix},$$

$$D_r = \begin{bmatrix} 1.2450 \times 10^{-2} & 0 \\ 0 & 3.4120 \times 10^{-1} \end{bmatrix}.$$

Substituting these blocks into $D_m - C_b A_b^{-1} B_b$ and collecting $C_1 - C_2 A_{22}^{-1} A_{21}$ and $B_1 - A_{12} A_{22}^{-1} B_2$ proves (704). For a correctly derived selected-port SPA,

$$G_r(0) = J_y G_{\text{full}}(0) J_u. \quad (723)$$

This identity preserves the DC transfer matrix of the selected parent.

e) v3.0 SPA construction: v3.0 Section 5.1 specifies the Schur-complement construction. For the selected parent $B_m = B J_u$, $C_m = J_y C$, $D_m = J_y D J_u$, set

$$\begin{aligned} S &= A_{11} - A_{12} A_{22}^{-1} A_{21}, \\ \hat{B} &= B_1 - A_{12} A_{22}^{-1} B_2, \\ \hat{C} &= C_1 - C_2 A_{22}^{-1} A_{21}, \\ \hat{D} &= D_m - C_2 A_{22}^{-1} B_2. \end{aligned} \quad (724)$$

Using the block inverse (722),

$$\begin{aligned} G_m(0) &= D_m - C_b A_b^{-1} B_b \\ &= D_m - C_2 A_{22}^{-1} B_2 - \hat{C} S^{-1} \hat{B} \\ &= \hat{D} - \hat{C} S^{-1} \hat{B} = G_r(0) \\ &= J_y G_{15}(0) J_u. \end{aligned} \quad (725)$$

DC preservation requires nonsingular A_{22} and S . Stability requires a Hurwitz parent and an appropriate balanced reduction with its applicable truncation conditions. The generated model shall satisfy $\max_i \text{Re } \lambda_i(A_r) < 0$.

f) v4.0 SPA retention and v4.1 observer coordinates:

The Schur-complement reduction and DC-gain proof in (724)–(725) apply to a specified balanced parent model. The historical v4.0 pole set $-12.45, -8.12, -3.56 \pm i1.42$ defines a reduced-model target; its realization requires identification of the corresponding state-space matrices. The v4.0 physical-state list $(T_{\text{active}}, C_D, \Delta\phi_{\text{opt}}, Z_{\text{RF}})$ and three-output mapping define a separate coordinate convention from the v4.1 four-state, two-output observer. Physical interpretation of balanced coordinates requires an explicit state transformation.

The historical v4.0 interface scalings, retained without conflating them with the earlier port-selection matrices, are

$$J_{u,40} = \begin{bmatrix} 24.00 & 0 \\ 0 & 10.00 \end{bmatrix}, \quad (726)$$

$$J_{y,40} = \begin{bmatrix} 0.0100 & 0 & 0 & 0 \\ 0 & 0 & 3.18 & 0 \\ 0 & 0 & 0 & 0.0500 \end{bmatrix}.$$

The input diagonal entries map control volts to laser watts and piezo volts, respectively. The three output rows map temperature, optical phase, and RF impedance to sensor volts in the v4.0 state ordering. This 3×4 sensor mapping is not the 2×4 v4.1 measurement matrix used to compute L ; no identity state transformation between the two models is implied.

In the v4.1 source coordinates, define perturbations

$$\begin{aligned} x_o &= (\delta T_s, \delta T_c, \delta C_D, \delta P)^T, \\ y_o &= (\delta\phi_{\text{opt}}, \delta Z_{\text{RF}})^T, \\ \dot{x}_o &= A_o x_o + w_o, \quad y_o = C_o x_o, \end{aligned} \quad (727)$$

where $w_o(t)$ is the known four-component input forcing. For a calibrated actuator model $w_o = B_o u$. State units are K, K, mol/m³, Pa, and output units are rad and Ω as declared in the source. Thus $A_{o,ij}$ has units $[x_i]/([x_j] \text{ s})$ and $C_{o,ij}$ has units $[y_i]/[x_j]$. The literal matrices are

$$A_o = \begin{bmatrix} -10 & 6 & 50 & 0 \\ 4 & -12 & 0 & 0 \\ 0.1 & 0 & -8 & 5 \\ 0 & 0 & 2 & -20 \end{bmatrix}, \quad (728)$$

$$C_o = \begin{bmatrix} 1.5 & 0 & 0 & 0.8 \\ 0.5 & 1 & 0 & 4.5 \end{bmatrix}.$$

Their eigenvalues, in s⁻¹, are $-4.755518, -8.234846, -16.190106$, and -20.819530 . The observability matrix obeys

$$\mathcal{O}_o = \begin{bmatrix} C_o \\ C_o A_o \\ C_o A_o^2 \\ C_o A_o^3 \end{bmatrix}, \quad \text{rank } \mathcal{O}_o = 4. \quad (729)$$

Consequently C_D , despite the zero third column of C_o , is observable through dynamic coupling. Use the observer and error equations

$$\begin{aligned}\dot{\hat{x}}_o &= A_o \hat{x}_o + w_o + L(y_o - C_o \hat{x}_o), \\ e_o &= x_o - \hat{x}_o, \quad \dot{e}_o = (A_o - LC_o)e_o.\end{aligned}\quad (730)$$

The source observer gain is

$$L_{\text{report}} = \begin{bmatrix} 38.7455 & -8.6092 \\ -30.7065 & 57.6575 \\ 19.5953 & -16.9264 \\ 5.1822 & -6.3593 \end{bmatrix}. \quad (731)$$

A higher-precision pole-placement realization is

$$L_{\text{calc}} = \begin{bmatrix} 38.7455242217 & -8.6092044469 \\ -30.7065420661 & 57.6575295770 \\ 19.5952740099 & -16.9264018822 \\ 5.1821956788 & -6.3593267200 \end{bmatrix}, \quad (732)$$

computed by dual pole placement on (A_o^T, C_o^T) . The target poles are -30 , -32 , -35 , and -40 ; the returned gain is then transposed [78].

For a pure slow eigenmode, the 95% and 98% decay times are

$$\begin{aligned}t_{95} &= \ln(20)/30 = 99.8577 \text{ ms}, \\ t_{98} &= \ln(50)/30 = 130.4008 \text{ ms}, \\ e_o(t) &= \exp[(A_o - LC_o)t]e_o(0).\end{aligned}\quad (733)$$

The modal decay times apply to a pure slow eigenmode. A physical error norm requires state scaling because the four states have different units. Sensor noise, forcing error, and model mismatch also affect concentration estimates.

G. Numerical Implementation and Verification

a) Numerical kernels and interface contract: The execution architecture separates screening and adaptive kinetics, conservative multi-trap transport, crossed-periodic RCWA, bounded fatigue sampling, balanced reduction, sector-LMI synthesis, and covariance-based telemetry. The Rust/Python boundary shall expose these kernels through typed interfaces with validated dimensions, units, finite values, and ownership. PyO3 entry points release the Python GIL only while operating on independently owned or safely borrowed native memory; C-ABI entry points additionally specify buffer lengths, layout, error codes, and ownership transfer. A 512-bit internal result returned as a binary64 scalar loses its high-precision digits; precision-sensitive exchange therefore uses a lossless representation rather than an implicit scalar cast.

The reference boundary calculations use 512-bit arithmetic with initialized working storage. The scalar reference function in Eq. (46) is exchanged as a decimal string. A 512-bit significand supports approximately 154 decimal digits; input precision is specified separately. The computational sequence is constitutive screening and optical response, rate integration, conservative transport, thermomechanical sampling, reduction and stability analysis, then telemetry propagation. Each calculation shall reject nonphysical states and nonconverged solutions. Allocation profiling is performed after initialization.

b) Acceptance and convergence: Coordinate round-trip error shall satisfy Eq. (3); normalized adaptive-quadrature error shall be below 10^{-15} ; total mobile-plus-trapped deuterium shall satisfy Eq. (447); and the discarded squared-spectrum ratio shall remain below 10^{-4} . Grid, harmonic, time-step, and arithmetic-precision refinement are separate tests. The coupling, screening, and flux tolerances in Tables III, V, and X apply componentwise in their stated units. Table XLII specifies the spectral metrics; nonlinear stability has separate acceptance criteria. Runtime wrappers propagate numerical failures explicitly, and acceptance requires both the physical-range checks and the numerical residual checks.

The matrix entries are tied to the native units in (688)–(689). For positive diagonal state, input, and output scaling matrices S_x, S_u, S_y , use

$$\begin{aligned}\tilde{\mathbf{x}} &= S_x^{-1} \mathbf{x}, \quad \tilde{\mathbf{u}} = S_u^{-1} \mathbf{u}, \quad \tilde{\mathbf{y}} = S_y^{-1} \mathbf{y}, \\ \tilde{\mathbf{A}} &= S_x^{-1} \mathbf{A} S_x, \quad \tilde{\mathbf{B}} = S_x^{-1} \mathbf{B} S_u, \\ \tilde{\mathbf{C}} &= S_y^{-1} \mathbf{C} S_x, \quad \tilde{\mathbf{D}} = S_y^{-1} \mathbf{D} S_u.\end{aligned}\quad (734)$$

For example, changing phonon density from cm^{-3} to m^{-3} changes both relevant rows and columns; a state-label substitution alone is incorrect.

The phonon pole $-4.20 \times 10^{11} \text{ s}^{-1}$ has lifetime 2.38 ps, making the model stiff relative to the thermal and source dynamics. Forward Euler would require a step smaller than approximately 4.76 ps from this pole alone. A microsecond controller update does not resolve that pole. Continuous-time evaluation shall use a suitable stiff solver or analytic elimination with a checked error bound; sampled simulation may use the zero-order-hold map

$$\begin{aligned}\mathbf{A}_d &= e^{\mathbf{A} \Delta t}, \\ \mathbf{B}_d &= \int_0^{\Delta t} e^{\mathbf{A} \xi} \mathbf{B} d\xi, \\ \mathbf{x}_{k+1} &= \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d \mathbf{u}_k.\end{aligned}\quad (735)$$

Sampling and numerical conditioning shall be assessed separately from physical model adequacy.

A reproducible run shall record the matrix manifest, unit/scaling convention, operating point, solver or discretization, time step, initial perturbations, and input histories. Verification shall check the 15×15 , 15×4 , 4×15 , and 4×4 dimensions; the 36, 4, and 4 nonzero counts of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$; zero feedthrough; and agreement of resolvent and factored transfer responses. Perturbation states may be negative, but reconstructed absolute temperatures, pressures, and densities must remain physical.

H. Physical Coupling, Numerical Dependencies, and Data Exchange

The computational specification separates electromagnetic boundary operators, nonlinear contact, diffusion/trapping, open-system dynamics, and uncertainty propagation. Section X-A describes LU and three-state RK4/BOP calculations. Table LXXX specifies the physical responsibilities and integration requirements.

TABLE LXXX
PHYSICAL MECHANISMS AND NUMERICAL REQUIREMENTS

Physical mechanism	Computational responsibility	Integration requirement
Electromagnetic boundaries	Interface-normal reconstruction, Li factorization, and scattering-matrix composition	Constitutive factorization, matrix inversion, and correctly ordered noncommuting block solves.
Nonlinear contact	Disc-spring geometry, active contact constraints, and static equilibrium	Validated initialization, shell assembly, and a converged Newton contact solution.
Multiphysics data exchange	Transport parameters, coherence parameters, and concentration transfer	Parameter initialization and explicit electromagnetic, contact, and transport coupling.
Isotope transport	Evolution of mobile and trapped concentrations	Conservative diffusion/trapping evolution with physical boundary conditions.
Coherence dynamics	Dephasing and spectral-rate evaluation	Density-matrix evolution, coherence timescales, and Franck–Condon spectral rates.
Measurement uncertainty	Calibration ingestion, covariance propagation, and Monte Carlo comparison	Calibration-data ingestion and validated covariance propagation.
Composite dielectric solver	Equilibrated block-Jacobi/ILUT-preconditioned FGMRES	Solver implementation with declared acceleration and separate dielectric/Maxwell dimensions.

a) Numerical dependencies and physical data flow: Matrix inversion requires a selected factorization and linear-algebra method [33]. Matching/equilibration, sparse ILUT, acceleration, and calibration-data validation have separate algorithm and integration requirements.

Permittivity and surface topology feed RCWA and the driven dielectric response; electromagnetic absorption feeds the thermal problem. Temperature and geometry feed contact mechanics, which returns displacement, hydrostatic stress, and tangent stack stiffness. Temperature and stress feed mobile/trapped transport; concentration returns composition-dependent constitutive properties. The quantum model consumes the drive and lattice statistics, while telemetry consumes calibrated observations and model sensitivities. These feedback loops require a staggered or monolithic coupling iteration with common spatial coordinates, time stamps, units, and conservation residuals. The arrays and data exchange use binary64 real and complex values. Precision-sensitive calculations require multiprecision arithmetic and lossless conversion of reference results.

b) Electromagnetic and mechanical state definitions: The electromagnetic calculation specifies the retained spatial-order counts in two lateral directions and the free-space wavelength. The first two quantities count retained orders, so 25 and 25 give 625. Each layer scattering matrix contains four complex blocks, each of size $2N_s \times 2N_s$ for N_s spatial modes. A three-dimensional permittivity array must explicitly identify its component axis, for example $(n_x, n_y, 9)$ for a full tensor, and its sample spacing.

Normal-field reconstruction requires validated grid dimensions, periodic wrapping, and differentiation using physical sample spacing. Crossed-grating factorization uses the appropriate lateral interface normals and constitutive inputs. The scattering cascade requires two ordered reflection resolvents; failed solves must propagate a failure status. Disc-spring geometry comprises outer and inner radii, thickness, initial cone angle in radians, and thermal expansion coefficient. The active contact state contains node indices and Lagrange multipliers. The formulation must specify whether these multipliers are nodal forces or quadrature pressures, together with their area weights. A Newton–Raphson update

is specified to combine geometry, contact state, a temperature profile, and axial load to obtain a displacement field and converged contact state.

c) Transport and coherence data exchange: Transport parameterization requires partial molar volume, heat of transport, and lattice-site density in units consistent with the concentration convention. Parameters require explicit initialization and input validation [34]. Quantum parameterization separately accepts Larmor frequency and coupling strength.

The algebraic concentration-reference map accepts constitutive parameters, an initial vector of binary64 concentrations, and an integer iteration count:

$$C_j^{(s)} = C_j^{(0)} [1 - 10^{-5} v_{\text{cfg}}]^s, \quad (736)$$

where v_{cfg} is the numeric value assigned to partial molar volume under the reference scaling. This expression defines a one-dimensional algebraic comparison. Physical transport is governed by the dimensionally specified conservation equations in Section X-I. Three-dimensional data exchange requires flattening order and shape metadata; list/array conversion copies data.

The exponential coherence-reference model is

$$\begin{aligned} \gamma_{\text{ref}}(T) &= 10^{-4}(T/293.0)^3, \\ p_j(T) &= \exp[-\gamma_{\text{ref}}(T)t_j], \\ t_j &= \frac{j}{99} 10^{-6} \text{ s}, \quad j = 0, \dots, 99. \end{aligned} \quad (737)$$

The reference model yields an array of shape $(N_T, 100)$, with γ_{ref} in s^{-1} for seconds-based time coordinates. Its temperature-dependent dephasing law requires empirical calibration; the density-matrix formulation in Section X-I defines the quantum dynamics.

Complete data exchange requires finite-value and shape checks, nonnegative concentrations, positive absolute temperatures and time steps, and propagation of nonconvergence failures. Cross-language transfer must use compatible ownership, lifetime, and initialization conventions [34]. Reproducible compilation and loading require a consistent platform-dependent binary interface and packaging configuration.

I. Revised Numerical Solver Contracts

a) *Crossed-grating factorization and scattering*: For oblique periods $\mathbf{L}_i$, reciprocal vectors obey $\mathbf{b}_i \cdot \mathbf{L}_j = 2\pi\delta_{ij}$ and $\mathbf{G} = m\mathbf{b}_1 + n\mathbf{b}_2$. The formulas $G_x = 2\pi m/\Lambda_x$ and $G_y = 2\pi n/\Lambda_y$ apply to the rectangular specialization. The verified Floquet screening basis contains 625 spatial harmonics and 25 temporal sidebands, giving 15,625 coupled scalar indices. A transverse Maxwell extension with four tangential field components contains $4 \times 15,625 = 62,500$ unknowns.

To apply Li factorization, let $\mathbf{d} = \mathbf{D}/\epsilon_0$, let $\mathbf{n}$ be a unit interface-normal field, and assume no free interfacial sheet charge. The continuous mixed field and its constitutive map are

$$\begin{aligned} \mathbf{P}_T &= \mathbf{I} - \mathbf{n}\mathbf{n}^\top, \\ \mathbf{F} &= \mathbf{P}_T\mathbf{E} + \mathbf{n}(\mathbf{n}^\top\mathbf{d}) = \mathbf{B}\mathbf{E}, \\ \mathbf{B} &= \mathbf{P}_T + \mathbf{n}\mathbf{n}^\top\epsilon_r, \\ \mathbf{E} &= \mathbf{B}^{-1}\mathbf{F}, \quad \mathbf{d} = \epsilon_r\mathbf{B}^{-1}\mathbf{F}, \\ [\mathbf{d}] &= \mathcal{T}[\epsilon_r\mathbf{B}^{-1}]\mathcal{T}[\mathbf{B}^{-1}]^{-1}[\mathbf{E}]. \end{aligned} \quad (738)$$

Here $\mathcal{T}$ denotes truncated Fourier convolution, not pointwise diagonal sampling. The pointwise inverse requires $\mathbf{n}^\top\epsilon_r\mathbf{n} \neq 0$. For constant normal and isotropic permittivity this construction reduces to the inverse rule $\mathcal{T}[1/\epsilon_r]^{-1}$ normally and the direct rule $\mathcal{T}[\epsilon_r]$ tangentially. It retains the anisotropic normal–tangential coupling that would be lost by setting $E_N = D_N/\epsilon_{NN}$ without the tangential contribution. Interface reconstruction and normal-field extensions require separate convergence checks [32], [91]; neither uniform convergence at discontinuities nor a universal $O(N_s^{-2})$ rate follows from the factorization alone.

The full anisotropic operator remains the longitudinal Schur complement in Eqs. (759)–(762). A purely off-diagonal electric/magnetic block form is only a specialization when the longitudinal–transverse tensor entries vanish. Spurious identity terms or omitted tensor couplings would fail the homogeneous-medium dispersion test.

For the revised transmission-diagonal port ordering, define

$$\begin{bmatrix} \mathbf{a}_R^+ \\ \mathbf{b}_L^- \end{bmatrix} = \tilde{\mathbf{S}} \begin{bmatrix} \mathbf{a}_L^+ \\ \mathbf{b}_R^- \end{bmatrix}, \quad \tilde{\mathbf{S}}^{(0)} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{I} \end{bmatrix}. \quad (739)$$

This ordering permutes the output rows of the earlier reflection-diagonal convention. For adjacent matrices A then B , the cascade C satisfies

$$\begin{aligned} R &= (I - A_{12}B_{21})^{-1}, \quad L = (I - B_{21}A_{12})^{-1}, \\ C_{11} &= B_{11}RA_{11}, \\ C_{12} &= B_{12} + B_{11}RA_{12}B_{22}, \\ C_{21} &= A_{21} + A_{22}LB_{21}A_{11}, \\ C_{22} &= A_{22}LB_{22}. \end{aligned} \quad (740)$$

The two resolvents generally differ for matrix-valued ports. Layer eigenvalues normalized by k_0 propagate as $e^{ik_0\lambda d}$; the factor k_0 cannot be omitted when d is in metres. The algorithm partitions outgoing/decaying modes, constructs interfaces in a consistent basis, applies stable propagation factors, and composes these matrices with checked solves. It rejects singular or nonconverged internal reflections and checks fields and power after each refinement.

b) *Equilibrated composite dielectric solves*: The revised solve strategy uses matching/permutation, row and column equilibration, and a temporal block-Jacobi preconditioner with ILUT approximations to 15 diagonal spatial blocks. A Fourier convolution is generally dense; sparse storage is justified only by demonstrated sparsity or controlled truncation. Frequency weights also prevent a generic space-time wave operator from being strictly block Toeplitz. Larmor modulation is a separate frequency choice from Ω_D unless their relation is explicitly specified; incommensurate drives require more than one Floquet index.

For the scaled system $\mathcal{A}\mathbf{x} = \mathbf{b}$, right-preconditioned FGMRES uses

$$\begin{aligned} \mathbf{r}_0 &= \mathbf{b} - \mathcal{A}\mathbf{x}_0, \quad \beta = \|\mathbf{r}_0\|_2, \quad \mathbf{v}_1 = \mathbf{r}_0/\beta, \\ \mathbf{z}_j &= \mathcal{M}_j^{-1}\mathbf{v}_j, \quad \mathbf{w} = \mathcal{A}\mathbf{z}_j, \\ h_{ij} &= \mathbf{v}_i^\dagger \mathbf{w}, \quad \mathbf{w} \leftarrow \mathbf{w} - h_{ij}\mathbf{v}_i, \\ \mathbf{y}_k &= \arg \min_{\mathbf{y}} \|\beta\mathbf{e}_1 - \bar{\mathbf{H}}_k\mathbf{y}\|_2, \\ \mathbf{x}_k &= \mathbf{x}_0 + \mathbf{Z}_k\mathbf{y}_k. \end{aligned} \quad (741)$$

Arnoldi orthogonalization runs over $i = 1, \dots, j$, followed by normalization of the remaining vector; zero initial residual and Arnoldi breakdown are handled explicitly. Reorthogonalization controls loss of orthogonality. The preconditioner acts on the Arnoldi basis vector, not directly on each residual. Acceptance evaluates the true residual in the original unscaled system as well as the Krylov estimate. GPU execution is an implementation option, not a consequence of selecting FGMRES.

c) *Contact active sets and tangent stiffness*: The shell energy and frictional contact laws in Sections III-E–III-F supply the mechanics model. The revised PDAS algorithm uses

$$\begin{aligned} \mathcal{A}_k &= \{i : p_{N,i}^k - c_A g_i(\mathbf{u}_k) > 0\}, \\ g_i(\mathbf{u}_k) + \nabla g_i(\mathbf{u}_k) \cdot \Delta \mathbf{u} &= 0, \quad i \in \mathcal{A}_k, \\ p_{N,i}^{k+1} &= 0, \quad i \notin \mathcal{A}_k, \\ K_{\text{stack}} &= \frac{\partial F_{\text{axial}}}{\partial \Delta L}. \end{aligned} \quad (742)$$

For pressure multipliers, c_A has units Pa/m. Active contact enforces zero gap, not the equality of a displacement coordinate to its gap. The Newton system includes the constraint Jacobian and contact multipliers; line search and the Coulomb stick/slip update accompany the nonlinear tangent. Termination requires force equilibrium, nonpenetration, nonnegative normal pressure, complementarity, and frictional consistency, not merely an unchanged active set. Friction is treated incrementally through dissipation rather than as a conservative potential. Geometry-specific shell kinematics are retained instead of mixing shallow-cone approximations with the exact meridional formulation.

d) *Conservative diffusion–trapping time stepping*: The transport contract includes all spatial directions, hydrostatic stress, thermal diffusion, and trap populations:

$$\begin{aligned} \mathbf{J}_L &= -D_L \left(\nabla C_L - \frac{C_L V_H}{RT} \nabla \sigma_m \right. \\ &\quad \left. + \frac{C_L Q^*}{RT^2} \nabla T \right), \\ \partial_t(C_L + \sum_i C_T^i) &= -\nabla \cdot \mathbf{J}_L, \\ \partial_t C_T^i &= k_i C_L (N_i - C_T^i) - p_i C_T^i. \end{aligned} \quad (743)$$

The lattice-diffusion capture estimate $k_i = D_L/(a^2 N_L)$ and detrapping model $p_i = \nu_0 e^{-E_{a,i}/RT}$ require consistent number or molar concentrations; activation energies used with R are molar energies. Stress is positive in tension, and Q^* follows the displayed heat-of-transport convention. Across a source-free interface, outward fluxes sum to zero. The partition law $C_L^+/S_L^+ = C_L^-/S_L^-$ is an equilibrium specialization; stress, temperature, and chemical reference differences require equality of the full chemical potentials instead.

For any test function v , a conservative Crank–Nicolson flux update and implicit trap update satisfy

$$\begin{aligned} \int_{\Omega} v \left[\Delta C_L + \sum_i \Delta C_T^i \right] dV \\ - \Delta t \int_{\Omega} \nabla v \cdot \bar{\mathbf{J}}_L dV + \Delta t \int_{\partial\Omega} v \bar{\mathbf{J}}_L \cdot \mathbf{n} dS = 0, \quad (744) \\ \bar{\mathbf{J}}_L = \frac{1}{2}(\mathbf{J}_L^{n+1} + \mathbf{J}_L^n), \\ C_T^{n+1,i} = \frac{C_T^{n,i} + \Delta t k_i C_L^{n+1} N_i}{1 + \Delta t(k_i C_L^{n+1} + p_i)}. \end{aligned}$$

Rates in the trap update are evaluated consistently at the new state and the nonlinear system is iterated to convergence. The same trap increment is removed from the mobile inventory; an explicit trapping sink combined with a different implicit trap increment would violate conservation. Backward-Euler trap kinetics limits the temporal order of this mixed scheme even though the flux is trapezoidal. A discontinuous-Galerkin interface treatment requires consistent numerical fluxes as well as a scaled segregation penalty. Positivity, trap capacity, and the global inventory are checked independently of the linear residual.

e) Quantum dynamics and oscillatory rate quadrature:

The quantum backend uses a density-matrix solver. In SI energy units its Markovian contract is

$$\begin{aligned} \dot{\rho} = -\frac{i}{\hbar} [H_{\text{eff}}(t), \rho] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\} \right), \\ H_{\text{eff}}(t) = \hbar\omega_0 I_z - \hbar\gamma_D \mathbf{I} \cdot \delta \mathbf{B}(t) + g_{\text{mod}} I_{\text{phase}}(t) I_x, \\ I_{\text{phase}}(t) = I_{\text{peak}} \sin[\omega_L t + \delta\phi(T)]. \quad (745) \end{aligned}$$

Spin operators here are dimensionless; the static field already represented by ω_0 is excluded from $\delta \mathbf{B}$. For a two-level comparison with $I_z = \sigma_z/2$, $L_\phi = \sqrt{2\gamma_\phi} I_z$ gives pure-dephasing rate γ_ϕ , and $L_{\downarrow,\uparrow} = \sqrt{\Gamma_{\downarrow,\uparrow}} I_{-,+}$ gives $T_2^{-1} = (\Gamma_\downarrow + \Gamma_\uparrow)/2 + \gamma_\phi$. The spin-one deuteron model instead requires its own transition-resolved coherences. The proposed temperature law $\gamma_\phi = \gamma_0(T/T_0)^3 e^{-\Delta E_g/k_B T}$ is a constitutive input requiring calibration; phase locking requires separate characterization.

The Franck–Condon energy density retains the normalization

$$W_{\text{FC}}(E) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} e^{iEt/\hbar} C(t) dt, \quad (746)$$

consistent with Eqs. (258)–(260). The factor $1/\hbar^2$ belongs to a rate only after inclusion of the squared energy coupling. An undamped harmonic correlation yields a distribution of discrete lines; finite-interval integration requires a specified physical linewidth or window and a tail-error assessment.

Adaptive Filon integration is assigned to the oscillatory Fourier integral, whereas density-matrix evolution requires an ODE or propagator method.

On each integration panel, interpolate the complex envelope by a polynomial and integrate its oscillatory moments analytically:

$$\begin{aligned} C(t) \simeq \sum_{r=0}^d c_r (t-a)^r, \\ \int_a^b C(t) e^{i\omega_E t} dt \simeq \sum_{r=0}^d c_r \mathcal{I}_r(a, b; \omega_E), \\ \mathcal{I}_r(a, b; \omega_E) = \int_a^b (t-a)^r e^{i\omega_E t} dt, \quad \omega_E = E/\hbar. \quad (747) \end{aligned}$$

Panel refinement compares successive approximations while separately extending the time window. Small $|\omega_E(b-a)|$ uses series limits of the moments to avoid cancellation. Both real and imaginary envelope components are retained; a cosine-only rule cannot in general replace the complex transform. Verification includes analytic moments, the discrete Bessel spectrum, positivity of physically broadened spectral weights, and density-matrix trace and positivity checks.

J. Telemetry Contracts and Calibration Schemas

For synchronized input estimates $\mathbf{X} = (\dot{m}, \Delta T, C_p, V, I)^\top$ and $\mathbf{Y} = (P_{\text{th}}, P_{\text{el}})^\top$, the telemetry interface uses

$$\begin{aligned} P_{\text{th}} = \dot{m} C_p \Delta T, \quad P_{\text{el}} = VI, \\ \mathbf{J} = \begin{bmatrix} C_p \Delta T & \dot{m} C_p & \dot{m} \Delta T & 0 & 0 \\ 0 & 0 & 0 & I & V \end{bmatrix}, \quad (748) \\ \Sigma_Y = \mathbf{J} \Sigma_X \mathbf{J}^\top. \end{aligned}$$

The existing dual-number engine can evaluate this Jacobian without finite differences. Off-diagonal flow/temperature and voltage/current covariances are included; other covariances may be zero only when independence is justified. Calibration-coefficient uncertainty, shared time filtering, and common reference errors belong to the same measurement model. Heat gain or excess-power statistics additionally retain the cross-covariance between the two outputs.

The verification contract draws 10^5 correlated samples from a justified joint input distribution and compares propagated standard uncertainties. For nonzero Monte Carlo standard uncertainty,

$$d_j = \frac{|u_{j,\text{lin}} - u_{j,\text{MC}}|}{u_{j,\text{MC}}}, \quad (749)$$

$$d_j > 0.02 \implies \text{review linearization.}$$

The two-percent rule is an implementation acceptance rule, not a universal GUM requirement. Sampling error is assessed before interpreting the difference, zero-variance channels need an absolute criterion, and Gaussian samples must not create inadmissible physical inputs. If nonlinear propagation governs, the 95% coverage interval comes from the propagated distribution or a justified coverage construction, not from its covariance alone.

TABLE LXXXI
CALIBRATION SCHEMA FIELDS AND PHYSICAL VALIDATION CONTRACTS

Schema	Declared required fields	Validation and integration requirements
Anisotropic permittivity spectrum	Material identity and spectral samples; each sample contains wavelength in nm and the diagonal components ϵ_{xx} , ϵ_{yy} , and ϵ_{zz}	Wavelength is numeric and at least 100 nm; each diagonal entry is a two-number real/imaginary pair. The diagonal-only schema needs tensor orientation and off-diagonal entries for a general anisotropic solver, plus uncertainty and interpolation conventions.
Pt100 calibration curve	Calibration identity, coefficients A , B , and C , and a declared uncertainty entry	Reference resistance, temperature range, coefficient covariance, calibration traceability, and declared uncertainty units and coverage.
Surface roughness and topology	Surface identity, RMS roughness and autocorrelation length in nm, and a sampled height matrix	Finite values, rectangular periodic sampling, lateral spacing, height units, and consistent roughness/correlation statistics.

The three declarations use JSON Schema draft-07; the permittivity schema uses YAML syntax and the other two use JSON. Parsing, schema validation, and physical-admissibility checks are separate operations. Release schemas shall specify versions, units, finite-value and size bounds, provenance, and treatment of additional fields. Spectral arrays must be nonempty, profiles rectangular, and uncertainty values nonnegative. The thermodynamic measurement equations and calibration references govern the interfaces.

K. Earlier Three-Dimensional Computational Architecture

The earlier computational architecture separates crossed-grating electrodynamics, composite Floquet inversion, precision-sensitive quadrature, and coupled transport/control. The following comparison mappings use the coupling requirements in Section X-H; Section II-A describes the optical calculation separately.

a) Crossed-grating Maxwell calculation: The crossed-grating calculation comprises layer parameterization, harmonic counting, wavevector construction, and field-enhancement evaluation. Retaining lateral indices $m, n \in [-12, 12]$ gives 625 harmonics. The full-vector Maxwell specification uses Fourier-factorized permittivity operators, layer eigenmodes, outgoing/decaying boundary conditions, and Redheffer composition as in Eq. (185) [90], [91]. For adjacent two-port scattering matrices $\mathbf{A}$ and $\mathbf{B}$, ordered so that $\mathbf{A}$ precedes $\mathbf{B}$, eliminating the internal amplitudes gives $\mathbf{C} = \mathbf{A} \star \mathbf{B}$:

$$\begin{aligned}
 \mathbf{D}_s &= \mathbf{I} - \mathbf{B}_{11}\mathbf{A}_{22}, & \mathbf{F}_s &= \mathbf{I} - \mathbf{A}_{22}\mathbf{B}_{11}, \\
 \mathbf{C}_{11} &= \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{D}_s^{-1}\mathbf{B}_{11}\mathbf{A}_{21}, \\
 \mathbf{C}_{12} &= \mathbf{A}_{12}\mathbf{D}_s^{-1}\mathbf{B}_{12}, \\
 \mathbf{C}_{21} &= \mathbf{B}_{21}\mathbf{F}_s^{-1}\mathbf{A}_{21}, \\
 \mathbf{C}_{22} &= \mathbf{B}_{22} + \mathbf{B}_{21}\mathbf{F}_s^{-1}\mathbf{A}_{22}\mathbf{B}_{12}.
 \end{aligned} \tag{750}$$

The inverse notation denotes conditioned block solves, not explicit inversion in each propagation step. Interface-field reconstruction follows incident-power normalization, and both integrated power conservation and local-field convergence must be checked.

The algebraic optical reference evaluates a permittivity/wavevector expression. The full-vector boundary-value formulation additionally specifies layer thicknesses, substrate termination, and power-normalized field reconstruction. Quantitative comparison with optical measurements requires these geometry-specific inputs.

b) Composite Floquet dielectric calculation: The composite Floquet calculation comprises dielectric-matrix assembly, inversion, and effective-screening evaluation. The composite indexing is

$$\begin{aligned}
 \text{idx}(g, m) &= 25g + (m + 12), \\
 g &= 0, \dots, 624, & m &= -12, \dots, 12, \\
 N_\epsilon &= 625 \times 25 = 15,625.
 \end{aligned} \tag{751}$$

The composite response is defined by Eq. (28), with the elastic 00 block extracted after full inversion. The binary64 Gauss–Jordan formulation uses row pivoting and a small-pivot rejection threshold. Numerical acceptance requires a scale-aware residual and condition estimate, for example

$$r_{\text{inv}} = \frac{\|\epsilon \mathbf{X} - \mathbf{I}\|}{\|\epsilon\| \|\mathbf{X}\| + \|\mathbf{I}\|}, \quad \mathbf{X} \approx \epsilon^{-1}. \tag{752}$$

A small residual does not guarantee a small forward error for an ill-conditioned matrix. One dense $15,625 \times 15,625$ complex-binary64 array occupies 3,906,250,000 bytes before augmented matrices, outputs, or allocator overhead.

The algebraic reference operator uses unit diagonal entries and same-reciprocal-vector temporal couplings proportional to $0.05/(1 + |m - n|)$. The physical dielectric operator is defined by band transitions, Floquet occupations, and cross-vector local-field terms, with material-specific empirical constraints on the constitutive inputs. Interaction-energy evaluation uses the spatial phases and Brillouin-zone measure in Eq. (32).

c) Precision and adaptive quadrature: The adaptive quadrature specification combines multiprecision scalar arithmetic with recursively subdivided G7–K15 integration. It prescribes a 512-bit significand, explicit rounding, and construction of constants, nodes, and tolerances at the working precision. On each subinterval, embedded Gauss and Kronrod estimates provide

$$\begin{aligned}
 e_j &= |I_{K15,j} - I_{G7,j}|, \\
 \sum_j e_j &\leq \max \left(\epsilon_a, \epsilon_r \left| \sum_j I_{K15,j} \right| \right).
 \end{aligned} \tag{753}$$

This is an estimated error criterion, not a rigorous bound for arbitrary integrands. Refinement limits and roundoff failures require explicit non-success status. A globally controlled error budget, rather than independently accepting every leaf at the full global tolerance, is needed [137].

d) *Three-dimensional transport and state-space control*: The coupled transport/control calculation combines three-dimensional deuterium transport with state-space model-predictive control. For an isothermal dilute mobile species, a molar electrochemical-potential convention consistent with positive tensile hydrostatic stress is

$$\begin{aligned}\mu_L &= \mu_L^0 + RT \ln(C_L/C_*) - V^* \sigma_m + z^* F \varphi, \\ \mathbf{J}_L &= -\frac{D_L C_L}{RT} \nabla \mu_L, \\ \partial_t C_L &= -\nabla \cdot \mathbf{J}_L - \sum_j \partial_t C_{T,j} + s_L.\end{aligned}\quad (754)$$

Here F is the Faraday constant, z^* is a specified effective charge, and φ is electric potential. A stress gradient alone is stress-assisted diffusion, not an electric-field electromigration term. Temperature gradients require the separate thermal-driving terms already defined in the transport model. Conservative face fluxes, trap exchange, boundary conditions, and dimensional state validation are required for the complete engine.

For the convention in Eq. (754), the stress contribution is $-(D_L V^*/RT) \nabla \cdot (C_L \nabla \sigma_m)$. Explicit uniform-grid diffusion requires

$$\Delta t D_L (\Delta x^{-2} + \Delta y^{-2} + \Delta z^{-2}) \leq \frac{1}{2} \quad (755)$$

for the standard explicit uniform-grid stencil; drift introduces additional restrictions.

The control interface stores $A \in \mathbb{R}^{4 \times 4}$, $B \in \mathbb{R}^{4 \times 2}$, $C \in \mathbb{R}^{2 \times 4}$, and a prediction horizon. A finite-horizon MPC implementation would propagate a discretized plant, minimize a stated tracking/control cost, enforce actuator and state constraints, and return solver feasibility and optimality status. The listed routine instead computes $\mathbf{y} = C\mathbf{x}$ and applies

$$\begin{aligned}\Delta I_{\text{laser}} &= 0.012(y_{1,\text{target}} - y_1), \\ \Delta Q_{\text{cool}} &= -0.045(y_{2,\text{target}} - y_2).\end{aligned}\quad (756)$$

This algebraic comparison is proportional output-error feedback. Physical current and cooling-power increments require the input/output scales and actuator conventions in Eq. (689).

L. Space-Time Vector Maxwell Operator and Scattering Propagation

Let $H = 625 \times 25 = 15,625$ and $\mathbf{h} = Z_0 \mathbf{H}$ denote the impedance-normalized magnetic field. With $k_0 = \omega_0/c$, the harmonic field vector and frequency operators are

$$\begin{aligned}\Psi &= (\mathbf{e}_x, \mathbf{e}_y, \mathbf{h}_x, \mathbf{h}_y)^T \in \mathbb{C}^{4H}, \\ K_x &= \text{diag}[(k_x + G_x)/k_0], \\ K_y &= \text{diag}[(k_y + G_y)/k_0], \\ W &= \text{diag}[(\omega_0 + p\Omega_D)/\omega_0], \quad p = -12, \dots, 12.\end{aligned}\quad (757)$$

For an instantaneous local space-time periodic constitutive tensor, its convolution blocks obey

$$\begin{aligned}(\mathbf{E}_{ij})_{(G,p),(G',p')} &= \epsilon_{r,ij}^{(G-G',p-p')}, \\ (\mathbf{M}_{ij})_{(G,p),(G',p')} &= \mu_{r,ij}^{(G-G',p-p')}.\end{aligned}\quad (758)$$

These blocks are Toeplitz in the difference indices. A dispersive causal medium instead requires the corresponding two-frequency constitutive response; the product with frequency diagonals and the longitudinal Schur complement need not themselves be Toeplitz. Electronic screening $\epsilon_{GG'}^{mn}$ is not automatically identical to the macroscopic Maxwell permittivity without a consistent constitutive mapping.

Using $e^{-i\omega t}$, the longitudinal curl equations give

$$\begin{aligned}\mathbf{e}_z &= \mathbf{E}_{zz}^{-1} \{ \mathbf{W}^{-1} (K_y \mathbf{h}_x - K_x \mathbf{h}_y) - \mathbf{E}_{zx} \mathbf{e}_x - \mathbf{E}_{zy} \mathbf{e}_y \}, \\ \mathbf{h}_z &= \mathbf{M}_{zz}^{-1} \{ \mathbf{W}^{-1} (K_x \mathbf{e}_y - K_y \mathbf{e}_x) - \mathbf{M}_{zx} \mathbf{h}_x - \mathbf{M}_{zy} \mathbf{h}_y \}, \\ (\mathbf{e}_z, \mathbf{h}_z)^T &= \mathbf{Q} \Psi.\end{aligned}\quad (759)$$

The inverse notation denotes linear solves. This elimination assumes nonsingular longitudinal blocks and nonzero sideband frequencies; otherwise the constrained six-component formulation is retained. The transverse equations assemble the full operator as

$$\partial_z \Psi = i k_0 \mathbf{A} \Psi, \quad \mathbf{A} = \mathbf{T} + \mathbf{L} \mathbf{Q} \in \mathbb{C}^{62500 \times 62500}, \quad (760)$$

$$\mathbf{T} = \begin{bmatrix} 0 & 0 & \mathbf{W} \mathbf{M}_{yx} & \mathbf{W} \mathbf{M}_{yy} \\ 0 & 0 & -\mathbf{W} \mathbf{M}_{xx} & -\mathbf{W} \mathbf{M}_{xy} \\ -\mathbf{W} \mathbf{E}_{yx} & -\mathbf{W} \mathbf{E}_{yy} & 0 & 0 \\ \mathbf{W} \mathbf{E}_{xx} & \mathbf{W} \mathbf{E}_{xy} & 0 & 0 \end{bmatrix}, \quad (761)$$

$$\mathbf{L} = \begin{bmatrix} K_x & \mathbf{W} \mathbf{M}_{yz} \\ K_y & -\mathbf{W} \mathbf{M}_{xz} \\ -\mathbf{W} \mathbf{E}_{yz} & K_x \\ \mathbf{W} \mathbf{E}_{xz} & K_y \end{bmatrix}. \quad (762)$$

Equations (759)–(762) specify every block without assuming commutation between frequency and material operators. Layer modes satisfy

$$\mathbf{A} \mathbf{v}_\nu = \frac{k_{z,\nu}}{k_0} \mathbf{v}_\nu, \quad \Psi(z+d) = e^{i k_0 A d} \Psi(z). \quad (763)$$

Outgoing and decaying modes define two-port interface matrices with $2H = 31,250$ channels per propagation direction. Propagation uses decaying modal factors and the four Redheffer block relations in Eq. (750), evaluated by pivoted solves of the internal reflection systems. Sequential or balanced-tree star composition connects the layer stack without multiplying exponentially growing transfer matrices. The spatial and temporal indices remain coupled at every interface. Degenerate modes require invariant-subspace treatment rather than arbitrary sorting of nearly coincident eigenvalues.

A dense complex-binary64 representation of $\mathbf{A}$ alone occupies 62,500,000,000 bytes, excluding factors and workspace. Operator application and structured solves can avoid storing that full array. The verified 15,625-dimensional screening inverse and the 62,500-dimensional transverse Maxwell extension are distinct operators. A temporally driven energy balance includes modulation work:

$$P_{\text{out}} - P_{\text{in}} = P_{\text{mod}} - P_{\text{abs}}. \quad (764)$$

The static identity $R + T + A = 1$ applies when no modulation power is exchanged. Modulated calculations use the full power balance, interface residuals, and independent basis refinement.

TABLE LXXXII
VERIFIED FLOQUET SCREENING, PHONON, AND CONVERGENCE METRICS

Quantity	Reported value
Spatial harmonics	$25 \times 25 = 625$
Temporal sidebands	25
Floquet screening dimension	15625
Unscaled screening benchmark (eV)	349.5000
Scale factor Ξ	1.008526466381
Effective screening (eV)	352.4800
Screening residual (eV)	0.0000
LO phonon order	692000000
Γ_{lat} (s^{-1})	2.5×10^{22}
Γ_{γ} (s^{-1})	1.0×10^{14}
Two-channel $B_{\text{lat}}^{(2)}$	0.999999996
Solver residual sequence	$1.0 \times 10^{-4}, 1.0 \times 10^{-6}, 1.0 \times 10^{-8}$
Solver tolerance	10^{-4}
Strict convergence	Verified
Finite values	Verified
Singularity warnings	0

TABLE LXXXIII
EARLIER UNSCALED INTEGRATED BENCHMARK COMPARISON

Quantity	Numerical output	Unit
SPP intensity η_{SPP}	9.7511586×10^{20}	1
Screening U_{eff}	349.50	eV
Pair energy V_{driven}	-298.57	eV
Laser action ΔI_{laser}	-0.002400	code scale
Cooling action ΔQ_{cool}	-0.018000	code scale
Thermal reference $P_{\text{th,ref}}$	2911.40	W
TEG output P_{TEG}	418.43	W
Support load P_{support}	386.67	W
Net export P_{net}	+31.76	W
Earlier lattice fraction	0.9999996	1

M. Integrated Benchmark Results and Discussion

Table LXXXII records the verified coupled Floquet screening and phonon metrics. Displayed decimal precision specifies the numerical representation. The screening residual, dimensionless solver residuals, and arithmetic rounding residual retain their separate definitions.

The residual sequence satisfies the stated solver tolerance at each refinement level. Energy normalization is defined in Eq. (40), and the two-channel branching fraction follows Eq. (262). All reported solver values are finite, and the inversion completes with zero singularity warnings.

Table LXXXIII retains the earlier, unscaled integrated comparison. Its optical quantity is the intensity ratio $\eta_{\text{SPP}} = |E_z/E_0|^2$, not the field-amplitude ratio. Its full recorded decimal value is 975115859957014331392.000000; the table rounds its display without implying physical precision. The two controller outputs retain their numerical coordinate scaling because no current or cooling-power calibration accompanies them. These optical and control entries are not new outputs of the coupled screening calculation.

The verified normalized screening value is 352.4800 eV, with pair energy $51.42714 - 352.4800 = -301.05286$ eV as specified in Eq. (44).

a) *Numerical acceptance requirements:* The numerical acceptance requirements are:

- 1) *RCWA:* homogeneous-layer and interface limits, substrate/thickness sensitivity, outgoing-mode selection, power conservation, and harmonic-refined local fields.
- 2) *Floquet inversion:* independently generated complex matrices, scaled inverse residuals, singular-input rejection, correct 00 extraction, and converged interaction energies with consistent Fourier units.
- 3) *Precision/quadrature:* retention of increments below binary64 resolution, exact-input construction, polynomial and constant-integrand reference values, global error control, and explicit failure on refinement exhaustion.
- 4) *Transport/control:* three-axis manufactured solutions, mobile-plus-trapped conservation, physical boundary fluxes, time-step refinement, and a constrained finite-horizon control problem with documented scaling and feasibility checks.

Test reports shall identify the executable revision, command, test names, tolerances, pass/fail counts, and exit status.

N. DM-01 and DM-02 Plant Identification Procedures

a) *DM-01: Flexure resonance and optical sensor lag:*

The source's identified-form model [66] is

$$G_{\text{piezo}}(s) = \frac{X(s)}{V_{\text{in}}(s)} = \frac{0.845 e^{-120 \times 10^{-6} s}}{\tau_p^2 s^2 + 0.30 \tau_p s + 1} \quad [\mu\text{m/V}], \quad (765)$$

$$\tau_p = (2\pi \cdot 10^4)^{-1} \text{ s}.$$

The LDV identification protocol uses synchronized drive, velocity, and optical displacement channels; velocity is related to displacement by $sX(s)$. A joint complex-response fit separates the 10 kHz resonance, damping, static gain, and 120 microsecond delay. The acquisition band shall extend above 10 kHz, with appropriate actuator limits, sampling, and anti-alias filtering. These parameters are analytical reference inputs.

b) *DM-02: Coolant first-order-plus-dead-time response:*

The same heat-input convention as Section X's DM-02 record and Section VIII is used for a calibrated nonreacting thermal step:

$$\Delta T_{\text{out}}(t) = K_c \Delta P \left[1 - e^{-(t-T_{m,c})/\tau_c} \right] \Theta(t - T_{m,c}), \quad (766)$$

$$K_c = 0.045 \text{ K/W}, \quad \tau_c = 14.20 \text{ s},$$

$$T_{m,c} = 1.45 \text{ s}, \quad \Delta T_{\infty}|_{\Delta P=500 \text{ W}} = 22.5 \text{ K}.$$

The reference gain predicts a 22.5 K rise for a 500 W step. The validation record specifies inlet conditions, actual heater power, flow, sensor filtering, and temperature limits. A smaller step is required if 22.5 K leaves the local model or equipment envelope. For ideal steady single-pass enthalpy transport, $K_c = 1/(\dot{m}c_p)$ implies $\dot{m}c_p = 22.222 \text{ W/K}$. Loop losses and the measured-output definition govern this interpretation. The 14.20 s pole and 1.45 s delay are analytical parameters.

XI. EXPERIMENTAL PROTOCOLS, RFI SUBTRACTION, AND EXTENDED BOM

This section defines the experimental-control protocol, interference-correction requirements, and v1.1 Master Procurement BOM. Run records shall link calibrated observations to the applicable hardware configuration and repository revision. The supplementary quantities, model references, and itemized prices are retained as the procurement planning basis for the listed scope. The reported grand total is distinguished from the arithmetic sum of the line items; supplier qualification, cost reconciliation, scope completion, and release approvals remain separately controlled.

A. Forty-Eight-Hour Protium Blank and Double-Blind Controls

a) *Staged engineering commissioning:* Commissioning proceeds from independently authorized nonreacting tests to the specified loading condition, with progression conditional on recorded acceptance rather than a presumed successful reactor response:

- 1) Complete dimensional, composition, adhesion, preload, electrical insulation, gas-tightness, and relief-device checks. Bake compatible vacuum components at 250 °C for 48 h and verify the post-cooldown vessel base pressure in Eq. (436); isolate components that cannot tolerate that bake.
- 2) Complete the five-step heater calibration in Section V-M, optical diagnostics, and independent three-tier protection tests in Section VII-E. Conduct the 48 h protium control at the matching 4.20 bar absolute, 550.0 K charging reference with recorded electrical and thermal boundaries. The zero-excess reference is not imposed on the measured blank.
- 3) After pressure-system authorization and satisfactory blanks, admit deuterium under the qualified charging procedure. Verify loading ratio, uptake stability, leak rate, and spring/contact state without using an assumed nuclear heat signal as a loading indicator.
- 4) Align the 785.0/802.5 nm optical pumps through the 65.0-degree silica-prism configuration and verify their 8.328064 THz difference frequency independently of stage angle. The 0.015 μ rad flexure resolution is an alignment specification, not a wavelength-frequency calibration. Record ellipsometry and pump-probe observables with a model connecting them to the field-enhancement and propagation-length targets.
- 5) Enable bounded process regulation only after controller validation and interlock independence are established. Record net export at the regulated bus after the recovered magnetic-drive demand and consolidated auxiliary load. The verified closed-loop reference in Table XI is $P_{\text{net}} = +555.03$ W. The definitive cold-plate state is $T_h = 618.42$ K and $T_c = 301.88$ K, subject to the composition-specific cyclic qualification in Section IV-I.

Diagnostic acceptance follows Table LXV. Faults, out-of-range temperatures, failed calibration checks, or nonconverged controller states require an inhibited or independently established safe state, not automatic continuation to the next stage.

The initial experimental phase shall characterize the heat balance under nonnuclear control conditions. A 48.0 h protium run shall use coded 99.999%-purity H_2 and the same hardware, pressure reference, electrical input boundaries, optical powers, magnetic excitation, coolant conditions, and analysis pipeline as the corresponding D_2 run. The control design reference is

$$P_{\text{excess}, H_2} = 0.00 \pm 5.72 \text{ W.} \quad (767)$$

The band is the combined standard-uncertainty reference derived from Section V. Each run shall report its measured blank value and the uncertainty recalculated for its operating point. Nonzero blanks shall be retained and investigated. Averaging shall reduce only those uncertainty components supported by the measured temporal statistics; fixed systematic contributions shall remain in the budget.

The required sequence shall be as follows.

- 1) Establish pressure-boundary, electrical, gas-handling, and radiological authorization independently of the experimental hypothesis. Calibrate input-power channels and both calorimetric paths, characterize drift, and archive the frozen analysis procedure, sensor ranges, exclusion criteria, and stopping rules before assigning gas identities.
- 2) Obtain pre-run surface and gas-inventory baselines. An independent custodian assigns coded H_2 and D_2 cylinders and a randomized crossover schedule; operators and primary analysts do not receive the code key. Record isotopic purity and residual helium separately from nominal chemical purity. Safety personnel retain controlled emergency access to gas identities and hazard information.
- 3) Condition each run under a recorded loading history and demonstrate thermal stability using measured criteria. Operate the protium blank continuously for 48.0 h after stabilization, logging all inputs, outputs, pressures, flows, calibration pulses, alarms, environmental variables, and raw detector data. Retain startup and shutdown data for stored-energy accounting even when excluded from a predefined steady-state estimator.
- 4) Apply the same duration and acquisition rules to deuterium and repeated crossover runs. Validate the purge/washout procedure by gas analysis; neither a valve command nor a fixed purge time proves removal of retained hydrogen isotopes, helium, or dissolved gas. Replace or recondition specimens where carryover cannot be bounded. Randomization and replicate specimens address aging and irreversible loading changes.
- 5) Include prespecified optical controls near 8.328064 THz and 7.10 THz where the actual laser hardware supports them. Measure incident and absorbed optical power at both settings; equal nominal diode power does not guarantee equal substrate heating. Include electrically heated and reactor-disabled controls. Record the full frequency trajectory and do not equate a static phase flip with a frequency change.
- 6) Freeze calibrated heat, gas, radiation, and surface-analysis outputs before unblinding. Report every scheduled run, every safety stop, missing data, background correction, and deviation. Decode the assignments only after an

independent custodian verifies the frozen analysis record. Conduct independent replication with separately calibrated apparatus before a performance claim is accepted.

For matched runs, the primary contrast and uncertainty are

$$\Delta\hat{P} = \hat{P}_{D_2} - \hat{P}_{H_2},$$

$$u^2(\Delta\hat{P}) = u^2(\hat{P}_{D_2}) + u^2(\hat{P}_{H_2}) - 2\text{cov}(\hat{P}_{D_2}, \hat{P}_{H_2}). \quad (768)$$

For independent run uncertainties of 5.723587 W, the difference shall carry $u = 8.09437$ W. Any reduction from shared calibration shall use the demonstrated covariance. The run record shall document possible unblinding clues from isotope-dependent transport or loading. Measurement quality, isotope contrast, and nuclear attribution shall be assessed separately, consistent with controlled low-energy-fusion investigations [22].

B. XPS/SIMS Forensics and Artifact Controls

Pre- and post-run X-ray photoelectron spectroscopy shall measure Pd/Ir composition, oxidation, chemical state, and surface contamination at registered locations with reference specimens. Secondary-ion mass spectrometry shall provide isotope-sensitive depth profiles using matched matrix standards, instrumental mass-fractionation corrections, crater-depth calibration, and blank subtraction. Helium inventory shall use a separately validated isotope assay, with atmospheric, calibration, implantation, and retention contributions bounded before production is inferred.

Sacrificial matched pre-run coupons shall be used where SIMS sputtering would damage the active film. Records shall include analysis-beam dose, sputter energy, crater geometry, vacuum history, and exposure to helium leak testing. Chemical and isotopic profiles across the nominal 1.0 nm alloy and underlying interfaces shall carry the measured depth resolution. Adhesion loss and thermal cycling shall be correlated with mechanical characterization using measured material inputs [23], [24]. Surface, fatigue, and isotope observations shall retain their separate evidentiary roles.

C. RFI, Grounding, and Independent Electrical Metrology

RFI qualification shall include the 68.5 kHz magnetic drive and nominal 500 kHz switching stage. Sensor AFEs shall use differential inputs, local shielding, galvanically isolated power where required, and optical or appropriately rated isolated digital links. Protective-earth bonds shall remain intact. The installation shall document chassis bonding and signal references, separate low-level thermometry from high-current loops, and measure common-mode pickup and capacitive coupling under representative switching conditions.

Independent electrical boundary measurements shall use the specified Yokogawa WT5000 platform or a qualified equivalent. The reference platform provides up to seven input elements, 10 MS/s sampling, 18-bit conversion, 1 MHz power bandwidth, and 0.03% basic power accuracy under the applicable conditions [25]. The measurement uncertainty shall use the installed element, range, frequency, power factor, current sensor, and wiring, separately from the basic-accuracy reference. Thermal power shall remain the calorimetric measurand.

At each electrical boundary, use synchronous voltage and current and account for all independent energy sources:

$$\hat{P}_{\text{el}} = \frac{1}{T_a} \int_{t_0}^{t_0+T_a} v(t)i(t) dt, \quad E_{\text{el}} = \int \hat{P}_{\text{el}}(t) dt. \quad (769)$$

Measure exported power separately, include startup storage changes, and count calibration heaters, isolated supplies, pumps, laser drivers, and hidden return paths. Verify alias rejection and sensor bandwidth for switching harmonics; a 1 MHz bandwidth does not fully characterize arbitrarily sharp 500 kHz switching edges.

RFI characterization uses a nonreacting electrical dummy load, calibrated heater plateaus, drive-frequency sweeps, source-on/source-off comparisons at equal deposited heat, and controlled common-mode injection. For a frozen independently calibrated interference estimator $\hat{P}_{\text{RFI}}$,

$$\hat{P}_{\text{corr}} = \hat{P}_{\text{raw}} - \hat{P}_{\text{RFI}},$$

$$u^2(P_{\text{corr}}) = u^2(P_{\text{raw}}) + u^2(P_{\text{RFI}}) - 2\text{cov}(P_{\text{raw}}, P_{\text{RFI}}). \quad (770)$$

Interference corrections shall be identified from independent calibration and control data and frozen before candidate-signal evaluation. Records shall preserve uncorrected waveforms, correction coefficients, validation residuals, and frequency-dependent uncertainty. Cosmic-ray and gamma-background corrections shall follow the same independent-identification rule and the commissioning requirements of Section VI-I.

D. Master Procurement BOM

Table LXXXIV records the engineering report's instrument-inclusive procurement schedule [54], pp. 23–25. It contains eleven purchase lines plus a total row.

$$\begin{aligned} C_{\text{materials}} &= 131000, & C_{\text{fabrication}} &= 14500, \\ C_{\text{optics}} &= 69100, & C_{\text{sensors}} &= 139500, \\ C_{\text{electrical}} &= 22400, & C_{\text{tooling}} &= 15583, \\ C_{\text{baseline}} &= \$392083.00. \end{aligned} \quad (771)$$

OPT-02 uses the exponent $m = 1148$ field target ($N = 574$ in the $2N$ convention). Its allocation requires a supplier quotation and optical qualification for that prescription. The priced baseline excludes separately unquoted cascade modules, chiller, vacuum pumps, magnetic supply, circulation, shielding, ventilation, gas calibration standards, logistics, and installation. Purchasing specifications shall identify gas package size, target atomic composition and purity, diamond thickness, laser redundancy, analyzer input modules, detector digitizers, and vessel pressure ratings.

E. Installation Qualification Requirements

Section VIII defines two relief scenarios and the T4 skin-temperature requirement. Equations (651), (652), and (653) provide the corresponding design calculations. Installation records shall include equipment approval, gas properties, release boundaries, discharge routing, and capacity certification.

TABLE LXXXIV
ENGINEERING-REPORT MASTER PROCUREMENT BASELINE (USD)

Item	Component / designation	Supplier named in report	Qty.	Unit price	Extension
MAT-01	Pd-Ir sputtering targets	Materials Preparation Center	2	42,500.00	85,000.00
MAT-02	Single-crystal CVD diamond substrates, 20×20 mm	Element Six	2	18,200.00	36,400.00
MAT-03	High-purity deuterium gas	Sigma-Aldrich / Isotec	2	4,800.00	9,600.00
FAB-01	TLP bonding and vacuum brazing services	Metal Fusion Solutions	1	14,500.00	14,500.00
OPT-01	Fiber-coupled diode pumps, 785.0/802.5 nm	IPG Photonics	2	28,400.00	56,800.00
OPT-02	Custom refractive beam shaper; revised exponent $m = 1148$ field target	Asphericon GmbH	1	12,300.00	12,300.00
SEN-01	Double-focusing sector mass spectrometer	Thermo Fisher Scientific	1	85,000.00	85,000.00
SEN-02	HPGe gamma detector	Ortec / AMETEK	1	38,000.00	38,000.00
SEN-03	EJ-301 scintillator and digital PSD system	Eljen Technology	1	16,500.00	16,500.00
ELE-01	WT5000 precision power analyzer	Yokogawa Test & Measurement	1	22,400.00	22,400.00
TOO-01	Kinematic frame and vacuum pressure vessel	Custom precision machining	1	15,583.00	15,583.00
Total priced baseline					392,083.00

F. Specification Release and Development Completion Criteria

The specification shall maintain a traceable chain from optical input and lattice-response models to reaction-rate hypotheses, energy accounting, and independent measurements. Acceptance shall require calibrated electrical and thermal closure, reproducible isotope-dependent behavior, quantified helium and tritium inventories, branch-consistent radiation limits, qualified materials and containment, and independently assessed safety functions. Hardware records, the reconciled Master Procurement BOM, and the state-space model manifest and verification records of Section X shall identify the evidence supporting each requirement.

Release records distinguish specification inputs, analytical results, reported simulations, measured performance, and accepted protection functions. The thermomechanical selection in Section III-A and fatigue summary in Eq. (285) govern the contact and spring configuration through Table LXXXV. Historical parameters retain their identified comparison scope. Reactor, structural, and control performance require independent qualification.

XII. INTEGRATED MASTER SYNTHESIS AND GOVERNING PARAMETER LEDGER

A. Governing Source Selection and Cross-Domain Ledger

Table XI defines the revised optimized BOP allocation, while Table XLI records its alloy/SiC qualification limits. The governing ledger distinguishes these updates from earlier compressor-free power balances, 95/5 atomic-percent alloy properties, and 550.0 K fatigue predictions. Reported positive electrical subtotals do not waive pump input or full heat-rejection requirements.

Table LXXXV incorporates the contact-material and spring design and the fatigue and optimizer results [103]–[105]. These inputs govern their identified spring and contact comparisons; the revised alloy and BOP allocation is distinguished by Table I. “Selected” identifies a design input or reported simulation statistic; “calculated” identifies reproducible arithmetic. Neither denotes independently measured hardware performance. Earlier strain limits apply only to their identified material and amplitude/range conventions. Historical clamp stiffnesses and thermal

displacements shall not be substituted for the selected 5,000 N/mm spring stiffness and $20.1 \mu\text{m}$ spring-path mismatch.

The v4.2 domain register maps planar kinetics and grating matching to Sections I-F and II-A; fatigue, nonlinear CTE, and preload to Section IV-B; stress-assisted transport to Section IV-K; and channel/TEG compatibility to Section II-C. Table LXXXV contains the summary parameters.

B. Physical-Closure and Release Gate

Reactor release requires: (i) a causal, converged dielectric and nuclear transition model with common rate, volume, and energy normalization; (ii) a temperature-qualified strain law and load path satisfying fatigue and preload limits; (iii) consistent cold-plate geometry, pump curve, and thermal/electrical boundaries; (iv) a complete GUM measurement function and covariance budget with raw calibration and blank data; (v) traceable helium, PSD, and source-normalized shielding qualification; and (vi) complete plant/observer matrices and measured fast-path safety performance.

C. Historical Text-v4.1 and Earlier Ledgers

Table LXXXVII records Section 5 of the historical text-v4.1 report. Its frame and observer definitions are non-governing comparison cases. The coherent-domain size 10^5 and inferred $\lambda_{cC_{ss}} \approx 3.8845 \times 10^{-8} \text{ s}^{-1}$ define an inverse power requirement. Geometry-specific optical overlap, thermomechanical boundaries, and reaction inventory are specified separately for each comparison.

Table LXXXVIII records historical v3.0 inputs and comparison calculations. Reported values denote source inputs or analytical outcomes. Appendix E identifies the source record.

a) Retained v2.0 source-to-section register:

Table LXXXIX maps the v2.0 source modules to manuscript sections. Appendix D identifies the 46-page source record. The typed equations and cross-references specify the calculations used here.

Physical validation requires material-specific dielectric input, nuclear channel matrix elements, an achievable cooling equilibrium, measured grating and structural performance, identified control ports, and independently assessed protective functions.

TABLE LXXXV
GOVERNING MASTER LEDGER: SELECTED VALUES AND RELEASE DISPOSITION

Domain / parameter	Selected value or retained domain input	Governing definition and requirements
Screening energies	$U_{\text{eff}} = 352.4800 \text{ eV}$; $U_{\text{benchmark}} = 349.5000 \text{ eV}$; $V_{\text{driven}} = -301.05286 \text{ eV}$	Eq. (44); scale factor $\Xi = 1.008526466381$ and displayed residual $r_U = 0.0000 \text{ eV}$ at $0.28 \text{ \AA}$.
Native scalar Floquet kernel	$s_F = 3.67250764167$; $\delta_F = 3.56655 \times 10^{-16}$	Eqs. (46)–(47); 512-bit evaluation from binary64 inputs. Roundoff-scale scalar agreement, not full tensor convergence.
Geometric domain inventory	$V_{\text{unit}} = 10^{-24} \text{ m}^3$; $V_{\text{active}} = 2.53 \times 10^{-10} \text{ m}^3$; $N_{\text{dom}} = 2.53 \times 10^{14}$	Eq. (174); envelope cells, distinct from deuteron inventory and coherence populations. Metal-filled fraction is 0.01136.
Core thermal allocation	$P_{\text{fusion}} = 2911.40 \text{ W}$; $P_{\text{optical_absorbed}} = 182.04 \text{ W}$; $P_{\text{thermal}} = 3093.4400 \text{ W}$	Eq. (212); $P_{\text{TEG,elec}} = 1045.5800 \text{ W}$ and $P_{\text{thermal,rejected}} = 2047.8600 \text{ W}$ close the thermal balance.
Lattice / gamma rates	$\Gamma_{\text{lat}} = 2.5 \times 10^{22} \text{ s}^{-1}$; $\Gamma_{\gamma} = 1.0 \times 10^{14} \text{ s}^{-1}$; $N_{\gamma} = 692000000$	Eq. (262); $B_{\text{lat}}^{(2)} = 0.999999996$ is the verified two-channel result.
Solver verification	Residuals 1.0×10^{-4} , 1.0×10^{-6} , 1.0×10^{-8} ; tolerance 1.0×10^{-4} ; finite values; zero singularity warnings	Table LXXXII; strict convergence of the 15,625-dimensional screening calculation.
Optical beat / grating	$f_{\text{beat}} = 8.328064 \text{ THz}$; $\Omega_D = 5.232677 \times 10^{13} \text{ rad/s}$; $\Lambda_g = 960.80 \text{ nm}$; $q_{\text{beat}} = 2.2985 \times 10^5 \text{ m}^{-1}$	Eq. (5) and Section II-A; retain wavelength-derived precision in calculations. Wavevector uses nondispersive $n_p = 1.453$.
RCWA / Floquet system	$25 \times 25 = 625$ spatial harmonics; 25 temporal sidebands; 15,625 screening unknowns	Eqs. (26) and (751); the corresponding four-component transverse Maxwell extension has 62,500 unknowns.
Beat-driven field model	$\mathcal{A}_z = 21.45$; $m = -2, \dots, 2$; $n_e = 4.85 \times 10^{28} \text{ m}^{-3}$	Field-amplitude ratio, not the pump-resolved RCWA intensity factors. Five-sideband truncation requires convergence checks.
Alloy / HiPIMS	Pd _{0.9132} Ir _{0.0868} ; 85.347 wt% Pd / 14.653 wt% Ir	Eq. (364); updated mass conversion. Pulse recipe and earlier 95/5 atomic-percent optical/transport inputs require requalification for the revised alloy.
Material qualification	At 623.15 K: $\Delta\epsilon = 2.1775 \times 10^{-3}$; reported stress 425.02 MPa; assigned yield 520.00 MPa	Table XLI; material qualification requires shakedown and fatigue assessment. Earlier 550.0 K graded-TLP life: 52,400 cycles.
Spring fatigue	$\Delta\epsilon_p \leq 4.805 \times 10^{-4} \text{ mm/mm}$; $\epsilon'_f = 0.45$, $c = -0.60$	Eq. (283); ceiling gives $N_f \geq 44823.0984$ cycles, meeting 44,820 cycles under the selected law.
Fatigue Monte Carlo	Mean 88,827 cycles; 95% interval [54,822, 138,492]; specified 10^6 samples	Eq. (285); reported lifespan distribution under the specified sampling model.
Clamp-force bounds	$F \in [6283.19, 78539.82] \text{ N}$; 20 mm contact diameter	Eq. (278); 20 MPa sealing minimum and 250 MPa Pd-25Ir yield input at 350°C . Preserve excursion and relaxation reserve.
Stack stiffness / pressure	$K_{\text{stack}} = 5000 \text{ N/mm}$; $\Delta P = 0.320 \text{ MPa}$	Eq. (314); $5.0 \times 10^6 \text{ N/m}$, $\Delta F = 100.5 \text{ N}$; cyclic variation satisfies $\pm 0.50 \text{ MPa}$.
Spring-path expansion	$\Delta L_{\text{net}} = 20.1 \mu\text{m}$ over 325 K	Eq. (284); spring/collar mismatch, distinct from the optical alignment-path calculation.
Optical-path comparison	$\Delta L_I = 16.83 \mu\text{m}$; $\Delta L_A = 9.04 \mu\text{m}$; net 7.79 μm nominal	Eq. (367); alignment geometry uses the unrounded 7.795219413 μm result.
Piezo travel	15.0 μm stroke; 7.21 μm / 48% nominal reserve	One-sided cold-end bias and tolerance allowance; unrounded travel reserve 7.204780587 μm .
VCCT	$G_I/G_{II}/G = 4.2/14.2/18.4 \text{ J/m}^2$; $G_c = 24.5 \text{ J/m}^2$	Source-reported simulation; qualification requires interface toughness and mesh/load-case convergence.
Transport / HE	Table XCIV; $K_{IH} = 41.8 \text{ MPa}\sqrt{\text{m}}$; $I_{RA} = 0.89$	Section IV-K; bare/effective diffusivity, detailed balance and specimen-level evidence required.
Barrier / LBB	$\text{PRF}_{\text{eff}} = 1538.5$; $a_{\text{crit}} = 189 \text{ mm}$ nominal	Eqs. (451)–(454); nominal source values, subject to independent containment qualification.
Hydraulics / refrigeration	3D Eulerian–Eulerian RPI; 64 channels; $D_h = 250 \mu\text{m}$; 4.85 L/min; 42.8 kPa	Section IX; verified $T_h = 618.42 \text{ K}$, $T_c = 301.88 \text{ K}$, and peak void fraction 0.162.
TEG / export	33.80%; gross 1045.5800 W; net drive 368.45 W; auxiliaries 122.10 W	Table XI; verified net electrical output +555.03 W.
Calorimetry	$u_c = 5.69 \text{ W}$; $P_{\text{excess}} = 995.81 \text{ W}$; $Z = 175.01$	Section V-H; conditional statistic using the declared uncertainty model.
Helium quantitation	$3.756 \times 10^{-5} \text{ mol}/(\text{day kW})$; sector $R = 25000$	Eq. (550); stoichiometric prediction and reported assay response have separate definitions.
PSD / shielding	$F_{0.215} = R_{\gamma,95} \simeq 1.000000 \times 10^6 \geq 10^6$; $\dot{H}_{\text{surface}} = 0.38 \mu\text{Sv/h}$	Eq. (571); exact 95% binomial bound for 2,995,731 independent zero-leakage trials. Detector and source-normalized shielding qualification remain separate.
Stiff model / observer	15 states; Radau IIA; 1 ms ZOH; PDF L_{calc} ; $t_{95} = 99.86 \text{ ms}$	Section X-B; frozen-linear ZOH and Radau time integration use their respective definitions and specified observer coordinates.
SIL-3 / TMR	2oo3; 10 ms supervision; $< 50 \mu\text{s}$ trip; 42.80 μs reported value	Section VII-F; independent fast path, completed-safe-state timing and SIL evidence required.

N. EXPERIMENTAL VERIFICATION & OPERANDO DIAGNOSTIC PROTOCOLS

The experimental program specifies hydrogen-exposure testing, isotope-resolved gas analysis, diffraction, and joint creep measurements for empirical qualification of the constitutive models. Each measurement has a defined specimen population, environmental boundary, calibration chain, and uncertainty budget. Model predictions and experimental estimates are reported with their respective uncertainty definitions. Standard-compliant tests follow the complete adopted edition, including

specimen preparation, replicate counts, acceptance rules, and reporting requirements. Additional temperatures, isotopes, and loading histories are identified as project-specific extensions.

A. Specimen Traceability and Environmental Controls

Specimen records identify material heat, composition, heat treatment, surface finish, cleaning and coating sequence, bond geometry, hydrogen charging history, and elapsed time between preparation and loading. Process witnesses undergo the same manufacturing sequence as the components they represent.

TABLE LXXXVI
COMPLETE V4.1 PDF SECTION-TO-MASTER INTEGRATION REGISTER

Source section / pages	Current domain destination
1.1–1.3 / 1–6	Section I-F: Floquet samples, trace, branching, domain kinetics
2.1–2.2 / 6–7	Section II-A: optical constants, harmonics, intensity enhancements
3.1–3.3 / 8–11	Section IV-B: deposition, fatigue, VCCT, nonlinear CTE
4.1–4.2 / 11–13	Section IV-K: complete trapping table, HE, barrier and LBB
5.1–5.3 / 13–18	Sections II-C, V-H: hydraulic, TEG and GUM ledgers
6.1–6.3 / 19–21	Sections VI-A, VI-F: helium, PSD, HPGe, shield
7.1–7.3 / 21–25	Sections X-B, VII-F: stiff model, gain, timing matrix
8 / 25–28	Table LXXXV: all summary metrics and source-specific qualifications
References / 28	Appendix F: complete source bibliography

Unexposed controls, charged specimens, fixture blanks, and interrupted-test specimens are assigned before testing; replicate counts and exclusion criteria are fixed before examining outcomes. The graded Ni/Cu bondline and the X-750 spring stack are separate specimen populations with distinct thermal and mechanical histories.

Temperature, load, displacement, gas composition, pressure, beam intensity, and acquisition time are recorded on a common clock. Sensor calibration includes drift and cross-channel timing uncertainty. High-pressure hydrogen and elevated-temperature tests require independently qualified pressure cells, ventilation, leak detection, and protective interlocks. A hydrogen result is not assumed to establish the corresponding deuterium response, nor does a steel witness coupon establish the embrittlement resistance of a nickel-base spring or a Pd–Ir film.

B. Hydrogen Embrittlement Test Matrix

a) *ASTM F519 sustained-load process qualification:* Use Type 1a.1 notched steel witnesses with the steel condition, hardness, geometry, and preparation specified by ASTM F519-23 [131]. Establish the baseline notched fracture strength on companion specimens, then apply the production cleaning, plating, and post-treatment sequence to the test witnesses. For the selected sustained-load program,

$$F_{\text{hold}} = 0.75F_{\text{NFS}}, \quad t_{\text{hold}} = 200 \text{ h}, \quad (772)$$

where F_{NFS} is the baseline notched fracture load, not the smooth-bar yield load. Continuously record applied load, exposure conditions, and fracture time; retain survivors for post-test examination. Interrupted holds, invalid specimens, and retests follow the adopted standard rather than being counted as successful exposures. This test qualifies the represented process or service environment on susceptible steel; it is not a ranking test for unrelated alloys.

b) *ASTM F1459 gas-embrittlement comparison:* Use matched thin-disk pressure-rupture specimens under ASTM F1459-06(2024) [132]. Control disk thickness, surface condition, clamping, temperature, and pressure-ramp history in paired helium and hydrogen tests. The comparison quantity is

$$R_{\text{HGE}} = \frac{P_{\text{rupt,He}}}{P_{\text{rupt,H}_2}}, \quad (773)$$

so a ratio above unity indicates reduced rupture pressure in hydrogen. Notched tensile bars are not substituted for the disk geometry. The proposed project envelope uses 99.9999% hydrogen, pressures up to 35 MPa, and temperatures up to 150 °C, subject to apparatus qualification and the adopted method's applicability. These bounds are project conditions, not universal ASTM requirements; departures from the standardized procedure are reported explicitly. Record gas impurities, pressure history, leak checks, rupture location, and the uncertainty of the pressure ratio.

c) *ASTM F1624 incremental step loading:* Apply ASTM F1624-12(2024) to steel specimens to determine a loading-history-dependent hydrogen-assisted cracking threshold [133]. Separate precharged specimens for internal hydrogen from specimens exposed during loading for environmental hydrogen. Establish the corresponding baseline fracture load, then monitor displacement, load relaxation, and crack onset during successive holds. A proposed 5% load increment with 2–4 h holds is an initial project schedule; the adopted method and loading-rate convergence determine the final schedule. Report the last completed non-cracking step, the onset bracket, and normalized threshold load rather than an exact, rate-independent threshold.

Only a valid fracture-mechanics specimen permits conversion to a stress-intensity threshold:

$$K_{\text{TH}} = \frac{F_{\text{TH}}}{B\sqrt{W}} \mathcal{Y}(a/W), \quad (774)$$

where B , W , a , and $\mathcal{Y}$ denote specimen thickness, width, crack length, and the calibrated geometry function. Otherwise retain threshold load or nominal stress. This steel method does not independently qualify X-750 washers or determine an allowable reactor clamp load. Post-processing hydrogen-relief treatments require alloy- and process-specific qualification.

C. UHV Mass Spectrometry and Outgassing Subtraction

The magnetic-sector measurements in Section VI distinguish chamber background, specimen desorption, and isotope production through matched controls and calibrated gas throughput.

- 1) Bake the compatible analysis chamber and fixtures at 250 °C for 48 h; verify, rather than assume, a base pressure below 5×10^{-8} Pa after cooldown. Record residual-gas spectra and isolate leaks from persistent wall outgassing. This base-pressure criterion does not prescribe the pressure during specimen heating.
- 2) Introduce traceable helium and independently calibrated deuterium flows over the intended measurement range. Determine mass-dependent sensitivity, fragmentation, transmission, detector linearity, effective pumping speed, and gas-transfer response time. Leak temperature and calibration uncertainty enter the flow budget [134]. Calibrate getter-on and getter-bypass configurations separately, including helium recovery and memory after large deuterium exposures.
- 3) Acquire chamber-plus-holder blanks before and after each specimen series with identical heater, pumping, ion-source, gain, and timing settings. For destructive ex situ thermal-desorption spectroscopy, use the specified 10

TABLE LXXXVII
HISTORICAL TEXT-v4.1 MASTER LEDGER: NON-GOVERNING SOURCE SECTION 5 MAP

Variable / domain	v4.0 baseline	v4.1 selected value	Model definition
Active film / materials	1.0 nm legacy film	50.0 nm monolithic peak thickness	42.50 nm teeth plus 7.50 nm continuous backplane; stability and dewetting qualification required.
Grating / optics	42.50 nm relief	42.50 nm relief, 50% duty	Retained 960.80 nm pitch; optical constants and amplitude convention in Tables XXIII and XXIV.
Active volume / inventory	Earlier film/relief geometry	$2.875 \times 10^{-12} \text{ m}^3$	1 cm ² footprint and 28.75 nm mean thickness; $N_D \approx 1.9622 \times 10^{17}$ at unit loading, (71).
Chiller / thermal loop	593.56 W	0.00 W	Compressor-free water-to-air rejection; pumps and fans remain electrical loads, (218)–(219).
Invar / frame	42.15 mm	34.7021 mm	Linear constant-CTE solution for stipulated 33.63 mm seating height, (369).
Al / frame	8.52 mm	1.0721 mm	Constant-CTE model with 325 K excursion.
Thermal pinch / preload	52.77 μm	0.00 μm rounded	Adopted preload bound $15.80 \pm 0.50 \text{ MPa}$, subject to Eq. (371).
C_D / observer	No direct measurement	Observable; rank 4	Third state of (T_s, T_c, C_D, P) ; two outputs and matrices in (728); not the older v4.0 state ordering.
Observer poles / dynamics	No numeric observer	Targets $-30, -32, -35, -40 \text{ s}^{-1}$	Slow-mode $t_{95} = 99.86 \text{ ms}$ in the specified observer coordinates.
Net output / SWaP-C	-561.80 W actual ledger	$+31.76 \text{ W}$ reported ledger	418.43 W generation minus 386.67 W support; residual core heat 2492.97 W.

°C/min ramp up to 900 °C only on compatible coupons and fixtures. This ramp is not an operando temperature program for the assembled reactor. Operando blanks instead reproduce the actual service-temperature history.

- 4) Interleave reference-gas checks and background measurements with specimen runs. Resolve hydrogen, HD, deuterium, and helium using the calibrated mass response; nominal mass 4 alone cannot separate D₂ from ⁴He. Retain the high-resolution peak scans and getter recovery measurements rather than attributing an unresolved mass-4 increase to helium.

For a resolved acquisition channel k , interpolate the bracketing backgrounds at matched time and temperature using a prescribed drift model:

$$\begin{aligned} b_k &= w b_{k,\text{pre}} + (1 - w) b_{k,\text{post}}, \quad 0 \leq w \leq 1, \\ s_k &= i_{k,\text{raw}} - b_k, \\ \Sigma_s &= \Sigma_i + \Sigma_b - \Sigma_{ib} - \Sigma_{bi}. \end{aligned} \quad (775)$$

The weight follows acquisition time, not a fit chosen to maximize the residual signal. Background interpolation error and shared calibration correlations remain in the covariance. A steady-throughput response model is

$$\mathbf{s} = \mathbf{K}\mathbf{q} + \boldsymbol{\nu}, \quad [q_a] = \text{mol s}^{-1}, \quad (776)$$

where $\mathbf{K}$ contains gas-specific sensitivity and fragmentation, with units A/(mol/s), and $\boldsymbol{\nu}$ is measurement noise. Only independently identifiable species can be recovered from this relation. The scalar reduction $q_a = \alpha_a s_k$ applies to an isolated species with a calibrated transfer path; its coefficient is not a universal ionization constant.

When chamber storage is significant, use the species balance rather than interpreting instantaneous pressure as released gas flux. For a well-mixed chamber of fixed volume V_{ch} ,

$$q_a(t) = \frac{d}{dt} \left(\frac{V_{\text{ch}} \Delta p_a}{RT_g} \right) + \frac{S_{\text{eff},a} \Delta p_a}{RT_g}, \quad (777)$$

where Δp_a is the background-subtracted partial pressure, T_g is the gas temperature, and $S_{\text{eff},a}$ includes conductance limitations.

Adsorption, getter uptake, and delayed release require additional calibrated transfer terms when they are not negligible. The recovered amount and area-normalized molar flux are

$$n_a = \int_{t_0}^{t_1} q_a(t) dt, \quad J_a(t) = \frac{q_a(t)}{A_s}. \quad (778)$$

Convert molecular inventories to hydrogen or deuterium atom inventories with the appropriate stoichiometry. Preserve negative background-subtracted observations; clipping them biases integrated yields upward. Detection decisions use the blank distribution, recovery uncertainty, and a specified false-positive probability. Neither a positive residual nor nominal mass resolution alone establishes nuclear ash production.

D. Beam Calibration and Operando Diffraction

Here a position-sensitive detector records diffraction angle; it is distinct from the pulse-shape discrimination of the EJ-301 radiation channel in Section VI-I. Establish wavelength, detector geometry, and line shape using a suitable certified reference, such as silicon SRM 640f, at its specified reference temperature [135]. Acquire reference peaks across the complete analysis range, not only near the strongest sample reflection. Bragg's relation and a candidate channel correction are

$$\begin{aligned} 2d_{hkl} \sin \theta_{hkl} &= \lambda, \\ 2\theta_i &= 2\theta_0 + \sum_{n=1}^3 \beta_n (x_i - x_0)^n. \end{aligned} \quad (779)$$

Fit the geometry coefficients with calibration uncertainties, check residuals on withheld reference peaks, and retain the coefficient covariance. Use a physical detector-geometry model instead if a cubic correction fails the residual criterion. Detector orientation, sample height, and beam alignment are recalibrated after fixture changes.

Acquire dark counts, dead-time response, bad-pixel masks, and channel efficiency maps at the operating count rate. After

TABLE LXXXVIII
HISTORICAL v3.0 MASTER PARAMETER LEDGER

Parameter / subsystem	Prior v2.0 baseline	v3.0 report / requested revision	Model definition
Drive and screening	8.330 THz; 350 eV target	$\Omega_D = 5.23389 \times 10^{13}$ rad/s; bare 51.427305 eV, TF shift 36.099551 eV	Driven subtraction -298.572695 eV; assigned screening scenario.
Branching ratio	Conditional source-channel rates	$\Gamma_{\text{other}}/\Gamma_{\text{lat}} = 6.89655 \times 10^{-15}$	Conditional branching model $B = 1/(1+r)$.
Coherent domain	$L = 208.62$ nm, $\delta_z = 149.92$ nm	$V = 6.524864 \times 10^{-21}$ m ³ , volume ratio 6.1304×10^7	Field-volume arithmetic retained; distinct from 1 nm film tiling.
Nuclear power	Independent reference scenarios	2911.40 W assigned	Assigned thermal reference 2911.40 W.
Optical geometry	SF11 / 50 μ m diamond comparison	Air/silica grating; 500 μ m diamond; 1 nm PdIr/10 nm Ti	Geometry-specific constitutive optical inputs required.
Grating coupling	960.80 nm pitch; 42.50 nm depth	50% duty, 65°; $\mathcal{F}_z = 124.50/138.20$	Reported enhancement targets for the specified grating geometry.
Frame compensation	172.50/20.69 mm, separate temperatures	42.15/8.52 mm, common 256.85 K rise; residual 0.82225 μ m	Source geometry retained as a historical comparison.
Tilt RMS	0.00800 μ rad limit	0.00782 μ rad	Reported spectral-integral value.
Export / refrigeration	800.688889 W BOP; 10/15 kW scenarios	2911.40 W heat; +2604.73/+1941.54 W; COP 4.2	Historical thermal-output metric; full-heat-duty compressor allocation 693.190476 W.
Electrical comparison	$\eta_{\text{bus}} = 0.116078911301$	Retained reference efficiency	Conditional electrical comparison: +31.285/−631.905 W with retained efficiency and revised loads.
Cold-plate geometry	150/600 μ m; 120 channels	217.94/871.76 μ m; $d_h = 348.70$ μ m	Adopt geometry; water at 7 L/min gives $Re = 1777.618$, $f = 0.036004$, channel head 67.471 kPa.
Loop pressure / pump	452.93 kPa channel estimate	350 kPa; 116.67 W	Total-loop head and pump allocation; channel and minor losses require separate calculation.
Joint fatigue	9289.7 cycles at $\Delta\epsilon_p = 0.00223$	$K = 4.85 \times 10^8$ N/m, preload 42.5 MPa; 44,820 cycles	Required strain range 0.0983786284% under the selected fatigue law.
Barrier defects / PRF	Intrinsic 2850, cap 8.5×10^{-4} ; PRF 832.93	Intrinsic 25000, pore factor 1.42, cap 6.49×10^{-4} ; PRF 1091	Historical model inputs; Eq. (665) defines the effective barrier factor.
Transport / leakage	Material-specific legacy trap data	$N_t = 1.20 \times 10^{26}$ m ^{−3} , $V_H^* = 1.70 \times 10^{-6}$ m ³ /mol, $Q^* = 12.50$ kJ/mol; $J_D = 3.15 \times 10^{-8}$ mol/(m ² s)	Reported substrate-flux reference with separate transport fields and boundary conditions.
SPA / DC preservation	Historical reduced v2.0 model	Symbolic balanced SPA, 15 to 4 states	Symbolic Schur construction for a specified balanced parent.
SIS thresholds / latency	100 W/s, 15 GHz/50 μ s, dump < 500 ns	15/45/120/85/110 ns chain; total 375 ns	Arithmetic margin 125 ns; sensing window, thermal bandwidth and energy handling remain separately qualified.
Calorimetry	2911.40 W; $u_c = 5.701706$ W	New water inputs, 8.42 K, 12.30 W envelope; $u_c = 5.6956$ W	Reference operating point 2911.40 W, 8.402823 K, $u_c = 5.691919$ W.
Excess-heat statistic	Hypothetical 1000 W gives $Z = 174.7156$	Revised calorimeter allocation	Conditional $Z = 175.013715$ at the reference point with 0.50 W input uncertainty.
Shield transport	35/120/10 mm shield; 0.380 μ Sv/h target	90/10 wt% W–Cu, 5 wt% B–HDPE, SS316L; 0.312 μ Sv/h	Reported dose subject to source, transport, and dose-conversion qualification.
Procurement	11-line \$392,083.00 baseline	No replacement cost schedule	Historical priced baseline; revised hardware requires separate quotations.

correcting the count statistics for the instrument’s dead-time model, an intensity normalization is

$$I_{i,\text{corr}} = \frac{N_i - d_i t_{\text{live}}}{t_{\text{live}} \Phi_{\text{mon}} \epsilon_i \Delta\Omega_i}, \quad (780)$$

where d_i is dark-count rate, Φ_{mon} is incident flux in a fixed monitor normalization, ϵ_i is channel efficiency, and $\Delta\Omega_i$ is the pixel solid angle. Glassy-carbon or polymer scattering measurements may support flat-field calibration only after accounting for their angular scattering profile, background, and polarization; they are not assumed to be perfectly isotropic sources. Separate absorption and texture corrections from detector efficiency.

During thermal cycling, track an internal reference with a characterized temperature-dependent lattice spacing, for example a compatible tungsten reference. Its thermal expansion must not be removed by forcing its peak to remain at the room-temperature angle. Use its predicted peak motion and an

independent temperature measurement to distinguish specimen displacement from detector drift. For a sample reflection,

$$\epsilon_{hkl}(T, t) = \frac{d_{hkl}(T, t) - d_{hkl}^0(T)}{d_{hkl}^0(T)}, \quad (781)$$

where $d_{hkl}^0(T)$ is the stress-free spacing for the same phase and composition. Phase fractions, elastic strains, and line broadening require a stated refinement model with texture, peak overlap, and calibration covariance. Beam-on/beam-off controls quantify beam heating or irradiation changes, and diffraction frames are synchronized with gas release, load, and temperature.

a) *EJ-301 pulse-shape calibration*: Independently calibrate the radiation channel with characterized gamma-only and neutron exposures over the declared energy and count-rate range. Record baseline, gain, trigger efficiency, live time, and pile-up cuts; apply the fixed 24 ns and 240 ns integration

TABLE LXXXIX
COMPLETE V2.0 SOURCE-TO-MANUSCRIPT INTEGRATION REGISTER

Source pages	Retained v2.0 content	Governing integration / numerical disposition
1–8	Bloch–Floquet tensor, quiver expansion, screening, Gamow quadrature, overlaps, coherent domains	Section I; (85)–(86); distinguish field and film volumes.
8–11	Exact BOP, segmented TEG losses, operating export and rejection table	Section II; (243), Table XVIII; finite-temperature sink feasibility is separate.
11–16	Frame mechanics, GD&T, CF feedthroughs, isolation, grating, polarization and aperture	Section III; retained tolerance/feedthrough schedules, Table XXVIII, (348)–(349).
16–21	Five-layer HiPIMS stack, Voce/VCCCT, fatigue, transport, HE and coating	Section IV; Tables XLVII and XLVIII; (469), (663).
21–25	Covariance, excess heat, transitional hydraulics, air exchanger, five-step Nichrome calibration	Section V; Table LVI and Eq. (519).
26–28	Sector mass resolution, St707 enrichment, micro-dosing, helium inventory	Retained mass-assay and ash equations; 0.0001601 u resolution, 1.45×10^5 enrichment and 3.7562×10^{-5} mol/day/kW source benchmark.
28–30	EJ-301 gates/PSD, binomial trials, HPGe windows, three-layer shielding	Section VI-I; 24/240 ns gates, FOM 2.15, 2,995,731 zero-failure trials at 95% confidence; dose remains source- and geometry-dependent.
30–33	DM-01/02, 2003 truth table, surge/dump protection, TR-01–TR-06	Retained control and identification sections; response times and trip thresholds retain signal and hardware boundaries.
34–36	Two relief cases, classified locations, fast isolation, FM-01–FM-09	Retained process-safety and FMECA schedules; certified capacity, local plume and downstream inventory are not inferred from average dilution.
37–42	Fifteen-state coefficients, four-state matrices, Schur/port maps	Section X; full-model coefficients and historical reduced realization in Eqs. (718)–(721).
42–44	Forty-eight-hour blinded H ₂ blank, WT5000 RFI subtraction, 11-line BOM	Section XI; 0.00 $\pm$ 5.72 W blank reference and itemized \$392,083.00 baseline.
44–46	Synthesis and source references	This section and Appendix D; source record and bibliography.

windows in Eq. (569). Select the discrimination boundary on calibration data, then evaluate a separate exposure without retuning the cut. Report neutron acceptance and gamma leakage together, by energy bin, along with the figure of merit in Eq. (581). The $R_{\gamma,95} \geq 10^6$ criterion requires the independent zero-leakage trial count and confidence construction in Eq. (571); a fitted separation of the pulse populations alone is insufficient. A diffraction position calibration cannot validate this rejection ratio.

E. TLP Double-Shear Creep and Phase Evolution

Test the actual micro-alloyed TLP bondline and graded Ni/Cu interlayer alongside separate Cu–Sn and Ag–Sn comparison joints. The comparison program uses OFHC Cu adherends and Ag-metallized silicon where compatible with the fixture, with either a 20 μ m Sn interlayer or porous Ag infiltrated by Sn–3.5Ag. Porous-metal infiltration is a distinct bonding route, not evidence that all Sn-containing joints solidify on the same schedule [136]. Symmetric double-shear specimens are the primary creep geometry; lap joints provide a separate mixed-mode comparison with bending explicitly characterized.

The proposed processing matrix spans 250–350 °C with an initial 10 min hold, followed by aging at 300 °C for intervals up to 500 h. Use a qualified reducing atmosphere, such as 95% N₂/5% H₂, or a qualified formic-acid process. A vacuum base-pressure target of 10^{-4} Pa describes a separate evacuated stage, not the pressure of an admitted forming-gas atmosphere. Record the actual gas pressure and thermal history. Determine completion of isothermal solidification using phase analysis,

cross-sectional microscopy, and residual-liquid or residual-Sn assessment; neither a fixed hold time nor aging temperature proves completion. Quantify bond thickness, bonded area, porosity, phase distribution, and interfacial reaction layers before mechanical testing.

Use a calibrated dead-weight frame and non-contact local extensometry at 150, 250, and 300 °C over nominal shear stresses of 5–40 MPa. The 350 °C service condition requires additional tests at that temperature; these lower-temperature exposures do not qualify it. Measure thermal gradients and stabilize the fixture before loading. For two shear planes,

$$\begin{aligned} F_1 + F_2 &= F, & \tau_j &= F_j/A_j, \\ A_1 = A_2 &= A, & F_1 = F_2 &\implies \tau_1 = \tau_2 = \frac{F}{2A}. \end{aligned} \quad (782)$$

The equal-load expression requires symmetric compliance and negligible bending. Otherwise determine load partition and local stress with the measured geometry and a validated mechanical solution. Dead weight fixes force, not true stress if bonded area evolves.

For each bondline j , subtract independently characterized fixture, adherend, and thermal-motion contributions from the optical displacement:

$$\begin{aligned} \delta_{j,\text{bond}} &= \delta_{j,\text{opt}} - \delta_{j,\text{fixture}} \\ &\quad - \delta_{j,\text{adherend}} - \delta_{j,\text{thermal}}, \\ \gamma_j(t) &= \delta_{j,\text{bond}}(t)/h_j(T), \\ \dot{\gamma}_{j,\text{cr}} &= \frac{d\gamma_j}{dt} - \dot{\gamma}_{j,\text{el}}. \end{aligned} \quad (783)$$

Here $h_j(T)$ is the measured bond thickness at temperature. Resolve the loading transient and elastic response before estimating creep rate over a predefined time window; report derivative uncertainty and do not impose a steady-state plateau where none exists. Interrupted tests and post-fracture sections document phase evolution, grain structure, interfacial voiding, and crack paths.

Compare measured phase fractions and concentration profiles with Eq. (382), and compare strain-rate histories with the vacancy-diffusion and dislocation relations in Eqs. (398) and (401). Independently constrained diffusion, vacancy, and microstructural inputs are retained when testing predictions on withheld specimens. A stress-rate slope alone cannot uniquely distinguish dislocation climb from grain-boundary sliding. The 52,400-cycle calculation in Eq. (276) remains a separate strain-life prediction until the corresponding thermal-cycle experiments are completed.

F. Uncertainty, Acceptance, and Data Retention

The measurement vector includes pressure ratios, threshold loads, isotope inventories, diffraction strains, phase fractions, and creep rates. Propagate calibration, blank, geometry, temperature, and timing covariances through the measurement functions using Eq. (525). Shared references create correlations across specimens and time points; threshold brackets and non-smooth decisions require interval or distributional treatment rather than an unqualified linear Jacobian. Fix model-comparison tolerances and detection criteria before unblinding the test data.

Retain raw observations, calibration certificates, adopted standard editions, specimen histories, background runs, uncertainty budgets, excluded observations with reasons, and complete loading and exposure records. Acceptance is specific to the tested material, process, temperature, gas environment, and statistical population. Reactor power, nuclear reaction yields, and functional-safety performance require independent system-level qualification in addition to coupon tests and detector calibration.

APPENDIX A

FLOQUET INVERSION AND DISPLACED-STATE PROOFS

a) *Composite inverse and static-source projection:* For a retarded inverse kernel, screening is the convolution

$$W(\mathbf{r}, t) = \int dt' \int d^3r' \epsilon^{-1}(\mathbf{r}, \mathbf{r}'; t, t') V_{\text{ext}}(\mathbf{r}', t'). \quad (784)$$

Space/time Fourier transformation converts convolution into multiplication in the composite reciprocal-vector/Floquet indices. Writing $\epsilon = I - M$ with $M = v\chi$, the geometric-series identity is

$$(I - M)^{-1} = \sum_{k=0}^{\infty} M^k, \quad \rho(M) < 1. \quad (785)$$

The condition is on the spectral radius of the truncated operator. An invertible dielectric matrix need not satisfy it, particularly near a resonance; direct factorization and residual checks apply when the Neumann series does not converge. A static source occupies only the incoming zero harmonic. Reconstruction of its outgoing harmonics yields Eq. (31), and the time average selects its outgoing zero harmonic.

b) *Gaussian displaced-thermal overlap:* For $\bar{n}_{\text{ph}} > 0$, the thermal state has the coherent-state representation

$$\rho_{\text{th}} = \frac{1}{\pi \bar{n}_{\text{ph}}} \int d^2\gamma e^{-|\gamma|^2/\bar{n}_{\text{ph}}} |\gamma\rangle\langle\gamma|. \quad (786)$$

Displacement changes $|\gamma\rangle$ into $|\gamma + \alpha_{-}\rangle$ up to a phase that cancels in the projector. Since $|\langle 0|\gamma + \alpha_{-}\rangle|^2 = e^{-|\gamma + \alpha_{-}|^2}$,

$$P_{\text{ov}} = \frac{1}{\pi \bar{n}_{\text{ph}}} \int d^2\gamma e^{-|\gamma|^2/\bar{n}_{\text{ph}} - |\gamma + \alpha_{-}|^2},$$

$$\frac{|\gamma|^2}{\bar{n}_{\text{ph}}} + |\gamma + \alpha_{-}|^2 = \frac{1 + \bar{n}_{\text{ph}}}{\bar{n}_{\text{ph}}} \left| \gamma + \frac{\bar{n}_{\text{ph}} \alpha_{-}}{1 + \bar{n}_{\text{ph}}} \right|^2 + \frac{|\alpha_{-}|^2}{1 + \bar{n}_{\text{ph}}}. \quad (787)$$

The two real Gaussian integrals give Eq. (49); the continuous zero-temperature limit is $e^{-|\alpha_{-}|^2}$. This proof is restricted to the same oscillator vacuum. It does not calculate the overlap with an independently compressed nuclear wavefunction.

For completeness, the exponential Laguerre generating identity gives

$$e^{-S} \sum_{k=0}^{\infty} \frac{S^k}{k!} L_k(-z) = I_0(2\sqrt{Sz}), \quad S, z \geq 0. \quad (788)$$

Taking $z = |\alpha_{-}|^2/[\bar{n}_{\text{ph}}(1 + \bar{n}_{\text{ph}})]$ yields the value above unity in Eq. (51).

A. Native Arithmetic and Domain Acceptance Benchmarks

The native Rust scalar calculation in Eq. (46) uses exactly promoted binary64 inputs. A rational-arithmetic reference is therefore obtained by writing the two binary64 values as exact integer ratios before evaluating

$$s_{F,\text{rat}} = \frac{u_{64}^2}{u_{64}^2 - v_{64}^2}. \quad (789)$$

The reference and the 512-bit calculation agree at the displayed precision, $s_F = 3.67250764167145629240511891246\dots$. Exact decimal inputs instead define a different reference:

$$s_{F,\text{dec}} = \frac{35000^2}{35000^2 - 29857^2} \quad (790)$$

$$= 3.67250764167145674086843941099\dots$$

The native benchmark uses the exactly promoted binary64 reference $s_{F,\text{rat}}$; $s_{F,\text{dec}}$ uses exact decimal inputs. A 512-bit significand has approximately $512 \log_{10} 2 = 154.13$ decimal digits of arithmetic resolution. Finite inputs, nonzero u_{64} , and a nonzero scalar denominator are required. Matrix conditioning and reciprocal-space convergence have separate acceptance criteria.

Table XC records the arithmetic, solver, material, and closed-loop power acceptance values. The 3093.4400 W thermal input includes 2911.40 W of fusion heat and 182.04 W of absorbed optical heat; Table XI defines the complete component ledger.

Table LXXXII records the verified Floquet screening and phonon benchmark. Its two-channel branching fraction, scalar precision difference δ_F , and spectral metrics in Table XLII retain their respective definitions. Precision comparisons use lossless decimal exchange of the 512-bit reference result.

TABLE XC
NATIVE NUMERICAL KERNEL AND DOMAIN ACCEPTANCE BENCHMARKS

Calculation	Numerical value	Acceptance scope
Scalar Floquet kernel	$s_F \simeq 3.67250764167$; $\delta_F \simeq 3.56655 \times 10^{-16}$	$\delta_{F,\text{rel}} \simeq 9.71148 \times 10^{-17} < 2^{-53}$; same binary64 inputs at both precisions
Floquet screening solver	$N_G = 625$; $N_T = 25$; $N = 15,625$	Residual sequence $10^{-4}, 10^{-6}, 10^{-8}$; finite values; zero singularity warnings
Gamma rejection	$R_{\gamma,95} \simeq 1000000.0748773175 \simeq 1.000000 \times 10^6$	$R_{\gamma,95} \geq 10^6$ at 95% confidence, conditional on 2,995,731 independent zero-leakage trials
Domain 1: fatigue	$\varepsilon_{\text{tot}} = \Delta\varepsilon_{\text{tot}}/2 \simeq 0.002137728$; $N_f = 52400$ cycles	Strain-life arithmetic and $N_f \geq 44820$ cycles for the 293.15–550.0 K bondline model
Domain 2: beat	$f_{\text{beat}} = 8.3280639958728$ THz; $\omega_{\text{beat}} \simeq 5.232677 \times 10^{13}$ rad/s	Wavelength-derived difference frequency and $\omega_{\text{beat}} = 2\pi f_{\text{beat}}$
Domain 2: screening	$U_{\text{benchmark}} = 349.5000$ eV; $\Xi = 1.008526466381$; $U_{\text{eff}} = 352.4800$ eV	Displayed residual 0.0000 eV; $V_{\text{driven}} = -301.05286$ eV at $r_d = 0.28$ Å
Domain 3: closed-loop power	$P_{\text{thermal}} = 3093.4400$ W; $P_{\text{TEG}} = 1045.5800$ W; $P_{\text{net}} = 555.03$ W	$P_{\text{drive,net}} = 368.45$ W and $P_{\text{aux}} = 122.10$ W close the electrical balance

APPENDIX B NUMERICAL ALGORITHMS, PRECISION, AND DATA EXCHANGE

The numerical descriptions of complex LU, three-state quantum dynamics, and balance-of-plant evaluation provide the algorithmic scope summarized in Table LXXVI [95].

A. Partial-Pivoted LU and Precision Boundary

The complex linear solver stores A_{ij} in row-major order at offset $in + j$ and performs partial-pivoted LU in supplied working storage. The pivot sequence represents the final row permutation, giving

$$PA = LU, \quad Ly = Pb, \quad Ux = y. \quad (791)$$

Matrix inversion repeats the triangular solution with unit right-hand sides using two supplied work vectors. Decomposition, solution, and inversion require no explicit heap allocation within their inner calculations. The singularity guard compares the squared complex pivot magnitude with 10^{-32} , equivalent to a magnitude threshold 10^{-16} for finite values. Matrix conditioning requires a scale-aware assessment. Dimension multiplication, permutation indices, finite values, and numerical error need validation in addition to slice lengths. A suitable dimensionless solve residual is

$$\eta_{LU} = \frac{\|Ax - b\|_\infty}{\|A\|_\infty \|x\|_\infty + \|b\|_\infty}, \quad (792)$$

with a defined zero-denominator case. An energy residual in eV additionally requires the dielectric assembly, integration weights, and potential extraction map.

The linear-system input and output comprise separate real and imaginary binary64 arrays for A , b , and x [96]. Each solve allocates a complex matrix, right-hand side, pivot vector, and solution. The calculation distinguishes invalid input storage, decomposition failure, solution failure, and ordinary completion. Because dimensions are not carried with the arrays, valid storage for n^2 matrix entries and n right-hand-side/output entries, consistent ownership and alignment, and checked size arithmetic must be established separately.

An alternative data-exchange scheme retains LU and pivot storage after initialization while allocating binary64 component,

working, and output vectors. The 512-bit specification requires lossless multiprecision inputs and outputs, selected rounding, allocation profiling after initialization, and residual and error estimates for the same matrix and potential integral [31].

B. Three-State RK4 Master Equation

The reduced quantum calculation propagates a row-major array of nine complex entries. Its frequency-valued Hamiltonian $h(t) = H(t)/\hbar$ is

$$h(t) = \begin{bmatrix} 0 & \omega_0 \cos(\omega_L t) & 0 \\ \omega_0 \cos(\omega_L t) & \Delta & g_{\text{nuc}} \\ 0 & g_{\text{nuc}} & \Delta_{\text{nuc}} \end{bmatrix}, \quad (793)$$

$$L_1 = |0\rangle\langle 1|, \quad L_2 = |1\rangle\langle 2|,$$

$$\dot{\rho} = -i[h(t), \rho] + \gamma_1 \mathcal{D}[L_1]\rho + \gamma_2 \mathcal{D}[L_2]\rho.$$

The entries of h have angular-frequency units, t and Δt are in seconds, and the nonnegative γ_j have units s^{-1} . The two explicit decay dissipators reproduce the Lindblad definition in Section II-K; the jump operators do not contain a second factor of the rate. RK4 evaluates four stages with the stage-specific time and density matrix. The time evolution initializes $\rho = |0\rangle\langle 0|$, advances the specified number of steps, then evaluates the BOP balance independently. Fixed arrays avoid explicit heap allocation in the displayed numerical loop. Step refinement and checks of trace, Hermiticity, and the minimum density-matrix eigenvalue remain necessary.

C. BOP Mapping and Electrical Completeness

The balance-of-plant calculation uses volumetric flow in m^3/s , pressure drop in Pa, lattice-loop heat in W, and dimensionless pump efficiency and COP. Its algebraic mapping is

$$P_{\text{pump}} = \dot{V} \Delta P / \eta_{\text{pump}},$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{lattice,loop}} + P_{\text{pump}}, \quad (794)$$

$$P_{\text{compressor}} = \dot{Q}_{\text{total}} / \text{COP}.$$

This is the all-pump-losses-to-fluid assumption in Eq. (208). With $\dot{V} = 2.0833 \times 10^{-4} m^3/s$, $\Delta P = 85000 \text{ Pa}$, $\eta_{\text{pump}} = 0.65$, $\dot{Q}_{\text{lattice,loop}} = 147.6600 \text{ W}$, and COP 3.0, its arithmetic reproduces $P_{\text{pump}} = 27.2432 \text{ W}$, $\dot{Q}_{\text{total}} = 174.9032 \text{ W}$, and $P_{\text{compressor}} = 58.3011 \text{ W}$ at the stated display precision.

The coupled output comprises nine real and nine imaginary density-matrix entries, pump input, total-loop heat, and compressor input. Input validation requires positive finite pump efficiency, COP, and time step; nonnegative decay rates; adequate storage capacity; and acceptable density-matrix diagnostics.

The electrical ledger includes separate pump and compressor inputs, optical absorption, thermoelectric conversion, fans, and auxiliaries. A coupled thermal solver must derive lattice heat from state-to-energy transfer and connect the full TEG rejection boundary.

APPENDIX C CONSTITUTIVE PARAMETERS AND COMPARISON CALCULATIONS

The dielectric entries, optical constants, transport coefficients, and fixed-shift Gamow comparison retain their own constitutive

TABLE XCI
 STATIC INVERSE-DIELECTRIC REFERENCE SAMPLES AT
 $|\mathbf{q} + \mathbf{G}| = 2.23 \text{ \AA}^{-1}$

G	G'	$\text{Re } \epsilon^{-1}$	$\text{Im } \epsilon^{-1}$	Contribution (eV)
(0, 0, 0)	(0, 0, 0)	-6.8082	0.0000	-350.00
(1, 1, 1)	(0, 0, 0)	0.1425	-0.0312	7.32
(2, 0, 0)	(0, 0, 0)	-0.0891	0.0104	-4.58
(2, 2, 0)	(1, 1, 1)	0.0314	0.0055	1.61
(3, 1, 1)	(2, 0, 0)	-0.0128	-0.0018	-0.66

TABLE XCII
 GAMOW ENHANCEMENT COMPARISON FOR THE ASSIGNED 350 eV SHIFT

E (eV)	$\ln \mathcal{E}_G$	Calculated ratio
10	261.641350	4.25985×10^{113}
100	52.482084	6.20411×10^{22}

definitions. They are distinct from the coupling, composition-screening, interface-flux, and Hankel-spectrum operating points.

TABLE XCIII
EARLIER OPTION B OPTICAL CONSTANTS AND ISOTROPIC RELATIVE-PERMITTIVITY ENTRIES

Layer	n_{785}	k_{785}	$\epsilon_{785} = (n + ik)^2$	$n_{802.5}$	$k_{802.5}$	$\epsilon_{802.5}$
Silica prism	1.4536	0	2.112953	1.4531	0	2.111500
Pd–Ir	1.8210	5.2140	$-23.869755 + 18.989388i$	1.8840	5.3420	$-24.987508 + 20.128656i$
Ti	2.7210	3.7840	$-6.914815 + 20.592528i$	2.7650	3.8420	$-7.115739 + 21.246260i$
CVD diamond	2.3980	0	5.750404	2.3965	0	5.743212

TABLE XCIV
EARLIER COMPOSITION-SPECIFIC MCNABB–FOSTER PARAMETERS AT 550 K

Layer	D_{eff} (m ² /s)	N_j (m ⁻³)	$E_{t,j}$ (eV)	k_j (m ³ /s)	p_j (s ⁻¹)	V_H^* (m ³ /mol)	Q^* (kJ/mol)
Pd _{0.95} Ir _{0.05}	3.42×10^{-9}	1.20×10^{26}	0.22	1.45×10^{-20}	1.82×10^3	1.70×10^{-6}	18.4
Ti	8.15×10^{-12}	4.50×10^{27}	0.65	3.10×10^{-21}	2.14×10^{-1}	1.40×10^{-6}	24.1
ϵ -Cu ₃ Sn	1.12×10^{-13}	8.80×10^{26}	0.48	8.40×10^{-22}	3.65×10^1	0.95×10^{-6}	12.8

APPENDIX D

COMPLETE SUPPLEMENTARY v2.0 SOURCE RECORD

This historical appendix identifies the previously integrated 46-page v2.0 source. If its PDF is present, the facsimile is included in its original form; Table [LXXXV](#) governs current selection. **Supplementary record.** The historical facsimile `SHBT Reactor supplementary.pdf` is not part of this document distribution. The typed integration is retained in the manuscript.

APPENDIX E

COMPLETE SUPPLEMENTARY v3.0 SOURCE RECORD

This historical appendix identifies the previously integrated 26-page v3.0 source. If present, its original PDF is included as non-governing provenance. **Supplementary record.** The historical v3.0 facsimile is not part of this document distribution. The typed integration is retained in the manuscript.

APPENDIX F

COMPLETE SUPPLIED V4.1 MASTER RESOLUTION SOURCE

This appendix identifies the 28-page source SHBT Reactor Master Resolution.pdf and the text-source identifier supplementary-4.2.txt. The PDF is included when available. Table LXXXV governs parameter selection.

Supplementary record. The master-resolution facsimile is not part of this document distribution. The typed integration is retained in the manuscript.

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